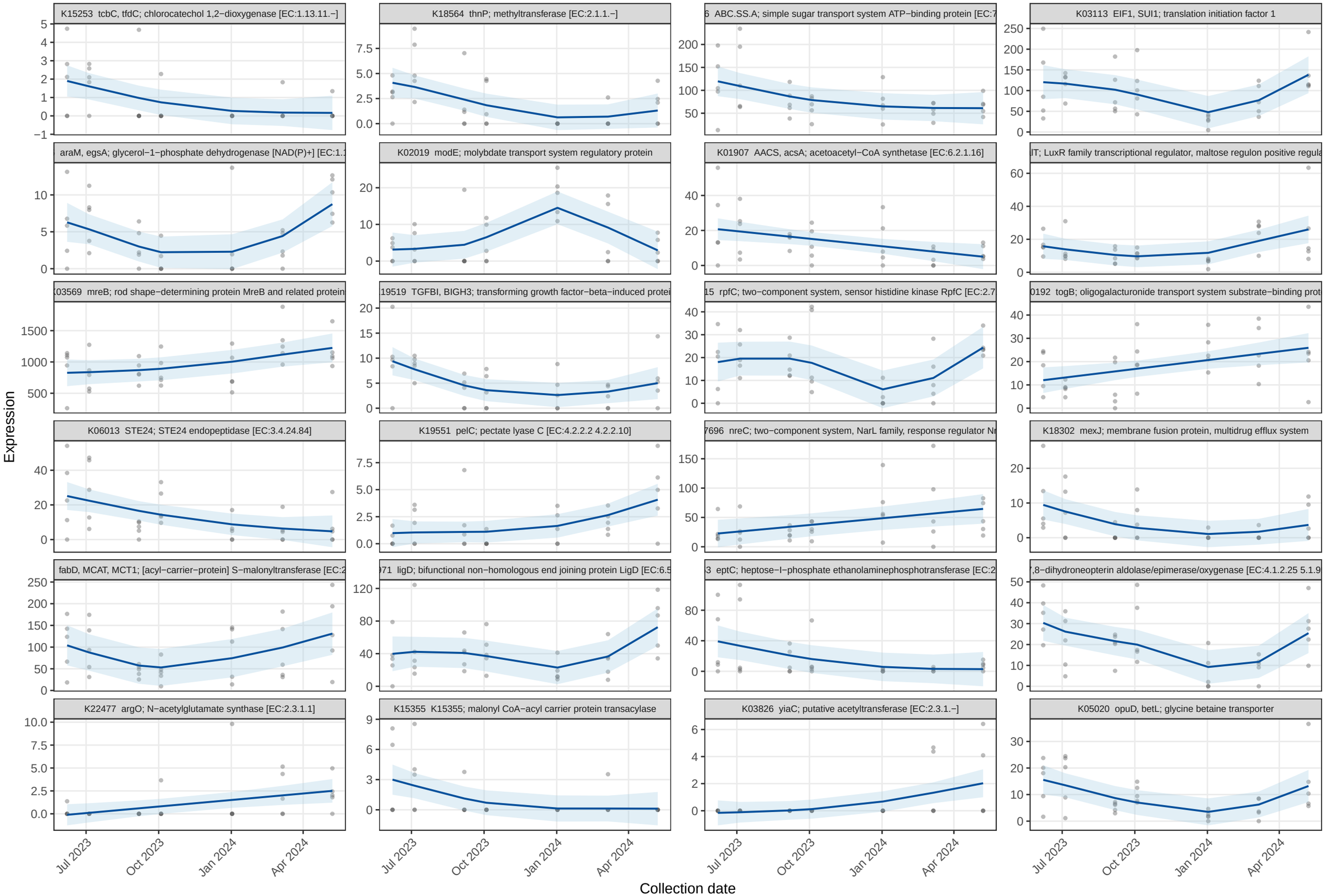
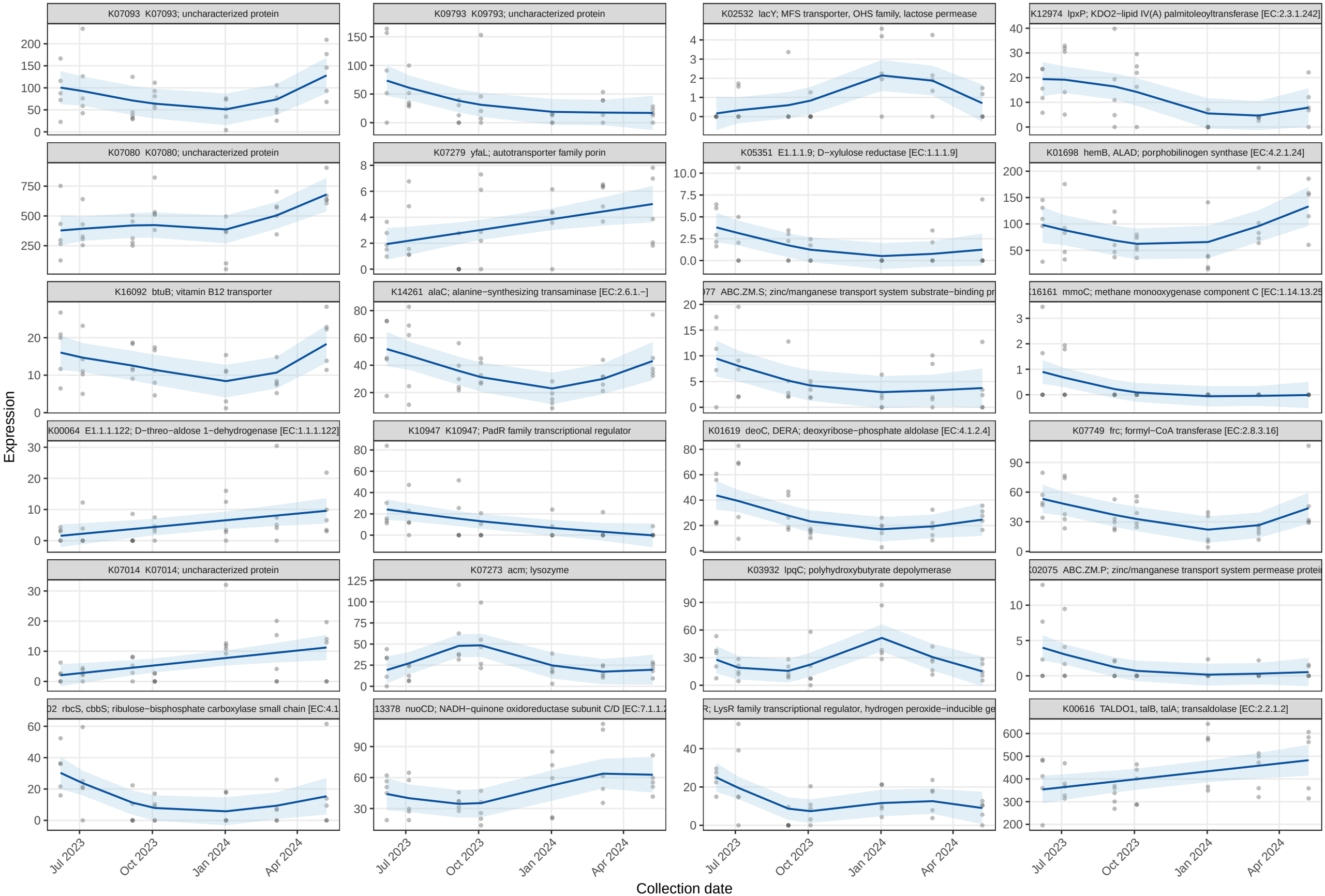
brite_D filtered temporal GAMs: page 1 of 303

All remaining BRITE classes after exclusions



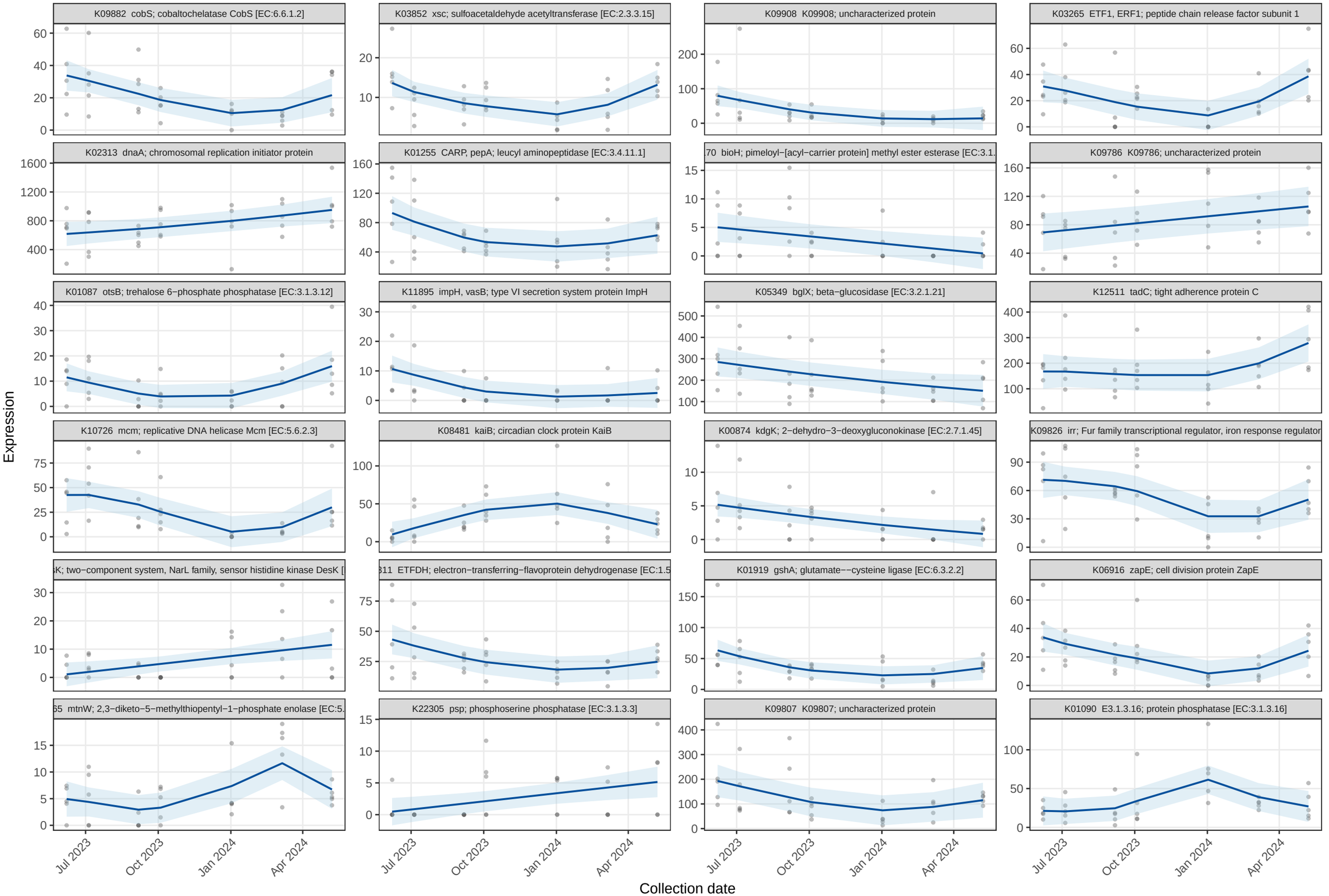
brite_D filtered temporal GAMs: page 2 of 303

All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 3 of 303

All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 4 of 303

All remaining BRITE classes after exclusions

idsA; geranylgeranyl diphosphate synthase, type I [EC:2.5.1.1 2.5.1.10

K20976 hptB; histidine phosphotransfer protein HptB

cmtAa; p−cumate 2,3−dioxygenase ferredoxin reductase component [EC

ndAa; anthranilate 1,2−dioxygenase ferredoxin reductase component [E

K09023 rutD; aminoacrylate hydrolase [EC:3.5.1.−]

561 ntcA; CRP/FNR family transcriptional regulator, global nitrogen regu

K18893 vcaM; ATP−binding cassette, subfamily B, multidrug efflux pump

K16080 mnoP; high affinity Mn2+ porin

03667 hslU; ATP−dependent HslUV protease ATP−binding subunit HslU

K11947 nidD; aldehyde dehydrogenase [EC:1.2.1.−]

K07403 nfeD; membrane−bound serine protease (ClpP class)

K10536 aguA; agmatine deiminase [EC:3.5.3.12]

Expression

K01857 pcaB; 3−carboxy−cis,cis−muconate cycloisomerase [EC:5.5.1.2

797 rpoBC; DNA−directed RNA polymerase subunit beta−beta' [EC:2.7

K06714 rocR; arginine utilization regulatory protein

06925 tsaE; tRNA threonylcarbamoyladenosine biosynthesis protein TsaE

K17893 AOX1, AOX2; ubiquinol oxidase [EC:1.10.3.11]

3; quinol−−−cytochrome c reductase cytochrome b subunit, bacillus type

712 tagE; poly(glycerol−phosphate) alpha−glucosyltransferase [EC:2.4.1

K01843 kamA; lysine 2,3−aminomutase [EC:5.4.3.2]

taA; S−adenosylmethionine−diacylglycerol 3−amino−3−carboxypropyl tr

K09471 puuB, ordL; gamma−glutamylputrescine oxidase [EC:1.4.3.−]

enylpropionate/trans−cinnamate dioxygenase ferredoxin reductase com

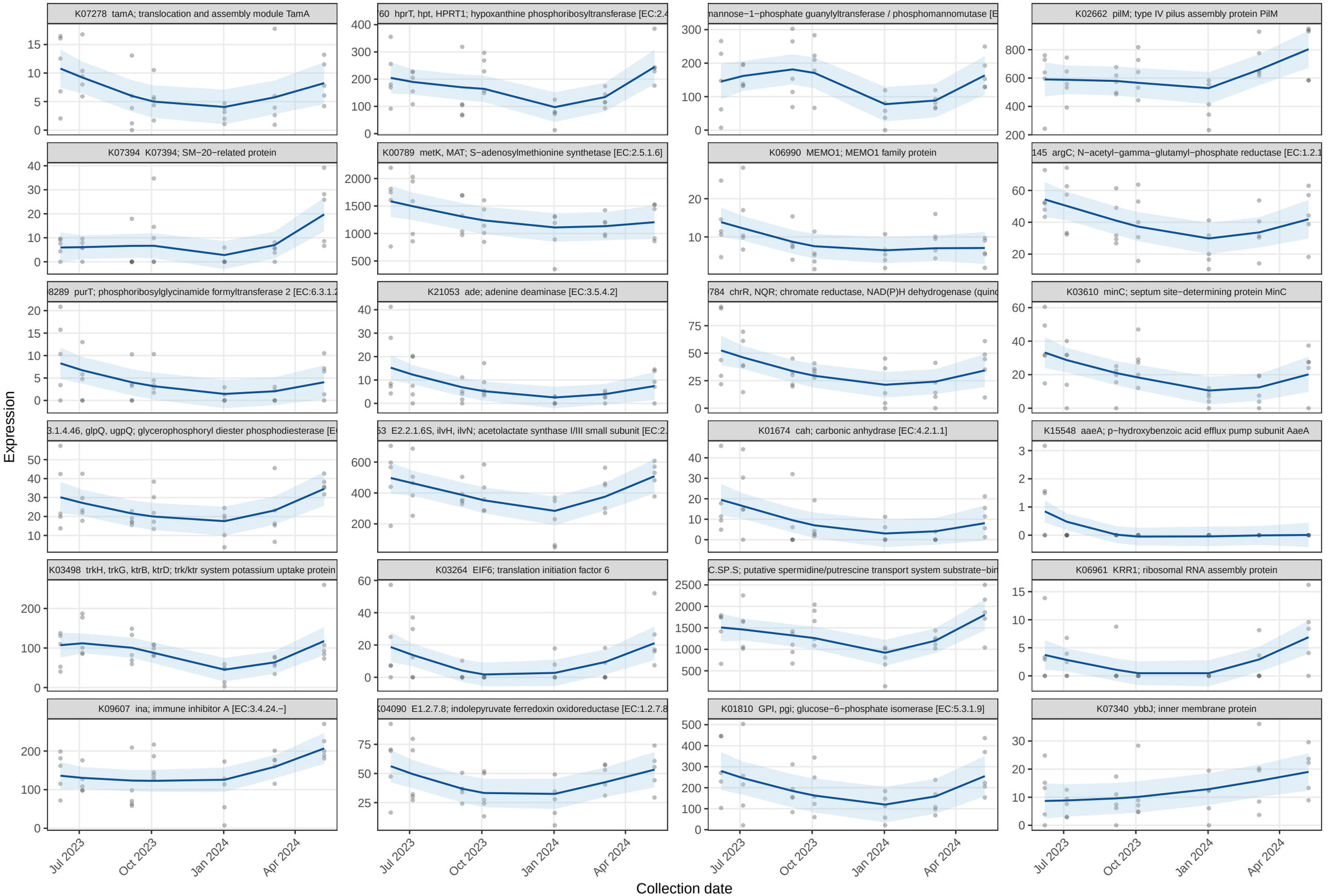
K02012 afuA, fbpA; iron(III) transport system substrate−binding protein

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All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

K07045 K07045; uncharacterized protein

pucR; PucR family transcriptional regulator, purine catabolism regulator

K03301 TC.AAA; ATP:ADP antiporter, AAA family

cynR; LysR family transcriptional regulator, cyn operon transcriptional

ator, regulatory protein of adaptative response / methylated−DNA−[prot

K06945 K06945; uncharacterized protein

K06199 crcB, FEX; fluoride exporter

K02192 bfd; bacterioferritin−associated ferredoxin

K16559 exoK; endo−1,3−1,4−beta−glycanase ExoK [EC:3.2.1.−]

13 argT; lysine/arginine/ornithine transport system substrate−binding p

K10014 hisJ; histidine transport system substrate−binding protein

K03406 mcp; methyl−accepting chemotaxis protein

Expression

K12516 bigA; putative surface−exposed virulence protein

K00362 nirB; nitrite reductase (NADH) large subunit [EC:1.7.1.15]

K20920 vpsM; polysaccharide biosynthesis protein VpsM

K00263 E1.4.1.9; leucine dehydrogenase [EC:1.4.1.9]

imidazole−4−carboxamide−1−(beta)−D−ribofuranosyl 5'−monophospha

K07761 5MMP; metalloendoproteinase 5−MMP [EC:3.4.24.−]

0 pfeR, pirR; two−component system, OmpR family, response regulato

K13722 trf2_3; tricorn protease interacting factor F2/3 [EC:3.4.11.−]

nsrAB; peptide methionine sulfoxide reductase msrA/msrB [EC:1.8.4.11

K09007 folE2; GTP cyclohydrolase IB [EC:3.5.4.16]

wo−component system, OmpR family, alkaline phosphatase synthesis r

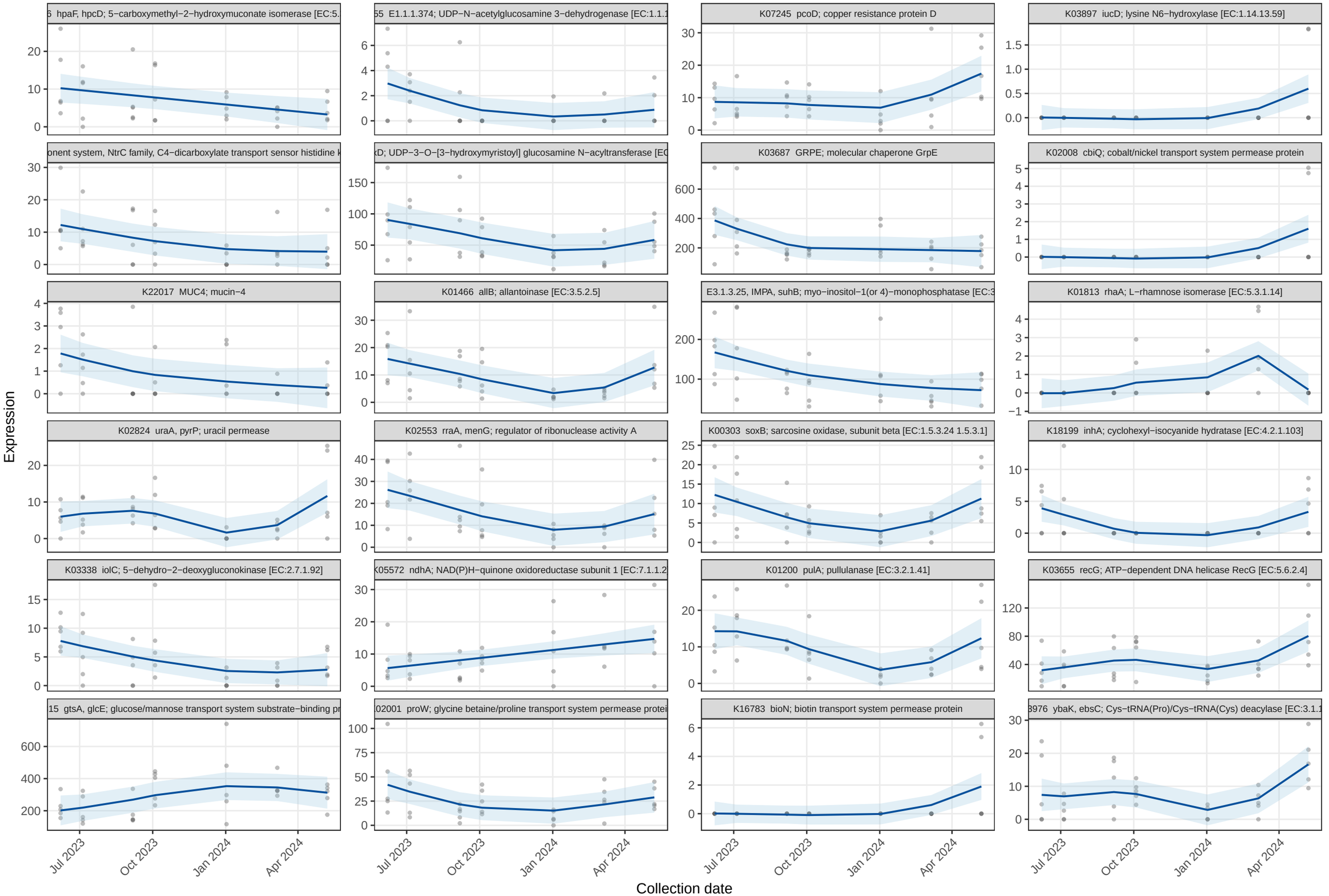
K01669 phr, PHR1; deoxyribodipyrimidine photo−lyase [EC:4.1.99.3]

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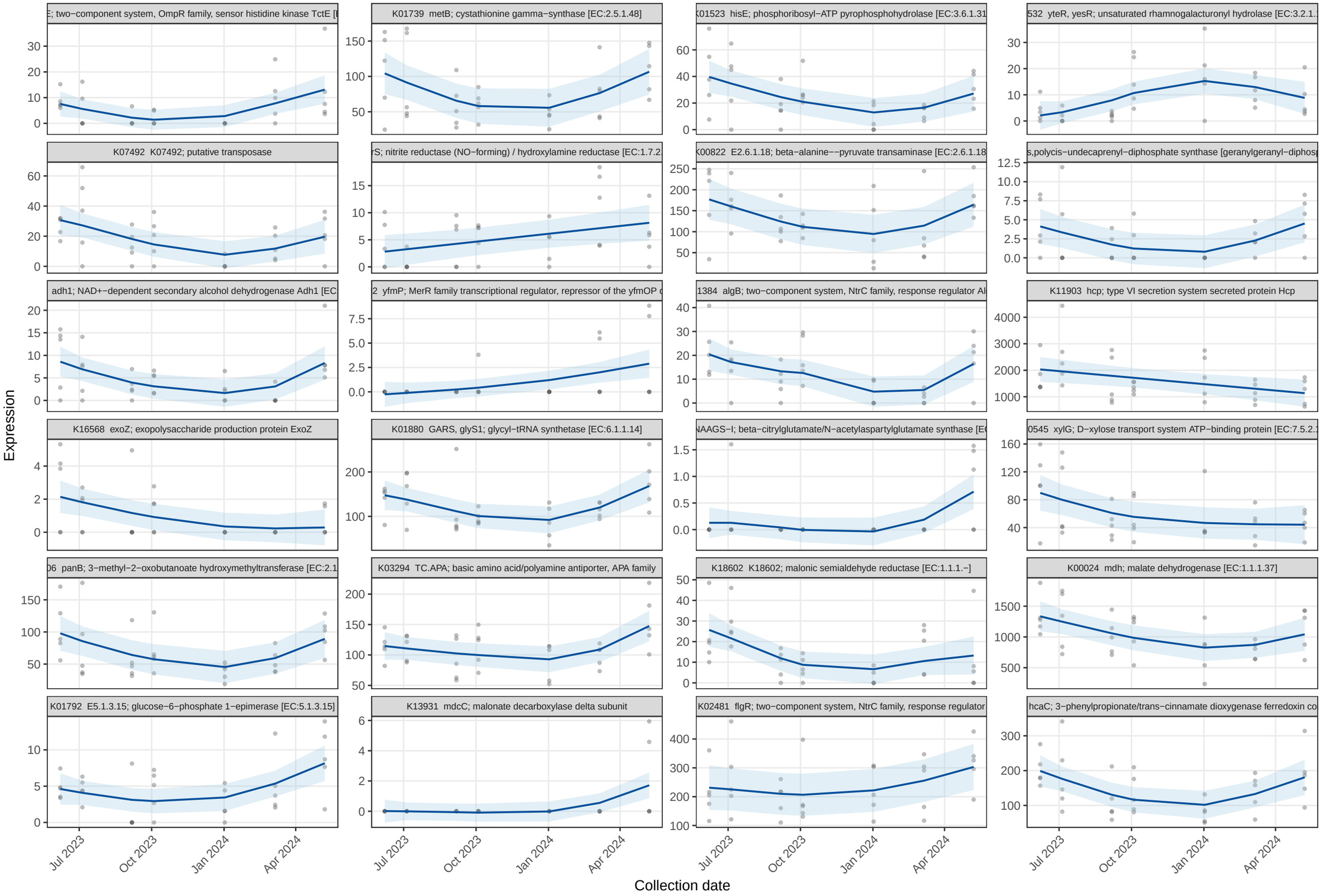
brite_D filtered temporal GAMs: page 7 of 303

All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 8 of 303

All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions

K00299 ssuE, msuE; FMN reductase [EC:1.5.1.38]

K05303 K05303; O−methyltransferase [EC:2.1.1.−]

K02279 cpaB, rcpC; pilus assembly protein CpaB

K01993 ABC−2.TX; HlyD family secretion protein

K18902 bpeF; multidrug efflux pump

K01887 RARS, argS; arginyl−tRNA synthetase [EC:6.1.1.19]

K03470 rnhB; ribonuclease HII [EC:3.1.26.4]

K02663 pilN; type IV pilus assembly protein PilN

K05582 ndhK; NAD(P)H−quinone oxidoreductase subunit K [EC:7.1.1.2

6 pdtaS; two−component system, sensor histidine kinase PdtaS [EC:2.

K03442 mscS; small conductance mechanosensitive channel

K08301 rng, cafA; ribonuclease G [EC:3.1.26.−]

K06910 PEBP, TFS1; phosphatidylethanolamine−binding protein

08 hcrA, hbaC; 4−hydroxybenzoyl−CoA reductase subunit alpha [EC:1.

K09902 K09902; uncharacterized protein

K01875 SARS, serS; seryl−tRNA synthetase [EC:6.1.1.11]

K21084 yegE; diguanylate cyclase [EC:2.7.7.65]

K03970 pspB; phage shock protein B

ibosyl−AMP cyclohydrolase / phosphoribosyl−ATP pyrophosphohydrola

K11737 cycA; D−serine/D−alanine/glycine transporter

K11928 putP; sodium/proline symporter

K06999 K06999; phospholipase/carboxylesterase

K03969 pspA; phage shock protein A

K10716 kch, trkA, mthK, pch; voltage−gated potassium channel

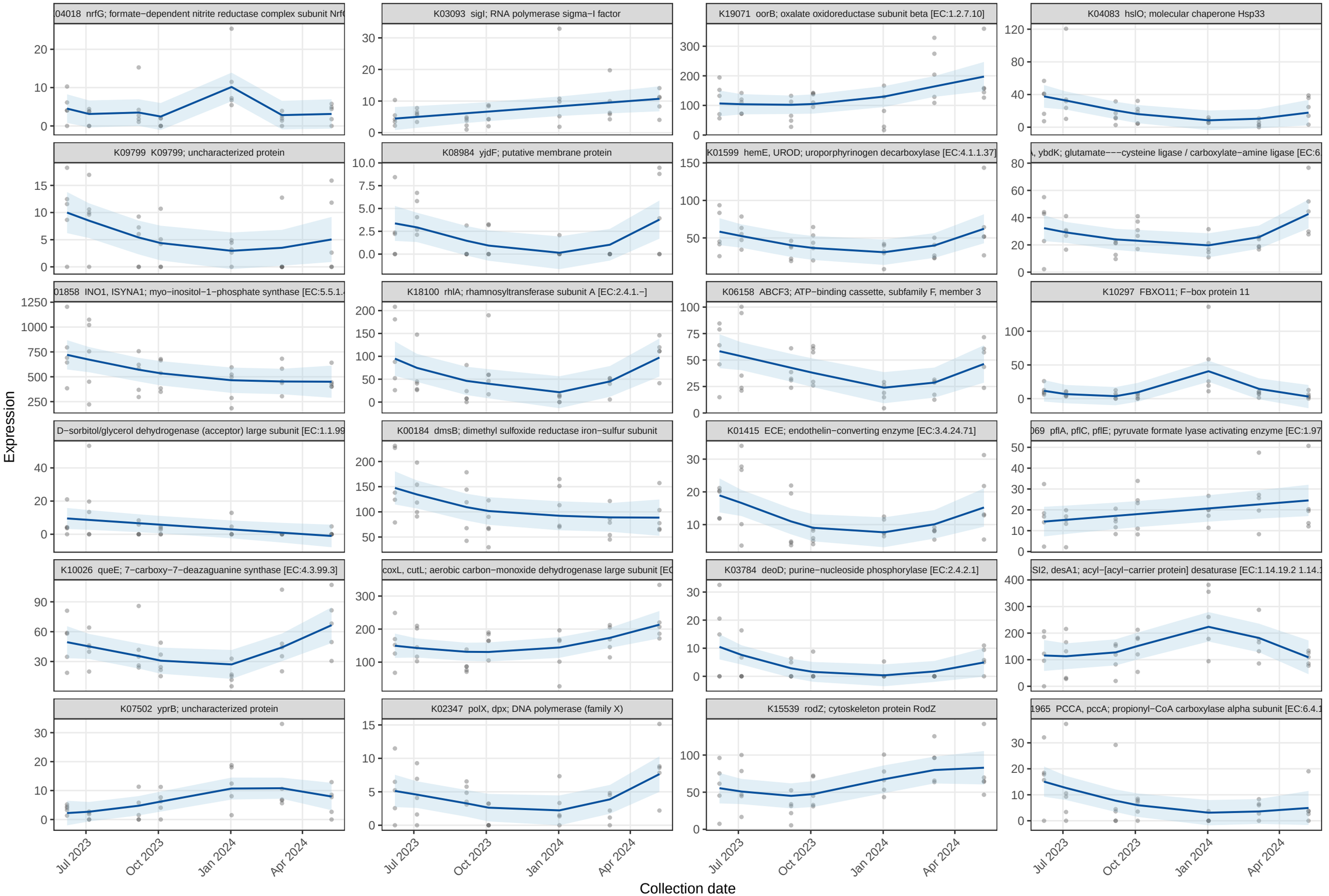
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Expression

Collection date

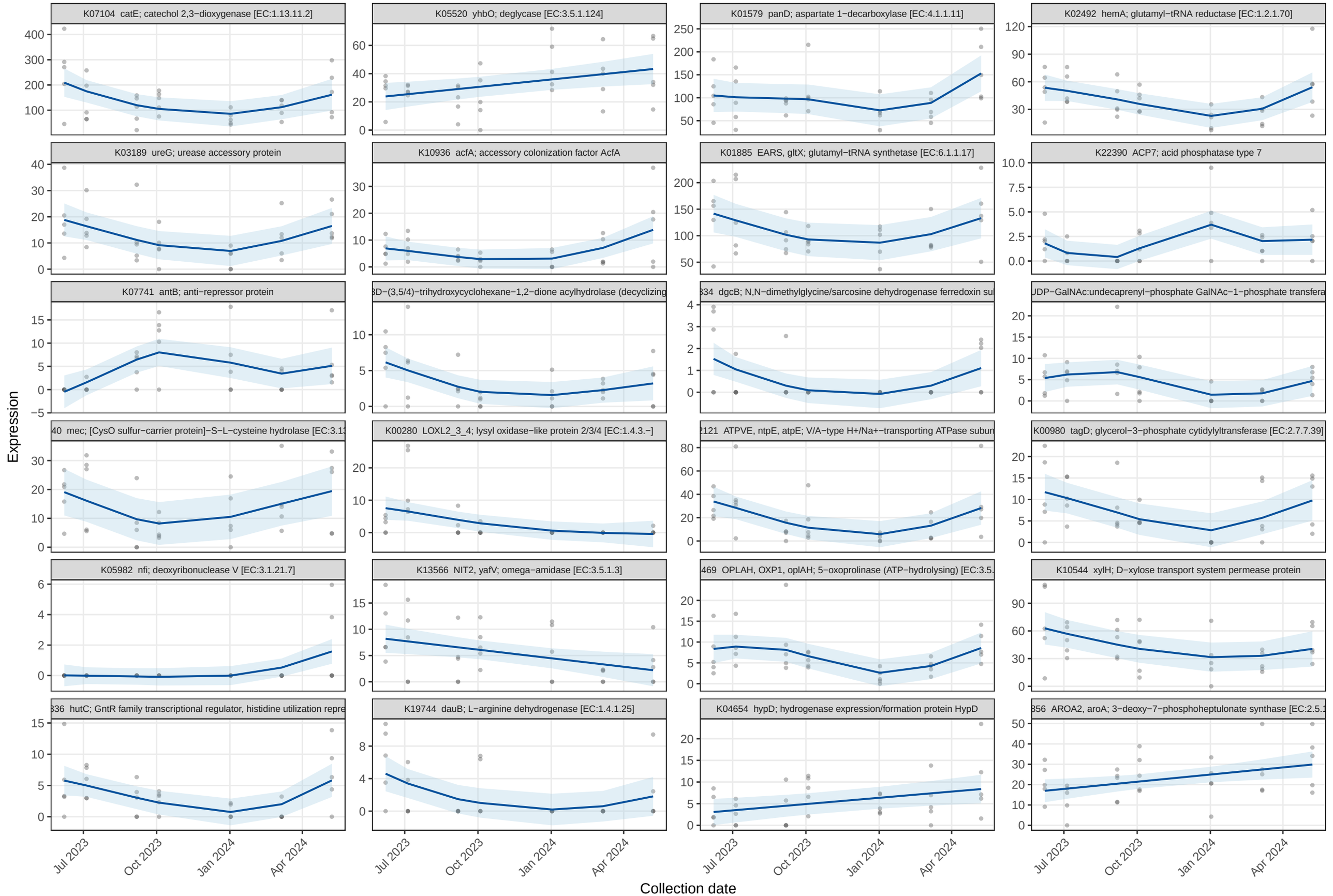
brite_D filtered temporal GAMs: page 10 of 303

All remaining BRITE classes after exclusions



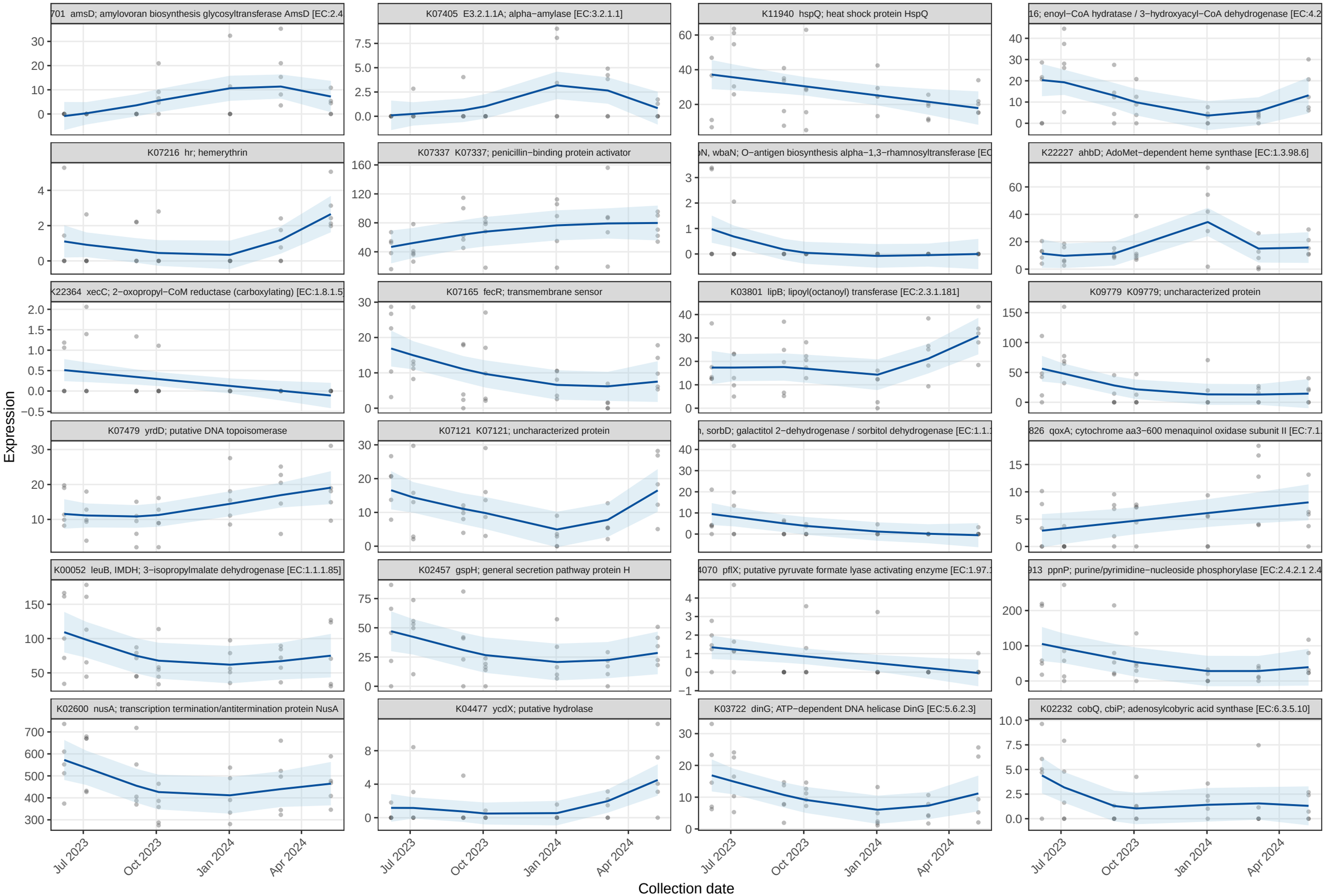
brite_D filtered temporal GAMs: page 11 of 303

All remaining BRITE classes after exclusions



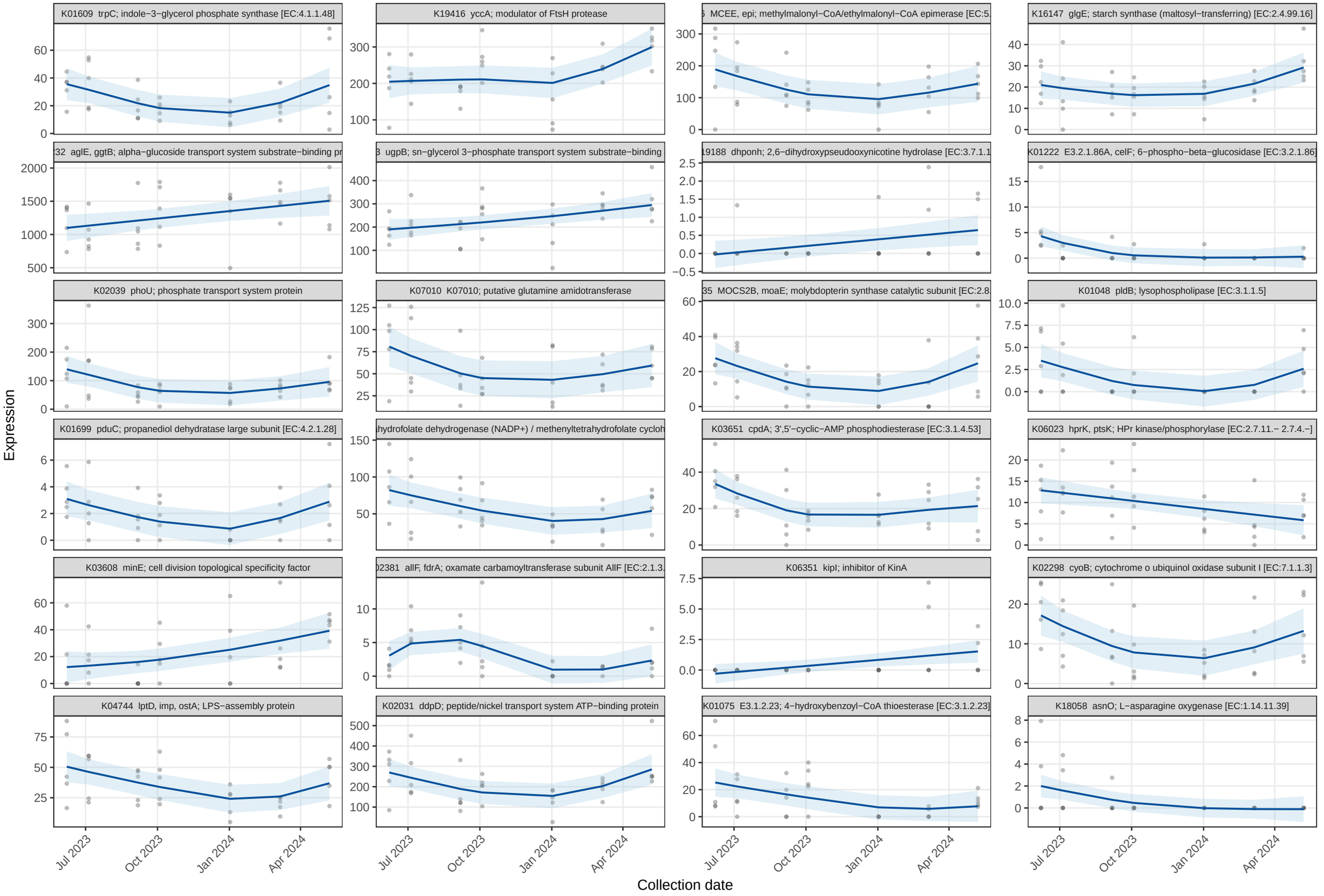
brite_D filtered temporal GAMs: page 12 of 303

All remaining BRITE classes after exclusions



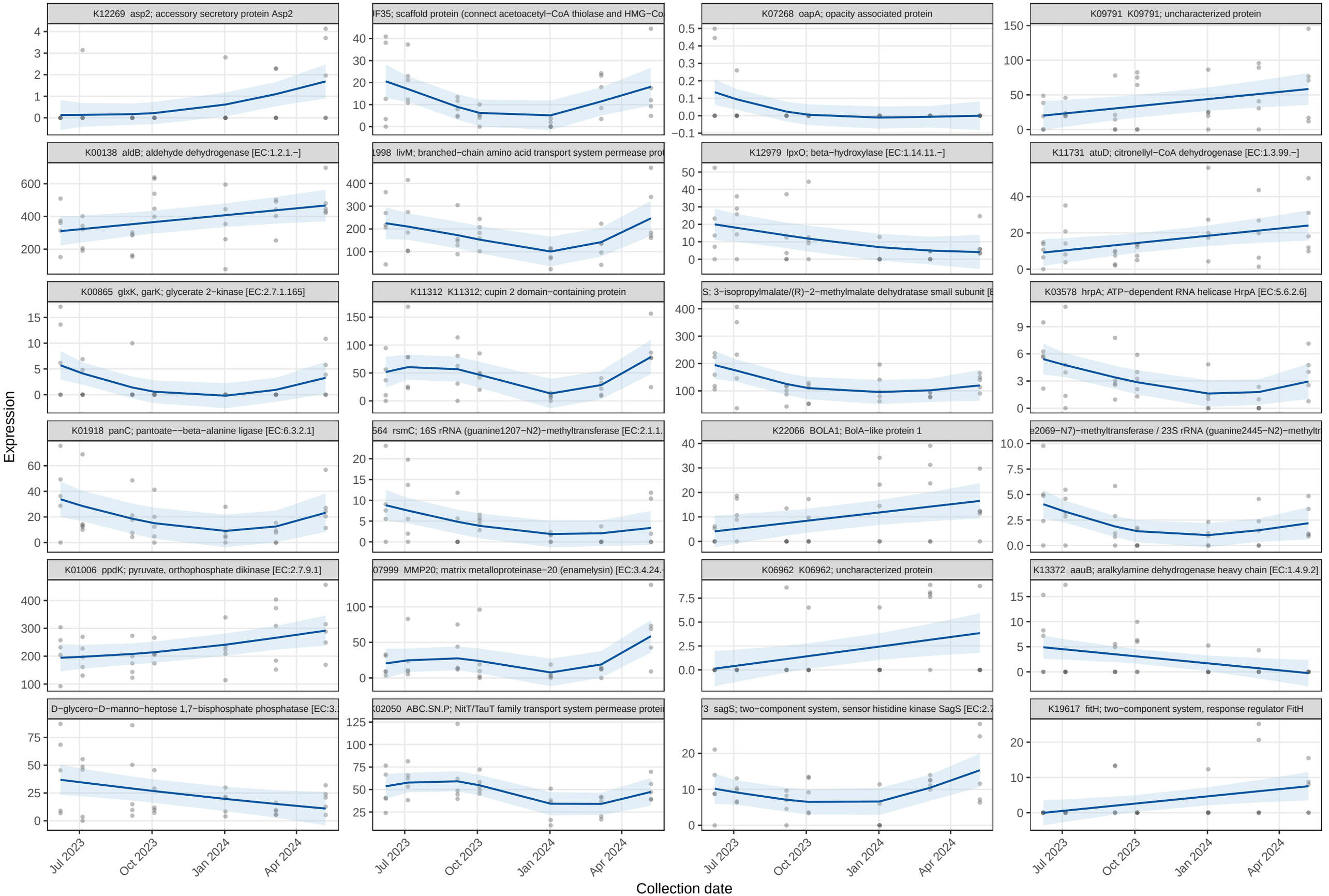
brite_D filtered temporal GAMs: page 13 of 303

All remaining BRITE classes after exclusions



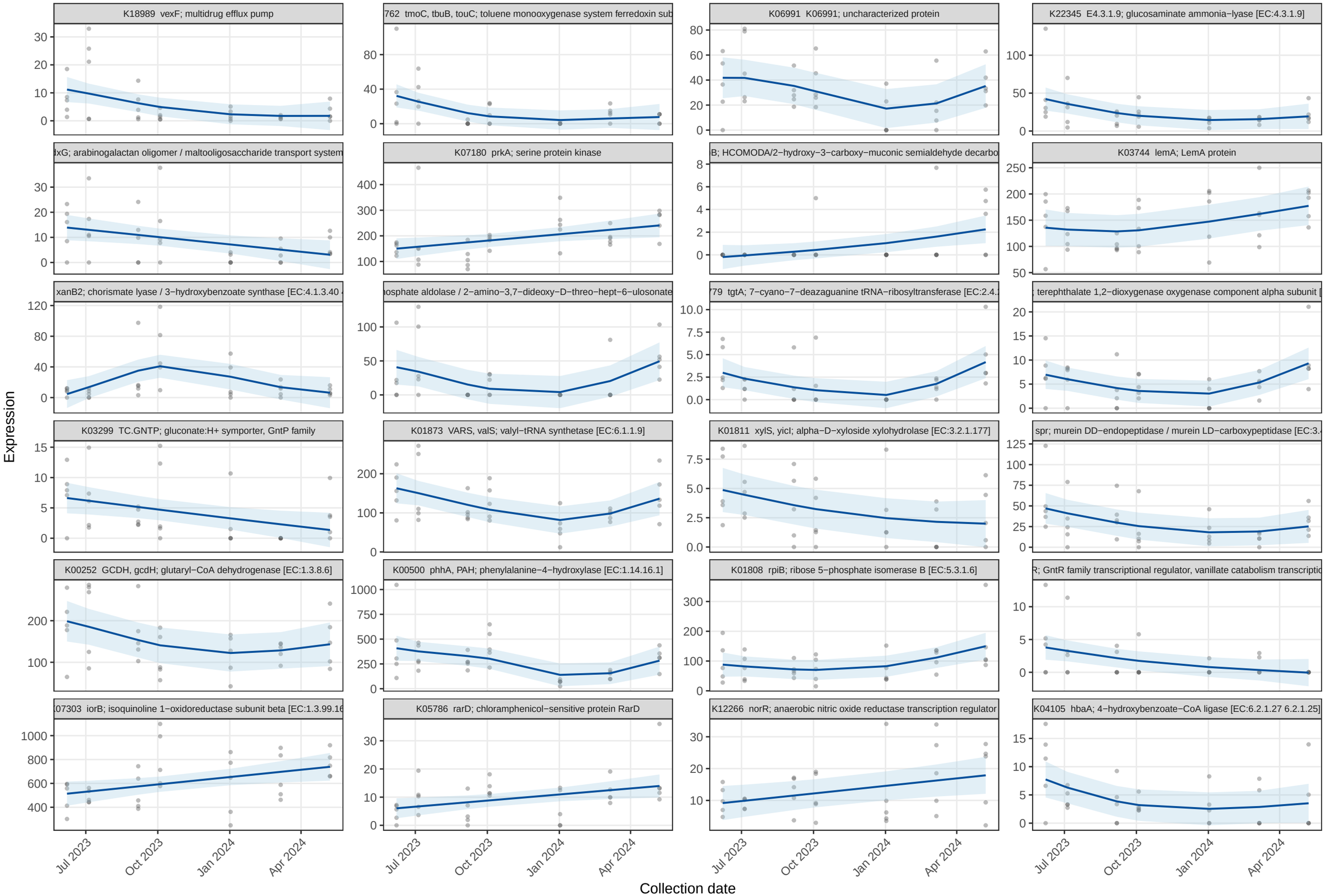
brite_D filtered temporal GAMs: page 14 of 303

All remaining BRITE classes after exclusions



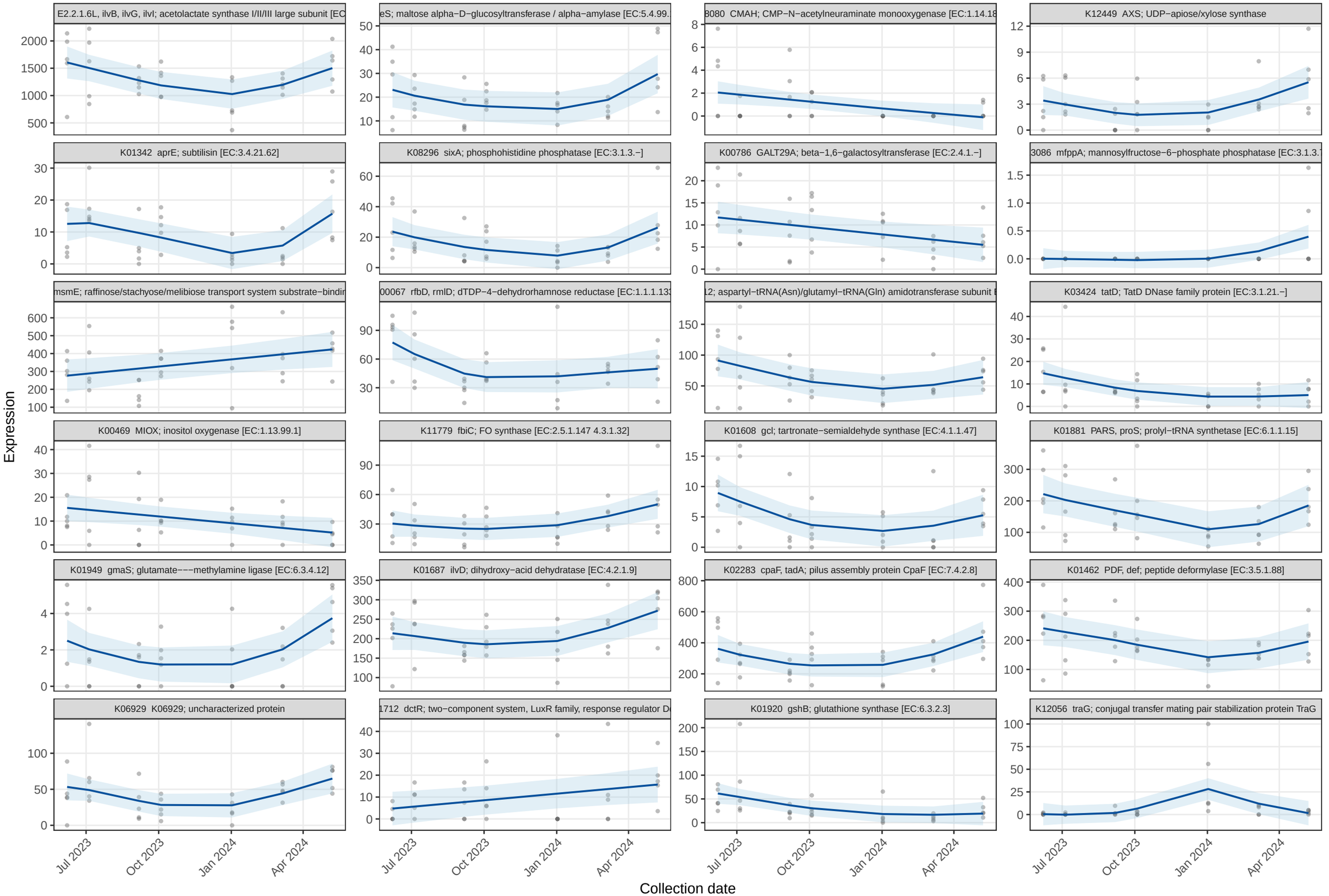
brite_D filtered temporal GAMs: page 15 of 303

All remaining BRITE classes after exclusions



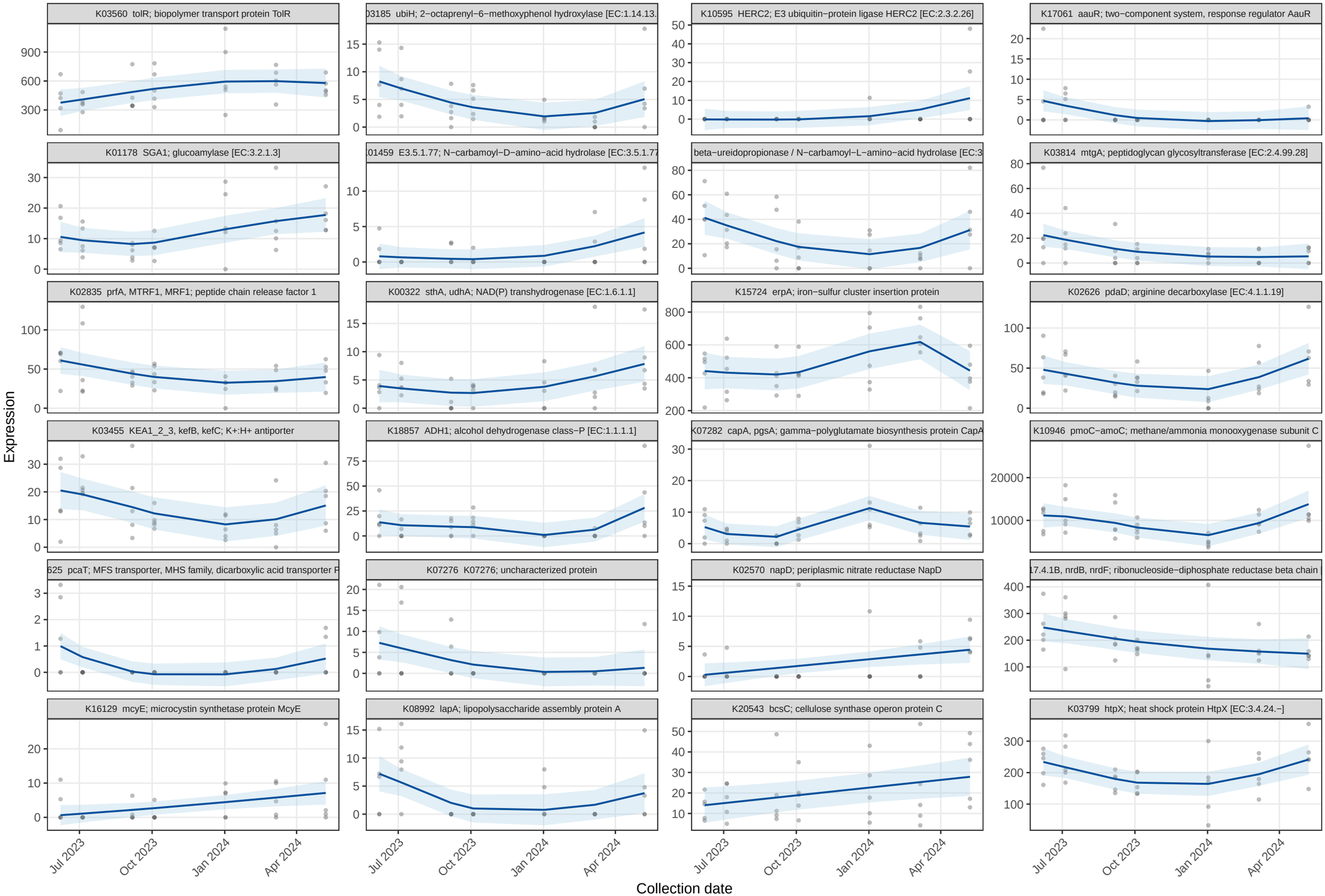
brite_D filtered temporal GAMs: page 16 of 303

All remaining BRITE classes after exclusions



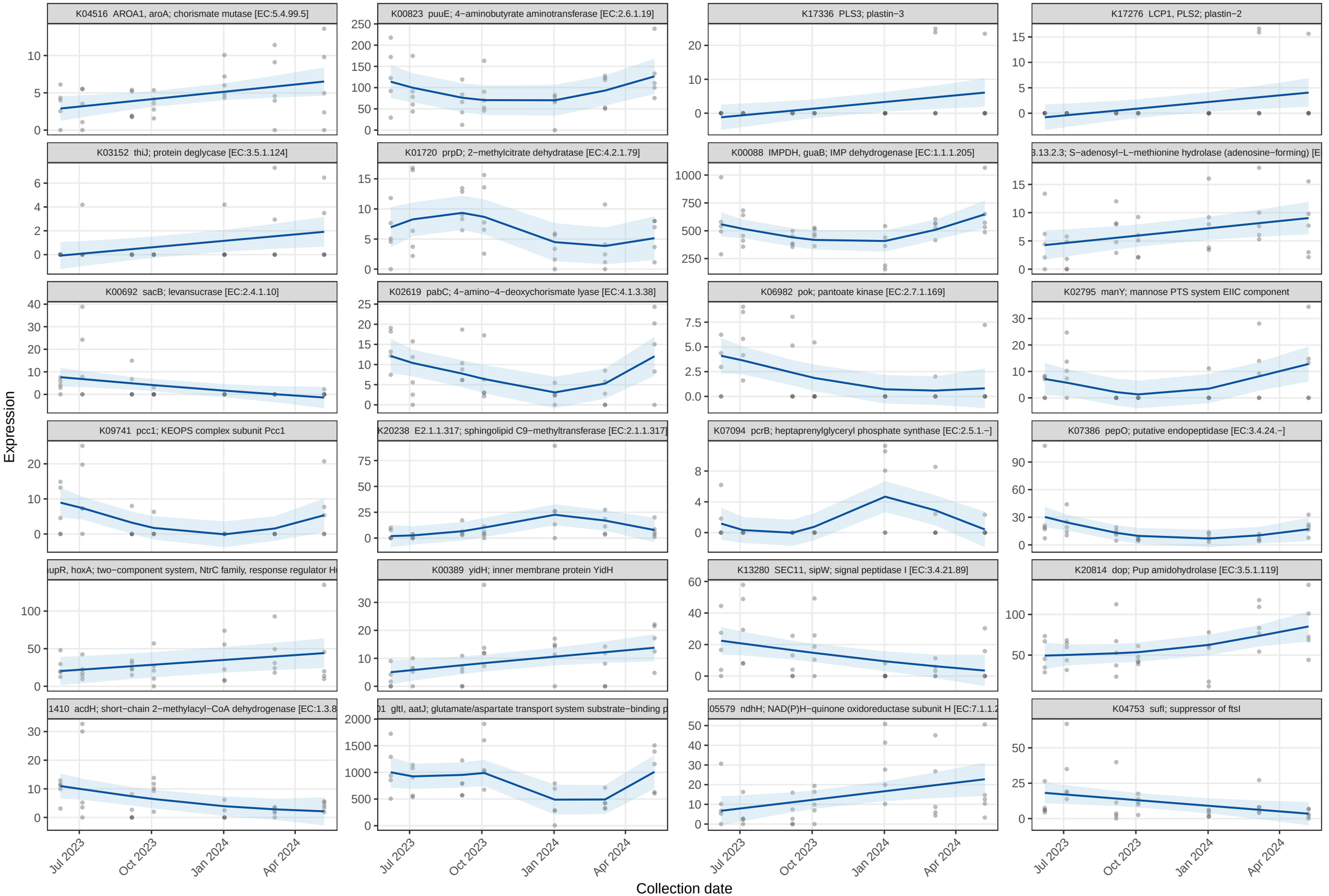
brite_D filtered temporal GAMs: page 17 of 303

All remaining BRITE classes after exclusions



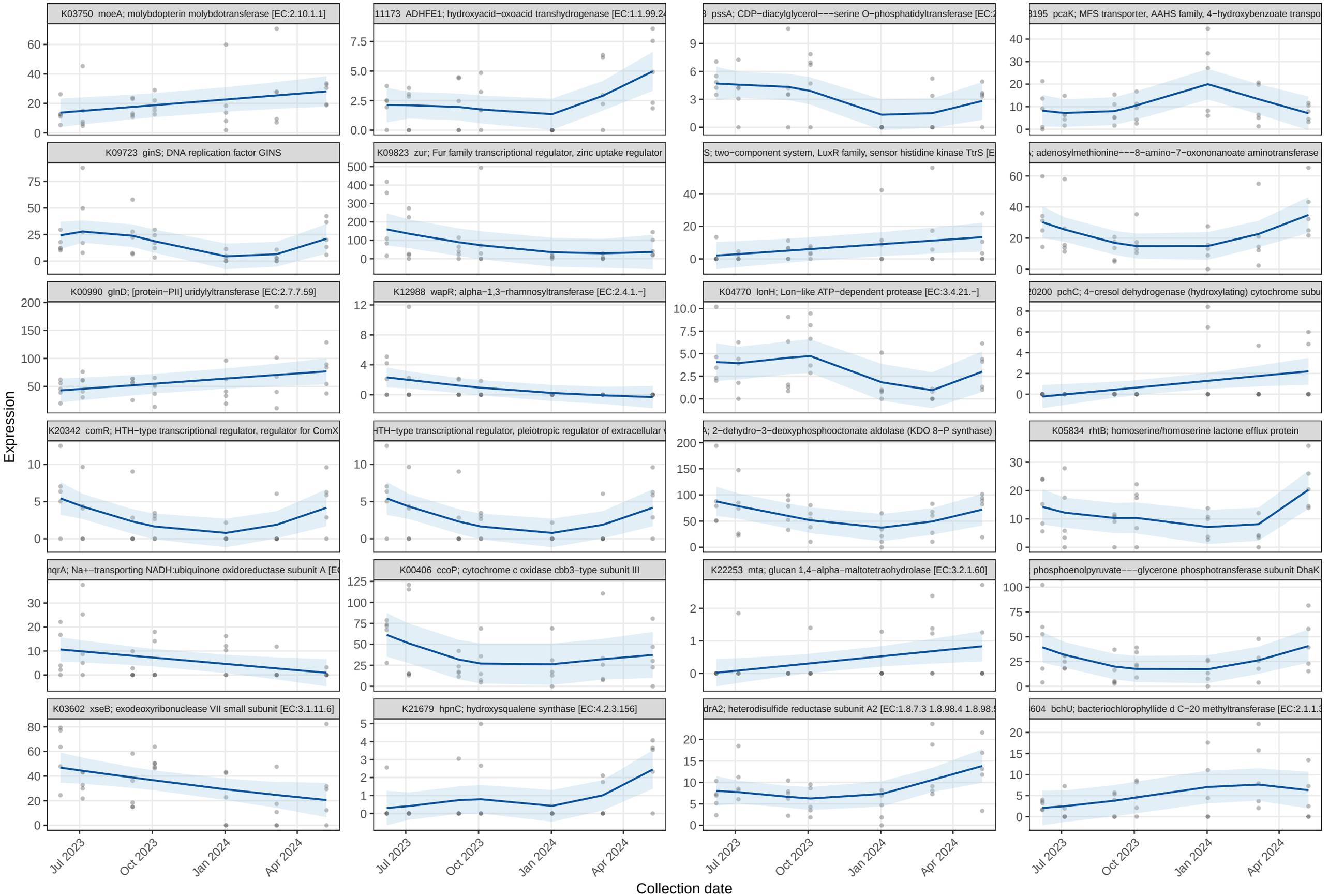
brite_D filtered temporal GAMs: page 18 of 303

All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 19 of 303

All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions

folX; D−erythro−7,8−dihydroneopterin triphosphate epimerase [EC:5

K07390 grxD, GLRX5; monothiol glutaredoxin

K05973 phaZ; poly(3−hydroxybutyrate) depolymerase [EC:3.1.1.75]

K15229 mauB; methylamine dehydrogenase heavy chain [EC:1.4.9.1]

K02032 ddpF; peptide/nickel transport system ATP−binding protein

K02096 apcE; phycobilisome core−membrane linker protein

K07448 mrr; restriction system protein

dauC, aknG; O−methyltransferase / aklanonic acid methyltransferase

K07734 paiB; transcriptional regulator

K15828 hycC; formate hydrogenlyase subunit 3

K07583 PUS10; tRNA pseudouridine synthase 10 [EC:5.4.99.25]

9761 rsmE; 16S rRNA (uracil1498−N3)−methyltransferase [EC:2.1.1.1

K03210 yajC; preprotein translocase subunit YajC

K00122 FDH; formate dehydrogenase [EC:1.17.1.9]

K10235 aglK; alpha−glucoside transport system ATP−binding protein

2,3−trimethyl−5−oxocyclopent−3−enyl)acetyl−CoA 1,5−monooxygenas

K05916 hmp, YHB1; nitric oxide dioxygenase [EC:1.14.12.17]

K12950 ctpC; manganese−transporting P−type ATPase C [EC:7.2.2.22

K19158 yoeB; toxin YoeB [EC:3.1.−.−]

05845 opuC; osmoprotectant transport system substrate−binding prote

rsaD, prsD, eexD; ATP−binding cassette, subfamily C, type I secretion

K08979 K08979; putative membrane protein

K19064 lysDH; lysine 6−dehydrogenase [EC:1.4.1.18]

2424 tcyA, tcyJ, fliY; L−cystine transport system substrate−binding prot

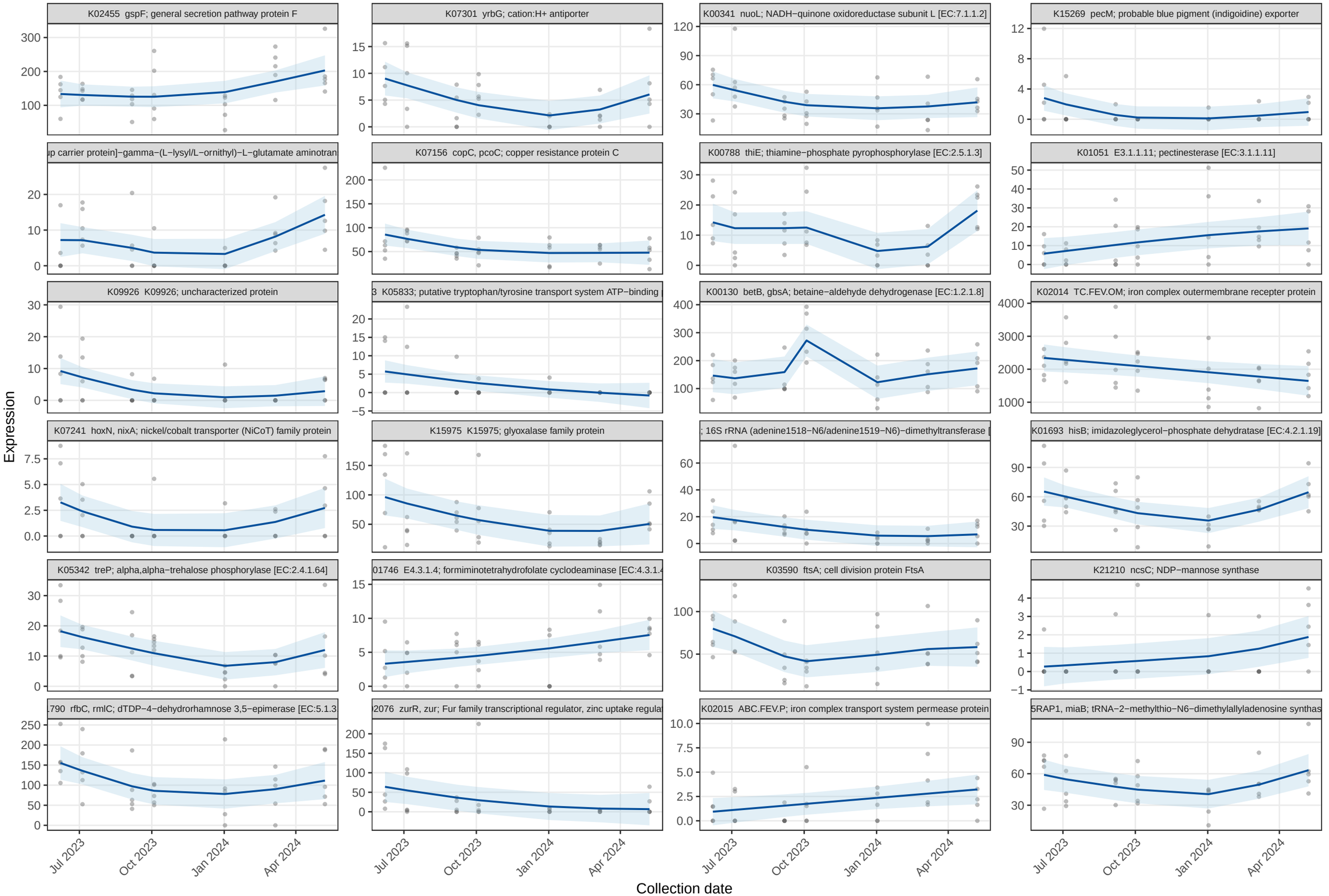
Expression

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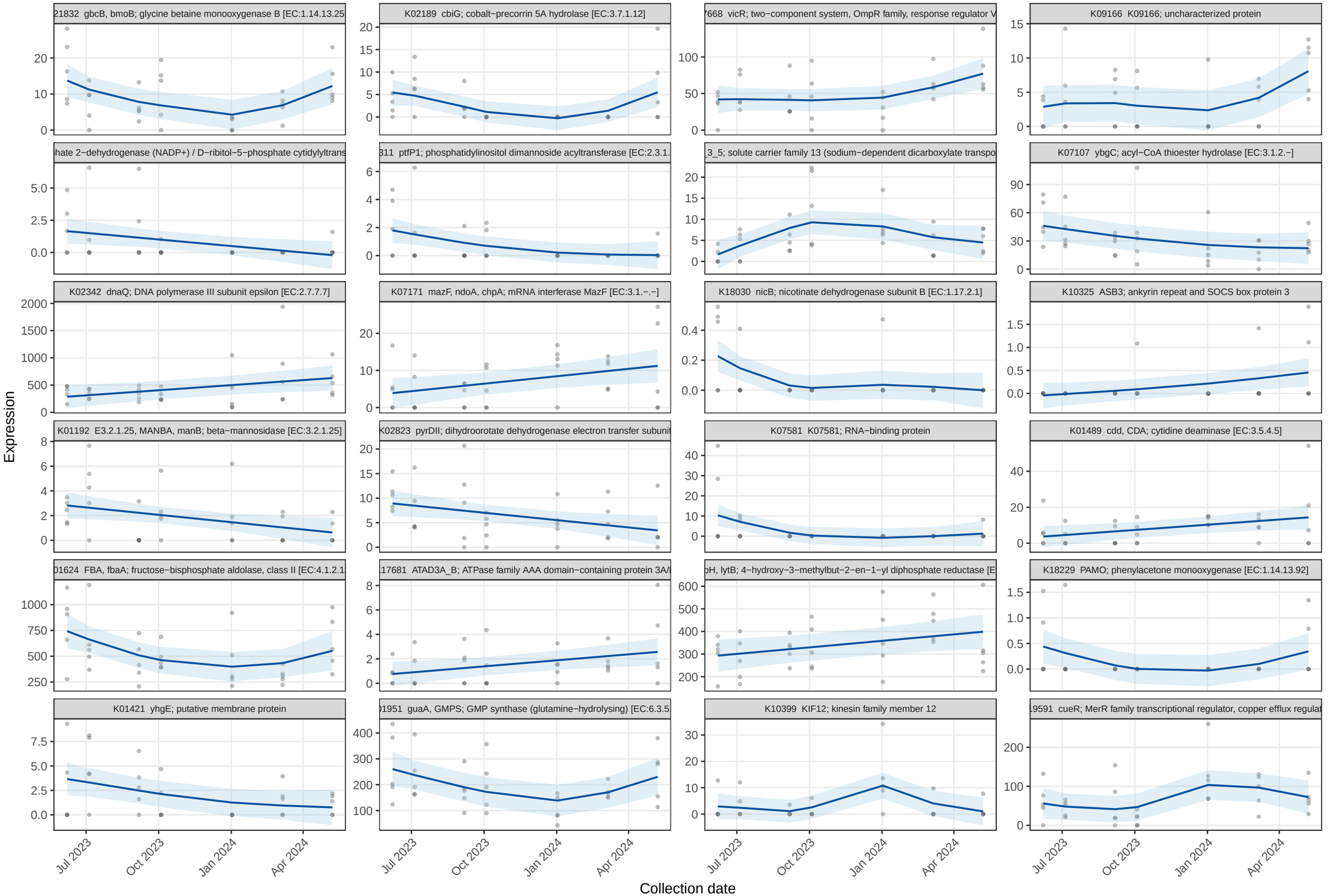
brite_D filtered temporal GAMs: page 21 of 303

All remaining BRITE classes after exclusions



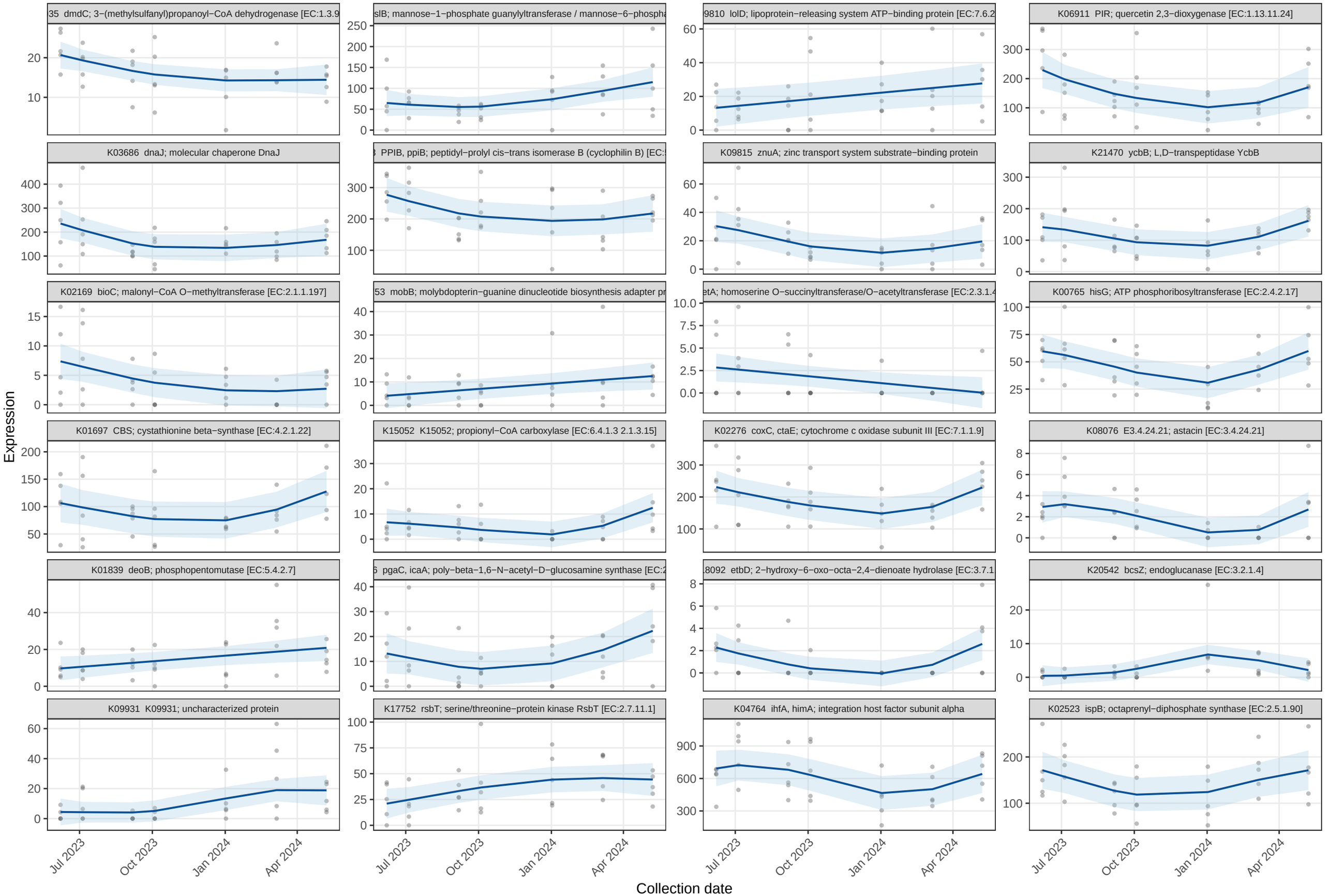
brite_D filtered temporal GAMs: page 22 of 303

All remaining BRITE classes after exclusions

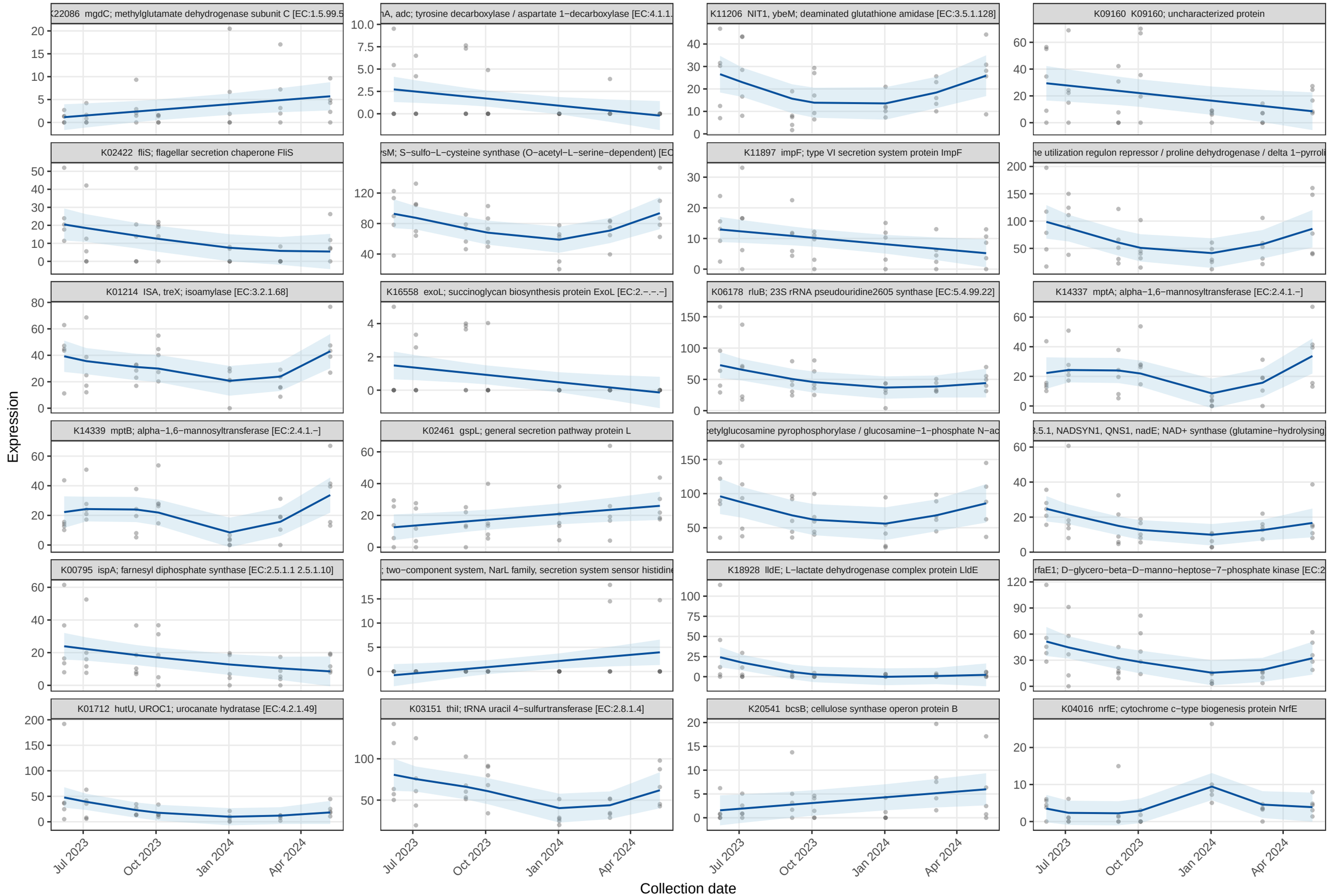


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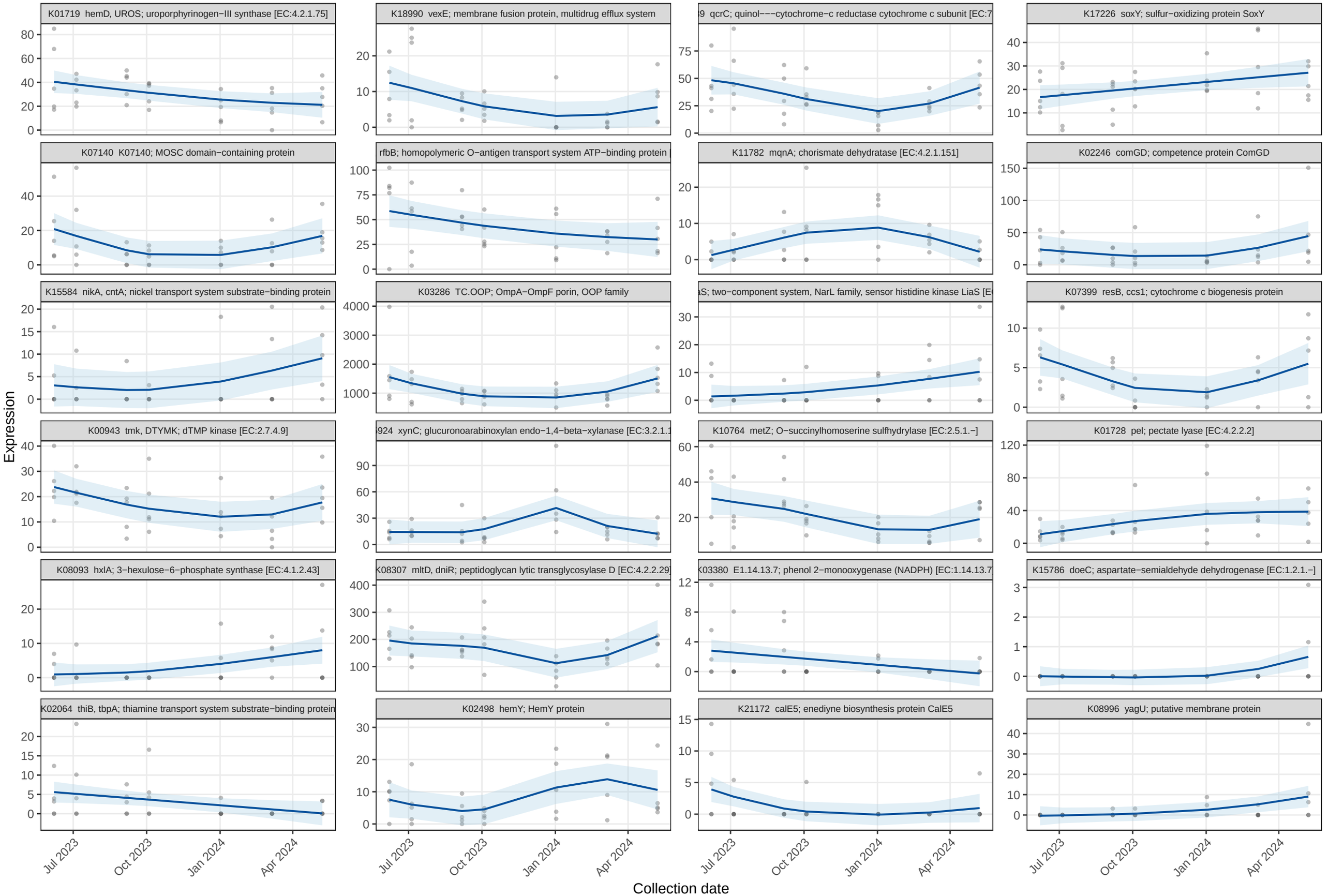
All remaining BRITE classes after exclusions



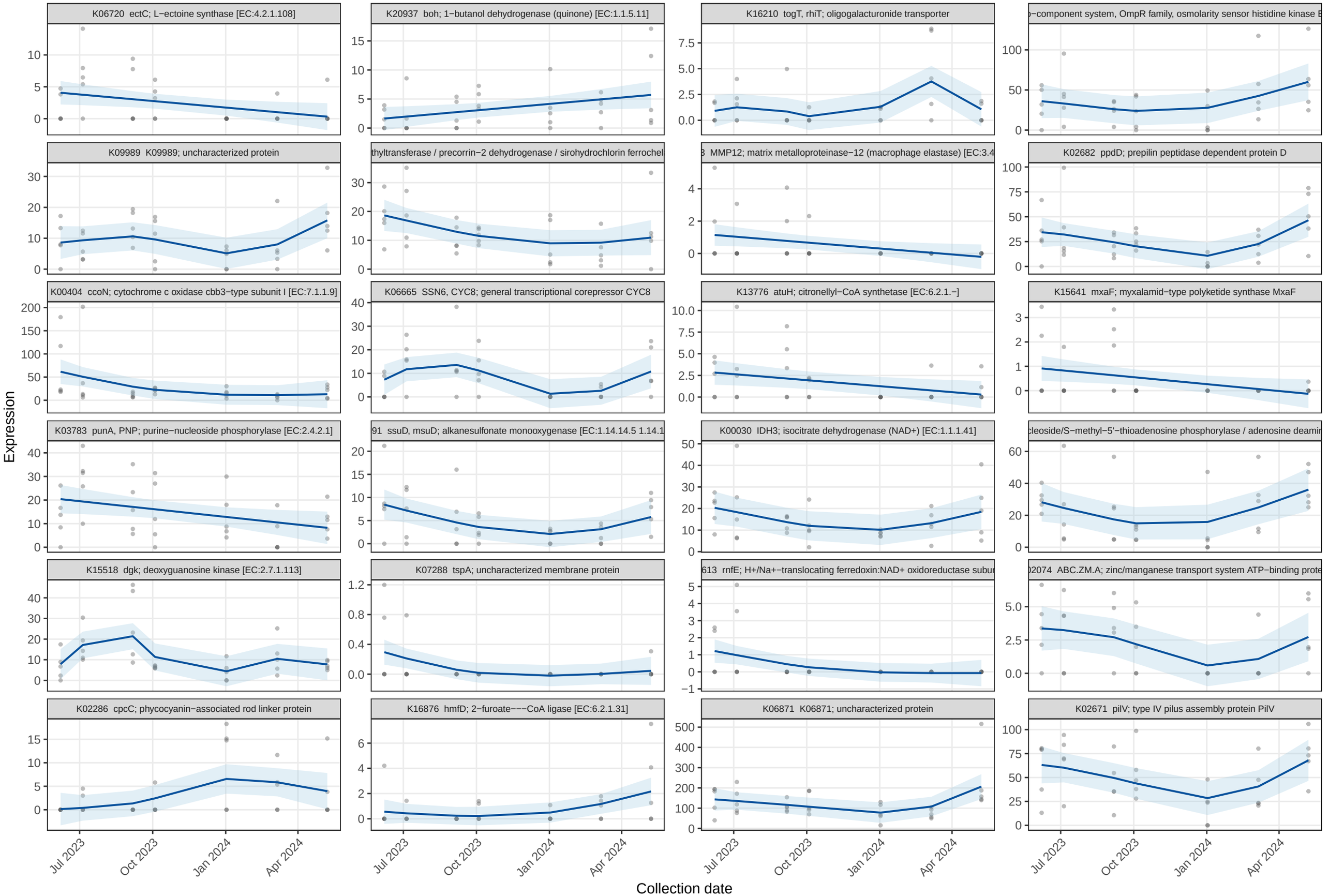
All remaining BRITE classes after exclusions



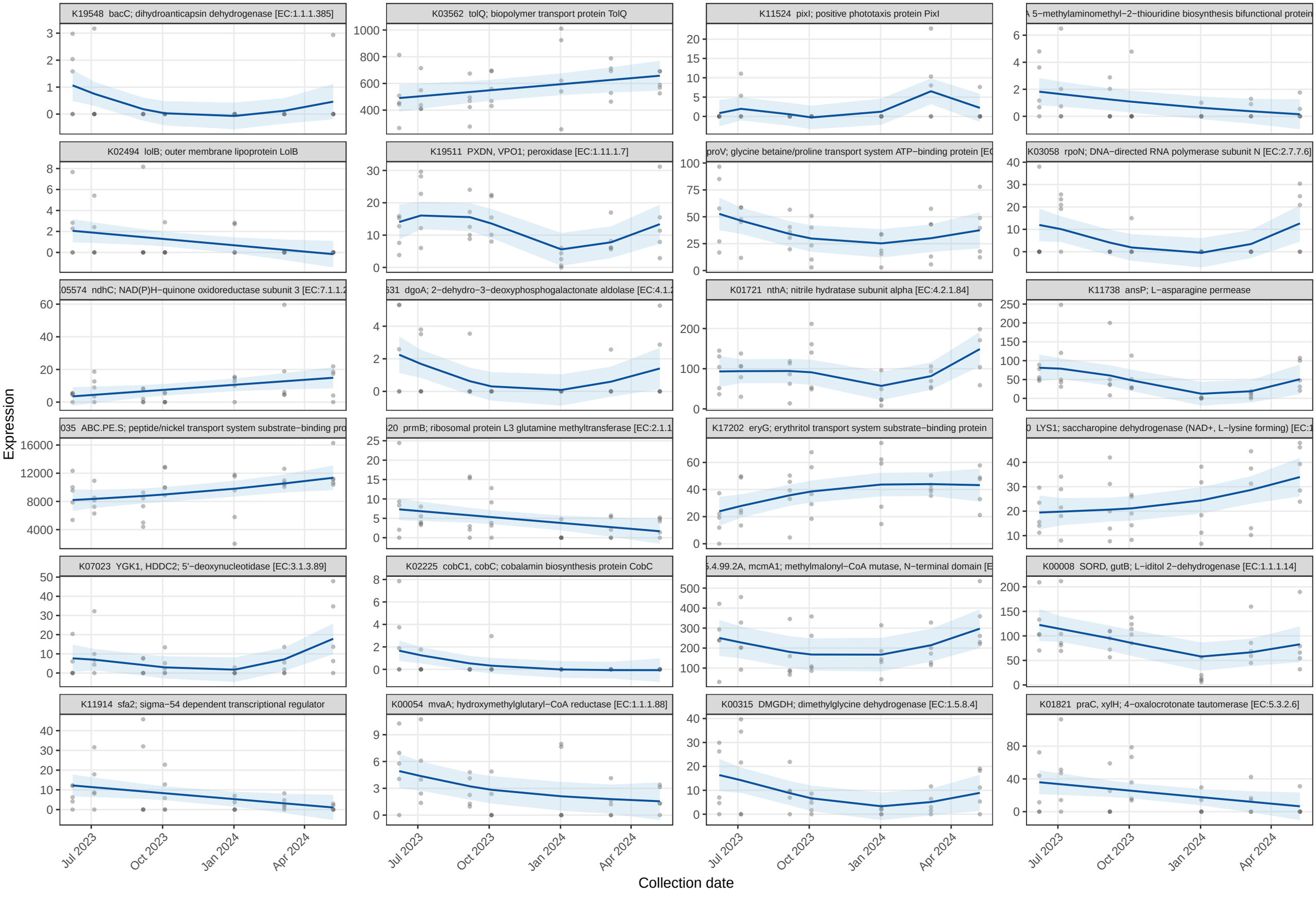
All remaining BRITE classes after exclusions



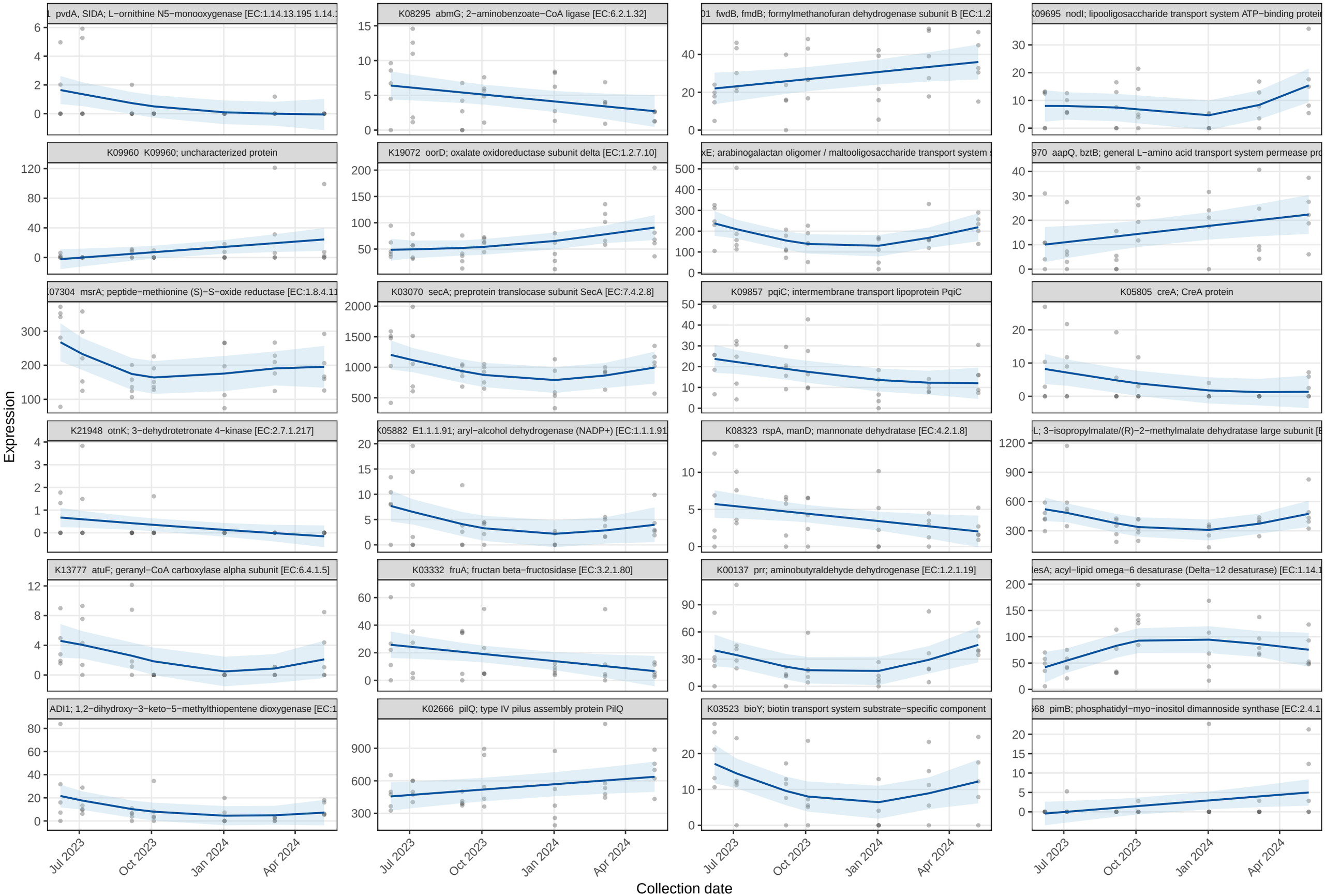
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

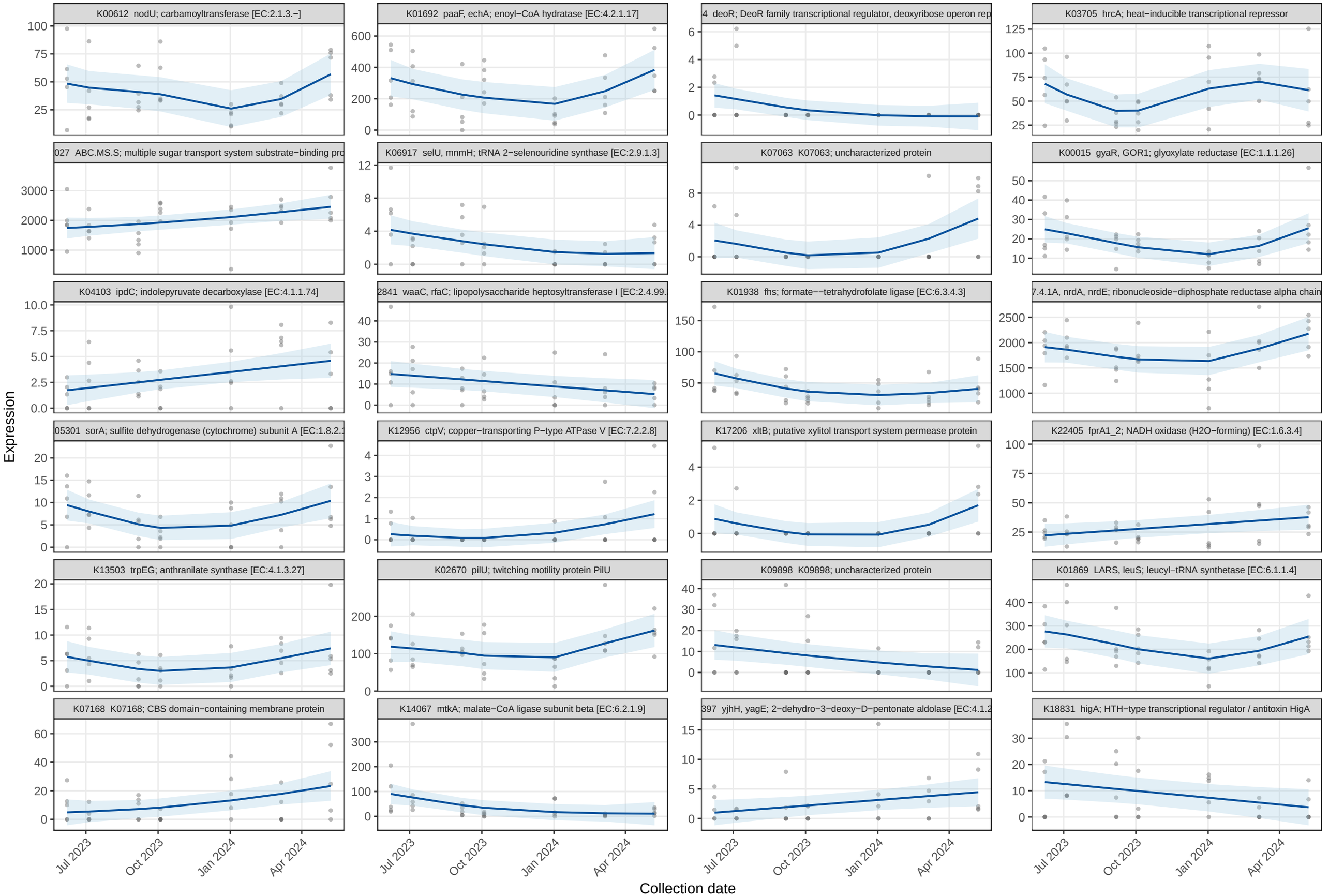


All remaining BRITE classes after exclusions

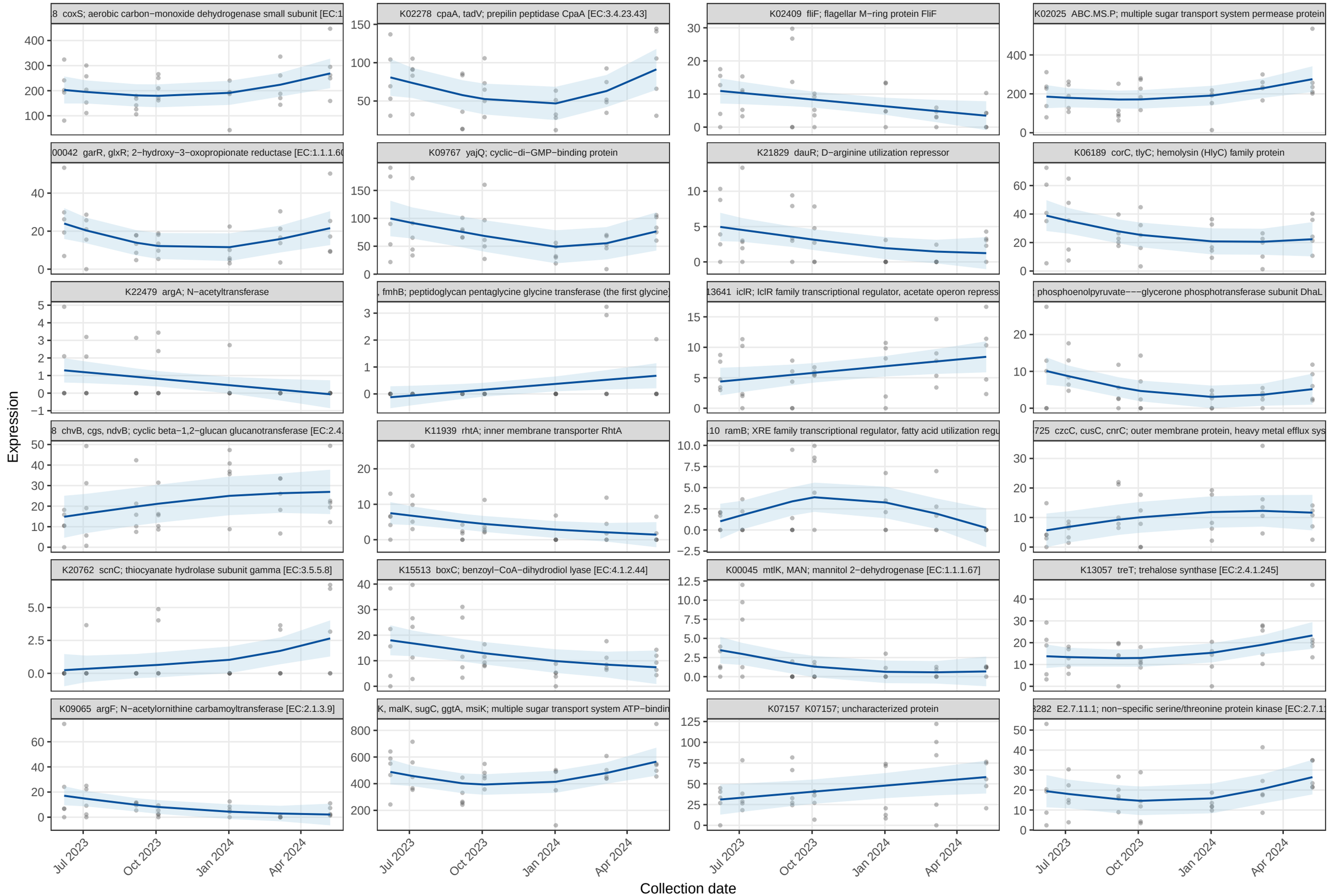


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All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

K16165 nagK; fumarylpyruvate hydrolase [EC:3.7.1.20]

K04021 eutE; aldehyde dehydrogenase

) ispF; 2−C−methyl−D−erythritol 2,4−cyclodiphosphate synthase [EC:4

1997 livH; branched−chain amino acid transport system permease prot

K08312 nudE; ADP compounds hydrolase [EC:3.6.1.−]

diphosphokinase / guanosine−3',5'−bis(diphosphate) 3'−diphosphatase

K15270 sam; S−adenosylmethionine uptake transporter

slV, clpQ; ATP−dependent HslUV protease, peptidase subunit HslV [EC

K22319 oleC; olefin beta−lactone synthetase [EC:6.1.3.1]

K03325 ACR3, arsB; arsenite transporter

K03703 uvrC; excinuclease ABC subunit C

K08602 pepF, pepB; oligoendopeptidase F [EC:3.4.24.−]

Expression

K17947 wbiB; dTDP−L−rhamnose 4−epimerase [EC:5.1.3.25]

K03087 rpoS; RNA polymerase nonessential primary−like sigma factor

treB, treP; trehalose PTS system EIIBC or EIIBCA component [EC:2.7

pntB; proton−translocating NAD(P)+ transhydrogenase subunit beta [EC

K15037 K15037; biotin carboxyl carrier protein

K18605 K18605; biotin carboxyl carrier protein

01845 hemL; glutamate−1−semialdehyde 2,1−aminomutase [EC:5.4.3.

K02247 comGE; competence protein ComGE

.1.77, pcm; protein−L−isoaspartate(D−aspartate) O−methyltransferase

K19070 oorA; oxalate oxidoreductase subunit alpha [EC:1.2.7.10]

K01801 nagL; maleylpyruvate isomerase [EC:5.2.1.4]

K11065 tpx; thioredoxin−dependent peroxiredoxin [EC:1.11.1.24]

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iA; MFS transporter, MHS family, shikimate and dehydroshikimate trans

K15545 mlc; transcriptional regulator of PTS gene

ICMT, STE14; protein−S−isoprenylcysteine O−methyltransferase [EC:2

0540 mglB; methyl−galactoside transport system substrate−binding pro

K03074 secF; preprotein translocase subunit SecF

K01874 MARS, metG; methionyl−tRNA synthetase [EC:6.1.1.10]

K00394 aprA; adenylylsulfate reductase, subunit A [EC:1.8.99.2]

K01520 dut, DUT; dUTP diphosphatase [EC:3.6.1.23]

K02024 lamB; maltoporin

K18122 cat2, abfT; 4−hydroxybutyrate CoA−transferase [EC:2.8.3.−]

1 gidB, rsmG; 16S rRNA (guanine527−N7)−methyltransferase [EC:2.1.

K11060 entB; probable enterotoxin B

Expres

K09159 cptB; antitoxin CptB

21572 susD; starch−binding outer membrane protein, SusD/RagB fami

6969 rlmI; 23S rRNA (cytosine1962−C5)−methyltransferase [EC:2.1.1.1

K06287 yhdE; nucleoside triphosphate pyrophosphatase [EC:3.6.1.−]

K21003 pslJ; polysaccharide biosynthesis protein PslJ

722 nikR; CopG family transcriptional regulator, nickel−responsive regu

K00957 cysD; sulfate adenylyltransferase subunit 2 [EC:2.7.7.4]

S; two−component system, NarL family, sensor histidine kinase DegS [

; two−component system, chemotaxis family, response regulator WspF

K21023 mucR; diguanylate cyclase [EC:2.7.7.65]

H+/Na+−translocating ferredoxin:NAD+ oxidoreductase subunit C [EC:7

K14977 ylbA, UGHY; (S)−ureidoglycine aminohydrolase [EC:3.5.3.26]

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K06983 K06983; uncharacterized protein

K02454 gspE; general secretion pathway protein E [EC:7.4.2.8]

5.4.99.2B, mcmA2; methylmalonyl−CoA mutase, C−terminal domain [E

DPH1, dph2; 2−(3−amino−3−carboxypropyl)histidine synthase [EC:2.5

K16074 zntB; zinc transporter

K01879 glyS; glycyl−tRNA synthetase beta chain [EC:6.1.1.14]

K07025 K07025; putative hydrolase of the HAD superfamily

K06001 trpB; tryptophan synthase beta chain [EC:4.2.1.20]

TC.CITMHS; citrate−Mg2+:H+ or citrate−Ca2+:H+ symporter, CitMHS

00575 cheR; chemotaxis protein methyltransferase CheR [EC:2.1.1.80

K01662 dxs; 1−deoxy−D−xylulose−5−phosphate synthase [EC:2.2.1.7]

K09118 K09118; uncharacterized protein

Expression

K07137 K07137; uncharacterized protein

K15034 yaeJ; ribosome−associated protein

; Rrf2 family transcriptional regulator, iron−sulfur cluster assembly trans

60 yfiC, trmX; tRNA1Val (adenine37−N6)−methyltransferase [EC:2.1.1

K13583 gcrA; GcrA cell cycle regulator

snC; Lrp/AsnC family transcriptional regulator, regulator for asnA, asnC

K08675 PRSS15, PIM1; ATP−dependent Lon protease [EC:3.4.21.53]

K01029 scoB; 3−oxoacid CoA−transferase subunit B [EC:2.8.3.5]

K03404 chlD, bchD; magnesium chelatase subunit D [EC:6.6.1.1]

K02057 ABC.SS.P; simple sugar transport system permease protein

K13007 wbpL; glycosyltransferase WbpL [EC:2.4.1.−]

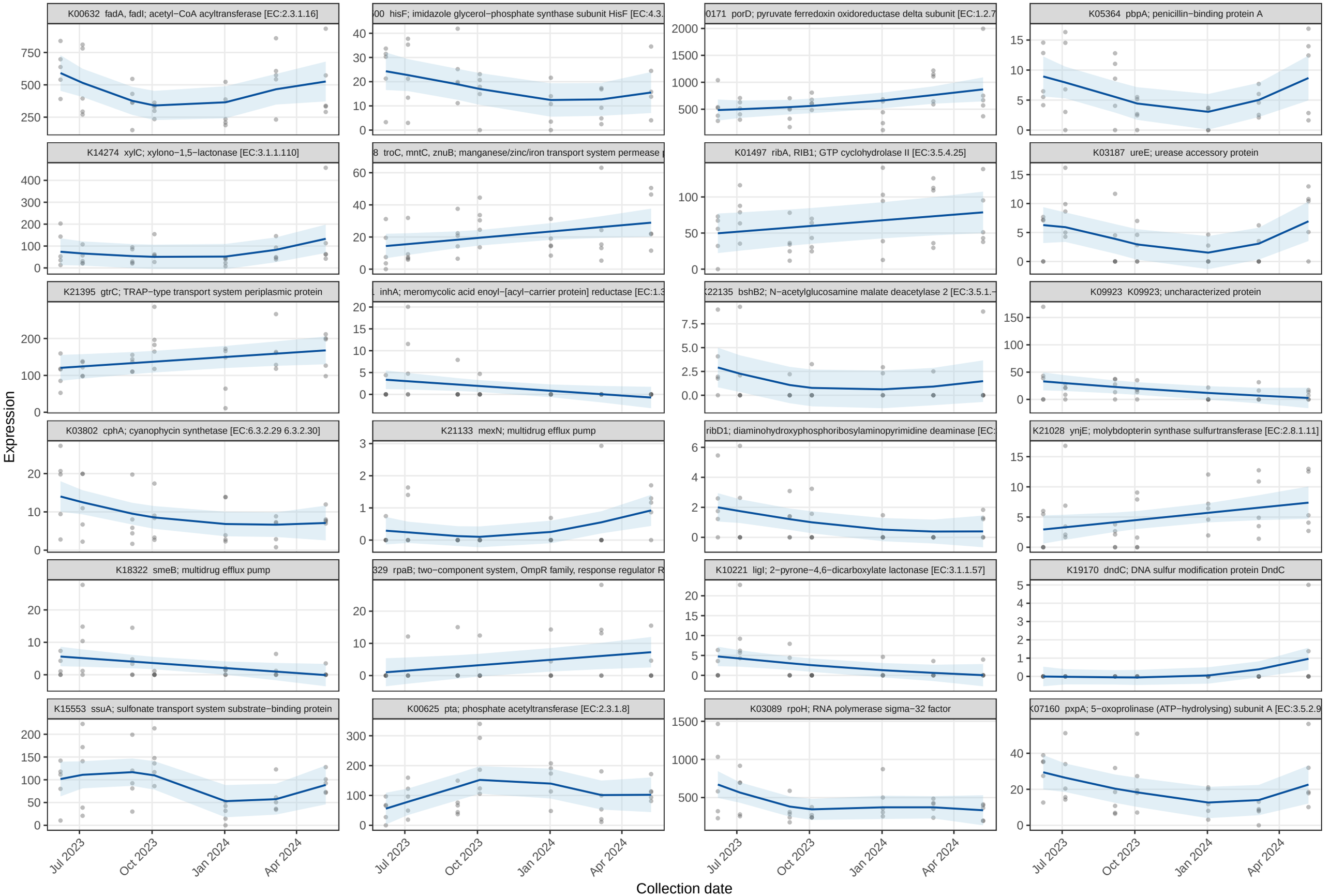
08223 fsr; MFS transporter, FSR family, fosmidomycin resistance prote

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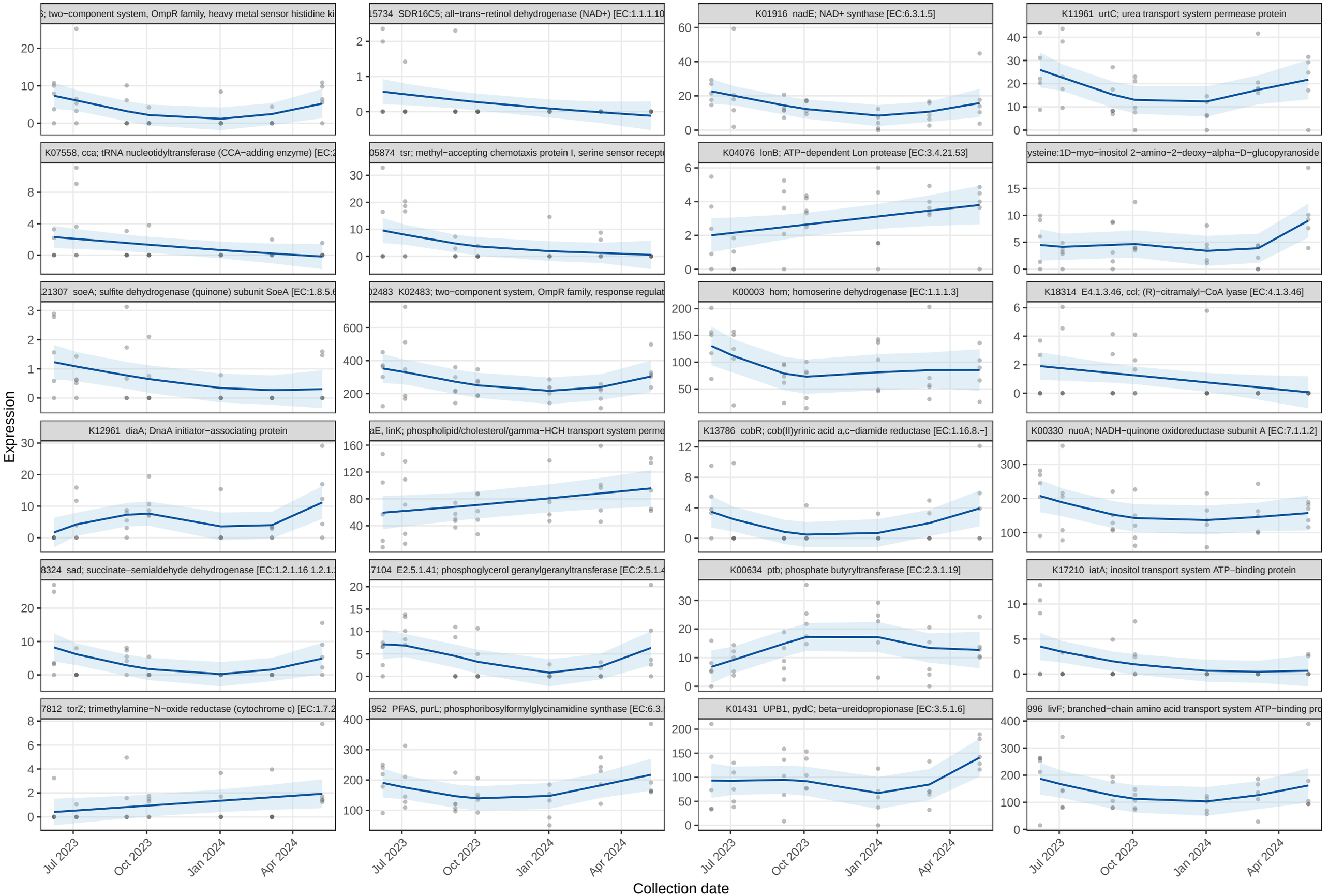
brite_D filtered temporal GAMs: page 34 of 303

All remaining BRITE classes after exclusions



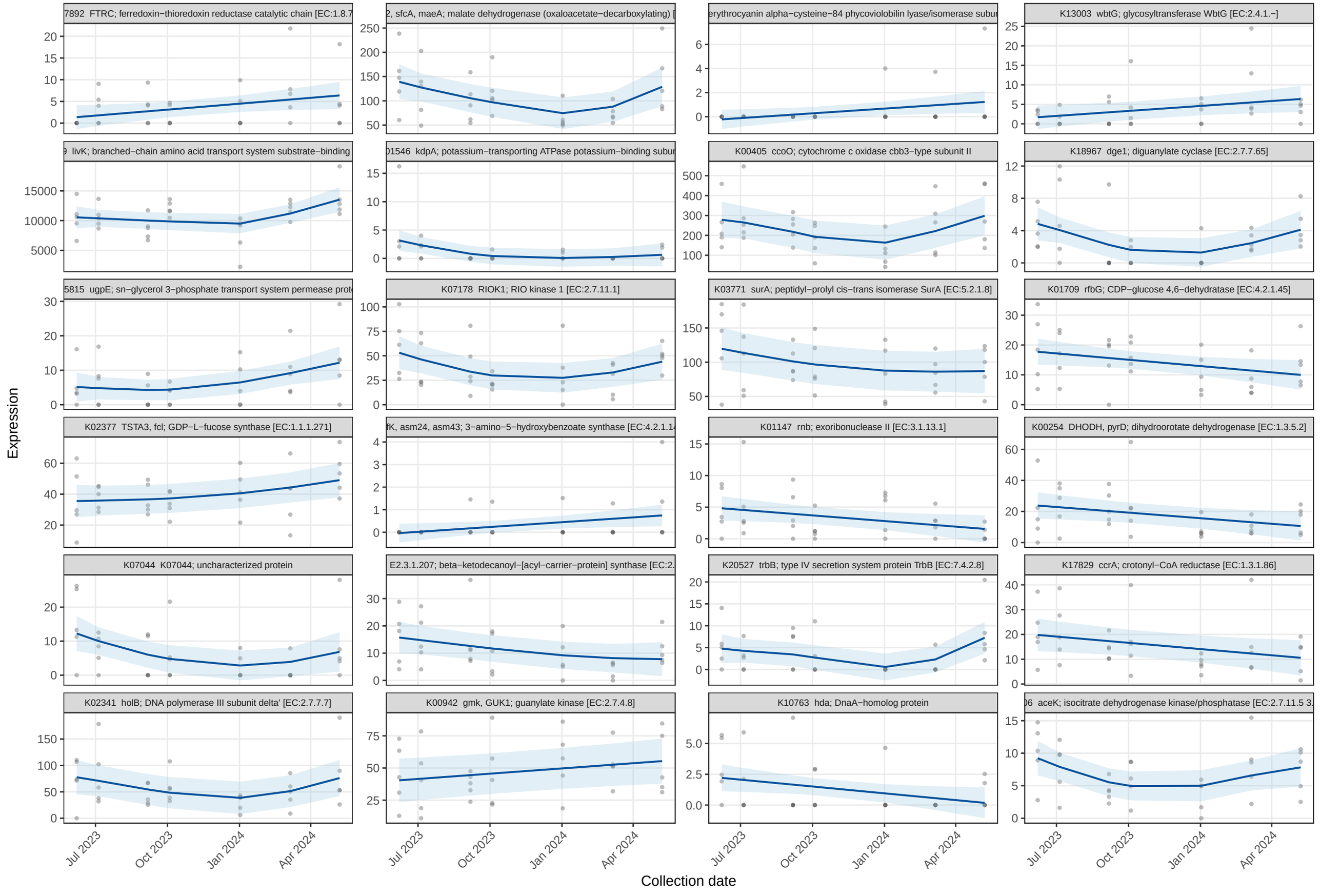
brite_D filtered temporal GAMs: page 35 of 303

All remaining BRITE classes after exclusions



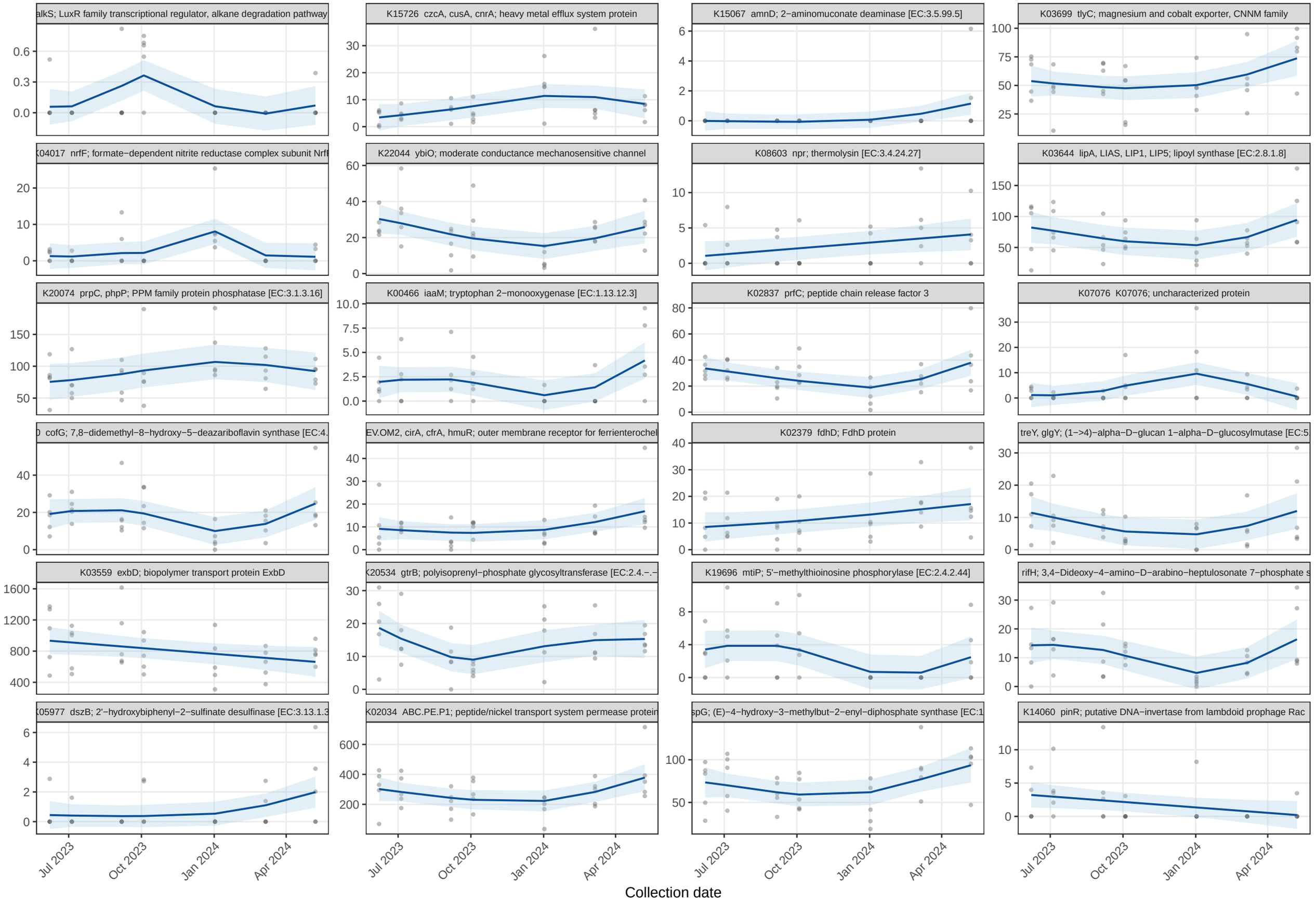
brite_D filtered temporal GAMs: page 36 of 303

All remaining BRITE classes after exclusions

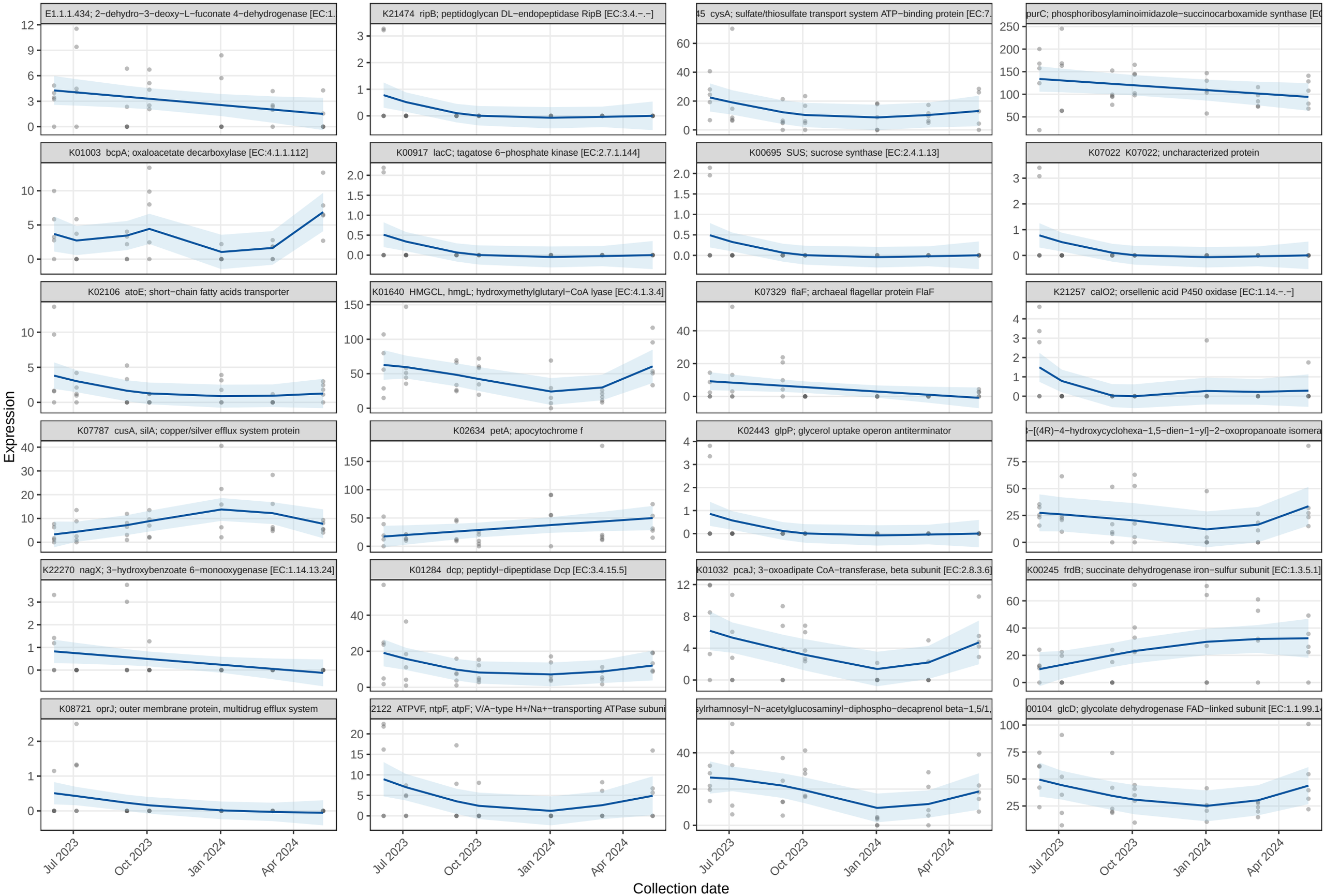


brite_D filtered temporal GAMs: page 37 of 303

All remaining BRITE classes after exclusions

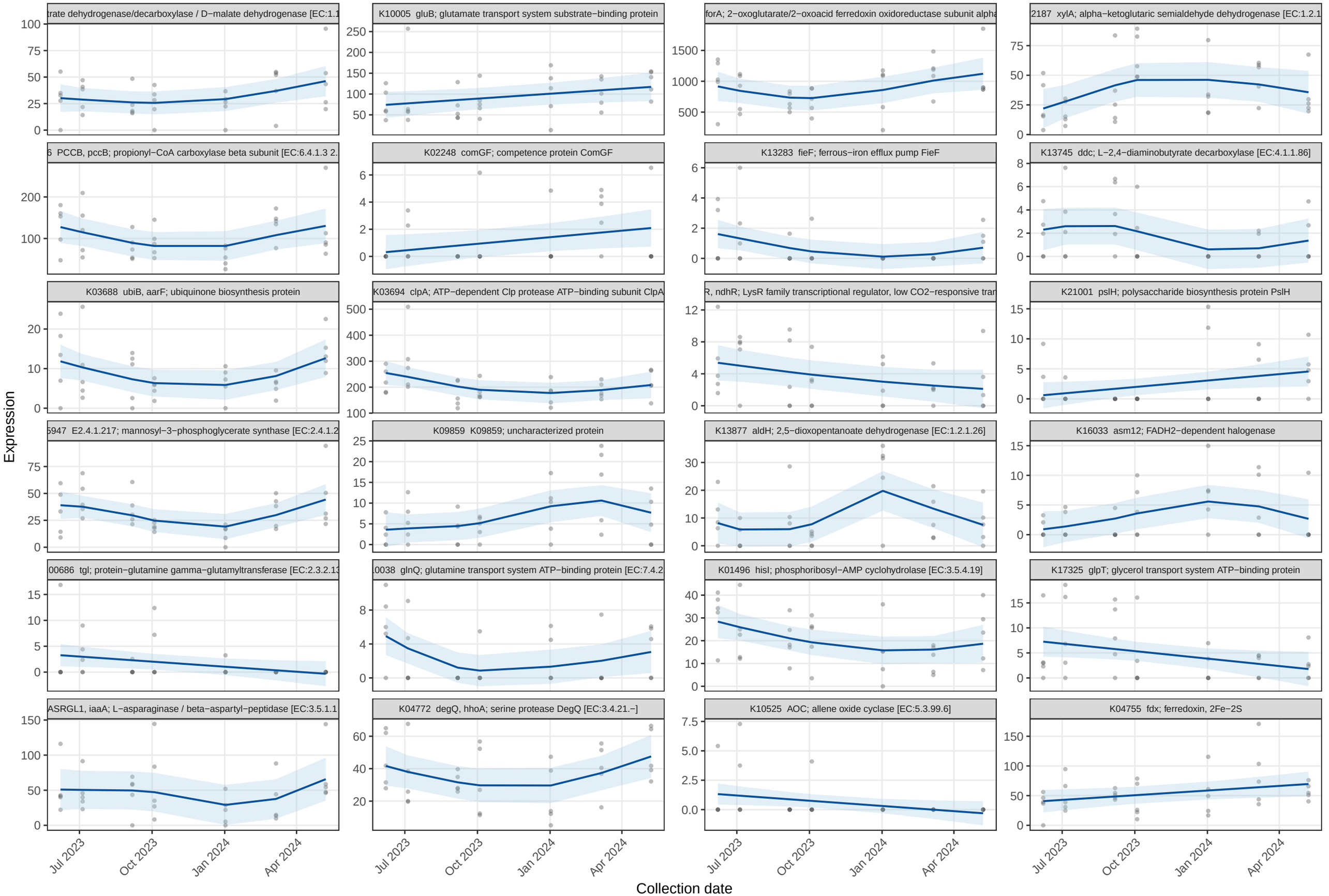


All remaining BRITE classes after exclusions



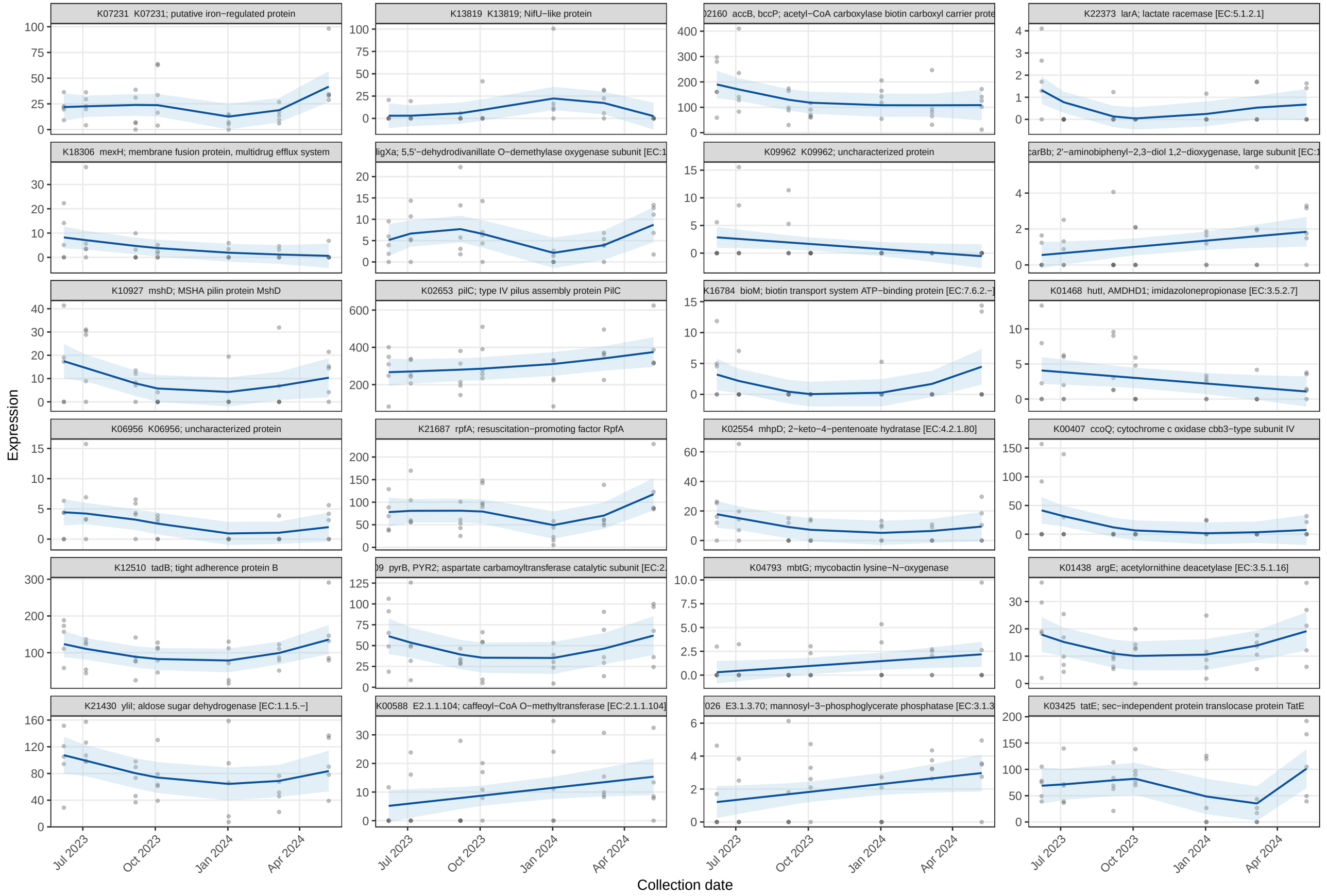
brite_D filtered temporal GAMs: page 39 of 303

All remaining BRITE classes after exclusions



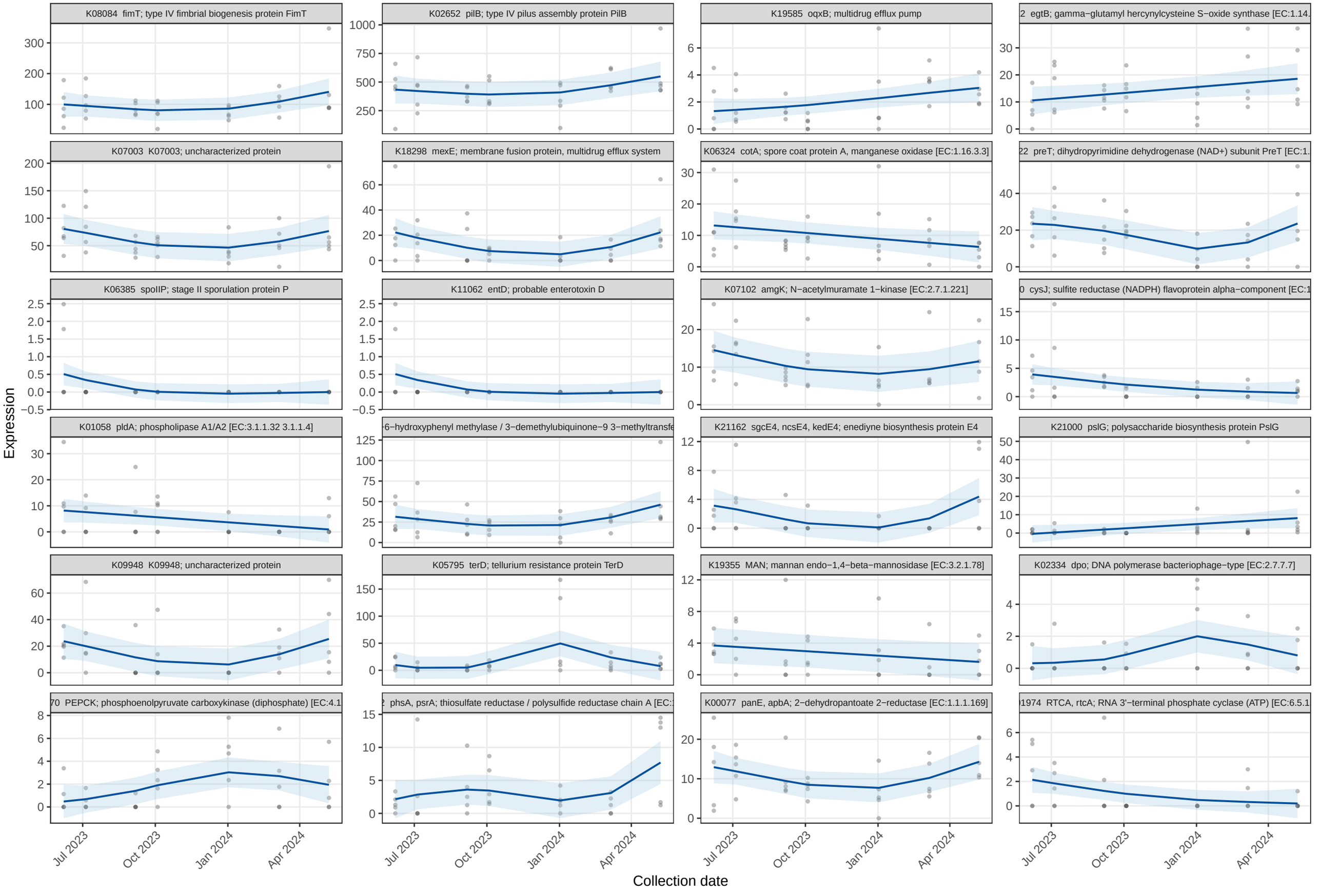
brite_D filtered temporal GAMs: page 40 of 303

All remaining BRITE classes after exclusions



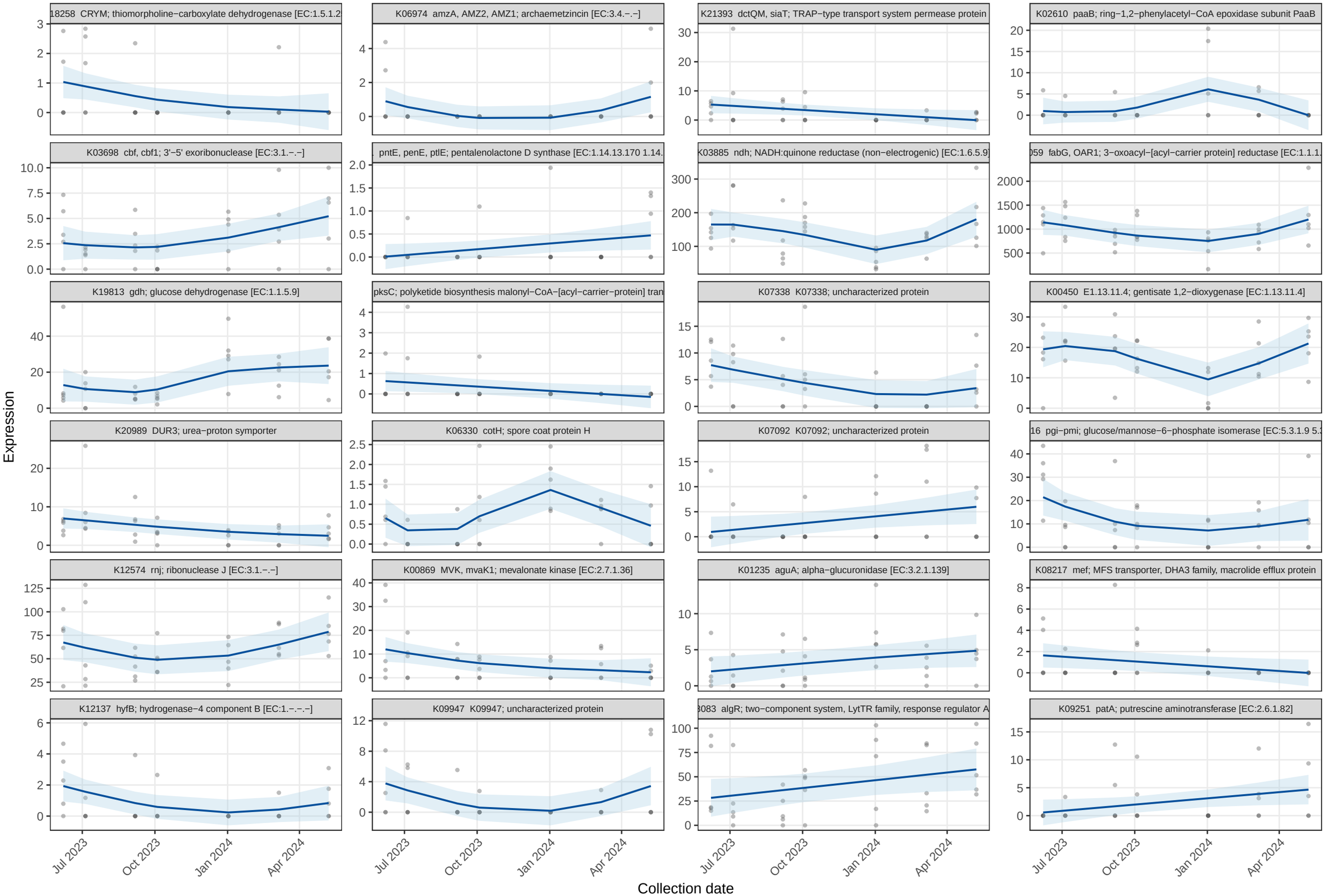
brite_D filtered temporal GAMs: page 41 of 303

All remaining BRITE classes after exclusions



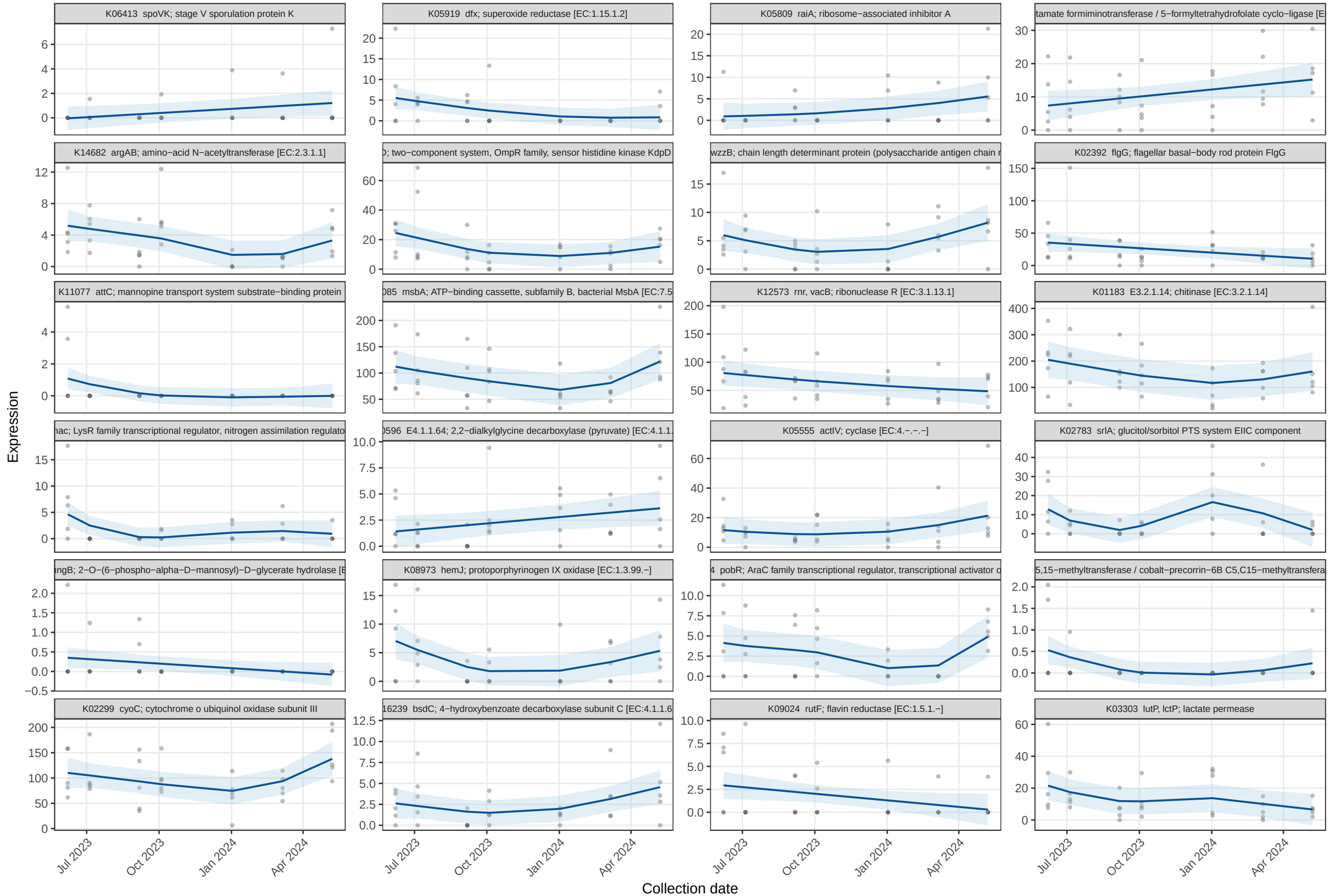
brite_D filtered temporal GAMs: page 42 of 303

All remaining BRITE classes after exclusions



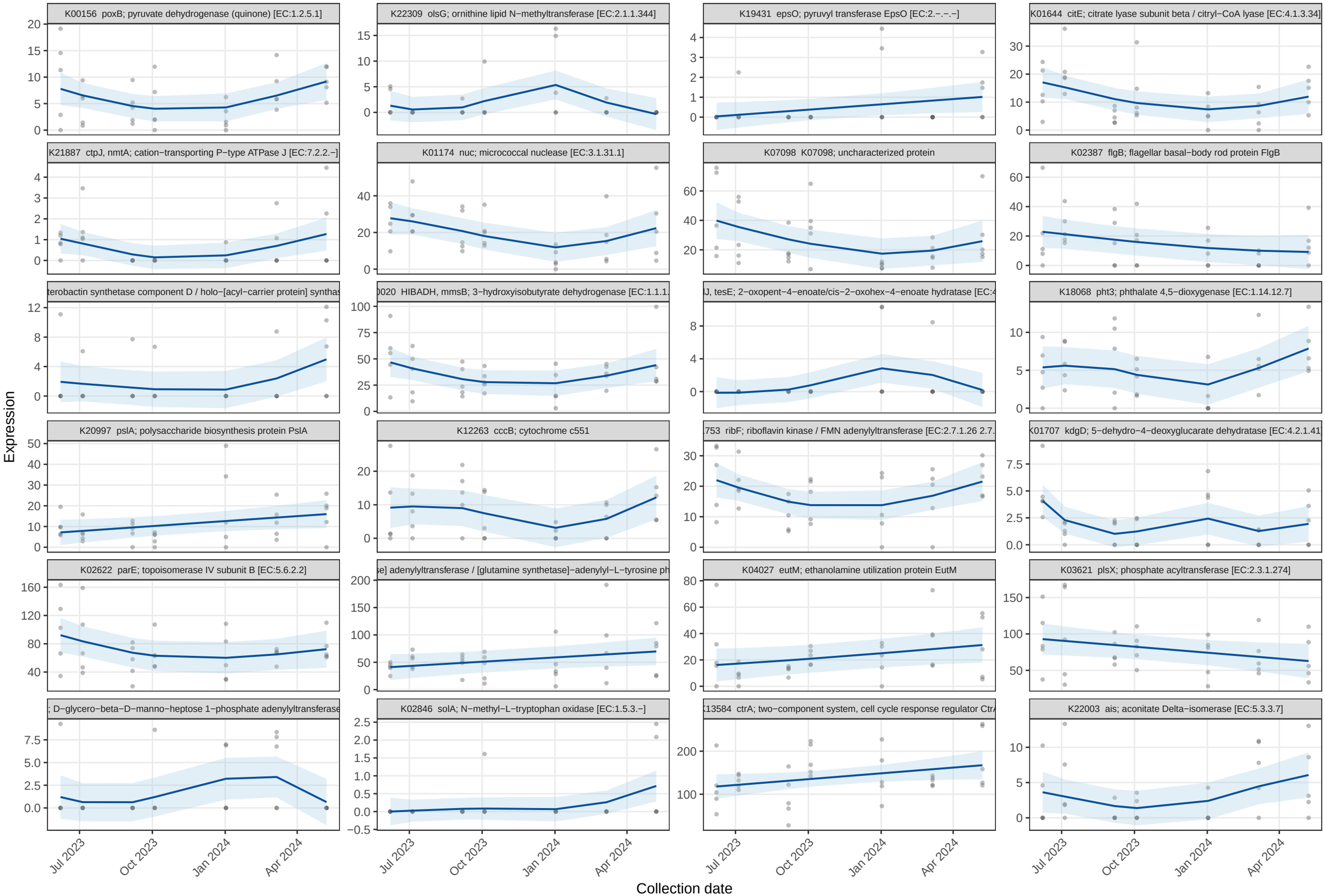
brite_D filtered temporal GAMs: page 43 of 303

All remaining BRITE classes after exclusions



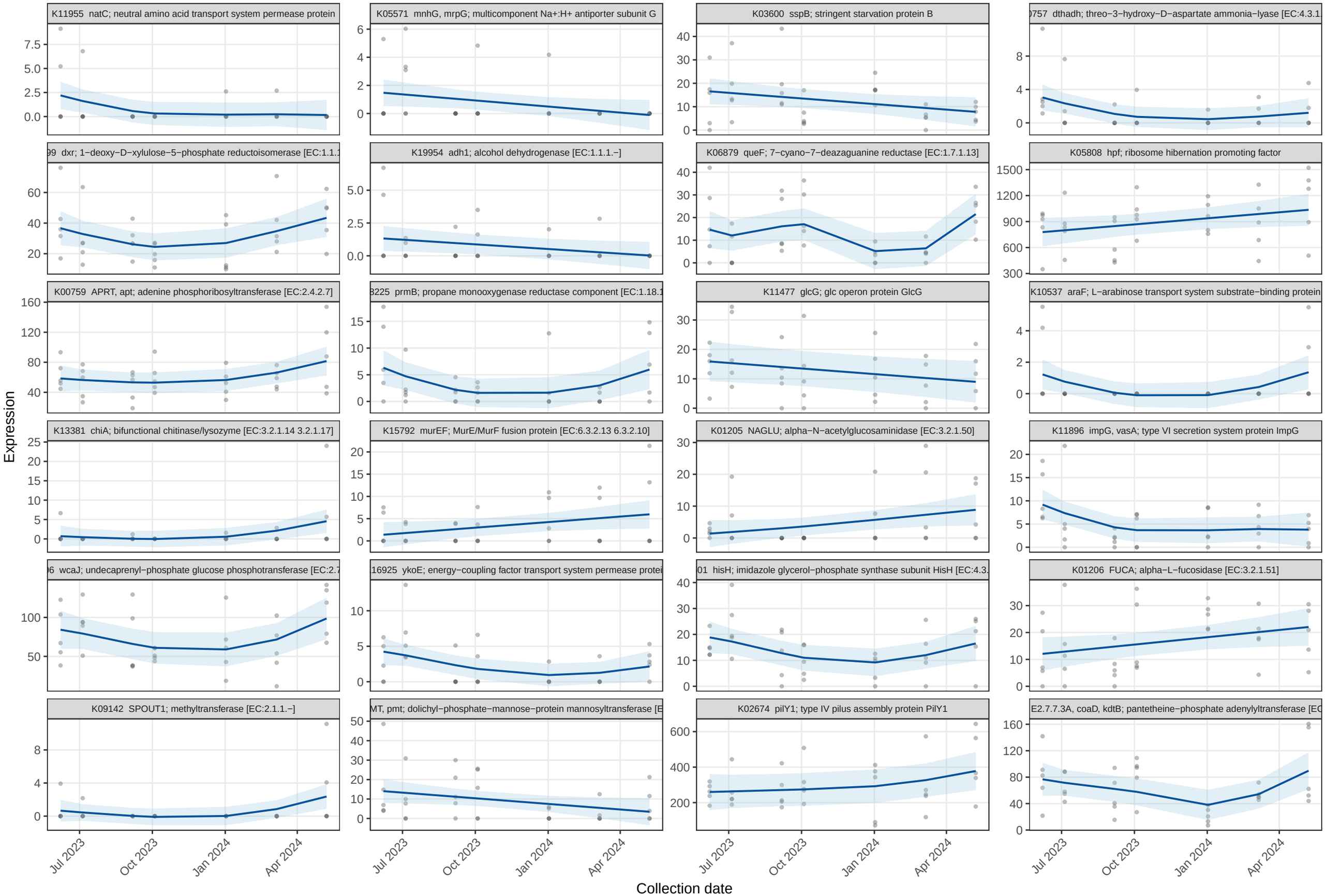
brite_D filtered temporal GAMs: page 44 of 303

All remaining BRITE classes after exclusions



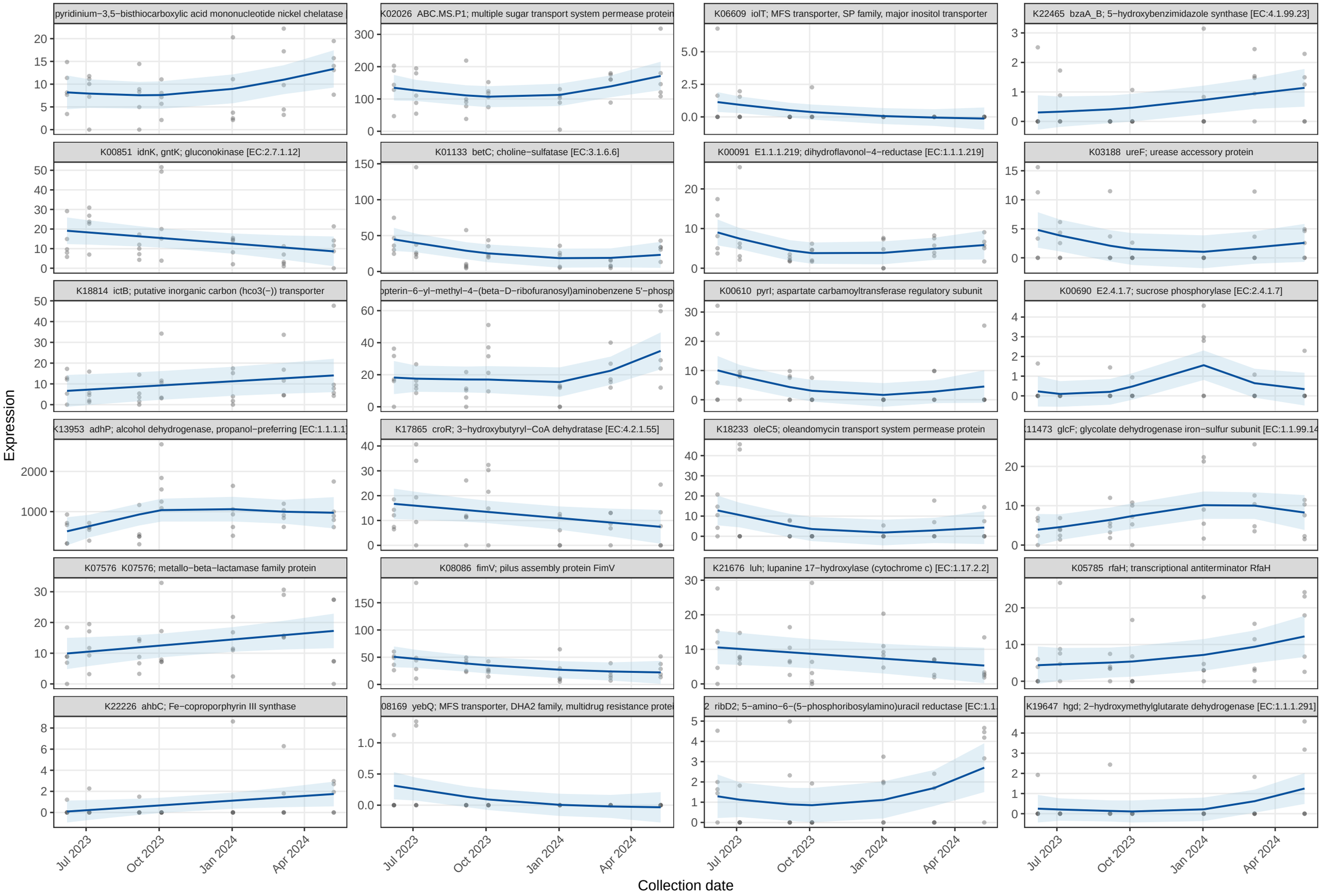
brite_D filtered temporal GAMs: page 45 of 303

All remaining BRITE classes after exclusions

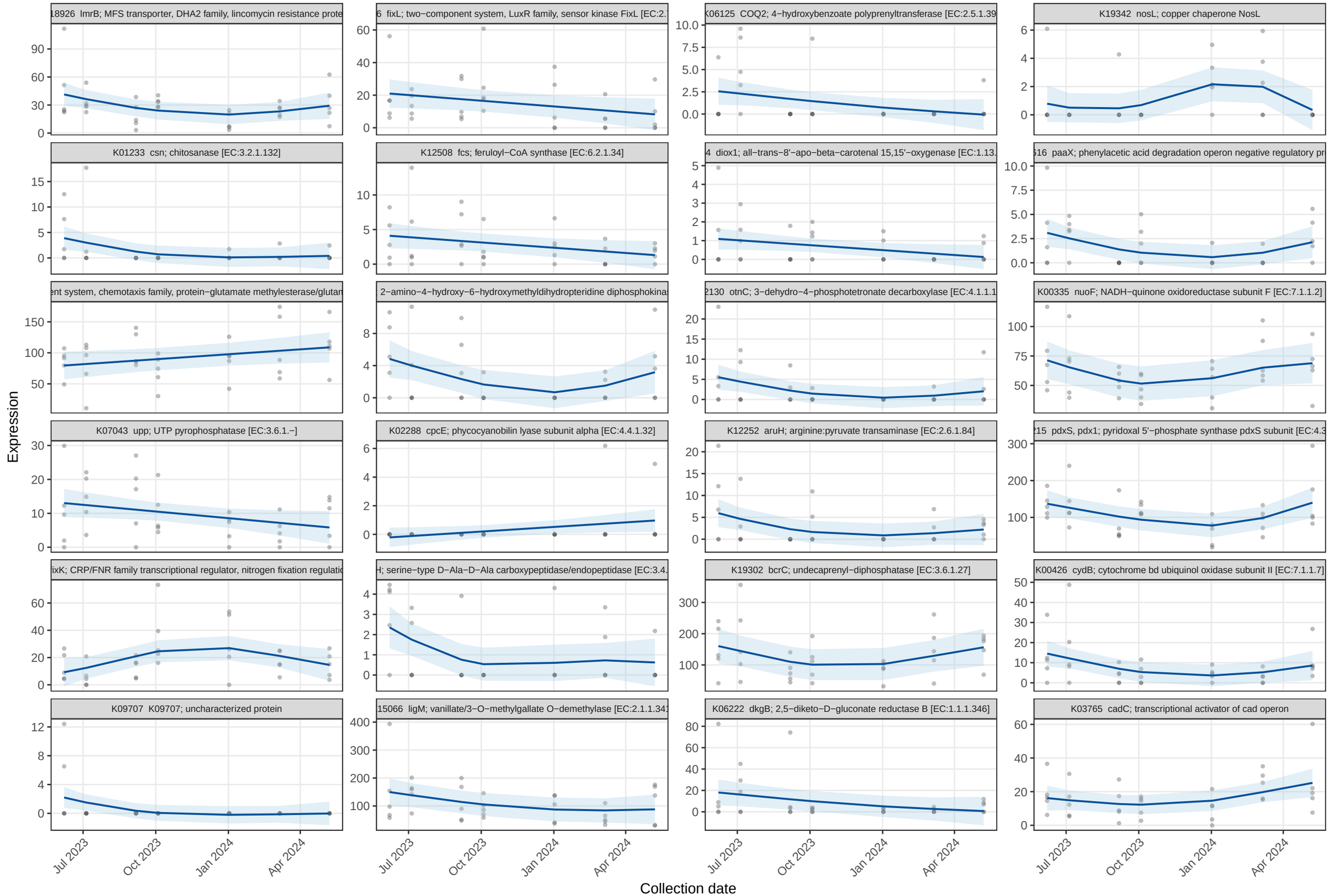


brite_D filtered temporal GAMs: page 46 of 303

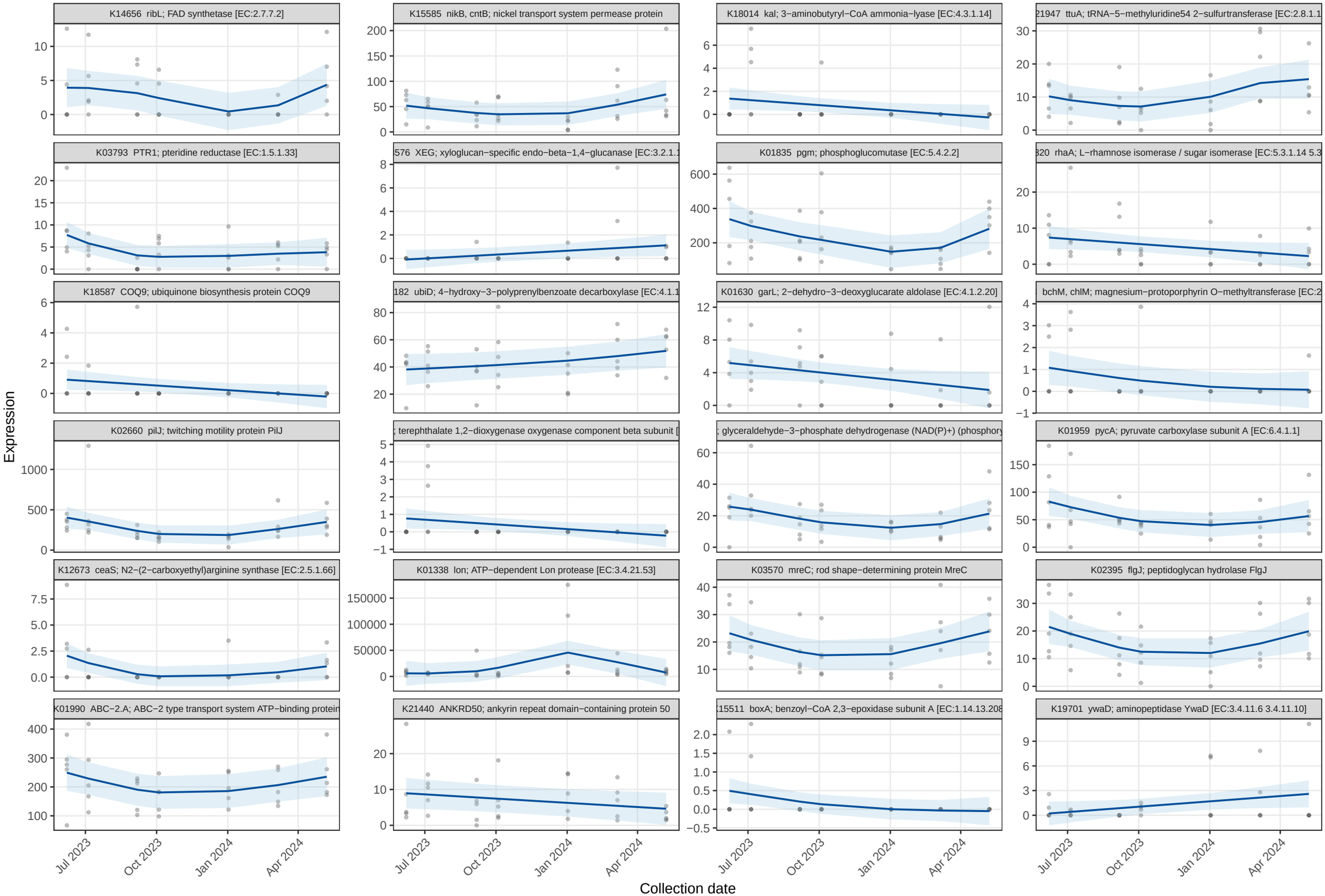
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

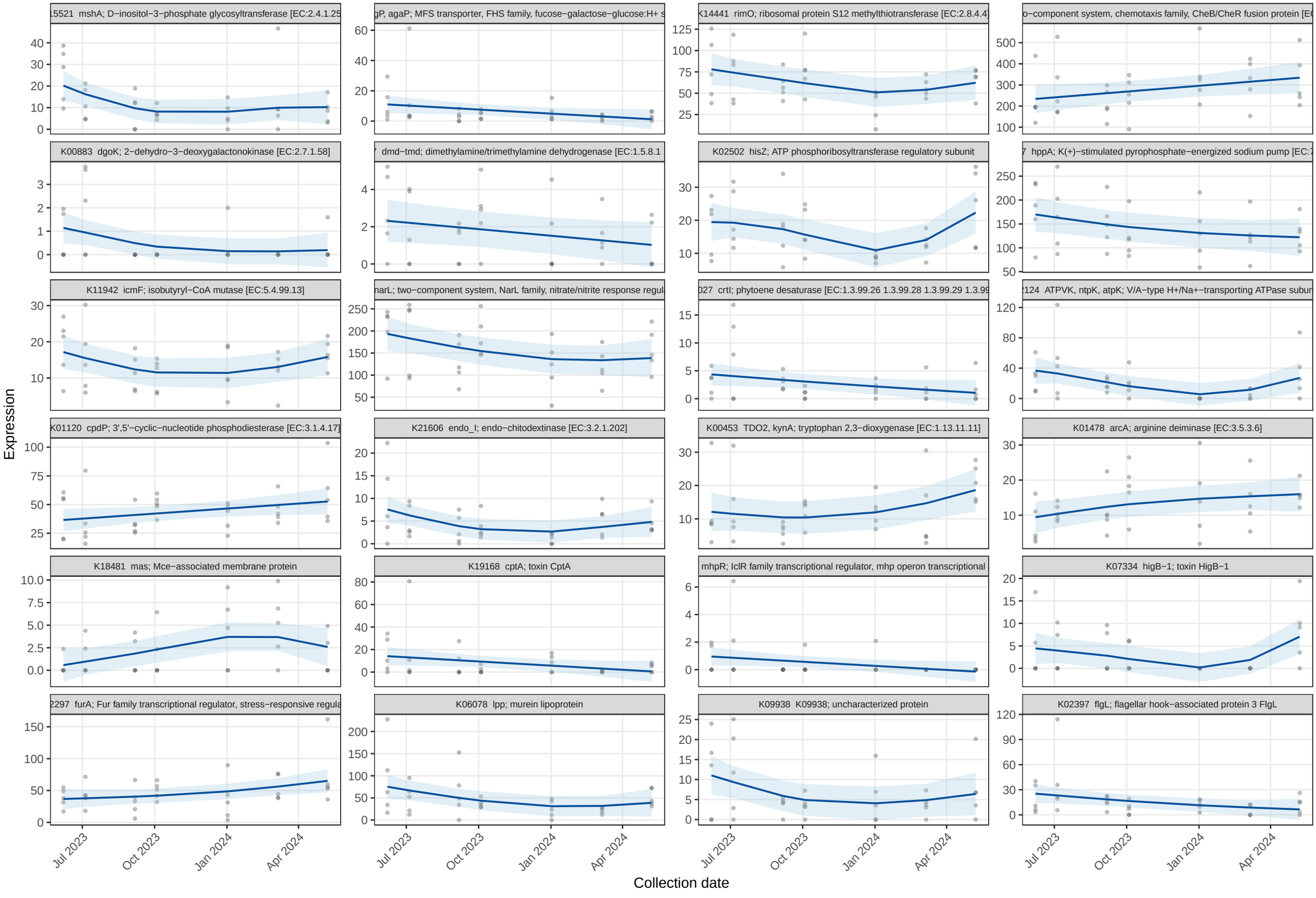


All remaining BRITE classes after exclusions



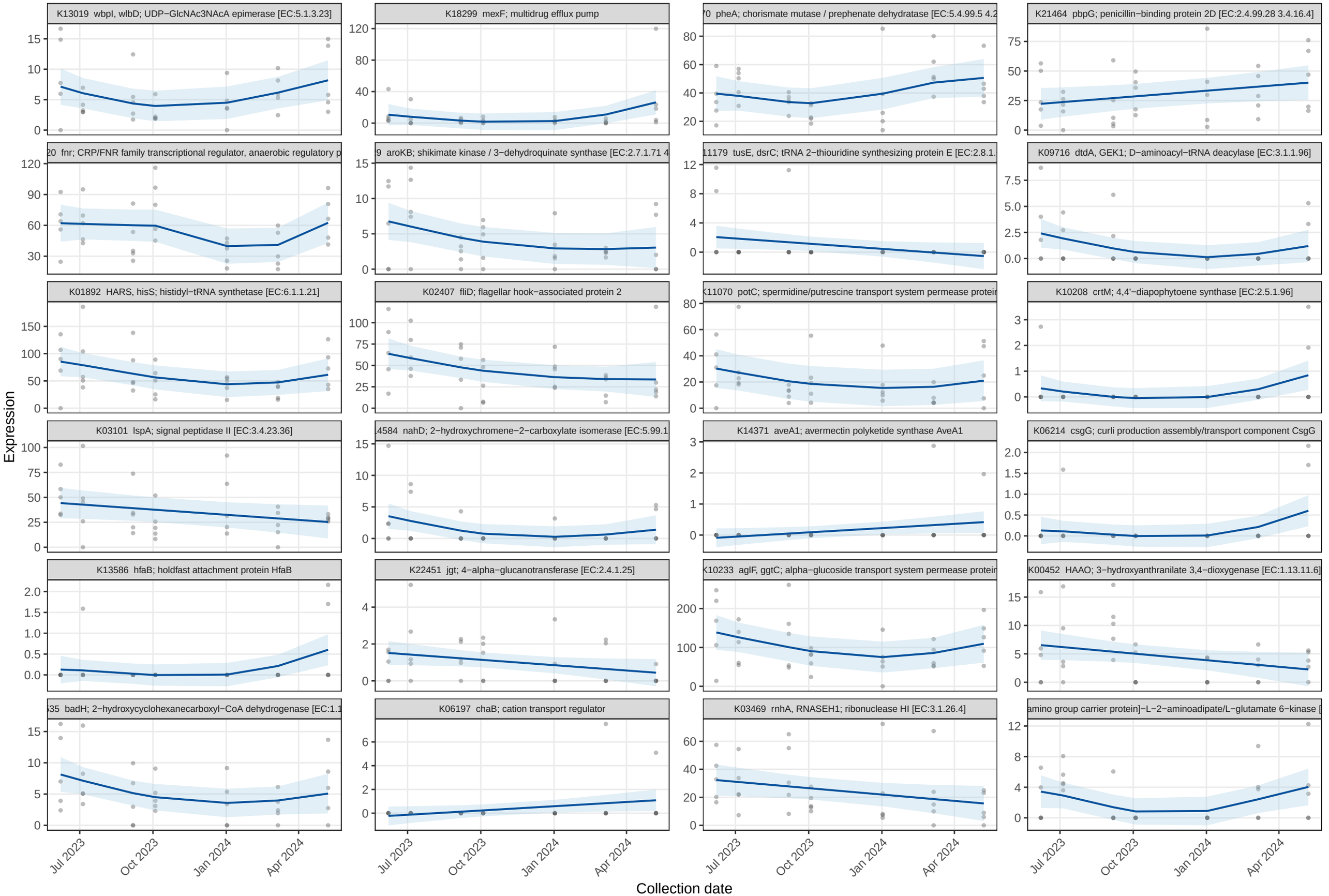
brite_D filtered temporal GAMs: page 49 of 303

All remaining BRITE classes after exclusions



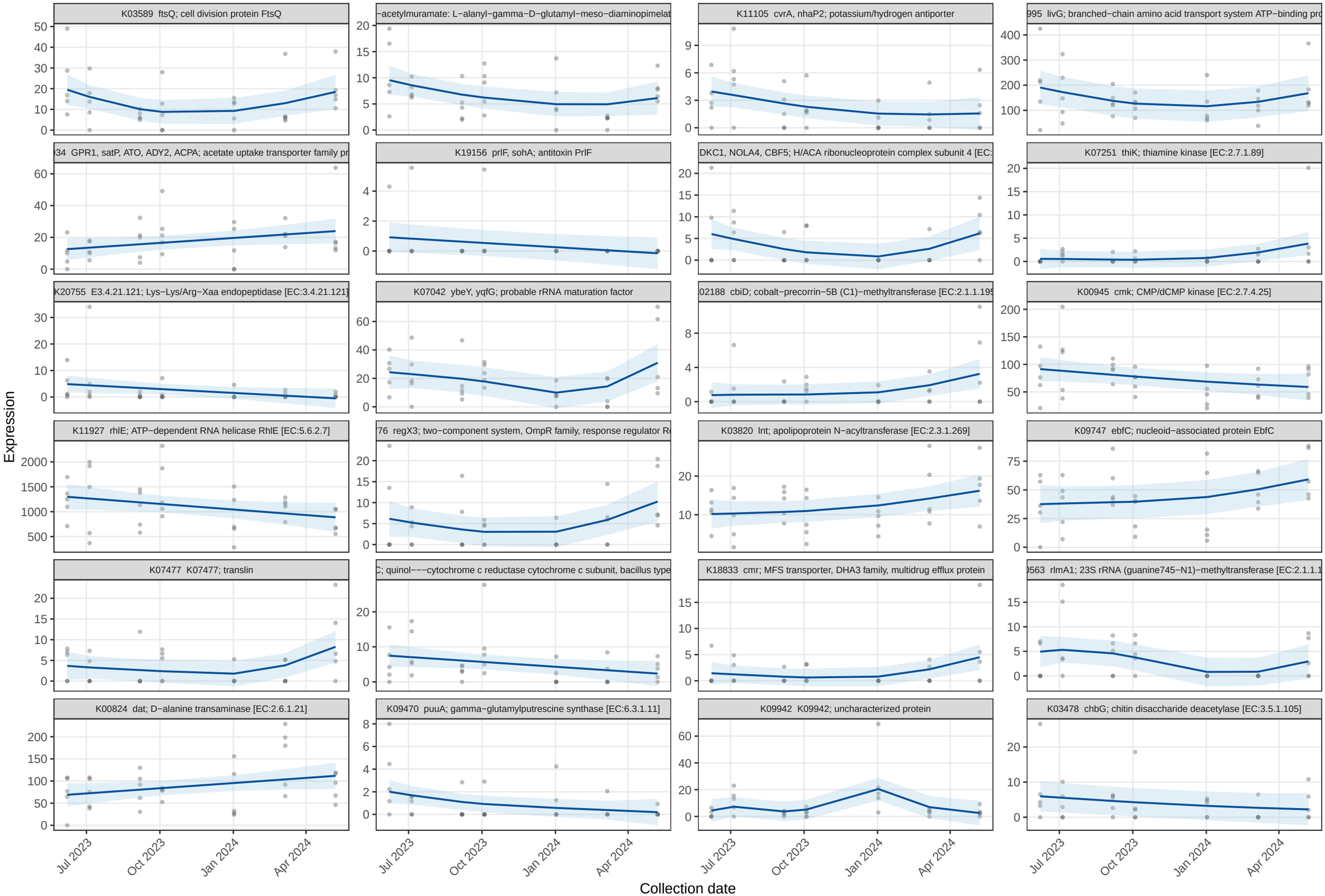
brite_D filtered temporal GAMs: page 50 of 303

All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

colanic acid/amylovoran/stewartan biosynthesis glycosyltransferase Wc

57 tupC, vupC; tungstate transport system ATP−binding protein [EC:7.1

K07569 gar1; RNA−binding protein

14267 dapC; N−succinyldiaminopimelate aminotransferase [EC:2.6.1.1

K01852 LSS, ERG7; lanosterol synthase [EC:5.4.99.7]

K14749 etbAb; ethylbenzene dioxygenase subunit beta [EC:1.14.12.−]

K01464 DPYS, dht, hydA; dihydropyrimidinase [EC:3.5.2.2]

K21005 pslL; polysaccharide biosynthesis protein PslL

K03385 nrfA; nitrite reductase (cytochrome c−552) [EC:1.7.2.2]

K09011 cimA; (R)−citramalate synthase [EC:2.3.3.21]

6844 comC; (2R)−3−sulfolactate dehydrogenase (NADP+) [EC:1.1.1.3

K06138 pqqD; pyrroloquinoline quinone biosynthesis protein D

K03770 ppiD; peptidyl−prolyl cis−trans isomerase D [EC:5.2.1.8]

K02314 dnaB; replicative DNA helicase [EC:5.6.2.3]

K14162 dnaE2; error−prone DNA polymerase [EC:2.7.7.7]

K09804 K09804; uncharacterized protein

K09889 yjgA; ribosome−associated protein

K15554 ssuC; sulfonate transport system permease protein

K00855 PRK, prk; phosphoribulokinase [EC:2.7.1.19]

108 malE; maltose/maltodextrin transport system substrate−binding pro

0824 nikE, cntF; nickel transport system ATP−binding protein [EC:7.2.2

K07465 K07465; putative RecB family exonuclease

G; R−phycocyanin alpha−cysteine−84 phycourobilin lyase/isomerase [E

10017 hisP; histidine transport system ATP−binding protein [EC:7.4.2.1

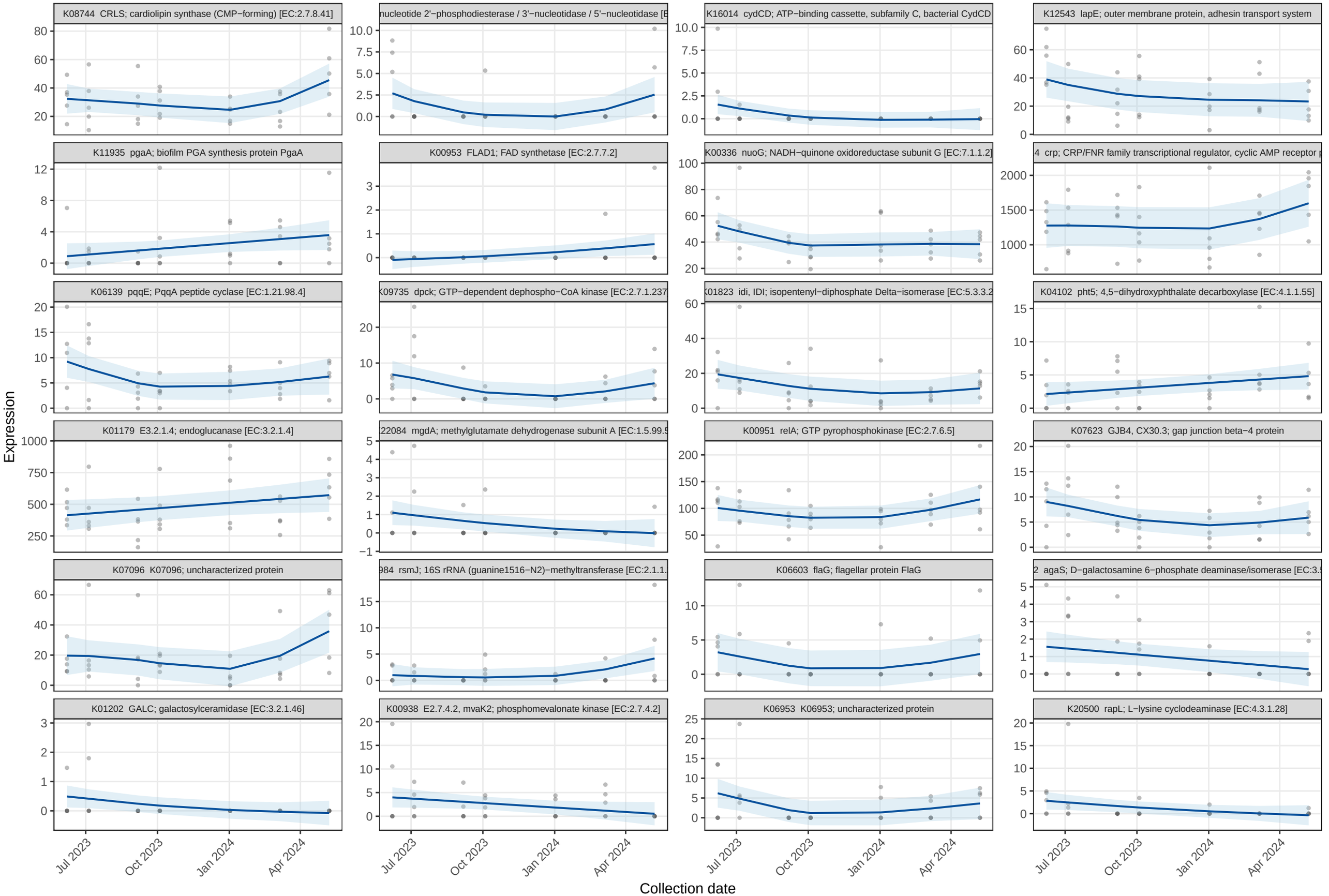
Expression

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All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

K00554 trmD; tRNA (guanine37−N1)−methyltransferase [EC:2.1.1.228]

K09800 tamB; translocation and assembly module TamB

K01197 hya; hyaluronoglucosaminidase [EC:3.2.1.35]

K00366 nirA; ferredoxin−nitrite reductase [EC:1.7.7.1]

−hydroxy acid dehydrogenase / malonic semialdehyde reductase [EC:1.

K00872 thrB; homoserine kinase [EC:2.7.1.39]

B; two−component system, OmpR family, sensor histidine kinase MtrB [

K09792 K09792; uncharacterized protein

K02195 ccmC; heme exporter protein C

K02458 gspI; general secretion pathway protein I

K11930 torT; periplasmic protein TorT

fabH; mycobacterial beta−ketoacyl−[acyl−carrier protein] synthase III [EC

K14677 ACY1; aminoacylase [EC:3.5.1.14]

K07175 phoH2; PhoH−like ATPase

K09990 K09990; uncharacterized protein

K09895 K09895; uncharacterized protein

K06934 K06934; uncharacterized protein

K18285 mqnE; aminodeoxyfutalosine synthase [EC:2.5.1.120]

039 ychN; uncharacterized protein involved in oxidation of intracellular s

K01247 alkA; DNA−3−methyladenine glycosylase II [EC:3.2.2.21]

21731 camP; 2,5−diketocamphane 1,2−monooxygenase [EC:1.14.14.10

K03978 engB; GTP−binding protein

paaG; 2−(1,2−epoxy−1,2−dihydrophenyl)acetyl−CoA isomerase [EC:5

K10234 aglG, ggtD; alpha−glucoside transport system permease protein

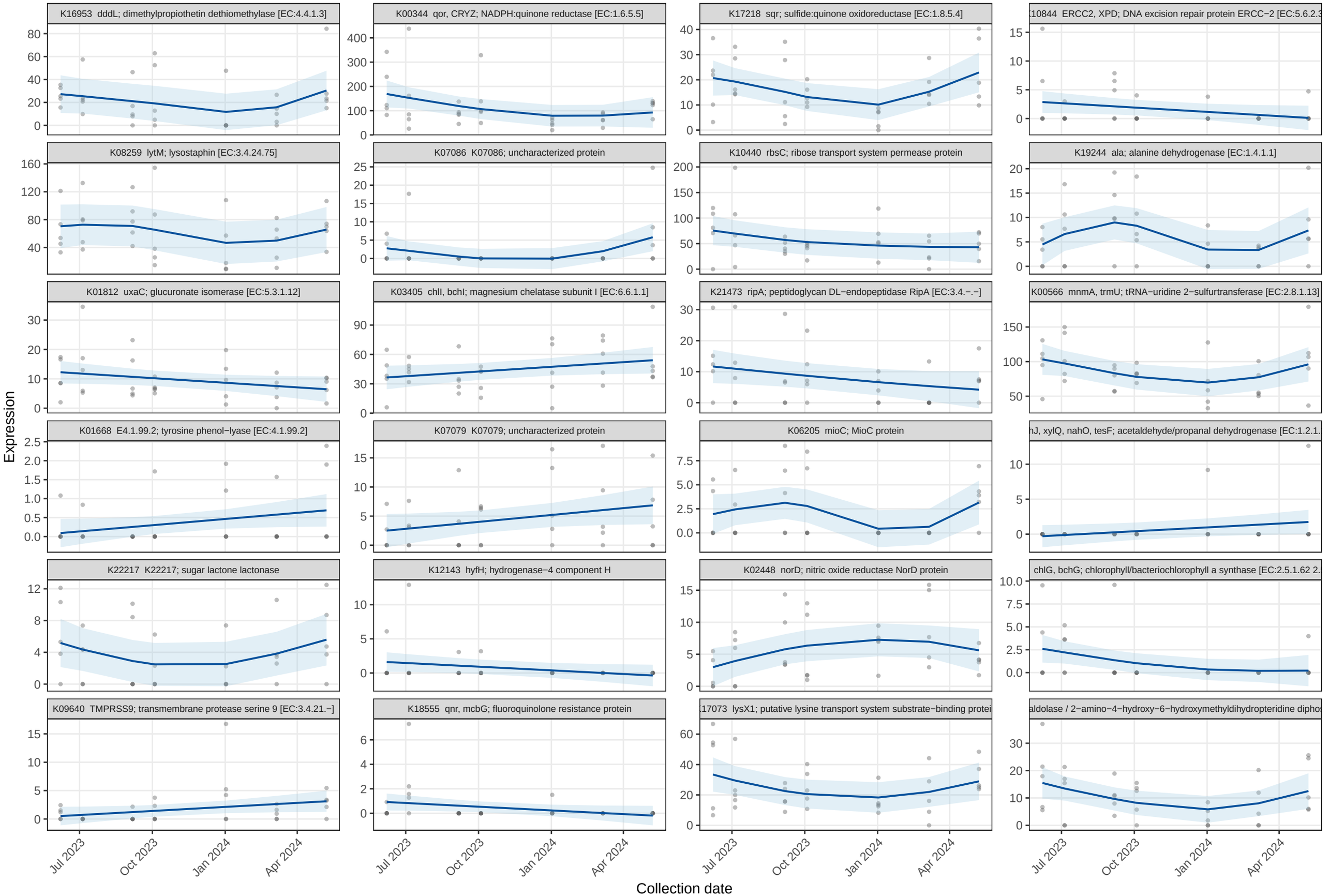
Expression

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Collection date

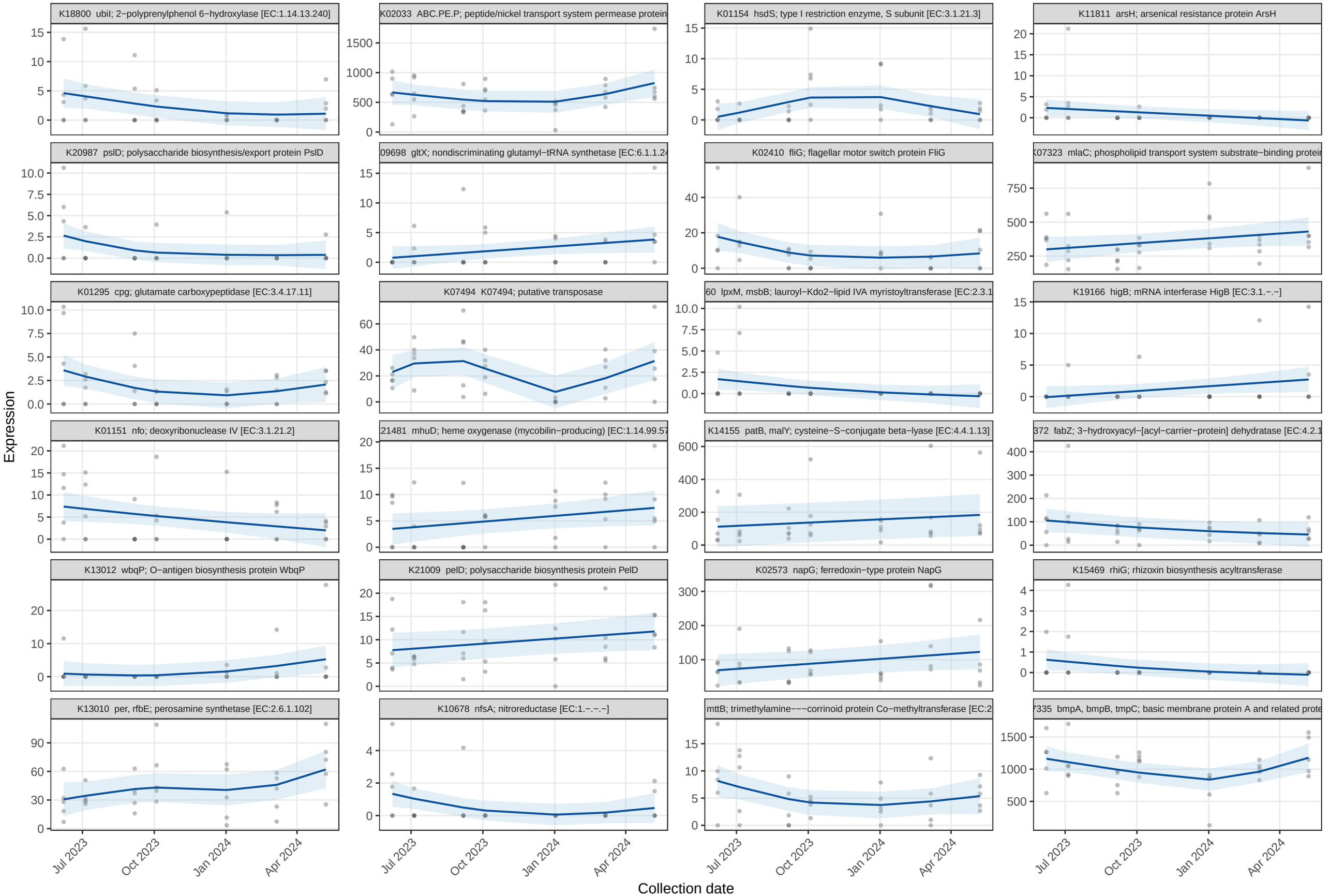
brite_D filtered temporal GAMs: page 55 of 303

All remaining BRITE classes after exclusions



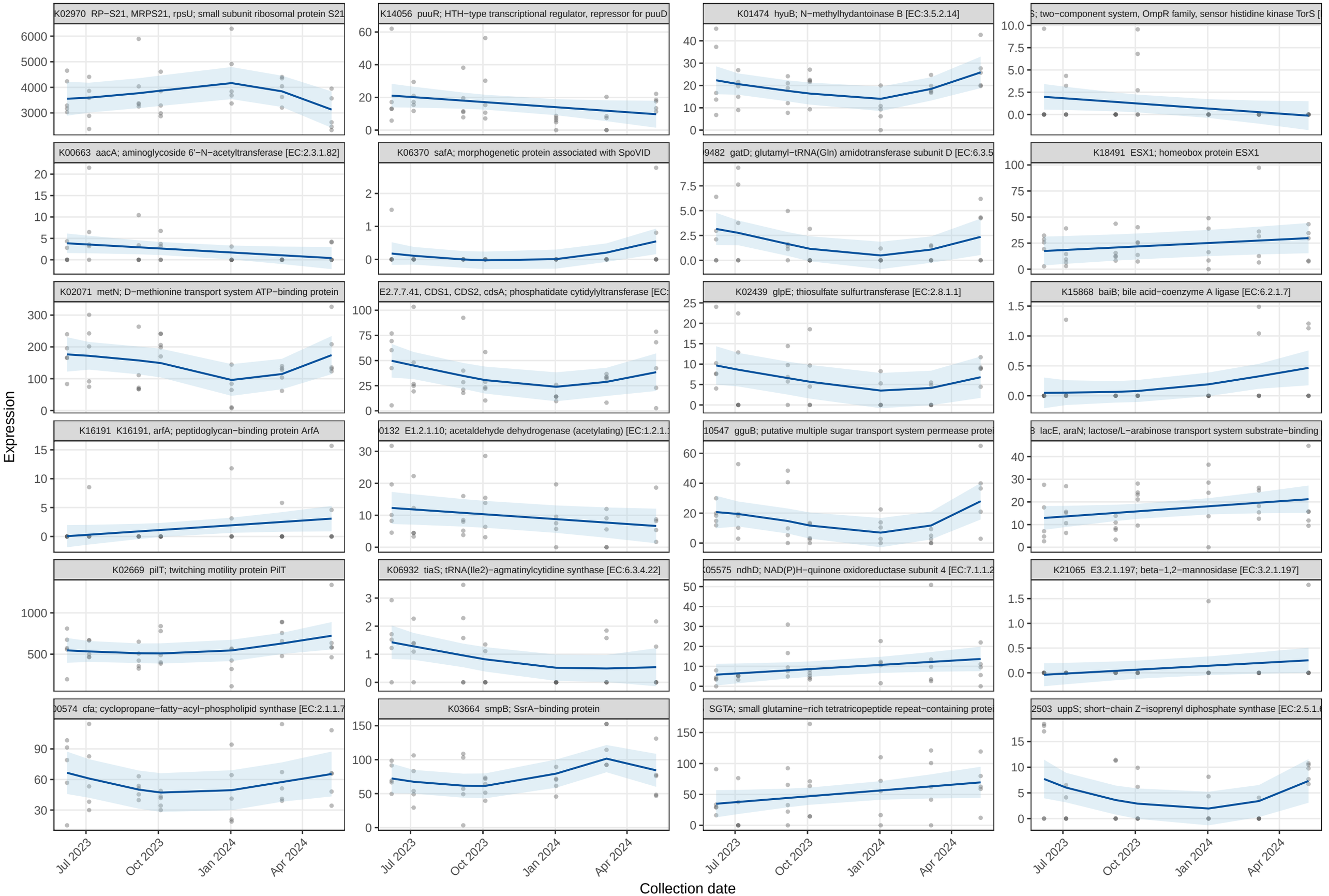
brite_D filtered temporal GAMs: page 56 of 303

All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 58 of 303

All remaining BRITE classes after exclusions

hemK, prmC, HEMK; release factor glutamine methyltransferase [EC:2.

K11894 impI, vasC; type VI secretion system protein ImpI

K00009 mtlD; mannitol−1−phosphate 5−dehydrogenase [EC:1.1.1.17]

K20274 zmpB; zinc metalloprotease ZmpB

ra5; 2−dehydro−3−deoxy−L−rhamnonate dehydrogenase (NAD+) [EC:1

K15945 snoaL2; C−1 hydroxylase

K20972 retS; two−component system, sensor histidine kinase RetS

K11739 nfrA; bacteriophage N4 adsorption protein A

K09817 znuC; zinc transport system ATP−binding protein [EC:7.2.2.20]

K07483 K07483; transposase

K03683 rnt; ribonuclease T [EC:3.1.13.−]

057 gpsA; glycerol−3−phosphate dehydrogenase (NAD(P)+) [EC:1.1.1

K09711 lipS1; lipoyl synthase

pmm−pgm; phosphomannomutase / phosphoglucomutase [EC:5.4.2.8

K01341 KEX2; kexin [EC:3.4.21.61]

03 fwdD, fmdD; formylmethanofuran dehydrogenase subunit D [EC:1.2.

K00997 acpS; holo−[acyl−carrier protein] synthase [EC:2.7.8.7]

24 pcaR; IclR family transcriptional regulator, pca regulon regulatory pr

K03226 yscR, sctR, hrcR, ssaR; type III secretion protein R

K00756 pdp; pyrimidine−nucleoside phosphorylase [EC:2.4.2.2]

K08994 yneE, BEST; ion channel−forming bestrophin family protein

K07152 SCO1; protein SCO1

; phosphoenolpyruvate−−−glycerone phosphotransferase subunit DhaM

K07120 K07120; uncharacterized protein

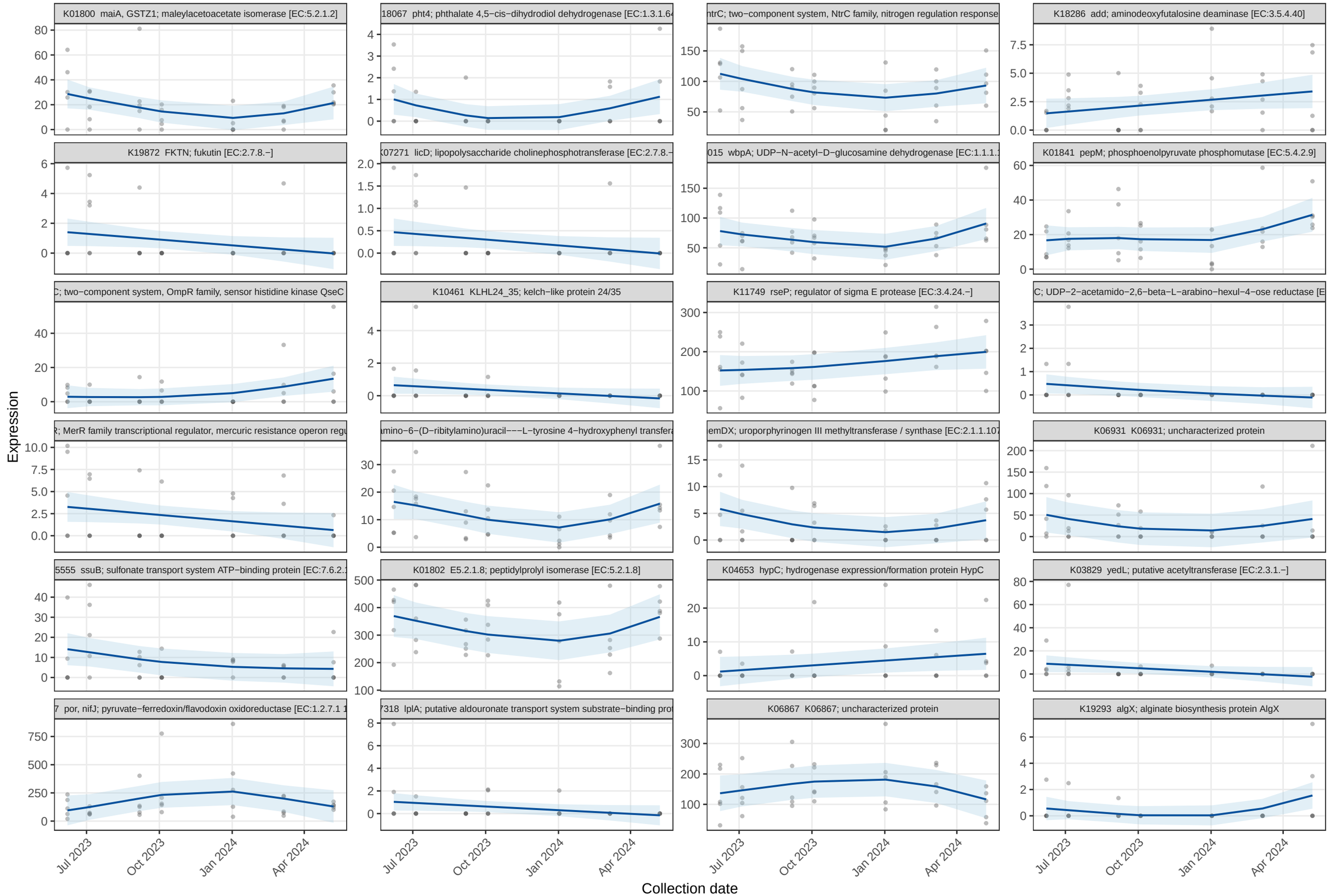
Expres

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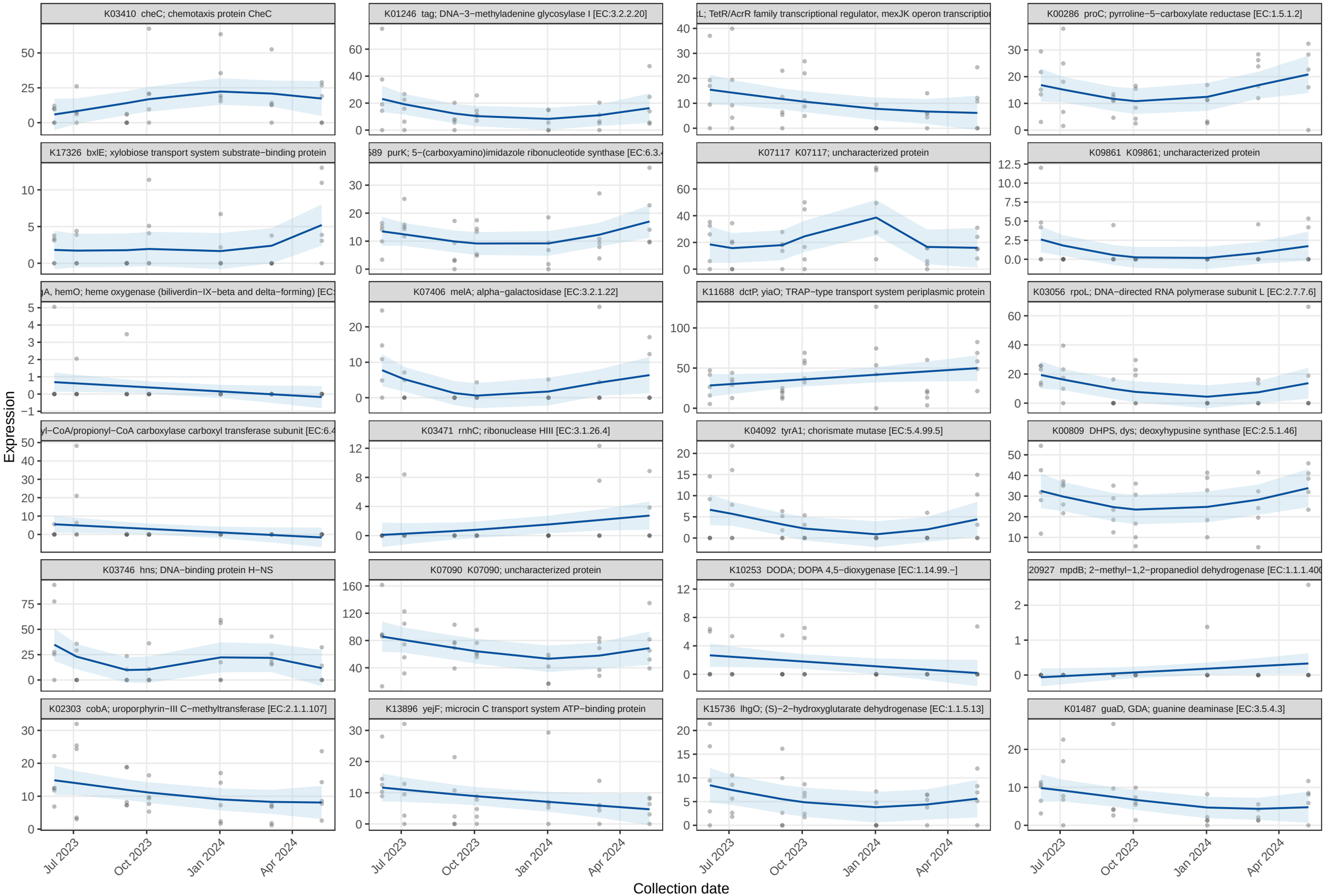
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All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions

K03118 tatC; sec−independent protein translocase protein TatC

K01694 TRP; tryptophan synthase [EC:4.2.1.20]

G, curD, aprE, corE; 3−carboxymethyl−3−hydroxy−acyl−[acp] synthase

K01465 URA4, pyrC; dihydroorotase [EC:3.5.2.3]

0 aauS; two−component system, sensor histidine kinase AauS [EC:2.7

formylmethanofuran−−tetrahydromethanopterin N−formyltransferase [E

A−dnaQ; ribonuclease HI / DNA polymerase III subunit epsilon [EC:3.1.

UDP−2−acetamido−3−amino−2,3−dideoxy−glucuronate N−acetyltrans

K00631 plsB; glycerol−3−phosphate O−acyltransferase [EC:2.3.1.15]

K01185 E3.2.1.17; lysozyme [EC:3.2.1.17]

K03641 tolB; TolB protein

asA; meromycolic acid 3−oxoacyl−[acyl−carrier protein] synthase I [EC:

K11210 fosB; metallothiol transferase [EC:2.5.1.−]

K01305 iadA; beta−aspartyl−dipeptidase (metallo−type) [EC:3.4.19.−]

glyoxylate/hydroxypyruvate/2−ketogluconate reductase [EC:1.1.1.79 1.1

K01512 acyP; acylphosphatase [EC:3.6.1.7]

K05581 ndhJ; NAD(P)H−quinone oxidoreductase subunit J [EC:7.1.1.2

K09977 K09977; uncharacterized protein

psT; capsular polysaccharide transport system ATP−binding protein [EC

K08884 K08884; serine/threonine protein kinase, bacterial [EC:2.7.11.1

K03321 TC.SULP; sulfate permease, SulP family

K19341 nosY; Cu−processing system permease protein

K02496 hemX; uroporphyrin−III C−methyltransferase [EC:2.1.1.107]

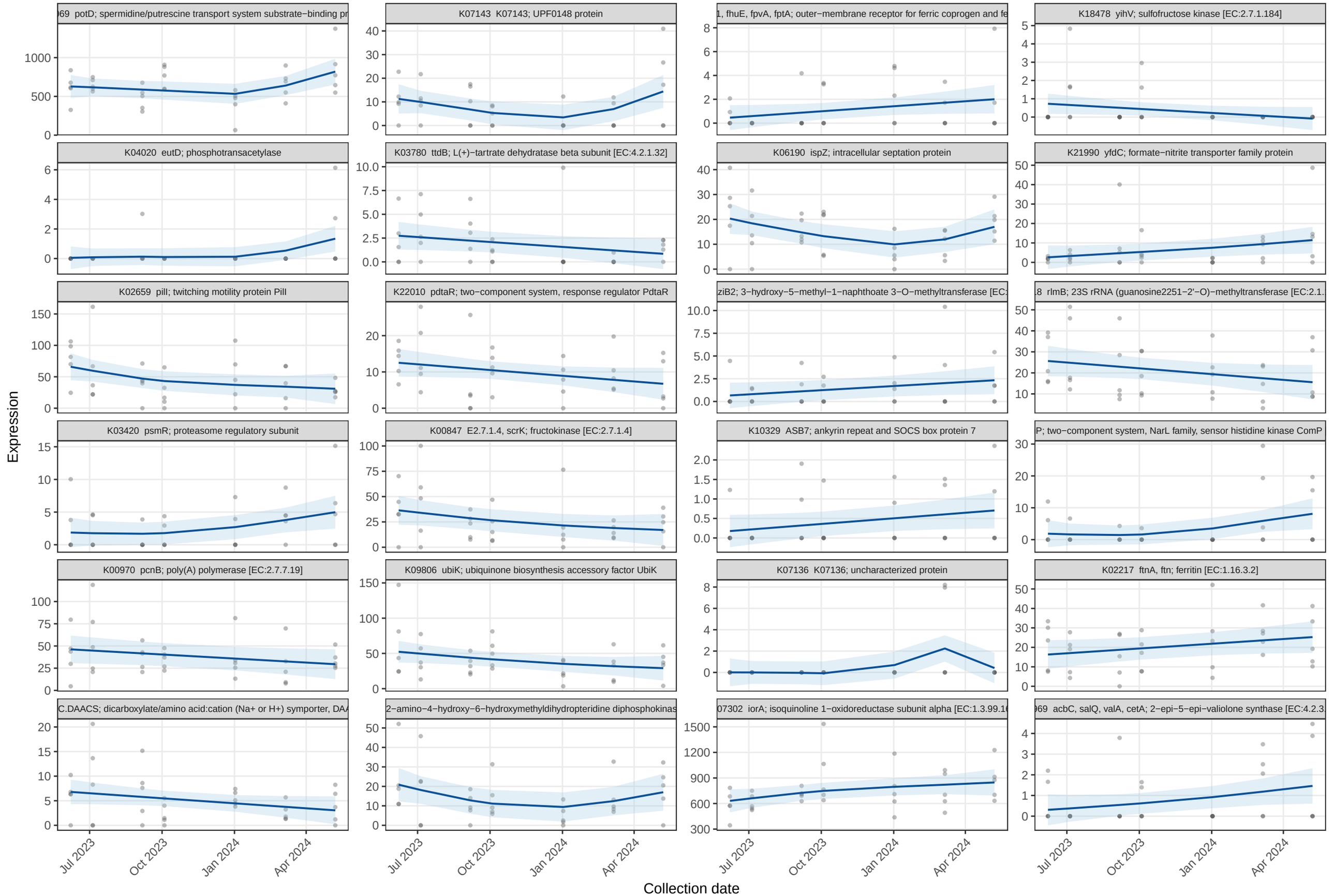
K02470 gyrB; DNA gyrase subunit B [EC:5.6.2.2]

Expression

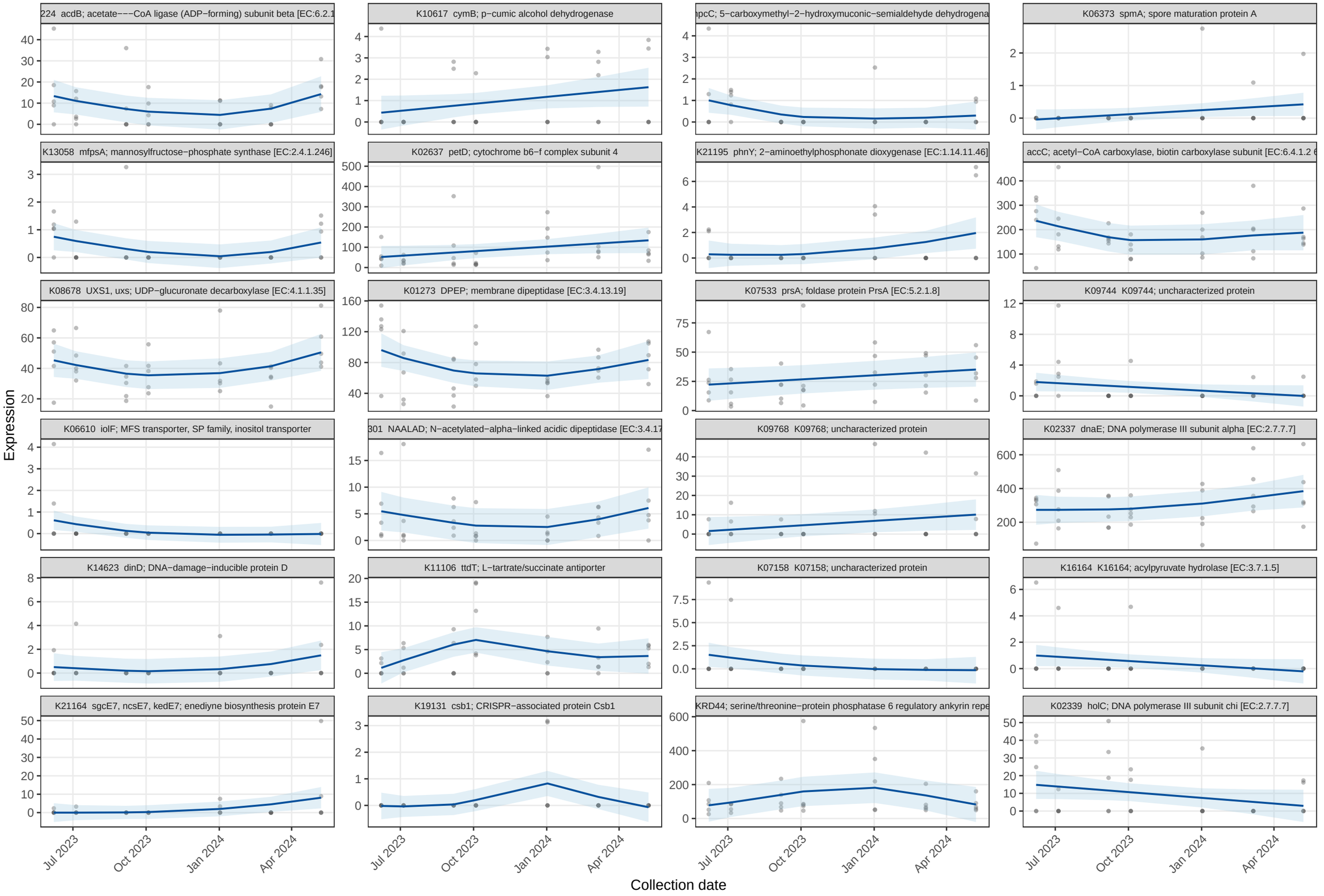
Jul 2023 Oct 2023 Jan 2024 Apr 2024

Collection date

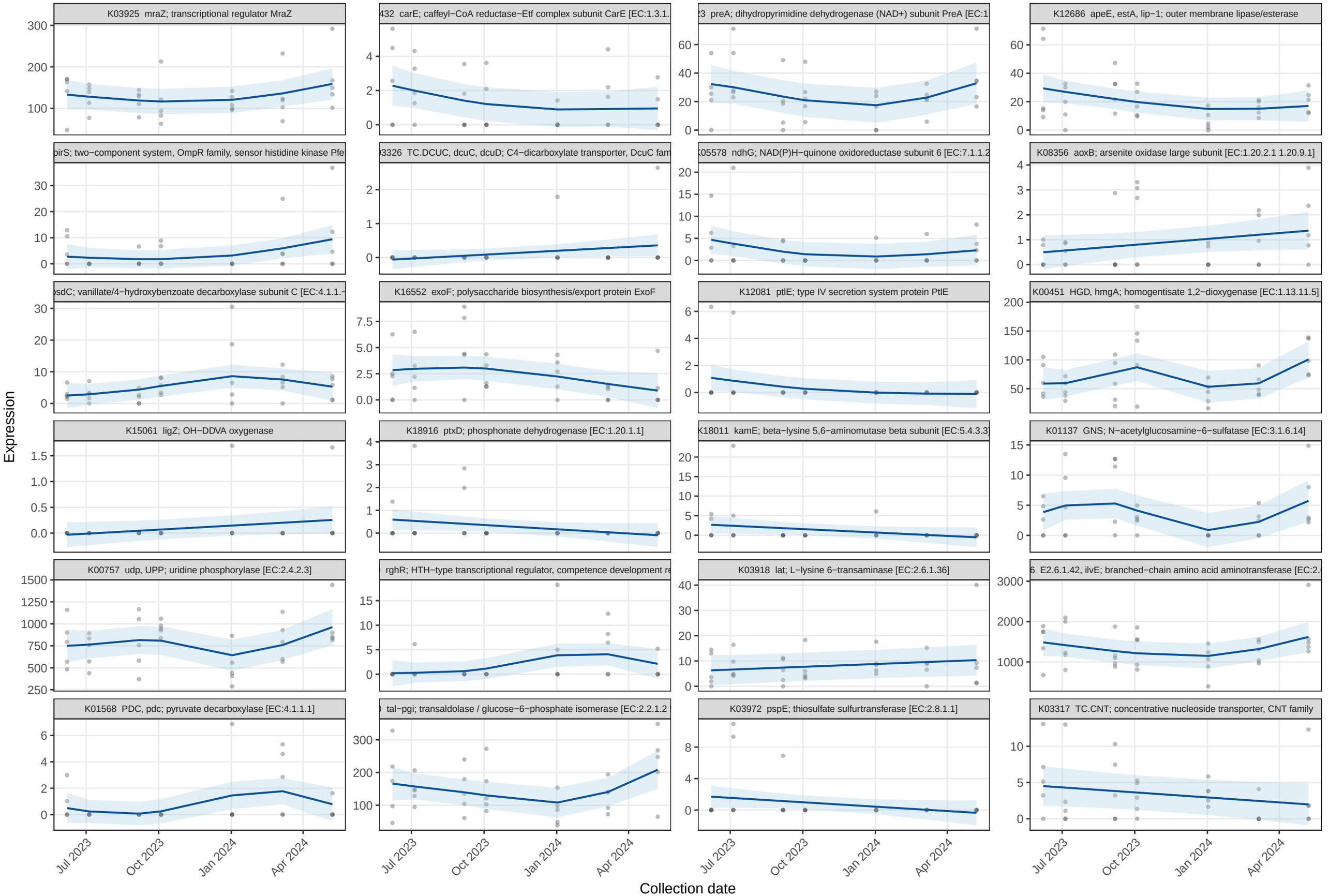
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

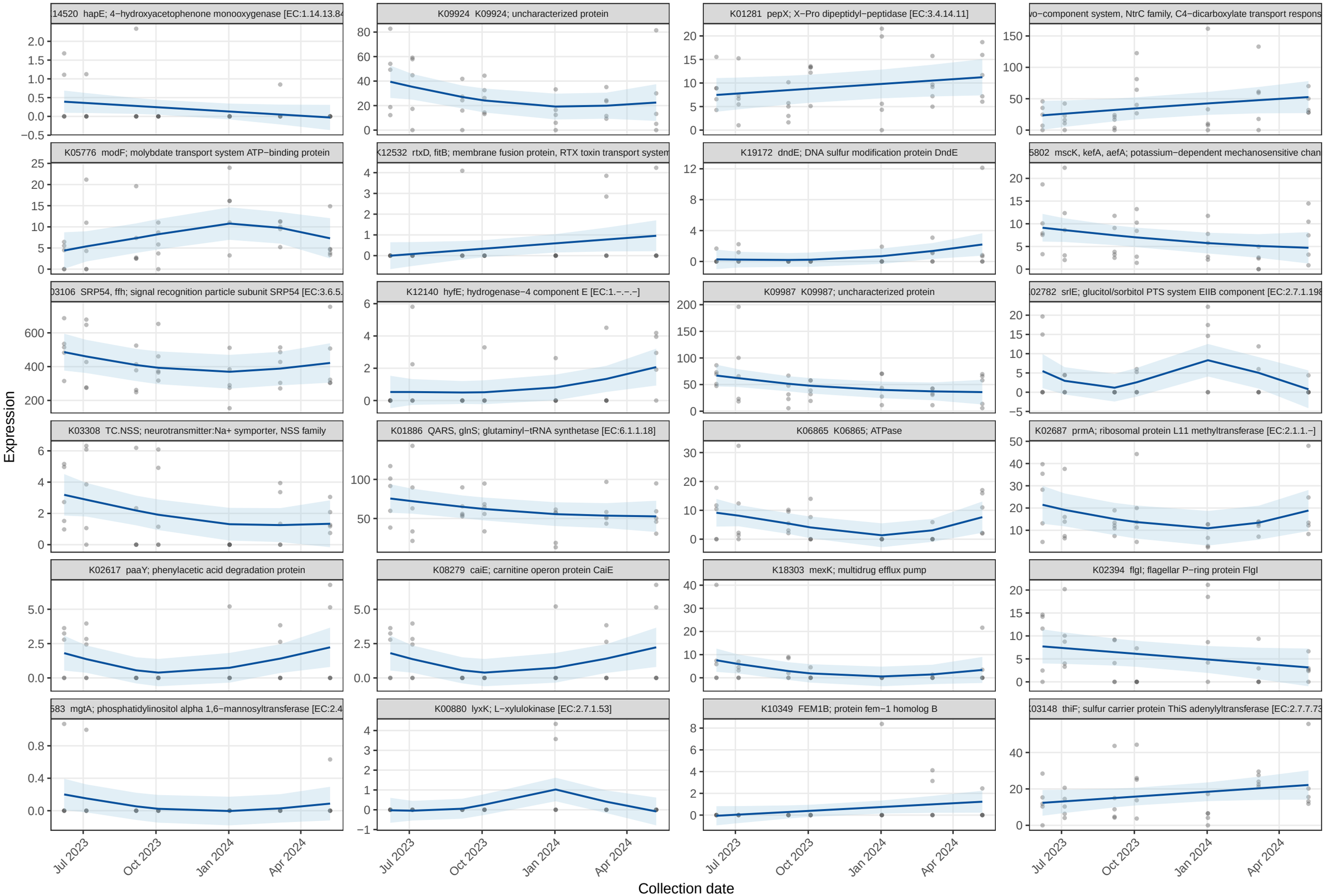


All remaining BRITE classes after exclusions



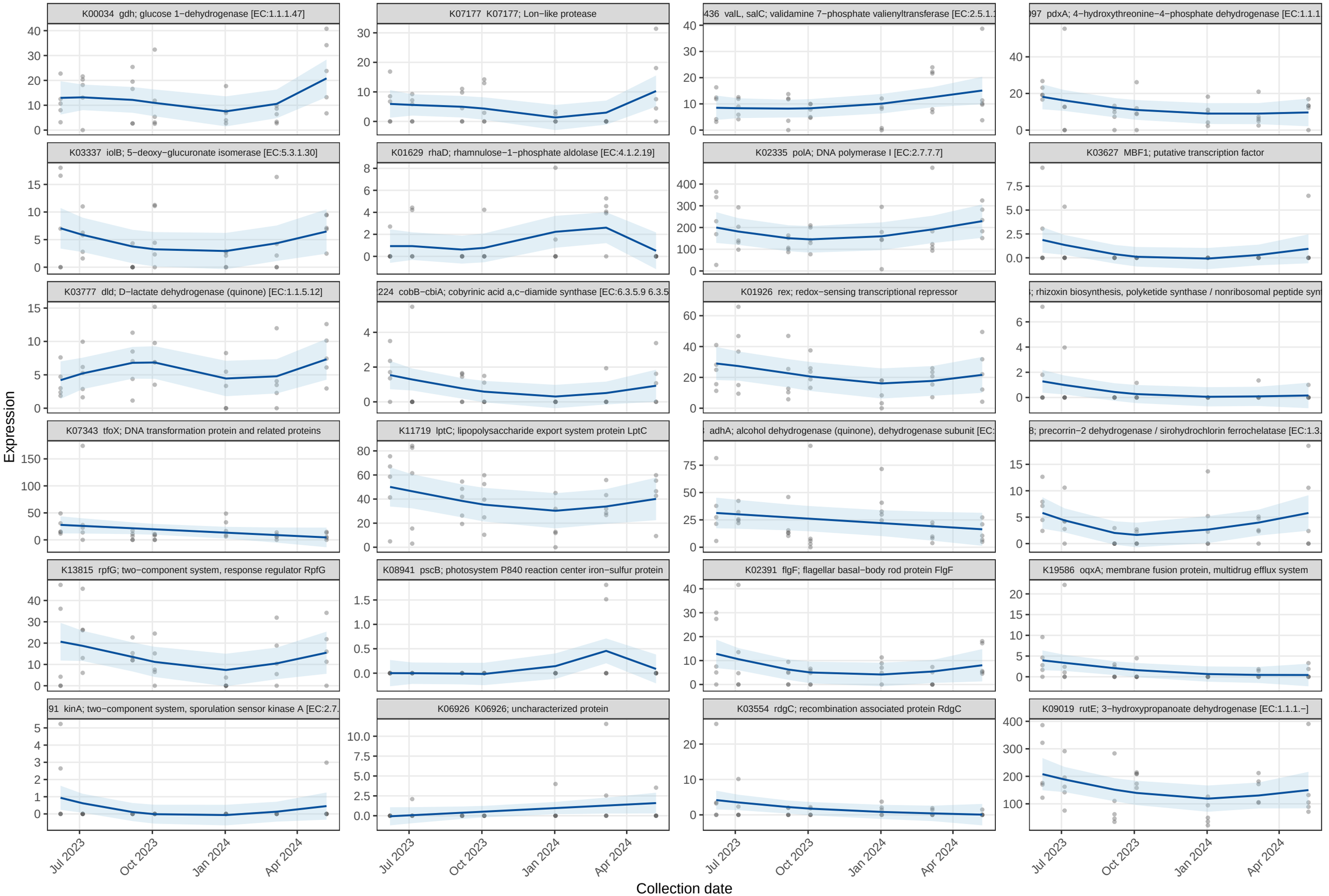
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All remaining BRITE classes after exclusions

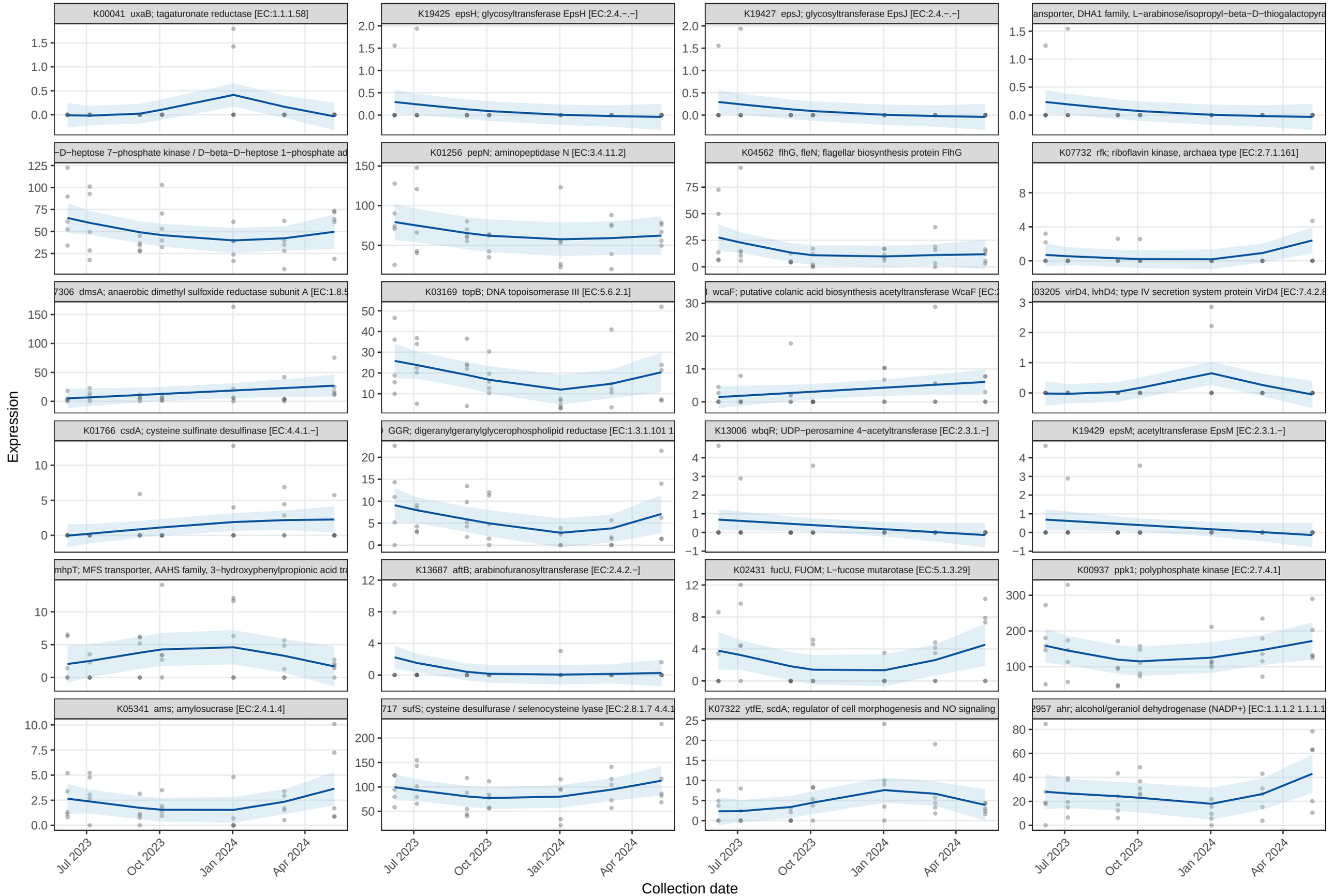


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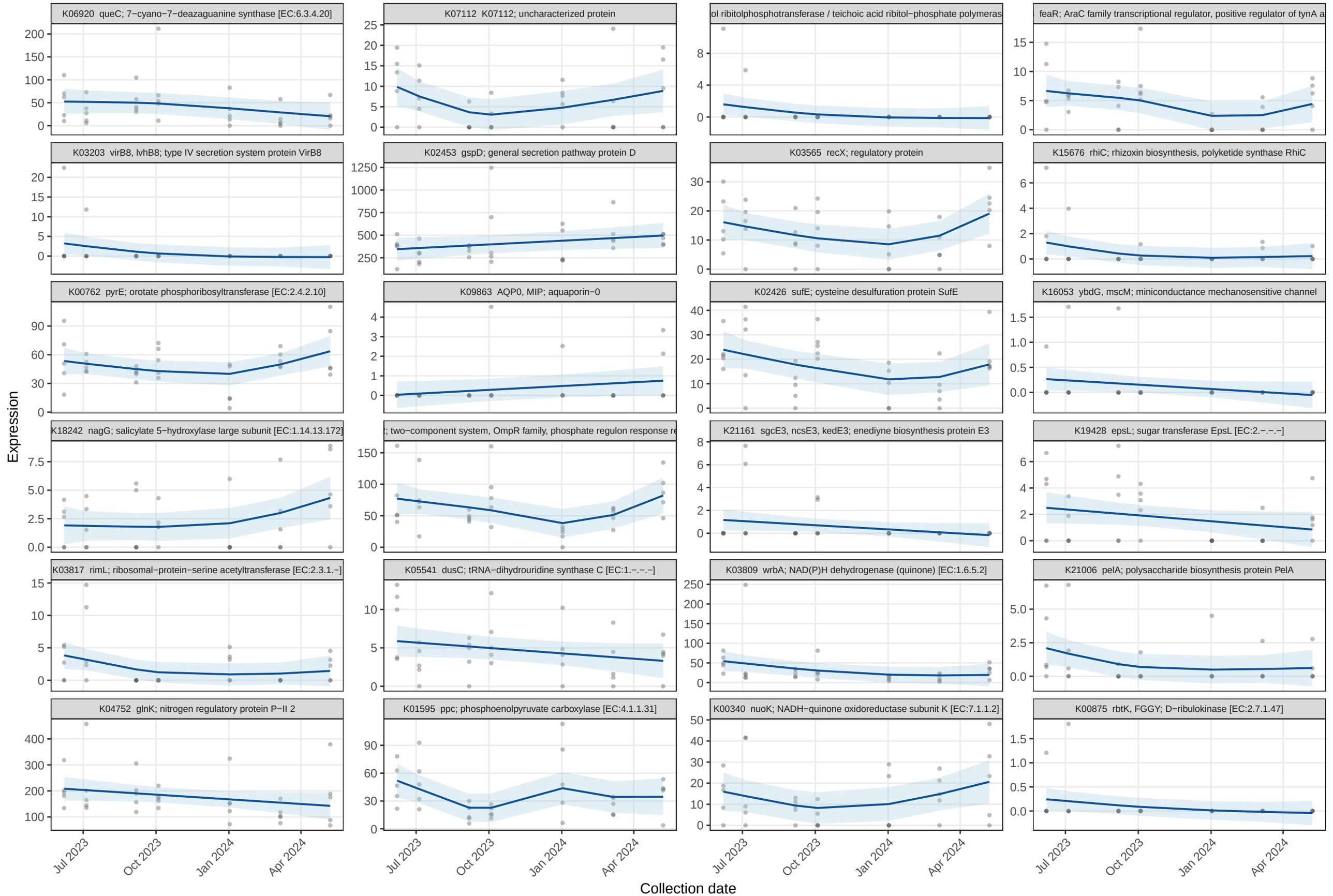
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

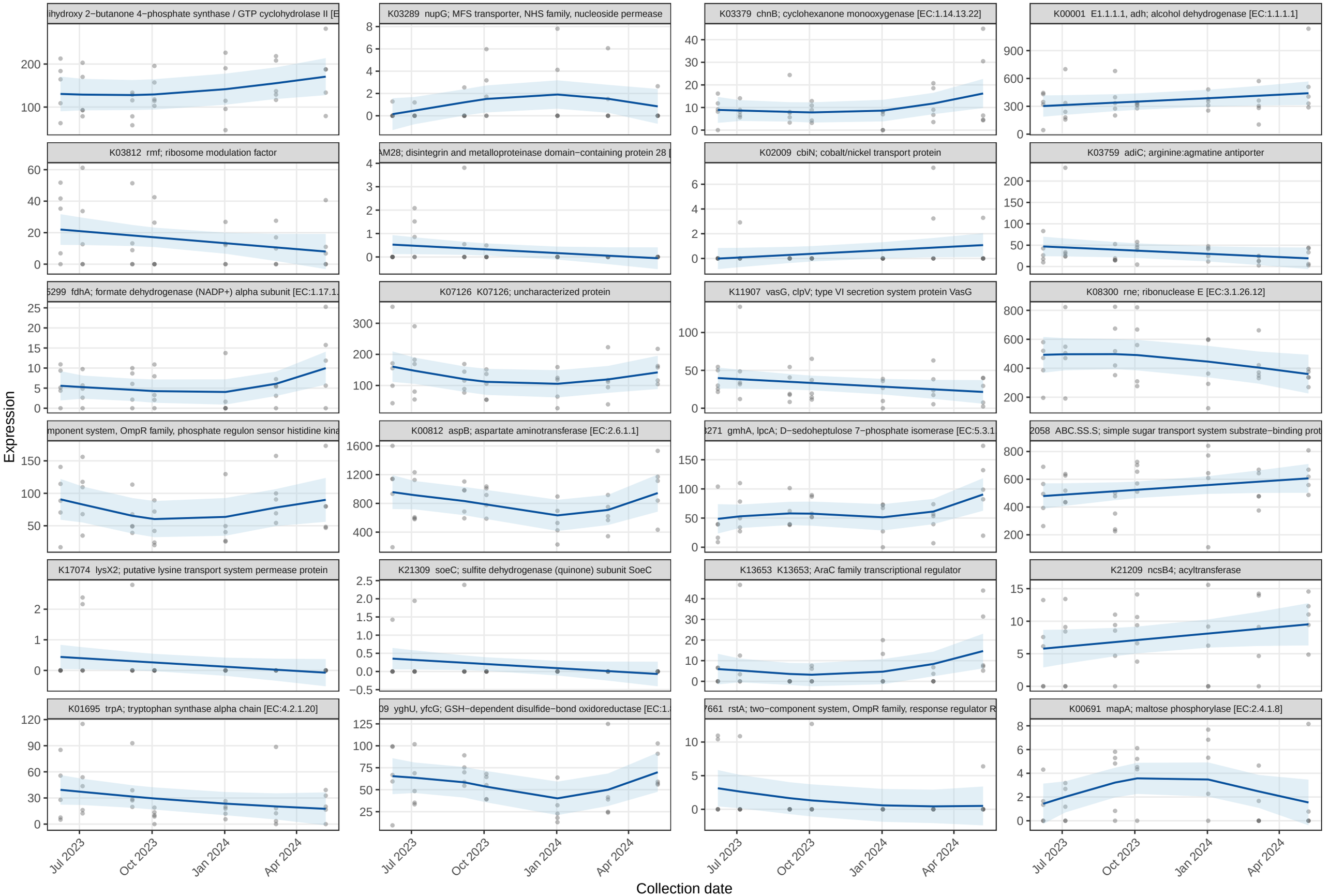


All remaining BRITE classes after exclusions

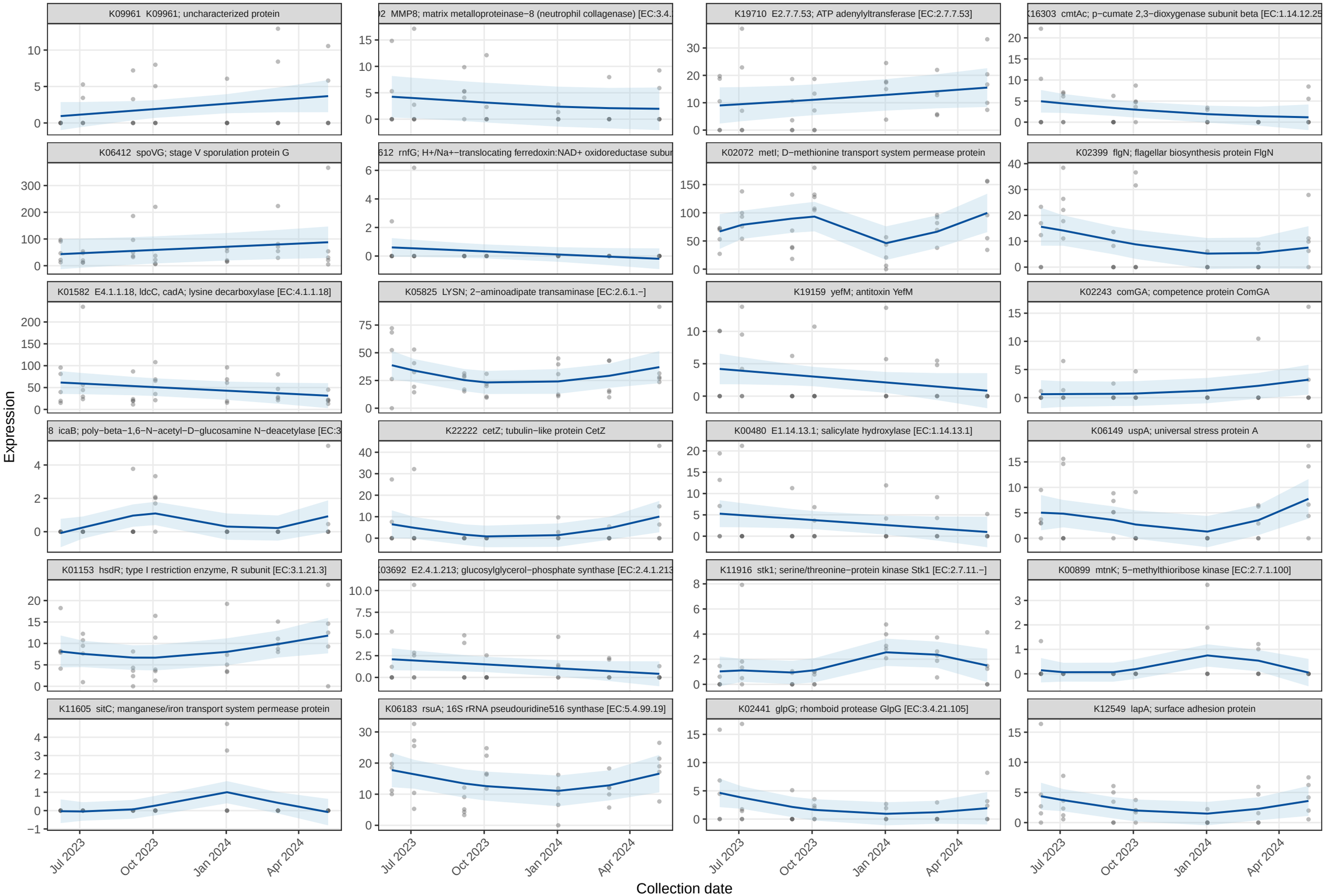


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All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

K18446 ygiF; triphosphatase [EC:3.6.1.25]

hB; formate dehydrogenase (coenzyme F420) beta subunit [EC:1.17.98

K03980 murJ, mviN; putative peptidoglycan lipid II flippase

525 apaH; bis(5'−nucleosyl)−tetraphosphatase (symmetrical) [EC:3.6.1

K12251 aguB; N−carbamoylputrescine amidase [EC:3.5.1.53]

K13657 gumH, aceA; alpha−1,3−mannosyltransferase [EC:2.4.1.252]

K09919 K09919; uncharacterized protein

?−N−acetylglucosamine:(glucosyl)LPS alpha−1,2−N−acetylglucosaminy

K00960 rpo13; DNA−directed RNA polymerase subunit 13 [EC:2.7.7.6]

K08985 K08985; putative lipoprotein

K09937 K09937; uncharacterized protein

K16922 yydH; putative peptide zinc metalloprotease protein

Expres

K03654 recQ; ATP−dependent DNA helicase RecQ [EC:5.6.2.4]

K02229 cobG; precorrin−3B synthase [EC:1.14.13.83]

K00073 allD; ureidoglycolate dehydrogenase (NAD+) [EC:1.1.1.350]

K06350 kipA; antagonist of KipI

K06877 K06877; DEAD/DEAH box helicase domain−containing protein

K18642 creS; crescentin

K17763 rsbR; rsbT co−antagonist protein RsbR

nB−xylY; benzoate/toluate 1,2−dioxygenase subunit beta [EC:1.14.12.1

K22051 mscM, bspA; miniconductance mechanosensitive channel

cprA; 3−chloro−4−hydroxyphenylacetate reductive dehalogenase [EC:

K00857 tdk, TK; thymidine kinase [EC:2.7.1.21]

K12954 ctpG; cation−transporting P−type ATPase G [EC:7.2.2.−]

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All remaining BRITE classes after exclusions

K15640 K15640, phoE; uncharacterized phosphatase

gE; N−acetylglucosamine PTS system EIICBA or EIICB component [EC

K02742 sprT; SprT protein

K01274 pepV; beta−Ala−Xaa dipeptidase [EC:3.4.13.−]

K07272 rgpF; rhamnosyltransferase [EC:2.4.1.−]

K17213 K17213; inositol transport system substrate−binding protein

K14287 ybdL; methionine transaminase [EC:2.6.1.88]

; two−component system, chemotaxis family, response regulator WspR

K01657 trpE; anthranilate synthase component I [EC:4.1.3.27]

K04795 flpA; fibrillarin−like pre−rRNA processing protein

K08085 fimU; type IV fimbrial biogenesis protein FimU

05 acdA; acetate−−−CoA ligase (ADP−forming) subunit alpha [EC:6.2.

Expression

K13775 atuG; citronellol/citronellal dehydrogenase

K14673 whiEII; putative monooxygenase

K22443 cntA; carnitine monooxygenase subunit [EC:1.14.13.239]

K19286 nfrA2; FMN reductase [NAD(P)H] [EC:1.5.1.39]

21 GLYR; glyoxylate/succinic semialdehyde reductase [EC:1.1.1.79 1.1

xpA; mannitol−1−/sugar−/sorbitol−6−phosphatase [EC:3.1.3.22 3.1.3.23

K03553 recA; recombination protein RecA

K01060 cah; cephalosporin−C deacetylase [EC:3.1.1.41]

K00946 thiL; thiamine−monophosphate kinase [EC:2.7.4.16]

16319 andAc; anthranilate 1,2−dioxygenase large subunit [EC:1.14.12.

11747 kefB; glutathione−regulated potassium−efflux system protein Ke

01118 acpD, azoR; FMN−dependent NADH−azoreductase [EC:1.7.1.1

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All remaining BRITE classes after exclusions

queA; S−adenosylmethionine:tRNA ribosyltransferase−isomerase [EC:2

K03575 mutY; A/G−specific adenine glycosylase [EC:3.2.2.31]

; chemosensory pili system protein ChpB (putative protein−glutamate m

K01212 sacC, levB; levanase [EC:3.2.1.65]

K06882 K06882; uncharacterized protein

K01046 lip, TGL2; triacylglycerol lipase [EC:3.1.1.3]

K02568 napB; nitrate reductase (cytochrome), electron transfer subunit

01359 PCSK1; proprotein convertase subtilisin/kexin type 1 [EC:3.4.21.9

98 PPCDC, coaC; phosphopantothenoylcysteine decarboxylase [EC:4.1

7774 tctD; two−component system, OmpR family, response regulator Tc

K21459 cypM; cypemycin N−terminal methyltransferase [EC:2.1.1.301]

K02462 gspM; general secretion pathway protein M

Expres

K19234 ynhG; L,D−transpeptidase YnhG

K03395 aac3−I; aminoglycoside 3−N−acetyltransferase I [EC:2.3.1.60]

K20938 ladA; long−chain alkane monooxygenase [EC:1.14.14.28]

K18372 acmB; methyl acetate hydrolase [EC:3.1.1.114]

K01457 atzF; allophanate hydrolase [EC:3.5.1.54]

R family transcriptional regulator, repressor of blue light− and temperature

1089 mlrA; MerR family transcriptional regulator, activator of the csg ger

K12513 tadE; tight adherence protein E

ABC.PA.A; polar amino acid transport system ATP−binding protein [EC

K01436 yhaA; amidohydrolase [EC:3.5.1.−]

K22132 tcdA; tRNA threonylcarbamoyladenosine dehydratase

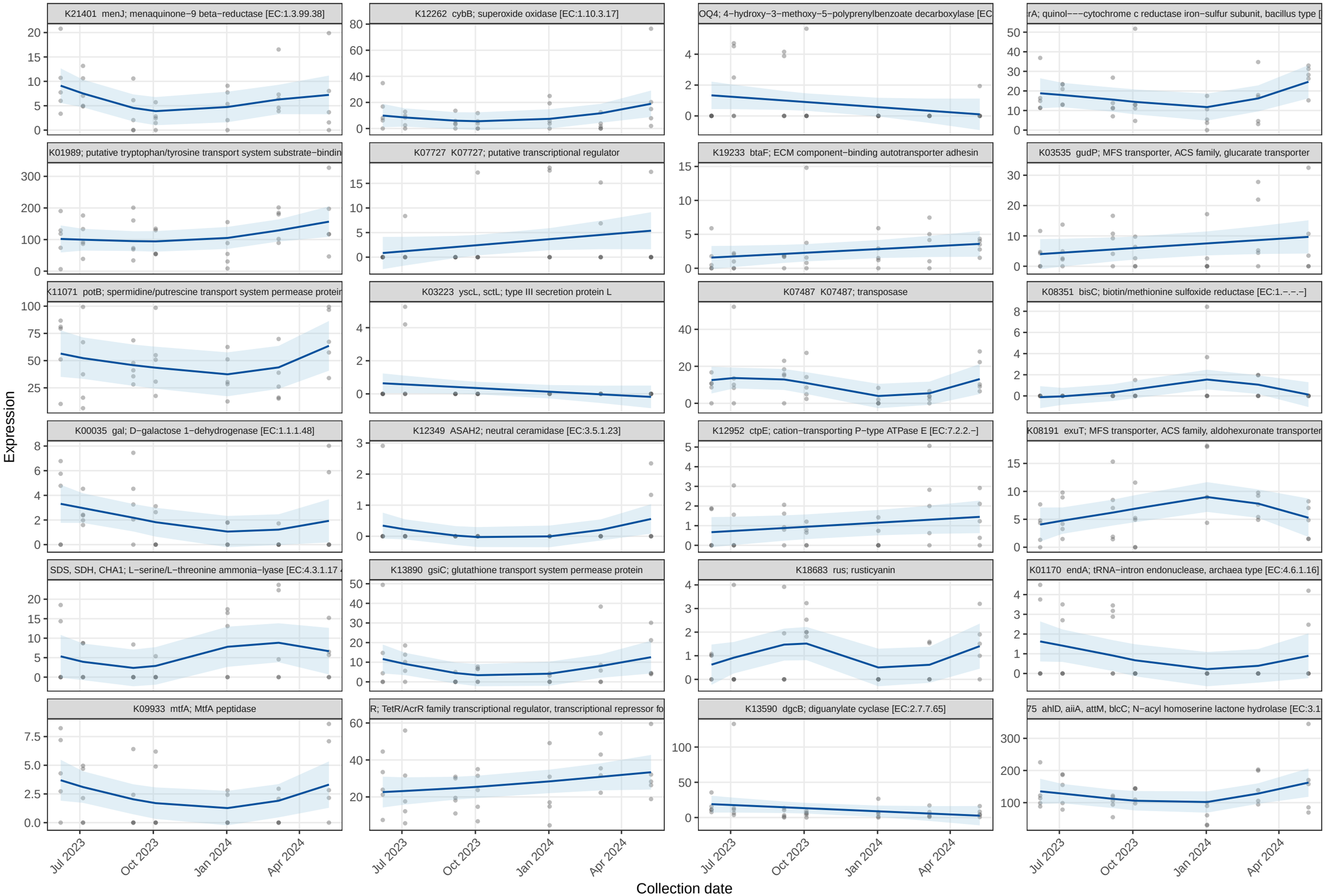
9 wcaB; putative colanic acid biosynthesis acetyltransferase WcaB [EC:2

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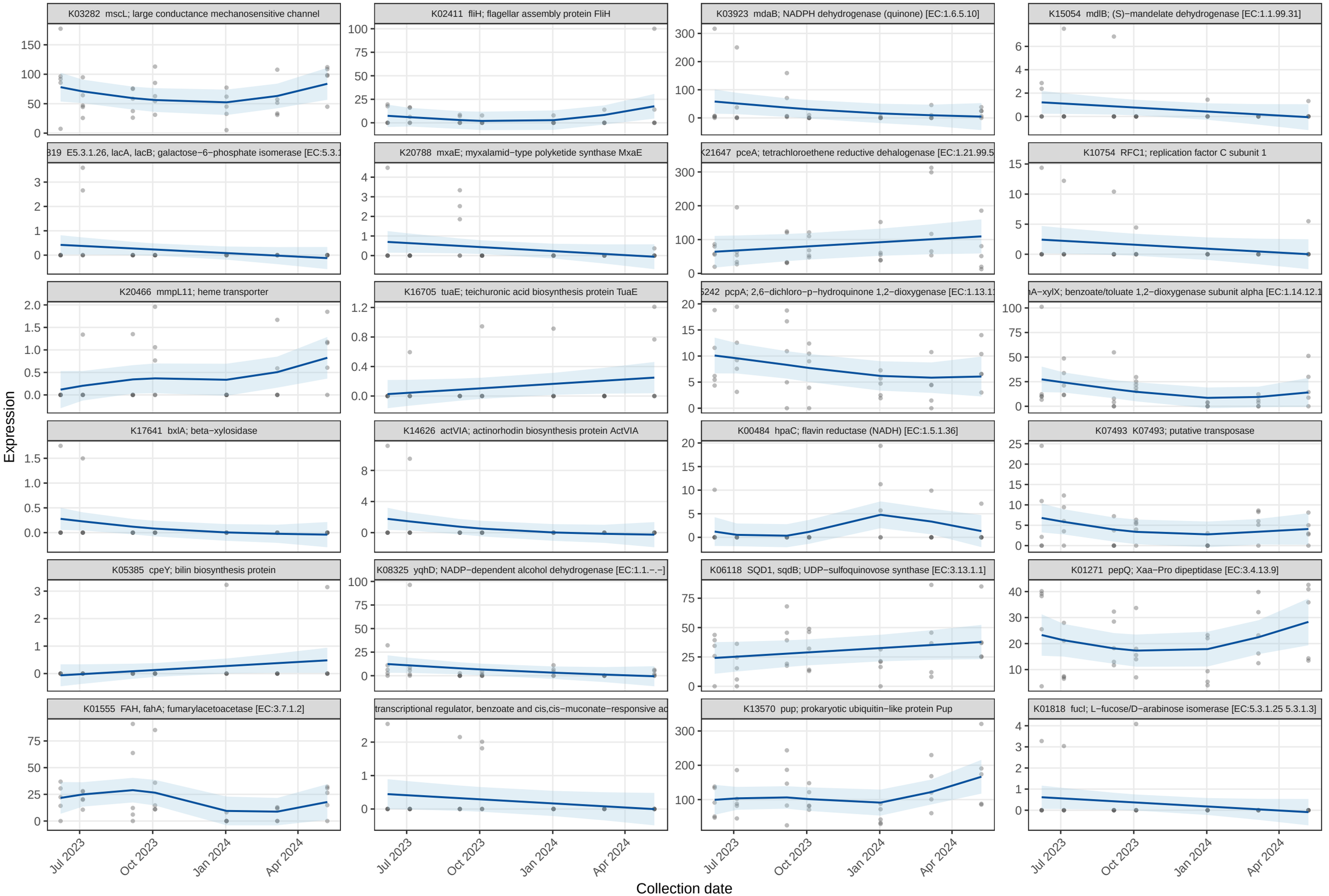
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All remaining BRITE classes after exclusions

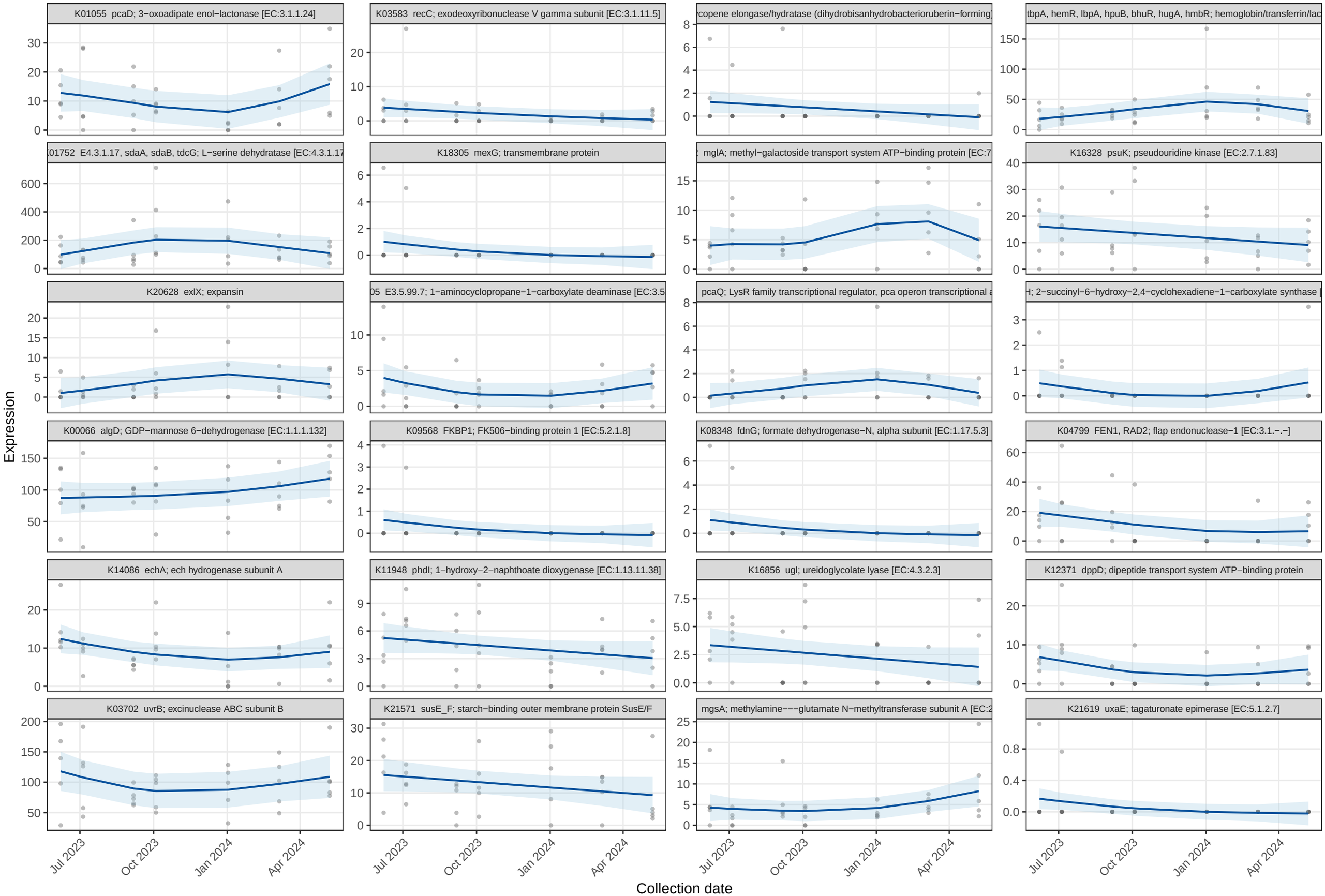


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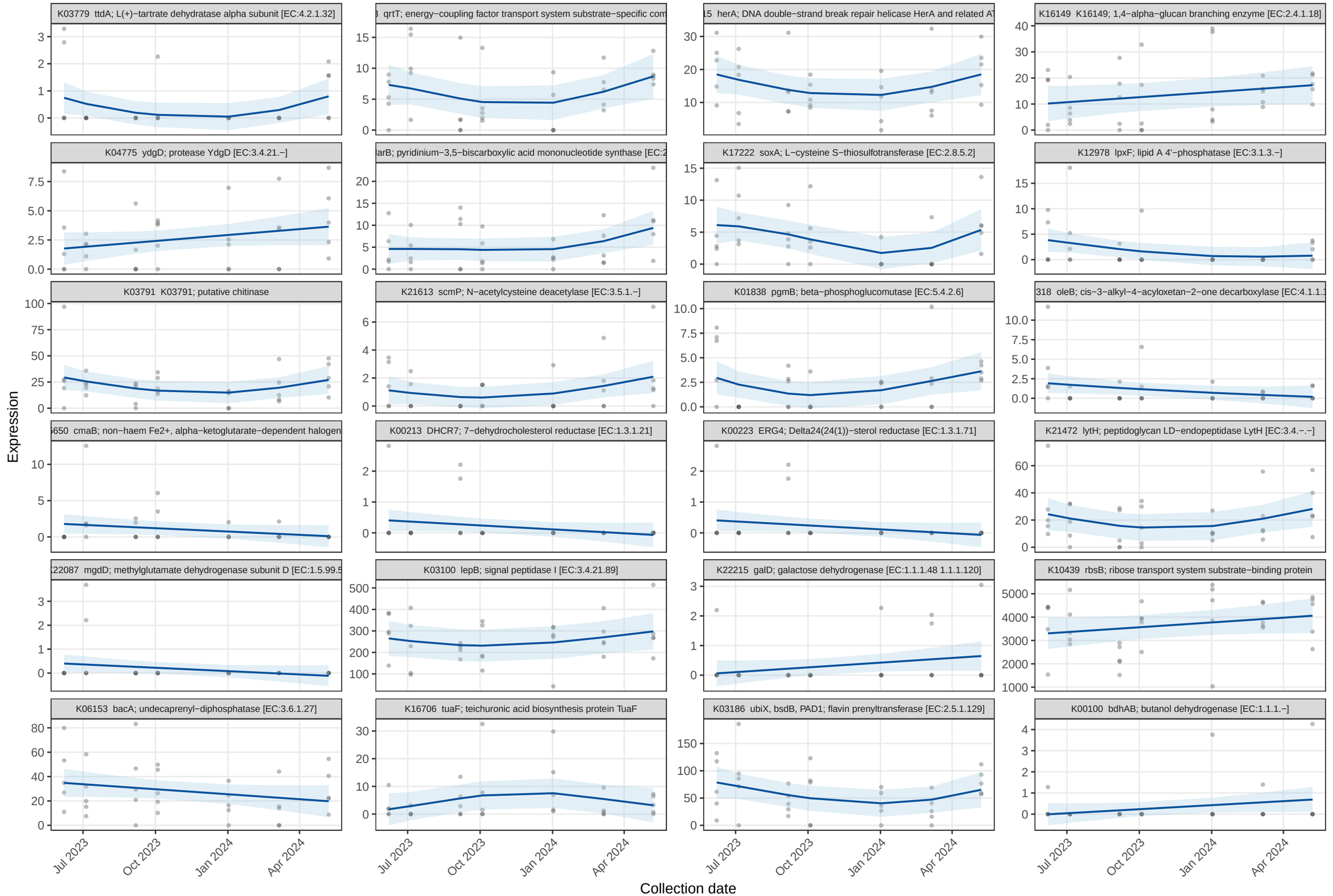
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

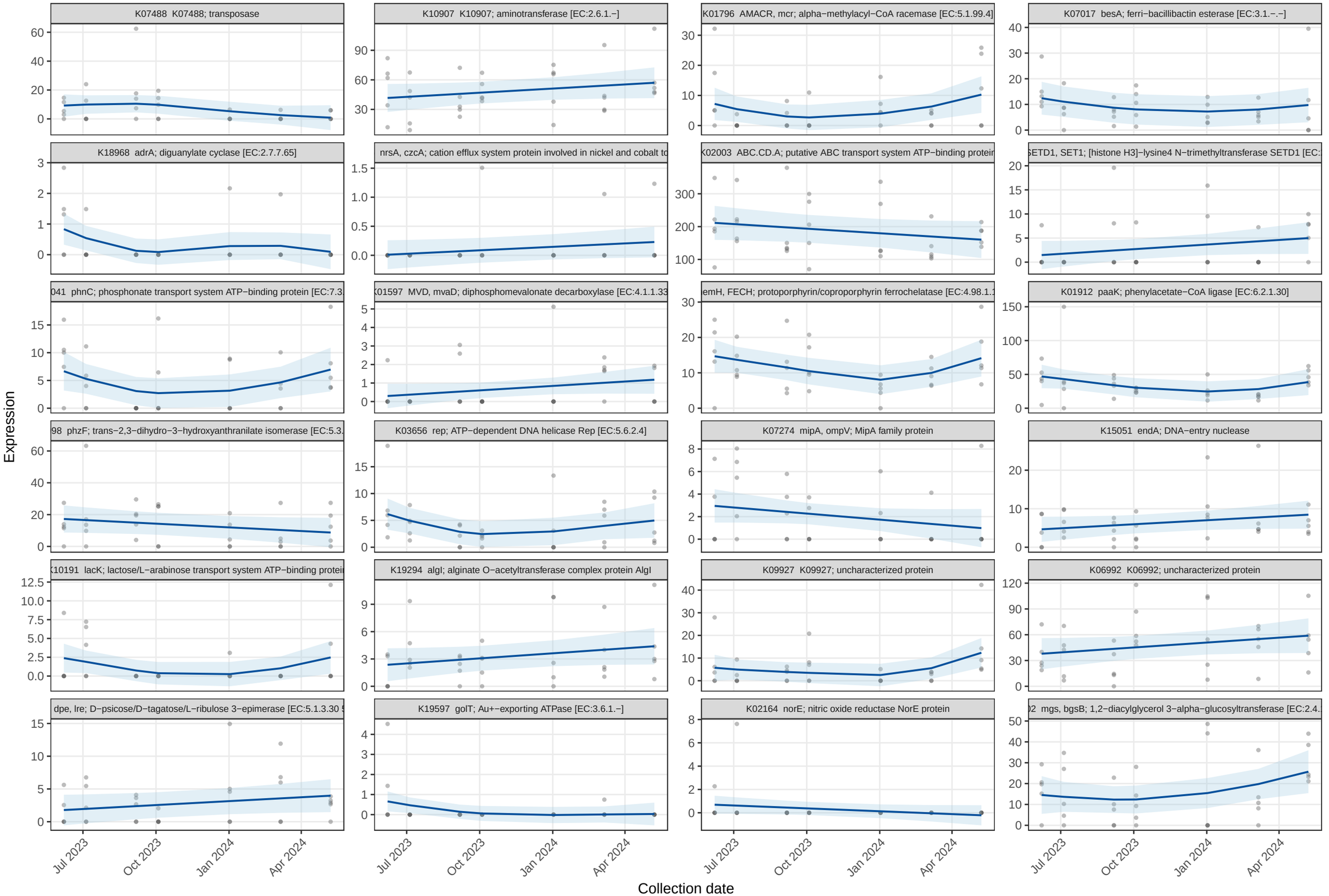


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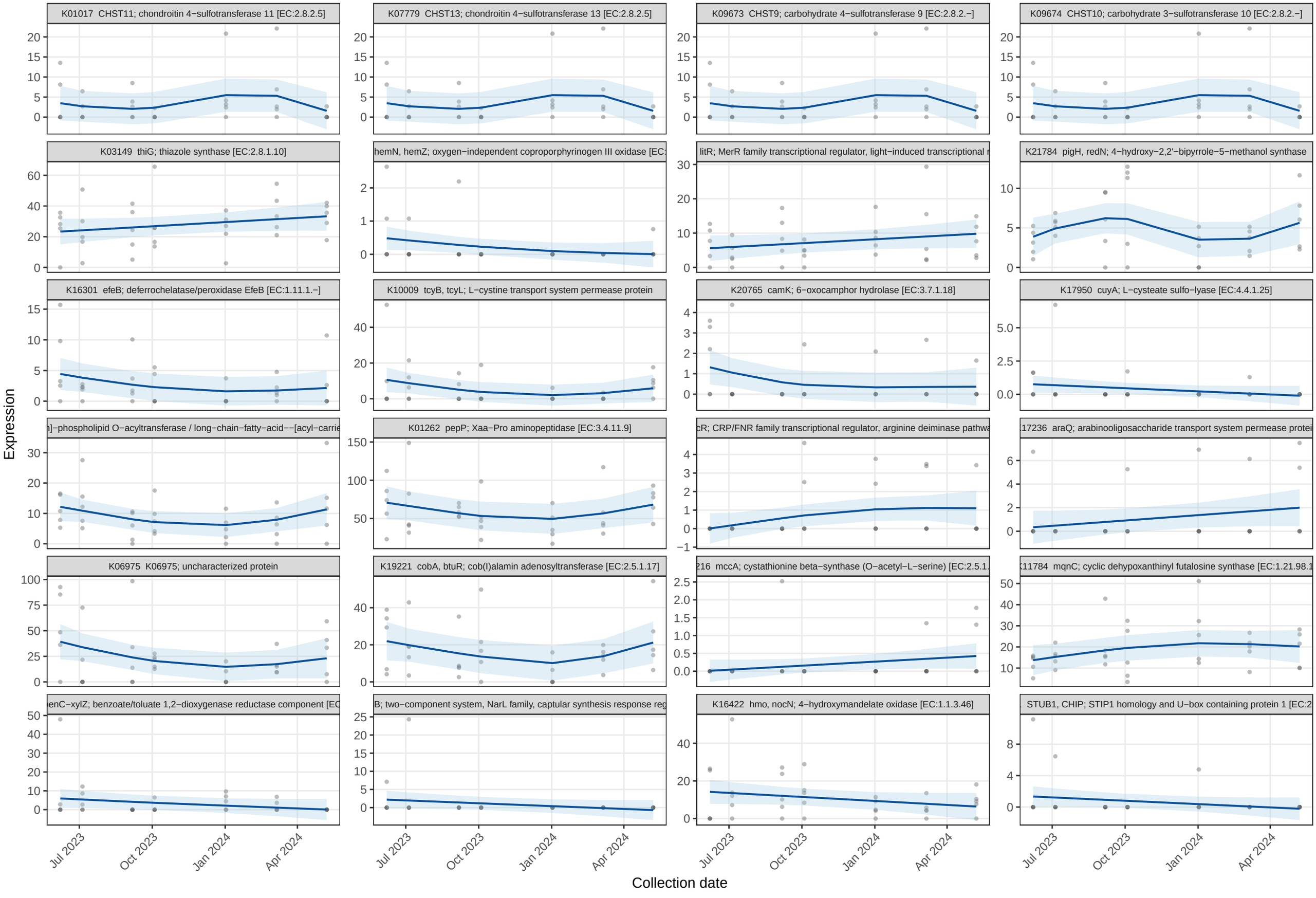
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All remaining BRITE classes after exclusions



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K12735 PPIL4; peptidyl−prolyl cis−trans isomerase−like 4 [EC:5.2.1.8]

K06996 K06996; uncharacterized protein

K03618 hyaF; hydrogenase−1 operon protein HyaF

K12139 hyfD; hydrogenase−4 component D [EC:1.−.−.−]

K09706 K09706; uncharacterized protein

K03747 smg; Smg protein

K07153 hflD; high frequency lysogenization protein

K06904 GP4; Escherichia/Staphylococcus phage prohead protease

onent system, OmpR family, aerobic respiration control sensor histidine k

K13612 pksL, baeL; bacillaene polyketide synthase PksL/BaeL

K07019 K07019; uncharacterized protein

K11720 lptG; lipopolysaccharide export system permease protein

Expres

22342 dmmA; dimethylamine monooxygenase subunit A [EC:1.14.13.23]

hidazoleglycerol−phosphate dehydratase / histidinol−phosphatase [EC:4

K13274 wprA; cell wall−associated protease [EC:3.4.21.−]

K21087 ycgR; flagellar brake protein

K06188 aqpZ, aqpM, AQP9; aquaporin

K03453 TC.BASS; bile acid:Na+ symporter, BASS family

K12369 dppB; dipeptide transport system permease protein

K07575 MCTS, TMA20; malignant T−cell−amplified sequence

K03168 topA; DNA topoisomerase I [EC:5.6.2.1]

02049 ABC.SN.A; NitT/TauT family transport system ATP−binding prote

K00784 rnz; ribonuclease Z [EC:3.1.26.11]

e / enoyl−CoA hydratase / 3−hydroxybutyryl−CoA epimerase / enoyl−Co

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K06902 UMF1; MFS transporter, UMF1 family

oF, tbuC, touF; toluene monooxygenase electron transfer component [E

K12980 lpxQ; lipid A oxidase

flrB, fleS, flaL; two−component system, sensor histidine kinase FlrB [EC

bacillaene biosynthesis, polyketide synthase / nonribosomal peptide sy

K16398 epoE; epothilone polyketide synthase E

K16417 mxaC; myxalamid−type polyketide synthase MxaC

K21678 hpnD; presqualene diphosphate synthase [EC:2.5.1.103]

K00853 araB; L−ribulokinase [EC:2.7.1.16]

K17266 MVP; major vault protein

K08095 CUT1_2_3; cutinase [EC:3.1.1.74]

K09021 rutC; aminoacrylate peracid reductase

Expression

K09117 K09117; uncharacterized protein

K19265 gpr; L−glyceraldehyde 3−phosphate reductase [EC:1.1.1.−]

K01816 hyi, gip; hydroxypyruvate isomerase [EC:5.3.1.22]

K08355 aoxA; arsenite oxidase small subunit [EC:1.20.2.1 1.20.9.1]

acI family transcriptional regulator, gluconate utilization system Gnt−II tr

K01494 dcd; dCTP deaminase [EC:3.5.4.13]

K09822 K09822; uncharacterized protein

K03732 rhlB; ATP−dependent RNA helicase RhlB [EC:5.6.2.7]

K01039 gctA; glutaconate CoA−transferase, subunit A [EC:2.8.3.12]

K16079 omp31; outer membrane immunogenic protein

ntA; proton−translocating NAD(P)+ transhydrogenase subunit alpha [E

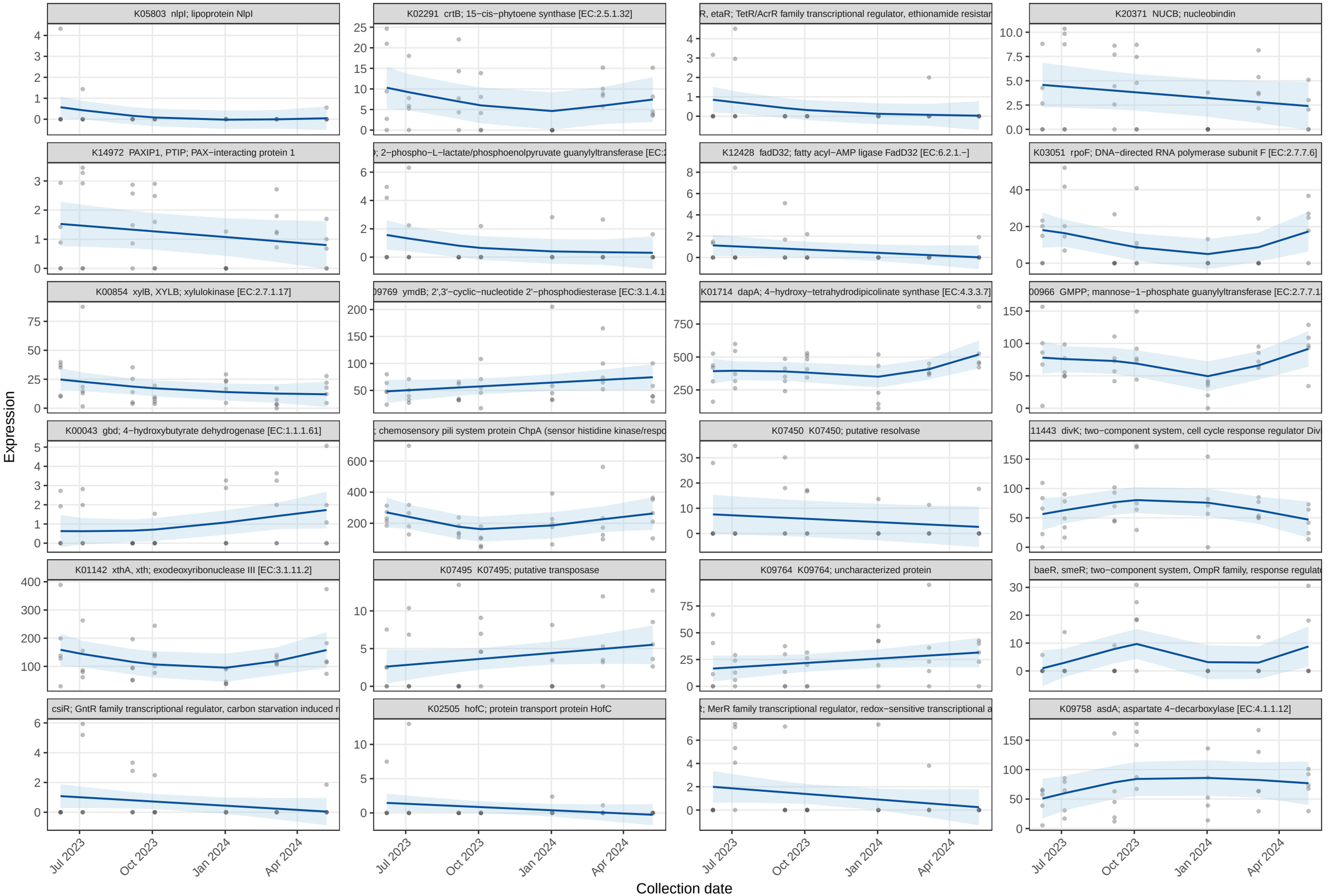
4 K02484; two−component system, OmpR family, sensor kinase [EC:2.

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All remaining BRITE classes after exclusions



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pehS; two−component system, NtrC family, sensor histidine kinase PilS

K16692 etk−wzc; tyrosine−protein kinase Etk/Wzc [EC:2.7.10.3]

K03499 trkA, ktrA, ktrC; trk/ktr system potassium uptake protein

O, gid; methylenetetrahydrofolate−−tRNA−(uracil−5−)−methyltransferase

K03785 aroD; 3−dehydroquinate dehydratase I [EC:4.2.1.10]

2227 cbiB, cobD; adenosylcobinamide−phosphate synthase [EC:6.3.1.

K12976 pagL; lipid A 3−O−deacylase [EC:3.1.1.−]

K20988 vpsN; polysaccharide biosynthesis/export protein VpsN

K00791 miaA, TRIT1; tRNA dimethylallyltransferase [EC:2.5.1.75]

02552 menF; menaquinone−specific isochorismate synthase [EC:5.4.4.

K11956 natD; neutral amino acid transport system permease protein

399 addB; ATP−dependent helicase/nuclease subunit B [EC:5.6.2.4 3.1

oxy−6−hydroxymethyldihydropteridine diphosphokinase / dihydropteroat

67 pbuE; MFS transporter, DHA1 family, purine base/nucleoside efflux

15912 pglF; UDP−N−acetylglucosamine 4,6−dehydratase [EC:4.2.1.13

K01318 sspA; glutamyl endopeptidase [EC:3.4.21.19]

K14647 vpr; minor extracellular serine protease Vpr [EC:3.4.21.−]

K18981 udh; uronate dehydrogenase [EC:1.1.1.203]

K22250 phaZ; poly(3−hydroxyoctanoate) depolymerase [EC:3.1.1.76]

K13652 K13652; AraC family transcriptional regulator

082 cysQ, MET22, BPNT1; 3'(2'), 5'−bisphosphate nucleotidase [EC:3.1

07 norG; GntR family transcriptional regulator, regulator for abcA and nc

DRG1, RBG1; developmentally−regulated GTP−binding protein 1 [EC:

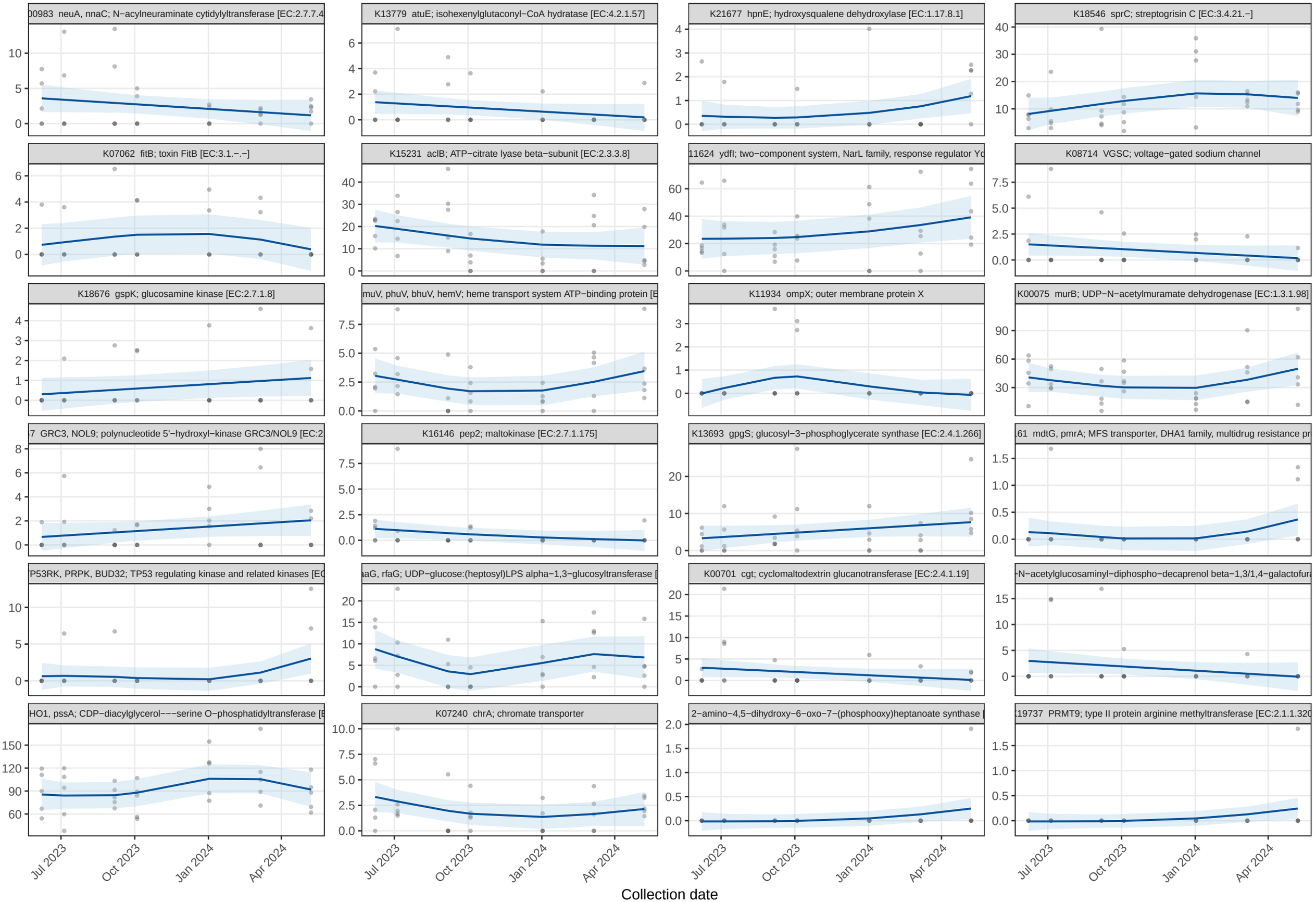
K16234 hutT; histidine transporter

Expression

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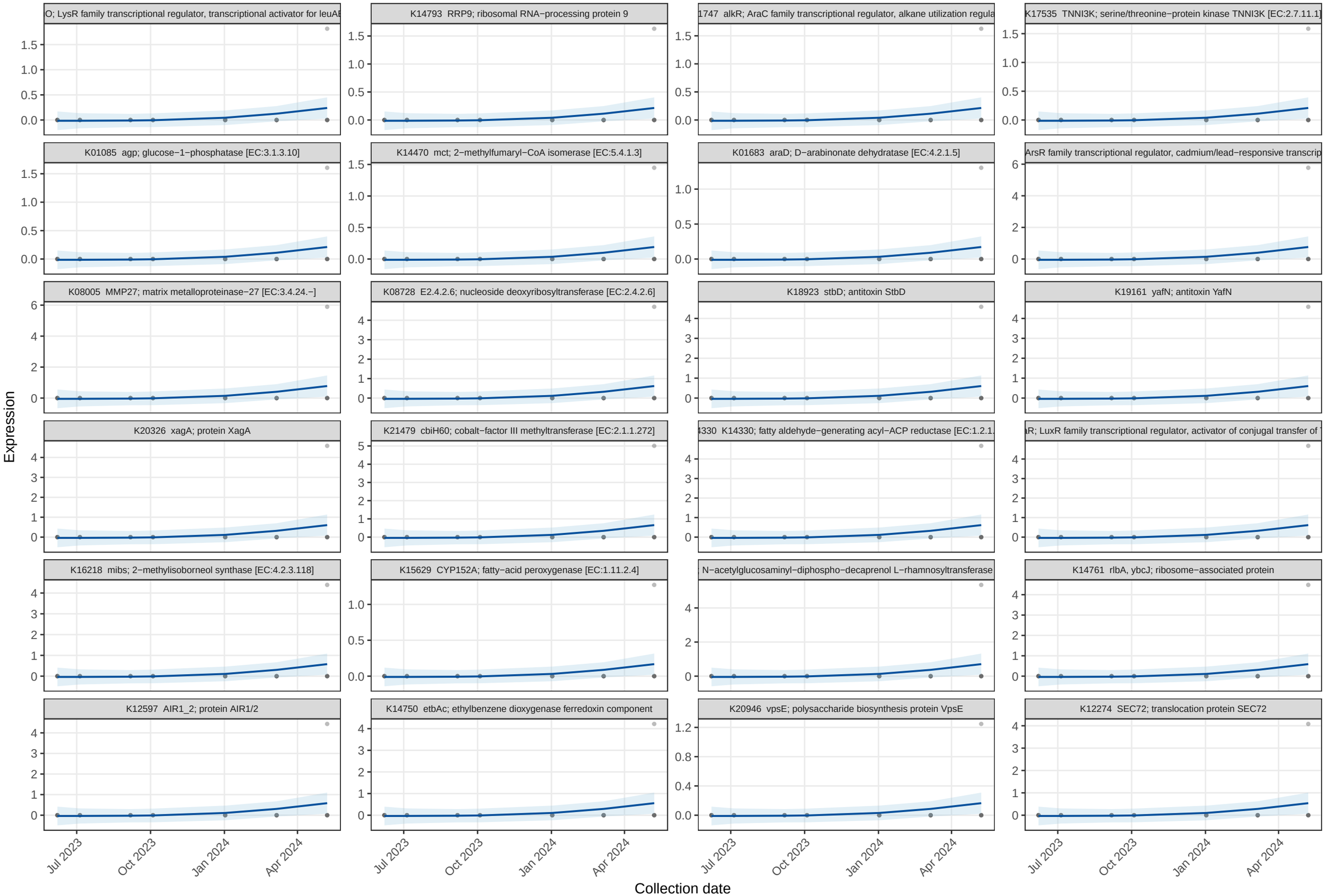
Collection date

All remaining BRITE classes after exclusions

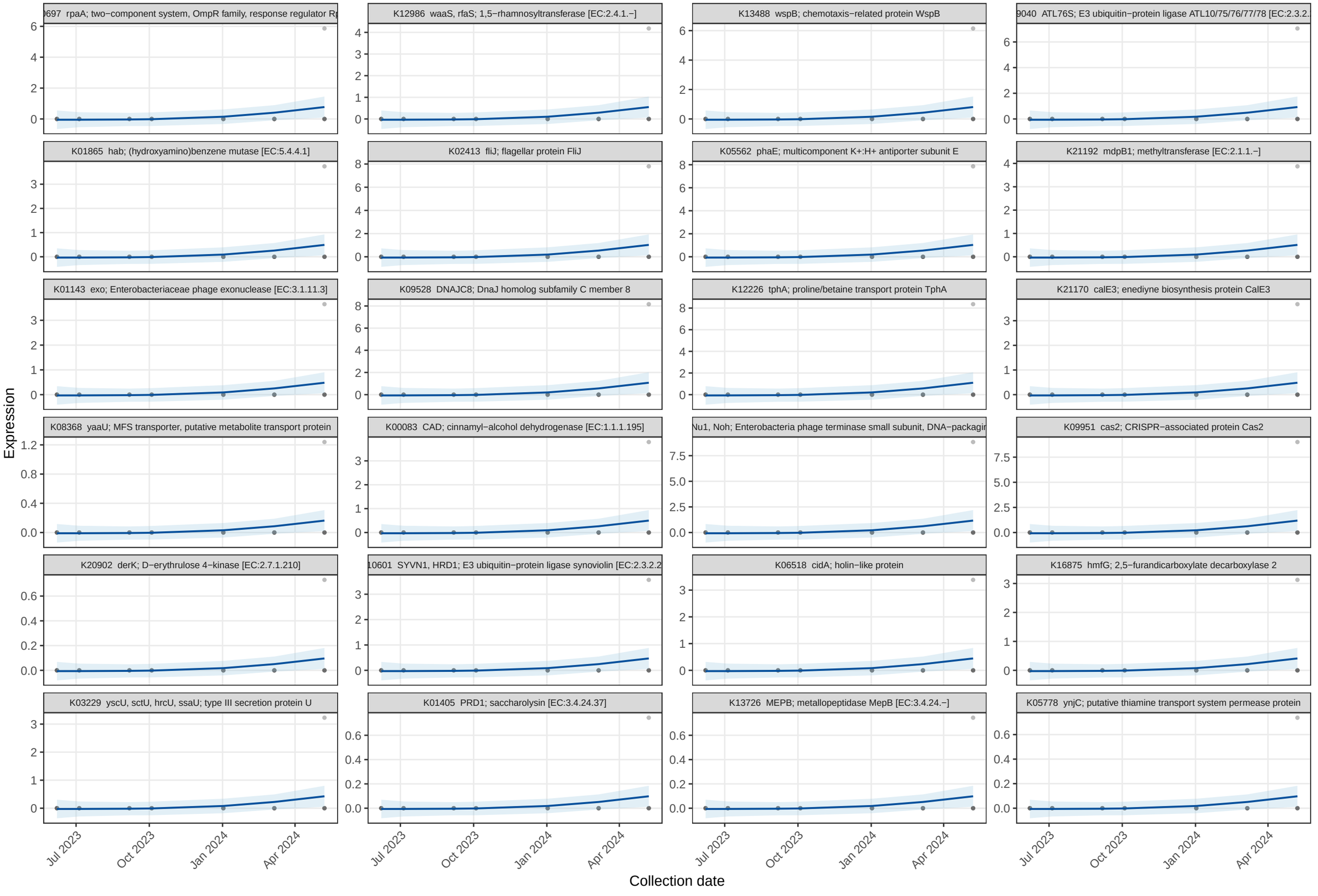


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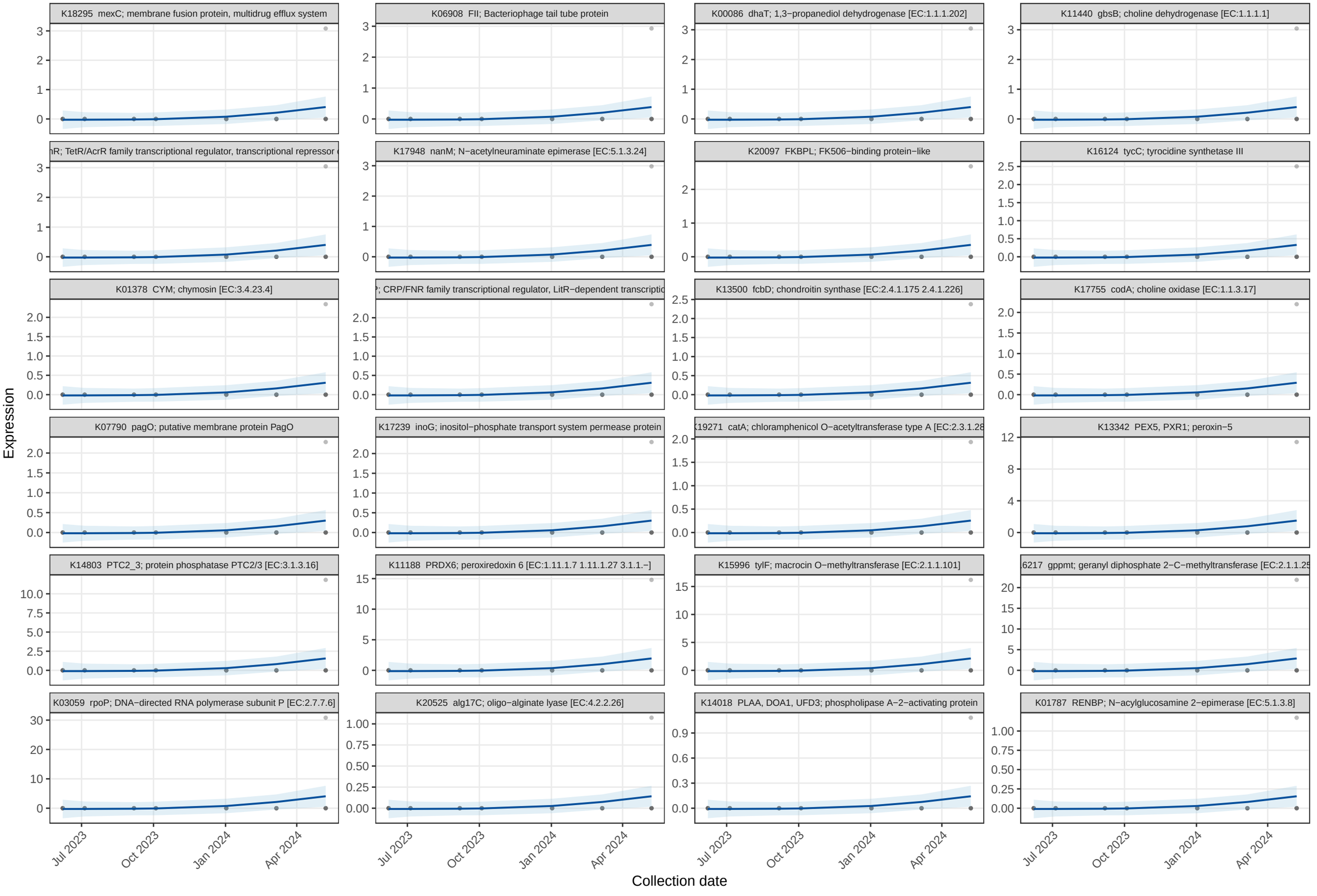
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

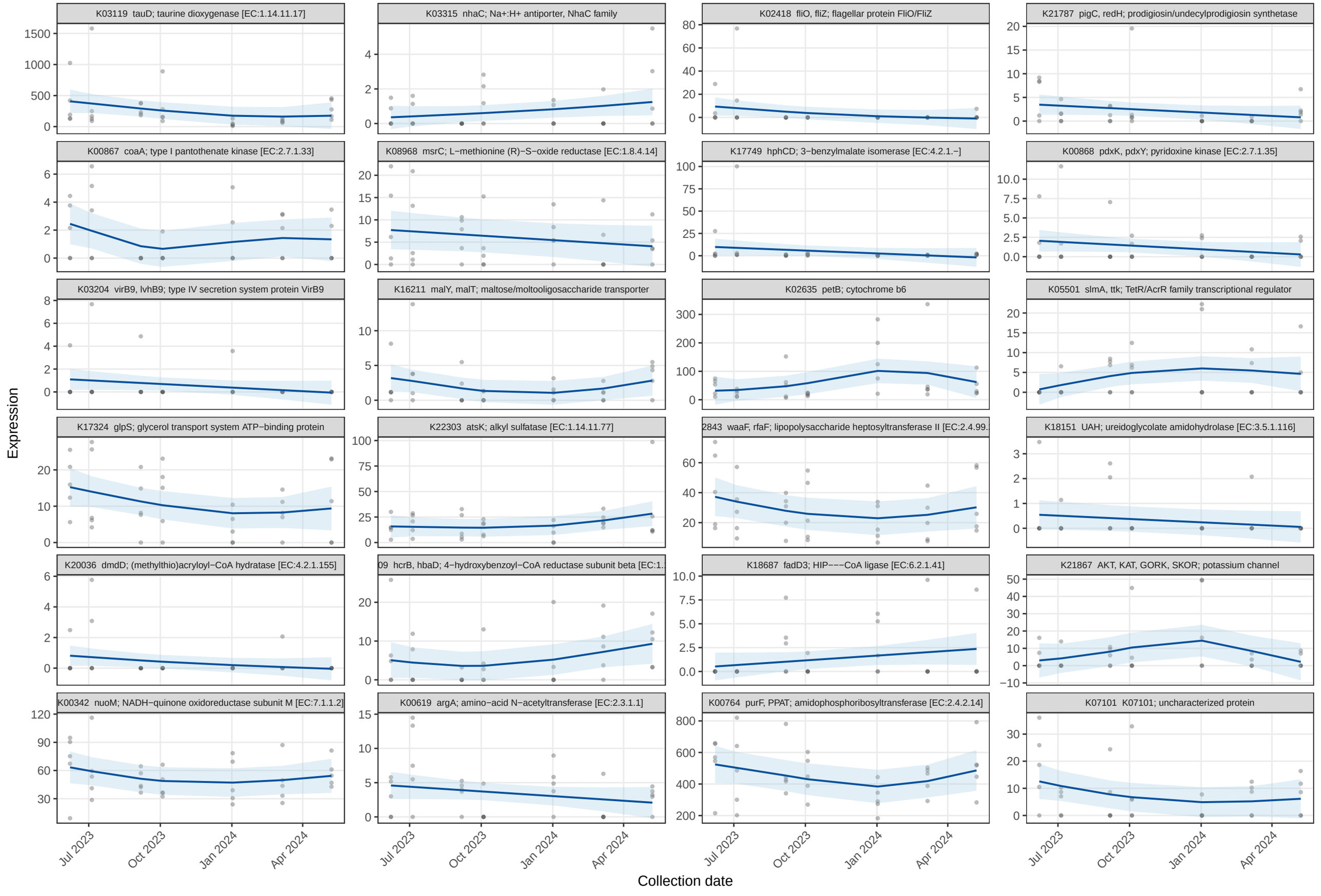


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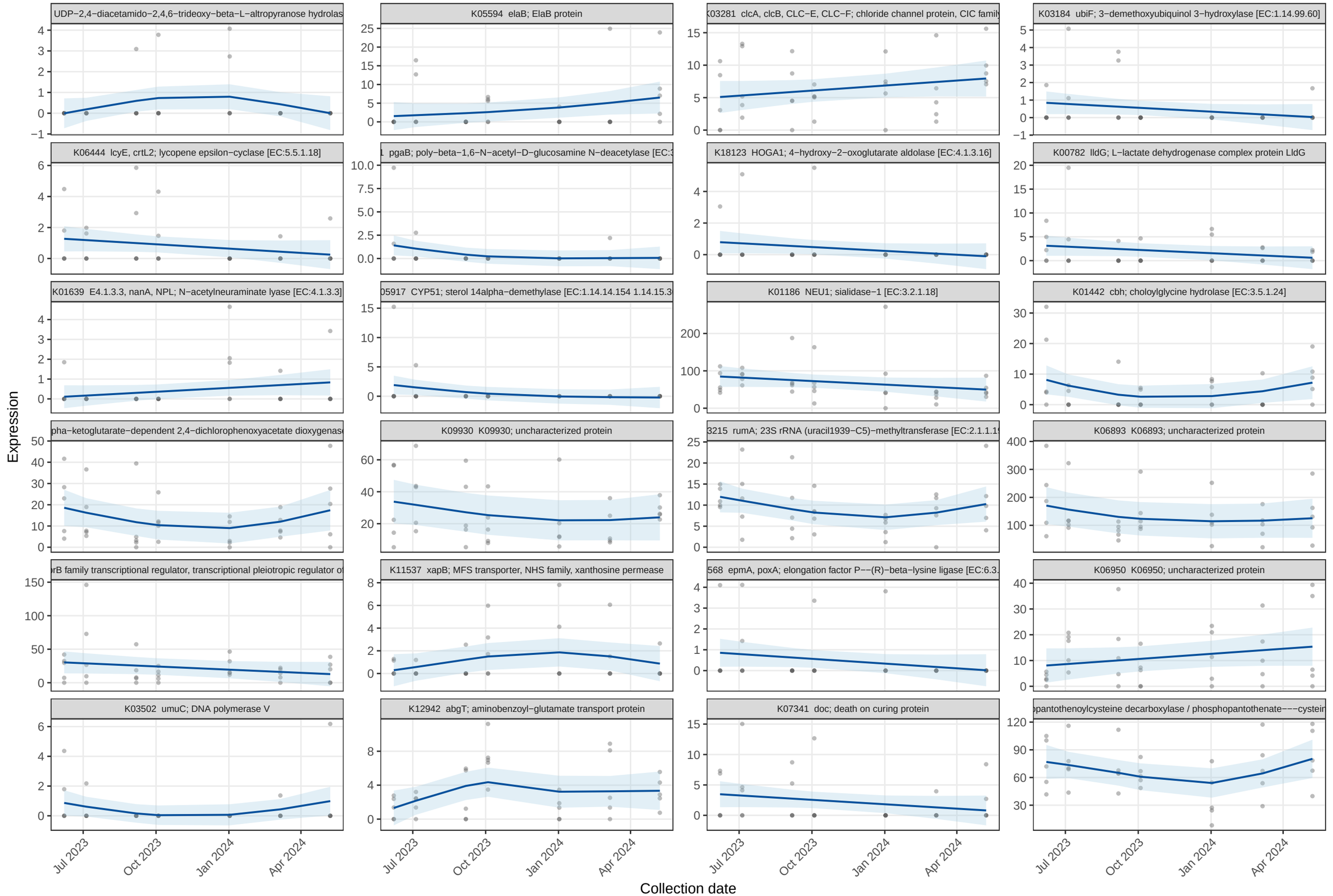


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All remaining BRITE classes after exclusions

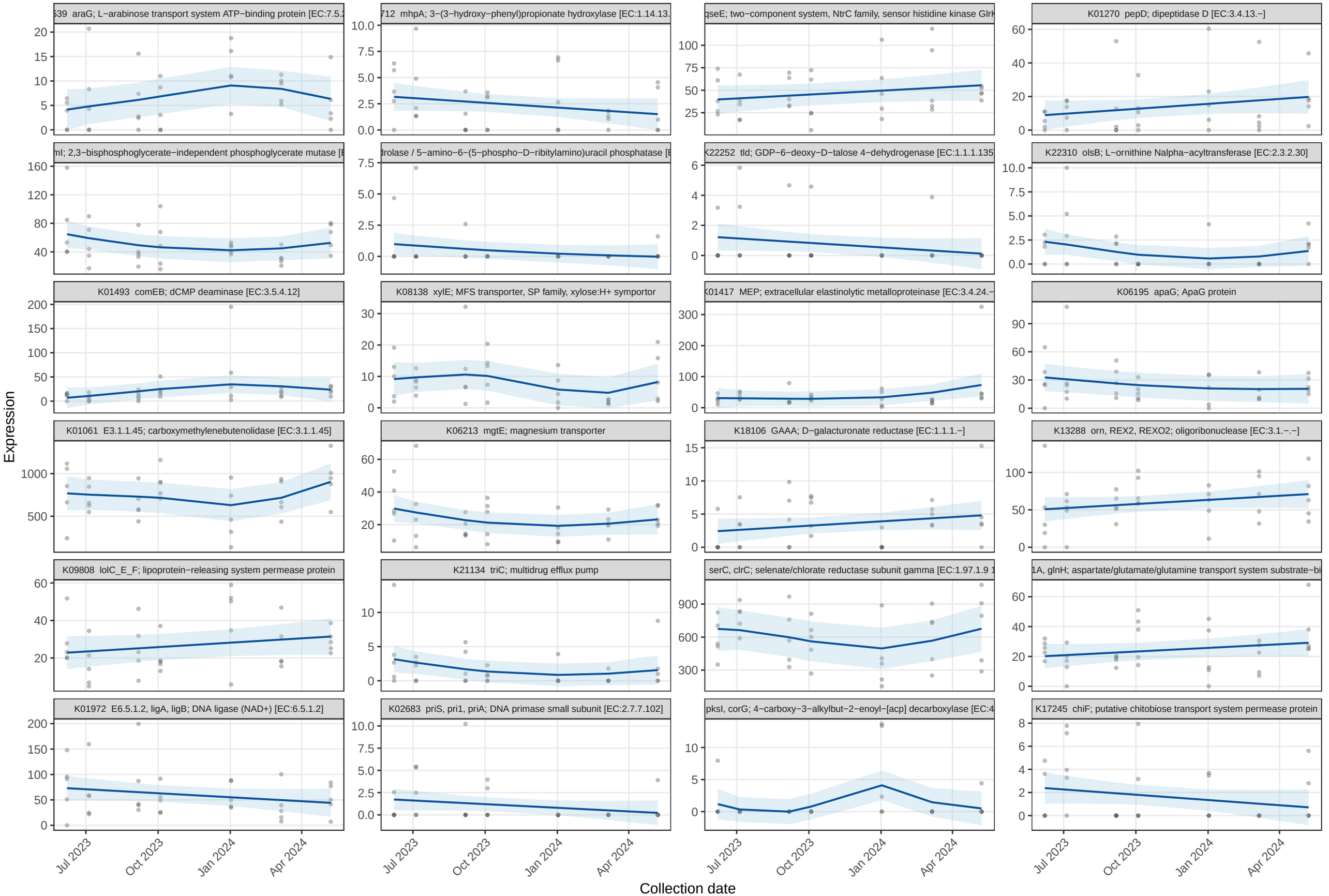


All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions

K08299 caiD; crotonobetainyl−CoA hydratase [EC:4.2.1.149]

K00127 fdoI, fdsG; formate dehydrogenase subunit gamma

K13959 KLHL38; kelch−like protein 38

K06887 K06887; uncharacterized protein

K02486 K02486; two−component system, sensor kinase [EC:2.7.13.3]

K07536 badI; 2−ketocyclohexanecarboxyl−CoA hydrolase [EC:3.1.2.−]

G; two−component system, OmpR family, sensor histidine kinase ChvG

rJ, narW; nitrate reductase molybdenum cofactor assembly chaperone

K01426 E3.5.1.4, amiE; amidase [EC:3.5.1.4]

K01685 uxaA; altronate hydrolase [EC:4.2.1.7]

arB, spo0J; ParB family transcriptional regulator, chromosome partitioni

engCP, engBF, endoEF; endo−alpha−N−acetylgalactosaminidase [EC:

K05368 fre, ubiB; NAD(P)H−flavin reductase [EC:1.5.1.41]

K07404 pgl; 6−phosphogluconolactonase [EC:3.1.1.31]

K06142 hlpA, ompH; outer membrane protein

K00467 E1.13.12.4; lactate 2−monooxygenase [EC:1.13.12.4]

K00761 upp, UPRT; uracil phosphoribosyltransferase [EC:2.4.2.9]

ispD; 2−C−methyl−D−erythritol 4−phosphate cytidylyltransferase [EC:2

K05967 K05967; uncharacterized protein

61 matB; malonyl−CoA/methylmalonyl−CoA synthetase [EC:6.2.1.76 6.

R; GntR family transcriptional regulator, hexuronate regulon transcriptior

K03701 uvrA; excinuclease ABC subunit A

K11159 K11159; carotenoid cleavage oxygenase [EC:1.13.11.−]

K07788 mdtB; multidrug efflux pump

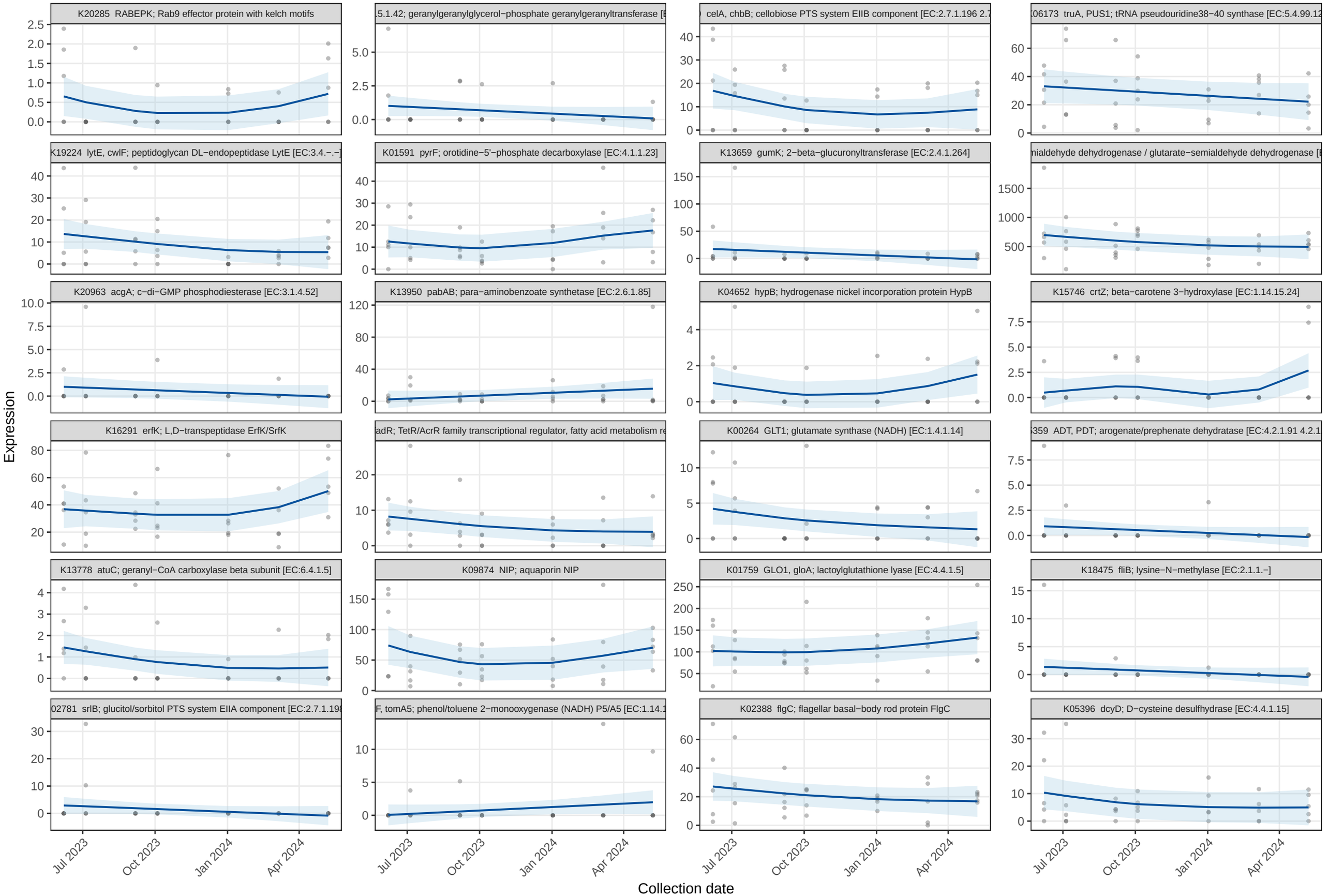
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K18120 4hbD, abfH; 4−hydroxybutyrate dehydrogenase [EC:1.1.1.61]

K07224 efeO; iron uptake system component EfeO

K18334 fucD; L−fuconate dehydratase [EC:4.2.1.68]

ndent phosphofructokinase / diphosphate−dependent phosphofructoki

K12021 TRIM45; tripartite motif−containing protein 45

K06077 slyB; outer membrane lipoprotein SlyB

K17831 lodA; L−lysine 6−oxidase [EC:1.4.3.20]

K12373 HEXA_B; hexosaminidase [EC:3.2.1.52]

K08364 merP; periplasmic mercuric ion binding protein

K06973 K06973; uncharacterized protein

K19421 epsC; polysaccharide biosynthesis protein EpsC

K09940 K09940; uncharacterized protein

Expression

K00537 arsC; arsenate reductase (glutaredoxin) [EC:1.20.4.1]

K07234 K07234; uncharacterized protein involved in response to NO

K00876 udk, UCK; uridine kinase [EC:2.7.1.48]

D; stage II sporulation protein D (peptidoglycan lytic transglycosylase)

K21034 virH2, CYP104; cytochrome P450 family 104

K17759 NAXE, nnrE; NAD(P)H−hydrate epimerase [EC:5.1.99.6]

K12541 lapB; ATP−binding cassette, subfamily C, bacterial LapB

2191 cbiT; cobalt−precorrin−6B (C15)−methyltransferase [EC:2.1.1.19

K07746 parD1_3_4; antitoxin ParD1/3/4

K16052 ynaI, mscMJ; MscS family membrane protein

430 carC; caffeyl−CoA reductase−Etf complex subunit CarC [EC:1.3.1.

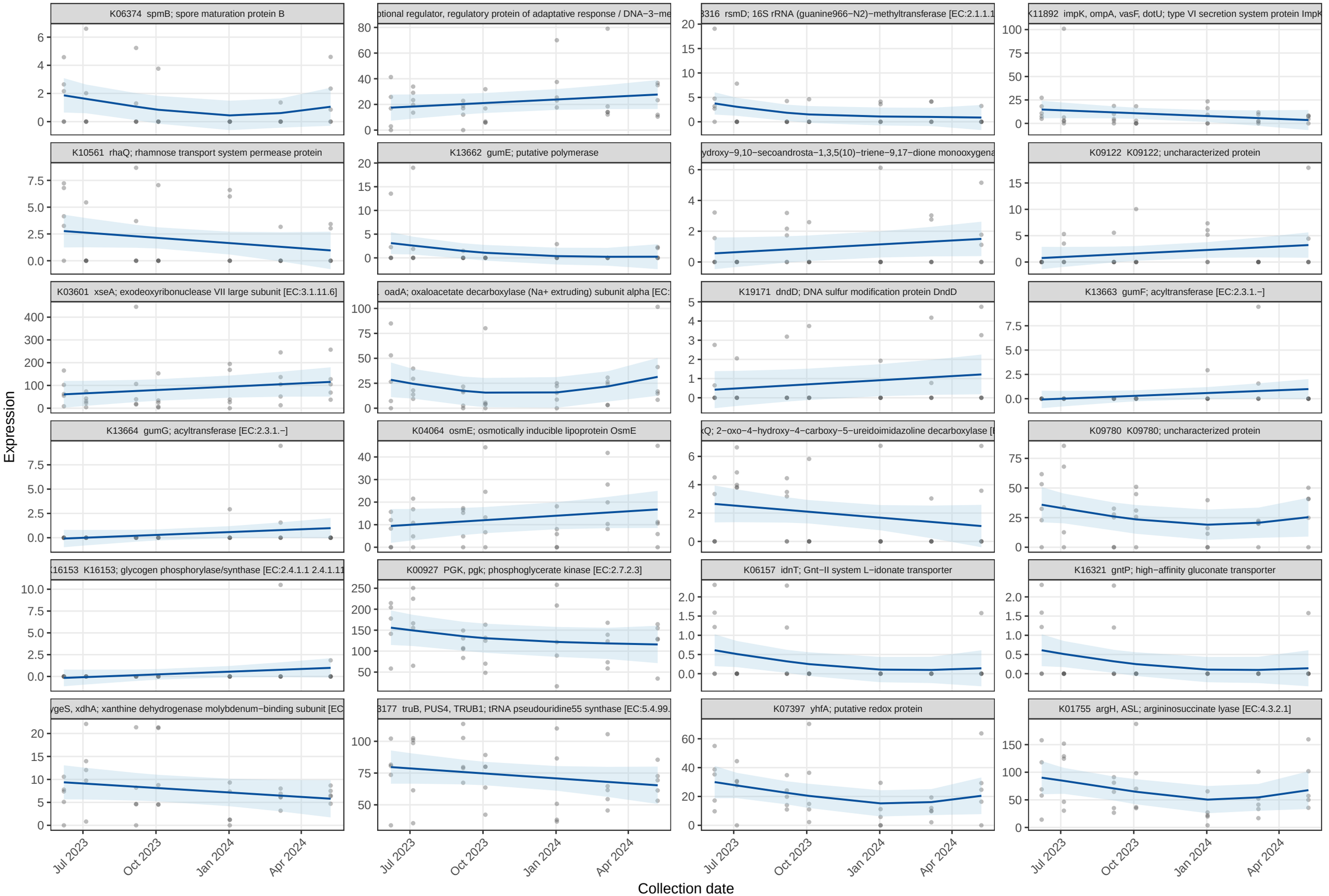
K07474 xtmA; phage terminase small subunit

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paaH, hbd, fadB, mmgB; 3−hydroxybutyryl−CoA dehydrogenase [EC:1.

wlbC; UDP−2−acetamido−2−deoxy−ribo−hexuluronate aminotransferas

paaA; ring−1,2−phenylacetyl−CoA epoxidase subunit PaaA [EC:1.14.

K03974 pspF; psp operon transcriptional activator

K03646 tolA; colicin import membrane protein

K01070 frmB, ESD, fghA; S−formylglutathione hydrolase [EC:3.1.2.12]

K19225 gluP; rhomboid protease GluP [EC:3.4.21.105]

K00126 fdsD; formate dehydrogenase subunit delta [EC:1.17.1.9]

K05787 hupA; DNA−binding protein HU−alpha

K02717 psbP; photosystem II oxygen−evolving enhancer protein 2

K00334 nuoE; NADH−quinone oxidoreductase subunit E [EC:7.1.1.2]

K21007 pelB; polysaccharide biosynthesis protein PelB

K09975 K09975; uncharacterized protein

K14743 mycP; membrane−anchored mycosin MYCP [EC:3.4.21.−]

K16850 uxaA2; altronate dehydratase large subunit [EC:4.2.1.7]

K01195 uidA, GUSB; beta−glucuronidase [EC:3.2.1.31]

K01095 pgpA; phosphatidylglycerophosphatase A [EC:3.1.3.27]

k; xylulose−5−phosphate/fructose−6−phosphate phosphoketolase [EC:4

K21498 higA−1; antitoxin HigA−1

C; ribosome biogenesis GTPase / thiamine phosphate phosphatase [EC

K14054 mpaA; murein peptide amidase A

K01483 allA; ureidoglycolate lyase [EC:4.3.2.3]

K15257 cmoB; tRNA (mo5U34)−methyltransferase [EC:2.1.1.−]

K08096 gch3; GTP cyclohydrolase IIa [EC:3.5.4.29]

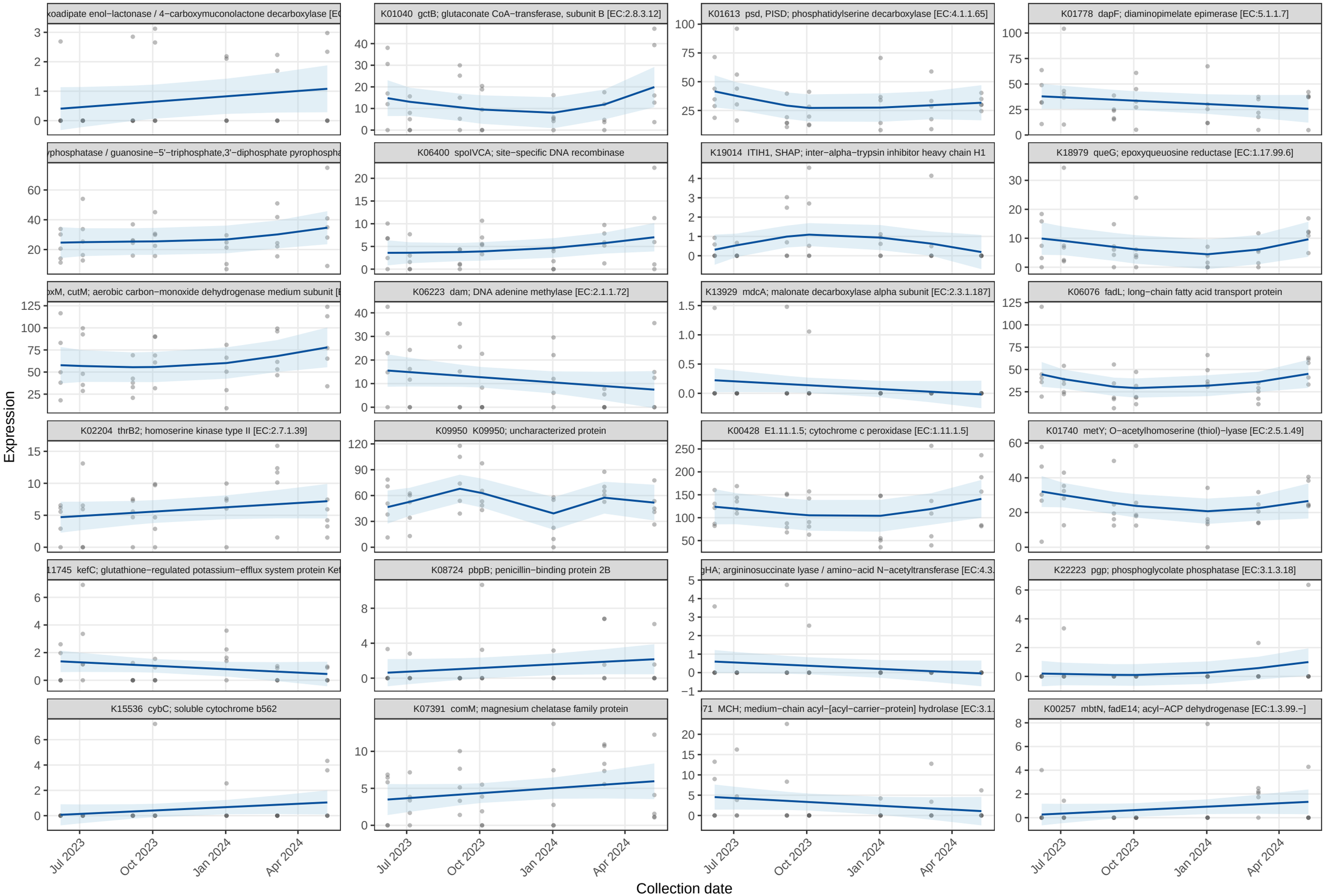
Expression

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All remaining BRITE classes after exclusions

o−inositol 2−dehydrogenase / D−chiro−inositol 1−dehydrogenase [EC:1.

20539 blcR; IclR family transcriptional regulator, blcABC operon repress

K19687 vapB28S; antitoxin VapB28/30/34/36/42

K02594 nifV; homocitrate synthase NifV [EC:2.3.3.14]

K01534 zntA; Zn2+/Cd2+−exporting ATPase [EC:7.2.2.12 7.2.2.21]

K01150 endA; deoxyribonuclease I [EC:3.1.21.1]

9973 mntA; manganese transport system ATP−binding protein [EC:7.2.2

K03842 ALG1; beta−1,4−mannosyltransferase [EC:2.4.1.142]

K01556 KYNU, kynU; kynureninase [EC:3.7.1.3]

K17925 SNX13; sorting nexin−13

K11531 lsrR; lsr operon transcriptional repressor

K07789 mdtC; multidrug efflux pump

Expression

K05844 rimK; ribosomal protein S6−−L−glutamate ligase [EC:6.3.2.−]

K16868 tehB; tellurite methyltransferase [EC:2.1.1.265]

96 rfbP; undecaprenyl−phosphate galactose phosphotransferase [EC:2.

K01659 prpC; 2−methylcitrate synthase [EC:2.3.3.5]

yrR; transcriptional regulator of aroF, aroG, tyrA and aromatic amino acid

K11963 urtE; urea transport system ATP−binding protein

K00796 folP; dihydropteroate synthase [EC:2.5.1.15]

K00891 aroK, aroL; shikimate kinase [EC:2.7.1.71]

K09949 lpxI; UDP−2,3−diacylglucosamine hydrolase [EC:3.6.1.54]

K07979 ytrA; GntR family transcriptional regulator

pchF; 4−cresol dehydrogenase (hydroxylating) flavoprotein subunit [EC:

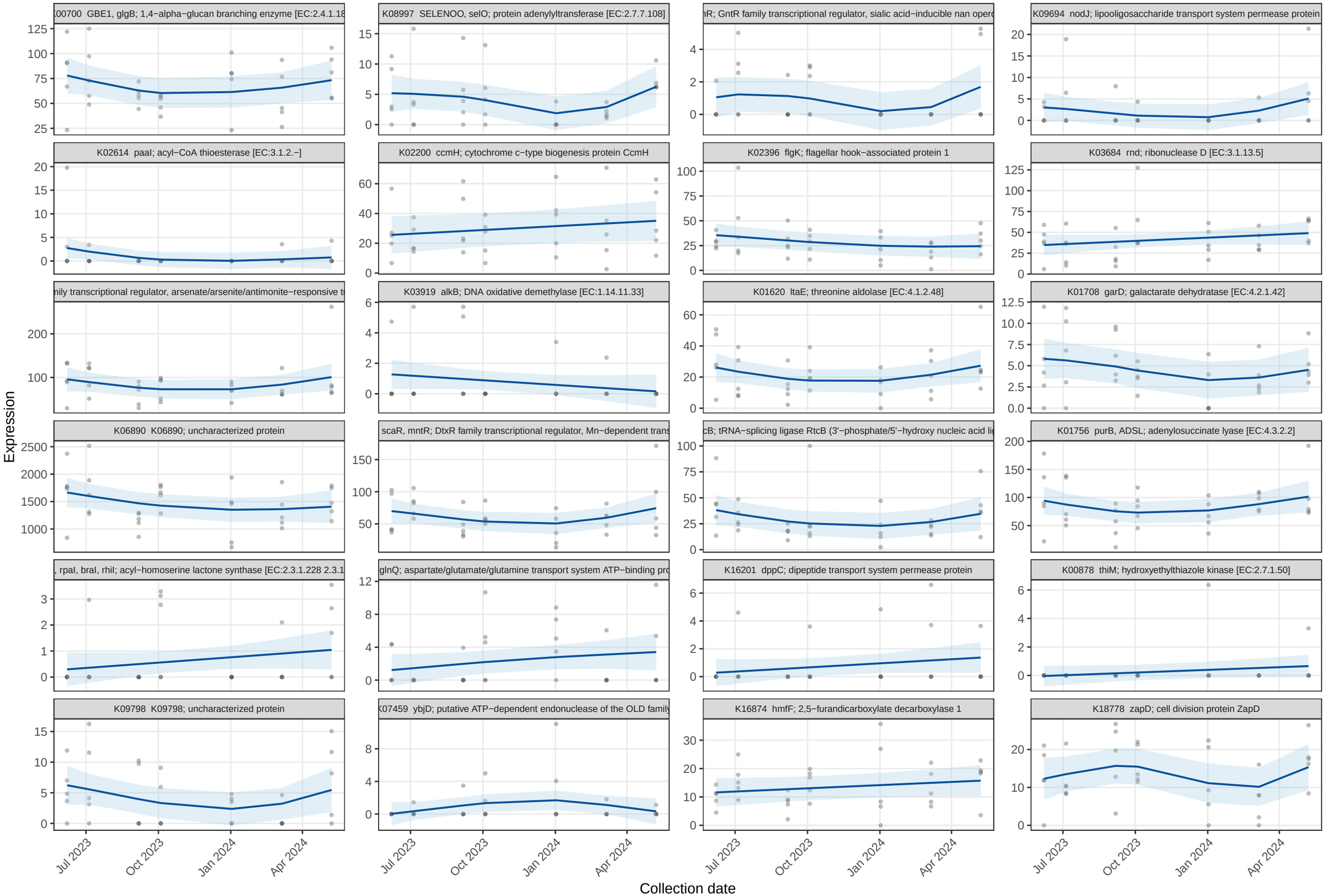
K22068 ISCU; iron−sulfur cluster assembly enzyme ISCU, mitochondria

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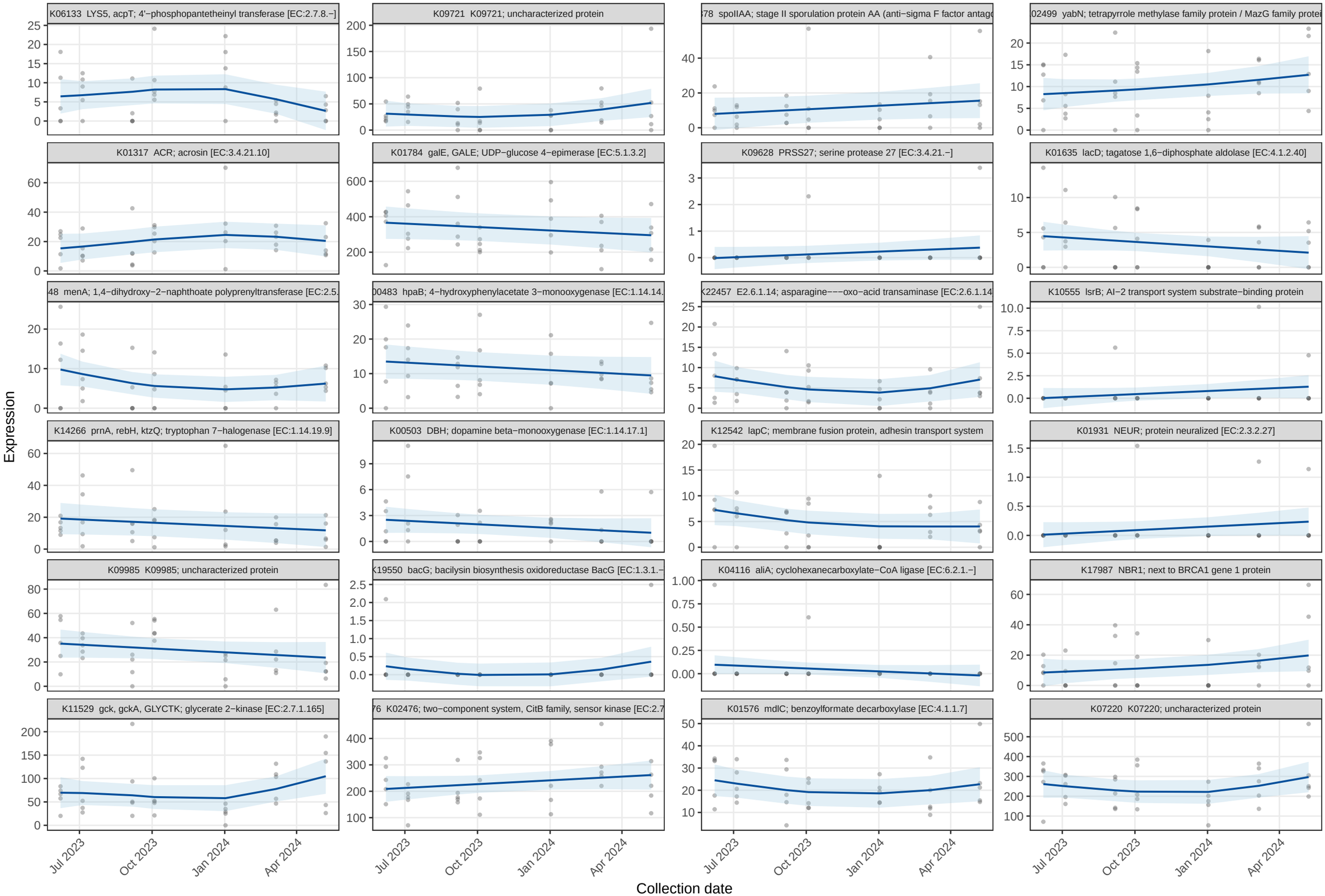
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All remaining BRITE classes after exclusions



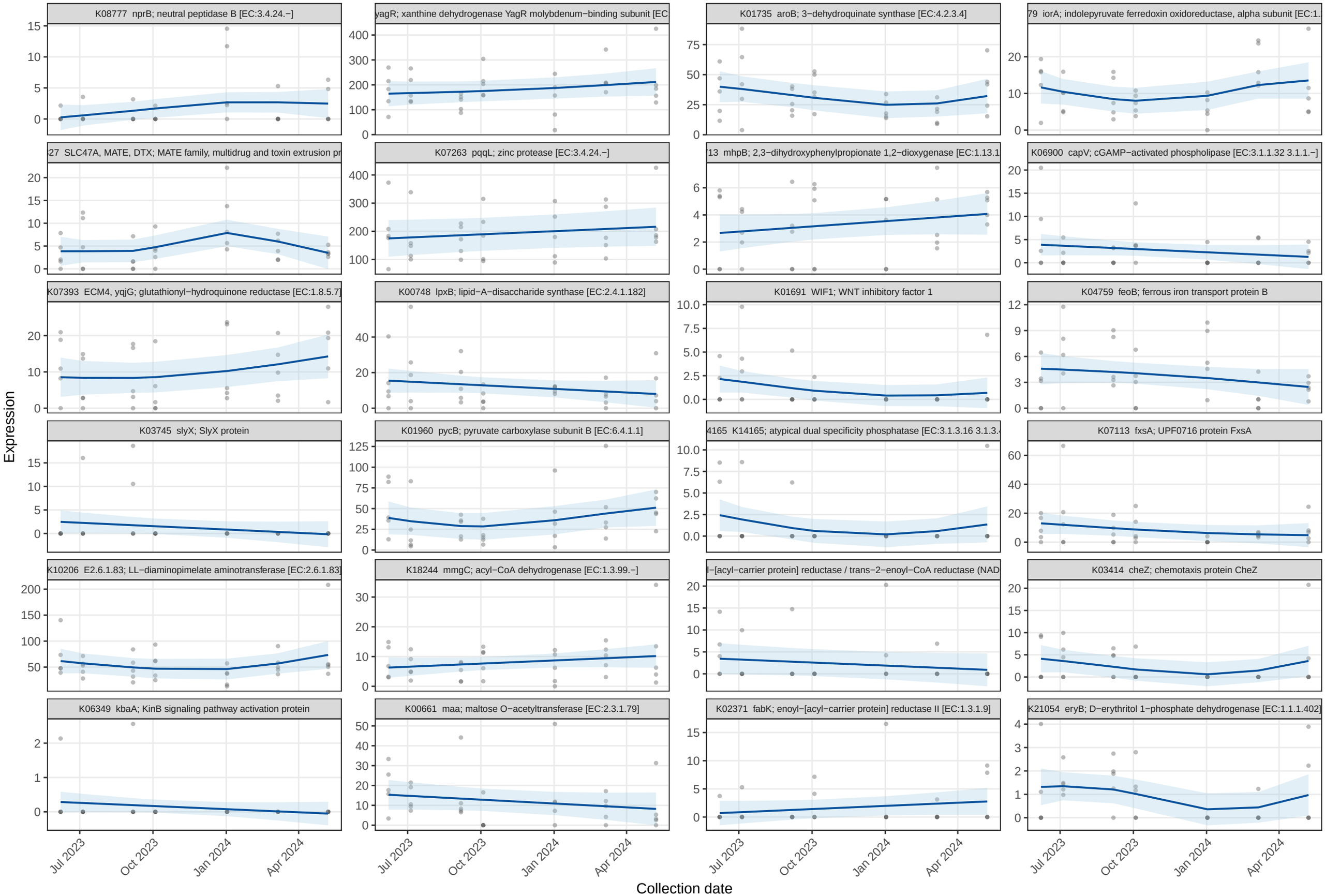
brite_D filtered temporal GAMs: page 99 of 303

All remaining BRITE classes after exclusions



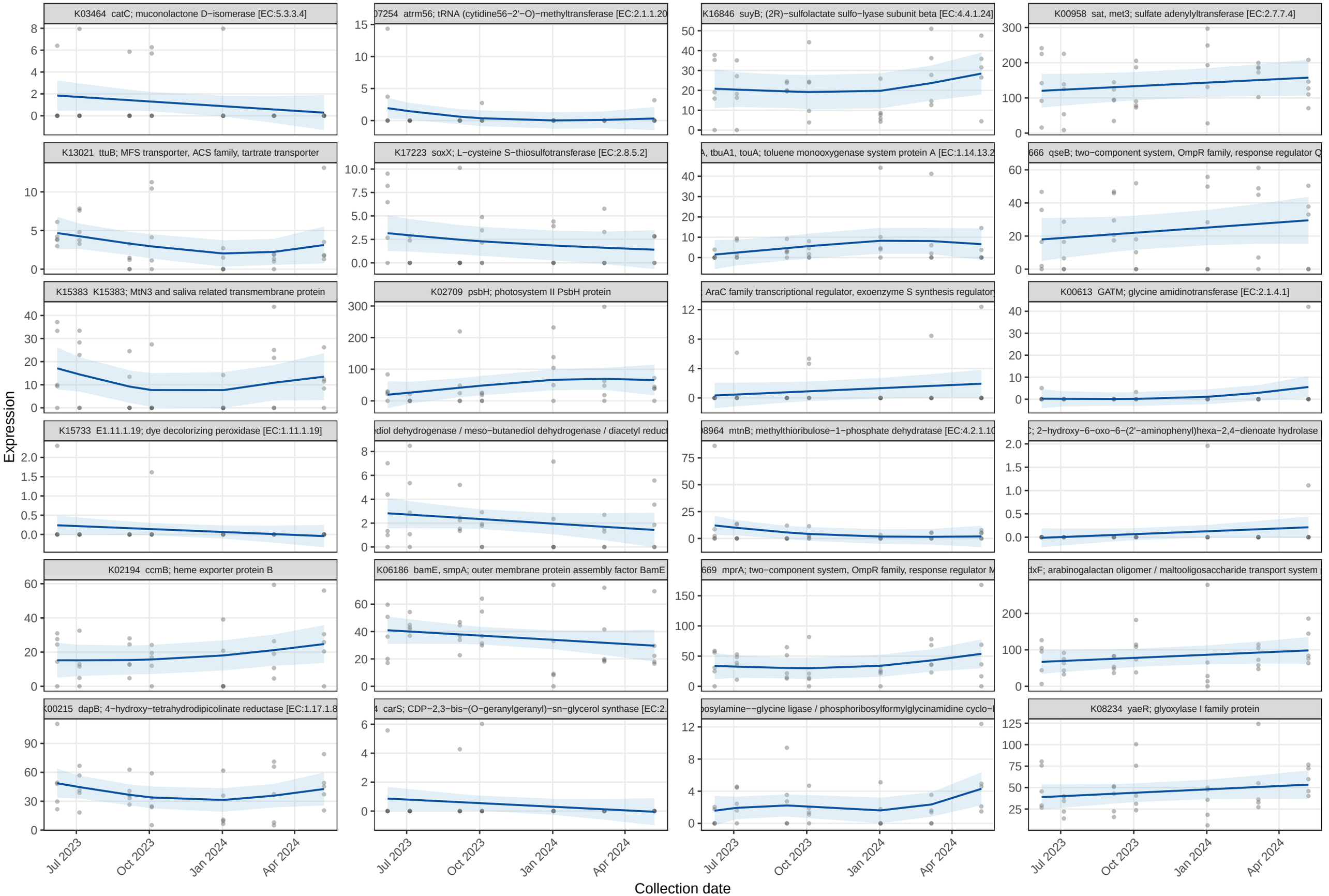
brite_D filtered temporal GAMs: page 100 of 303

All remaining BRITE classes after exclusions



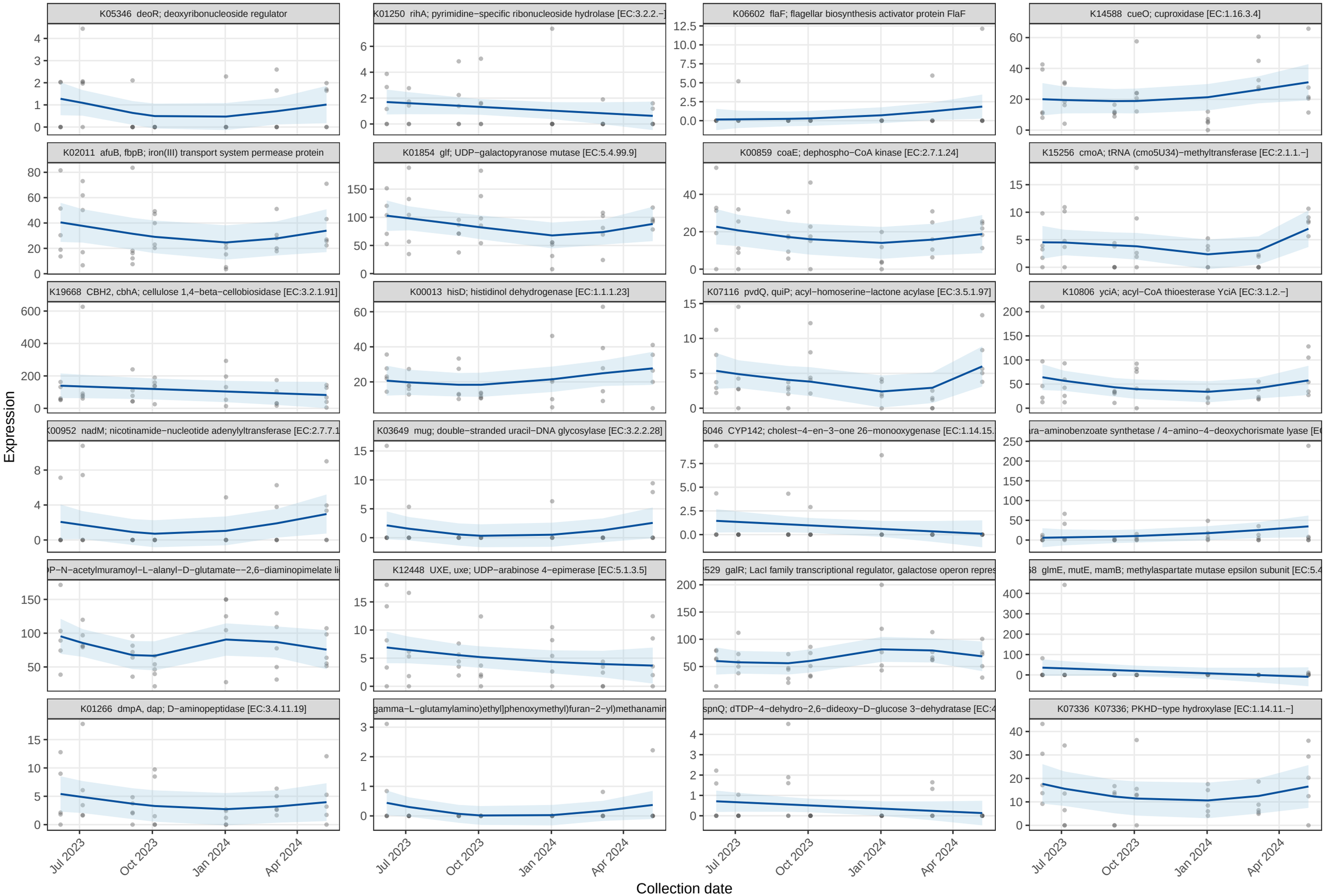
brite_D filtered temporal GAMs: page 101 of 303

All remaining BRITE classes after exclusions



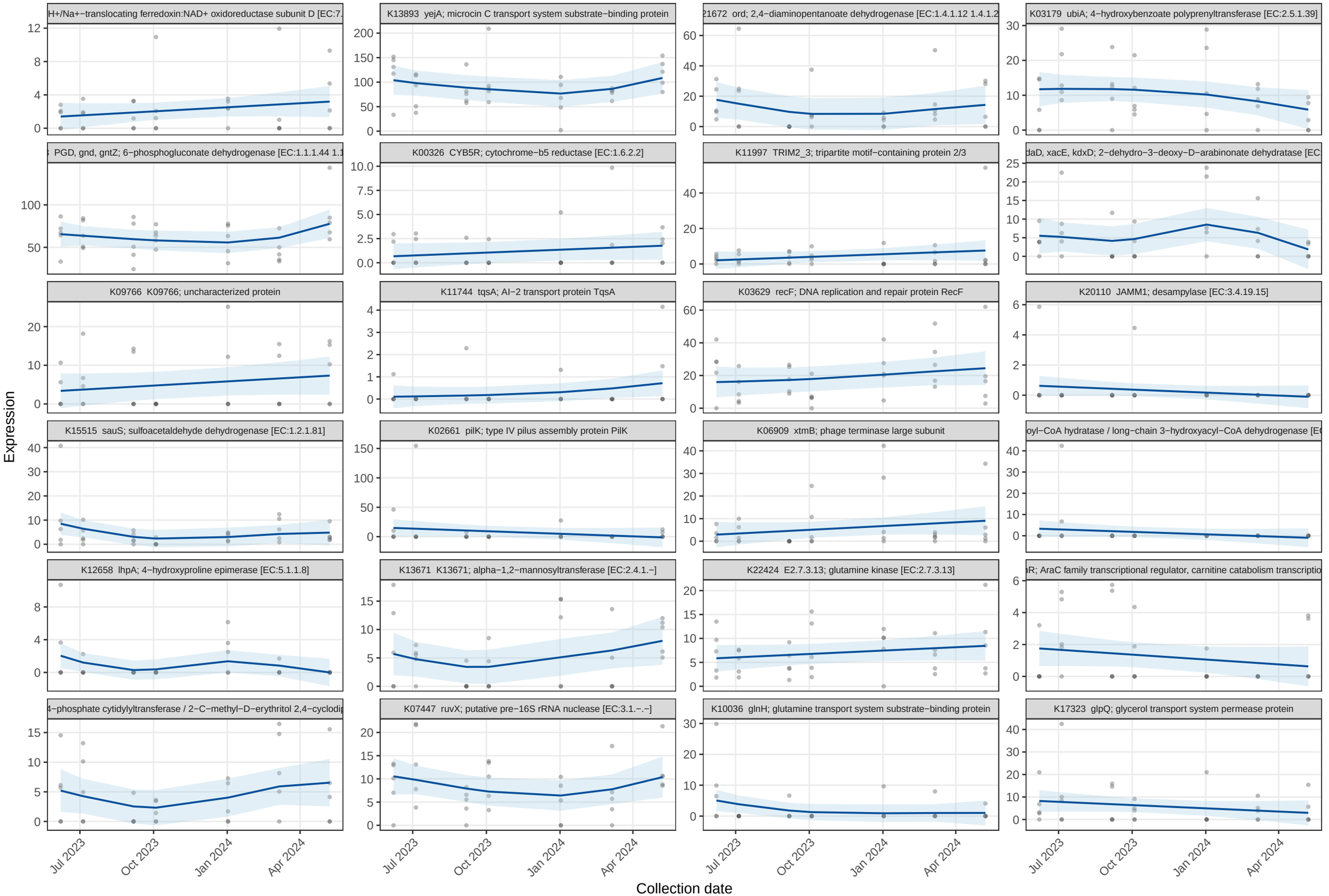
brite_D filtered temporal GAMs: page 102 of 303

All remaining BRITE classes after exclusions



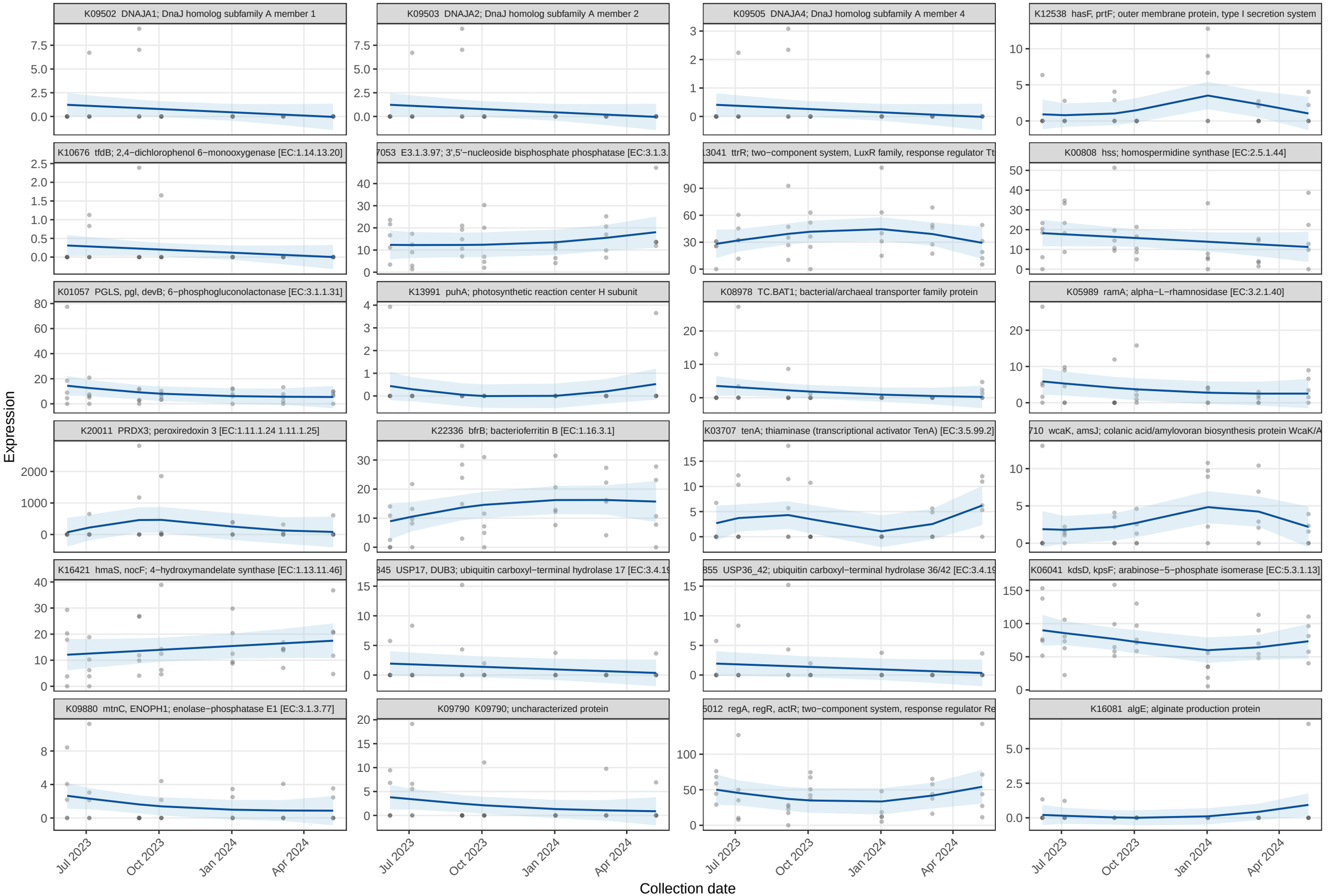
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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



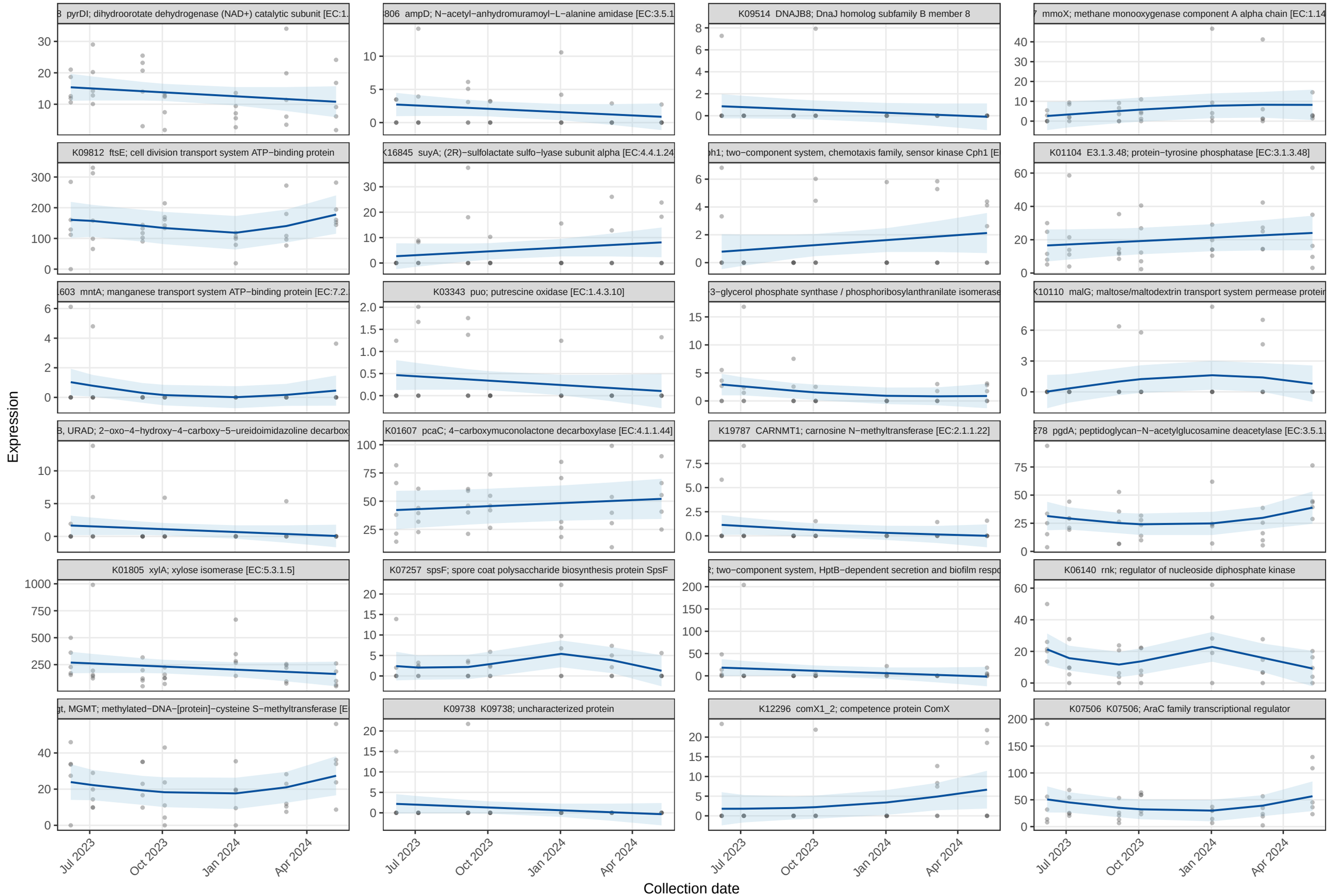
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All remaining BRITE classes after exclusions

Expression

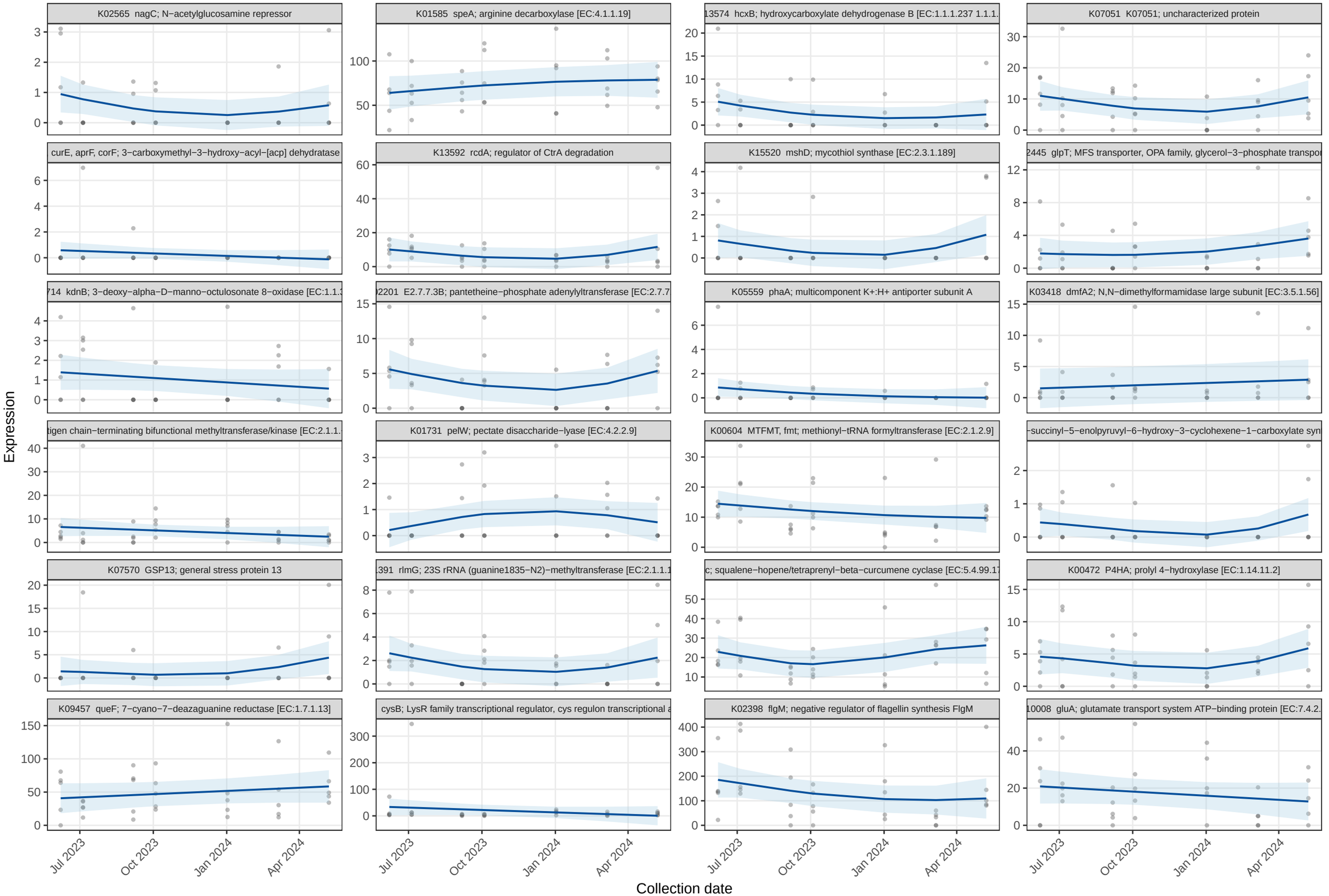
Collection date

All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions

mntB, znuC; manganese/zinc/iron transport system ATP− binding prote

K00226 pyrD; dihydroorotate dehydrogenase (fumarate) [EC:1.3.98.1]

K18701 arsC; arsenate−mycothiol transferase [EC:2.8.4.2]

K06989 nadX, ASPDH; aspartate dehydrogenase [EC:1.4.1.21]

35 DIS3, RRP44; exosome complex exonuclease DIS3/RRP44 [EC:3.1

K03668 hslJ; heat shock protein HslJ

36 K14136; decaprenyl−phosphate phosphoribosyltransferase [EC:2.4.

K17320 lplC; putative aldouronate transport system permease protein

K18028 nicX; 2,5−dihydroxypyridine 5,6−dioxygenase [EC:1.13.11.9]

K02584 nifA; Nif−specific regulatory protein

5 hpaD, hpcB; 3,4−dihydroxyphenylacetate 2,3−dioxygenase [EC:1.13.

K10794 prdB; D−proline reductase (dithiol) PrdB [EC:1.21.4.1]

Expression

vS; two−component system, NarL family, sensor histidine kinase DevS [

K06206 sfsA; sugar fermentation stimulation protein A

chbR, celD; AraC family transcriptional regulator, dual regulator of chb

K01008 selD, SEPHS; selenide, water dikinase [EC:2.7.9.3]

K08651 E3.4.21.66; thermitase [EC:3.4.21.66]

K07497 K07497; putative transposase

K20088 vioE; violacein biosynthesis protein VioE

K21185 sgcE6, mdpE6, kedE6; flavin reductase

K17717 pld; phospholipase D [EC:3.1.4.4]

−component system, NtrC family, nitrogen regulation sensor histidine kir

K09515 DNAJB9; DnaJ homolog subfamily B member 9

aapJ, bztA; general L−amino acid transport system substrate−binding

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sylcobinamide kinase / adenosylcobinamide−phosphate guanylyltransfe

K07222 K07222; putative flavoprotein involved in K+ transport

617 dmpH, xylI, nahK; 2−oxo−3−hexenedioate decarboxylase [EC:4.1.1

K07491 rayT; REP−associated tyrosine transposase

K00227 SC5DL, ERG3; Delta7−sterol 5−desaturase [EC:1.14.19.20]

01884 cysS1; cysteinyl−tRNA synthetase, unknown class [EC:6.1.1.16

K04044 hscA; molecular chaperone HscA

K06401 spoIVFA; stage IV sporulation protein FA

K17938 sbmA, bacA; peptide/bleomycin uptake transporter

nPP; phosphoribosyl 1,2−cyclic phosphate 1,2−diphosphodiesterase [E

K07313 pphA; serine/threonine protein phosphatase 1 [EC:3.1.3.16]

K20850 cgiA; iota−carrageenase [EC:3.2.1.157]

Expression

218 ligK, galC; 4−hydroxy−4−methyl−2−oxoglutarate aldolase [EC:4.1.3

07147 msrP; methionine sulfoxide reductase catalytic subunit [EC:1.8.−

K11959 urtA; urea transport system substrate−binding protein

K01470 E3.5.2.10; creatinine amidohydrolase [EC:3.5.2.10]

E; tRNA 2−(methylsulfanyl)−N6−isopentenyladenosine37 hydroxylase [E

K12433 pks5; polyketide synthase 5

5 gesA, mexP; membrane fusion protein, gold/copper resistance efflux

PTS; 6−pyruvoyltetrahydropterin/6−carboxytetrahydropterin synthase [E

K21551 tsrM; tryptophan 2−C−methyltransferase [EC:2.1.1.106]

K13660 gumM; beta−1,4−glucosyltransferase [EC:2.4.1.−]

K07346 fimC; fimbrial chaperone protein

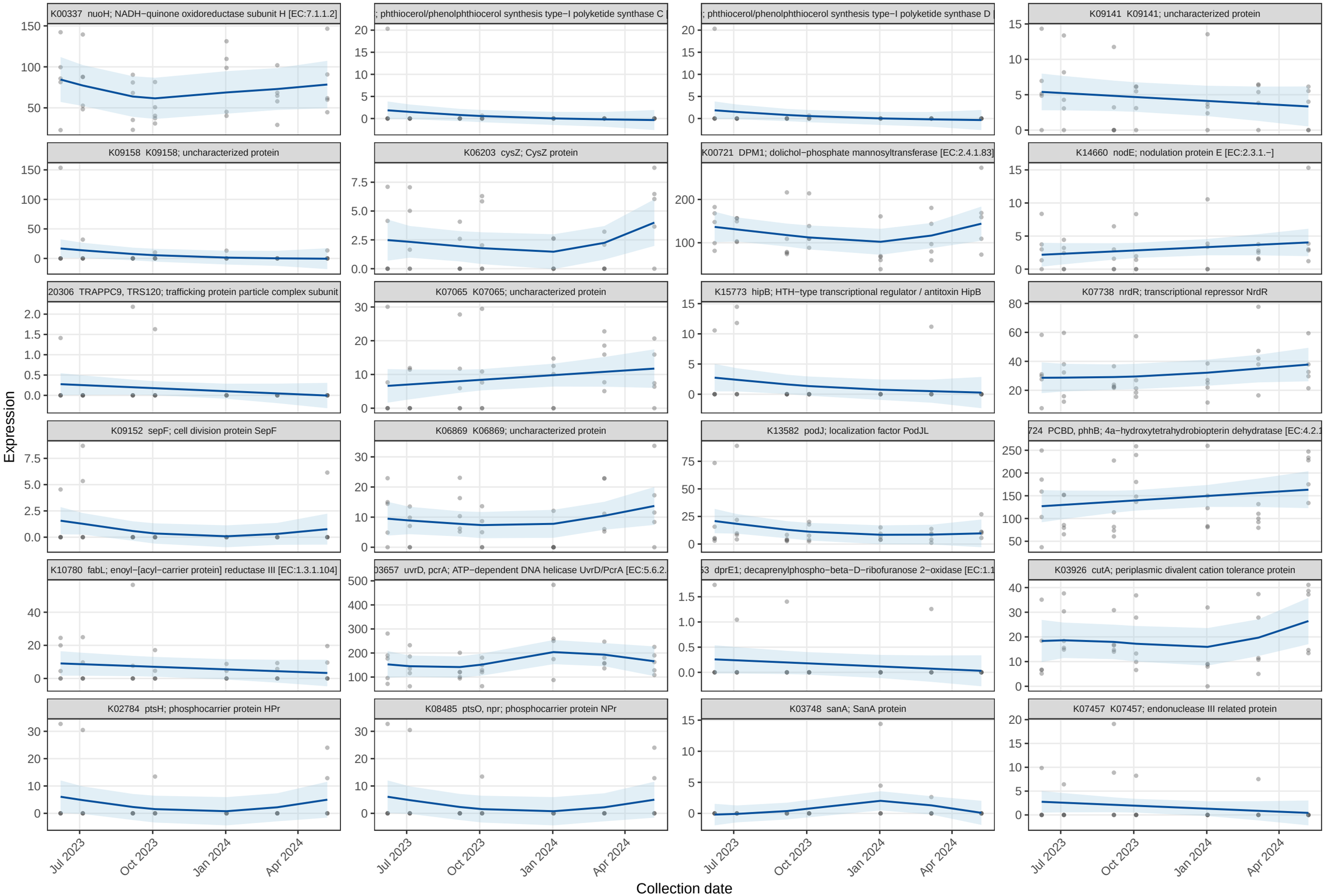
, gacA, varA; two−component system, NarL family, invasion response re

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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions

K03738 aor; aldehyde:ferredoxin oxidoreductase [EC:1.2.7.5]

CDIPT; CDP−diacylglycerol−−inositol 3−phosphatidyltransferase [EC:2

K10556 lsrC; AI−2 transport system permease protein

K15358 ena; enamidase [EC:3.5.2.18]

K19919 CRISP1_2_3; cysteine−rich secretory protein 1/2/3

K20412 PI16, CRISP9, CD364; peptidase inhibitor 16

mtsT; energy−coupling factor transport system substrate−specific com

araC; AraC family transcriptional regulator, arabinose operon regulatory

K12370 dppC; dipeptide transport system permease protein

K03086 rpoD; RNA polymerase primary sigma factor

K10775 PAL; phenylalanine ammonia−lyase [EC:4.3.1.24]

K17312 treT; trehalose transport system permease protein

Expres

scriptional regulator, biotin operon repressor / biotin−−−[acetyl−CoA−car

K00102 LDHD, dld; D−lactate dehydrogenase (cytochrome) [EC:1.1.2.4

K18189 TACO1; translational activator of cytochrome c oxidase 1

K03322 mntH; manganese transport protein

53 actI3, oxyC, tcmM, snoa3, aknD, mtmS; minimal PKS acyl carrier pr

K19424 epsF; glycosyltransferase EpsF [EC:2.4.−.−]

K06891 clpS; ATP−dependent Clp protease adaptor protein ClpS

K19706 FAH; dihydroceramide fatty acyl 2−hydroxylase [EC:1.14.18.7]

K03769 ppiC; peptidyl−prolyl cis−trans isomerase C [EC:5.2.1.8]

K13669 pimE; alpha−1,2−mannosyltransferase [EC:2.4.1.−]

D; LysR family transcriptional regulator, nod−box dependent transcriptio

mponent system, NarL family, capsular synthesis sensor histidine kinas

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All remaining BRITE classes after exclusions

K11208 yncG; GST−like protein

drolase / cobalt−factor III methyltransferase / precorrin−3B C17−methy

K07229 yqjH; ferric−chelate reductase (NADPH) [EC:1.16.1.9]

K21574 susB; glucan 1,4−alpha−glucosidase [EC:3.2.1.3]

1923 cueR; MerR family transcriptional regulator, copper efflux regula

K11945 phdF; extradiol dioxygenase [EC:1.13.11.−]

05 lytT, lytR; two−component system, LytTR family, response regulato

ffT; dTDP−N−acetylfucosamine:lipid II N−acetylfucosaminyltransferase

53 wzyE, rffT; enterobacterial common antigen polymerase [EC:2.4.9

K01563 dhaA; haloalkane dehalogenase [EC:3.8.1.5]

22 menI, DHNAT; 1,4−dihydroxy−2−naphthoyl−CoA hydrolase [EC:3.1

K01191 MAN2C1; alpha−mannosidase [EC:3.2.1.24]

purR; LacI family transcriptional regulator, purine nucleotide synthesis

K03658 helD; DNA helicase IV [EC:5.6.2.4]

K08955 YME1; ATP−dependent metalloprotease [EC:3.4.24.−]

K16842 hpxB; allantoinase [EC:3.5.2.5]

0 pigI, redM, pltF; L−proline−−−[L−prolyl−carrier protein] ligase [EC:6.

K12424 fadD22; 4−hydroxybenzoate adenylyltransferase [EC:6.2.1.50

E; carotenoid phi−ring synthase / carotenoid chi−ring synthase [EC:1.3

K14818 SQT1; ribosome assembly protein SQT1

K11866 STAMBP, AMSH; STAM−binding protein [EC:3.4.19.12]

K19979 acbM; 2−epi−5−epi−valiolone 7−kinase [EC:2.7.1.188]

K20433 valC; C7−cyclitol 7−kinase [EC:2.7.1.214]

K16839 hpxO; FAD−dependent urate hydroxylase [EC:1.14.13.113]

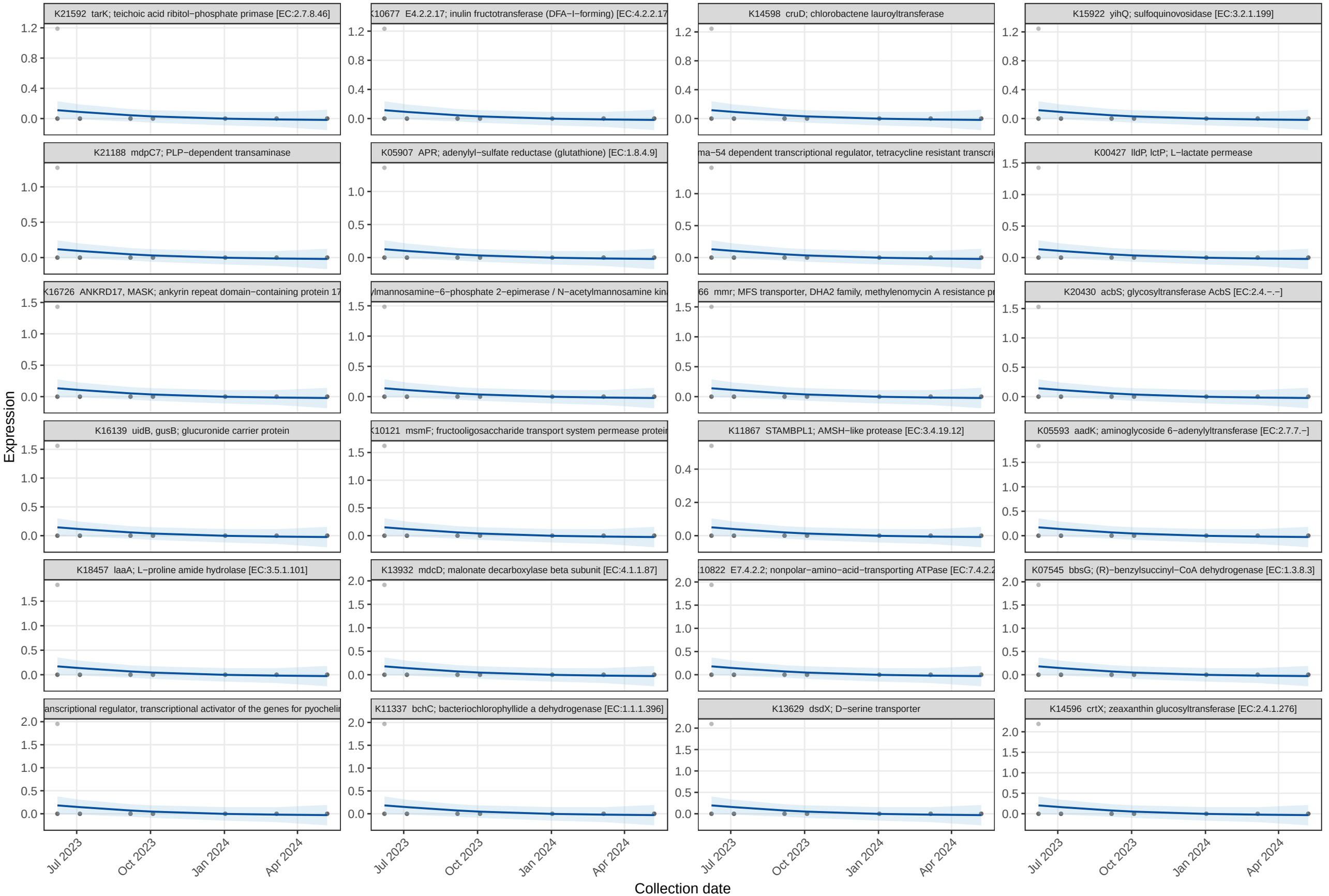
Expression

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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions

had; 6−hydroxycyclohex−1−ene−1−carbonyl−CoA dehydrogenase [EC:1

eryBIV, tylCIV, tylC2; dTDP−4−keto−6−deoxy−L−hexose 4−reductase [E

K13322 megDV; dTDP−4−keto−6−deoxyhexose 4−ketoreductase

K16439 evaE; reductase EvaE

K19857 aveBIV, avrE; dTDP−4−keto−6−deoxy−L−hexose 4−reductase

7 COQ5; 2−methoxy−6−polyprenyl−1,4−benzoquinol methylase [EC:2.1

K14591 aroM; protein AroM

K18569 thnH; 4'−phosphopantetheine phosphatase [EC:3.1.3.110]

K18598 EML6; echinoderm microtubule−associated protein−like 6

K20533 trbI; type IV secretion system protein TrbI

K12700 rihC; non−specific ribonucleoside hydrolase [EC:3.2.2.−]

14057 abgR; LysR family transcriptional regulator, regulator of abg opero

Expres

adaB; methylated−DNA−[protein]−cysteine S−methyltransferase [EC:2

ylC, oleB, carA, srmB; macrolide transport system ATP−binding/permeas

K02228 cobF; precorrin−6A synthase [EC:2.1.1.152]

781 TIM13; mitochondrial import inner membrane translocase subunit T

K21792 redL; polyketide synthase

K19870 SPAG1; sperm−associated antigen 1

K15361 WDR48, UAF1; WD repeat−containing protein 48

K09980 K09980; uncharacterized protein

K10776 LIG3; DNA ligase 3 [EC:6.5.1.1]

; phthiocerol/phenolphthiocerol synthesis type−I polyketide synthase B [

K20529 trbD; type IV secretion system protein TrbD

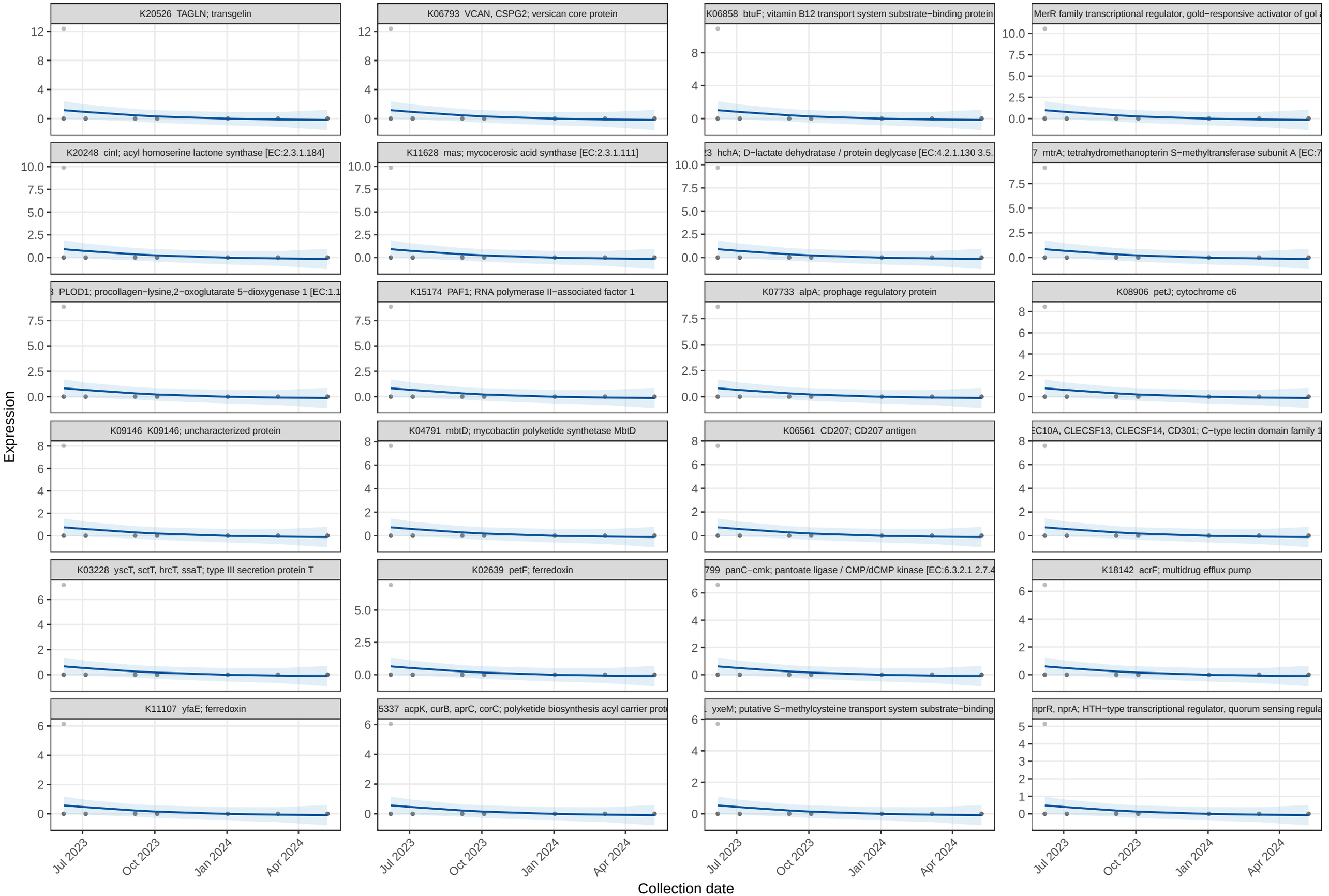
ysyl hydroxylase/galactosyltransferase/glucosyltransferase [EC:1.14.11.4

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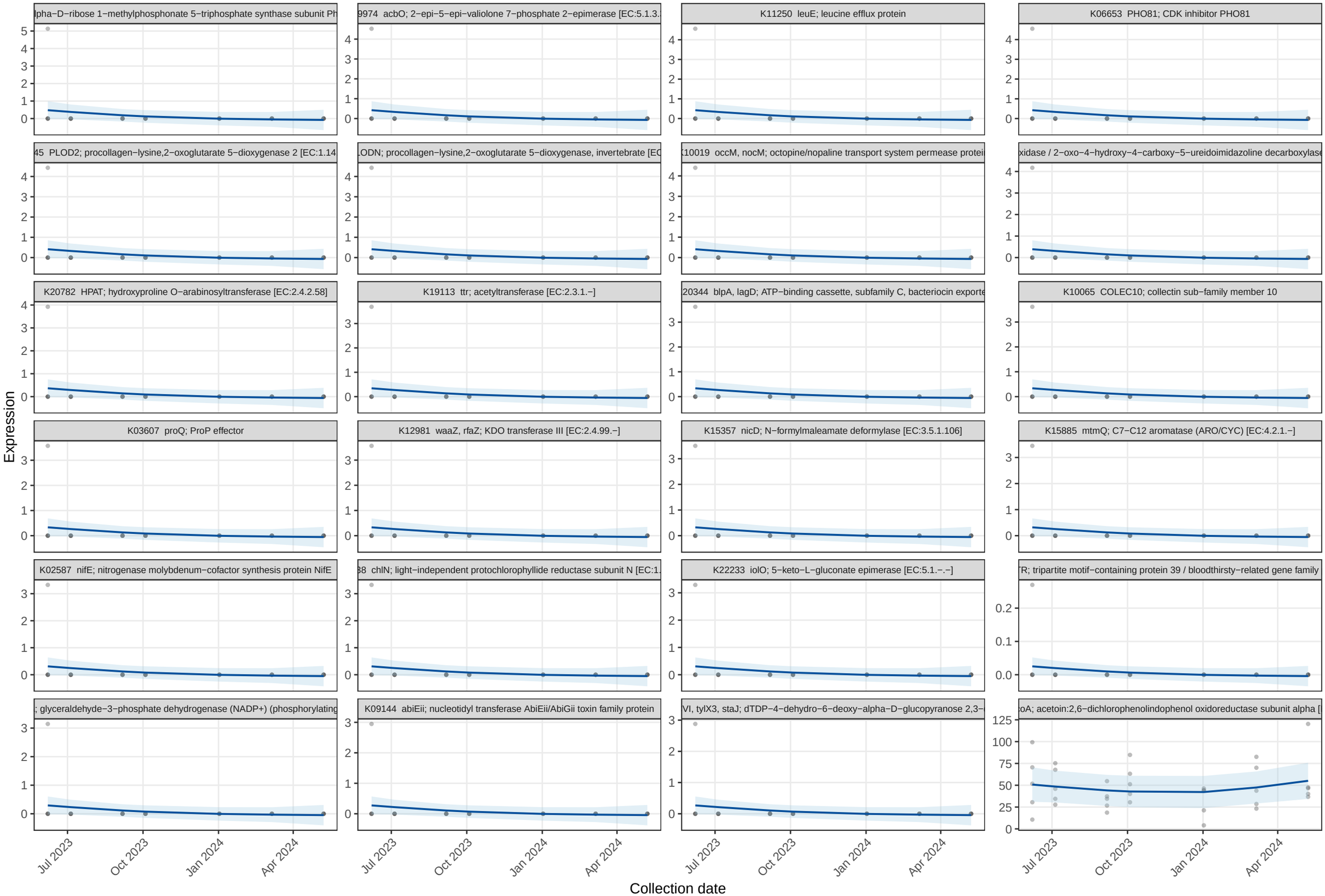
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All remaining BRITE classes after exclusions



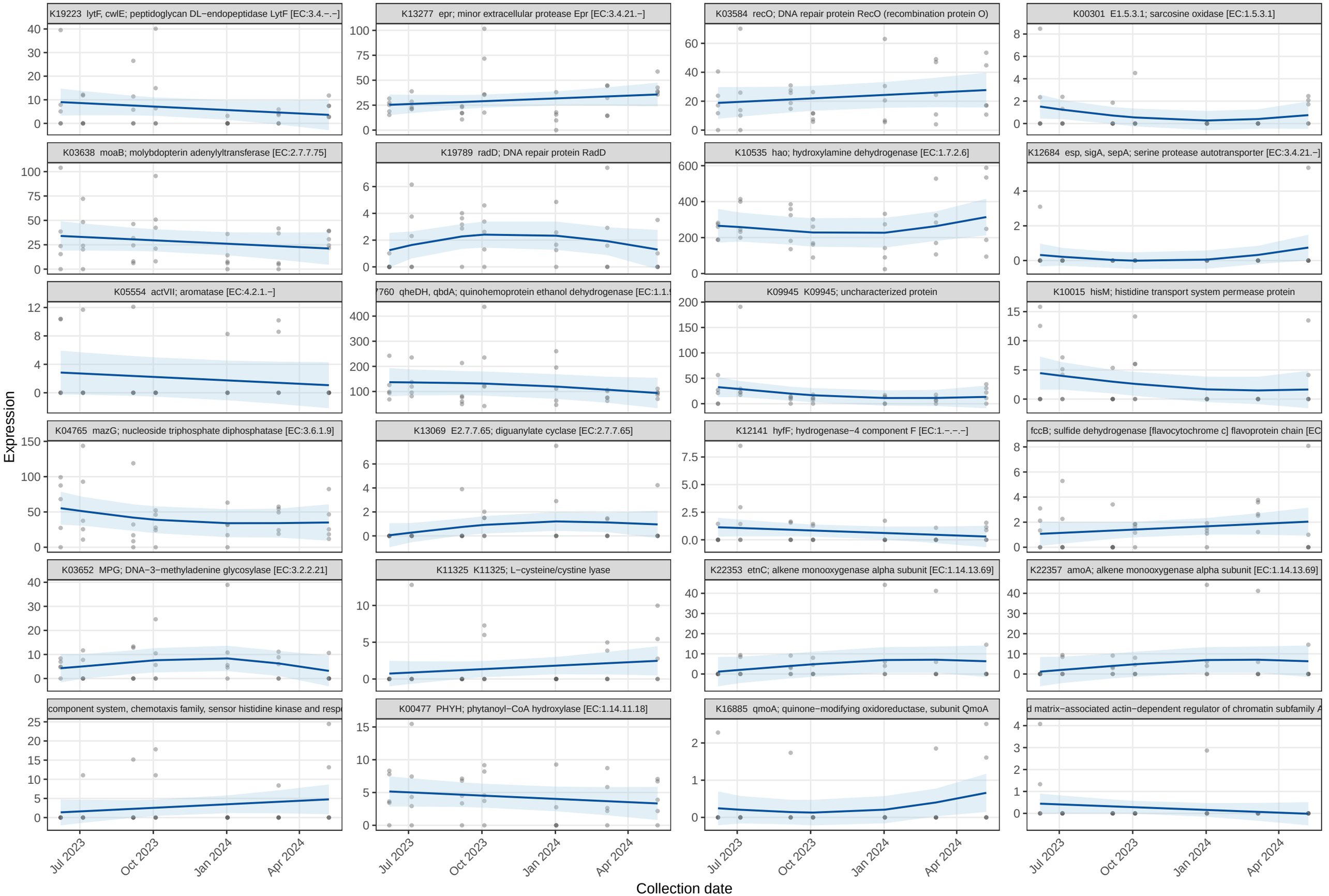
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All remaining BRITE classes after exclusions



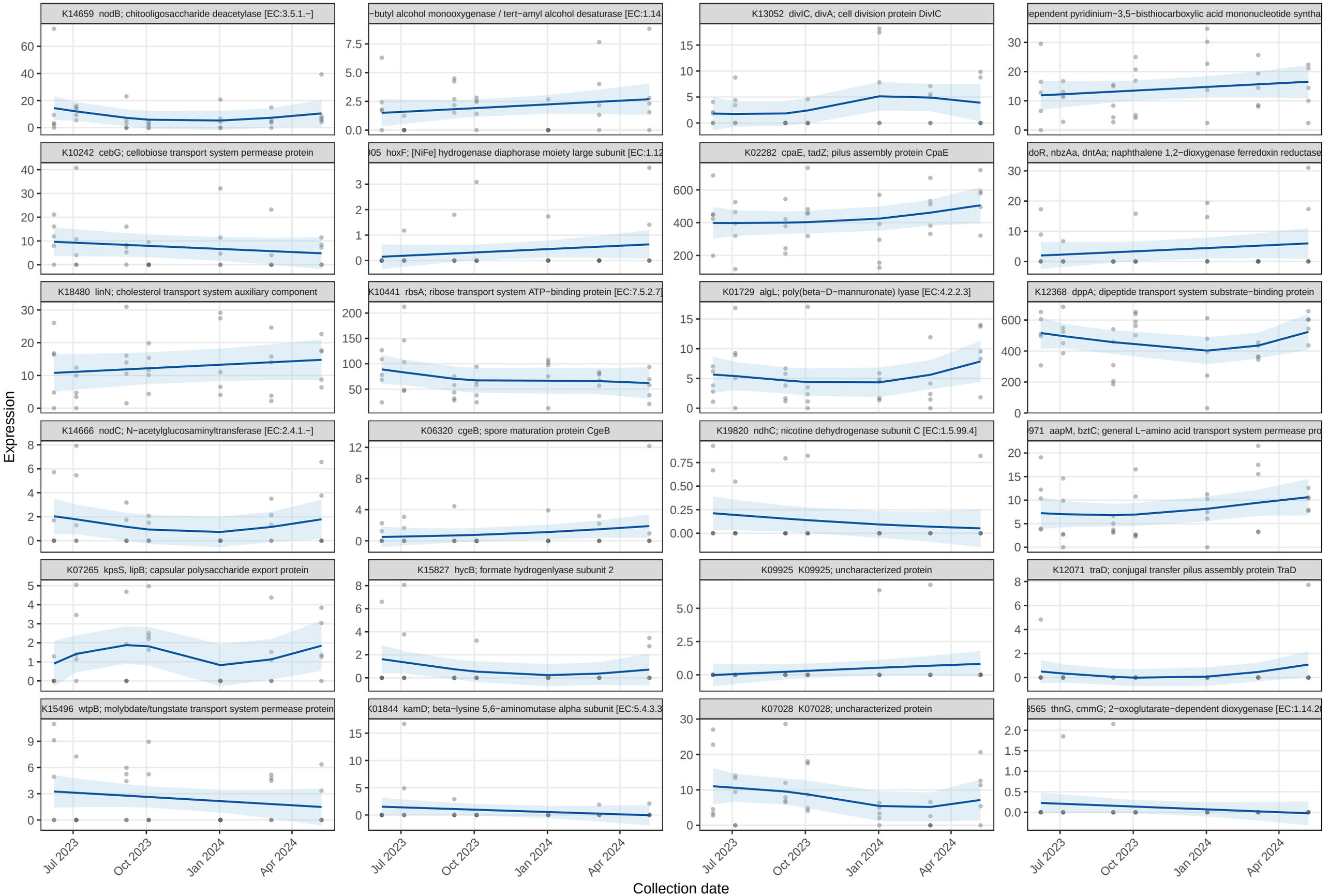
brite_D filtered temporal GAMs: page 117 of 303

All remaining BRITE classes after exclusions



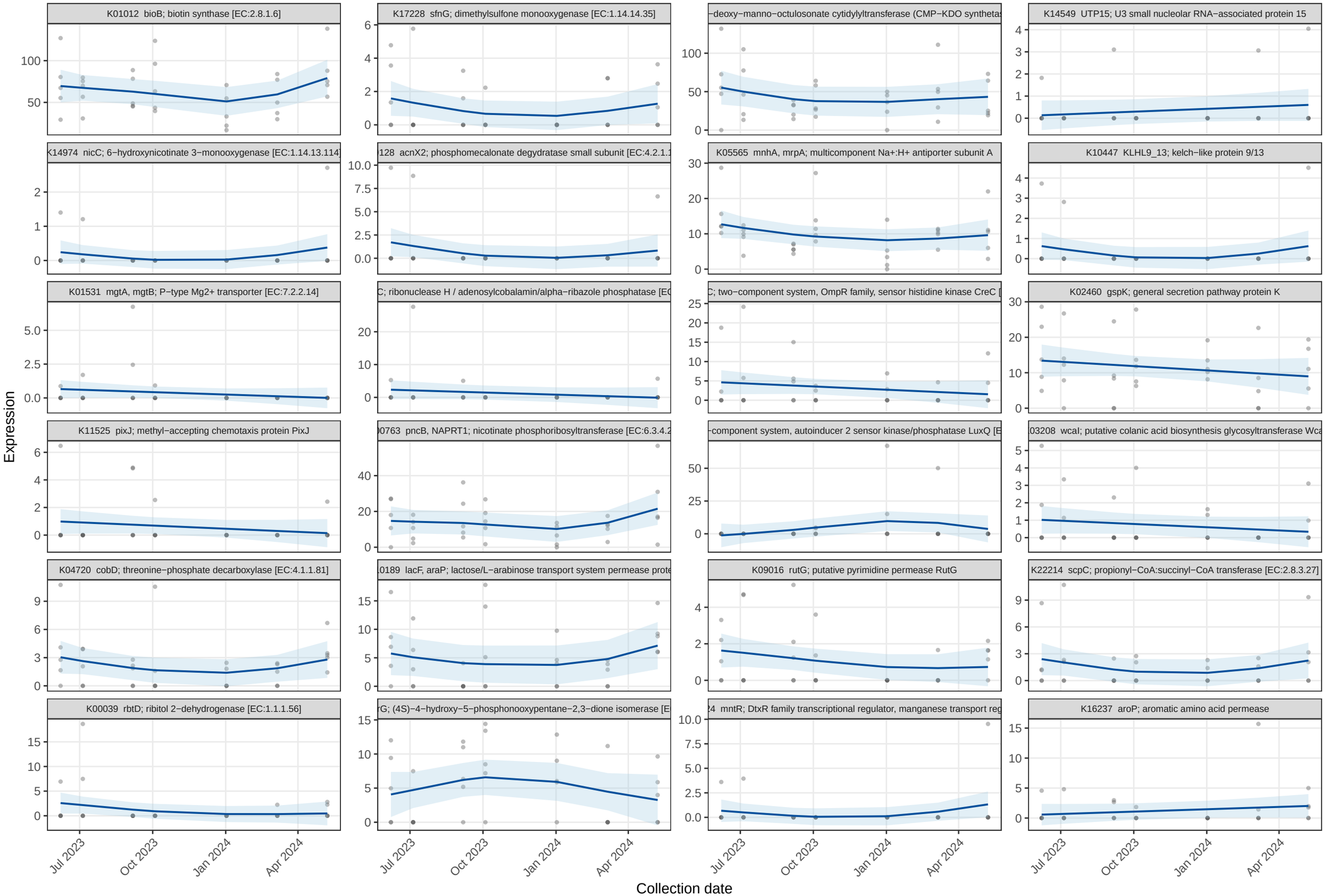
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All remaining BRITE classes after exclusions

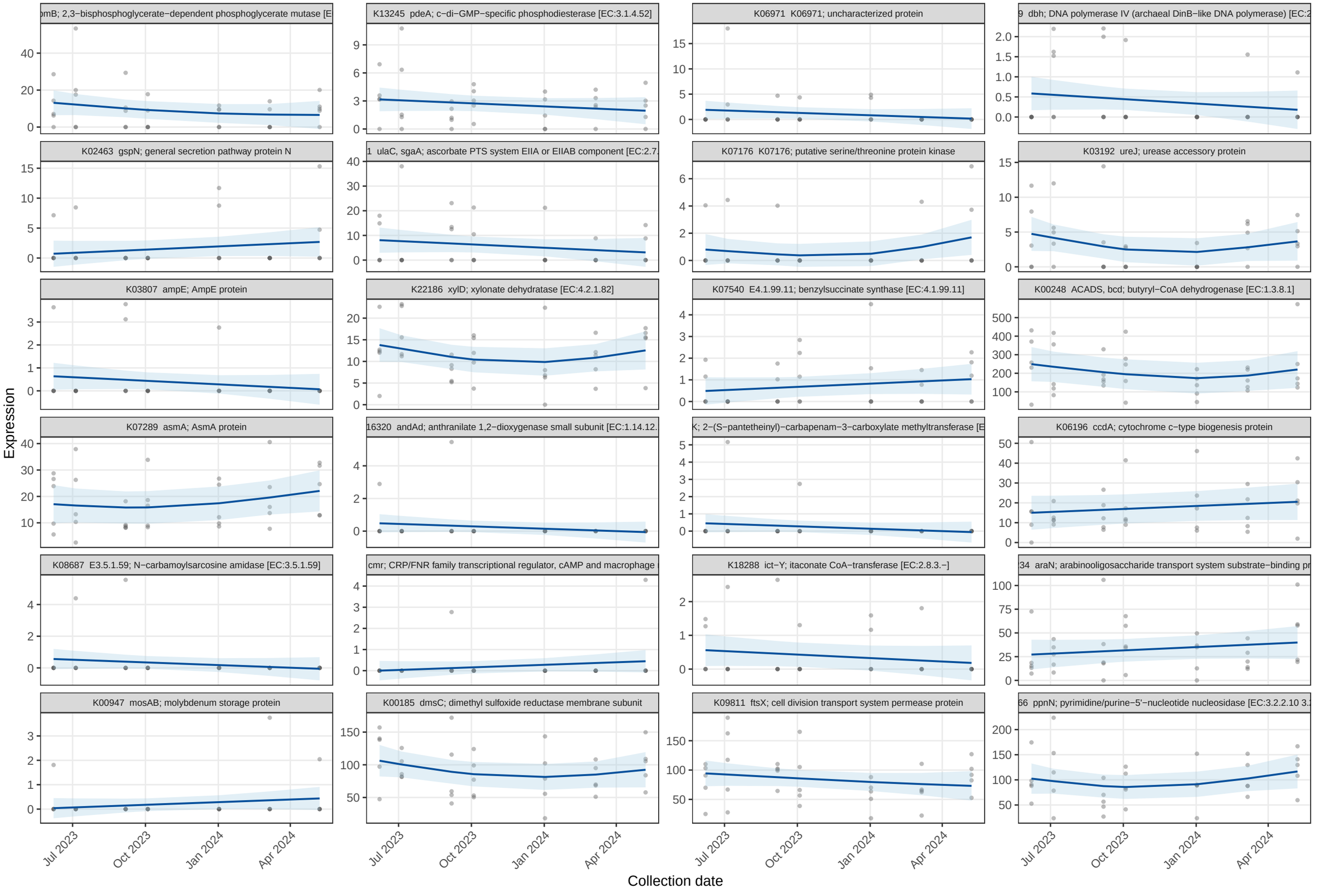


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All remaining BRITE classes after exclusions

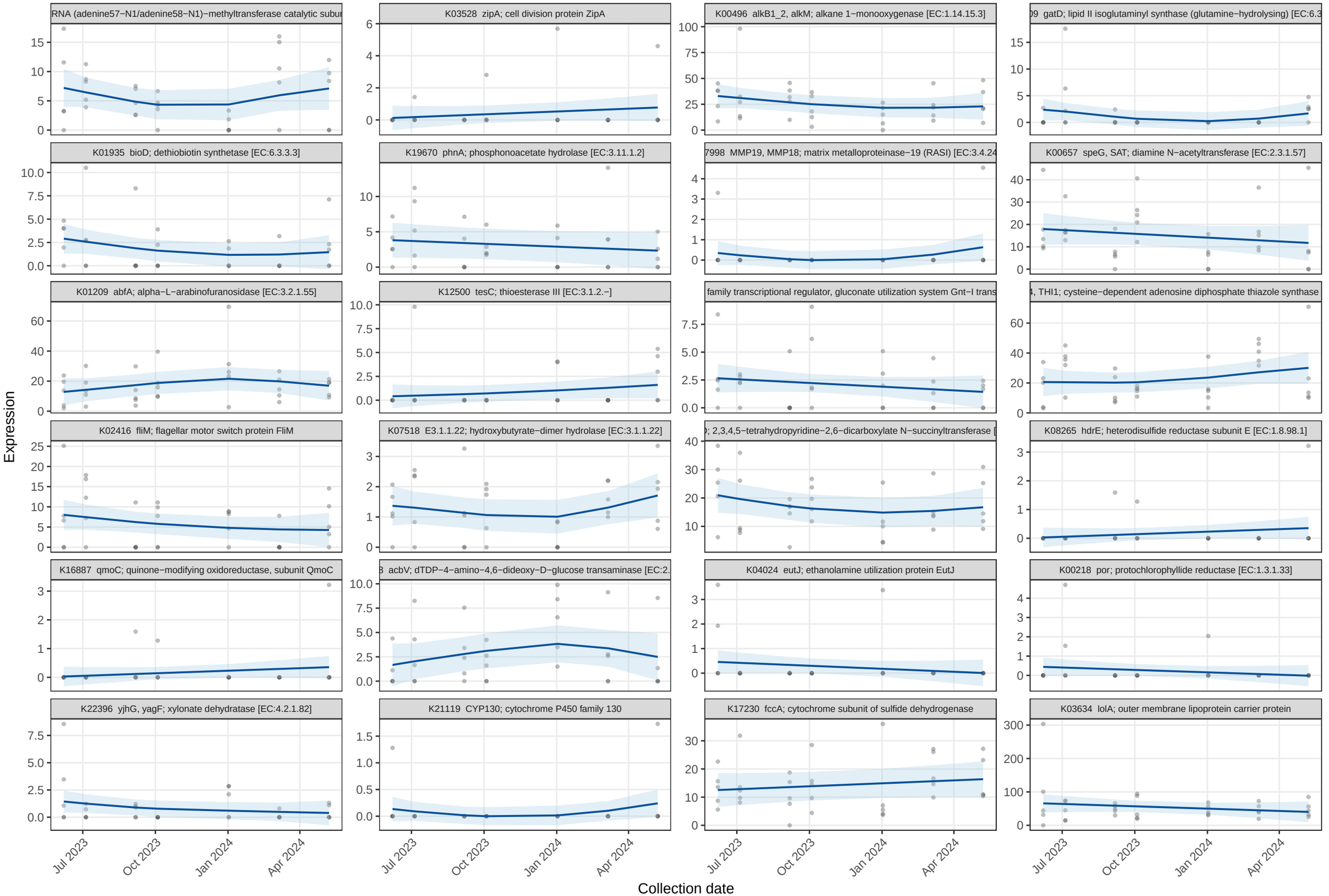


All remaining BRITE classes after exclusions



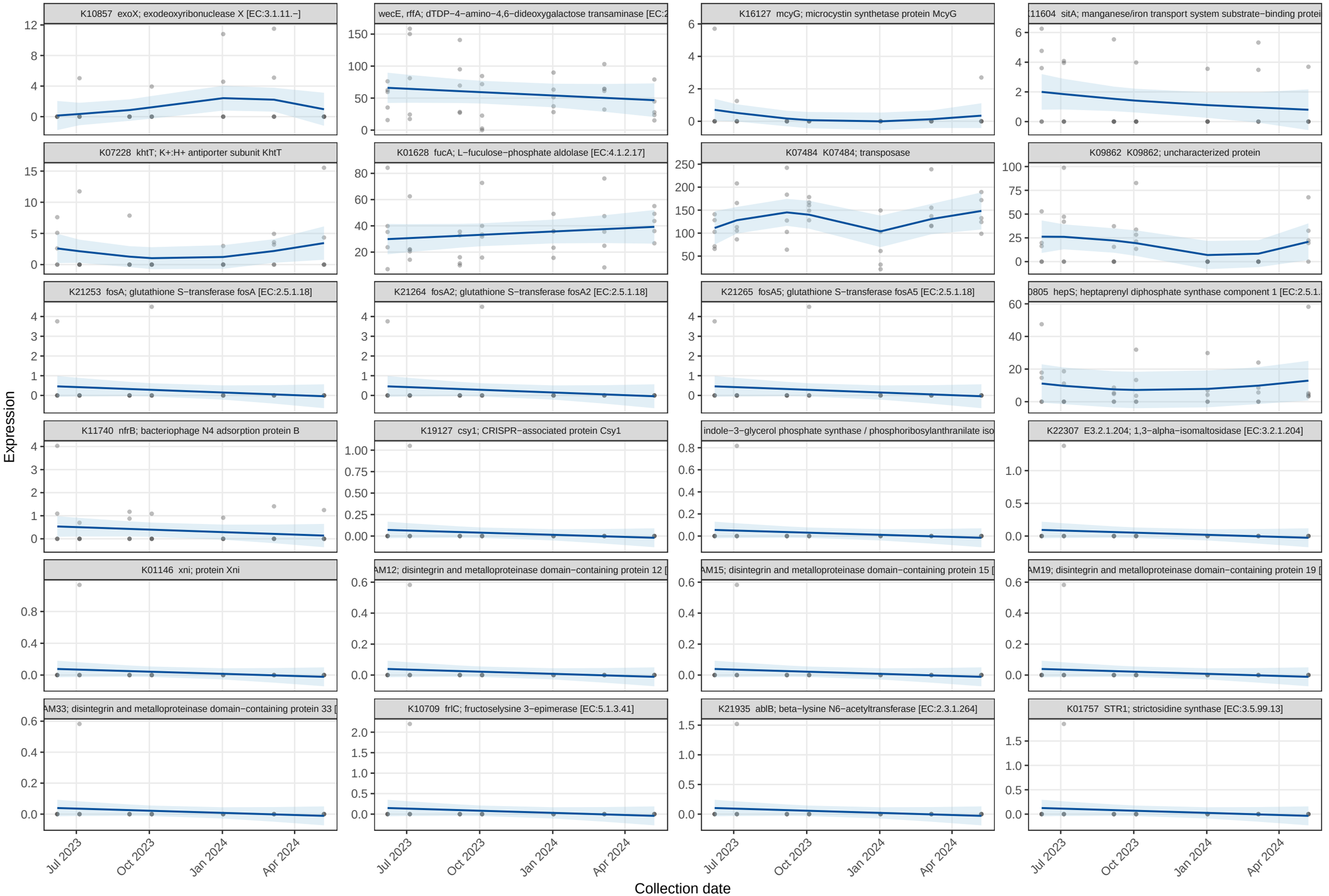
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All remaining BRITE classes after exclusions



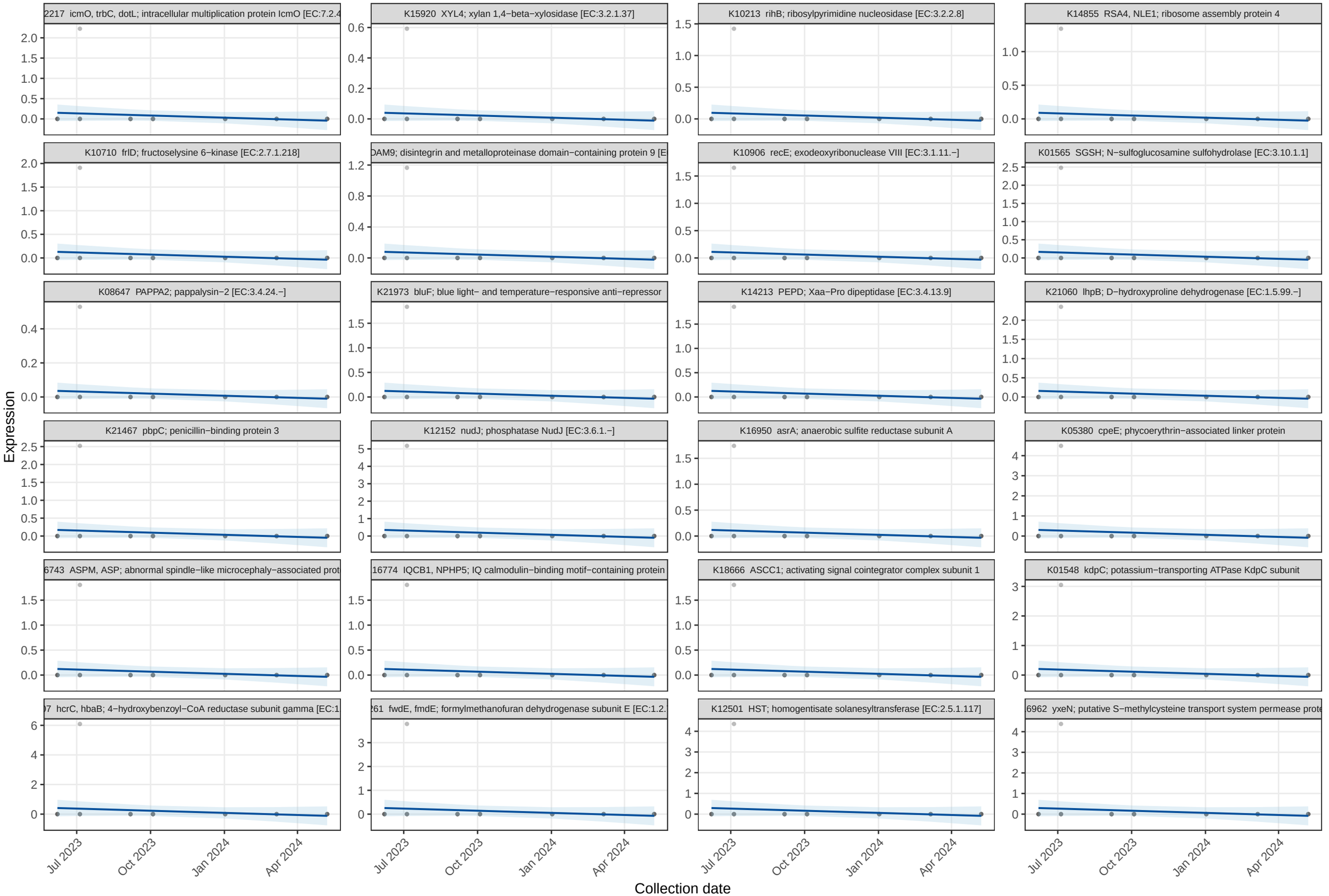
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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions

K10442 KLHL1_4_5; kelch−like protein 1/4/5

K03475 ulaA, sgaT; ascorbate PTS system EIIC component

K17486 dmdA; dimethylsulfoniopropionate demethylase [EC:2.1.1.269]

K08699 ccmN; carbon dioxide concentrating mechanism protein CcmN

K12231 HECTD1; E3 ubiquitin−protein ligase HECTD1 [EC:2.3.2.26]

K06826 NID; nidogen (entactin)

K20096 DDX28; ATP−dependent RNA helicase DDX28 [EC:5.6.2.7]

K13444 SUMF1, FGE; formylglycine−generating enzyme [EC:1.8.3.7]

K06963 TAN1, THUMPD1; tRNA acetyltransferase TAN1

mtB; ArsR family transcriptional regulator, zinc−responsive transcriptiona

K08786 HTRA4; HtrA serine peptidase 4 [EC:3.4.21.−]

K03340 dapdh; diaminopimelate dehydrogenase [EC:1.4.1.16]

K02300 cyoD; cyochrome o ubiquinol oxidase subunit IV

K01226 treC; trehalose−6−phosphate hydrolase [EC:3.2.1.93]

K21585 kat; 3−aminobutanoyl−CoA transaminase [EC:2.6.1.111]

K11517 HAO; (S)−2−hydroxy−acid oxidase [EC:1.1.3.15]

K07453 K07453; putative restriction endonuclease

K15679 rhiF; rhizoxin biosynthesis, polyketide synthase RhiF

K02641 petH; ferredoxin−−NADP+ reductase [EC:1.18.1.2]

K22144 TEME2; cell surface hyaluronidase [EC:3.2.1.35]

3 btaB; S−adenosylmethionine−diacylgycerolhomoserine−N−methlytrans

3896 iucB; acetyl CoA:N6−hydroxylysine acetyl transferase [EC:2.3.1.10

16444 gtfB, gtfE; vancomycin aglycone glucosyltransferase [EC:2.4.1.31

K09843 CYP707A; (+)−abscisic acid 8'−hydroxylase [EC:1.14.14.137]

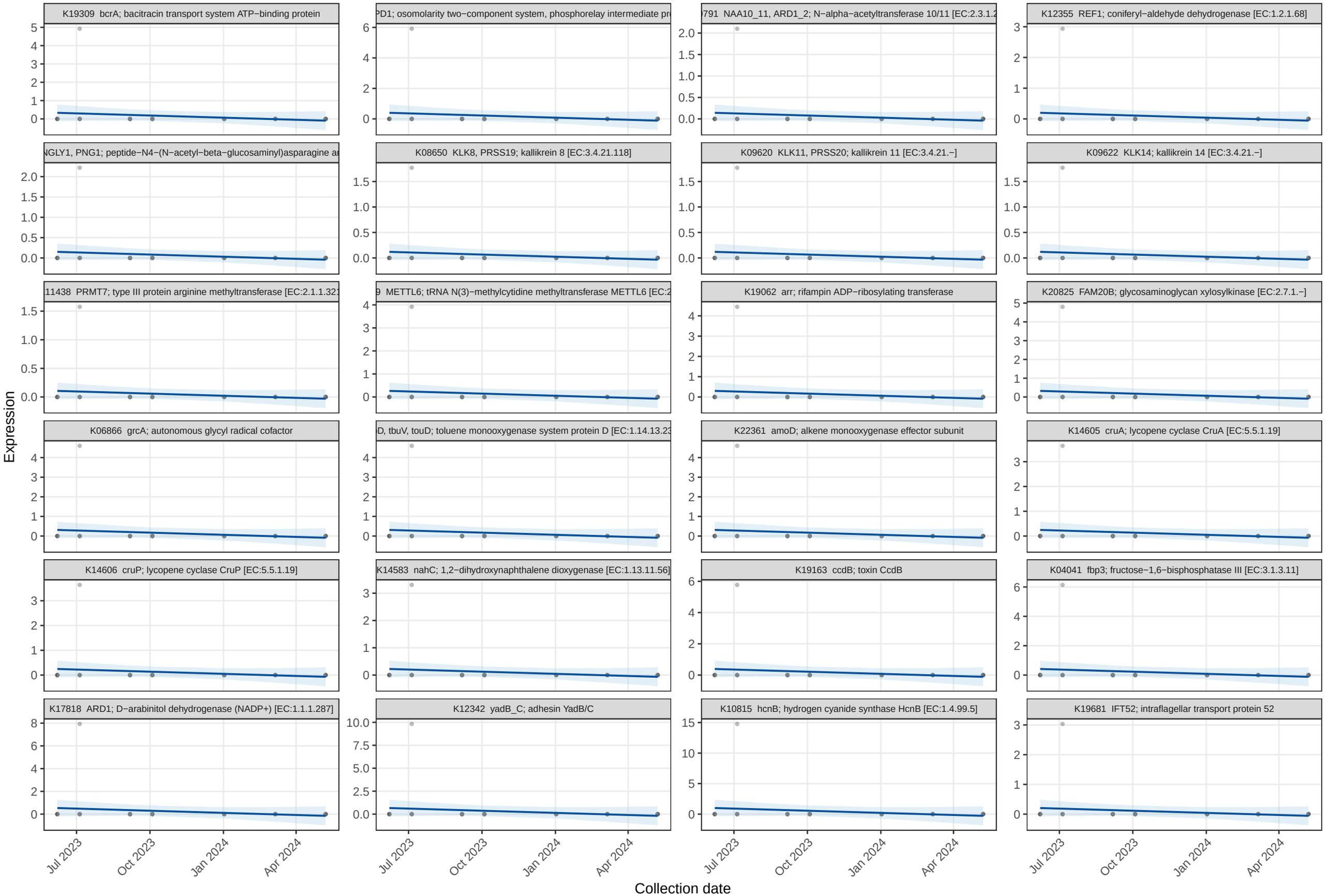
Expression

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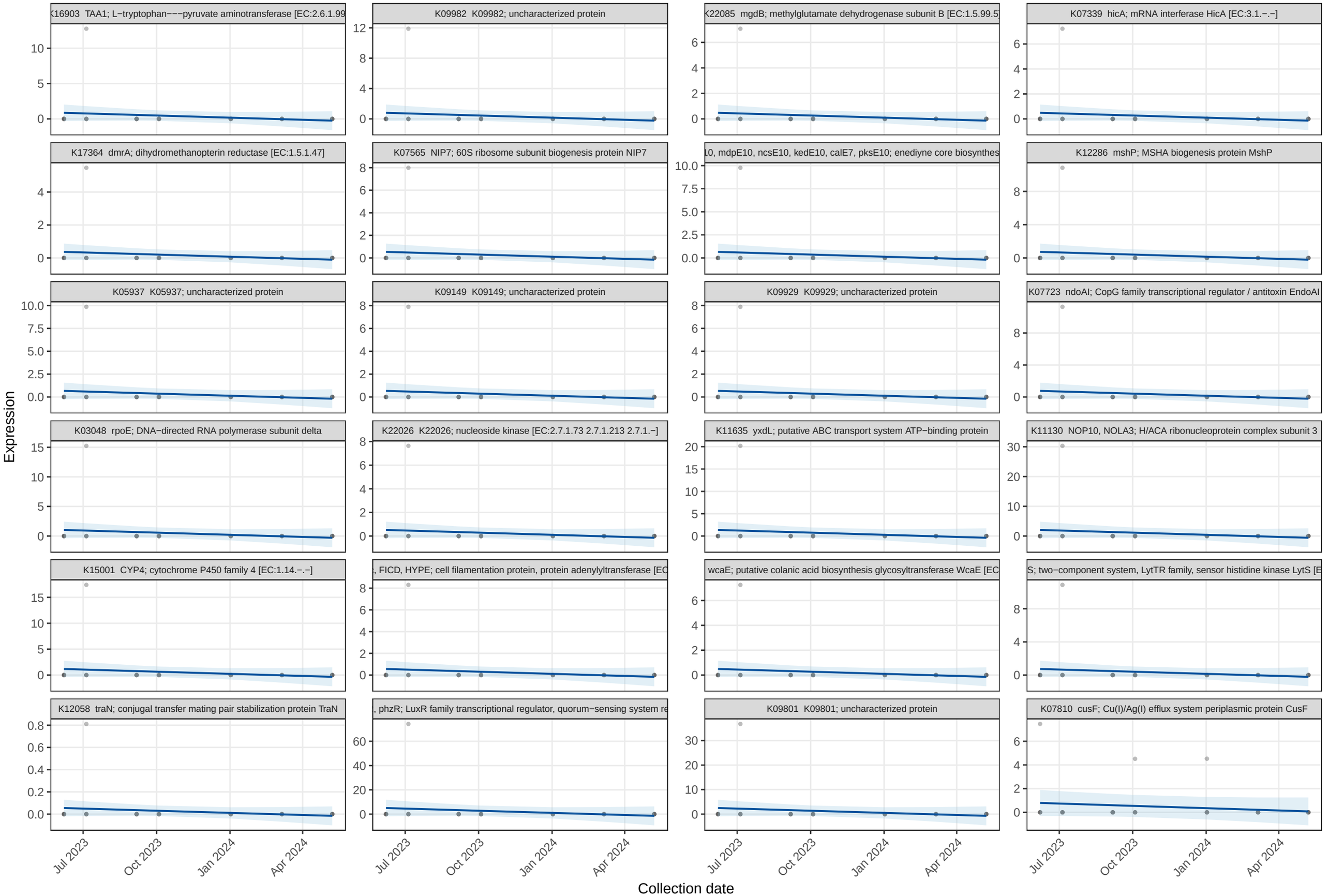
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All remaining BRITE classes after exclusions



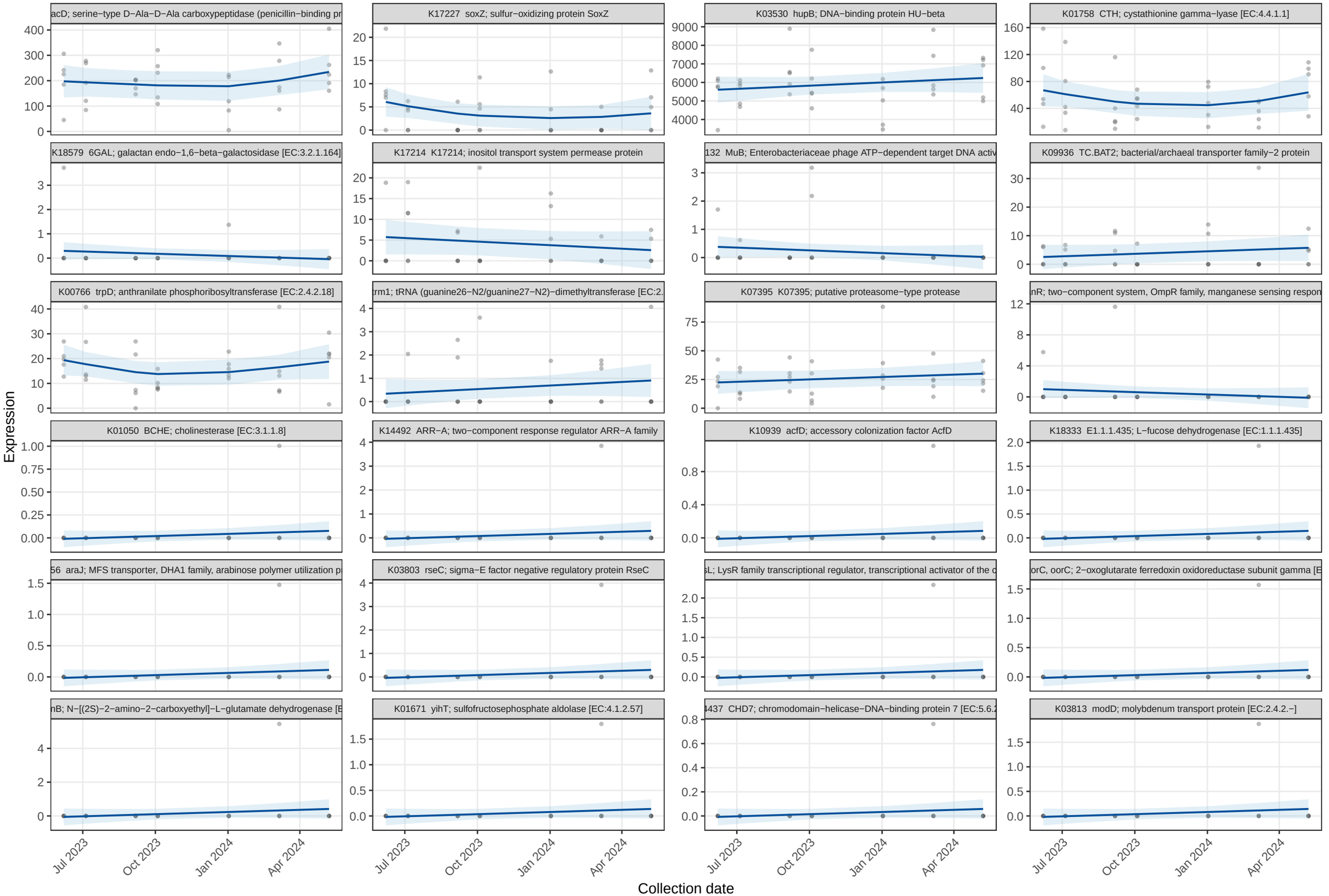
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All remaining BRITE classes after exclusions



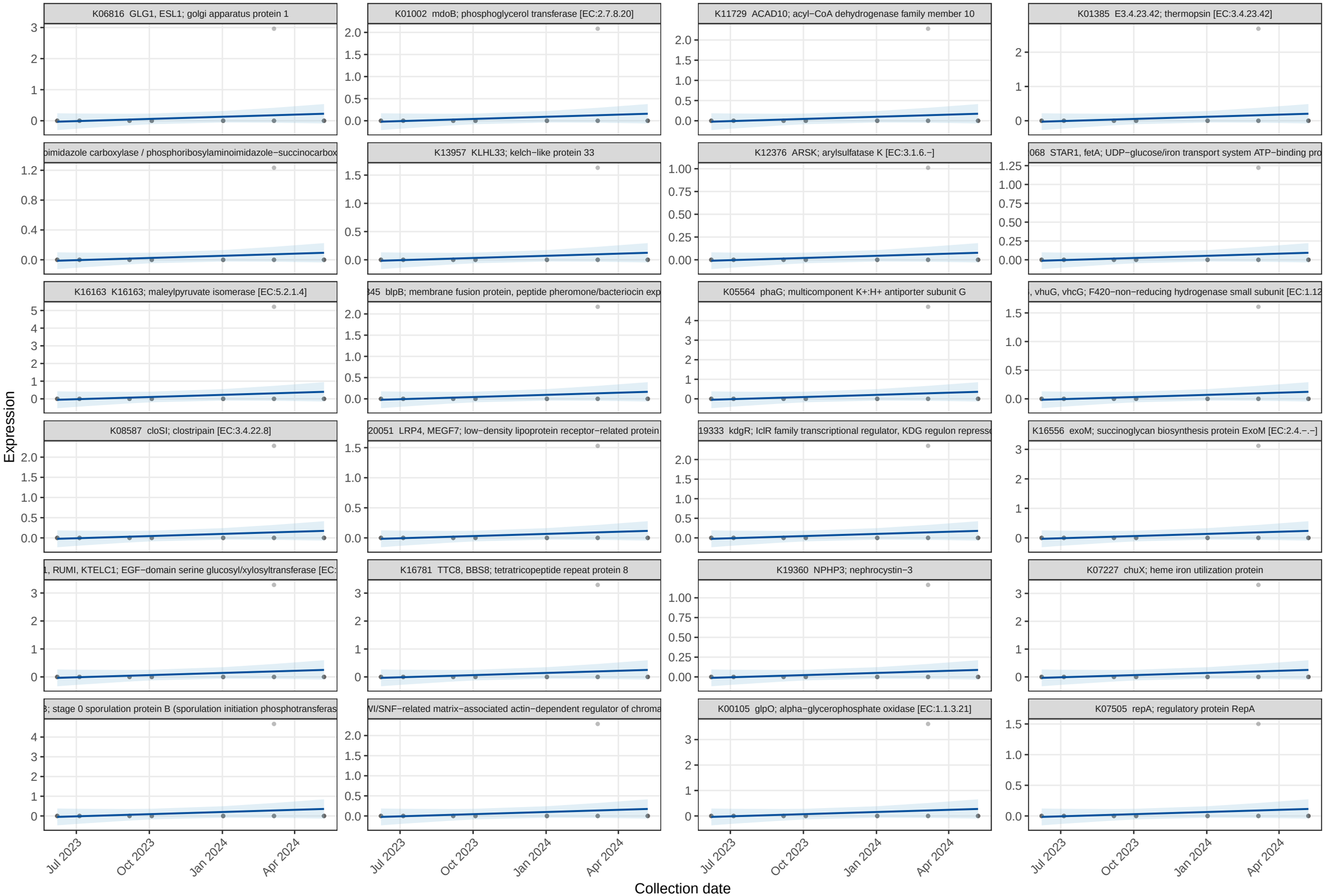
brite_D filtered temporal GAMs: page 127 of 303

All remaining BRITE classes after exclusions

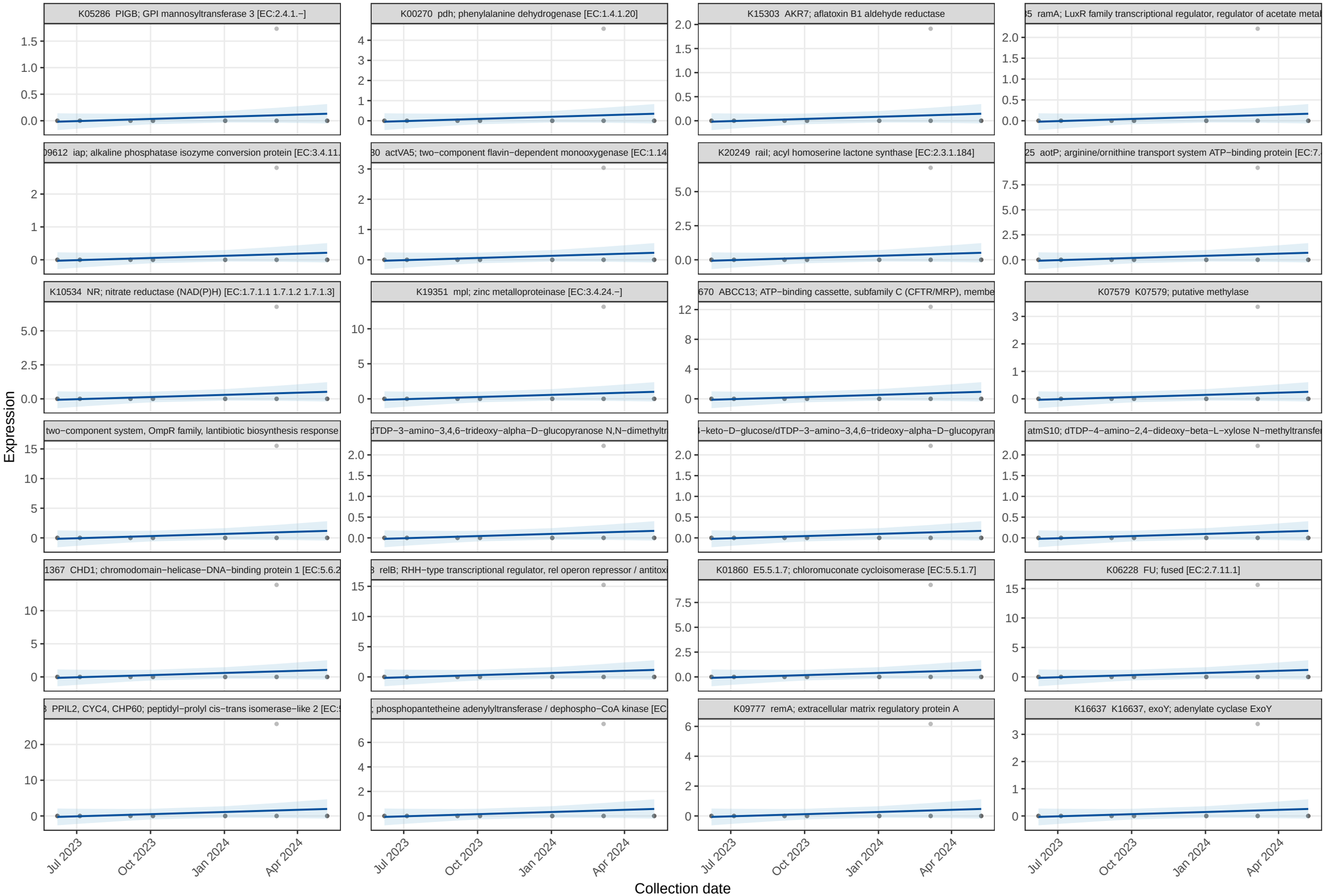


brite_D filtered temporal GAMs: page 128 of 303

All remaining BRITE classes after exclusions

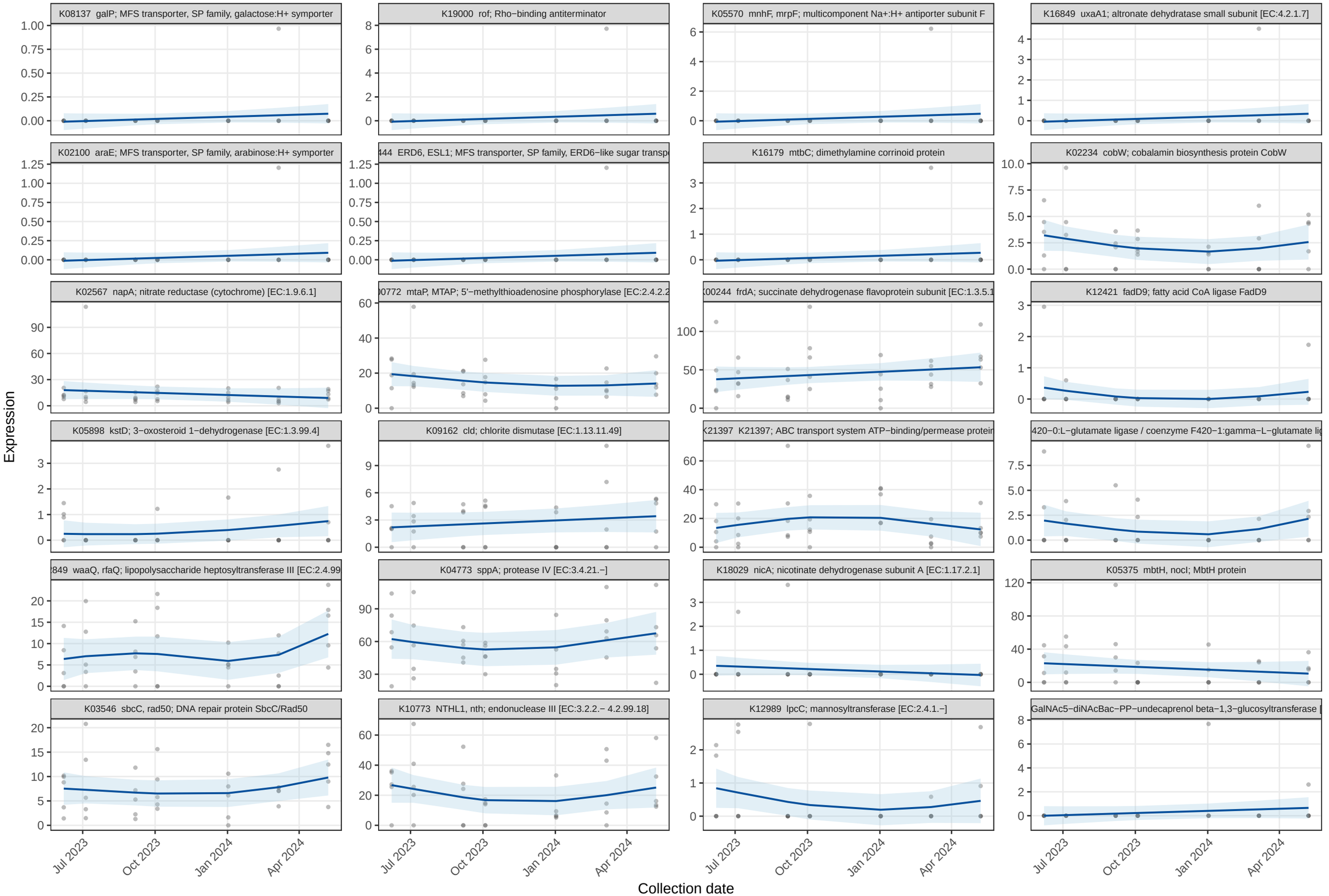


All remaining BRITE classes after exclusions



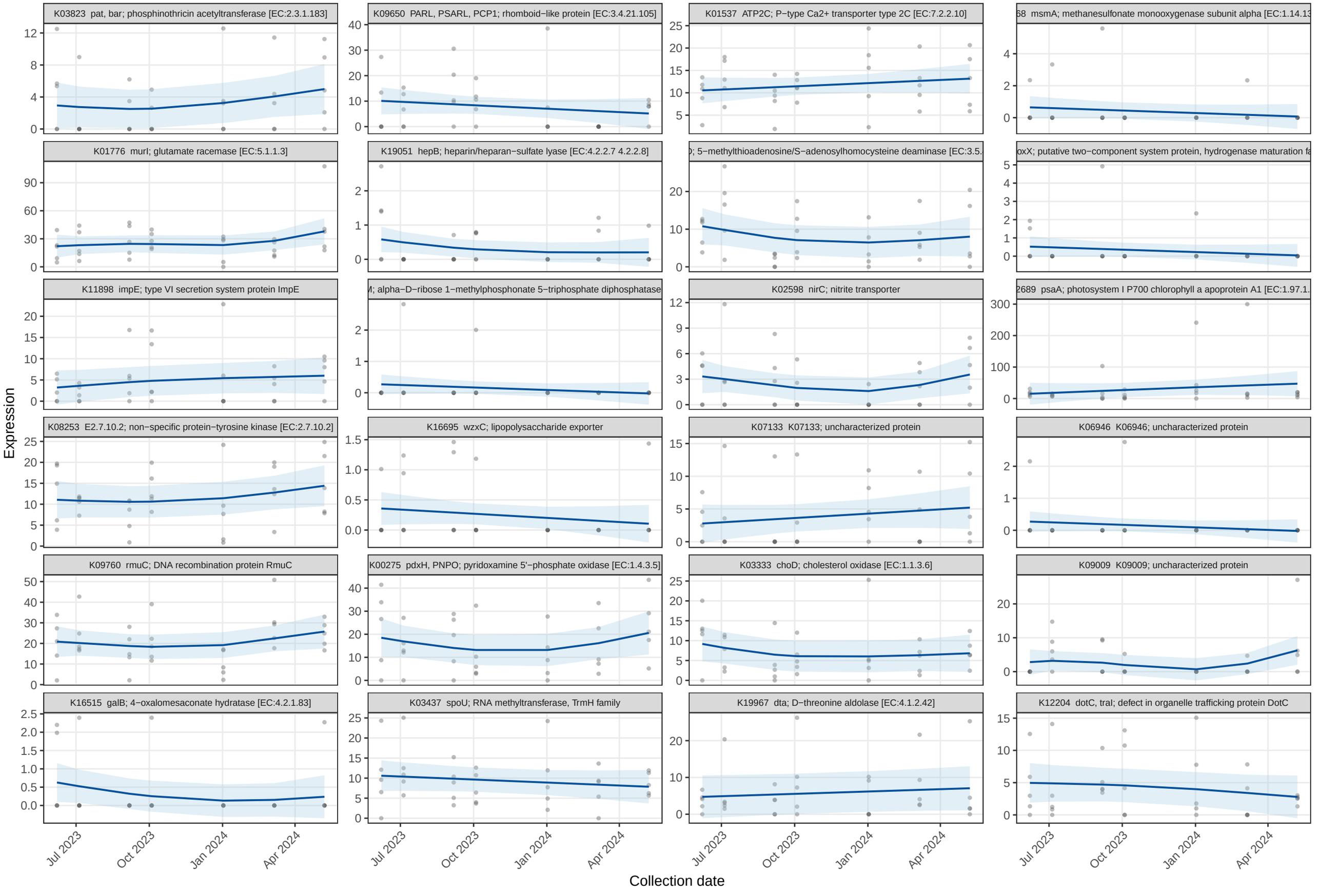
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All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 131 of 303

All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 132 of 303

All remaining BRITE classes after exclusions

K09157 K09157; uncharacterized protein

nqrE; Na+−transporting NADH:ubiquinone oxidoreductase subunit E [EC

K12136 hyfA; hydrogenase−4 component A [EC:1.−.−.−]

K10444 KLHL6; kelch−like protein 6

K19982 prnD; aminopyrrolnitrin oxygenase [EC:1.14.13.−]

K05384 cpeU, mpeU; bilin biosynthesis protein

K03522 fixB, etfA; electron transfer flavoprotein alpha subunit

K18545 sprB; streptogrisin B [EC:3.4.21.81]

K16148 glgM; alpha−maltose−1−phosphate synthase [EC:2.4.1.342]

K19707 rsbQ; sigma−B regulation protein RsbQ

K04342 strK; streptomycin−6−phosphatase [EC:3.1.3.39]

K06194 nlpD; lipoprotein NlpD

Expression

K03929 pnbA; para−nitrobenzyl esterase [EC:3.1.1.−]

K03808 pqiA, letA; intermembrane transport protein PqiA/LetA

K00423 E1.10.3.3; L−ascorbate oxidase [EC:1.10.3.3]

3484 scrR; LacI family transcriptional regulator, sucrose operon repress

18013 kce; 3−keto−5−aminohexanoate cleavage enzyme [EC:2.3.1.247

3715 MGD; 1,2−diacylglycerol 3−beta−galactosyltransferase [EC:2.4.1.

K00246 frdC; succinate dehydrogenase subunit C

10843 ERCC3, XPB; DNA excision repair protein ERCC−3 [EC:5.6.2.4

K02419 fliP; flagellar biosynthesis protein FliP

K15777 DODA; 4,5−DOPA dioxygenase extradiol [EC:1.13.11.−]

K02724 psbZ; photosystem II PsbZ protein

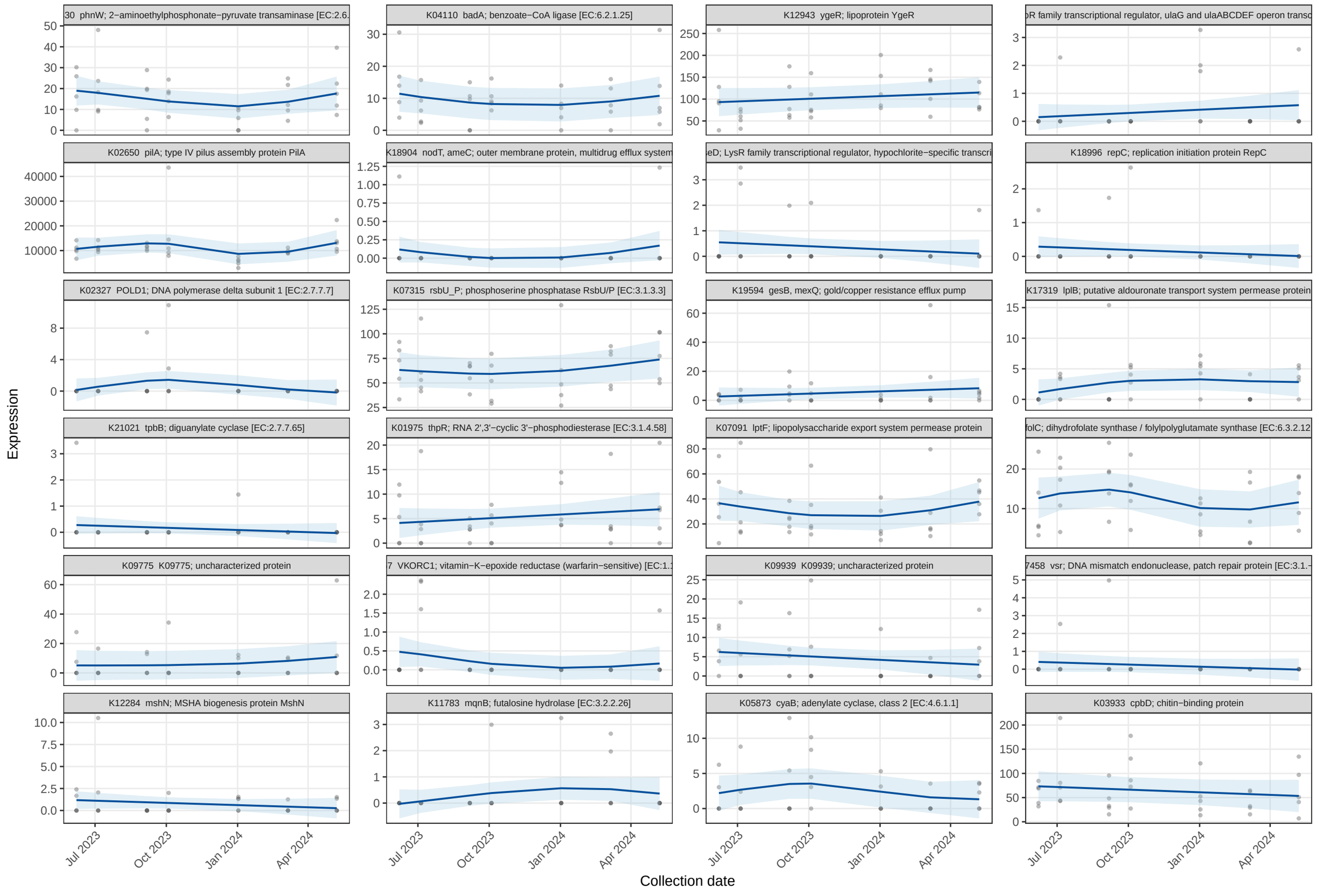
, treR; GntR family transcriptional regulator, trehalose operon transcripti

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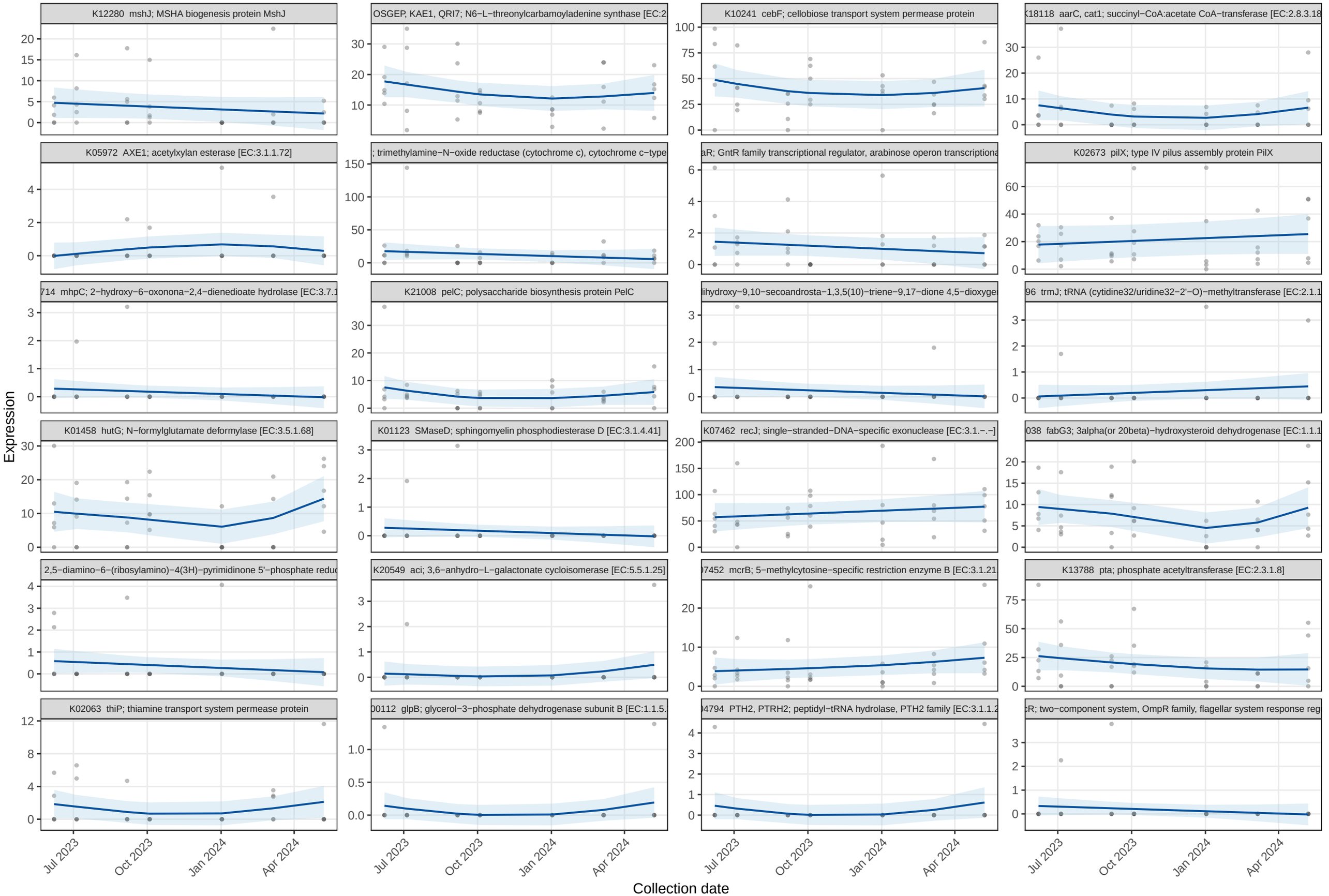
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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

; 2',3'−cyclic−nucleotide 2'−phosphodiesterase / 3'−nucleotidase [EC:3.1

K19092 parE1_3_4; toxin ParE1/3/4

K11901 impB; type VI secretion system protein ImpB

K01485 codA; cytosine/creatinine deaminase [EC:3.5.4.1 3.5.4.21]

K00633 lacA; galactoside O−acetyltransferase [EC:2.3.1.18]

K21014 stf0; trehalose 2−sulfotransferase [EC:2.8.2.37]

K09384 K09384; uncharacterized protein

K03324 yjbB; phosphate:Na+ symporter

K21801 iaaH; indoleacetamide hydrolase [EC:3.5.1.−]

tnV; L−glutamine−−−4−(methylsulfanyl)−2−oxobutanoate aminotransfera

K01219 E3.2.1.81; beta−agarase [EC:3.2.1.81]

K01575 alsD, budA, aldC; acetolactate decarboxylase [EC:4.1.1.5]

Expres

K07456 mutS2; DNA mismatch repair protein MutS2

mgsC; methylamine−−−glutamate N−methyltransferase subunit C [EC:2

K14160 imuA; protein ImuA

1567 fnr; ferredoxin/flavodoxin−−−NADP+ reductase [EC:1.18.1.2 1.19.1

K18847 E2.2.1.8; fluorothreonine transaldolase [EC:2.2.1.8]

)HB, bkdA2; 2−oxoisovalerate dehydrogenase E1 component subunit be

K20951 bap1; extracellular matrix protein

K03555 mutS; DNA mismatch repair protein MutS

K03328 TC.PST; polysaccharide transporter, PST family

K09795 K09795; uncharacterized protein

K21019 sadC; diguanylate cyclase [EC:2.7.7.65]

K07792 dcuB; anaerobic C4−dicarboxylate transporter DcuB

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olM; dihydromonapterin reductase / dihydrofolate reductase [EC:1.5.1.5

K18554 cpt; chloramphenicol 3−O phosphotransferase [EC:2.7.1.−]

02477 K02477; two−component system, LytTR family, response regulat

K14519 aldH; NADP−dependent aldehyde dehydrogenase [EC:1.2.1.4]

K04063 osmC, ohr; lipoyl−dependent peroxiredoxin [EC:1.11.1.28]

K03110 ftsY; fused signal recognition particle receptor

K01407 ptrA; protease III [EC:3.4.24.55]

K03734 apbE; FAD:protein FMN transferase [EC:2.7.1.180]

K11029 cyaA; anthrax edema toxin adenylate cyclase [EC:4.6.1.1]

0484 lanC, nisC, spaC, epiC; class I lanthipeptide synthase [EC:3.13.2

K15853 luxD; acyl transferase [EC:2.3.1.−]

10619 cmtAb; p−cumate 2,3−dioxygenase subunit alpha [EC:1.14.12.2

K16348 ecnB; entericidin B

GTP 3',8−cyclase / cyclic pyranopterin monophosphate synthase [EC:4

9 vioA; dTDP−4−amino−4,6−dideoxy−D−glucose transaminase [EC:2.6

K00060 tdh; threonine 3−dehydrogenase [EC:1.1.1.103]

K03529 smc; chromosome segregation protein

K05306 phnX; phosphonoacetaldehyde hydrolase [EC:3.11.1.1]

, fruR; DeoR family transcriptional regulator, fructose operon transcripti

K19246 loxA; arachidonate 15−lipoxygenase [EC:1.13.11.33]

K18691 mltF; peptidoglycan lytic transglycosylase F [EC:4.2.2.29]

1 plsY; acyl phosphate:glycerol−3−phosphate acyltransferase [EC:2.3.

K02386 flgA; flagellar basal body P−ring formation protein FlgA

K01045 PON; arylesterase / paraoxonase [EC:3.1.1.2 3.1.8.1]

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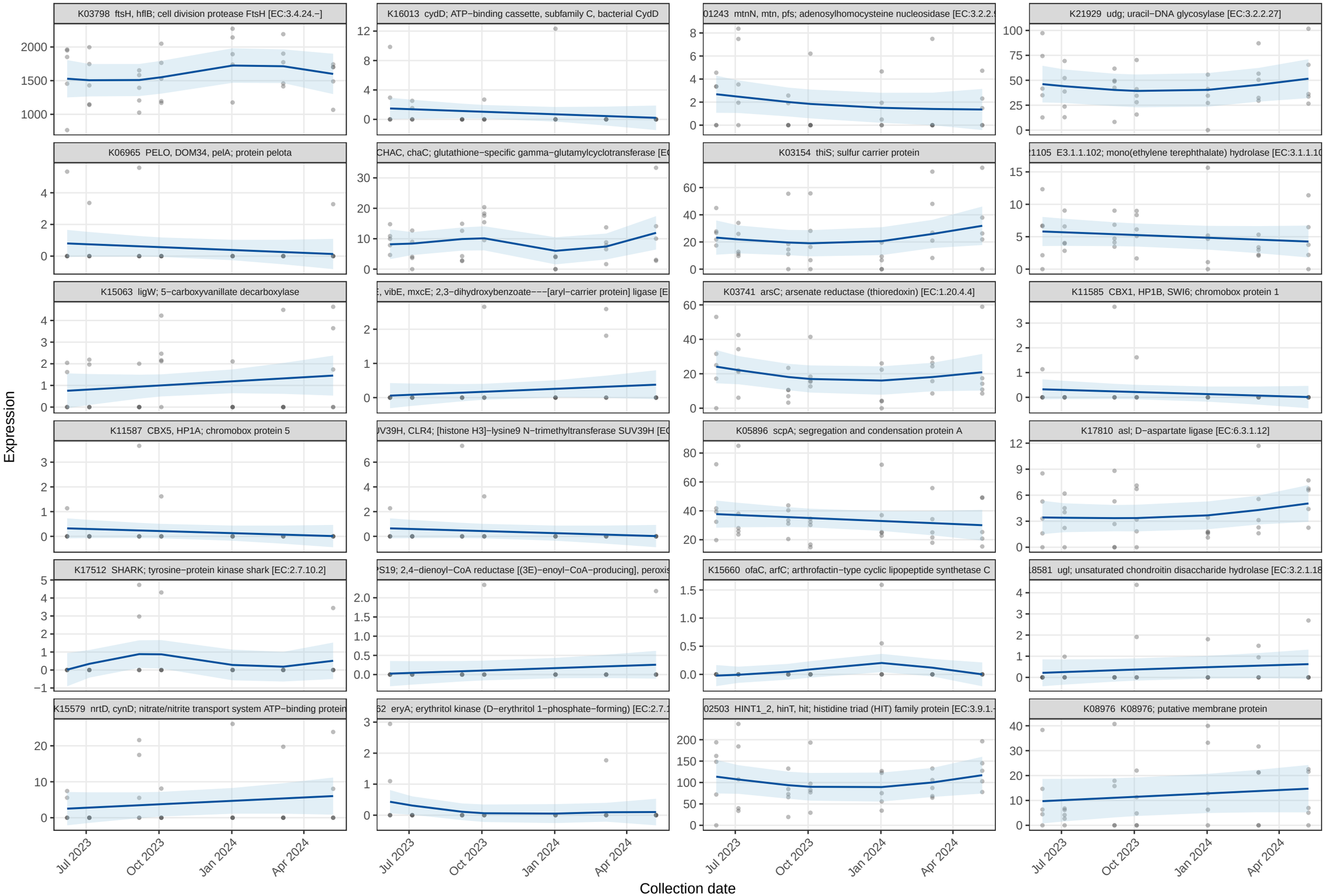
Collection date

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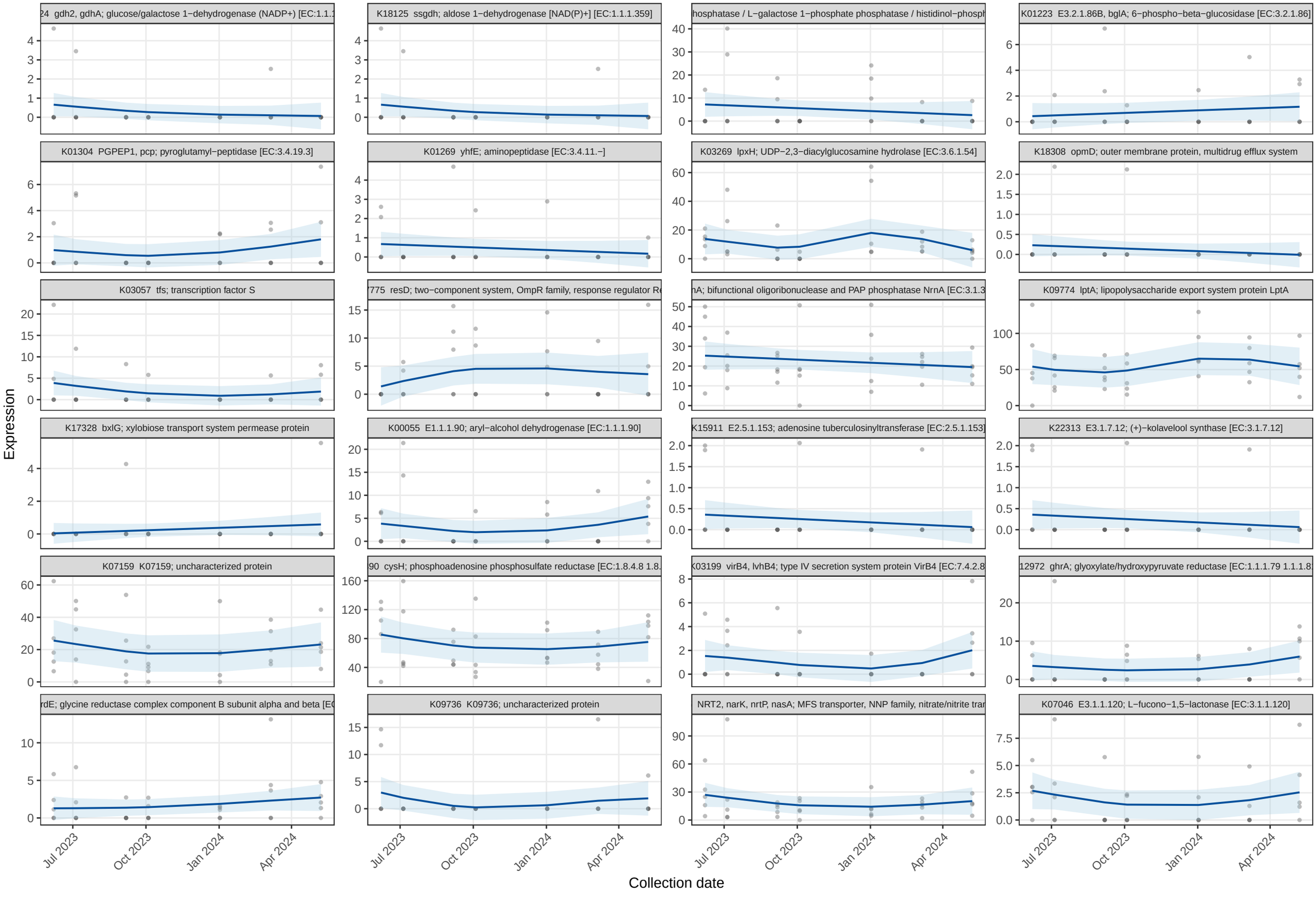
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All remaining BRITE classes after exclusions

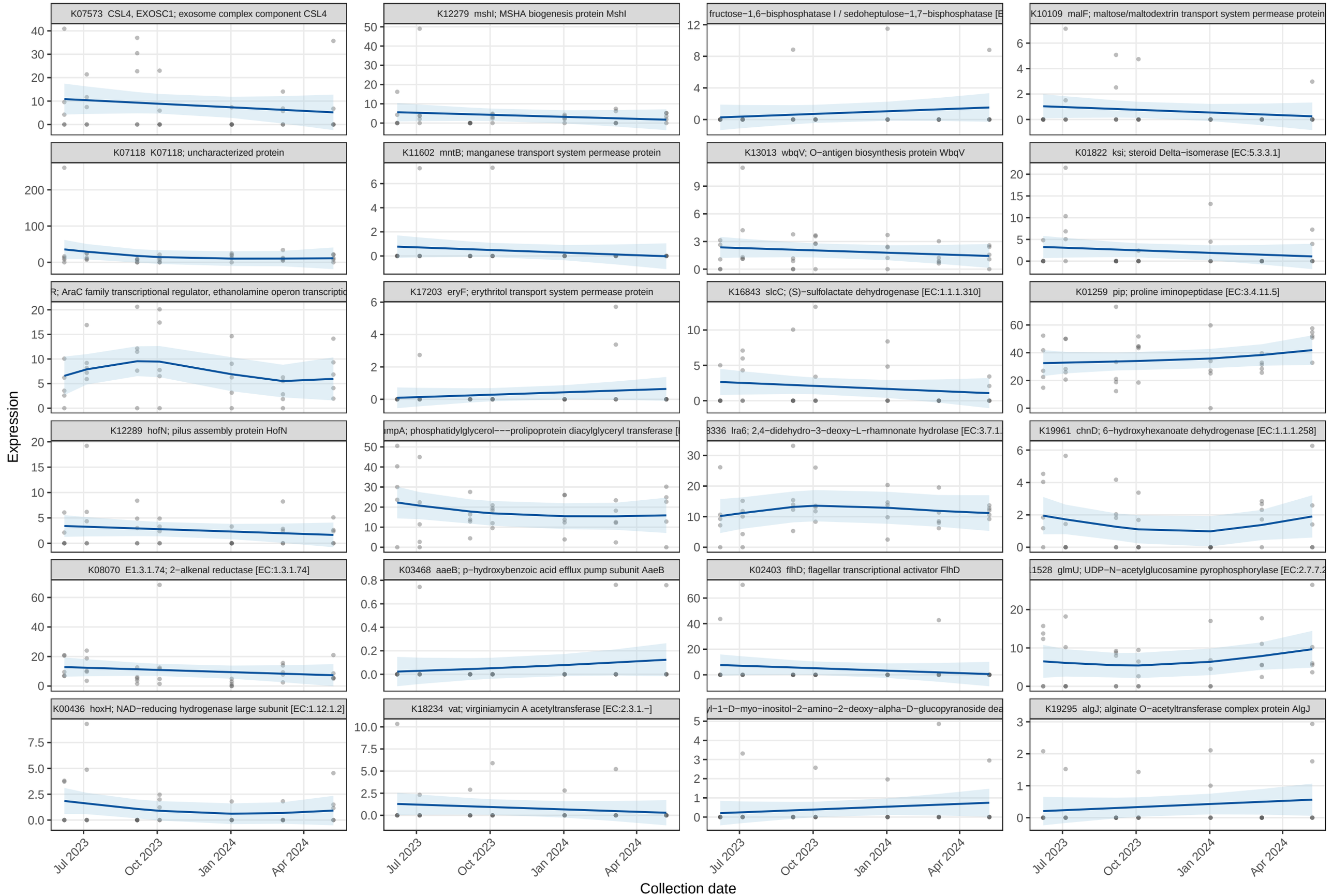


All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

PNKP; bifunctional polynucleotide phosphatase/kinase [EC:3.1.3.32 2

K04087 hflC; modulator of FtsH protease HflC

K21378 nphT7; acetoacetyl−CoA synthase [EC:2.3.1.194]

K07275 ompW; outer membrane protein

K12953 ctpF; cation−transporting P−type ATPase F [EC:7.2.2.−]

K11906 vasD, lip; type VI secretion system protein VasD

, IMDH3; 3−isopropylmalate/methylthioalkylmalate dehydrogenase [EC

K22382 WDR26; WD repeat−containing protein 26

16877 hmfA; 2−furoyl−CoA dehydrogenase large subunit [EC:1.3.99.8

-hydroxy−4−carboxymuconate semialdehyde hemiacetal dehydrogenas

K10007 gluD; glutamate transport system permease protein

73 potF, spuD, spuE; putrescine transport system substrate−binding pr

Expres

K10854 acxB; acetone carboxylase, alpha subunit [EC:6.4.1.6]

K16171 faaH; fumarylacetoacetate (FAA) hydrolase [EC:3.7.1.2]

K10926 mshC; MSHA pilin protein MshC

K02380 fdhE; FdhE protein

38 pleD; two−component system, cell cycle response regulator [EC:2.7

K07148 K07148; uncharacterized protein

K06936 raSEA; archaeosine synthase beta−subunit [EC:2.6.1.−]

05889 pvadh; polyvinyl alcohol dehydrogenase (cytochrome) [EC:1.1.2.

K06894 yfhM; alpha−2−macroglobulin

K22213 PATG; 6−methylsalicylate decarboxylase [EC:4.1.1.52]

K01706 gudD; glucarate dehydratase [EC:4.2.1.40]

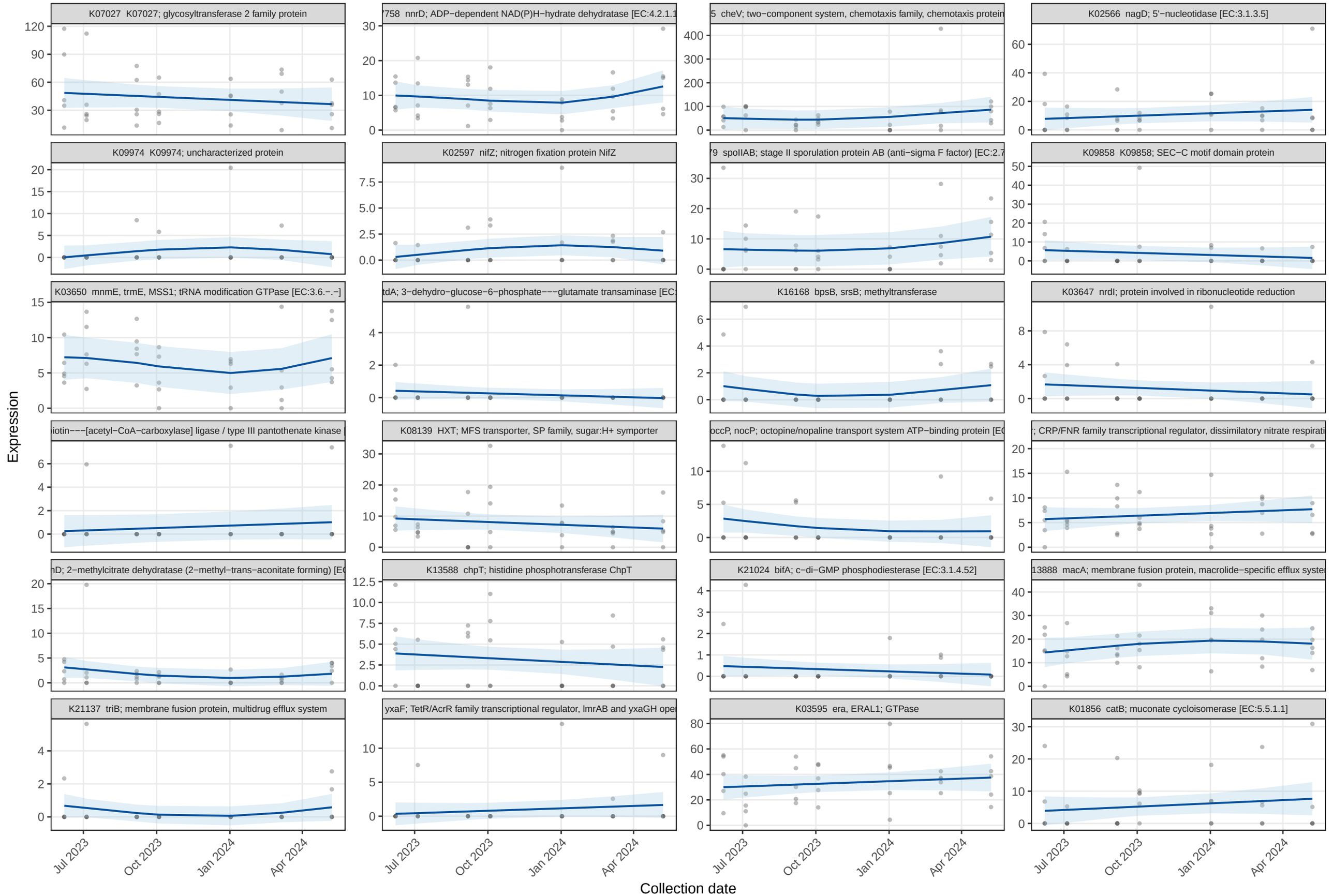
00 bpeT; LysR family transcriptional regulator, regulator for bpeEF and

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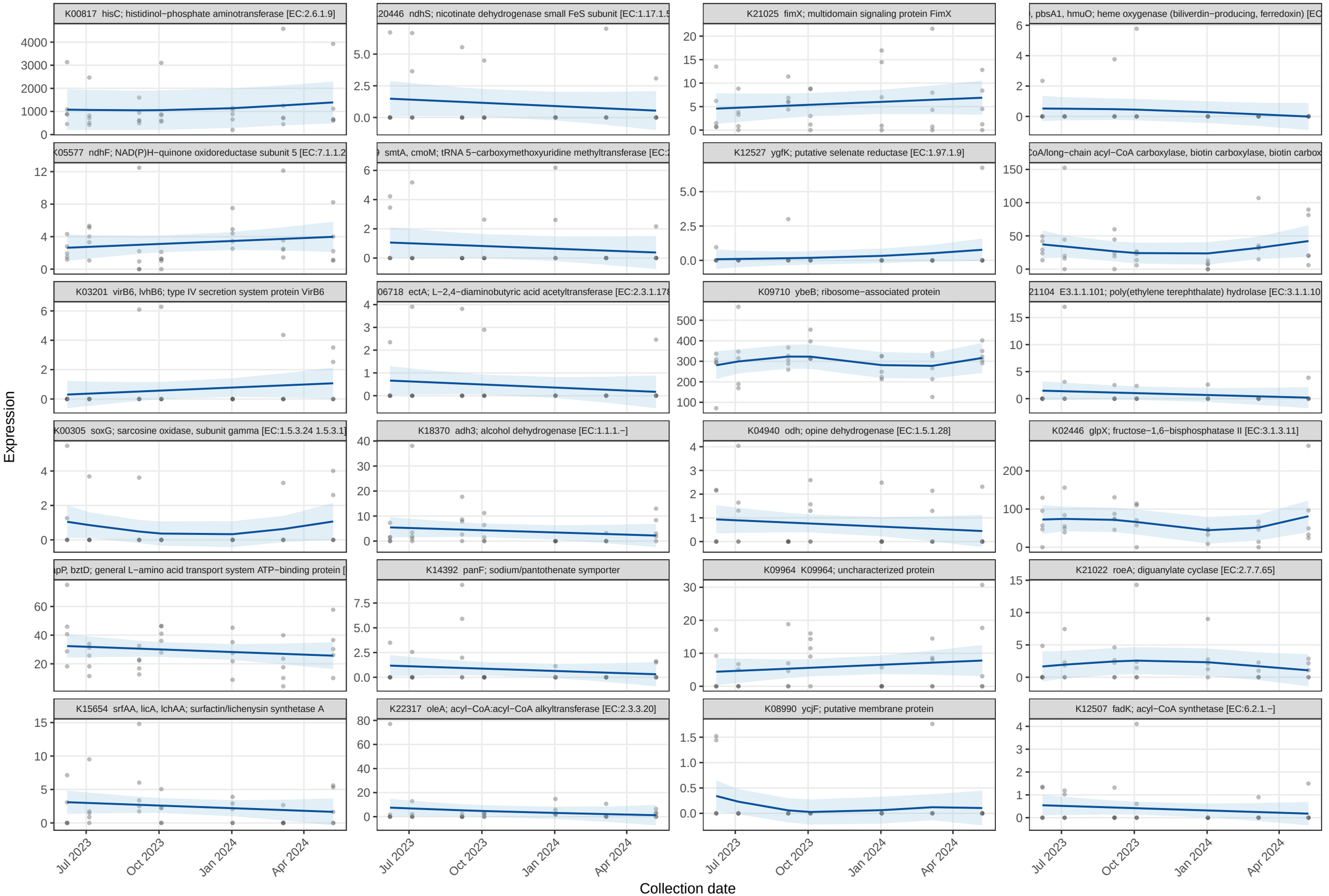
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All remaining BRITE classes after exclusions



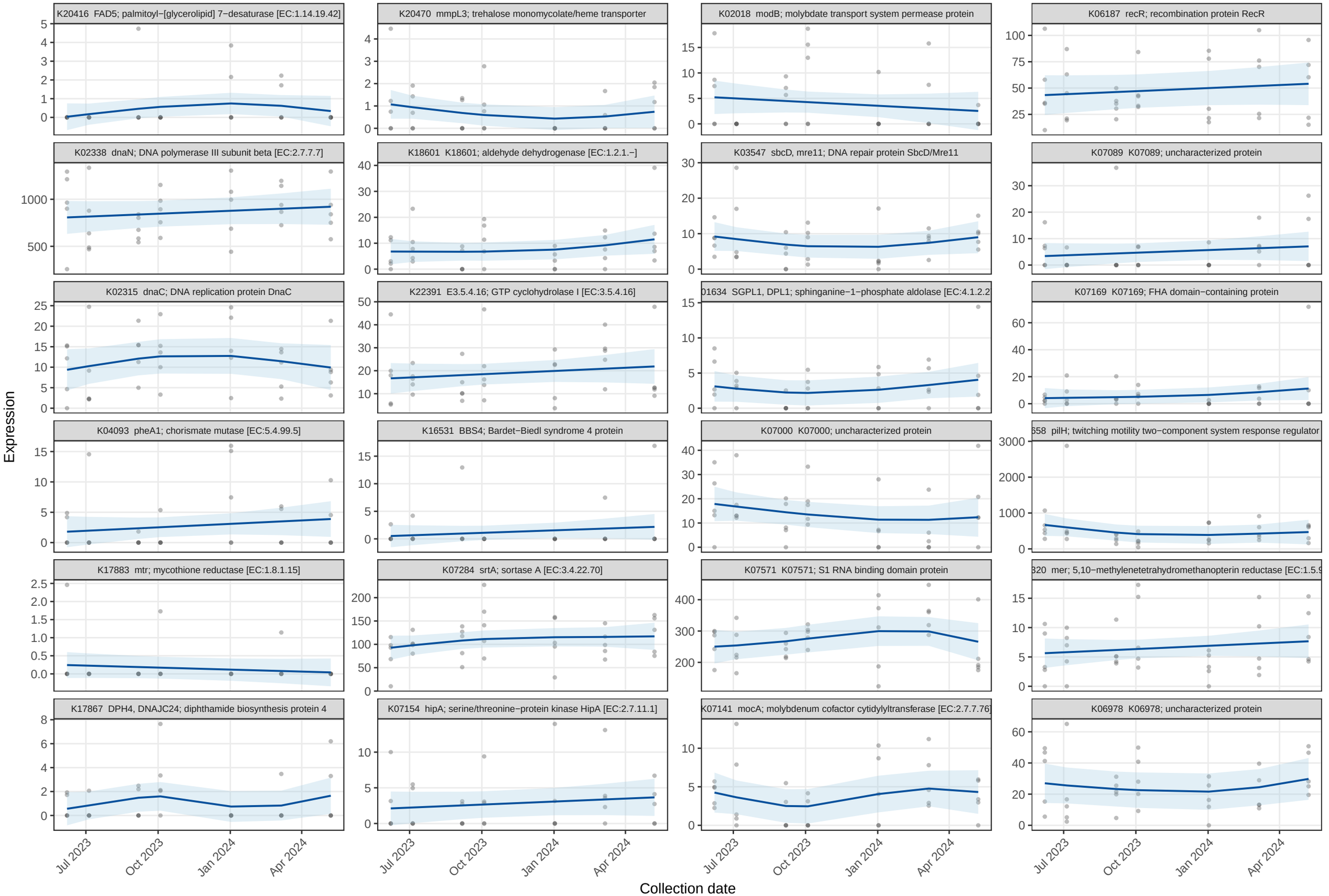
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All remaining BRITE classes after exclusions

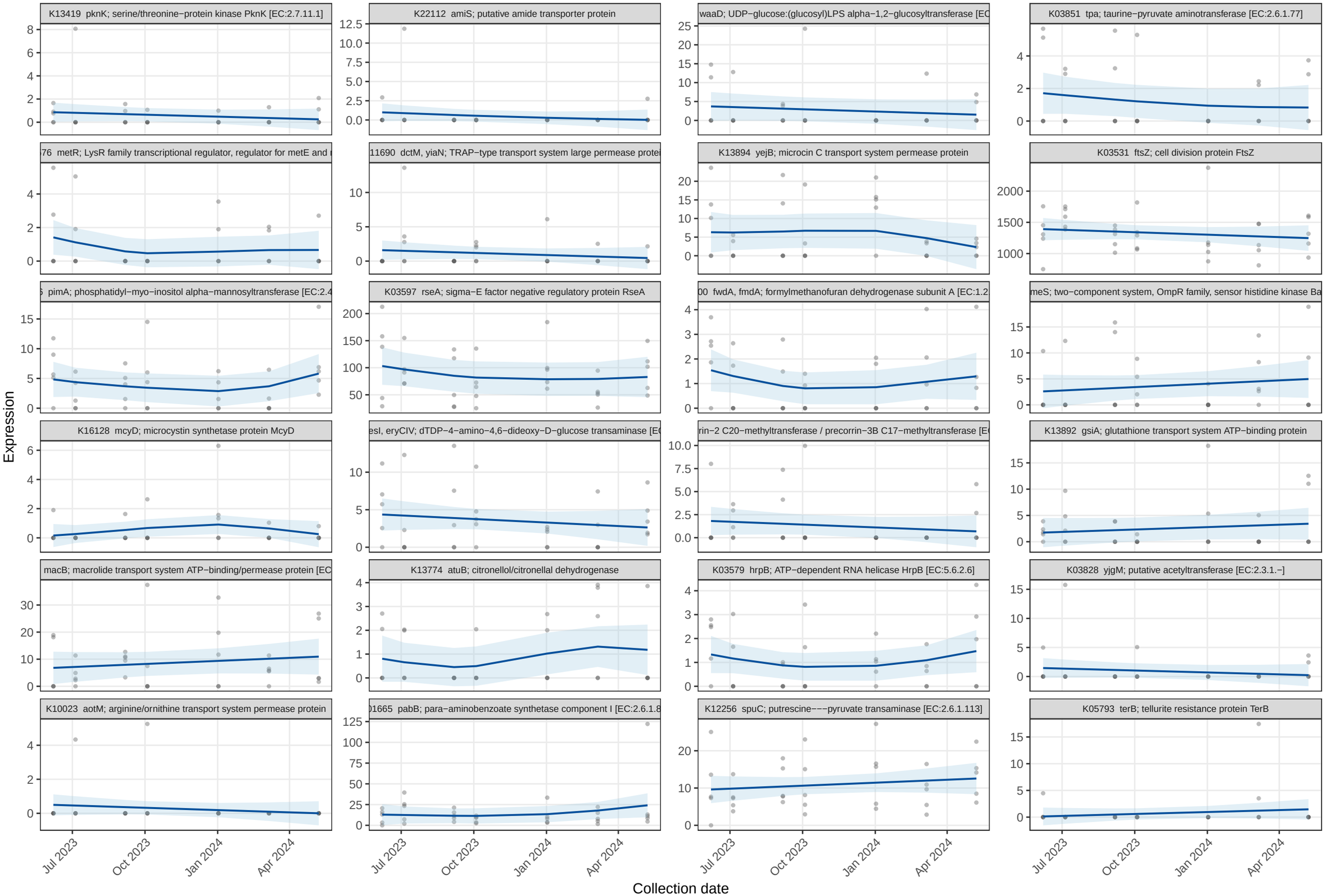


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All remaining BRITE classes after exclusions

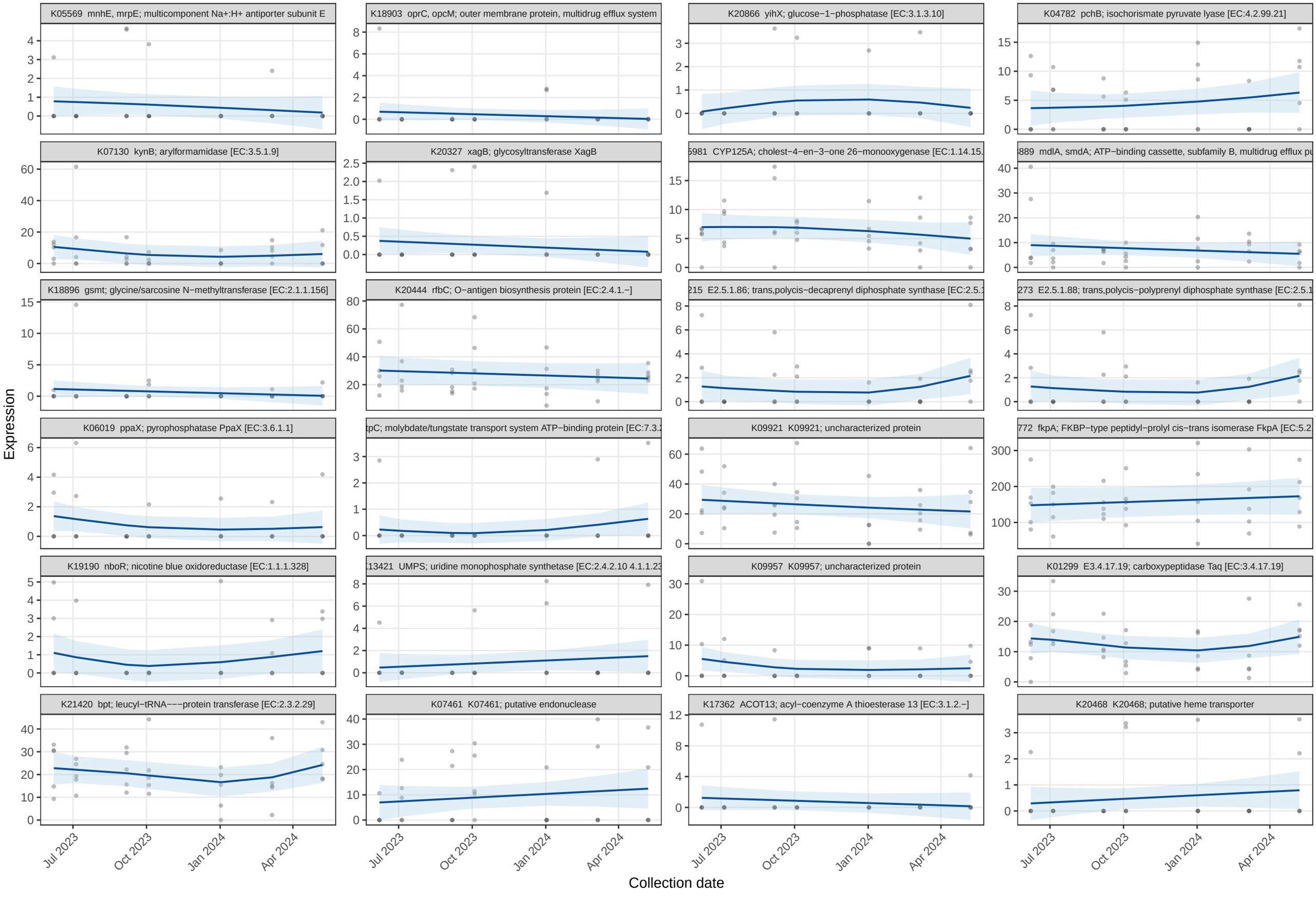


All remaining BRITE classes after exclusions



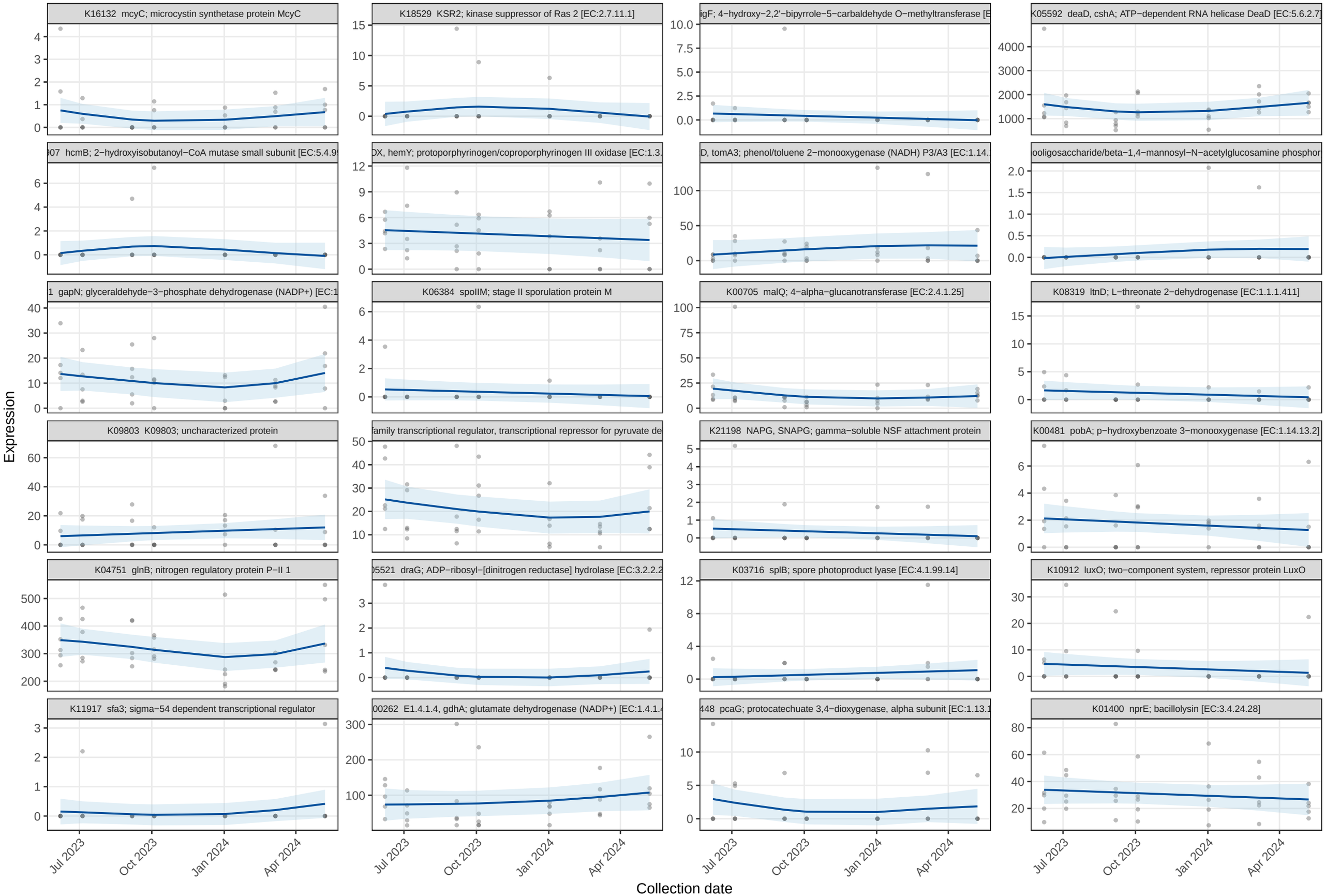
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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions

K19545 lnuA_C_D_E, lin; lincosamide nucleotidyltransferase A/C/D/E

K06601 flbT; flagellar biosynthesis repressor protein FlbT

K16261 YAT; yeast amino acid transporter

K01193 INV, sacA; beta−fructofuranosidase [EC:3.2.1.26]

K01753 dsdA; D−serine dehydratase [EC:4.3.1.18]

K09973 K09973; uncharacterized protein

K18824 sul2; dihydropteroate synthase type 2 [EC:2.5.1.15]

1915 stp1, pppA; serine/threonine protein phosphatase Stp1 [EC:3.1.3.

srtN, npdA; NAD−dependent protein deacetylase/lipoamidase [EC:2.3.1

K17209 iatP; inositol transport system permease protein

K01710 rfbB, rmlB, rffG; dTDP−glucose 4,6−dehydratase [EC:4.2.1.46]

K18232 oleC4; oleandomycin transport system ATP−binding protein

Expres

12 mgp; 4−O−beta−D−mannosyl−D−glucose phosphorylase [EC:2.4.1

R; GntR family transcriptional regulator, galactonate operon transcriptio

K14451 mcl2; (3S)−malyl−CoA thioesterase [EC:3.1.2.30]

K01713 pheC; cyclohexadienyl dehydratase [EC:4.2.1.51 4.2.1.91]

K11686 racA; chromosome−anchoring protein RacA

K13628 iscA; iron−sulfur cluster assembly protein

K02417 fliN; flagellar motor switch protein FliN

K02704 psbB; photosystem II CP47 chlorophyll apoprotein

K14189 dltE; uncharacterized oxidoreductase [EC:1.−.−.−]

K09946 K09946; uncharacterized protein

K09527 DNAJC7; DnaJ homolog subfamily C member 7

K10921 toxR; cholera toxin transcriptional activator

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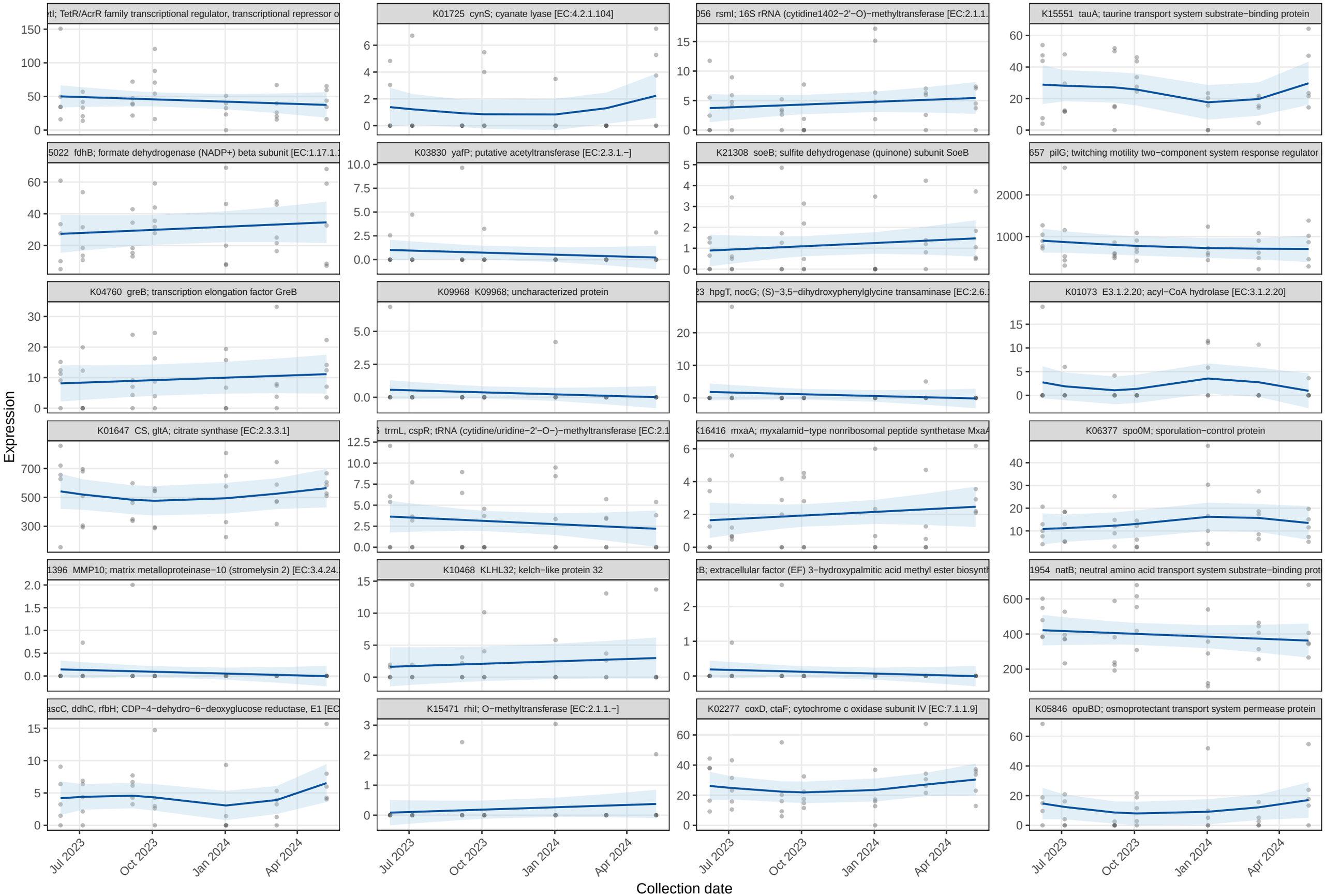
Jan 2024

Apr 2024

Collection date

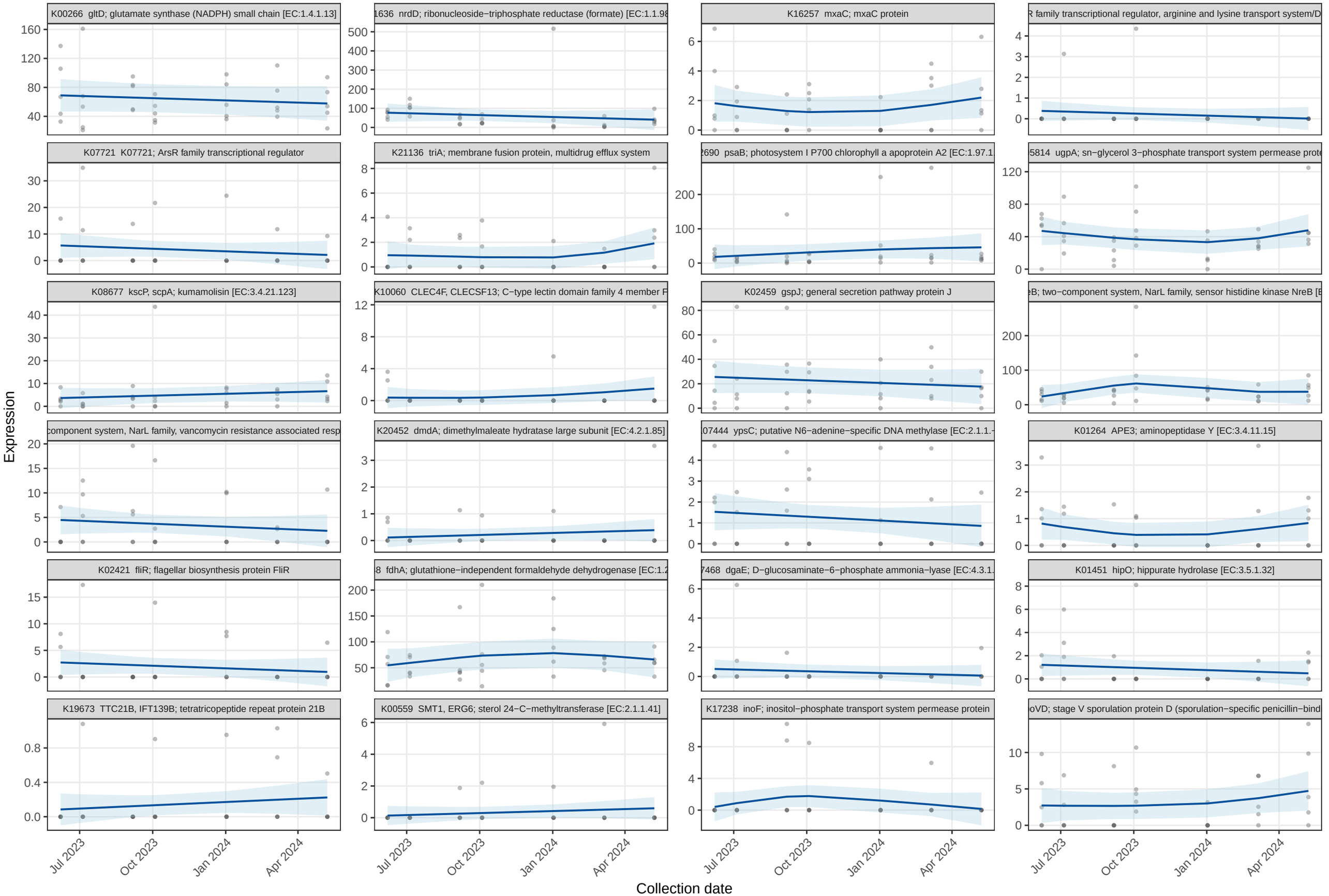
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All remaining BRITE classes after exclusions



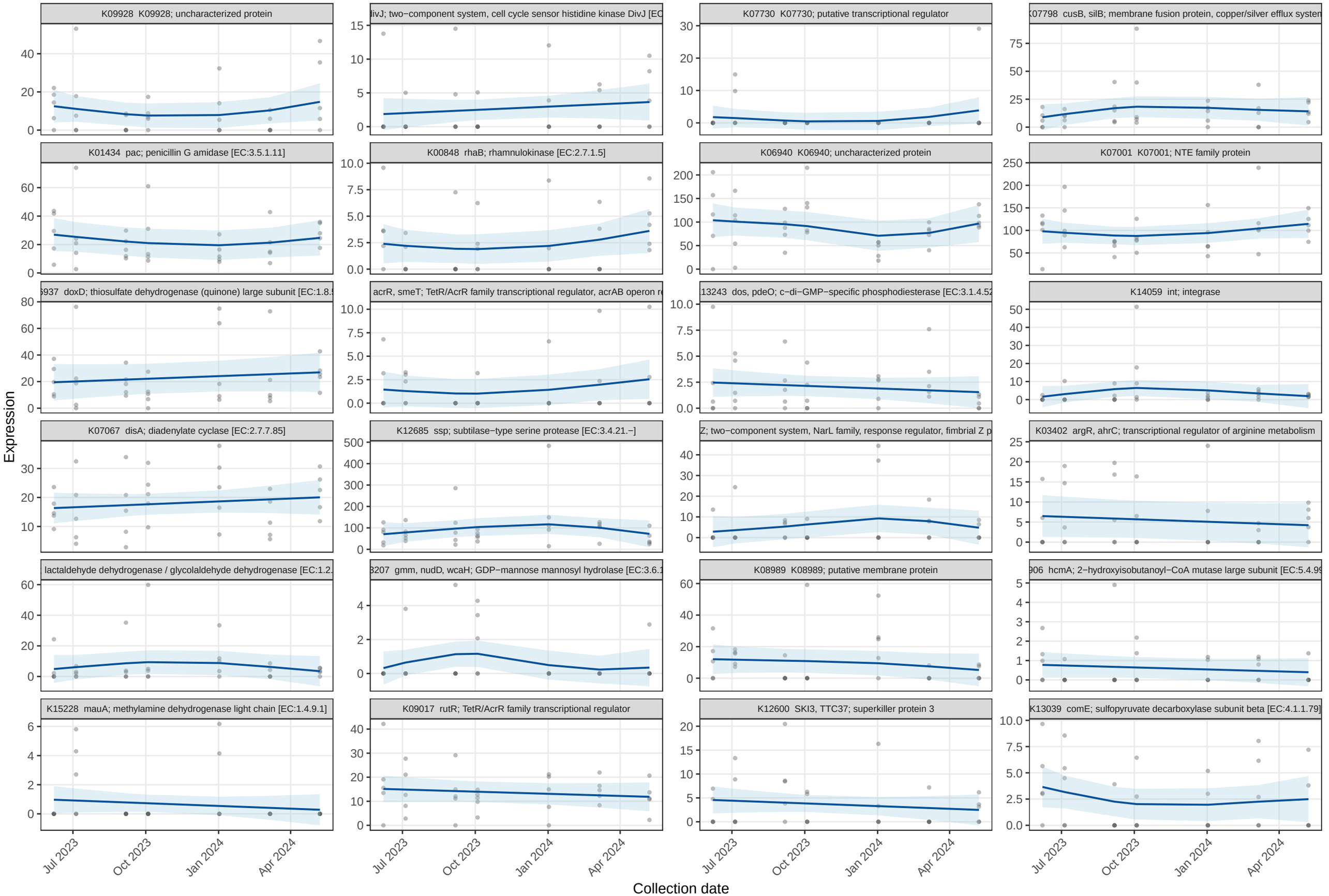
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All remaining BRITE classes after exclusions



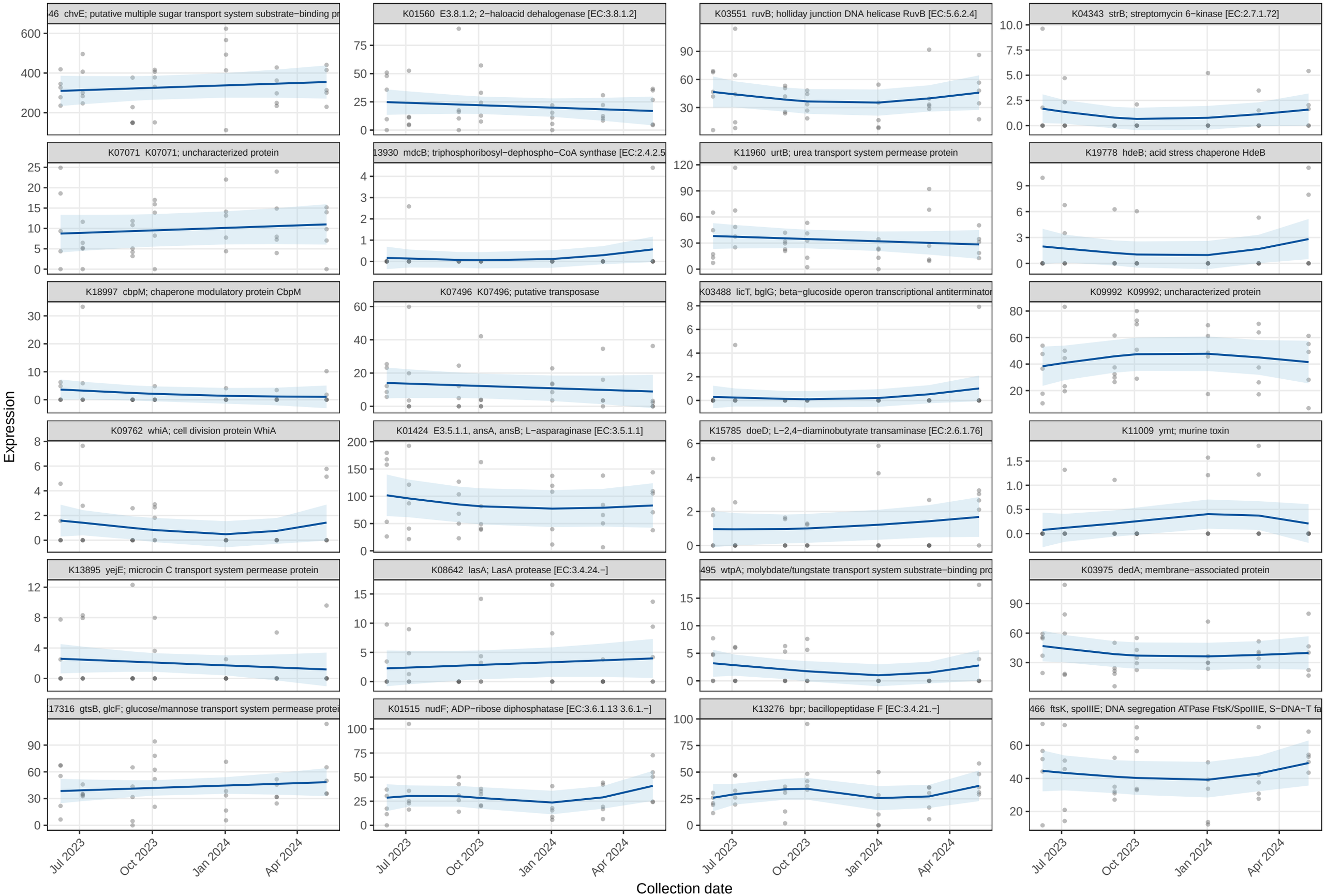
brite_D filtered temporal GAMs: page 150 of 303

All remaining BRITE classes after exclusions

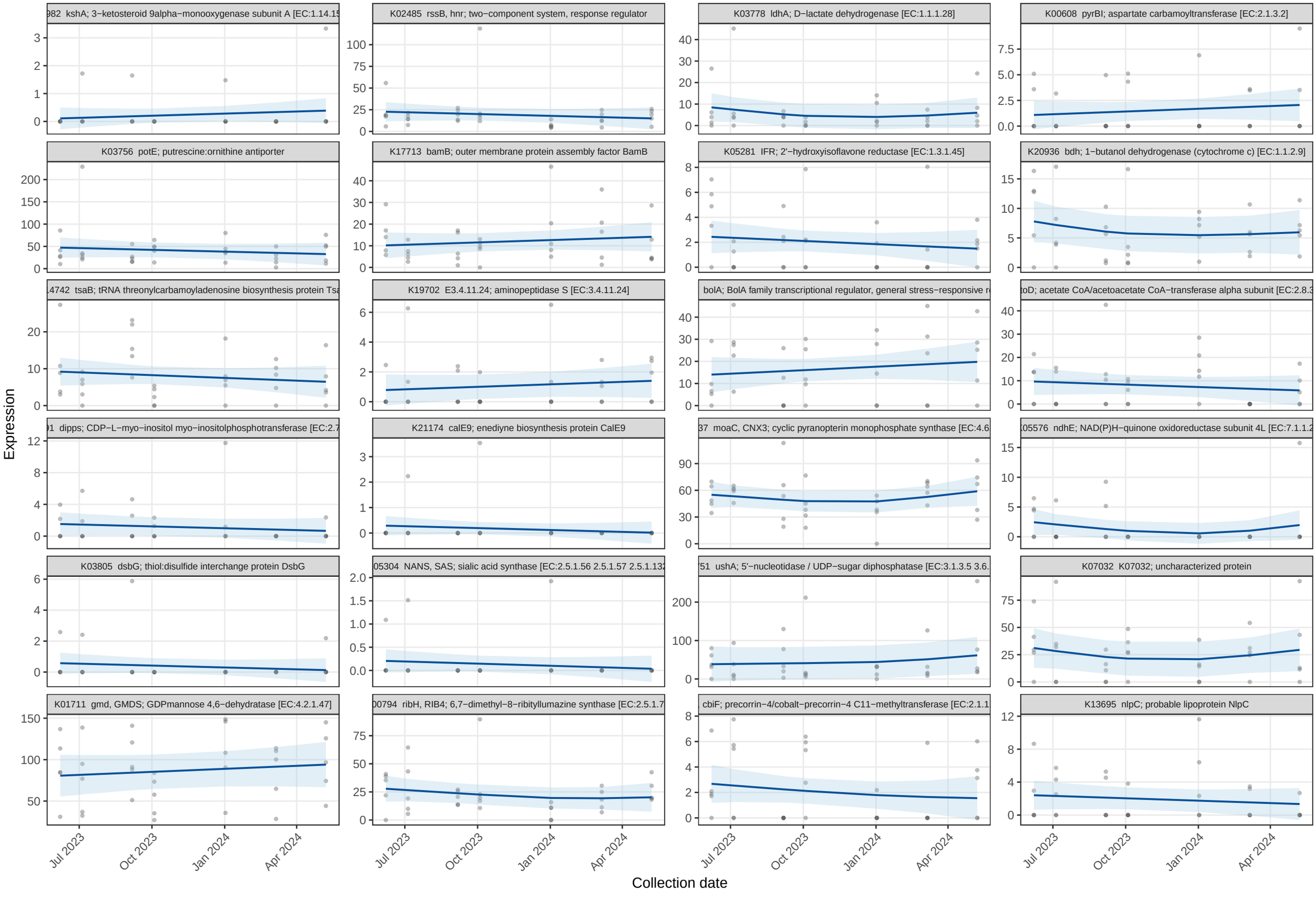


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All remaining BRITE classes after exclusions

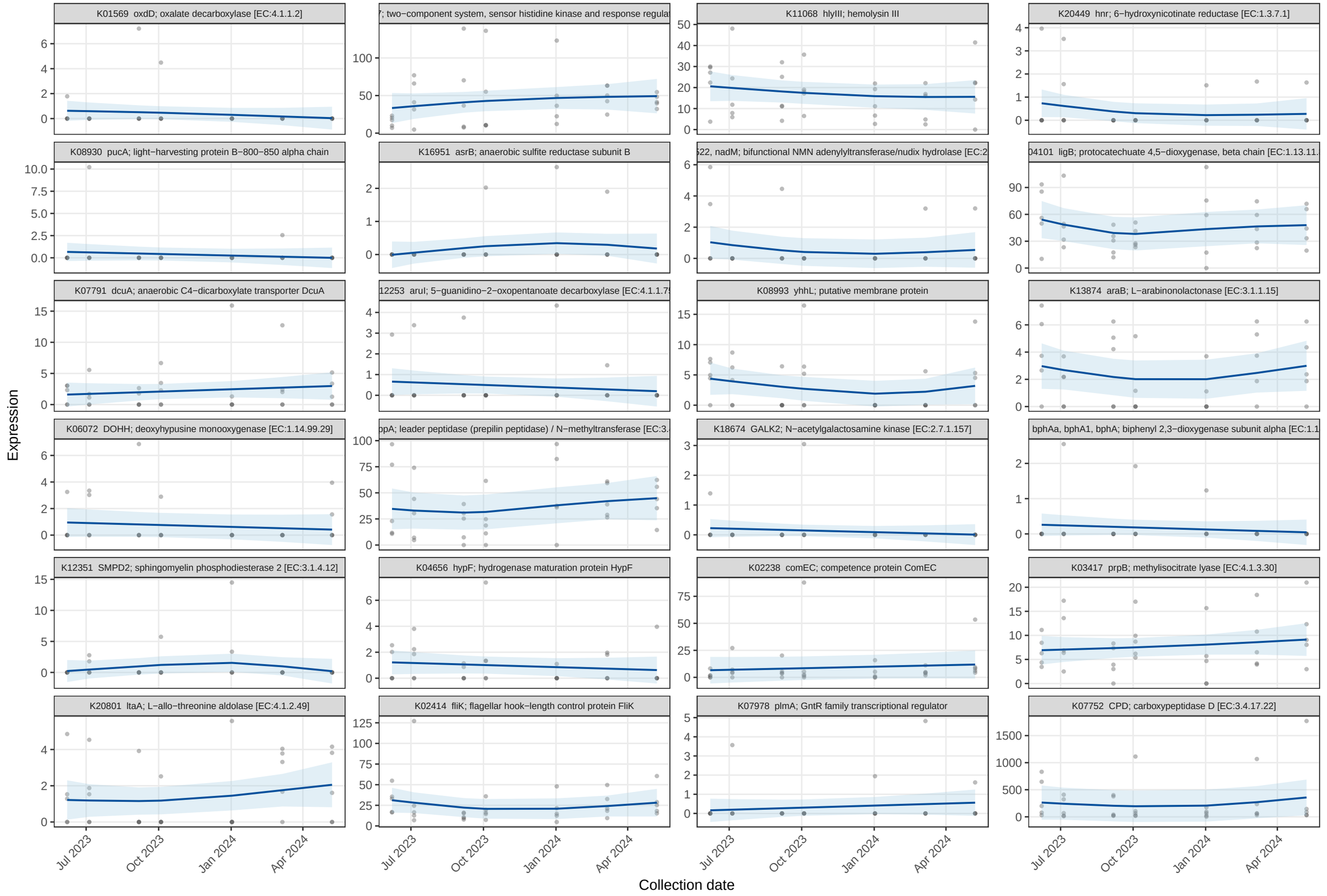


All remaining BRITE classes after exclusions

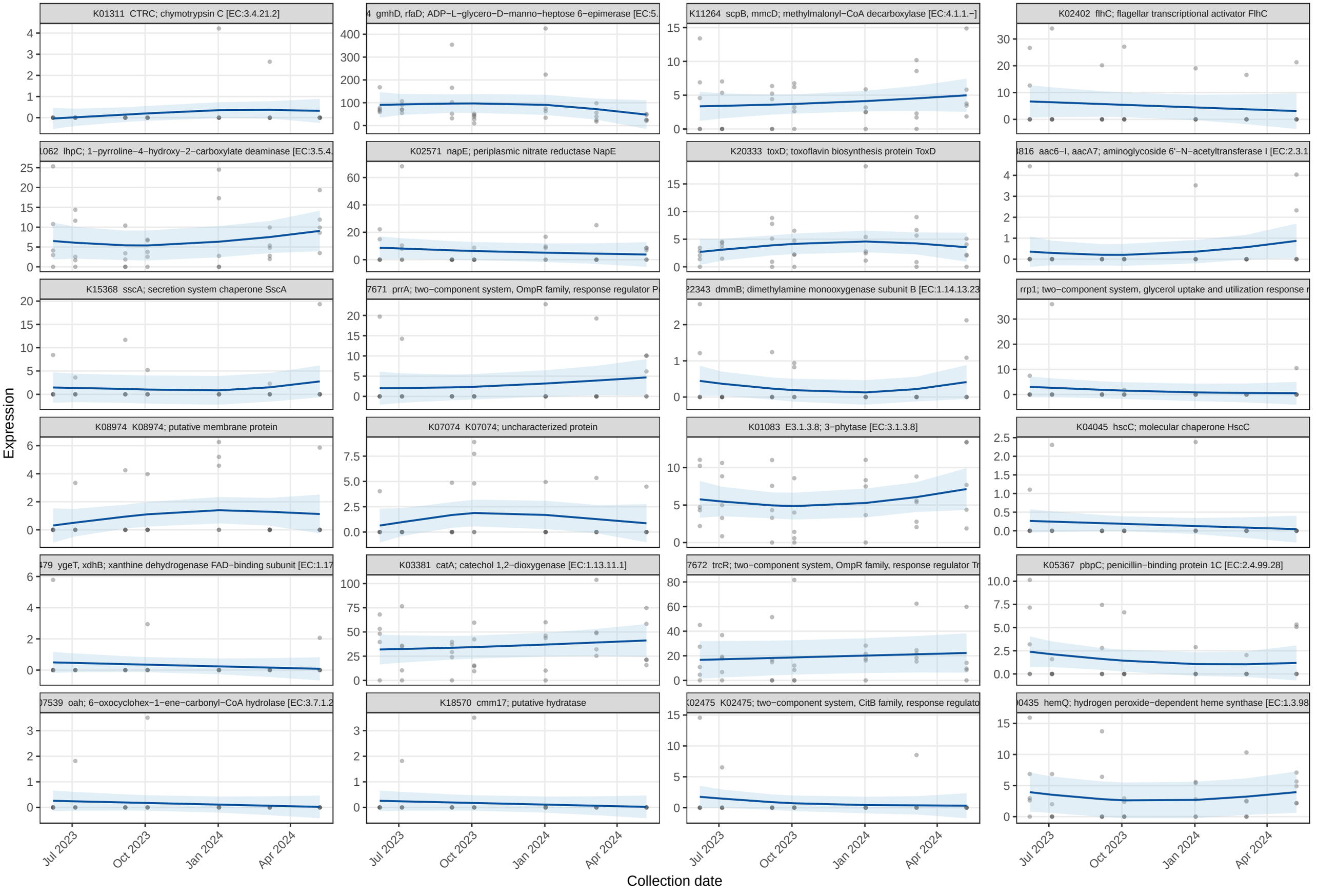


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All remaining BRITE classes after exclusions

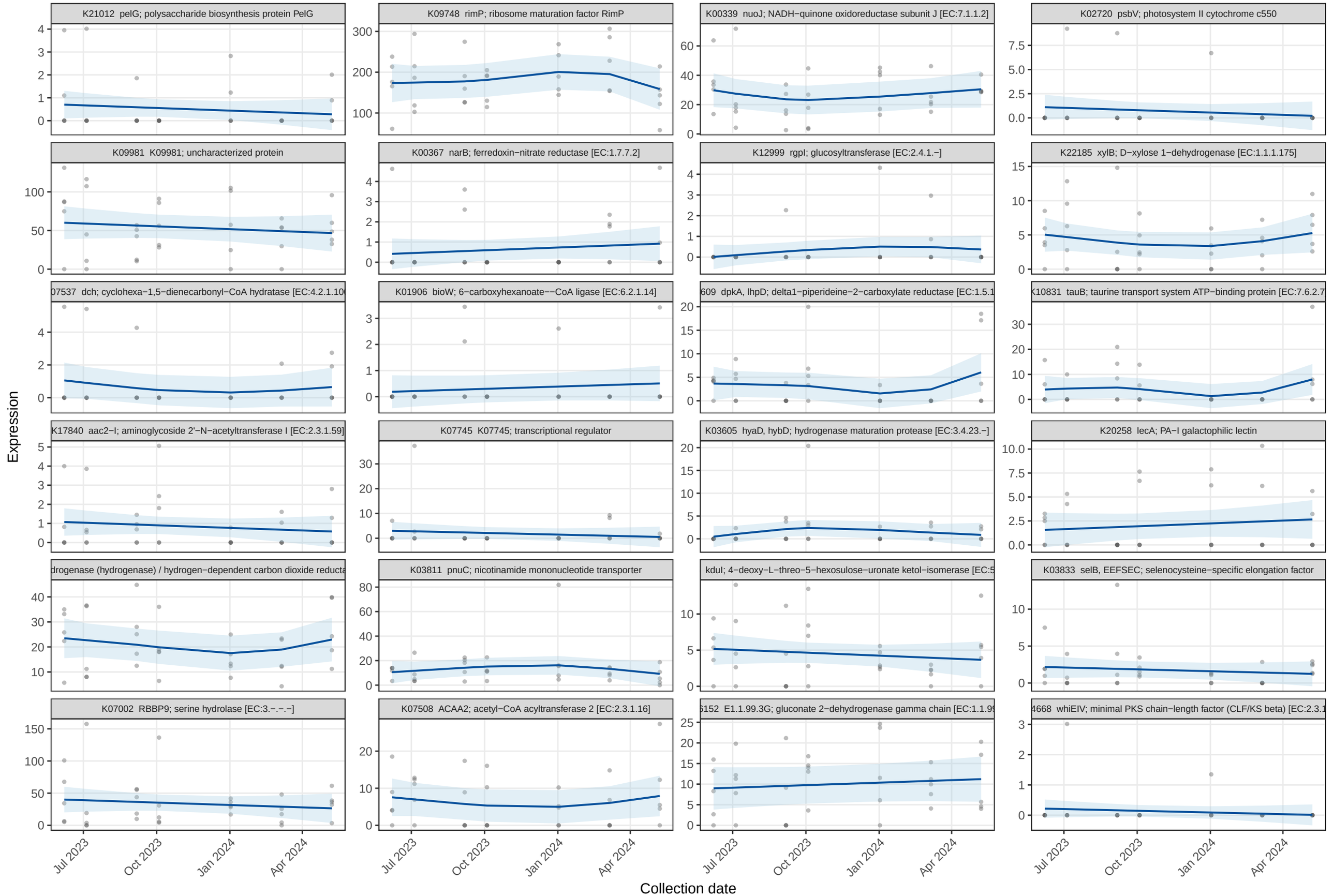


All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

1127 GPLD1; glycosylphosphatidylinositol phospholipase D [EC:3.1.4.

K01749 hemC, HMBS; hydroxymethylbilane synthase [EC:2.5.1.61]

osylaminopyrimidine deaminase / 5−amino−6−(5−phosphoribosylamino

K18547 sprD; streptogrisin D [EC:3.4.21.−]

K12285 mshO; MSHA biogenesis protein MshO

K09986 K09986; uncharacterized protein

1 sgcC4, mdpC4; MIO−dependent L−tyrosine 2,3−aminomutase [EC:5

K12065 traB; conjugal transfer pilus assembly protein TraB

A; acetyl−CoA carboxylase carboxyl transferase subunit alpha [EC:6.4.

K10195 togA; oligogalacturonide transport system ATP−binding protein

K07100 K07100; putative phosphoribosyl transferase

K07460 yraN; putative endonuclease

Expression

K13626 fliW; flagellar assembly factor FliW

69 msmB; methanesulfonate monooxygenase subunit beta [EC:1.14.13

17317 gtsC, glcG; glucose/mannose transport system permease prote

K07150 K07150; uncharacterized protein

K02415 fliL; flagellar protein FliL

ino−4,6−dideoxy−D−glucose/dTDP−4−amino−2,4−dideoxy−beta−L−xyl

K08173 ydfJ; MFS transporter, MHS family, metabolite:H+ symporter

803 AGPS, agpS; alkyldihydroxyacetonephosphate synthase [EC:2.5.1

K09788 prpF; 2−methylaconitate isomerase [EC:5.3.3.−]

K07238 TC.ZIP, zupT, ZRT3, ZIP2; zinc transporter, ZIP family

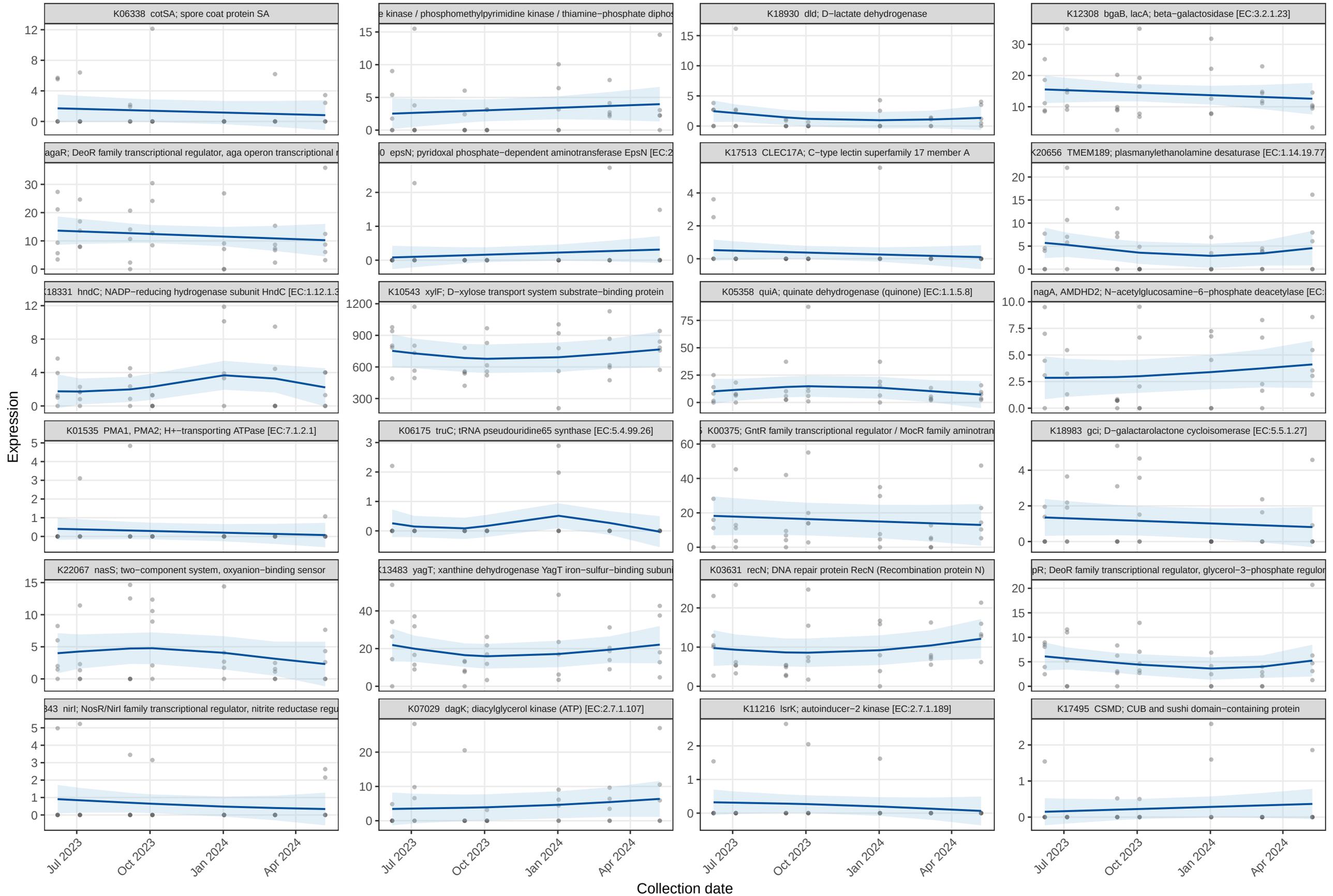
lomaltodextrinase / maltogenic alpha−amylase / neopullulanase [EC:3.2

K11946 phdG; hydratase−aldolase [EC:4.1.2.−]

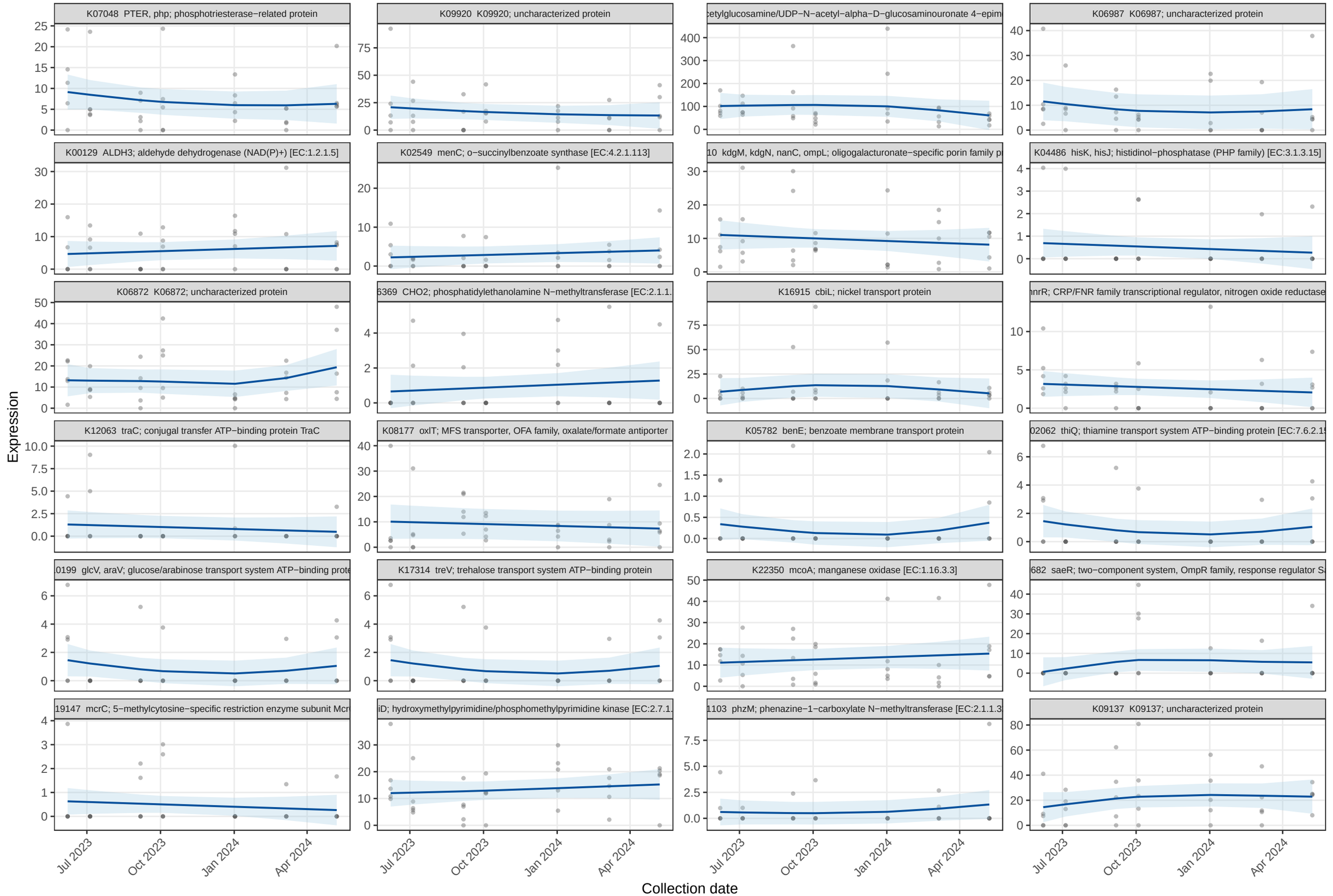
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All remaining BRITE classes after exclusions

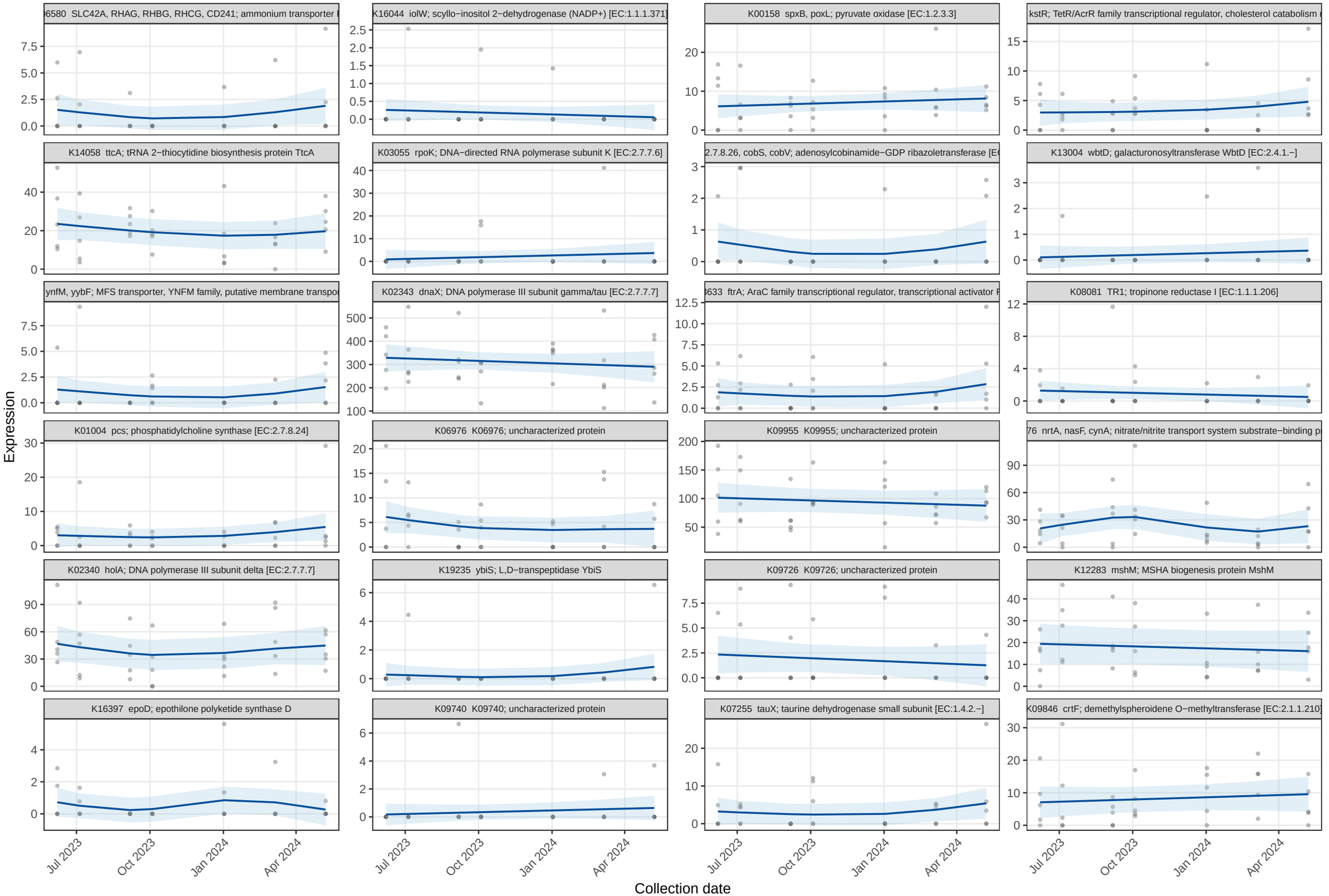


All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions

794 tcuR; LysR family transcriptional regulator, regulatory protein for tcu

K00462 bphC; biphenyl−2,3−diol 1,2−dioxygenase [EC:1.13.11.39]

K02638 petE; plastocyanin

K05791 terZ; tellurium resistance protein TerZ

yl−8−(2,3−dimethoxy−alpha−L−rhamnosyl)tetracenomycin−C 4'−O−meth

K03153 thiO; glycine oxidase [EC:1.4.3.19]

K00769 gpt; xanthine phosphoribosyltransferase [EC:2.4.2.22]

K12601 SKI8; superkiller protein 8

K12602 WDR61, REC14, SKI8; WD repeat−containing protein 61

K06202 cyaY; iron−sulfur cluster assembly protein CyaY

515 aviRa; 23S rRNA (guanine2535−N1)−methyltransferase [EC:2.1.1.2

K02294 crtR; beta−carotene hydroxylase [EC:1.14.13.−]

Expression

monooxygenase / 4−nitrocatechol 4−monooxygenase, oxygenase compo

K00098 idnD; L−idonate 5−dehydrogenase [EC:1.1.1.264]

588 purE; 5−(carboxyamino)imidazole ribonucleotide mutase [EC:5.4.99

K13670 pimF; putative glycosyltransferase [EC:2.4.−.−]

K22325 cndA; chloroacetanilide N−alkylformylase [EC:1.14.15.23]

ycR; LysR family transcriptional regulator, putative pyruvate carboxylase

K12472 EPS15; epidermal growth factor receptor substrate 15

tsG, glcA, glcB; glucose PTS system EIICBA or EIICB component [EC:2

180 tll; dTDP−6−deoxy−L−talose 4−dehydrogenase (NAD+) [EC:1.1.1.3

K05886 sdh; serine 3−dehydrogenase (NADP+) [EC:1.1.1.276]

R; LysR family transcriptional regulator, transcriptional activator for bauA

K21020 siaD; diguanylate cyclase [EC:2.7.7.65]

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dgcA; L−Ala−D/L−Glu epimerase / N−acetyl−D−glutamate racemase [E

K19515 bamB; benzoyl−CoA reductase subunit BamB [EC:1.3.−.−]

K17311 treS; trehalose transport system substrate−binding protein

K09705 K09705; uncharacterized protein

K14335 pimC; alpha−1,6−mannosyltransferase [EC:2.4.1.−]

K02836 prfB; peptide chain release factor 2

K06880 ereA_B; erythromycin esterase [EC:3.1.1.−]

K16886 qmoB; quinone−modifying oxidoreductase, subunit QmoB

K00886 ppgK; polyphosphate glucokinase [EC:2.7.1.63]

58 mmoY; methane monooxygenase component A beta chain [EC:1.14.

K10943 flrC, fleR, flaM; two−component system, response regulator FlrC

K02385 flbD; flagellar protein FlbD

Expression

1160 rusA; crossover junction endodeoxyribonuclease RusA [EC:3.1.21.

K03776 aer; aerotaxis receptor

0049 GRHPR; glyoxylate/hydroxypyruvate reductase [EC:1.1.1.79 1.1.1.

K01878 glyQ; glycyl−tRNA synthetase alpha chain [EC:6.1.1.14]

MarR family transcriptional regulator, 2−MHQ and catechol−resistance re

K12287 mshQ; MSHA biogenesis protein MshQ

K05603 hutF; formimidoylglutamate deiminase [EC:3.5.3.13]

A, sacP, sacX, ptsS; sucrose PTS system EIIBCA or EIIBC component [E

K06193 phnA; protein PhnA

K01904 4CL; 4−coumarate−−CoA ligase [EC:6.2.1.12]

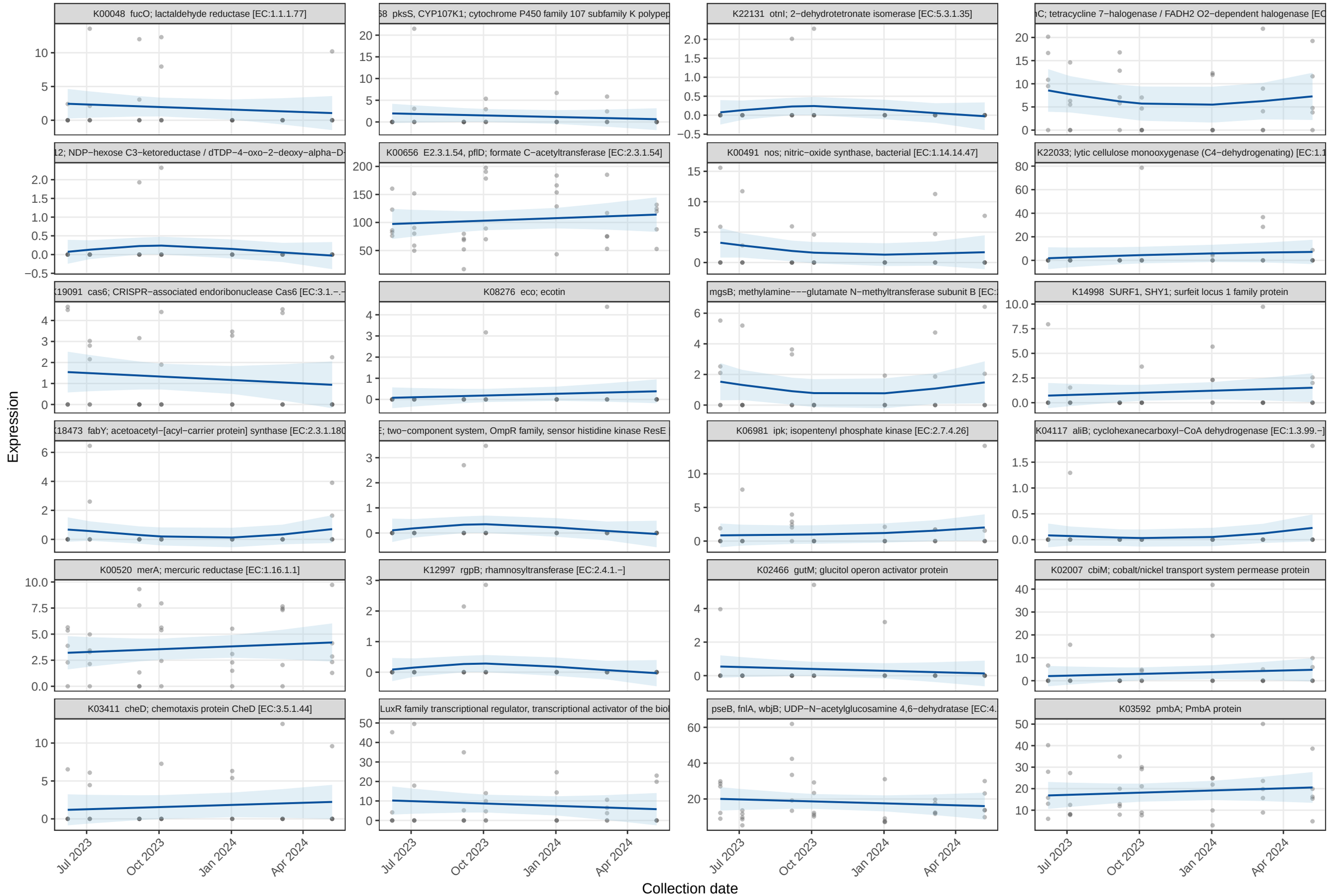
K01042 selA; L−seryl−tRNA(Ser) seleniumtransferase [EC:2.9.1.1]

K01678 E4.2.1.2AB, fumB; fumarate hydratase subunit beta [EC:4.2.1.2

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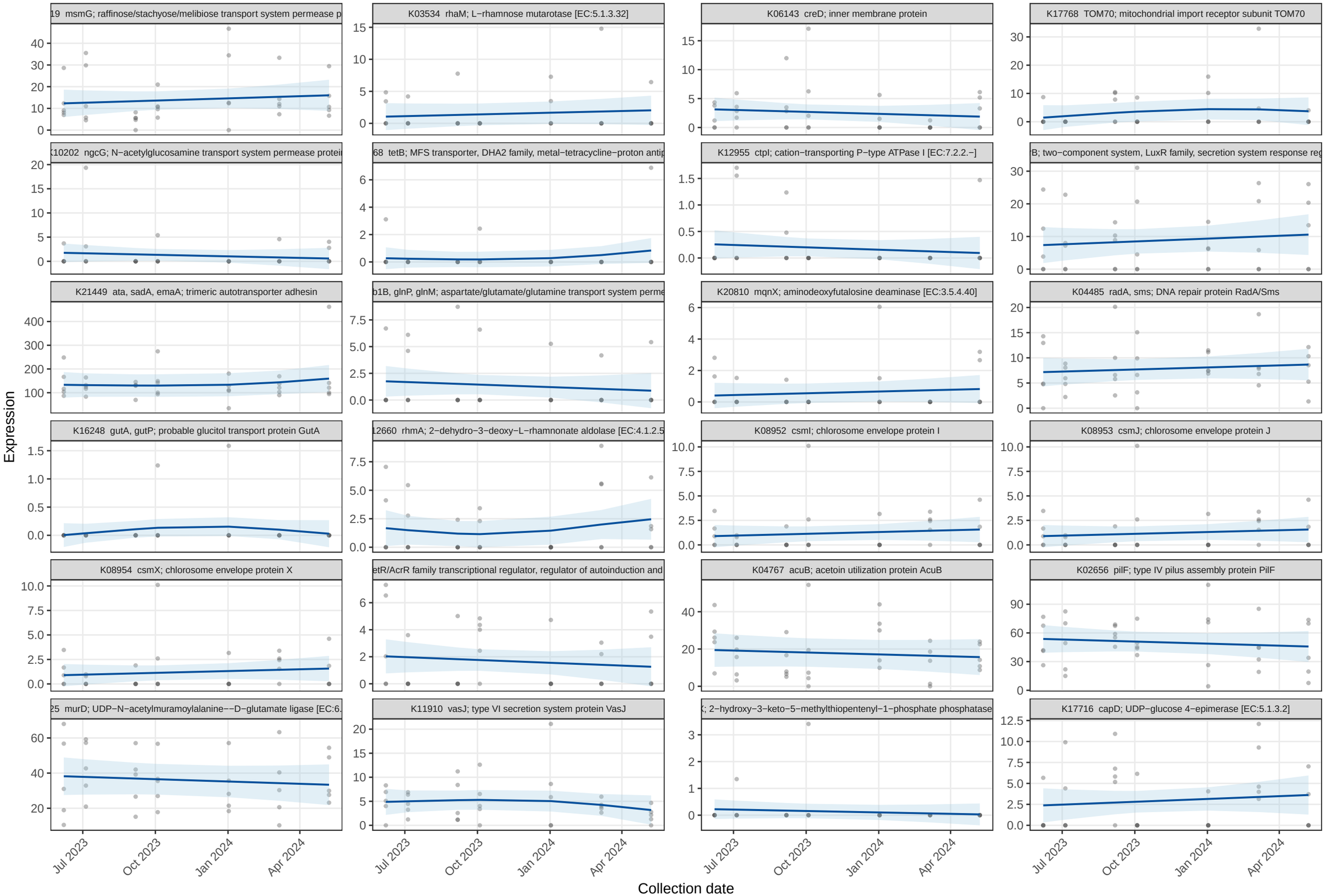
Collection date

All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



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079 ulaE, sgaU, sgbU; L−ribulose−5−phosphate 3−epimerase [EC:5.1.3

K00247 frdD; succinate dehydrogenase subunit D

K00926 arcC; carbamate kinase [EC:2.7.2.2]

K00849 galK; galactokinase [EC:2.7.1.6]

K08722 yfbR; 5'−deoxynucleotidase [EC:3.1.3.89]

K02420 fliQ; flagellar biosynthesis protein FliQ

K00552 GNMT; glycine N−methyltransferase [EC:2.1.1.20]

4696 SLC30A9, ZNT9; solute carrier family 30 (zinc transporter), membe

3−deoxy−D−manno−octulosonic−acid transferase [EC:2.4.99.12 2.4.99.1

K09991 K09991; uncharacterized protein

K12282 mshL; MSHA biogenesis protein MshL

K14449 mch, mcd; 2−methylfumaryl−CoA hydratase [EC:4.2.1.148]

Expression

K00243 K00243; uncharacterized protein

K01754 E4.3.1.19, ilvA, tdcB; threonine dehydratase [EC:4.3.1.19]

K01155 E3.1.21.4; type II restriction enzyme [EC:3.1.21.4]

K00624 E2.3.1.7; carnitine O−acetyltransferase [EC:2.3.1.7]

Z; two−component system, LytTR family, sensor histidine kinase AlgZ [E

K06913 K06913; uncharacterized protein

K10896 FANCM; fanconi anemia group M protein

K15232 ccsA; citryl−CoA synthetase large subunit [EC:6.2.1.18]

nurU; N−acetyl−alpha−D−muramate 1−phosphate uridylyltransferase [EC

K04019 eutA; ethanolamine utilization protein EutA

K00109 L2HGDH; 2−hydroxyglutarate dehydrogenase [EC:1.1.99.2]

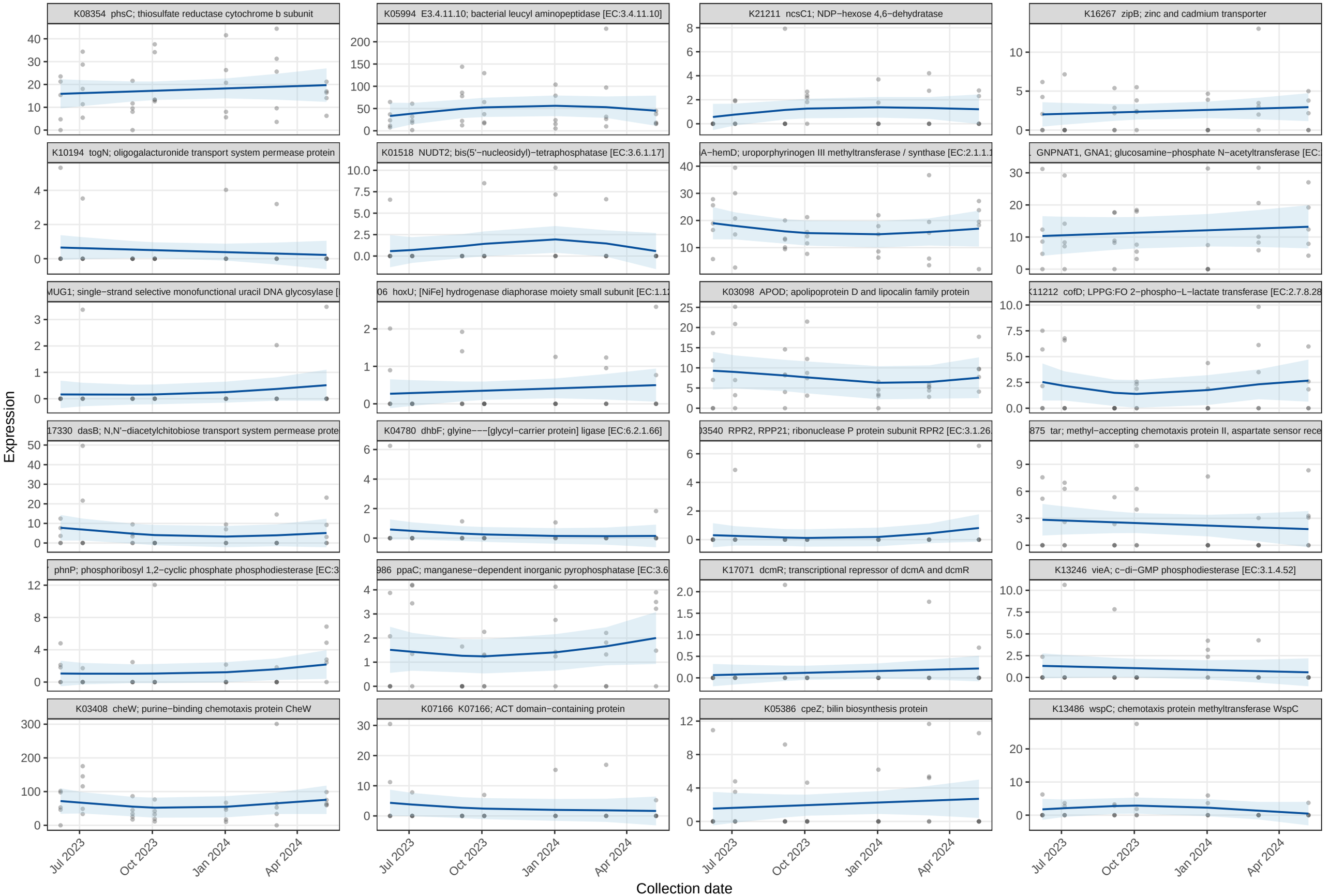
K05365 mrcB; penicillin−binding protein 1B [EC:2.4.99.28 3.4.16.4]

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−component system, NarL family, nitrate/nitrite sensor histidine kinase N

K14661 nodF; nodulation protein F [EC:2.3.1.−]

K21085 yedQ; diguanylate cyclase [EC:2.7.7.65]

K10747 LIG1; DNA ligase 1 [EC:6.5.1.1 6.5.1.6 6.5.1.7]

18430 legI, neuB2; N,N'−diacetyllegionaminate synthase [EC:2.5.1.101

K07270 K07270; glycosyl transferase, family 25

K19164 ccdA; antitoxin CcdA

K02556 motA; chemotaxis protein MotA

K00117 gcd; quinoprotein glucose dehydrogenase [EC:1.1.5.2]

K06182 rluF; 23S rRNA pseudouridine2604 synthase [EC:5.4.99.21]

K03796 bax; Bax protein

1499 mch; methenyltetrahydromethanopterin cyclohydrolase [EC:3.5.4.

Expression

K11187 PRDX5; peroxiredoxin 5 [EC:1.11.1.24]

K17208 ibpA; inositol transport system substrate−binding protein

K20469 K20469; putative heme transporter

K06208 aroH; chorismate mutase [EC:5.4.99.5]

19623 phcQ; two−component system, probable response regulator Phc

K15065 desZ; 3−O−methylgallate 3,4−dioxygenase [EC:1.13.11.−]

K05807 bamD; outer membrane protein assembly factor BamD

K09181 yfiQ; acetyltransferase

K07214 fes; iron(III)−enterobactin esterase [EC:3.1.1.108]

E, baeE; trans−AT polyketide synthase, acyltransferase and oxidoreduct

K16841 hpxA; allantoin racemase [EC:5.1.99.3]

K22363 etnE; 2−hydroxypropyl−CoM lyase [EC:4.4.1.23]

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POLRMT, RPO41; DNA−directed RNA polymerase, mitochondrial [EC:

K06034 comD; sulfopyruvate decarboxylase subunit alpha [EC:4.1.1.79

K05979 comB; 2−phosphosulfolactate phosphatase [EC:3.1.3.71]

K04073 mhpF; acetaldehyde dehydrogenase [EC:1.2.1.10]

K09388 K09388; uncharacterized protein

K16193 arfC; uncharacterized membrane protein ArfC

310 nudB, ntpA; dihydroneopterin triphosphate diphosphatase [EC:3.6.1

K07455 recT; recombination protein RecT

3 ybaZ; methylated−DNA−protein−cysteine methyltransferase related p

K13820 fliR−flhB; flagellar biosynthesis protein FliR/FlhB

K09386 K09386; uncharacterized protein

K08252 E2.7.10.1; receptor protein−tyrosine kinase [EC:2.7.10.1]

Expres

K07243 FTR, FTH1, efeU; high−affinity iron transporter

K09984 K09984; uncharacterized protein

63 mutM, fpg; formamidopyrimidine−DNA glycosylase [EC:3.2.2.23 4.2.

4 boxD; 3,4−dehydroadipyl−CoA semialdehyde dehydrogenase [EC:1.2

K09696 natB; sodium transport system permease protein

K18815 aac6−I; aminoglycoside 6'−N−acetyltransferase I [EC:2.3.1.82]

mdo; formaldehyde dismutase / methanol dehydrogenase [EC:1.2.98.1 1

1381 bkdA; 2−oxoisovalerate dehydrogenase E1 component [EC:1.2.4

K11021 tccC; insecticidal toxin complex protein TccC

K09510 DNAJB4; DnaJ homolog subfamily B member 4

itrophenol 2−monooxygenase / 4−nitrocatechol 4−monooxygenase, red

K04127 cefD; isopenicillin−N epimerase [EC:5.1.1.17]

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153 E1.1.1.306; S−(hydroxymethyl)mycothiol dehydrogenase [EC:1.1.1.

K03571 mreD; rod shape−determining protein MreD

K21010 pelE; polysaccharide biosynthesis protein PelE

K07013 K07013; uncharacterized protein

K19292 algK; alginate biosynthesis protein AlgK

roA, mntA, znuA; manganese/zinc/iron transport system substrate−bindi

K18482 ADCL; 4−amino−4−deoxychorismate lyase [EC:4.1.3.38]

K07507 mgtC; putative Mg2+ transporter−C (MgtC) family protein

922 hydB; quinone−reactive Ni/Fe−hydrogenase large subunit [EC:1.12.

04085 tusA, sirA; tRNA 2−thiouridine synthesizing protein A [EC:2.8.1.−

4 AGA, aspG; N4−(beta−N−acetylglucosaminyl)−L−asparaginase [EC:3.

sufR; DeoR family transcriptional regulator, suf operon transcriptional re

Expression

K00887 dgkA; undecaprenol kinase [EC:2.7.1.66]

K16203 dppA1; D−amino peptidase [EC:3.4.11.−]

K01387 colA; microbial collagenase [EC:3.4.24.3]

K01482 DDAH, ddaH; dimethylargininase [EC:3.5.3.18]

K01184 E3.2.1.15; polygalacturonase [EC:3.2.1.15]

856 rmd; GDP−4−dehydro−6−deoxy−D−mannose reductase [EC:1.1.1.1

05957 stsC; L−glutamine:scyllo−inosose aminotransferase [EC:2.6.1.50

K07485 K07485; transposase

K02647 cdaR; carbohydrate diacid regulator

72 nasC, nasA; assimilatory nitrate reductase catalytic subunit [EC:1.7.

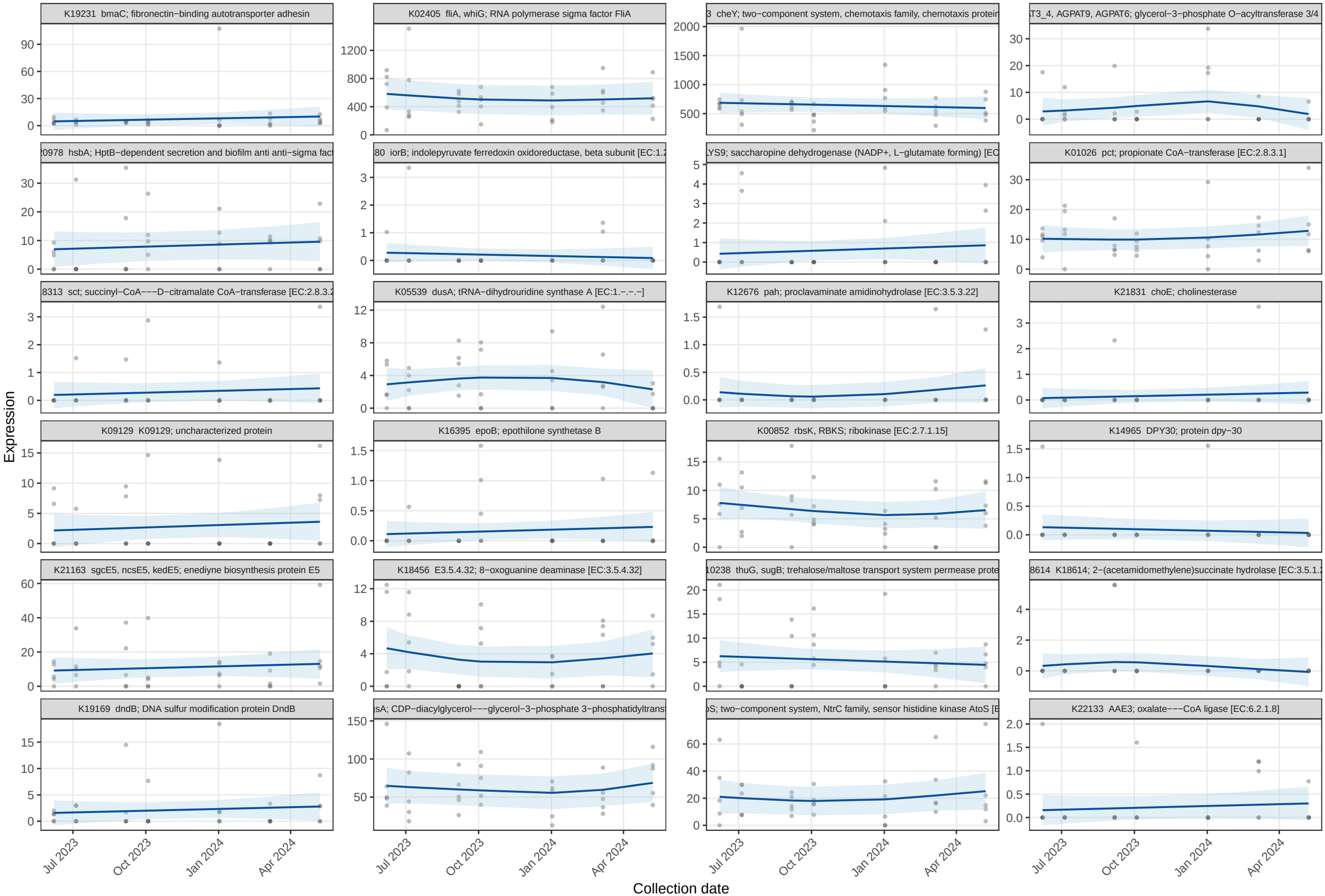
K00005 gldA; glycerol dehydrogenase [EC:1.1.1.6]

K00106 XDH; xanthine dehydrogenase/oxidase [EC:1.17.1.4 1.17.3.2]

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79 grdH; betaine reductase complex component B subunit beta [EC:1.2

2 DTWD2, tapT; tRNA−uridine aminocarboxypropyltransferase [EC:2.5

K03550 ruvA; holliday junction DNA helicase RuvA

12528 xdhD; putative selenate reductase molybdopterin−binding subur

40 TSR3; rRNA small subunit aminocarboxypropyltransferase [EC:2.5.1

88 MLL3; [histone H3]−lysine4 N−trimethyltransferase MLL3 [EC:2.1.1

K06972 PITRM1, PreP, CYM1; presequence protease [EC:3.4.24.−]

3 uhpC; MFS transporter, OPA family, sugar phosphate sensor protein

K03598 rseB; sigma−E factor negative regulatory protein RseB

K03090 sigB; RNA polymerase sigma−B factor

K21142 moaX; MoaE−MoaD fusion protein [EC:2.8.1.12]

K22320 oleD; 2−alkyl−3−oxoalkanoate reductase [EC:1.1.1.412]

Expression

K00989 rph; ribonuclease PH [EC:2.7.7.56]

655 plsC; 1−acyl−sn−glycerol−3−phosphate acyltransferase [EC:2.3.1

3772 rirA; Rrf2 family transcriptional regulator, iron−responsive regulat

K12242 pchC; pyochelin biosynthesis protein PchC

K07316 mod; adenine−specific DNA−methyltransferase [EC:2.1.1.72]

K18911 egtD; L−histidine Nalpha−methyltransferase [EC:2.1.1.44]

K12278 mshG; MSHA biogenesis protein MshG

K07445 K07445; putative DNA methylase

K15552 tauC; taurine transport system permease protein

949 phdJ; 4−(2−carboxyphenyl)−2−oxobut−3−enoate aldolase [EC:4.1.1

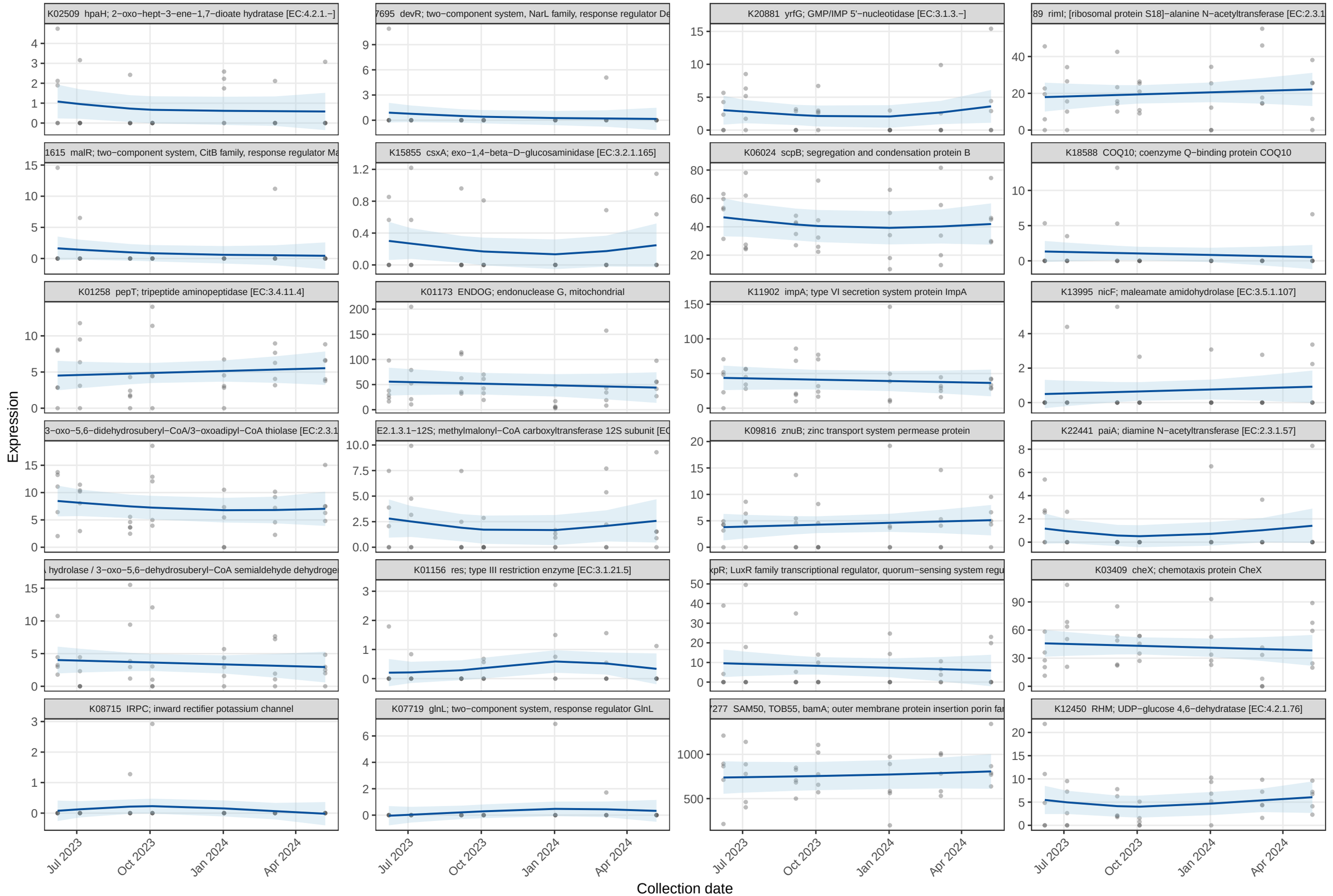
05966 citG; triphosphoribosyl−dephospho−CoA synthase [EC:2.4.2.52

K01130 atsA, aslA; arylsulfatase [EC:3.1.6.1]

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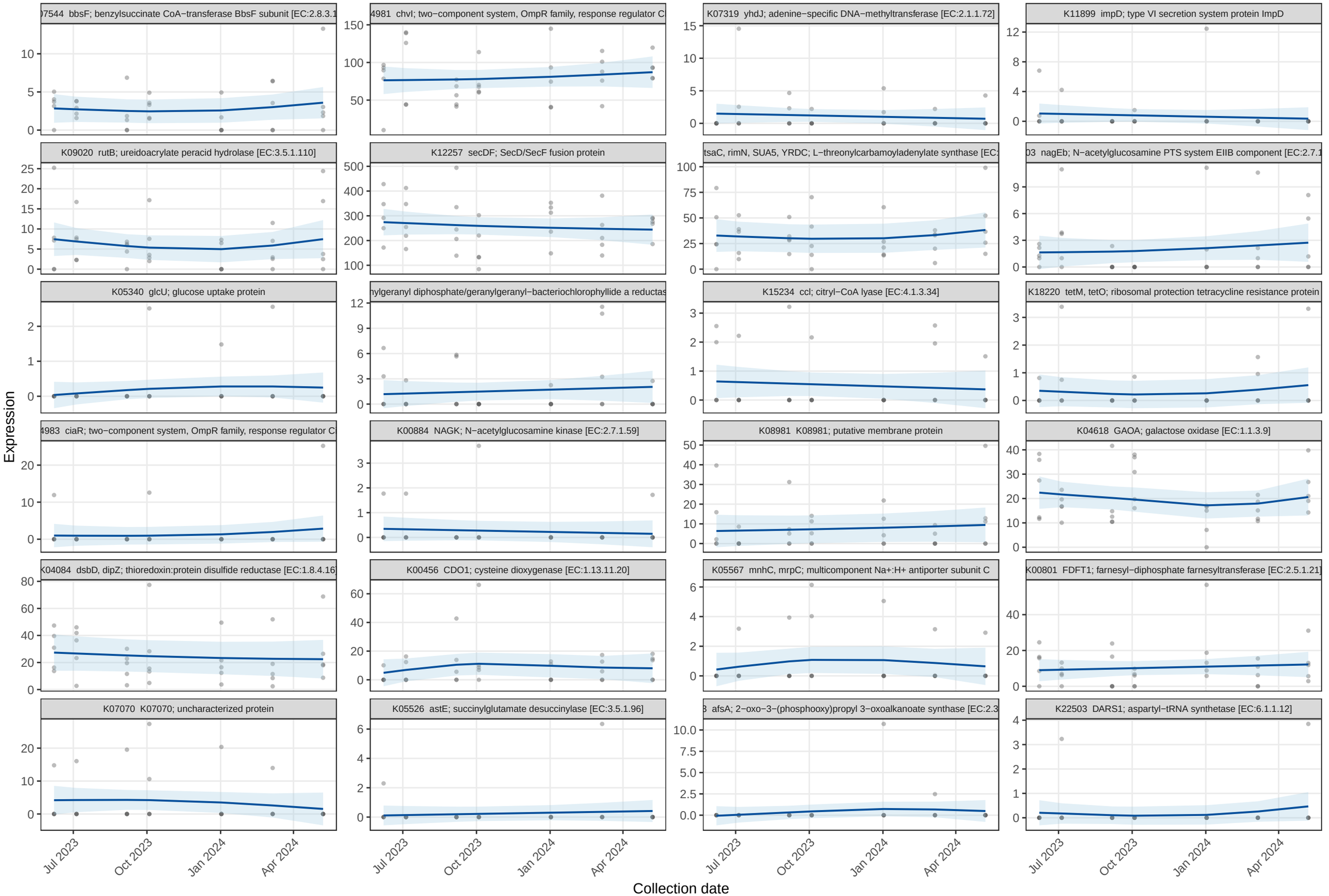
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K03383 atzD; cyanuric acid amidohydrolase [EC:3.5.2.15]

K19794 bar; barbiturase [EC:3.5.2.1]

7 cbrT; energy−coupling factor transport system substrate−specific com

K03981 dsbC; thiol:disulfide interchange protein DsbC [EC:5.3.4.1]

K06132 clsC; cardiolipin synthase C [EC:2.7.8.−]

K03653 K03653; N−glycosylase/DNA lyase [EC:3.2.2.− 4.2.99.18]

K06666 TUP1; general transcriptional corepressor TUP1

K19580 rppA; 1,3,6,8−tetrahydroxynaphthalene synthase [EC:2.3.1.233

ppB; UDP−N−acetyl−2−amino−2−deoxyglucuronate dehydrogenase [EC

K09692 tagG; teichoic acid transport system permease protein

,10−secoandrosta−1,3,5(10)−triene−9,17−dione monooxygenase reduc

yltransferase / trehalose O−mycolyltransferase / mycolyltransferase Ag8

K19033 PSRP4, RPS31, rpsU; 30S ribosomal protein S31/Thx

K17876 CYP105D; pentalenic acid synthase [EC:1.14.15.11]

K12544 rsaA; S−layer protein

87 cyc2; germacradienol/geosmin synthase [EC:4.2.3.22 4.2.3.75 4.1.9

K06889 K06889; uncharacterized protein

K11937 pgaD; biofilm PGA synthesis protein PgaD

K08306 mltC; peptidoglycan lytic transglycosylase C [EC:4.2.2.29]

K07300 chaA, CAX; Ca2+:H+ antiporter

K01053 gnl, RGN; gluconolactonase [EC:3.1.1.17]

K03743 pncC; nicotinamide−nucleotide amidase [EC:3.5.1.42]

K20273 zmpA; zinc metalloprotease ZmpA

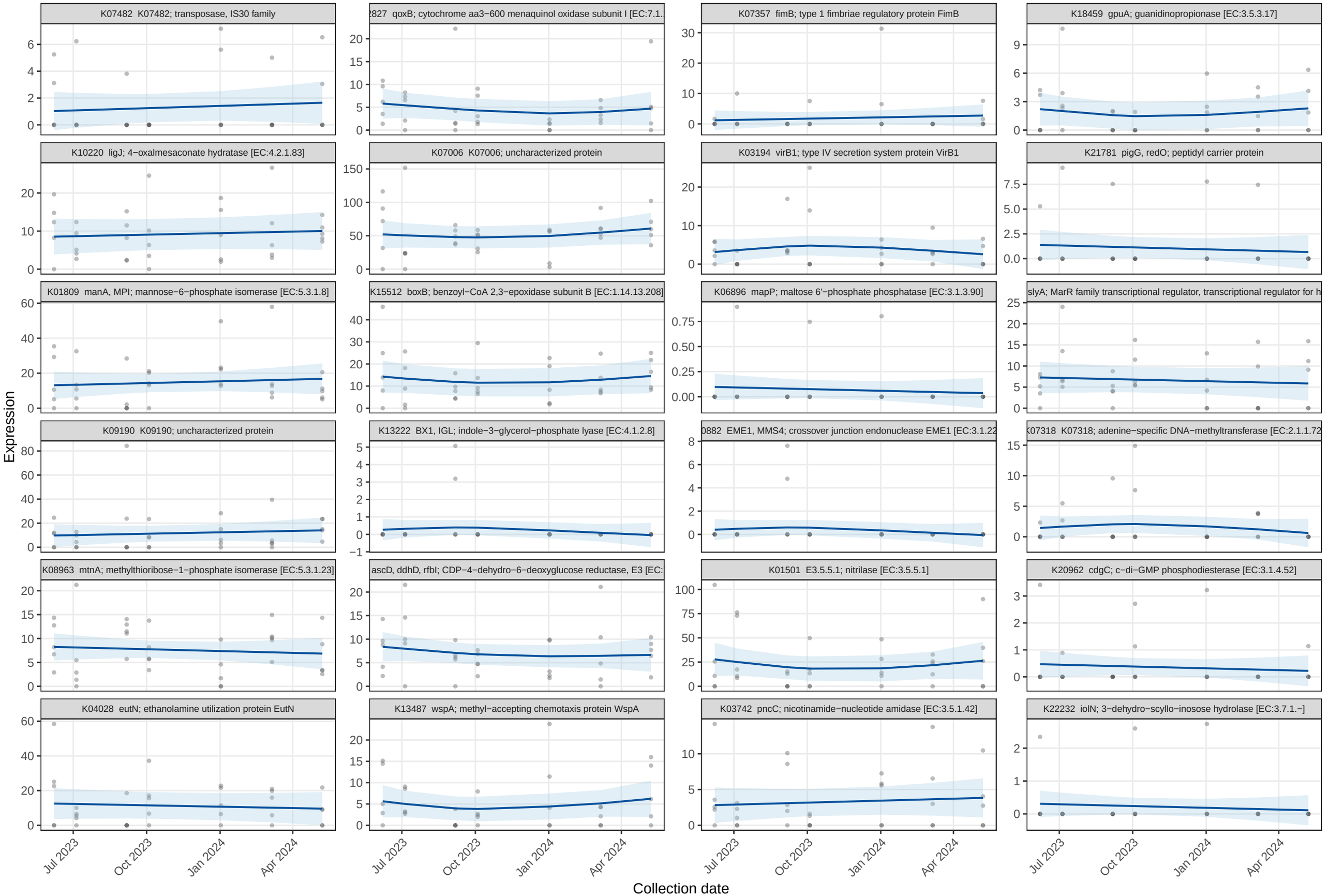
K15923 AXY8, FUC95A, afcA; alpha−L−fucosidase 2 [EC:3.2.1.51]

Expression

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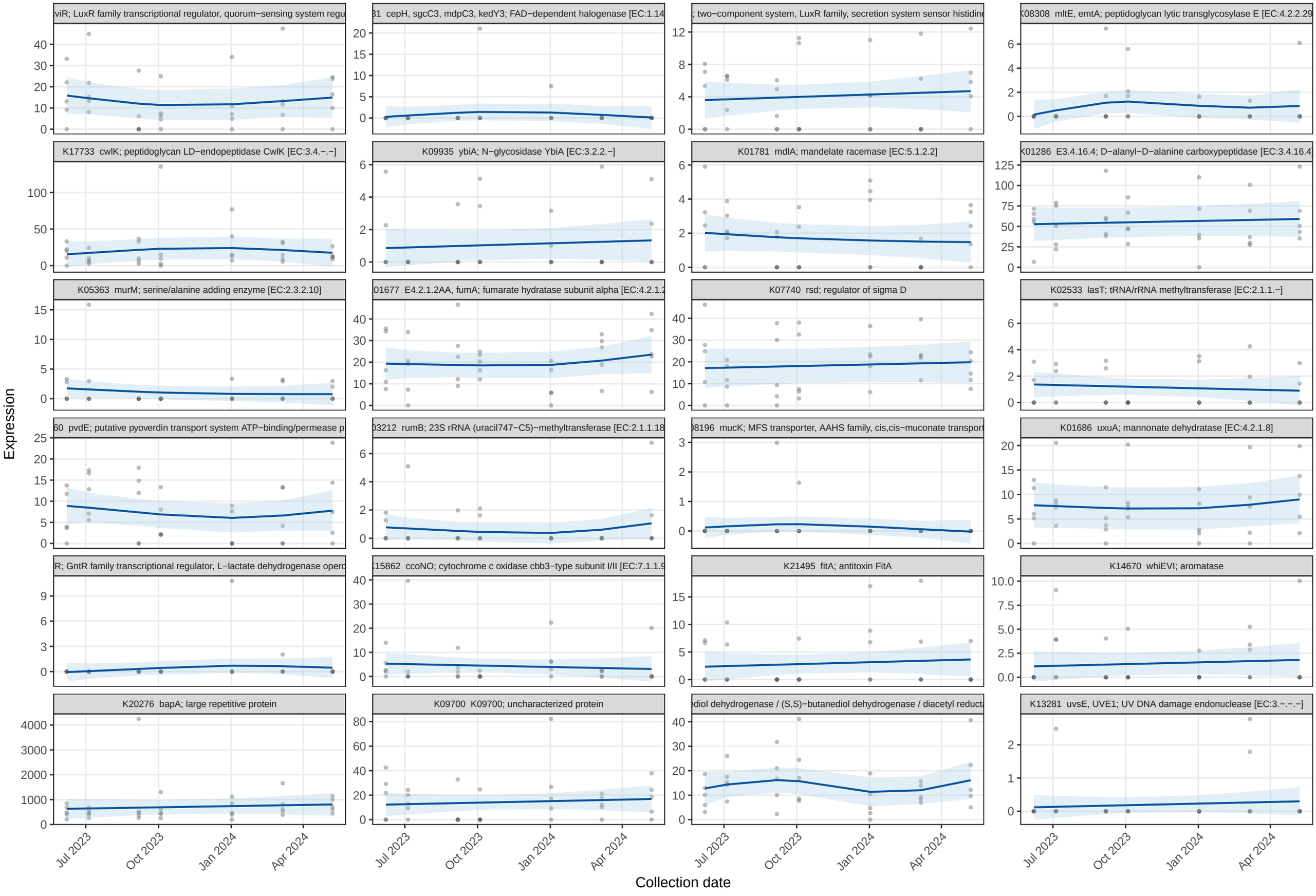
Collection date

All remaining BRITE classes after exclusions

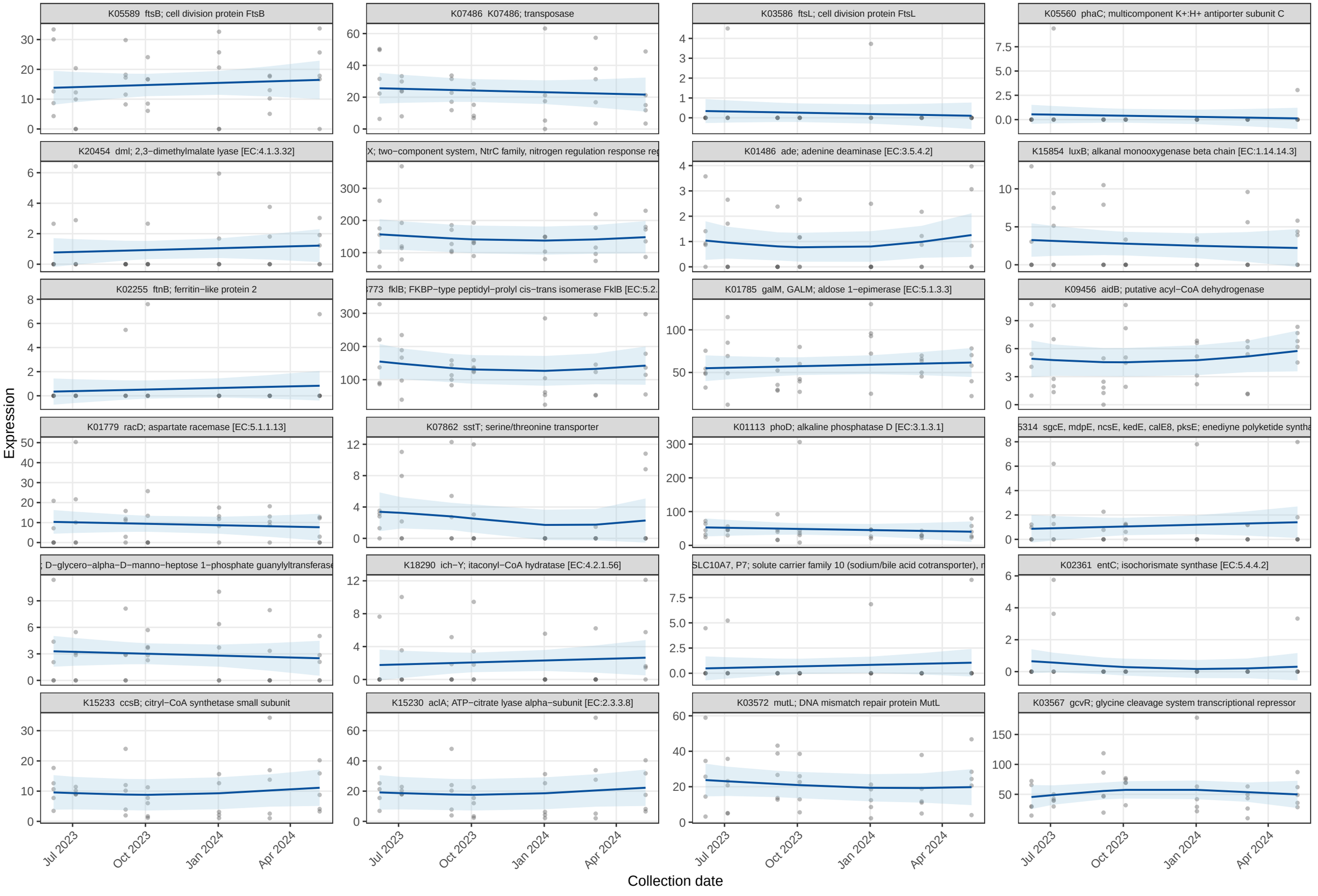


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All remaining BRITE classes after exclusions



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spR; GntR family transcriptional regulator, rspAB operon transcriptional

K05996 cpt; carboxypeptidase T [EC:3.4.17.18]

K14164 glyQS; glycyl−tRNA synthetase [EC:6.1.1.14]

K08344 scsB; suppressor for copper−sensitivity B

K14671 whiEVII; cyclase

K01992 ABC−2.P; ABC−2 type transport system permease protein

K07574 yhbY; RNA−binding protein

K02665 pilP; type IV pilus assembly protein PilP

K05997 sufA; Fe−S cluster assembly protein SufA

K00271 vdh; valine dehydrogenase (NAD+) [EC:1.4.1.23]

K03558 cvpA; membrane protein required for colicin V production

B; two−component system, OmpR family, sensor histidine kinase RstB [E

Expression

K20148 dke1; acetylacetone−cleaving enzyme [EC:1.13.11.50]

B, tomA1; phenol/toluene 2−monooxygenase (NADH) P1/A1 [EC:1.14.1

K09778 lpxJ; Kdo2−lipid IVA 3' secondary acyltransferase [EC:2.3.1.−]

21977 coaB; phosphopantothenate−−−cysteine ligase (CTP) [EC:6.3.2.

K19745 acuI; acrylyl−CoA reductase (NADPH) [EC:1.3.1.−]

01236 treZ, glgZ; maltooligosyltrehalose trehalohydrolase [EC:3.2.1.14

K09477 citT; citrate:succinate antiporter

E3.4.21.102, prc, ctpA; carboxyl−terminal processing protease [EC:3.4

K08685 qhpA; quinohemoprotein amine dehydrogenase [EC:1.4.9.1]

K13613 pksM, baeM; bacillaene polyketide synthase PksM/BaeM

06151 E1.1.99.3A; gluconate 2−dehydrogenase alpha chain [EC:1.1.99.

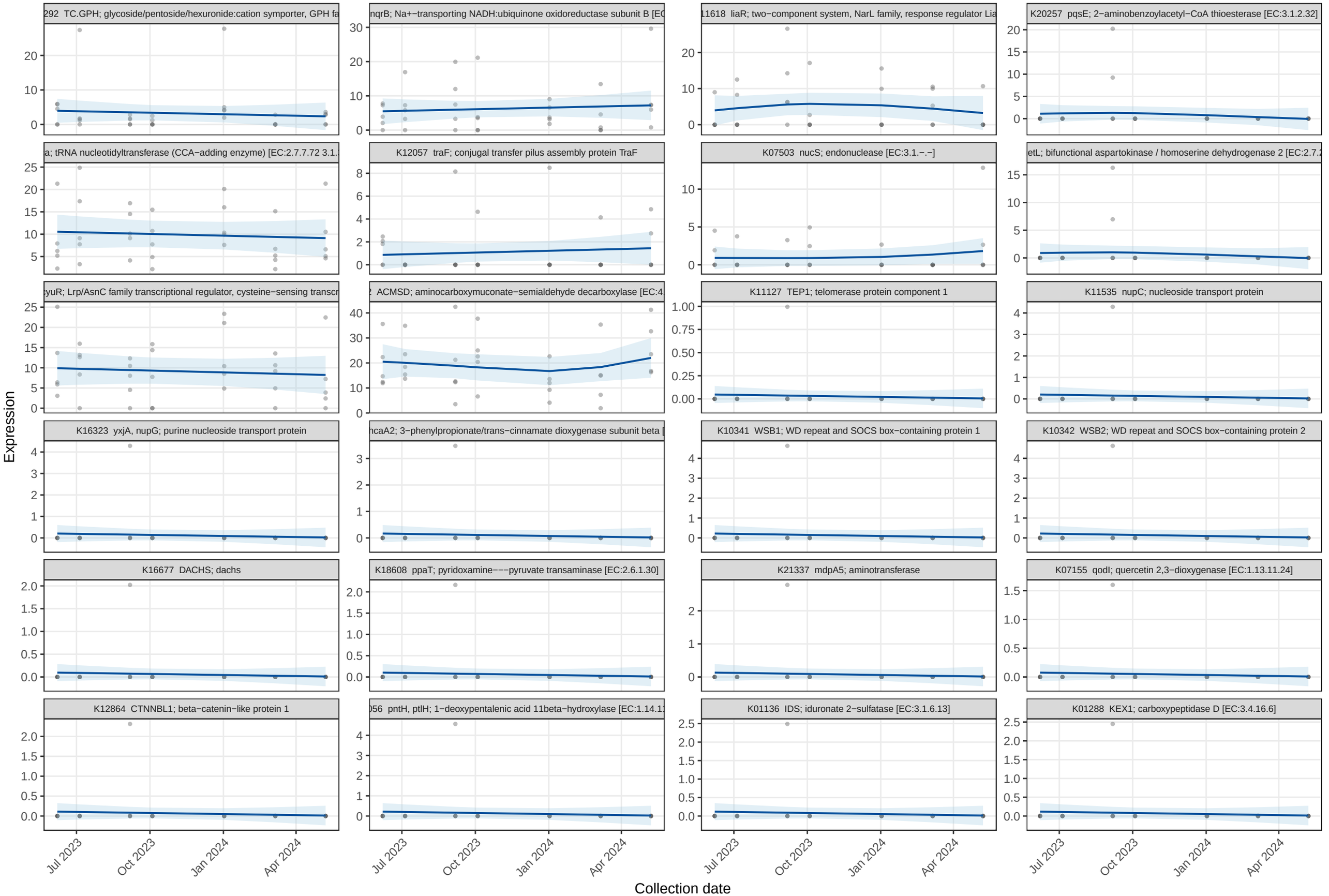
K20201 ywlE; protein arginine phosphatase [EC:3.9.1.2]

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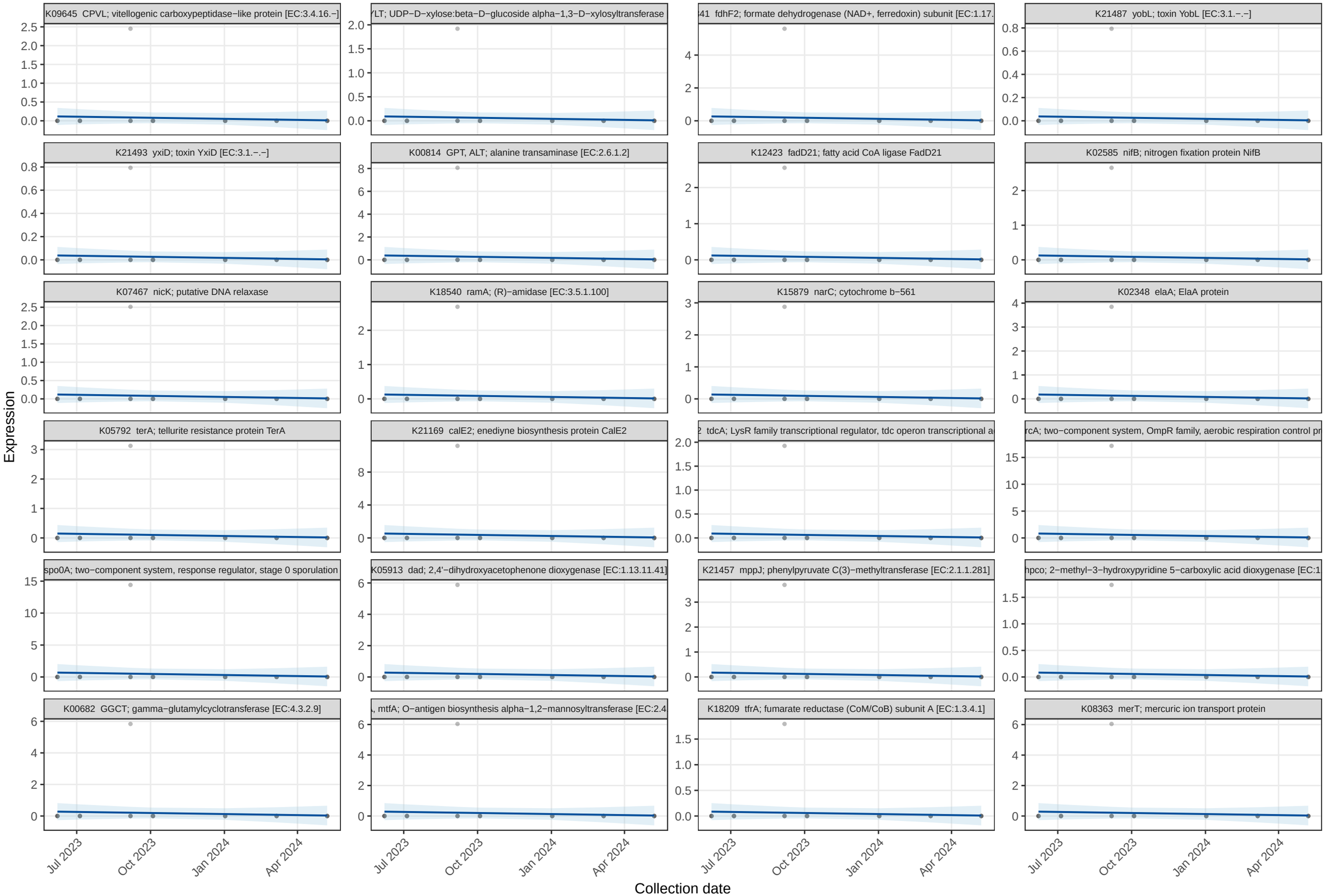
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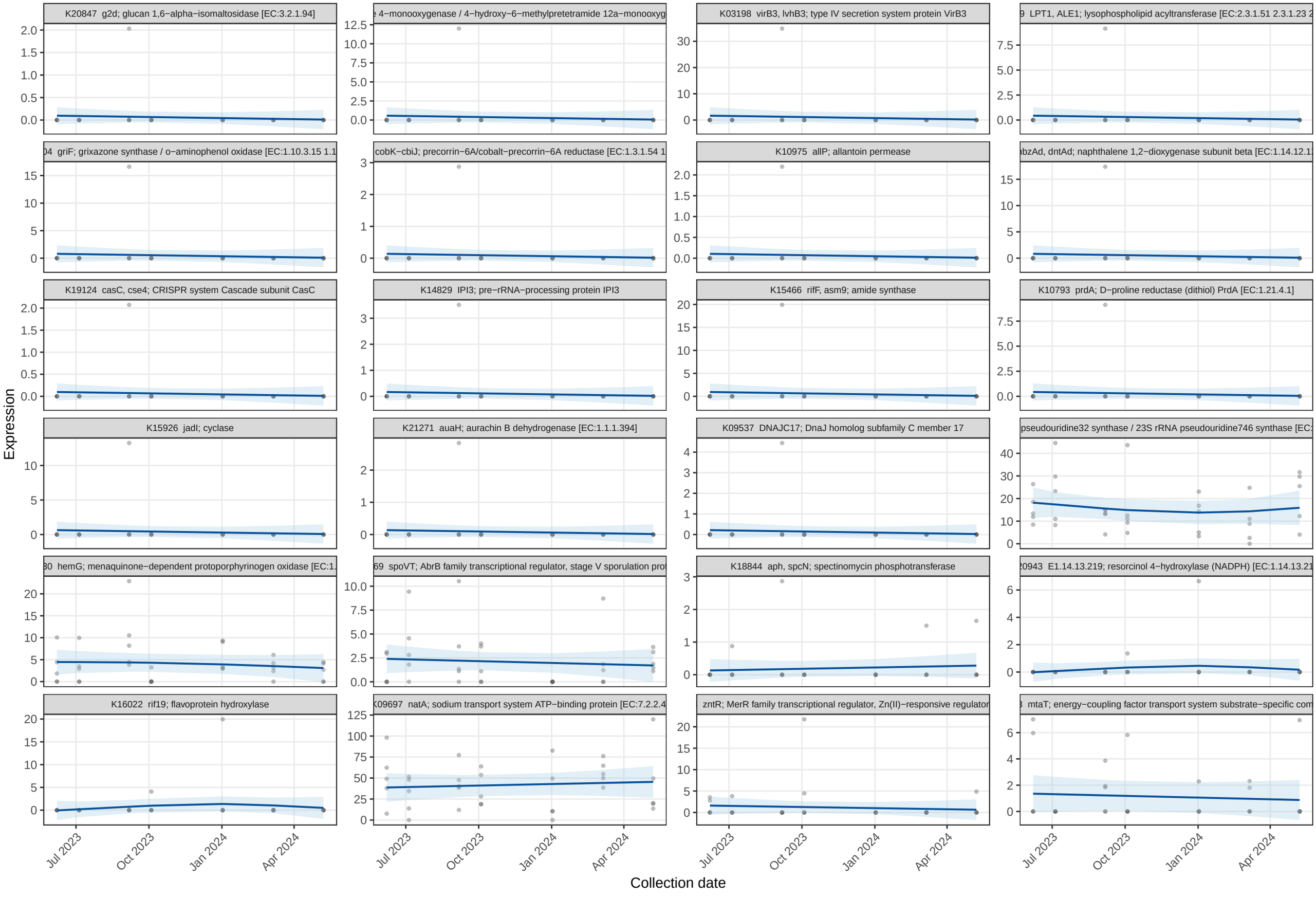
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K07501 K07501; 3'−5' exonuclease

hosR; NosR/NirI family transcriptional regulator, nitrous oxide reductase

K11075 potH, spuG; putrescine transport system permease protein

K11900 impC; type VI secretion system protein ImpC

K06192 pqiB, letB; intermembrane transport protein PqiB/LetB

l−CoA/propionyl−CoA carboxylase carboxyl transferase subunit [EC:6.4

K18842 chpS, chpBI; antitoxin ChpS

K11034 nheBC; non−hemolytic enterotoxin B/C

K00221 E4.99.1.2; alkylmercury lyase [EC:4.99.1.2]

00374 narI, narV; nitrate reductase gamma subunit [EC:1.7.5.1 1.7.99.

K20754 pstI; aqualysin 1 [EC:3.4.21.111]

K16394 epoA; epothilone polyketide synthase A

Expression

K01138 K01138; uncharacterized sulfatase [EC:3.1.6.−]

K07358 fimE; type 1 fimbriae regulatory protein FimE

01851 pchA; salicylate biosynthesis isochorismate synthase [EC:5.4.4.

K03403 chlH, bchH; magnesium chelatase subunit H [EC:6.6.1.1]

eA; two−component system, chemotaxis family, sensor kinase CheA [E

K03690 ubiJ; ubiquinone biosynthesis accessory factor UbiJ

K10455 KLHL18; kelch−like protein 18

04100 ligA; protocatechuate 4,5−dioxygenase, alpha chain [EC:1.13.11

bsR; sigma−54 dependent transcriptional regulator, positive regulator of

D; two−component system, NarL family, sensor histidine kinase RcsD [E

K03620 hyaC; Ni/Fe−hydrogenase 1 B−type cytochrome subunit

K07237 tusB, dsrH; tRNA 2−thiouridine synthesizing protein B

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63 polC; DNA polymerase III subunit alpha, Gram−positive type [EC:2.7

K00638 catB; chloramphenicol O−acetyltransferase type B [EC:2.3.1.28

arcosine/betaine reductase complex component C subunit alpha [EC:1.2

sarcosine/betaine reductase complex component C subunit beta [EC:1.2

983 kshB; 3−ketosteroid 9alpha−monooxygenase subunit B [EC:1.14.15

K03609 minD; septum site−determining protein MinD

K09154 K09154; uncharacterized protein

K08350 fdnI; formate dehydrogenase−N, gamma subunit

K09884 AQPN; aquaporin rerated protein, invertebrate

K10443 KLHL2_3; kelch−like protein 2/3

K16167 bpsA, srsA, pks11, pks10; alkylresorcinol/alkylpyrone synthase

K01771 plc; 1−phosphatidylinositol phosphodiesterase [EC:4.6.1.13]

Expres

K22136 bshC; bacillithiol synthase

l family transcriptional regulator, repressor for deo operon, udp, cdd, tsx

K16507 DCHS1_2, PCDH16_23; protocadherin−16/23

K16669 FAT4; protocadherin Fat 4

K08115 CSPG4; chondroitin sulfate proteoglycan 4

K19013 STAB2, HARE; stabilin−2

xy−D−manno−octulosonate 8−phosphate phosphatase (KDO 8−P phosp

K03319 TC.DASS; divalent anion:Na+ symporter, DASS family

K10240 cebE; cellobiose transport system substrate−binding protein

K00984 aadA; streptomycin 3"−adenylyltransferase [EC:2.7.7.47]

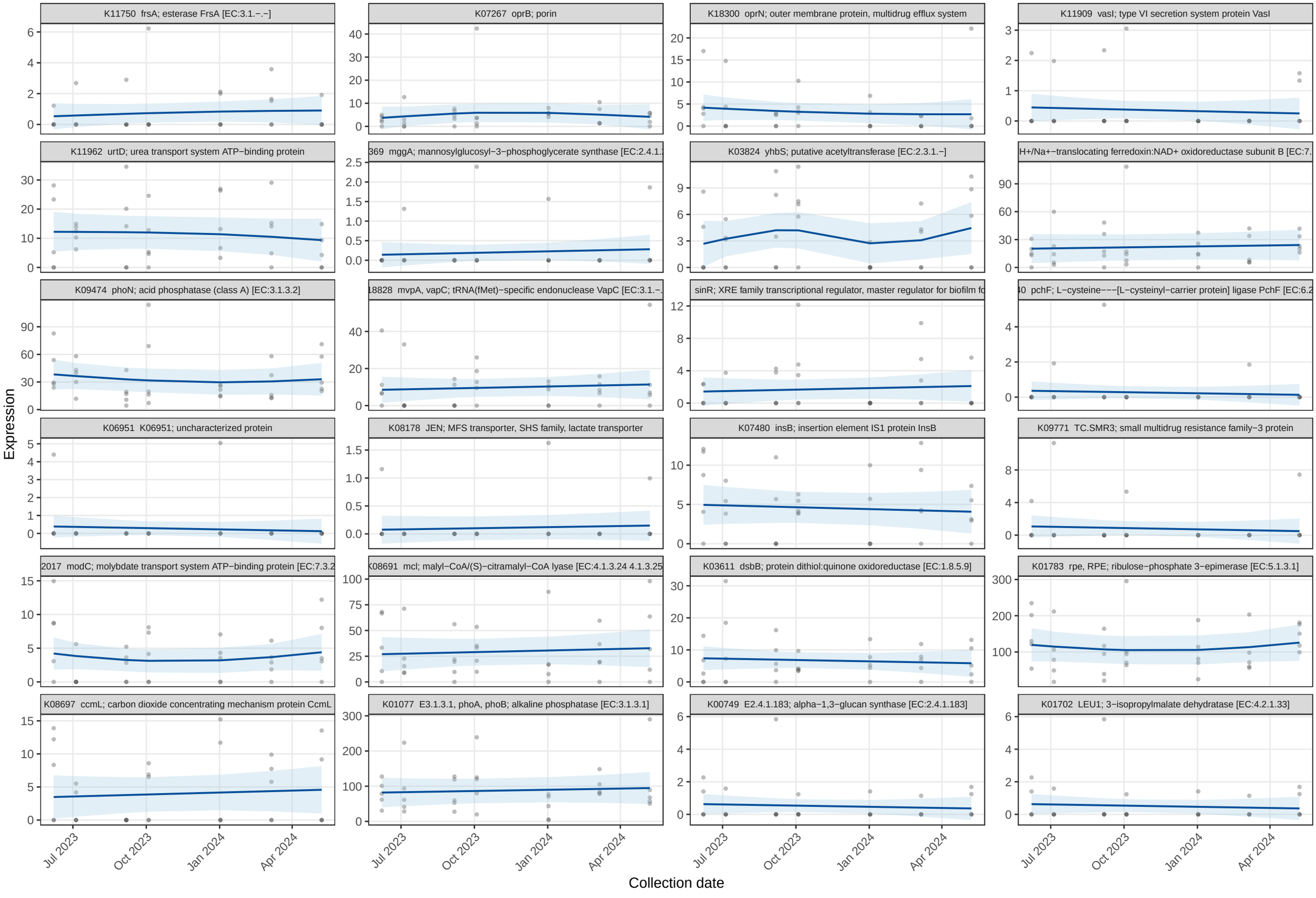
nqrD; Na+−transporting NADH:ubiquinone oxidoreductase subunit D [E

K20941 graF; gamma−resorcylate decarboxylase [EC:4.1.1.103]

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K08776 NPEPPS; puromycin−sensitive aminopeptidase [EC:3.4.11.14]

K00949 thiN, TPK1, THI80; thiamine pyrophosphokinase [EC:2.7.6.2]

K14733 limB; limonene 1,2−monooxygenase [EC:1.14.13.107]

K03810 mviM; virulence factor

K07407 E3.2.1.22B, galA, rafA; alpha−galactosidase [EC:3.2.1.22]

427 fadD28; long−chain fatty acid adenylyltransferase FadD28 [EC:6.2.1

17235 araP; arabinooligosaccharide transport system permease protei

gB, regS, actS; two−component system, sensor histidine kinase RegB [E

K03568 tldD; TldD protein

K12941 abgB; aminobenzoyl−glutamate utilization protein B

K01280 TPP2; tripeptidyl−peptidase II [EC:3.4.14.10]

K21224 kedN5; radical SAM C−methyltransferase

Expres

K17215 K17215; inositol transport system ATP−binding protein

paR, rhiR; LuxR family transcriptional regulator, quorum−sensing systen

K06113 abnA; arabinan endo−1,5−alpha−L−arabinosidase [EC:3.2.1.99

K03473 pdxB; erythronate−4−phosphate dehydrogenase [EC:1.1.1.290

K03563 csrA; carbon storage regulator

K07131 K07131; uncharacterized protein

K11244 WSC; cell wall integrity and stress response component

00684 aat; leucyl/phenylalanyl−tRNA−−−protein transferase [EC:2.3.2.6

K03831 mogA; molybdopterin adenylyltransferase [EC:2.7.7.75]

K18235 vgb; virginiamycin B lyase [EC:4.2.99.−]

K17762 rsbS; rsbT antagonist protein RsbS

K05880 dhaR; transcriptional activator for dhaKLM operon

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te 3−monooxygenase (FAD) / 4−hydroxyphenylacetate 3−monooxygena

21063; 5−amino−6−(5−phospho−D−ribitylamino)uracil phosphatase [EC

K09763 K09763; uncharacterized protein

K09966 K09966; uncharacterized protein

K04099 desB, galA; gallate dioxygenase [EC:1.13.11.57]

hD−xylL; dihydroxycyclohexadiene carboxylate dehydrogenase [EC:1.3.1

K14658 nodA; nodulation protein A [EC:2.3.1.−]

K02862 ctaG; putative membrane protein

rlmE, rrmJ, ftsJ; 23S rRNA (uridine2552−2'−O)−methyltransferase [EC:2

K17321 glpV; glycerol transport system substrate−binding protein

K11918 lip3; type VI secretion system protein

16871 POP2; 4−aminobutyrate−−−pyruvate transaminase [EC:2.6.1.96

Expression

vo−component system, OmpR family, copper resistance phosphate regu

K02569 napC; NapC/NirT family cytochrome c

K01667 tnaA; tryptophanase [EC:4.1.99.1]

noyl−CoA hydratase/3−hydroxyacyl−CoA dehydrogenase [EC:4.2.1.17 1

K03787 surE; 5'/3'−nucleotidase [EC:3.1.3.5 3.1.3.6]

3 ABC.FEV.A; iron complex transport system ATP−binding protein [EC:7

K13992 pufC; photosynthetic reaction center cytochrome c subunit

K18786 E2.4.1.321; cellobionic acid phosphorylase [EC:2.4.1.321]

K17811 K17811; halimadienyl−diphosphate synthase [EC:5.5.1.16]

J, cobT; nicotinate−nucleotide−−dimethylbenzimidazole phosphoribosylt

K12026 TRIM56; tripartite motif−containing protein 56 [EC:2.3.2.27]

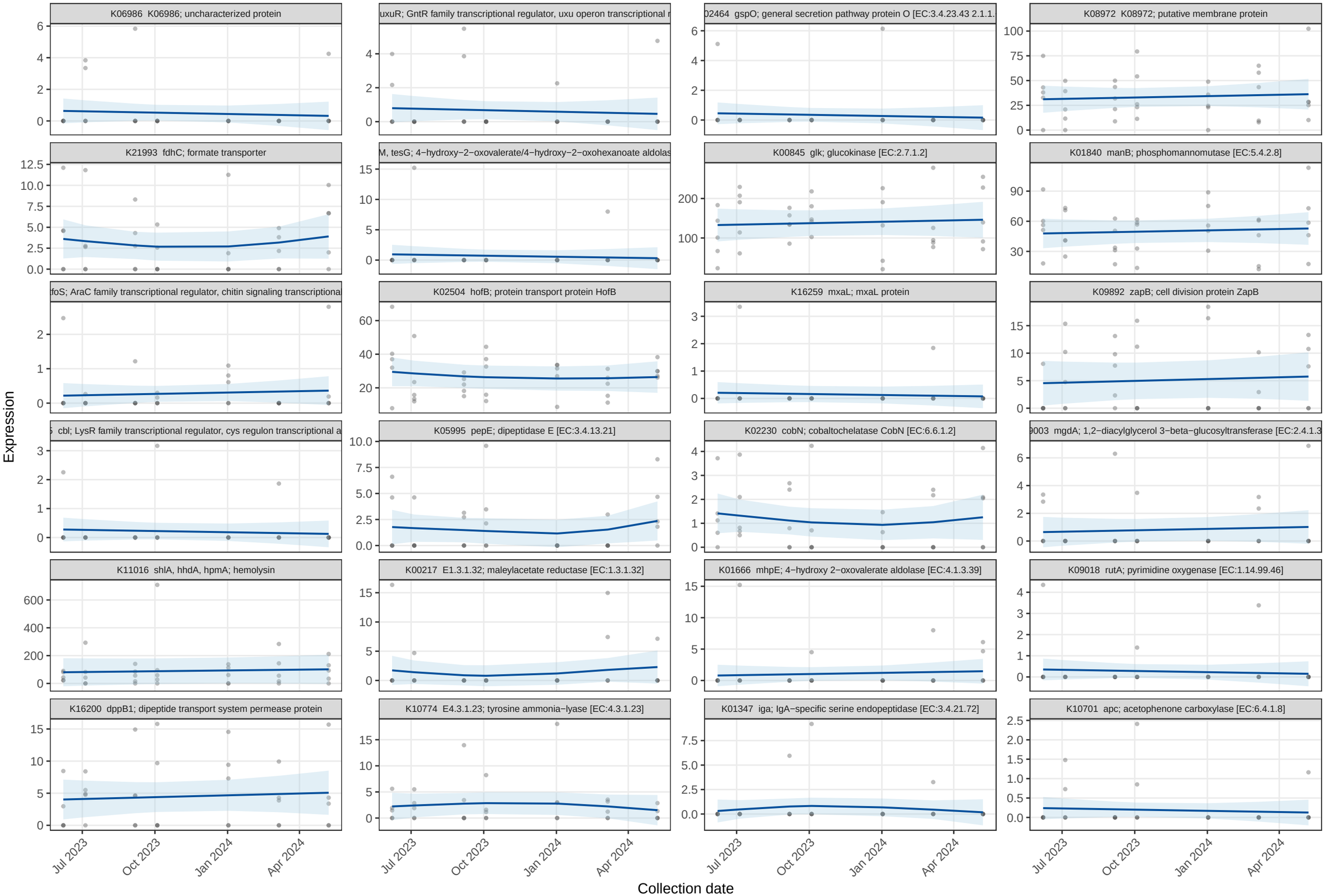
K01654 neuB, nnaB; N−acetylneuraminate synthase [EC:2.5.1.56]

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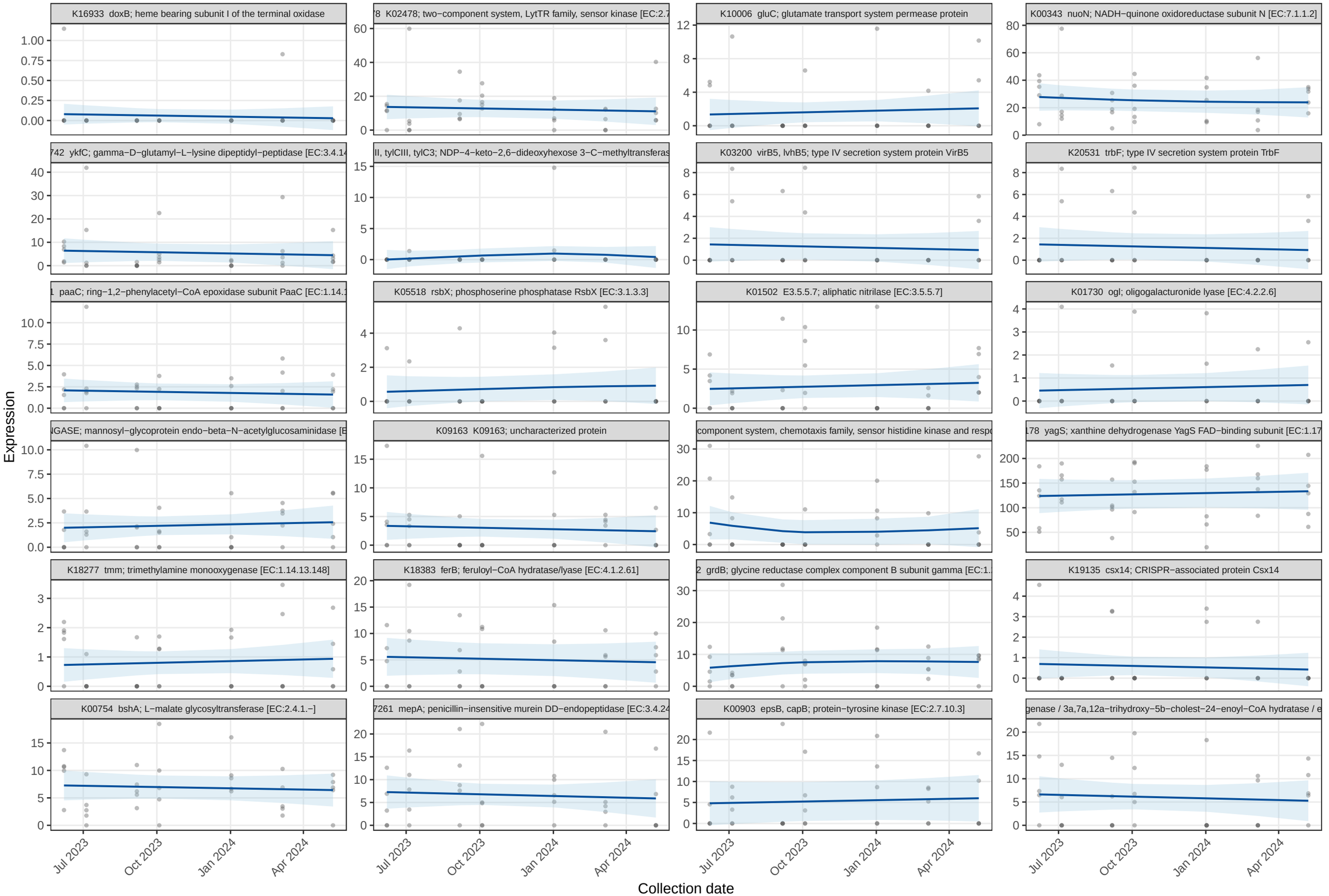
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All remaining BRITE classes after exclusions



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K16264 czcD, zitB; cobalt−zinc−cadmium efflux system protein

13589 cpdR; two−component system, cell cycle response regulator Cpd

07235 tusD, dsrE; tRNA 2−thiouridine synthesizing protein D [EC:2.8.1.

K07266 kpsC, lipA; capsular polysaccharide export protein

K03117 tatB; sec−independent protein translocase protein TatB

486 hilA; transcriptional regulator HilA, main transcriptional regulator of

K13875 araC; L−arabonate dehydrase [EC:4.2.1.25]

K14414 rtcR; transcriptional regulatory protein RtcR

K20087 vioB; violacein biosynthesis protein VioB

K08322 rspB; L−gulonate 5−dehydrogenase [EC:1.1.1.380]

K11913 fha1; type VI secretion system protein

K18425 dhaa, bhcC; 3−hydroxy−D−aspartate aldolase [EC:4.1.3.41]

Expression

K17327 bxlF; xylobiose transport system permease protein

tZ, hugZ; heme oxygenase (biliverdin−IX−beta and delta−forming) [EC:

2564 nagB, GNPDA; glucosamine−6−phosphate deaminase [EC:3.5.99

G1, NEP1; rRNA small subunit pseudouridine methyltransferase Nep1 [E

K09958 K09958; uncharacterized protein

rosA; 8−amino−8−demethylriboflavin N,N−dimethyltransferase [EC:2.1

K16034 asm7; 20−O−methyltransferase

SUGCT; succinate−−−hydroxymethylglutarate CoA−transferase [EC:2

K13661 gumC; GumC protein

25 entS; MFS transporter, ENTS family, enterobactin (siderophore) exp

K03483 mtlR; mannitol operon transcriptional activator

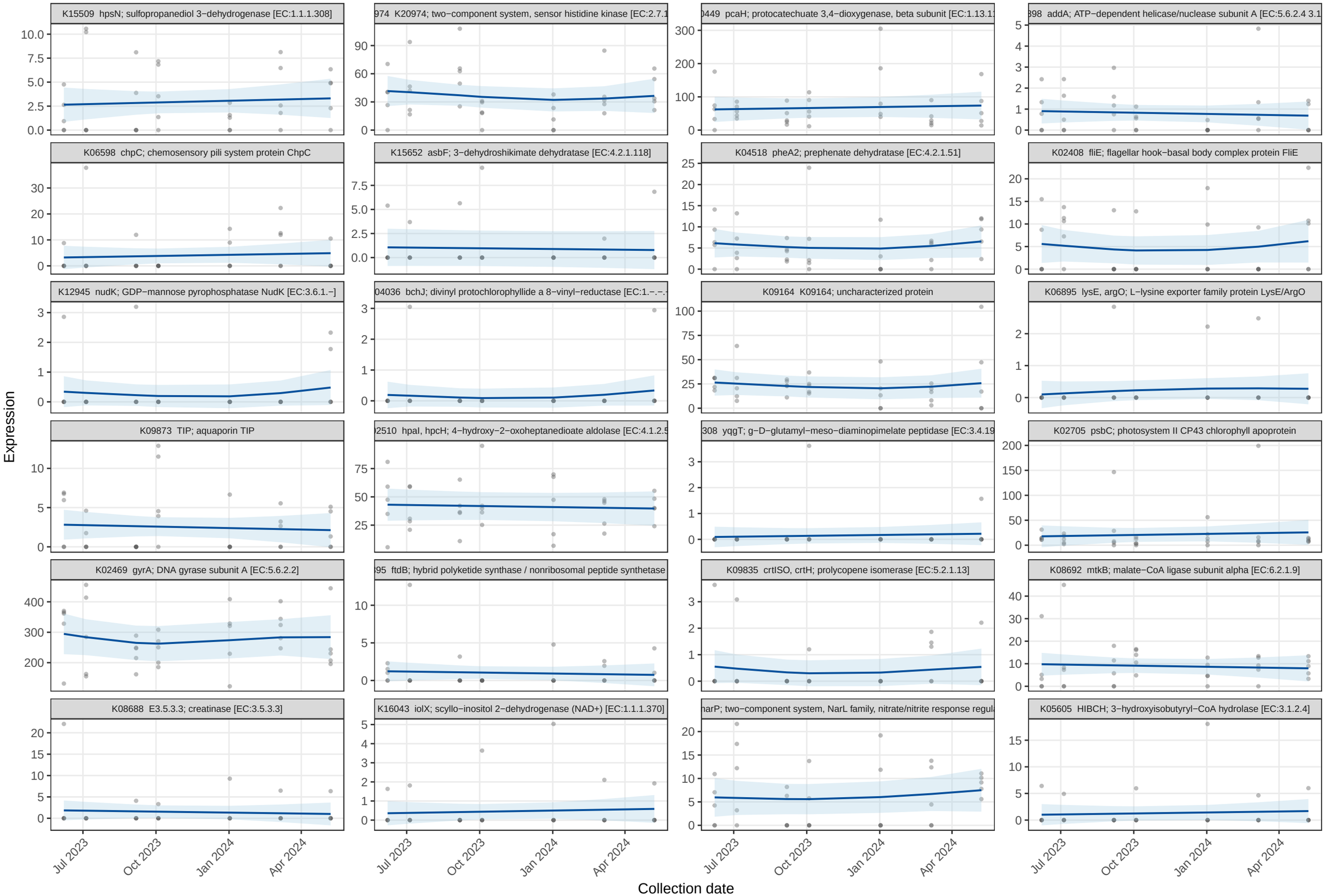
K04096 smf; DNA processing protein

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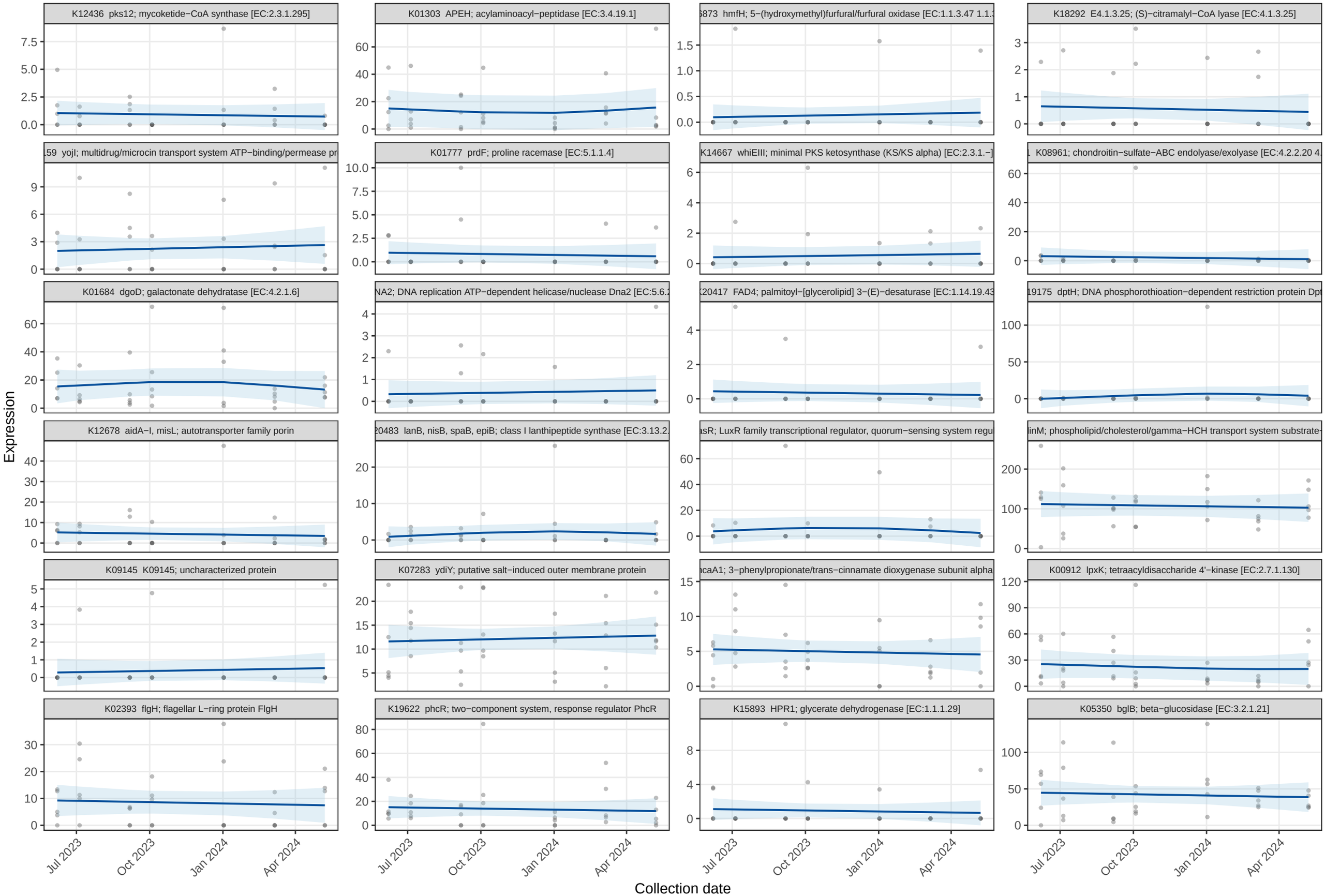
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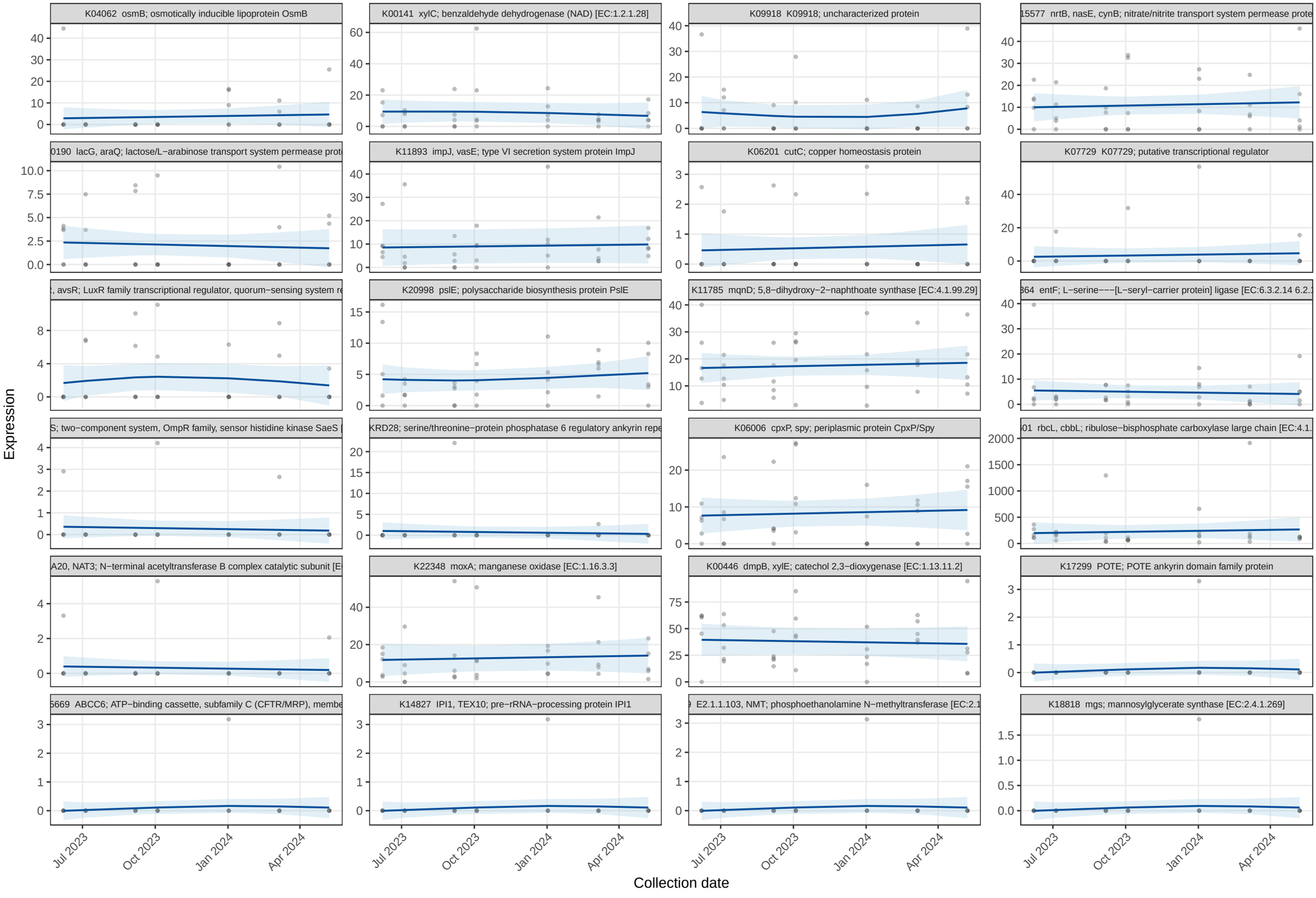
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All remaining BRITE classes after exclusions



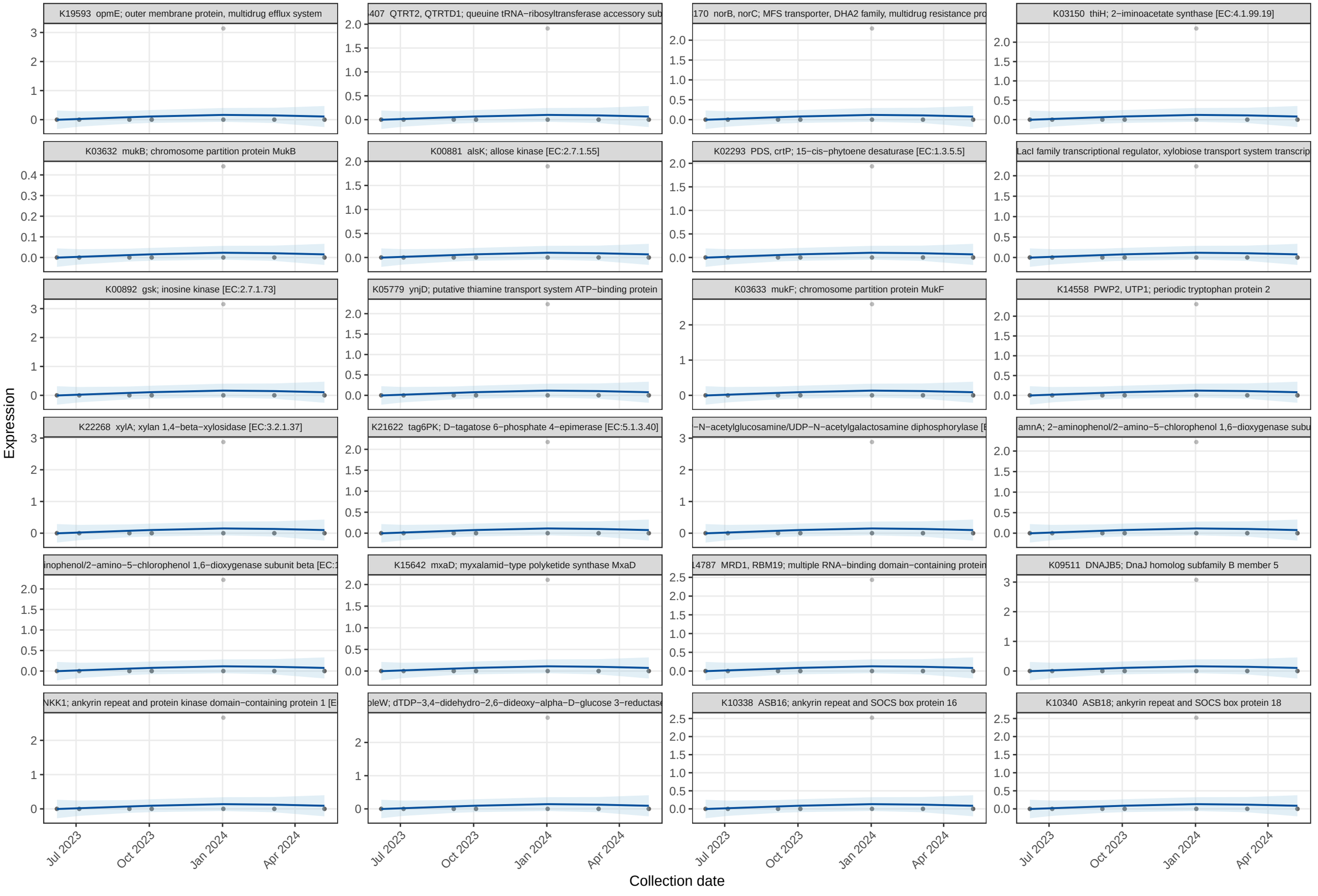
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All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



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02 gatDH; galactitol 2−dehydrogenase (L−tagatose−forming) [EC:1.1.1

N'−diacetylbacillosaminyl−diphospho−undecaprenol 4−alpha−N−acetyl

K19981 prnB; monodechloroaminopyrrolnitrin synthase

K08605 gelE; coccolysin [EC:3.4.24.30]

K21428 pigD; 3−acetyloctanal synthase [EC:2.2.1.12]

K00932 tdcD; propionate kinase [EC:2.7.2.15]

K21779 pigB; 2−methyl−3−n−amyl−dihydropyrrolel dehydrogenase

K17582 MKI67; antigen KI−67

5531 MNN10; mannan polymerase II complex MNN10 subunit [EC:2.4.1

K20552 ler; transcriptional regulator of LEE operons

6 gumD; undecaprenyl−phosphate glucose phosphotransferase [EC:2.7

03 METTL21B, EEF1AKMT3; EEF1A lysine methyltransferase 3 [EC:2.

Expression

79 DDX18, HAS1; ATP−dependent RNA helicase DDX18/HAS1 [EC:5.

K06795 BCAN; brevican

K10811 E2.5.1.2; thiamine pyridinylase [EC:2.5.1.2]

10798 PARP3_4; protein mono−ADP−ribosyltransferase 3/4 [EC:2.4.2.

K12451 UER1; 3,5−epimerase/4−reductase [EC:5.1.3.− 1.1.1.−]

K06794 NCAN, CSPG3; neurocan core protein

K16029 asmA; ansamitocin polyketide synthase A

K21741 rclC; reactive chlorine resistance protein C

K03400 luxC; long−chain−fatty−acyl−CoA reductase [EC:1.2.1.50]

mabA; meromycolic acid 3−oxoacyl−[acyl−carrier protein] reductase [EC

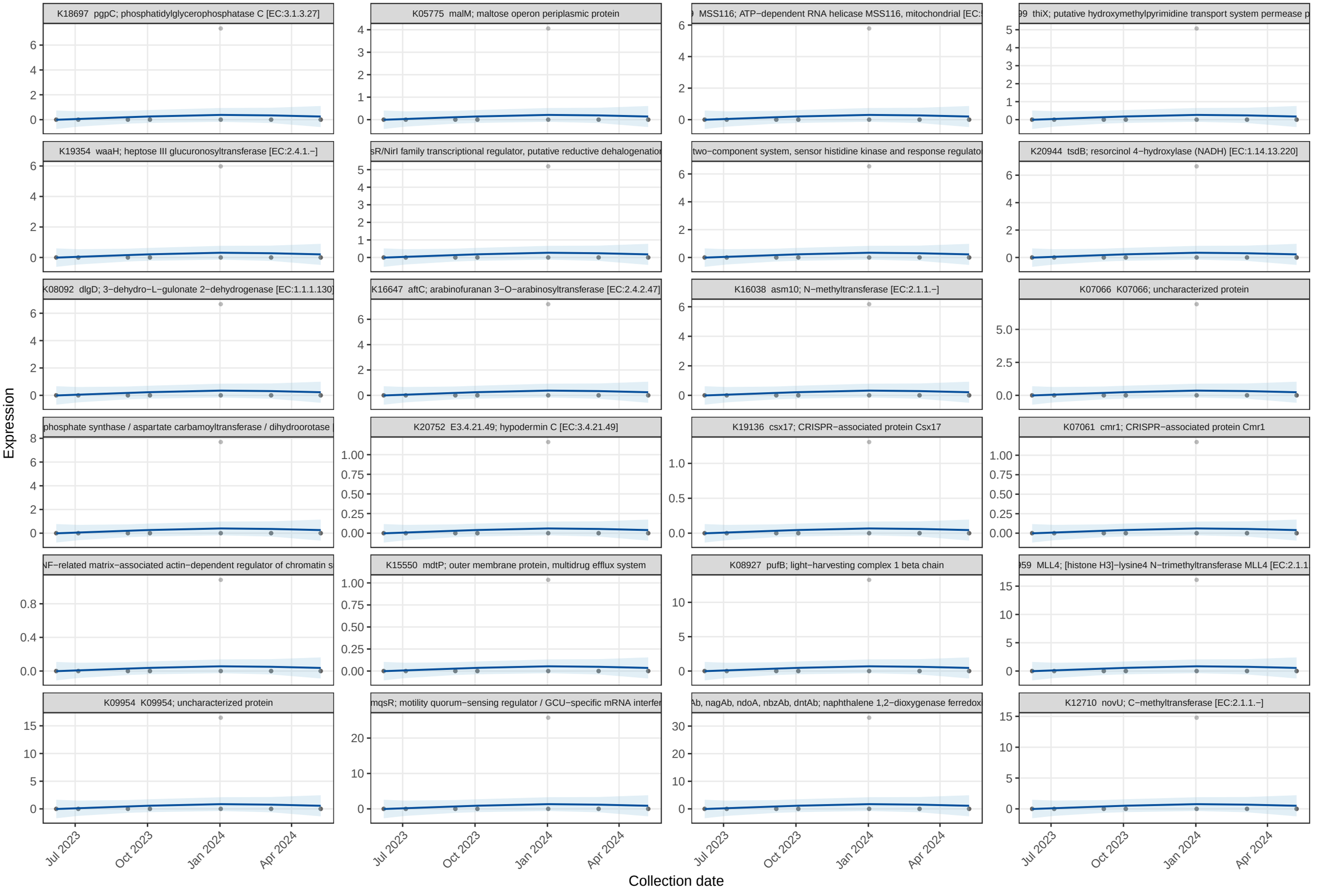
K12944 nudI; nucleoside triphosphatase [EC:3.6.1.−]

ciR; LuxR family transcriptional regulator, quorum−sensing system regu

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All remaining BRITE classes after exclusions



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K08982 K08982; putative membrane protein

C3, TFC4; general transcription factor 3C polypeptide 3 (transcription fac

LysR family transcriptional regulator, glycine cleavage system transcrip

K01023 assT; arylsulfate sulfotransferase [EC:2.8.2.22]

K15898 pseI, neuB3; pseudaminic acid synthase [EC:2.5.1.97]

K09922 K09922; uncharacterized protein

K16955 mtsB; methylated−thiol−−corrinoid protein

K10068 SFTPD, SFTP4; pulmonary surfactant−associated protein D

K08723 yjjG; pyrimidine 5'−nucleotidase [EC:3.1.3.−]

K05774 phnN; ribose 1,5−bisphosphokinase [EC:2.7.4.23]

; phthiocerol/phenolphthiocerol synthesis type−I polyketide synthase E [

K21297 ctcQ; flavin reductase (NADH) [EC:1.5.1.36]

K03105 SRP19; signal recognition particle subunit SRP19

K21802 vdh; vanillin dehydrogenase [EC:1.2.1.67]

200 ngcE; N−acetylglucosamine transport system substrate−binding pro

K02389 flgD; flagellar basal−body rod modification protein FlgD

K06980 ygfZ; tRNA−modifying protein YgfZ

K03219 yscC, sctC, ssaC; type III secretion protein C

K09956 K09956; uncharacterized protein

K16130 mcyA; microcystin synthetase protein McyA

D−Ala−D−Ala carboxypeptidase/endopeptidase (penicillin−binding prote

K07179 RIOK2; RIO kinase 2 [EC:2.7.11.1]

K07586 ygaC; uncharacterized protein

K07402 xdhC; xanthine dehydrogenase accessory factor

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Expres

sion

All remaining BRITE classes after exclusions

; LytTR family transcriptional regulator, CO−responsive transcriptional re

K11601 mntC; manganese transport system substrate−binding protein

K16564 exoU; succinoglycan biosynthesis protein ExoU [EC:2.4.−.−]

ristoyl] N−acetylglucosamine deacetylase / 3−hydroxyacyl−[acyl−carrie

K07081 K07081; putative periplasmic solute−binding protein

K11066 amiD; N−acetylmuramoyl−L−alanine amidase [EC:3.5.1.28]

0497 CYP124; methyl−branched lipid omega−hydroxylase [EC:1.14.15.

puuC, aldH; 4−(gamma−glutamylamino)butanal dehydrogenase [EC:1

lyU; ArsR family transcriptional regulator, virulence genes transcriptiona

K10201 ngcF; N−acetylglucosamine transport system permeate protei

; serine−type D−Ala−D−Ala endopeptidase (penicillin−binding protein 7

10−diseco−3−hydroxy−5,9,17−trioxoandrosta−1(10),2−diene−4−oate hy

K03786 aroQ, qutE; 3−dehydroquinate dehydratase II [EC:4.2.1.10]

K06878 K06878; tRNA−binding protein

K13049 PM20D1; carboxypeptidase PM20D1 [EC:3.4.17.−]

08151 tetA; MFS transporter, DHA1 family, tetracycline resistance prote

K00543 ASMT; acetylserotonin O−methyltransferase [EC:2.1.1.4]

K02777 crr; sugar PTS system EIIA component [EC:2.7.1.−]

73 acd; glutaryl−CoA dehydrogenase (non−decarboxylating) [EC:1.3.9

dsbE; cytochrome c biogenesis protein CcmG, thiol:disulfide interchang

K00316 spdH; spermidine dehydrogenase [EC:1.5.99.6]

spo0F; two−component system, response regulator, stage 0 sporulation

K07720 yesN; two−component system, response regulator YesN

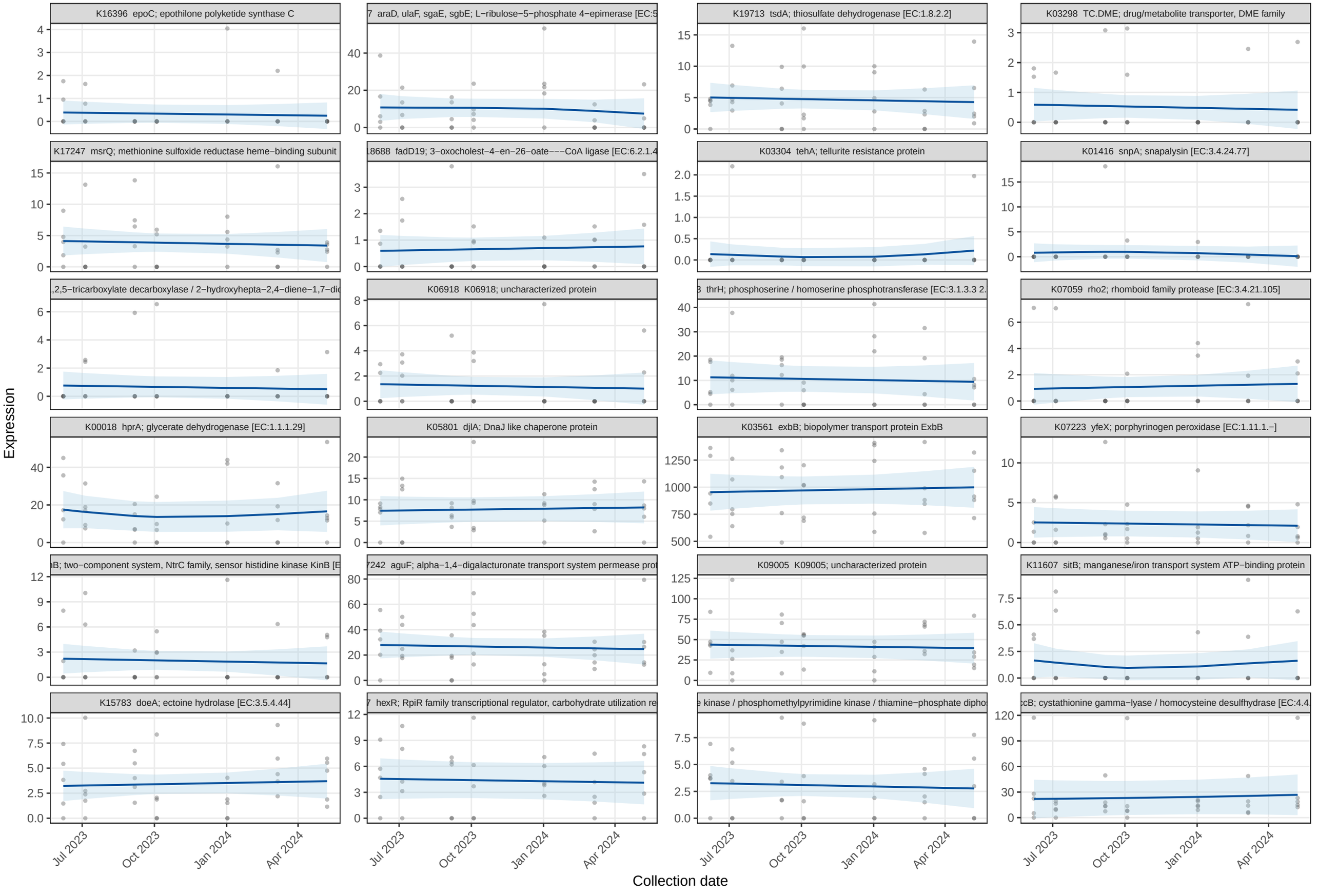
K01210 E3.2.1.58; glucan 1,3−beta−glucosidase [EC:3.2.1.58]

Expression

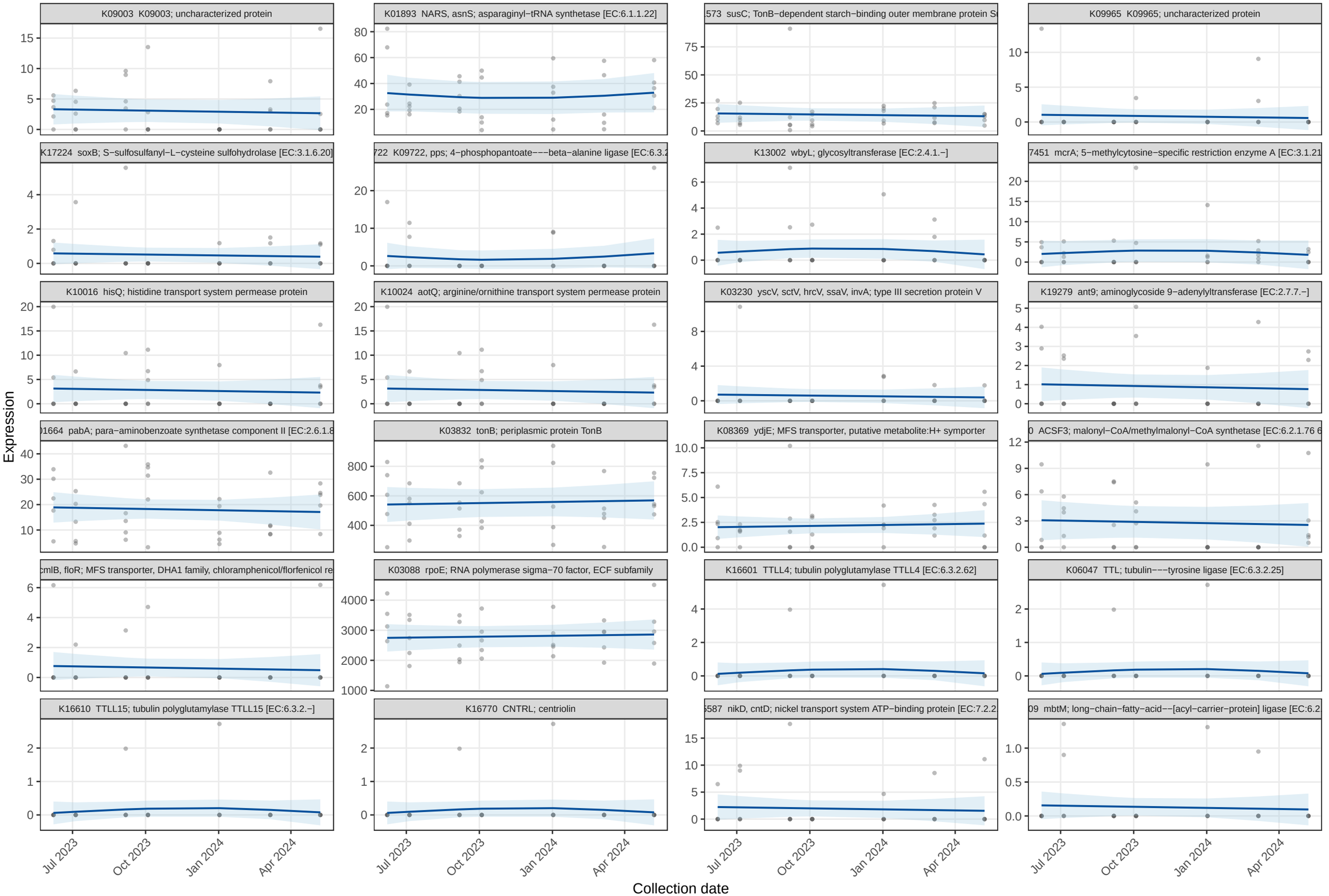
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All remaining BRITE classes after exclusions

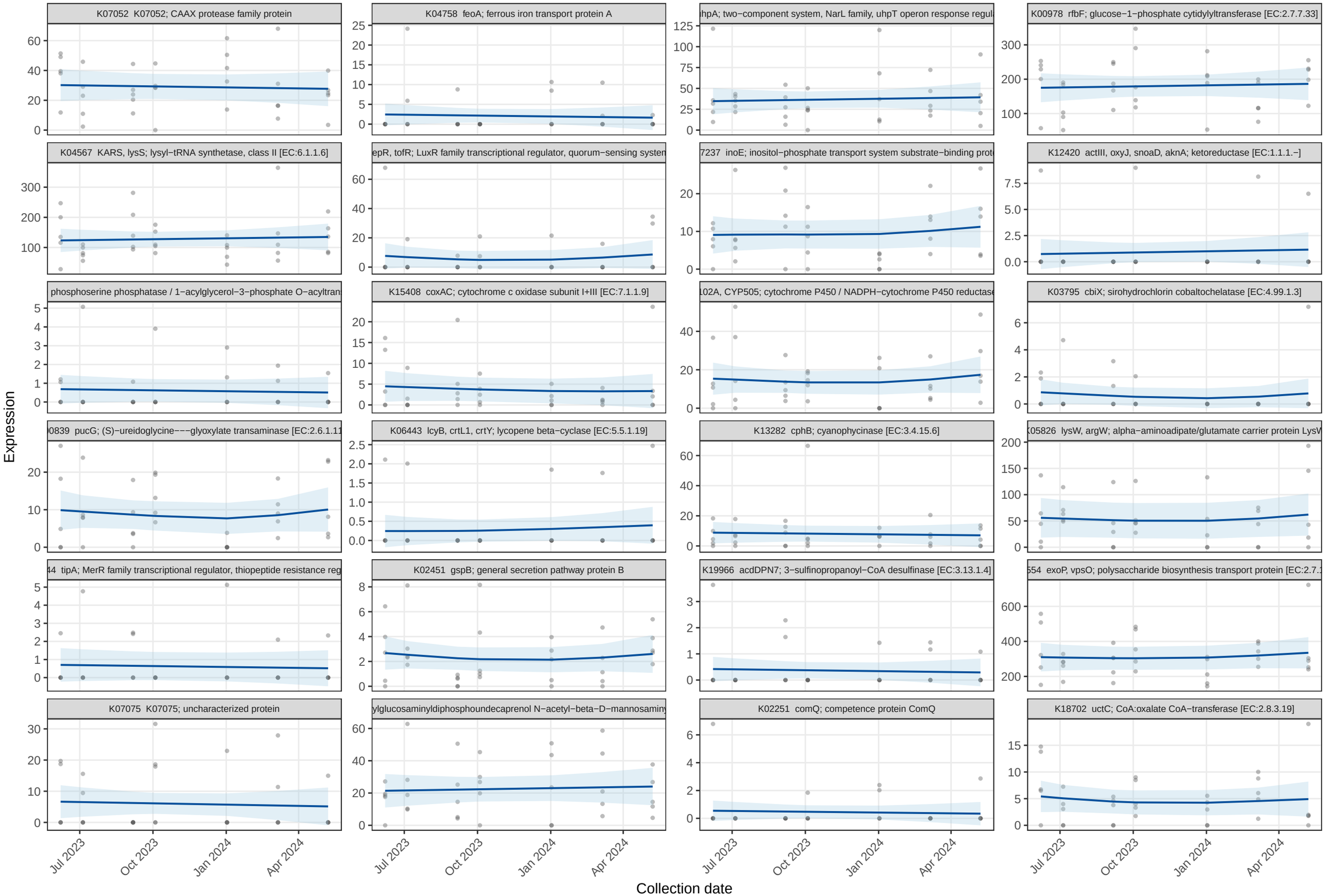


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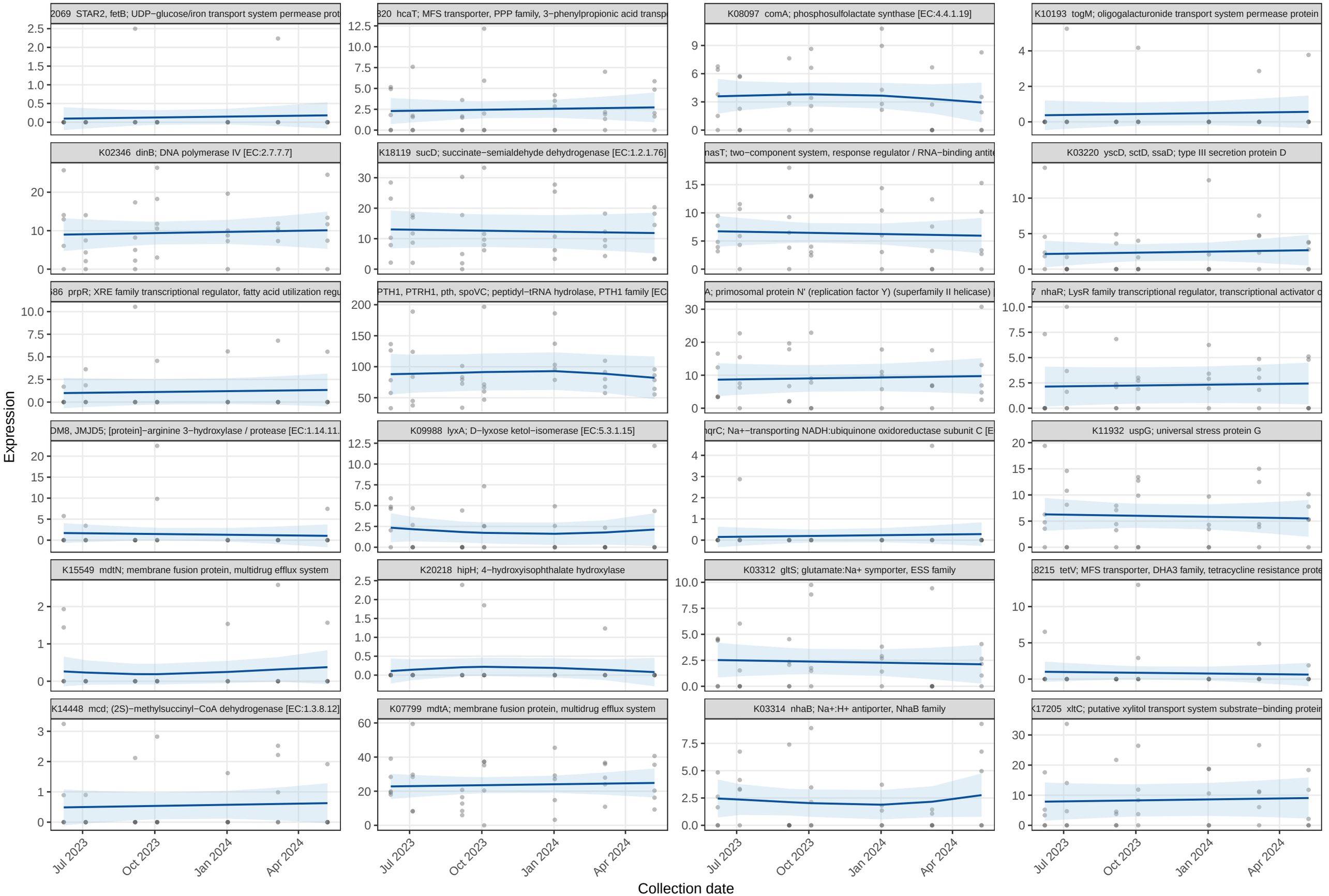
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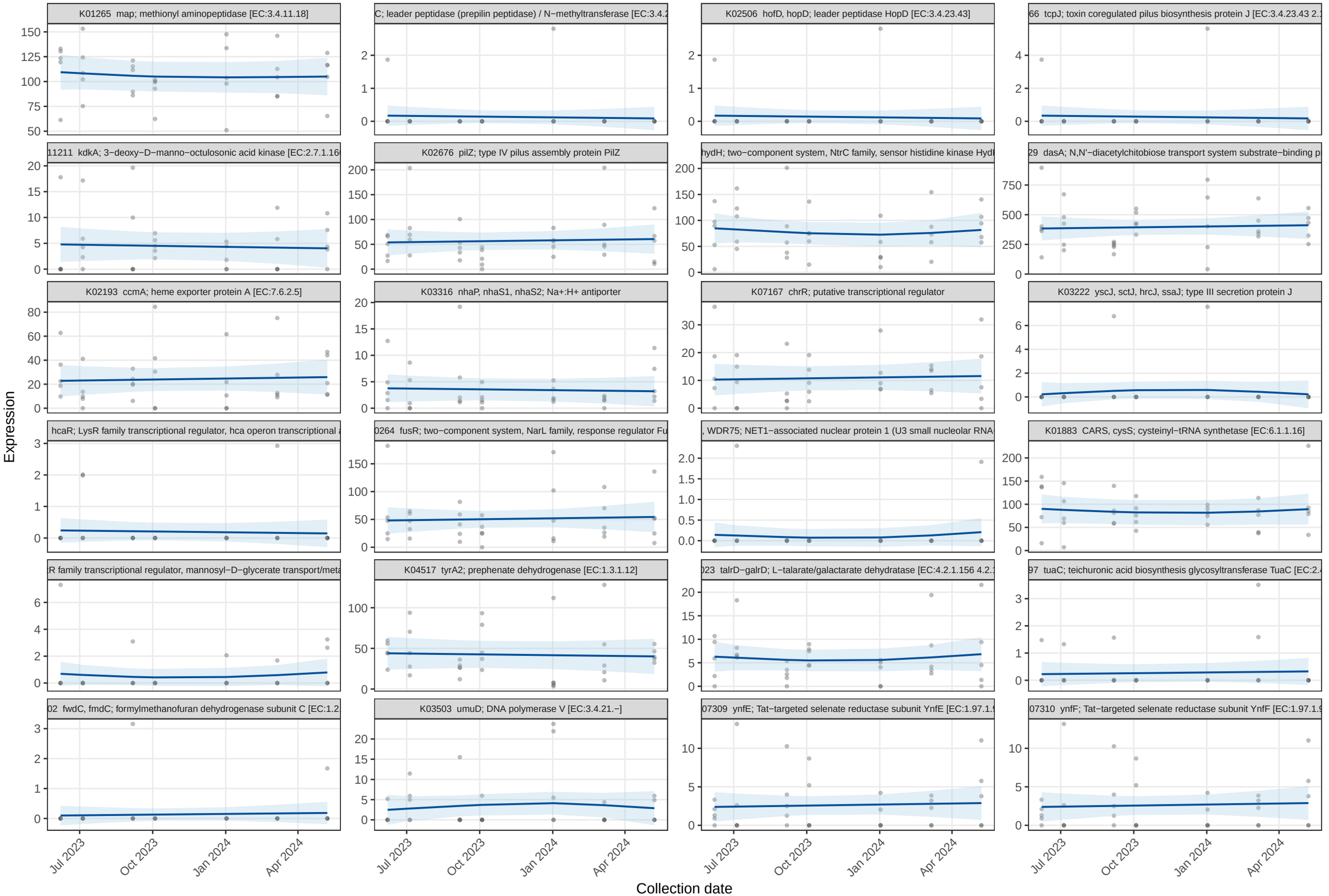
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All remaining BRITE classes after exclusions

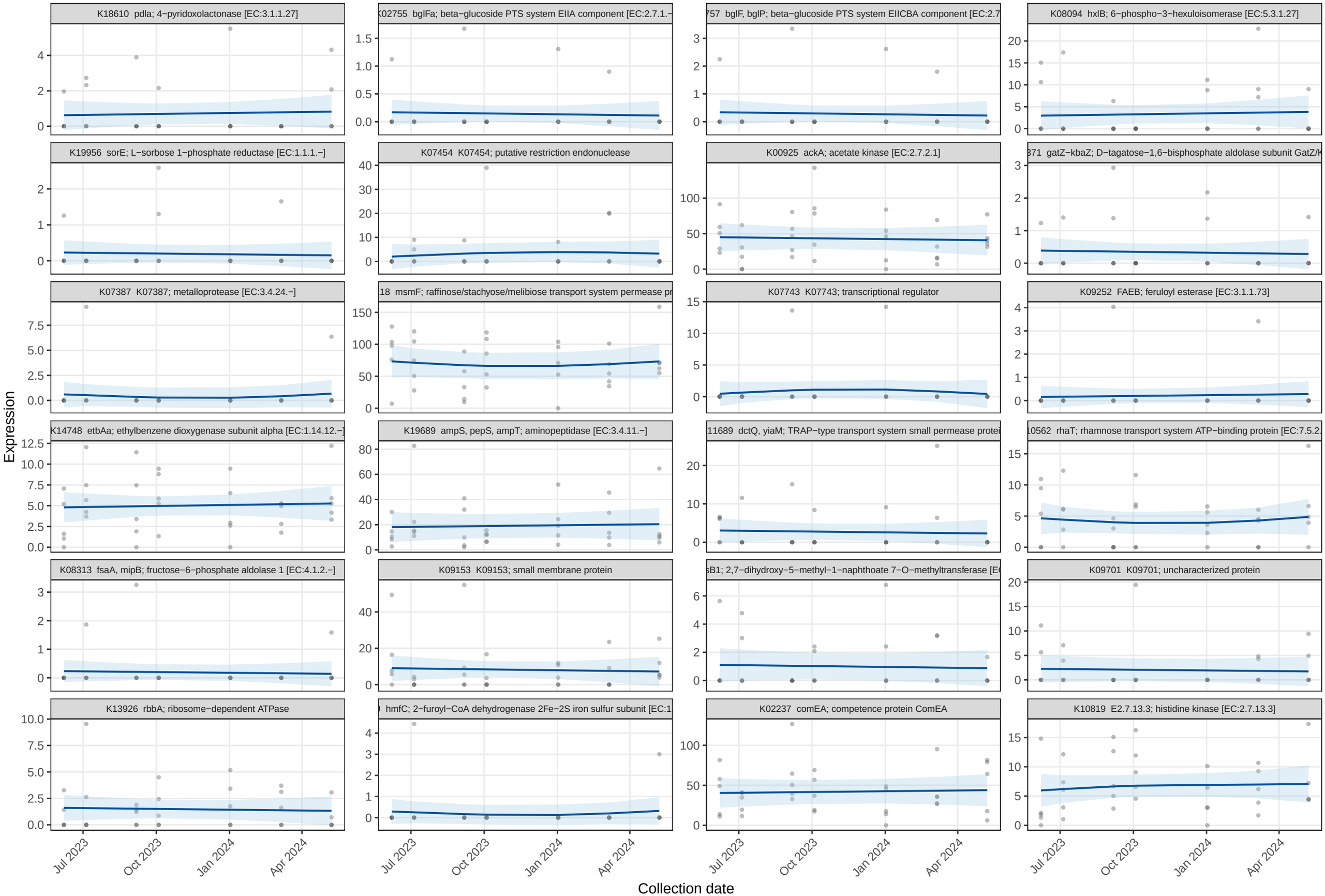


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All remaining BRITE classes after exclusions

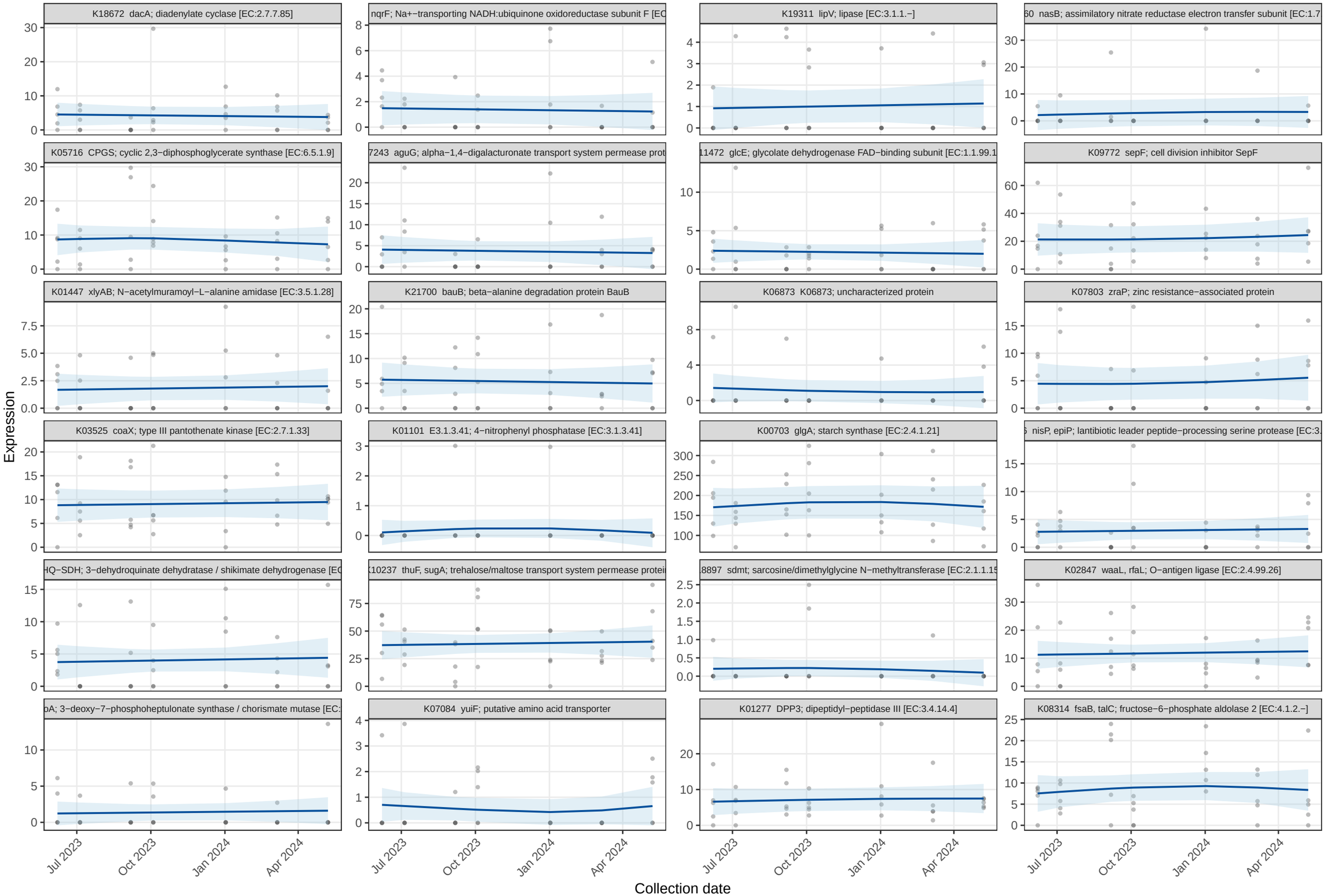


All remaining BRITE classes after exclusions



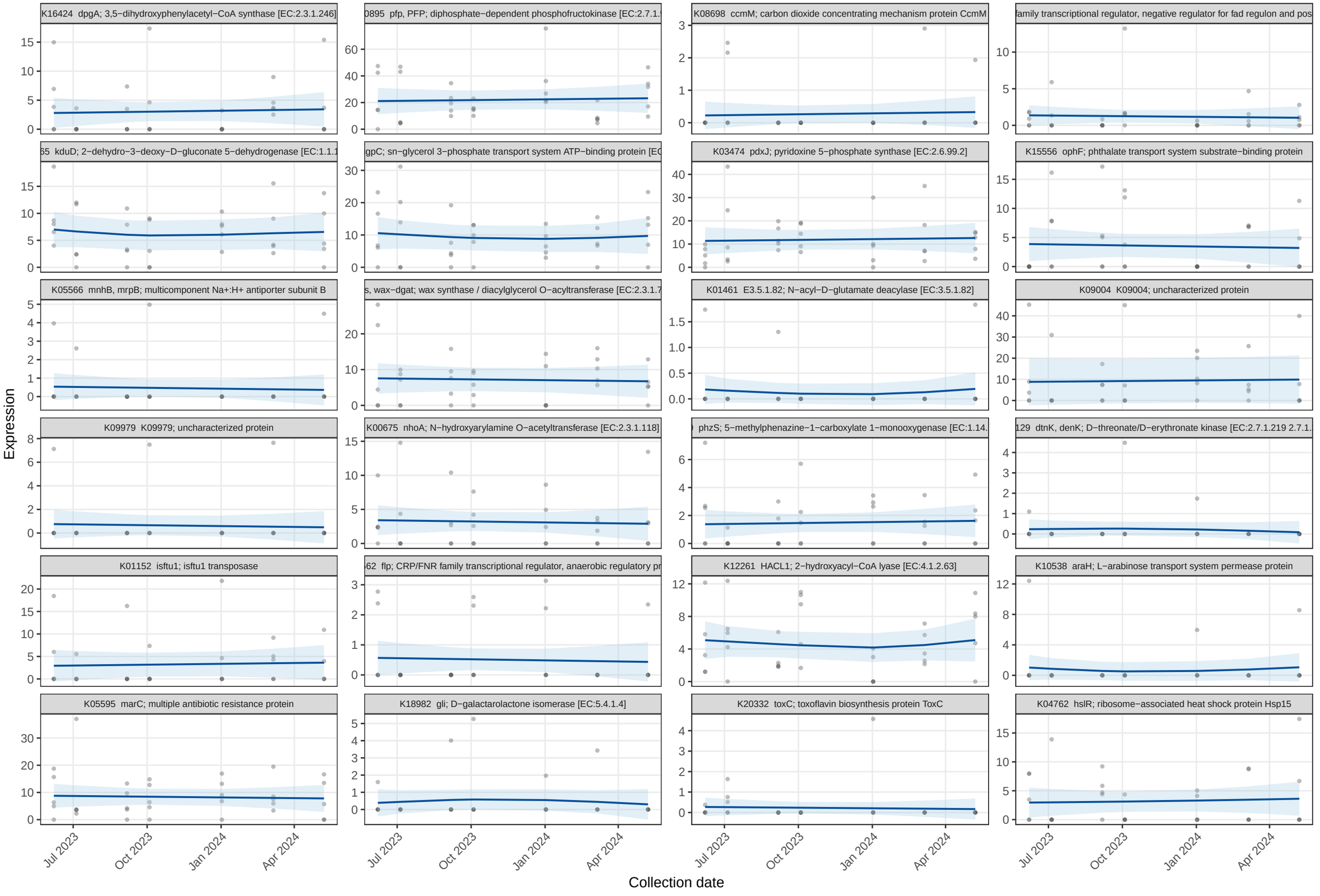
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All remaining BRITE classes after exclusions



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K09143 K09143; uncharacterized protein

K00076 hdhA; 7−alpha−hydroxysteroid dehydrogenase [EC:1.1.1.159]

K18775 lf2; levanbiose−producing levanase [EC:3.2.1.64]

K02507 hofQ; protein transport protein HofQ

K16255 mxaG; cytochrome c−L

ethanol/glycine betaine−specific corrinoid protein]:coenzyme M methyltra

K16329 psuG; pseudouridylate synthase [EC:4.2.1.70]

K04088 hflK; modulator of FtsH protease HflK

K21218 motY; sodium−type flagellar protein MotY

K09915 K09915; uncharacterized protein

K09934 K09934; uncharacterized protein

K00846 KHK; ketohexokinase [EC:2.7.1.3]

Expression

K10855 acxA; acetone carboxylase, beta subunit [EC:6.4.1.6]

K02636 petC; cytochrome b6−f complex iron−sulfur subunit [EC:7.1.1.6]

hddA; D−glycero−alpha−D−manno−heptose−7−phosphate kinase [EC:2

K05568 mnhD, mrpD; multicomponent Na+:H+ antiporter subunit D

727 czcB, cusB, cnrB; membrane fusion protein, heavy metal efflux sys

K03285 TC.GBP; general bacterial porin, GBP family

K21689 rpfC; resuscitation−promoting factor RpfC

K21690 rpfD; resuscitation−promoting factor RpfD

K02226 cobC, phpB; alpha−ribazole phosphatase [EC:3.1.3.73]

K13889 gsiB; glutathione transport system substrate−binding protein

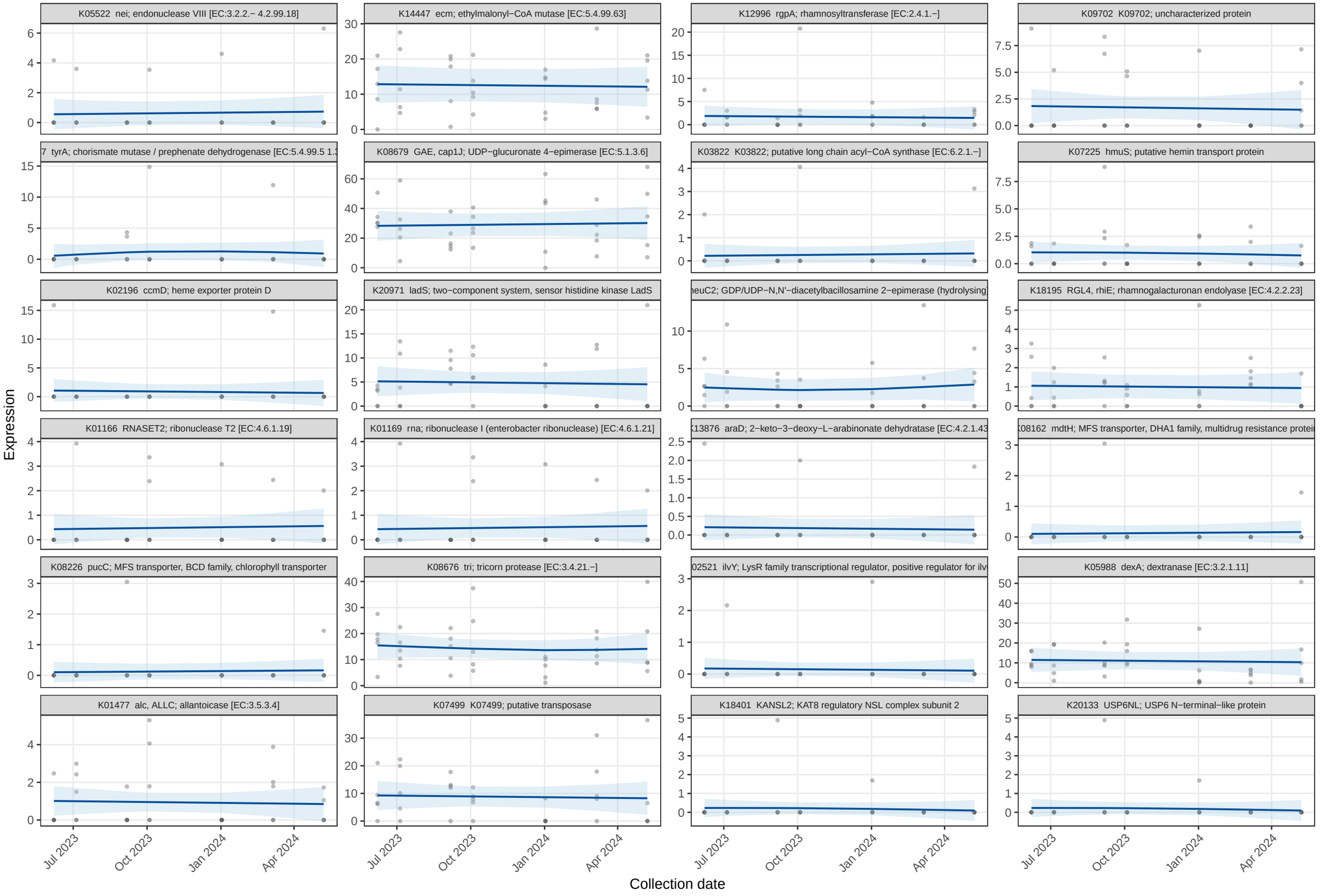
K18652 ntdC; glucose−6−phosphate 3−dehydrogenase [EC:1.1.1.361]

K08357 ttrA; tetrathionate reductase subunit A

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K08153 blt; MFS transporter, DHA1 family, multidrug resistance protein

K21256 calO3; flavin−dependent halogenase

K14161 imuB; protein ImuB

K02042 phnE; phosphonate transport system permease protein

17331 dasC; N,N'−diacetylchitobiose transport system permease prote

5538 MANEA; glycoprotein endo−alpha−1,2−mannosidase [EC:3.2.1.1

K14940 cofF; gamma−F420−2:alpha−L−glutamate ligase [EC:6.3.2.32]

K19997 gnu; GlcNAc−P−P−Und epimerase [EC:5.1.3.26]

478 ACADSB; short−chain 2−methylacyl−CoA dehydrogenase [EC:1.3.

K15020 K15020; acryloyl−coenzyme A reductase [EC:1.3.1.84]

K06921 K06921; uncharacterized protein

12203 dotB, traJ; defect in organelle trafficking protein DotB [EC:7.2.4.8

K21429 impA; immunomodulating metalloprotease

K13572 pafB; proteasome accessory factor B

K01732 E4.2.2.10; pectin lyase [EC:4.2.2.10]

K19540 frlA; fructoselysine transporter

K01337 E3.4.21.50; lysyl endopeptidase [EC:3.4.21.50]

14987 fixJ; two−component system, LuxR family, response regulator Fix

K02080 agaI; putative deaminase/isomerase [EC:3.5.99.−]

K11144 dnaI; primosomal protein DnaI

K05580 ndhI; NAD(P)H−quinone oxidoreductase subunit I [EC:7.1.1.2]

K13252 ptcA; putrescine carbamoyltransferase [EC:2.1.3.6]

07106 murQ; N−acetylmuramic acid 6−phosphate etherase [EC:4.2.1.12

K06995 K06995; uncharacterized protein

Expression

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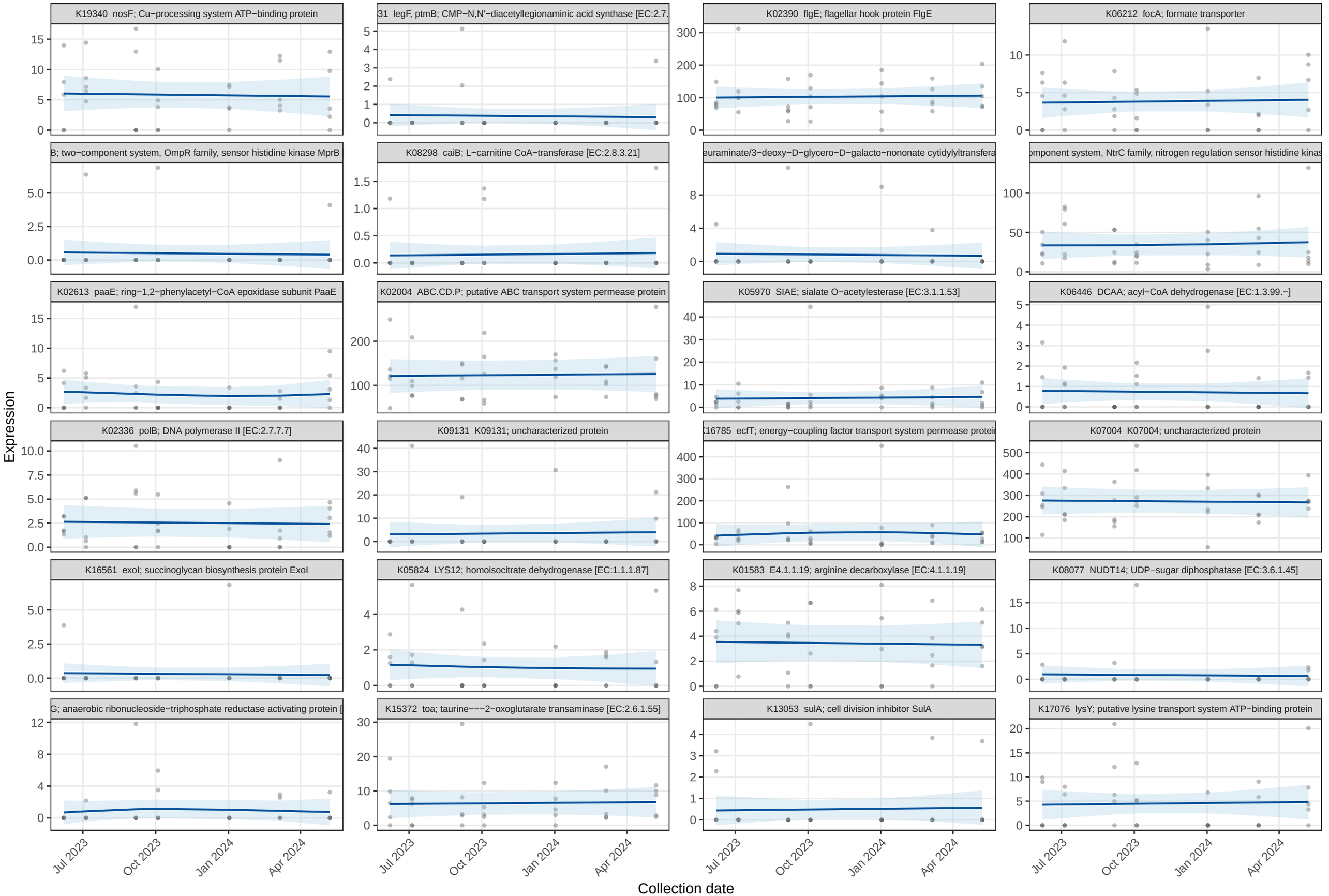
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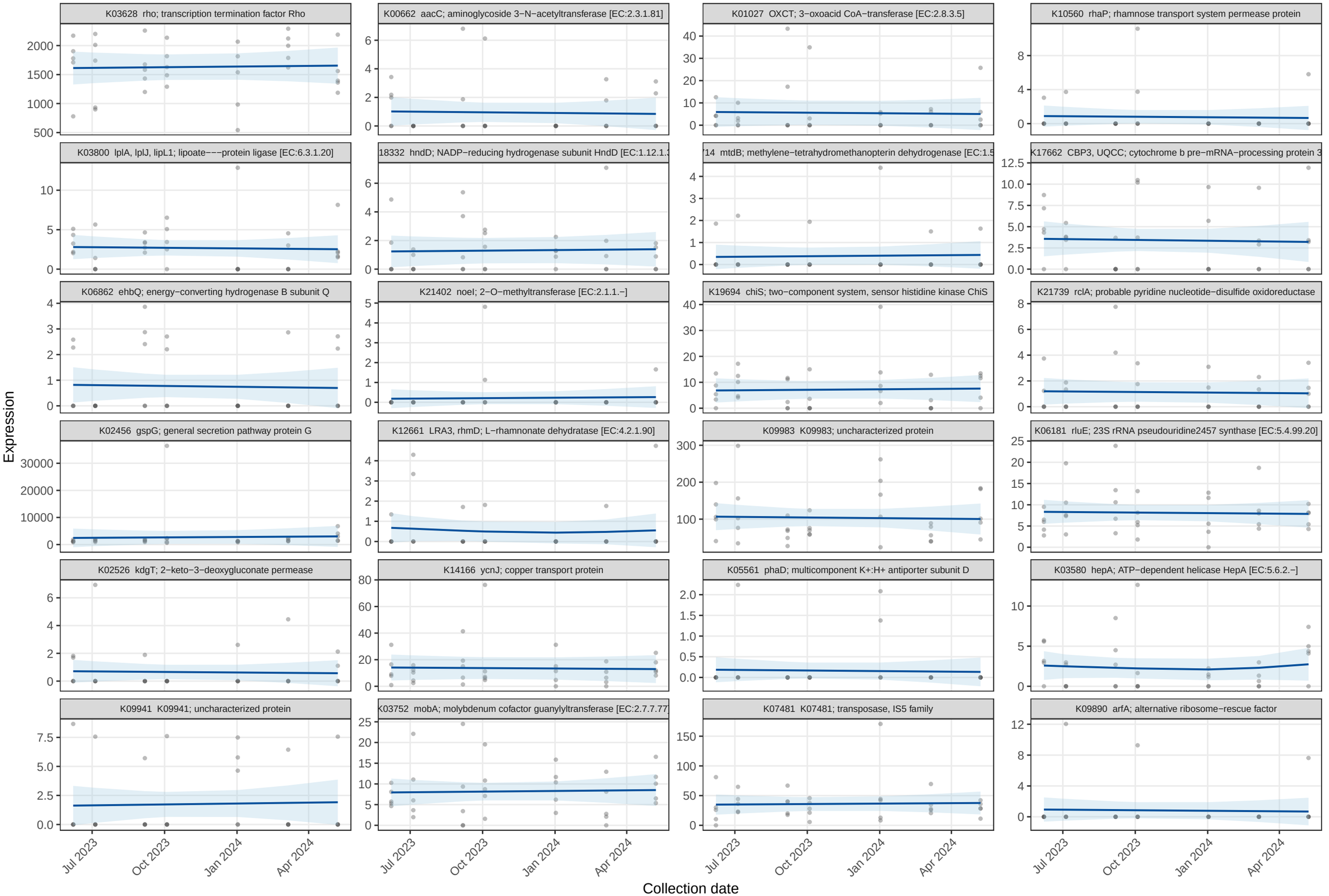
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K11957 natA; neutral amino acid transport system ATP−binding protein

K19125 casD, cse5; CRISPR system Cascade subunit CasD

08 DDX54, DBP10; ATP−dependent RNA helicase DDX54/DBP10 [EC:5.

09 troD, mntD, znuB; manganese/zinc/iron transport system permease p

K08326 ypdF; aminopeptidase [EC:3.4.11.−]

45 ndx1; diadenosine hexaphosphate hydrolase (ATP−forming) [EC:3.6.

0710 GALNT; polypeptide N−acetylgalactosaminyltransferase [EC:2.4.1.

K07473 dinJ; DNA−damage−inducible protein J

K19975 mntC; manganese transport system substrate−binding protein

K19976 mntB; manganese transport system permease protein

K19678 IFT80; intraflagellar transport protein 80

K21624 c3lhypd; cis−L−3−hydroxyproline dehydratase [EC:4.2.1.171]

Expression

K03145 TFIIS; transcription elongation factor S−II

533 kinE; two−component system, sporulation sensor kinase E [EC:2.7.1

K07559 kptA; putative RNA 2'−phosphotransferase [EC:2.7.1.−]

2073 E3.1.2.28; 1,4−dihydroxy−2−naphthoyl−CoA hydrolase [EC:3.1.2.2

002 pqsC; 2−heptyl−4(1H)−quinolone synthase subunit PqsC [EC:2.3.1.1

3087 bphAc, bphA3, bphF; biphenyl 2,3−dioxygenase ferredoxin compon

snoa2, aknC, mtmK; minimal PKS chain−length factor (CLF/KS beta) [EC

K07083 K07083; putative periplasmic solute−binding protein

8360 padA; phenylacetyl−CoA:acceptor oxidoreductase accessory prote

K03476 ulaG; L−ascorbate 6−phosphate lactonase [EC:3.1.1.−]

K12445 K12445; trans enoyl reductase [EC:1.3.1.−]

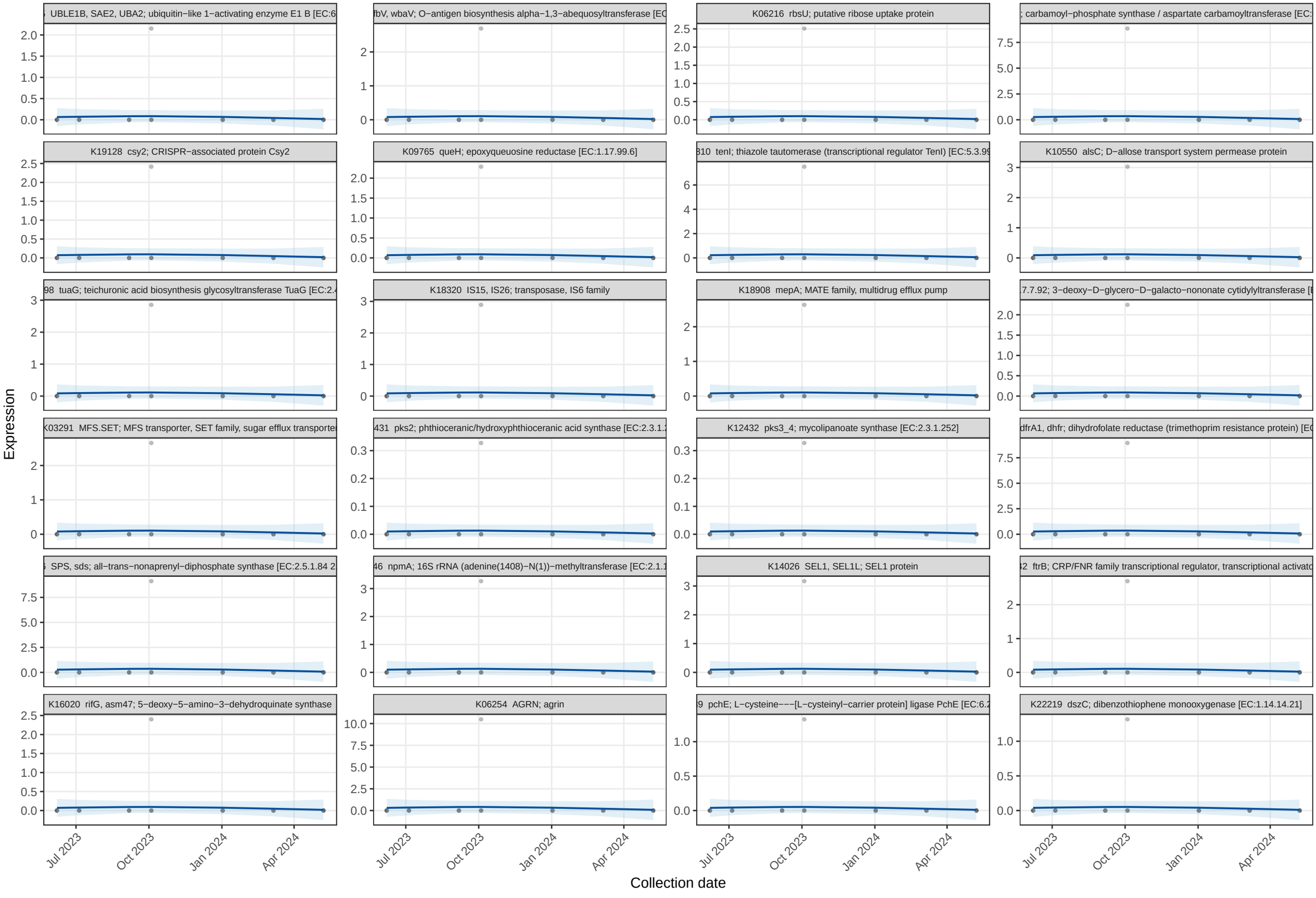
K20830 por; beta−porphyranase [EC:3.2.1.178]

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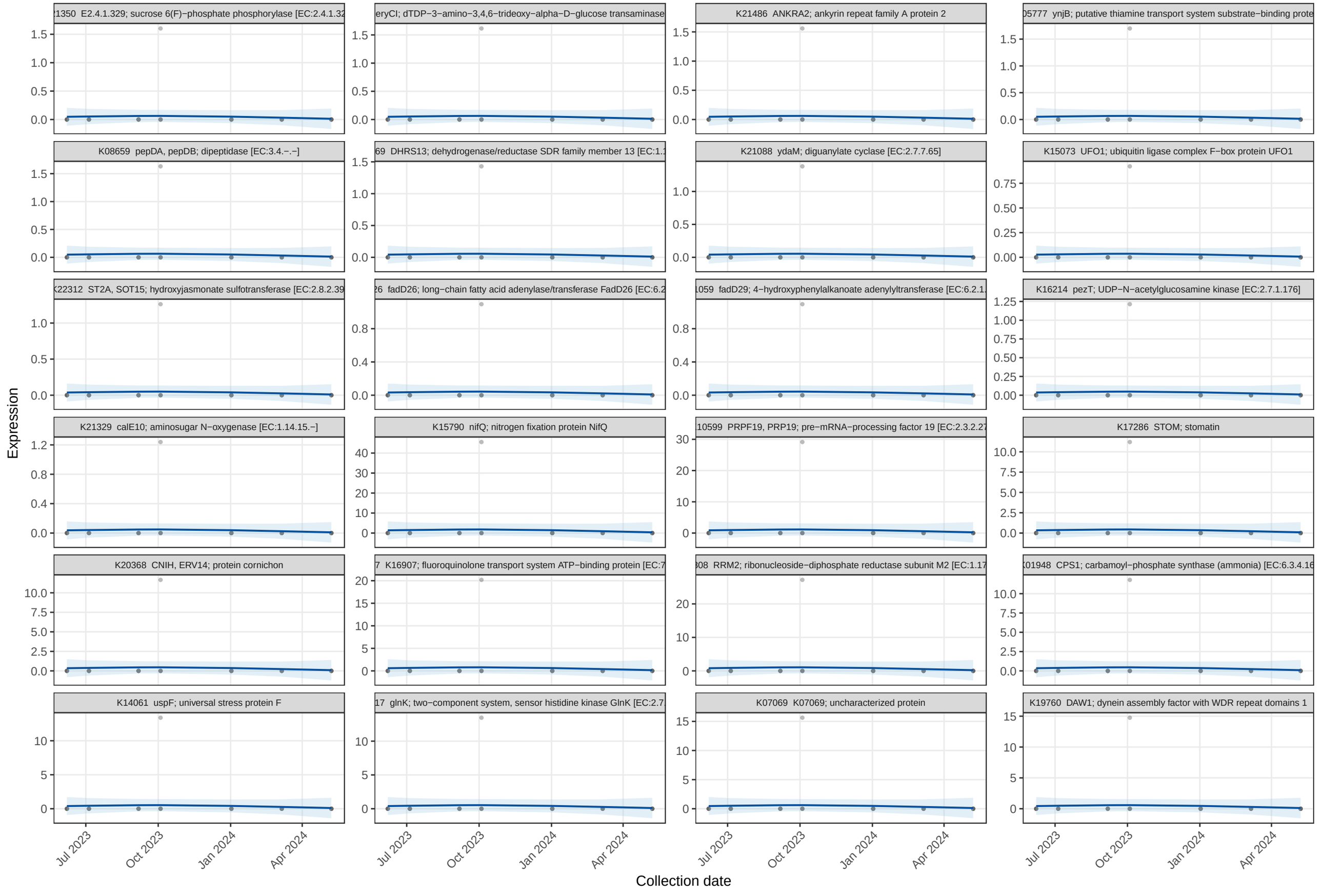
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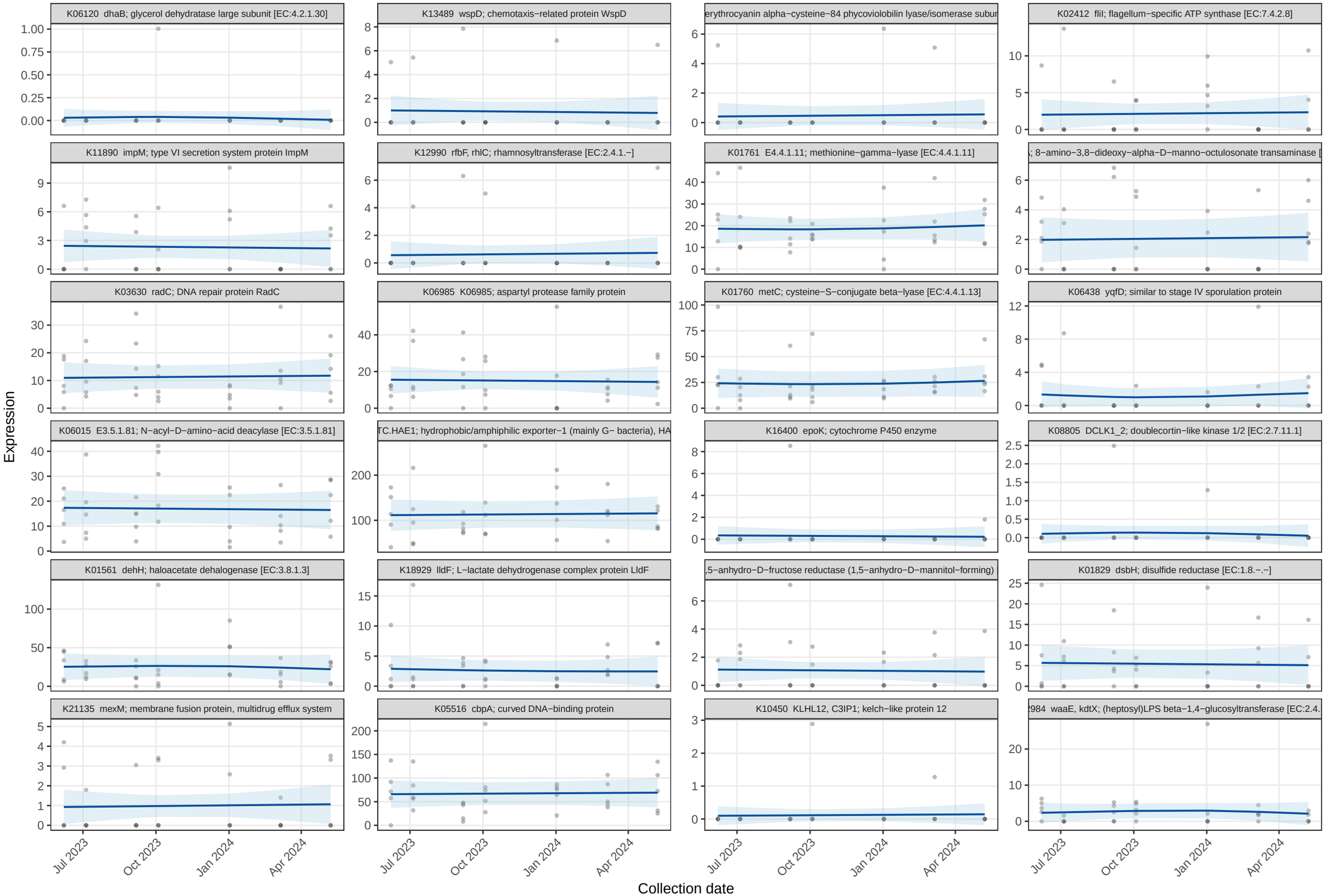


All remaining BRITE classes after exclusions



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K01067 E3.1.2.1, ACH1; acetyl−CoA hydrolase [EC:3.1.2.1]

K07085 K07085; putative transport protein

K03642 rlpA; peptidoglycan lytic transglycosylase [EC:4.2.2.29]

K04768 acuC; acetoin utilization protein AcuC

K09797 K09797; uncharacterized protein

K18316 cpmA, carA; carbapenam−3−carboxylate synthase [EC:6.3.3.6

K06218 relE, stbE; mRNA interferase RelE/StbE

29 TRM5, TRMT5; tRNA (guanine37−N1)−methyltransferase [EC:2.1.1

K00639 kbl, GCAT; glycine C−acetyltransferase [EC:2.3.1.29]

K16555 exoO; succinoglycan biosynthesis protein ExoO [EC:2.4.−.−]

32 ZDHHC13_17, HIP14; palmitoyltransferase ZDHHC13/17 [EC:2.3.1

K15176 CTR9; RNA polymerase−associated protein CTR9

Expression

K10712 ADO; cysteamine dioxygenase [EC:1.13.11.19]

K09978 K09978; uncharacterized protein

K06155 gntT; Gnt−I system high−affinity gluconate transporter

T, gabT; 5−aminovalerate/4−aminobutyrate aminotransferase [EC:2.6.1

K16437 evaC; methylation protein EvaC

K21336 mdpA4; C−methyltransferase [EC:2.1.1.−]

K14333 DHBD; 2,3−dihydroxybenzoate decarboxylase [EC:4.1.1.46]

K16557 exoA; succinoglycan biosynthesis protein ExoA [EC:2.4.−.−]

cfA1; energy−coupling factor transport system ATP−binding protein [E

2703 psbA; photosystem II P680 reaction center D1 protein [EC:1.10.3

K12146 hyfR; hydrogenase−4 transcriptional activator

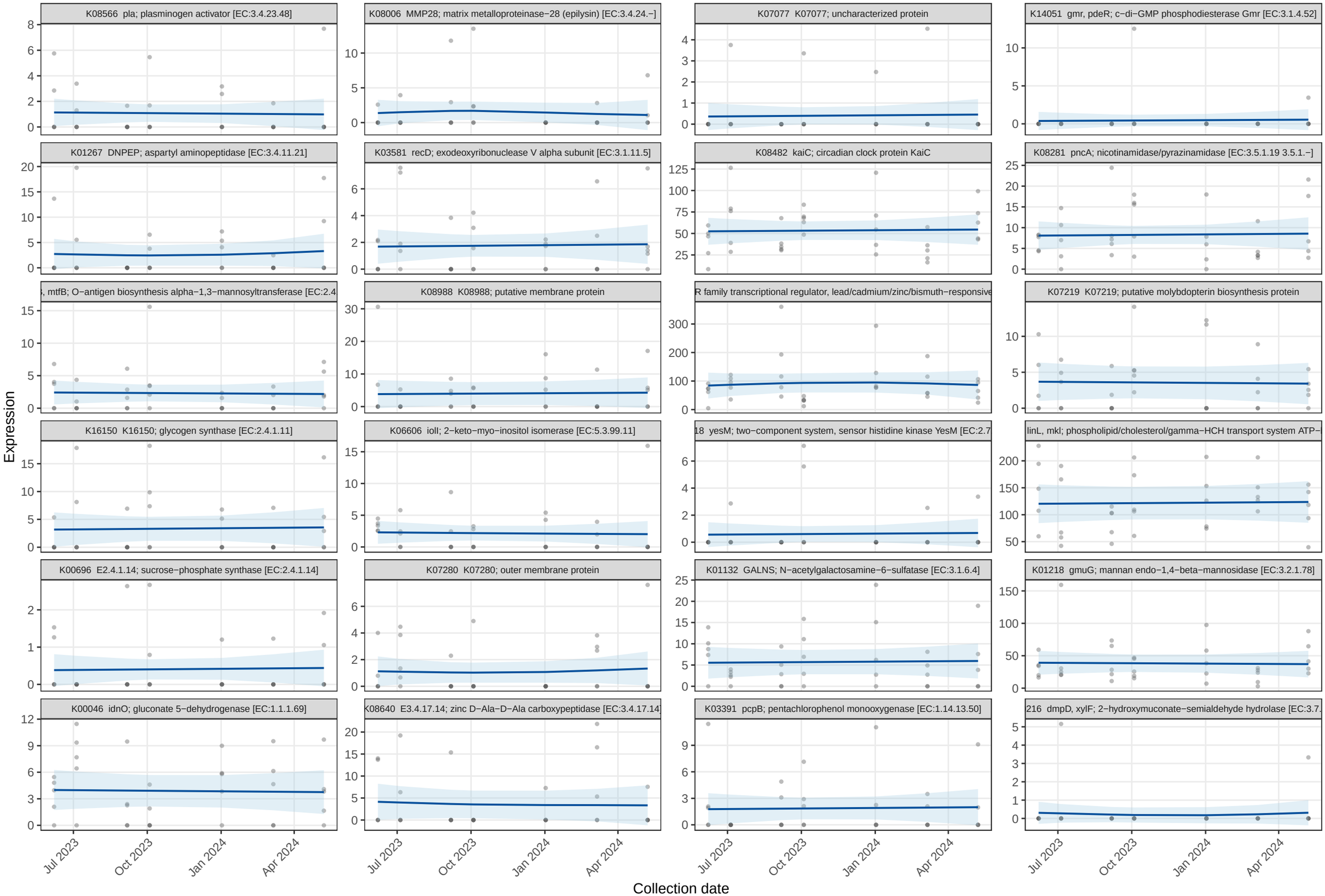
K01355 ompT; omptin [EC:3.4.23.49]

Jul 2023 Oct 2023 Jan 2024 Apr 2024

Collection date

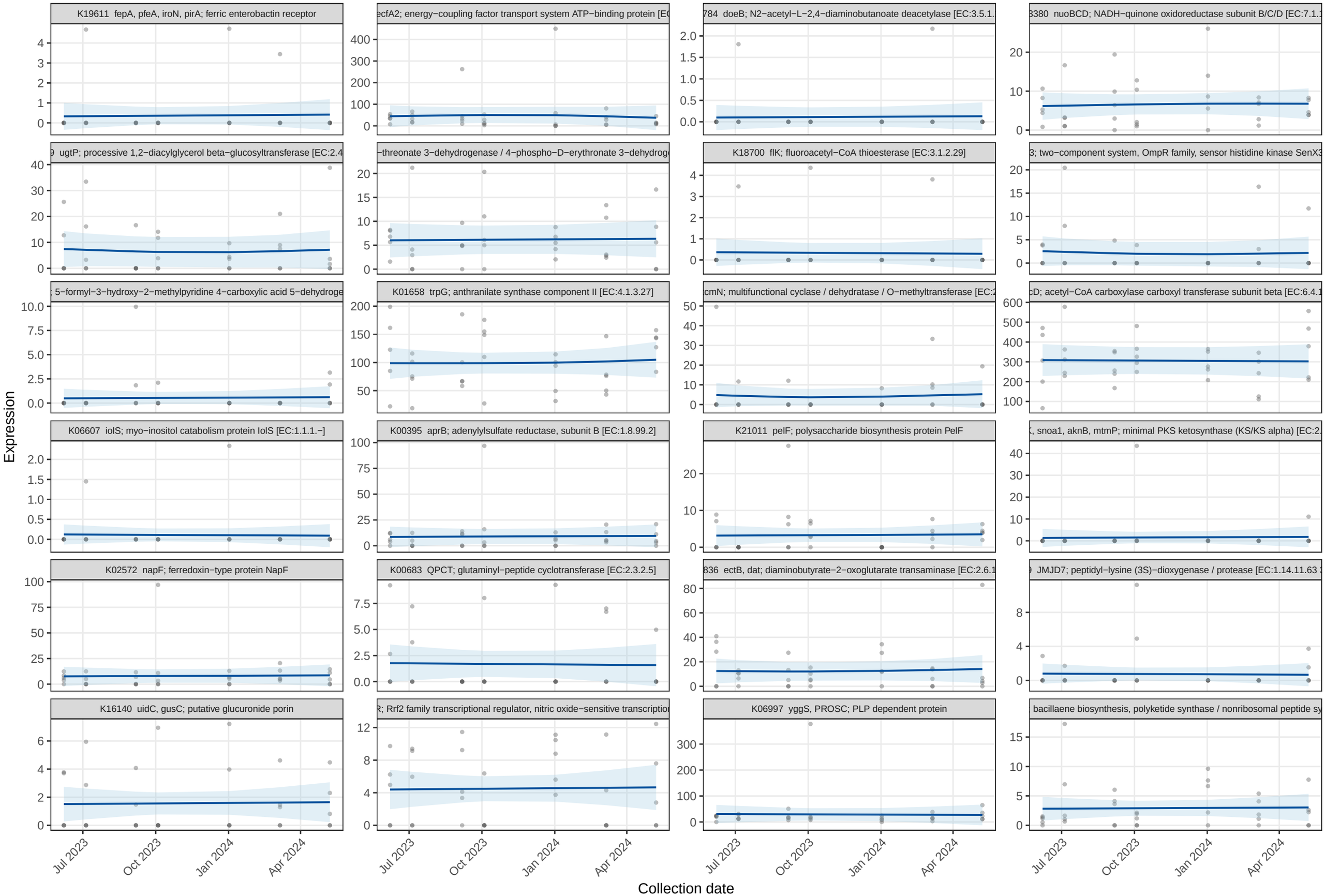
brite_D filtered temporal GAMs: page 215 of 303

All remaining BRITE classes after exclusions



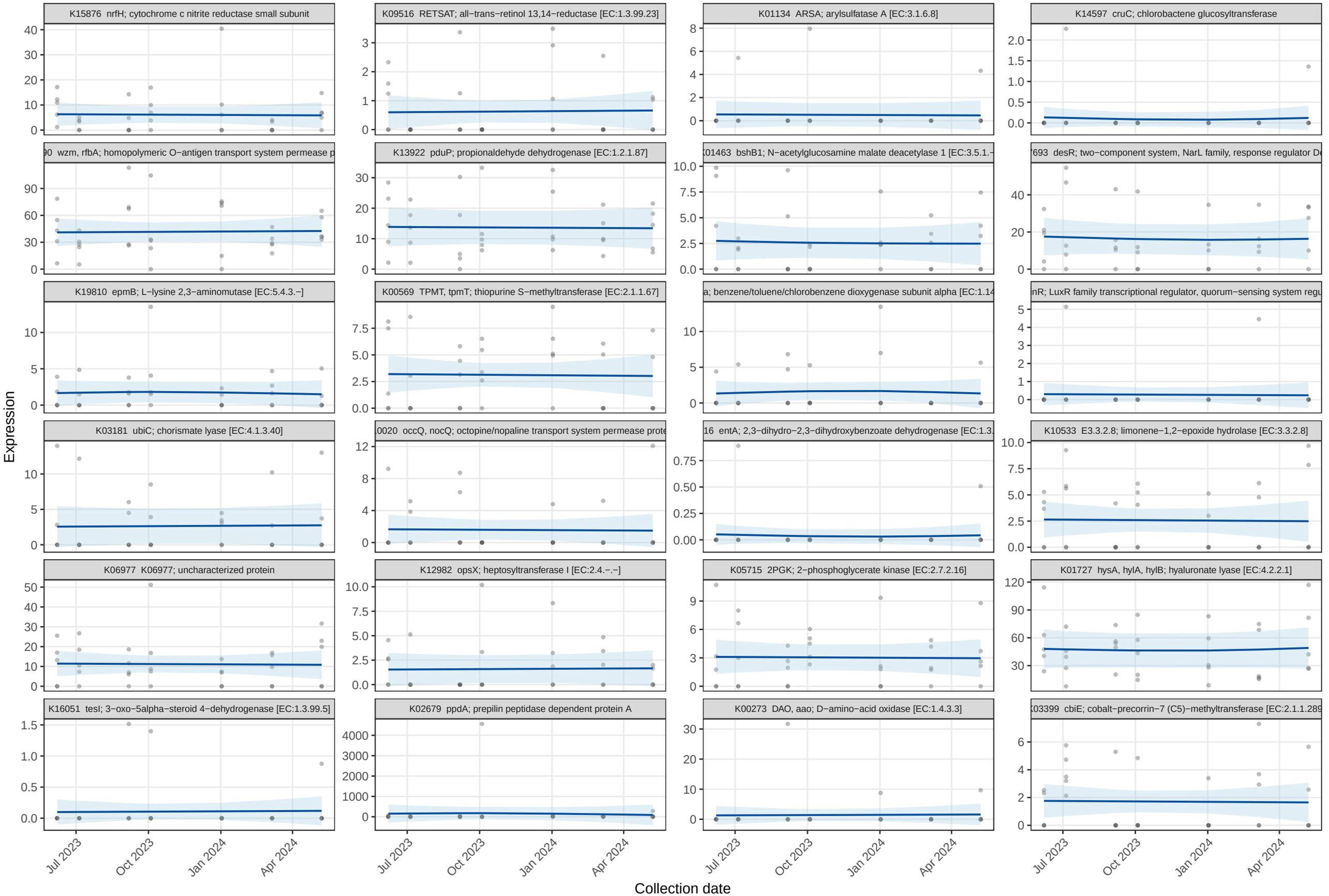
brite_D filtered temporal GAMs: page 216 of 303

All remaining BRITE classes after exclusions

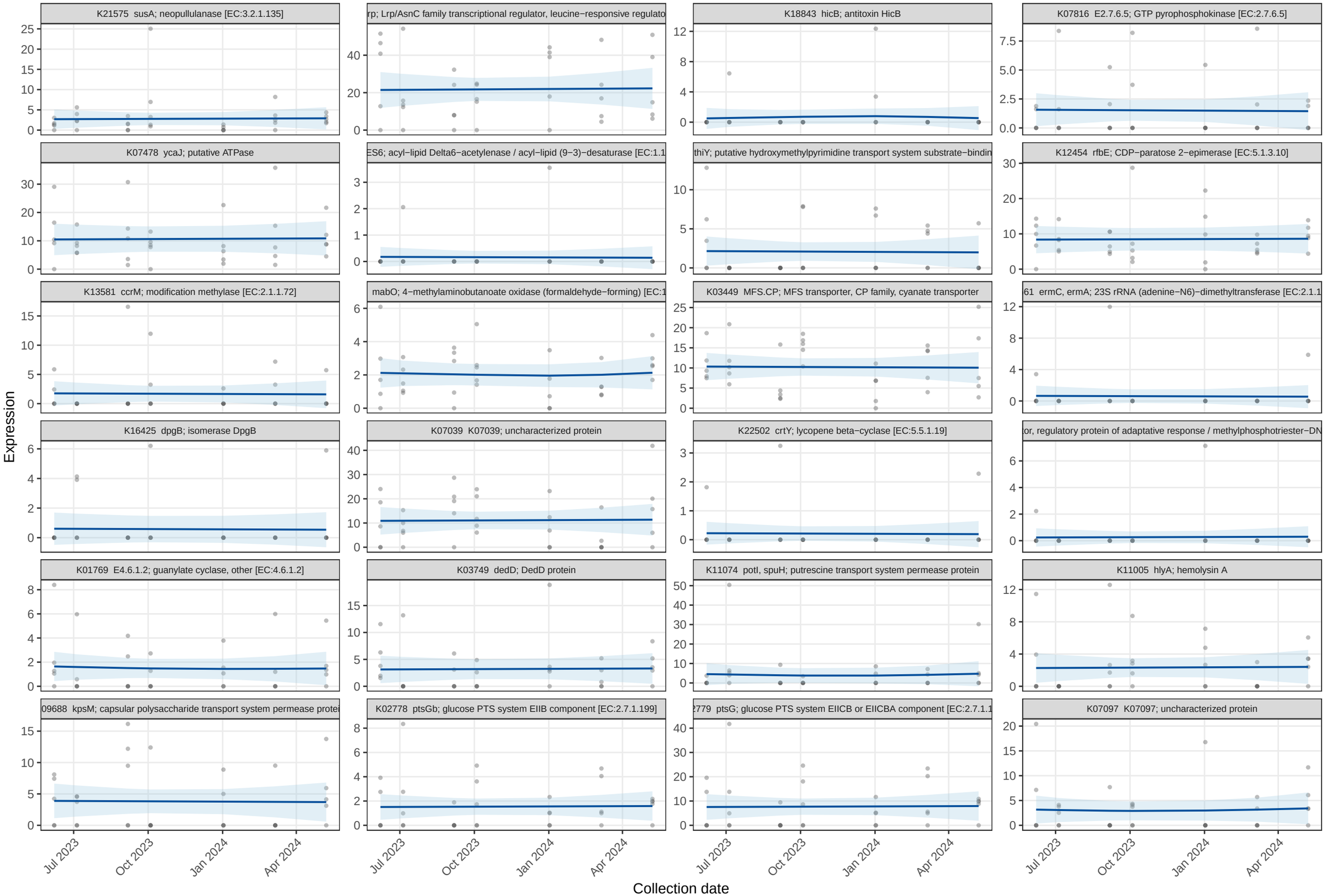


brite_D filtered temporal GAMs: page 217 of 303

All remaining BRITE classes after exclusions

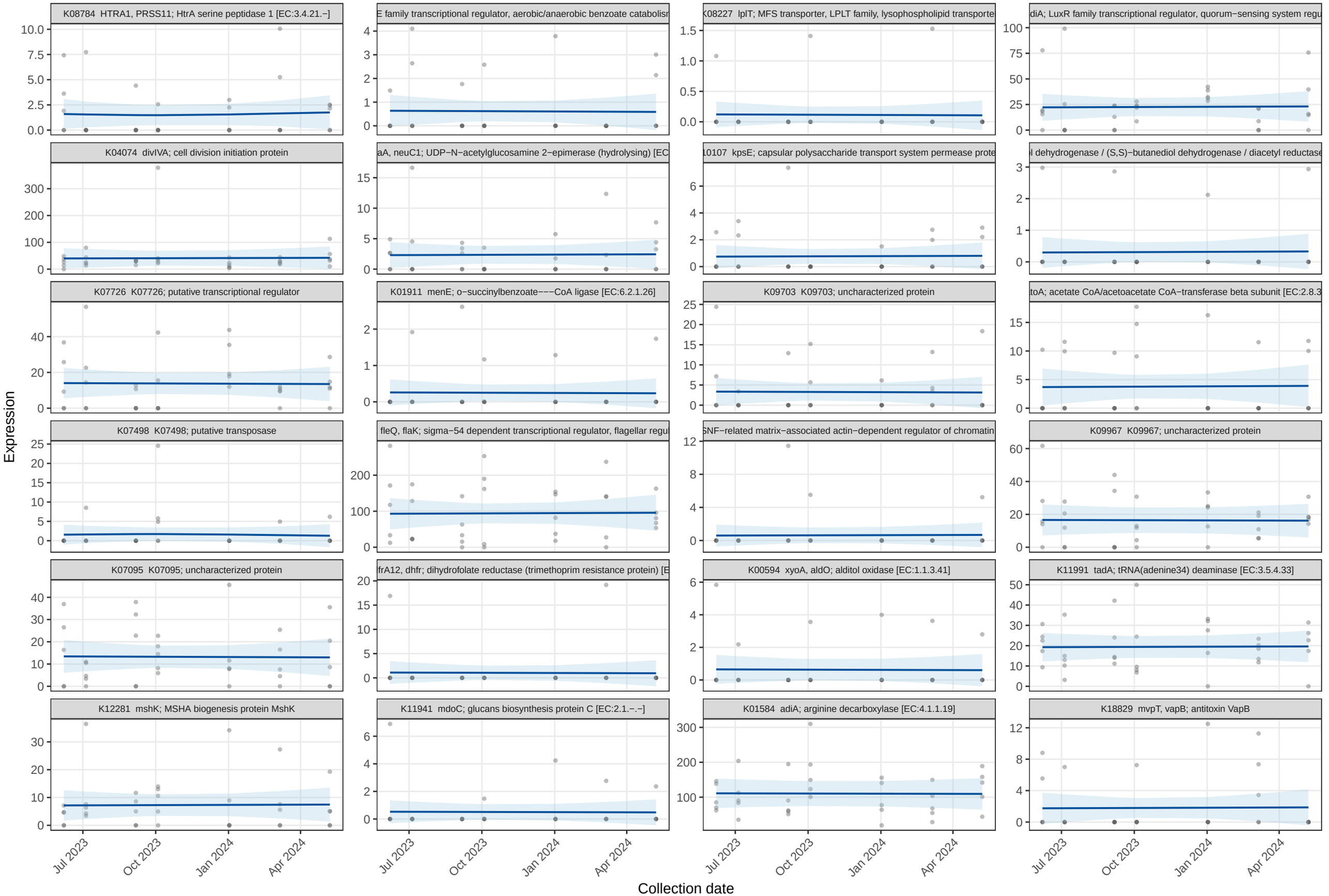


All remaining BRITE classes after exclusions



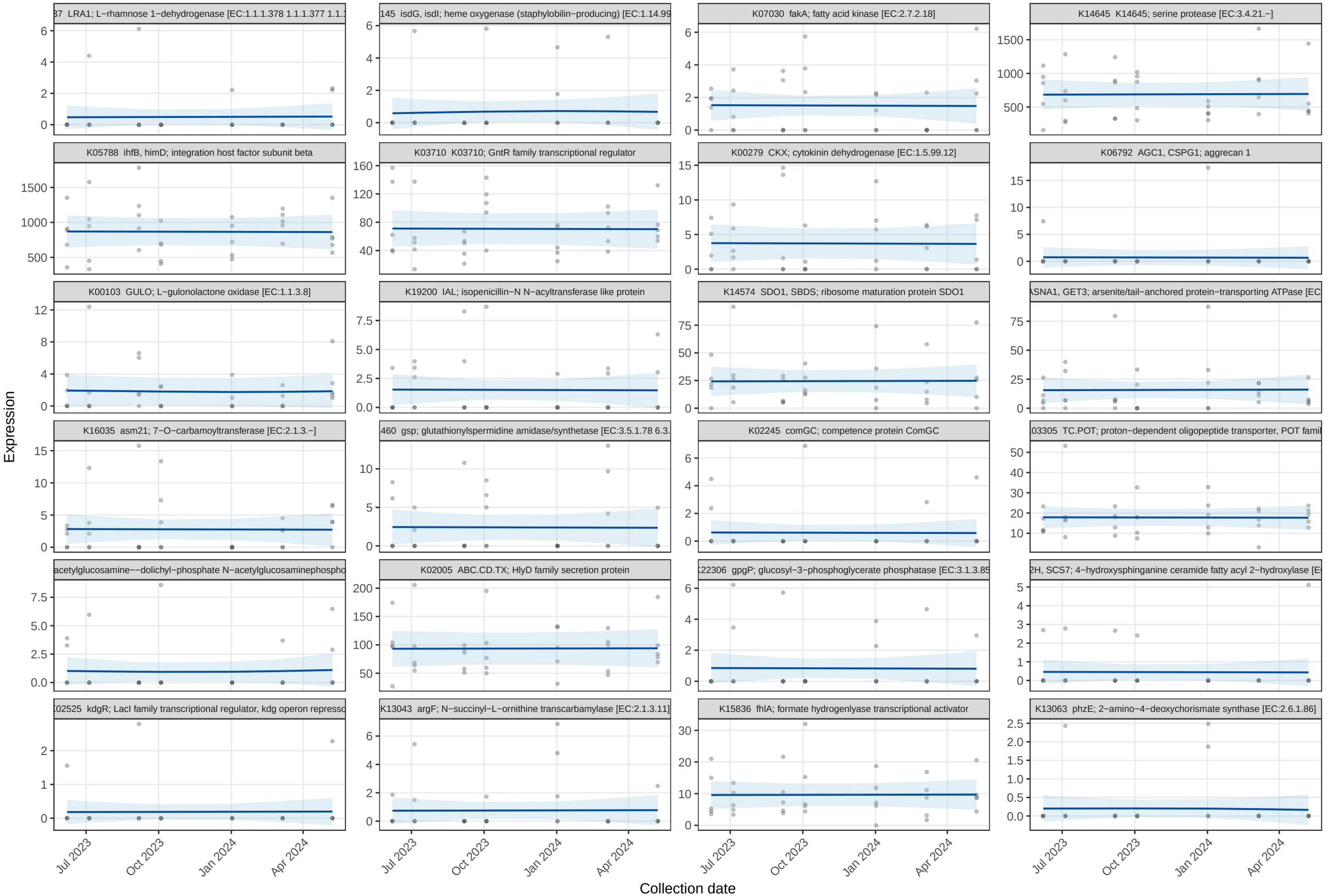
brite_D filtered temporal GAMs: page 219 of 303

All remaining BRITE classes after exclusions

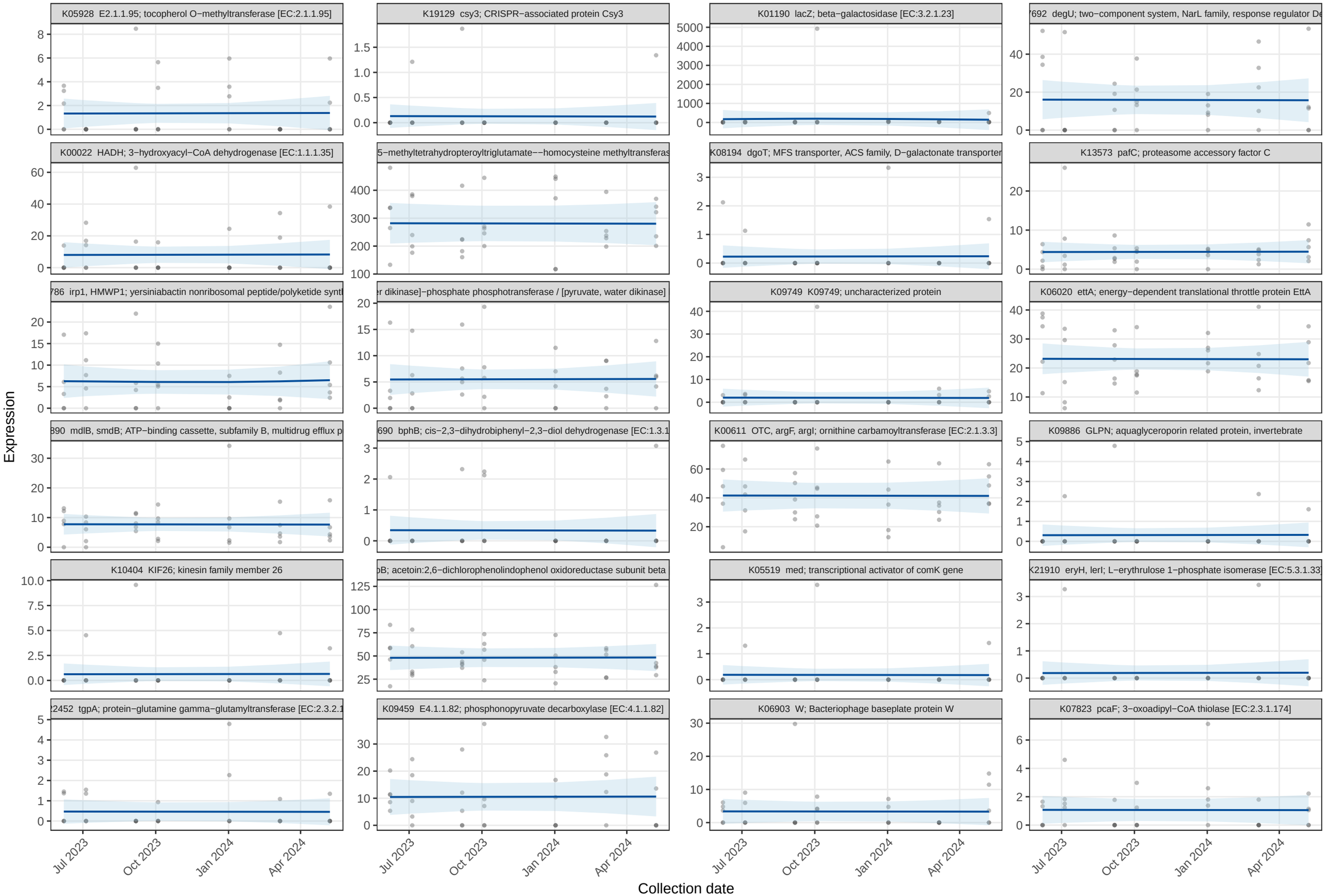


brite_D filtered temporal GAMs: page 220 of 303

All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

K13767 fadB; enoyl−CoA hydratase [EC:4.2.1.17]

K01734 mgsA; methylglyoxal synthase [EC:4.2.3.3]

K20157 mppO; enduracididine beta−hydroxylase [EC:1.14.11.40]

K02794 manX; mannose PTS system EIIAB component [EC:2.7.1.191]

K02401 flhB; flagellar biosynthesis protein FlhB

K16131 mcyB; microcystin synthetase protein McyB

32 K05832; putative tryptophan/tyrosine transport system permease pr

07317 K07317; adenine−specific DNA−methyltransferase [EC:2.1.1.72

K00068 srlD; sorbitol−6−phosphate 2−dehydrogenase [EC:1.1.1.140]

K13954 yiaY; alcohol dehydrogenase [EC:1.1.1.1]

K07172 mazE, chpAI; antitoxin MazE

00556 trmH; tRNA (guanosine−2'−O−)−methyltransferase [EC:2.1.1.34

Expression

K08983 K08983; putative membrane protein

7543 bbsE; benzylsuccinate CoA−transferase BbsE subunit [EC:2.8.3.1

711 fur, zur, furB; Fur family transcriptional regulator, ferric uptake regul

K04757 rsbW; serine/threonine−protein kinase RsbW [EC:2.7.11.1]

K18197 yesW; rhamnogalacturonan endolyase [EC:4.2.2.23]

K00422 E1.10.3.1; polyphenol oxidase [EC:1.10.3.1]

K00541 NNMT; nicotinamide N−methyltransferase [EC:2.1.1.1]

K00562 INMT, TEMT; methyltransferase [EC:2.1.1.49 2.1.1.96]

K08970 rcnA; nickel/cobalt transporter (NicO) family protein

K00441 frhB; coenzyme F420 hydrogenase subunit beta [EC:1.12.98.1

K14340 K14340; mannosyltransferase [EC:2.4.1.−]

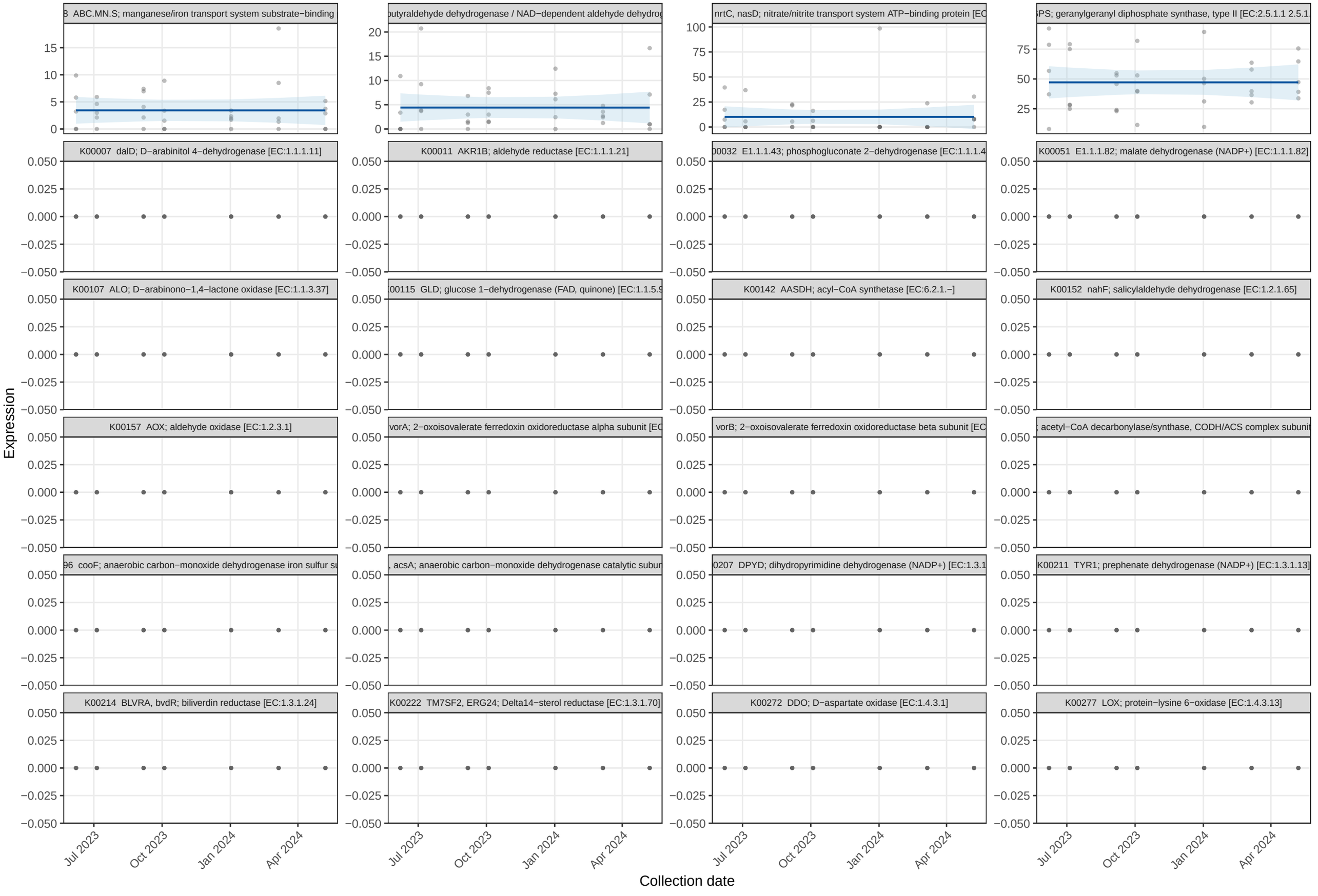
K16566 exoY; exopolysaccharide production protein ExoY

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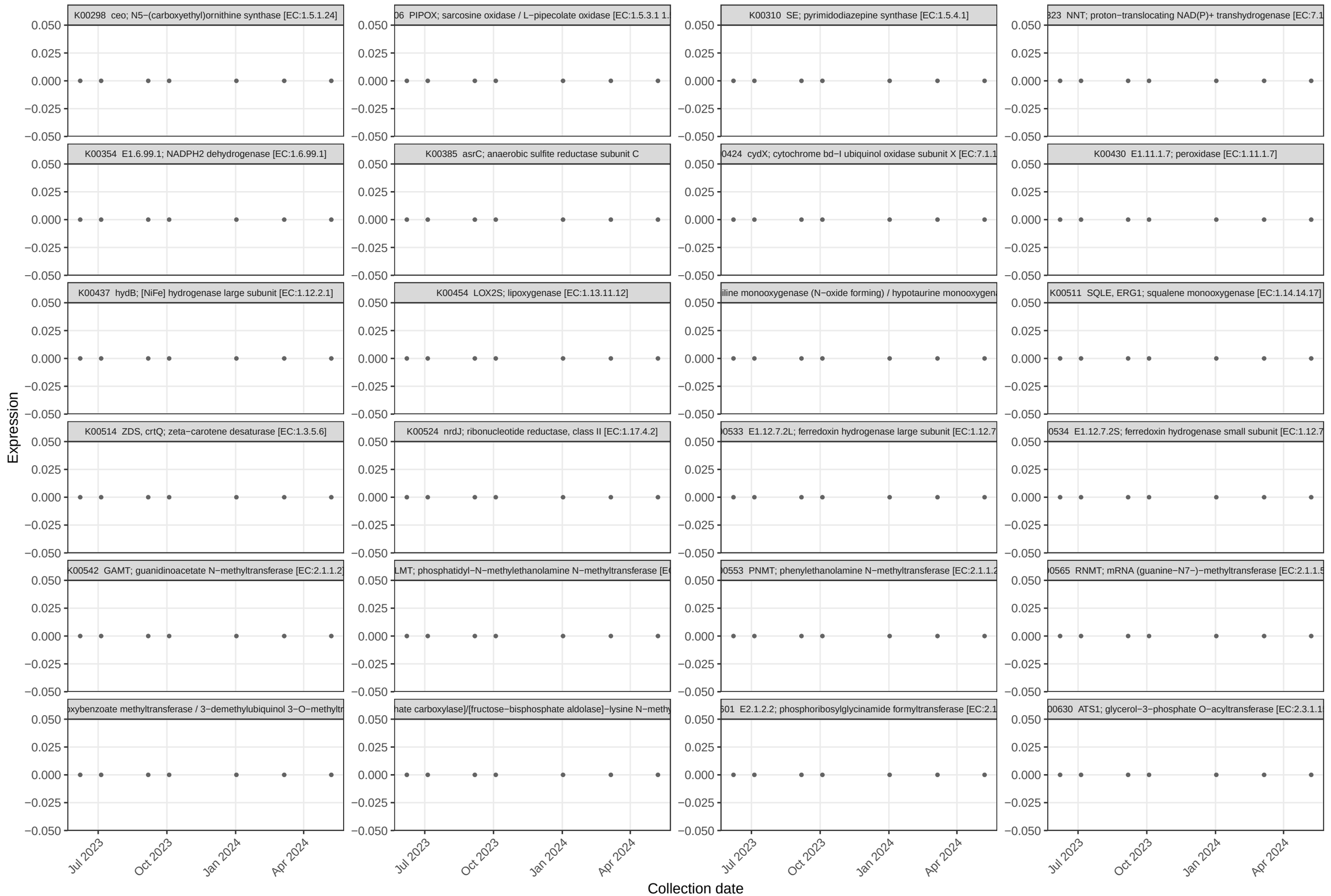
Collection date

brite_D filtered temporal GAMs: page 223 of 303

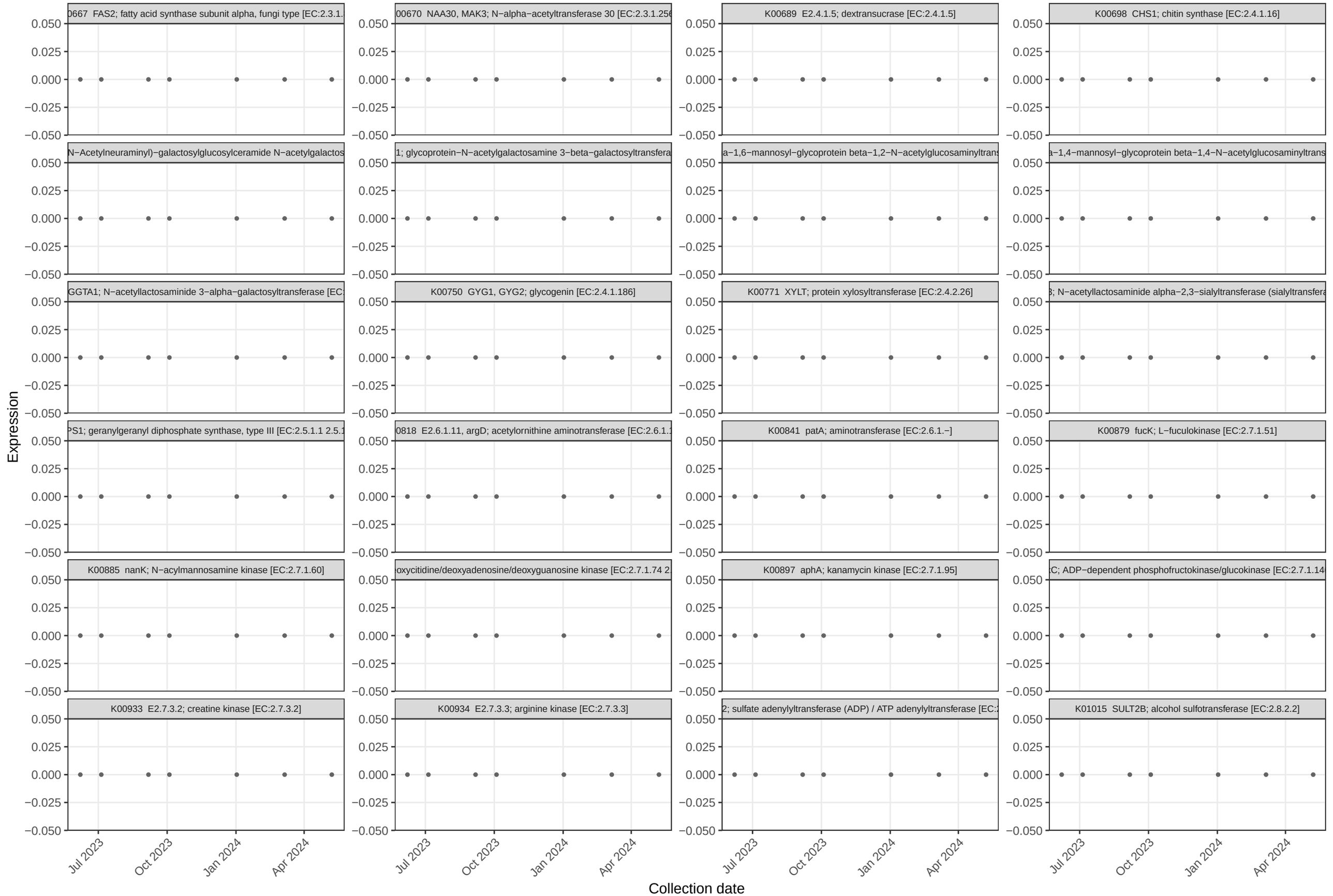
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 226 of 303

All remaining BRITE classes after exclusions

K01016 SULT1E1, STE; estrone sulfotransferase [EC:2.8.2.4]
K01021 TPST; protein−tyrosine sulfotransferase [EC:2.8.2.20]
K01025 SULT1; sulfotransferase [EC:2.8.2.−]
K01044 CES1; carboxylesterase 1 [EC:3.1.1.1]
K01063 JHE; juvenile−hormone esterase [EC:3.1.1.59]
K01093 appA; 4−phytase / acid phosphatase [EC:3.1.3.26 3.1.3.2]
pB; phosphatidylglycerophosphatase B [EC:3.1.3.27 3.6.1.75 3.1.3
; 6−phosphofructo−2−kinase / fructose−2,6−biphosphatase 3 [EC:2
K01117 sph; sphingomyelin phosphodiesterase [EC:3.1.4.12]
K01135 ARSB; arylsulfatase B [EC:3.1.6.12]
01161 E3.1.25.1; deoxyribonuclease (pyrimidine dimer) [EC:3.1.25.
K01175 ybfF; esterase [EC:3.1.−.−]
K01188 E3.2.1.21; beta−glucosidase [EC:3.2.1.21]
K01189 GLA; alpha−galactosidase [EC:3.2.1.22]
K01204 NAGA; alpha−N−acetylgalactosaminidase [EC:3.2.1.49]
K01215 dexB; glucan 1,6−alpha−glucosidase [EC:3.2.1.70]
K01217 IDUA; L−iduronidase [EC:3.2.1.76]
K01240 URH1; uridine nucleosidase [EC:3.2.2.3]
, vibB, mxcF; bifunctional isochorismate lyase / aryl carrier protein [
K01275 CTSC; cathepsin C [EC:3.4.14.1]
K01279 TPP1, CLN2; tripeptidyl−peptidase I [EC:3.4.14.9]
K01293 E3.4.17.4; Gly−Xaa carboxypeptidase [EC:3.4.17.4]
K01316 TMPRSS15, PRSS7; enterokinase [EC:3.4.21.9]
K01336 PRB1; cerevisin [EC:3.4.21.48 3.4.21.−]

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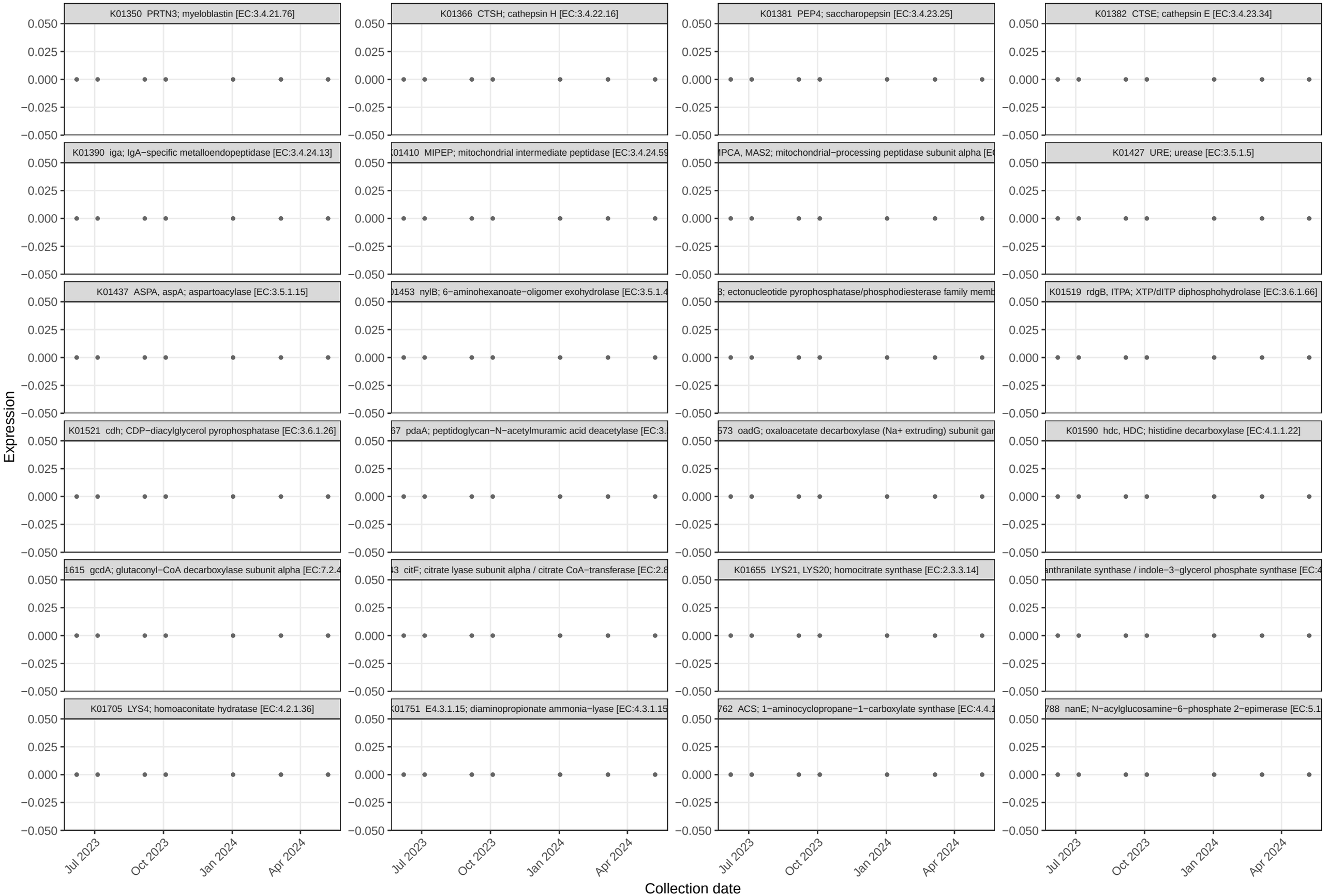
Expression

Jul 2023
Oct 2023
Jan 2024
Apr 2024

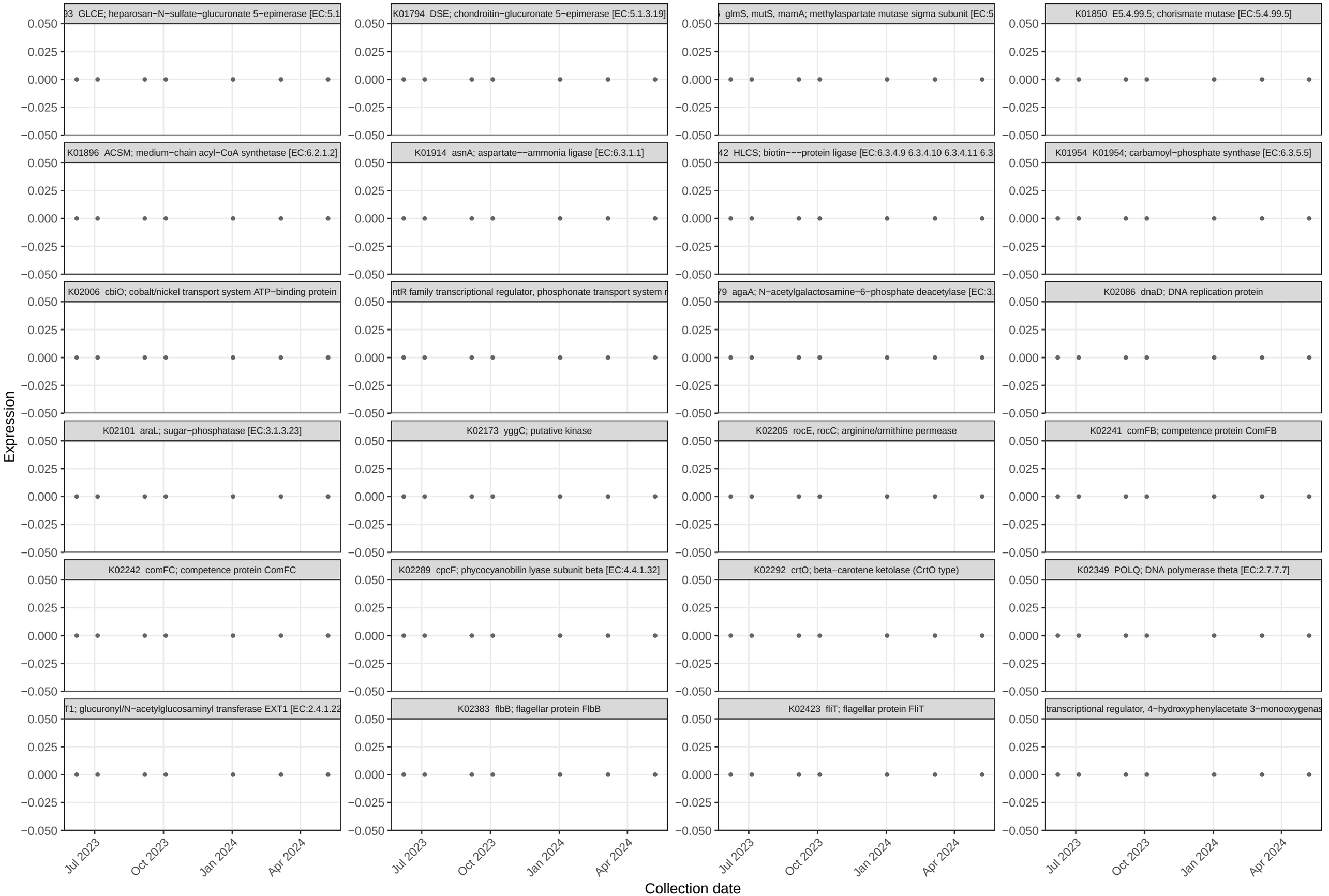
Collection date

brite_D filtered temporal GAMs: page 227 of 303

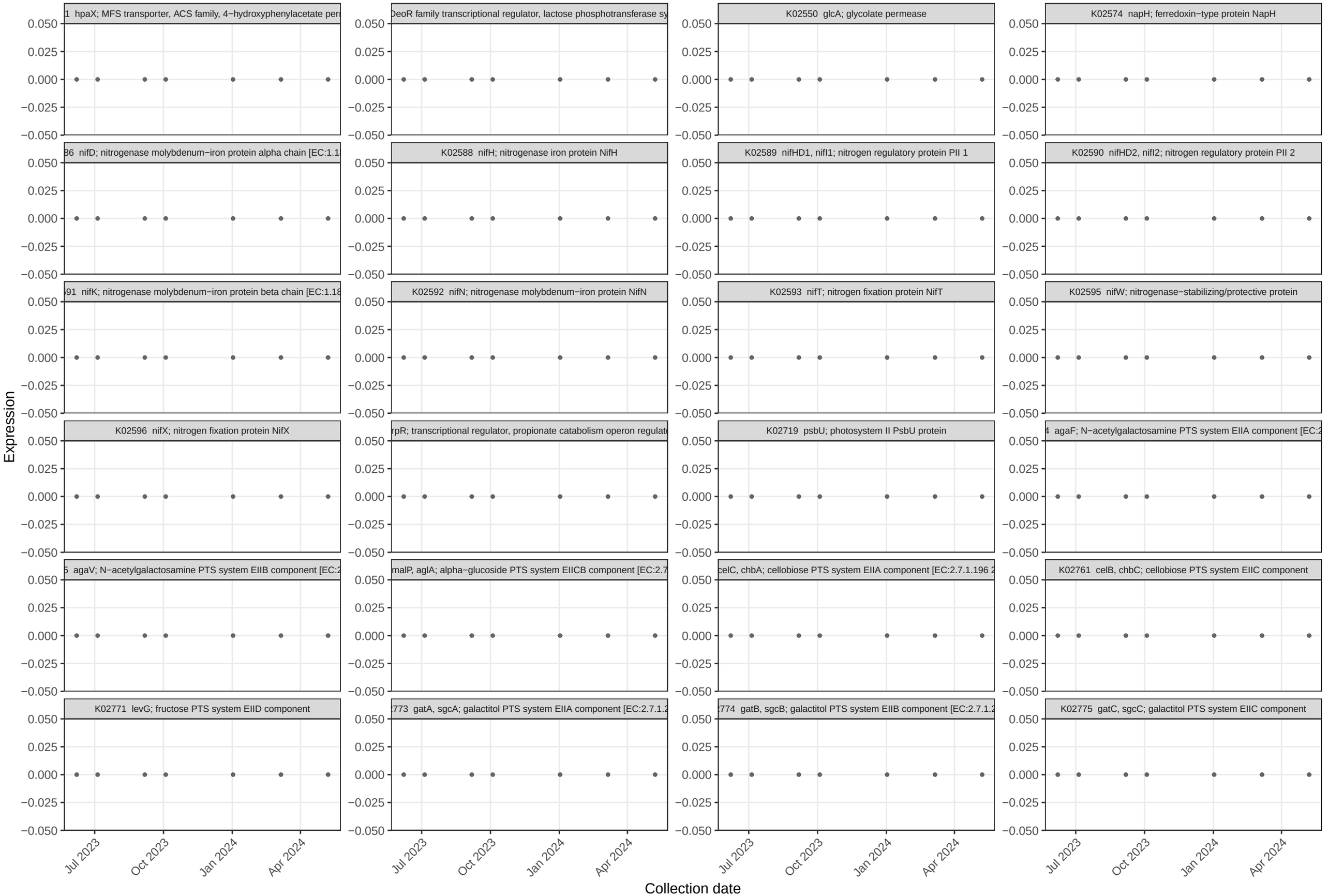
All remaining BRITE classes after exclusions



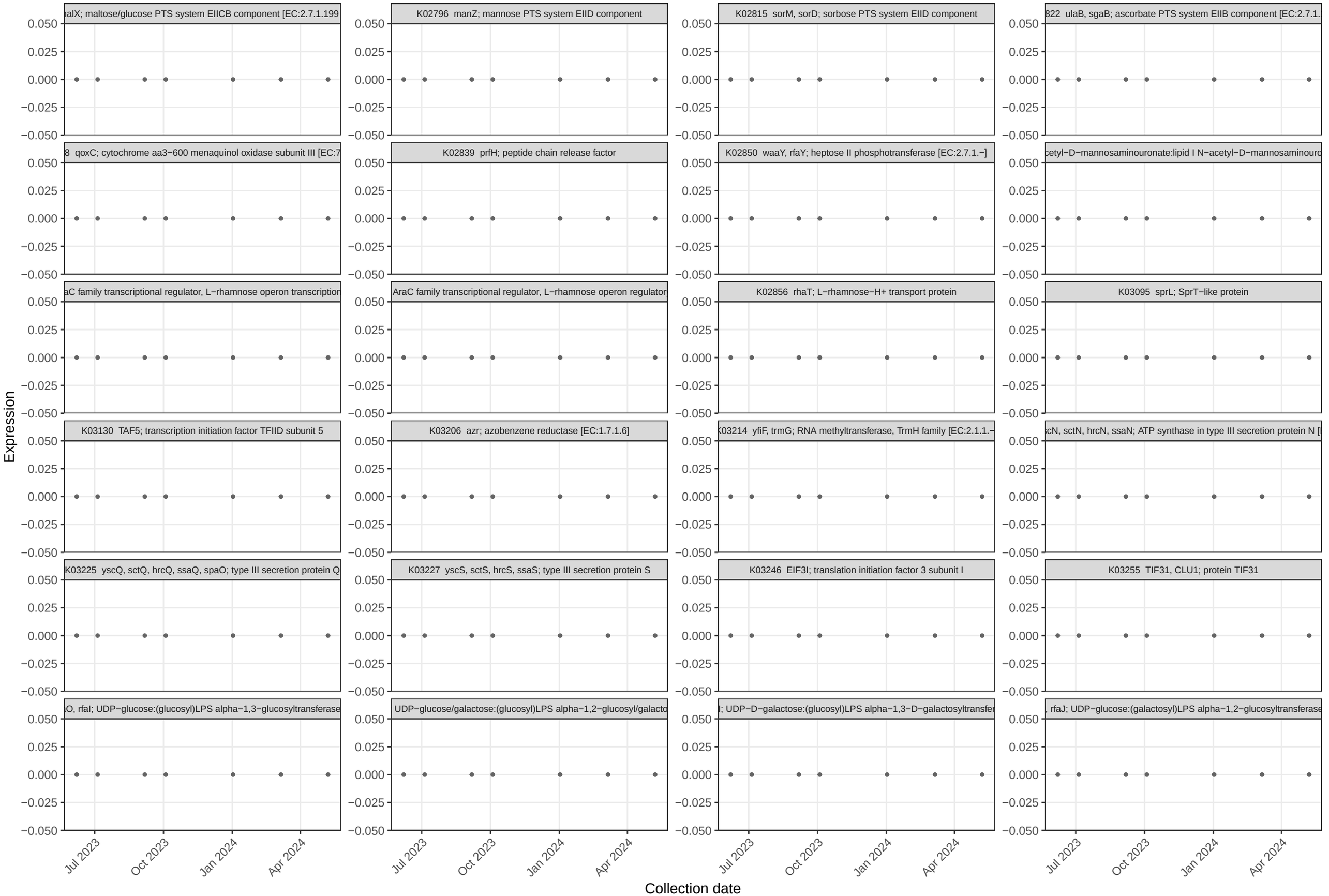
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

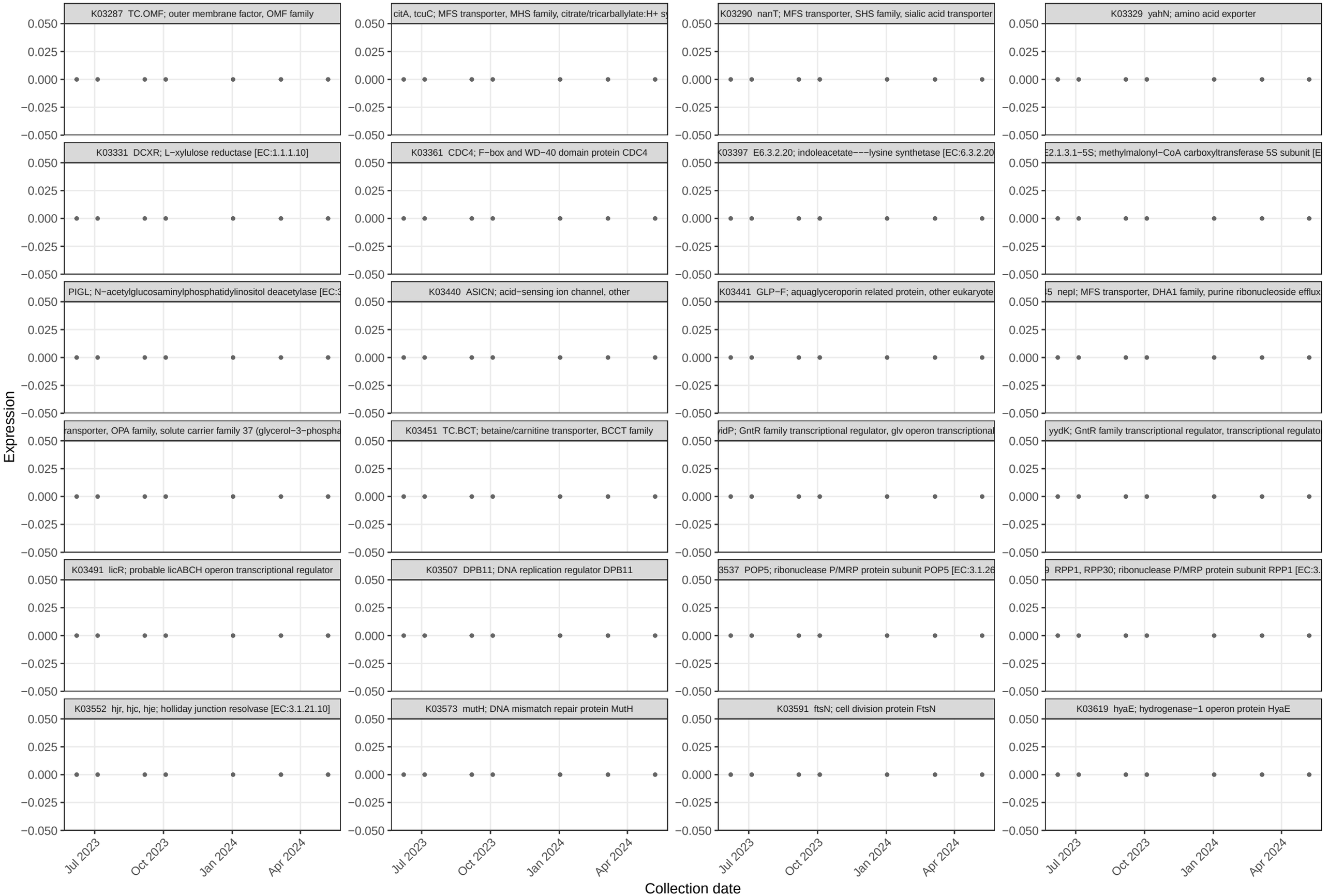


All remaining BRITE classes after exclusions

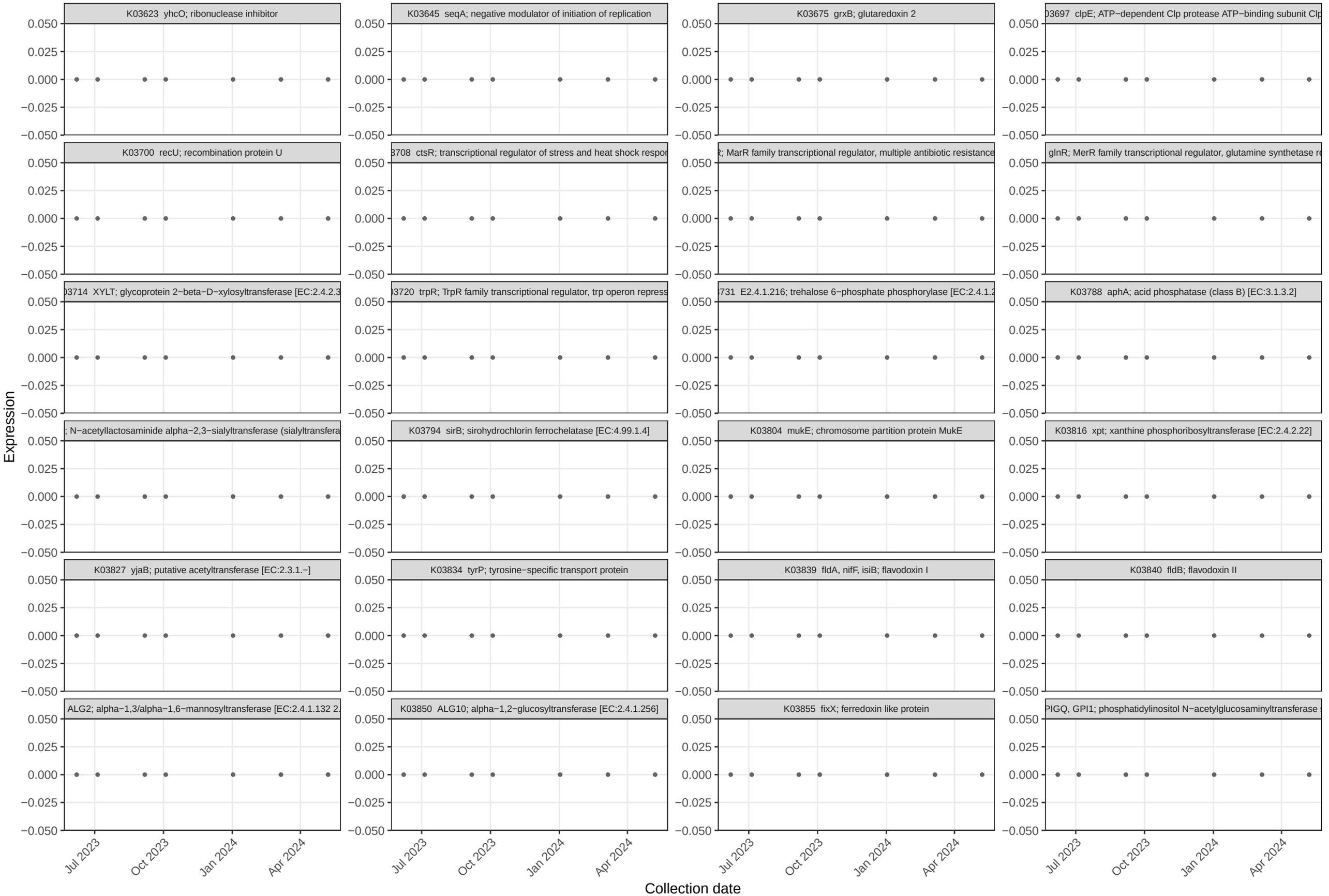


brite_D filtered temporal GAMs: page 231 of 303

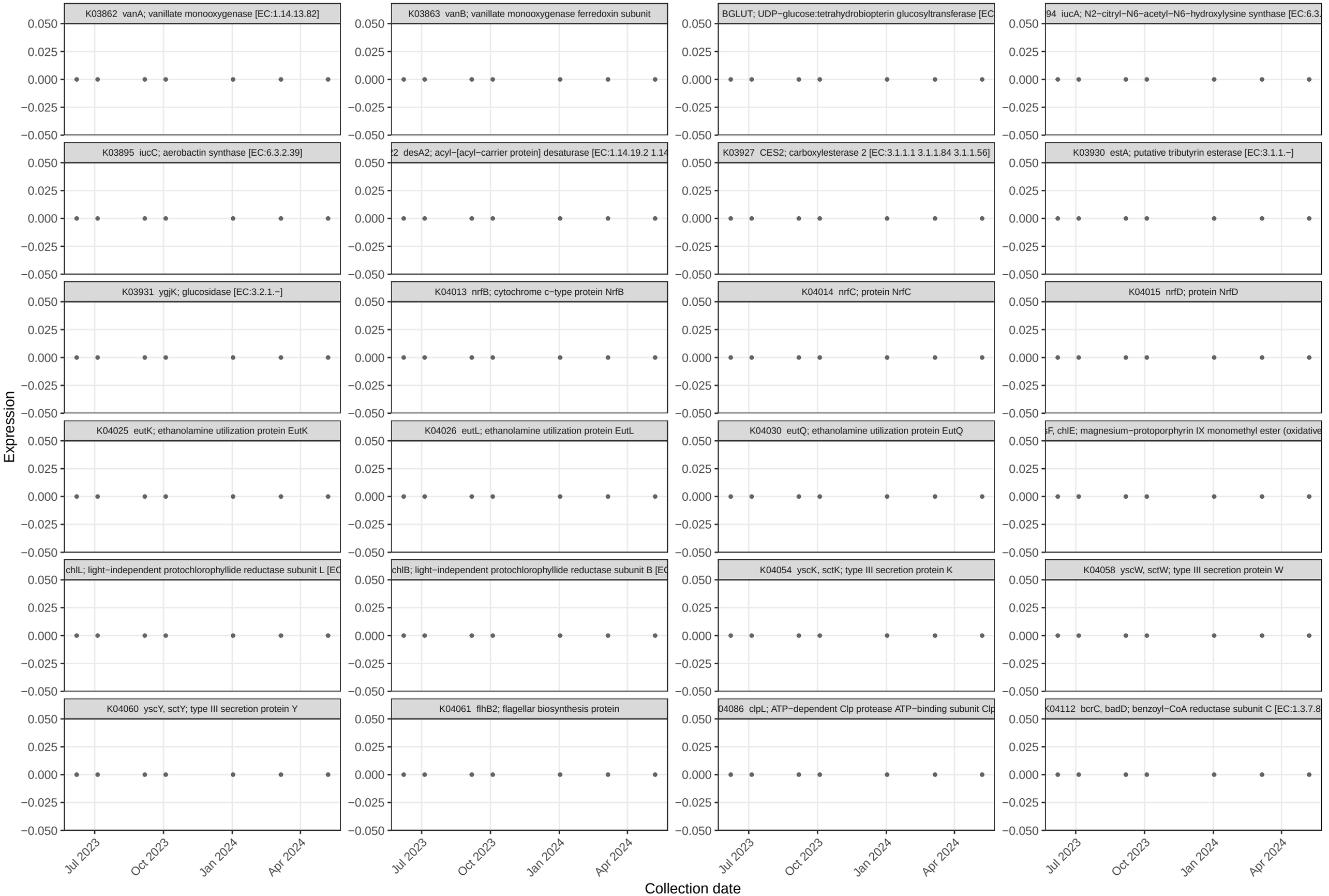
All remaining BRITE classes after exclusions



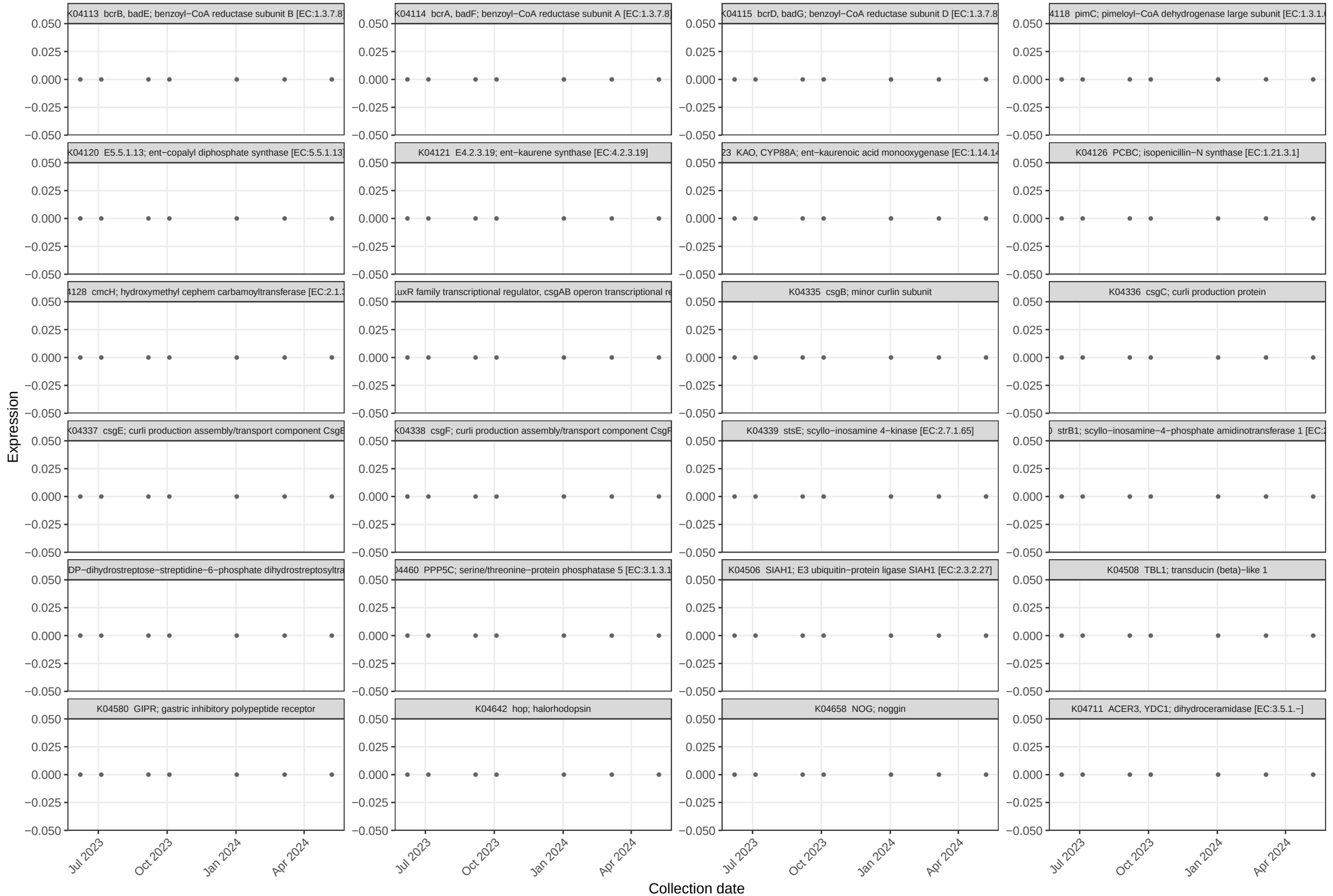
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

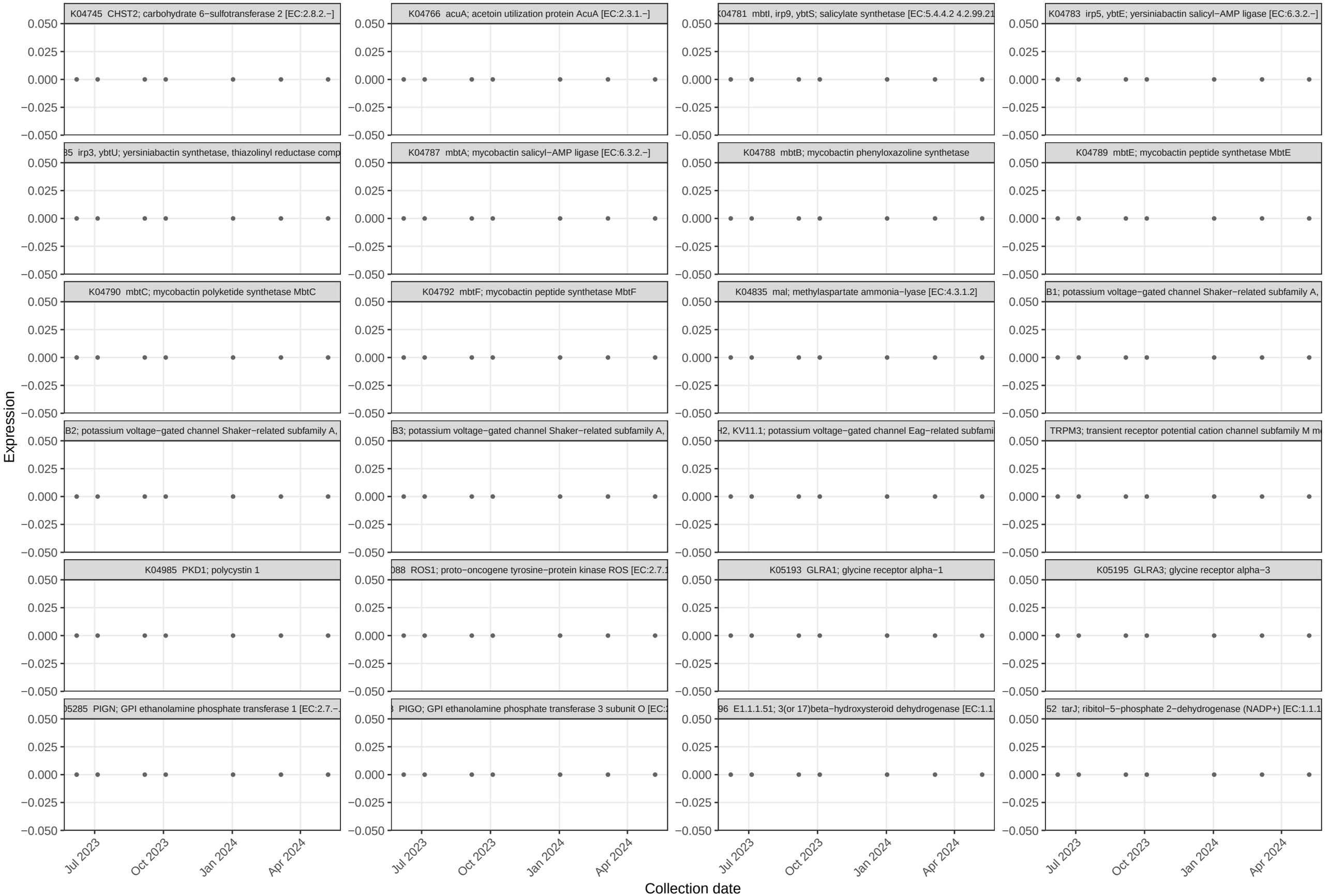


All remaining BRITE classes after exclusions

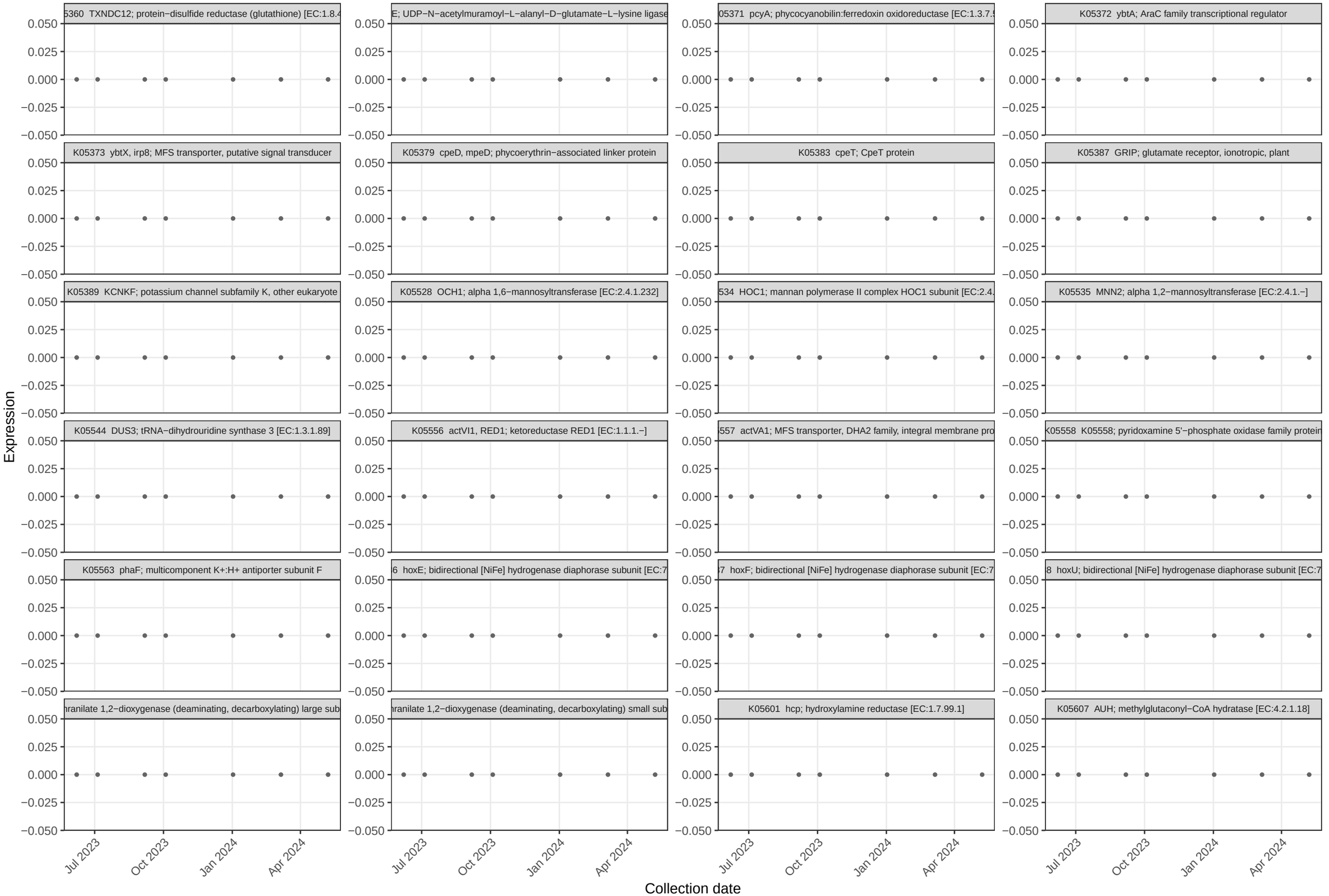


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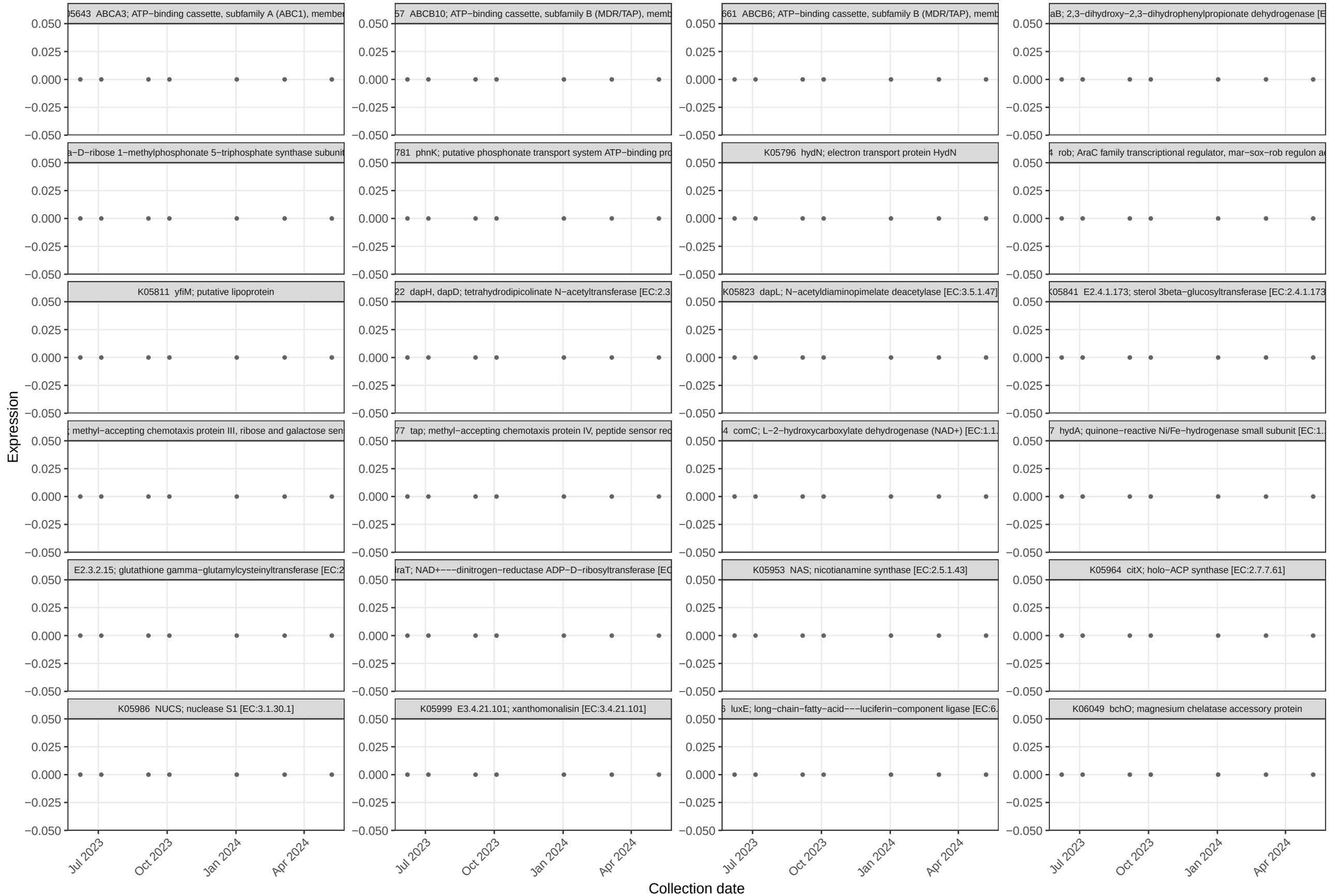
All remaining BRITE classes after exclusions



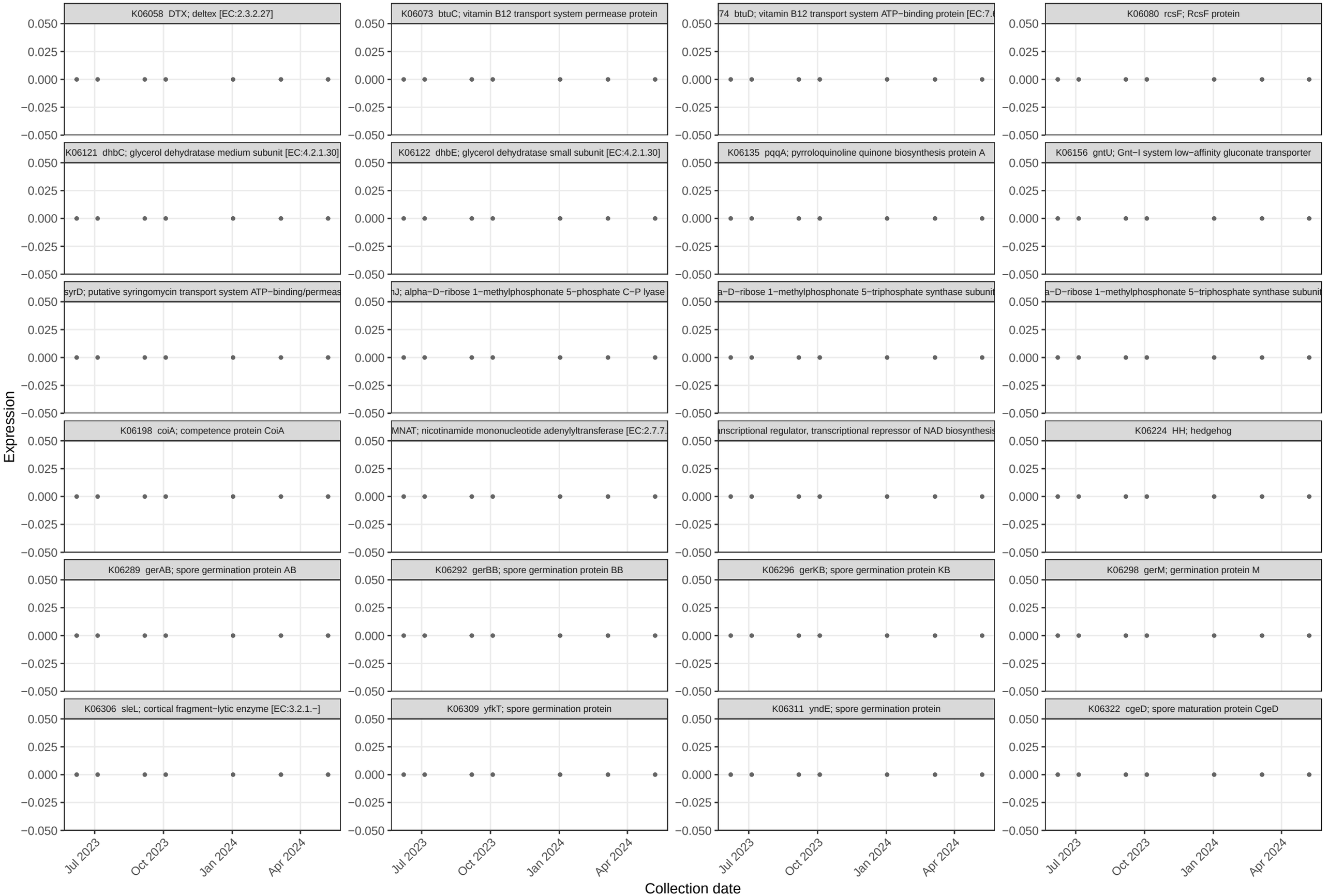
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

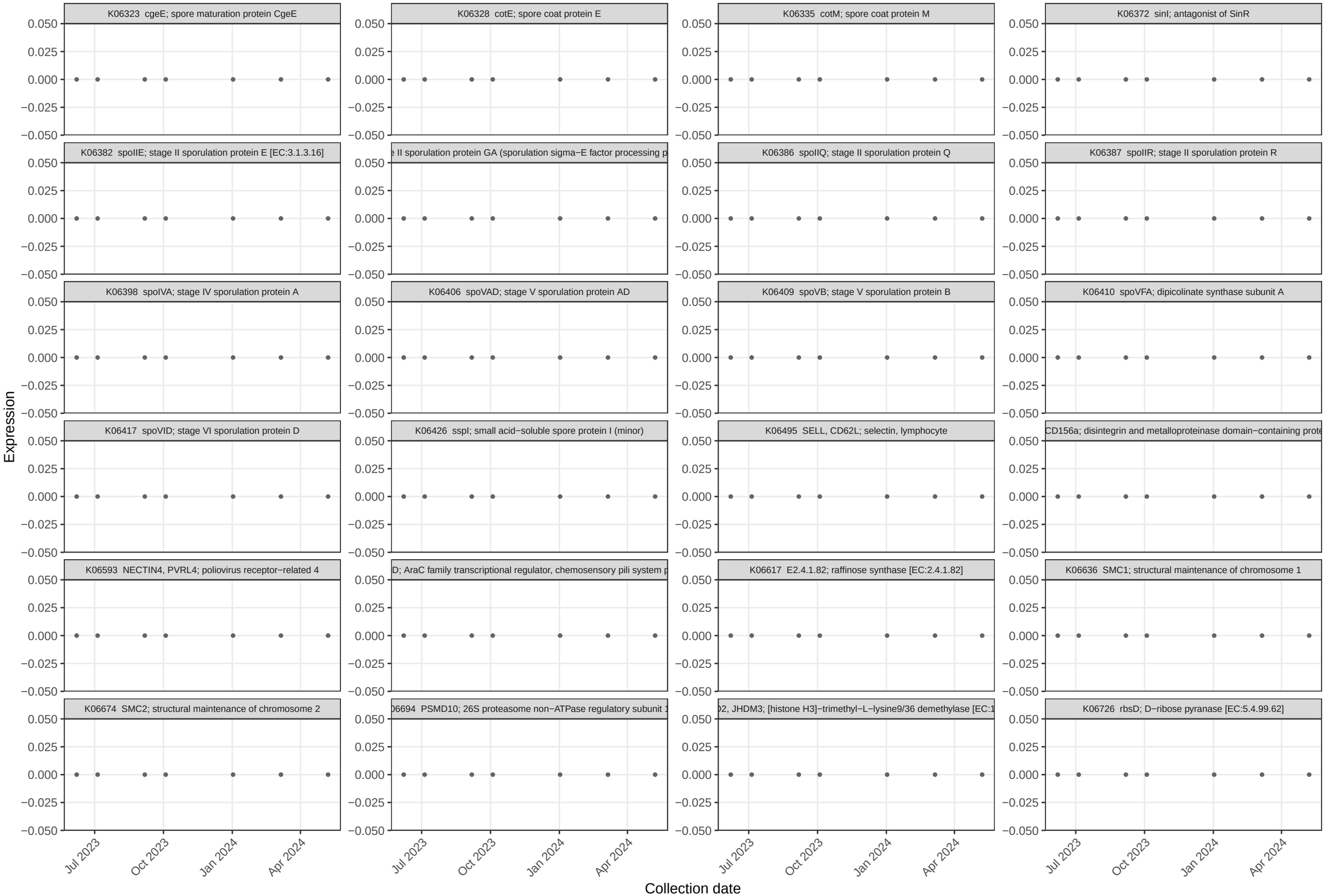


All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 239 of 303

All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 240 of 303

All remaining BRITE classes after exclusions

K06756 NRCAM; neuronal cell adhesion molecule
K06767 DSCAM; Down syndrome cell adhesion molecule
768 DSCAML1; Down syndrome cell adhesion molecule−like prote
7 PTPRD; receptor−type tyrosine−protein phosphatase delta [EC:3.
78 PTPRS; receptor−type tyrosine−protein phosphatase S [EC:3.1
06859 pgi1; glucose−6−phosphate isomerase, archaeal [EC:5.3.1.9
K06860 K06860; putative heme uptake system protein
K06876 phrB; (6−4)DNA photolyase [EC:4.1.99.13]
K06906 gpU; phage tail protein
K06914 mfnD; tyramine−−−L−glutamate ligase [EC:6.3.4.24]
K06928 NTPCR; nucleoside−triphosphatase [EC:3.6.1.15]
ype transcriptional regulator, bacterioopsin transcriptional activator
K06943 NOG1; nucleolar GTP−binding protein
K06957 tmcA; tRNA(Met) cytidine acetyltransferase [EC:2.3.1.193]
70 rlmF; 23S rRNA (adenine1618−N6)−methyltransferase [EC:2.1.1
K06979 mph; macrolide phosphotransferase
06984; beta−ribofuranosylaminobenzene 5'−phosphate synthase [E
egtC; gamma−glutamyl hercynylcysteine S−oxide hydrolase [EC:3.
K07015 yqeG; putative phosphatase [EC:3.1.3.−]
K07016 csm1, cas10; CRISPR−associated protein Csm1
07057 bpsA; N4−bis(aminopropyl)spermidine synthase [EC:2.5.1.12
K07060 nob1; endoribonuclease Nob1 [EC:3.1.−.−]
K07099 K07099; uncharacterized protein
K07105 K07105; uncharacterized protein

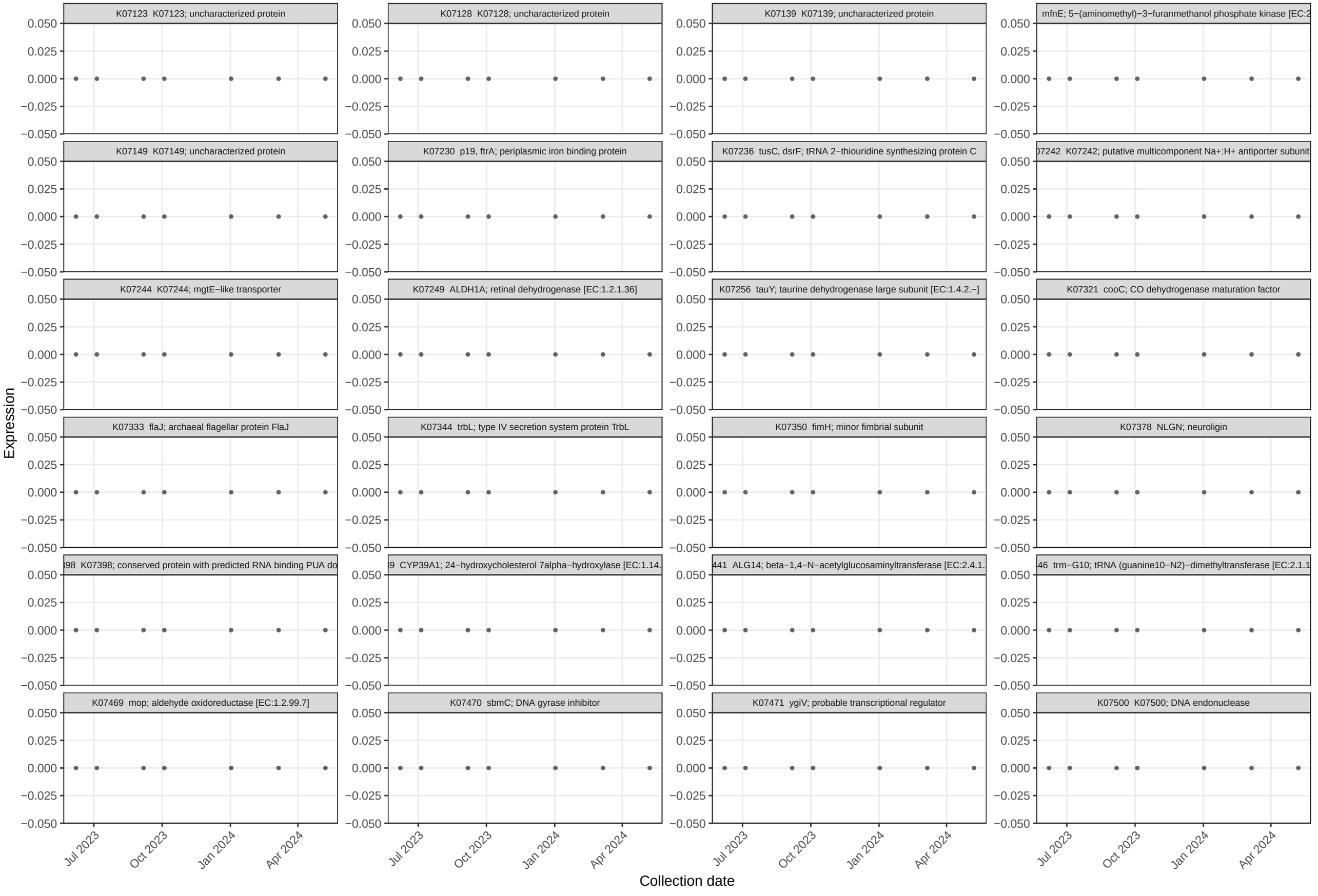
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Expression

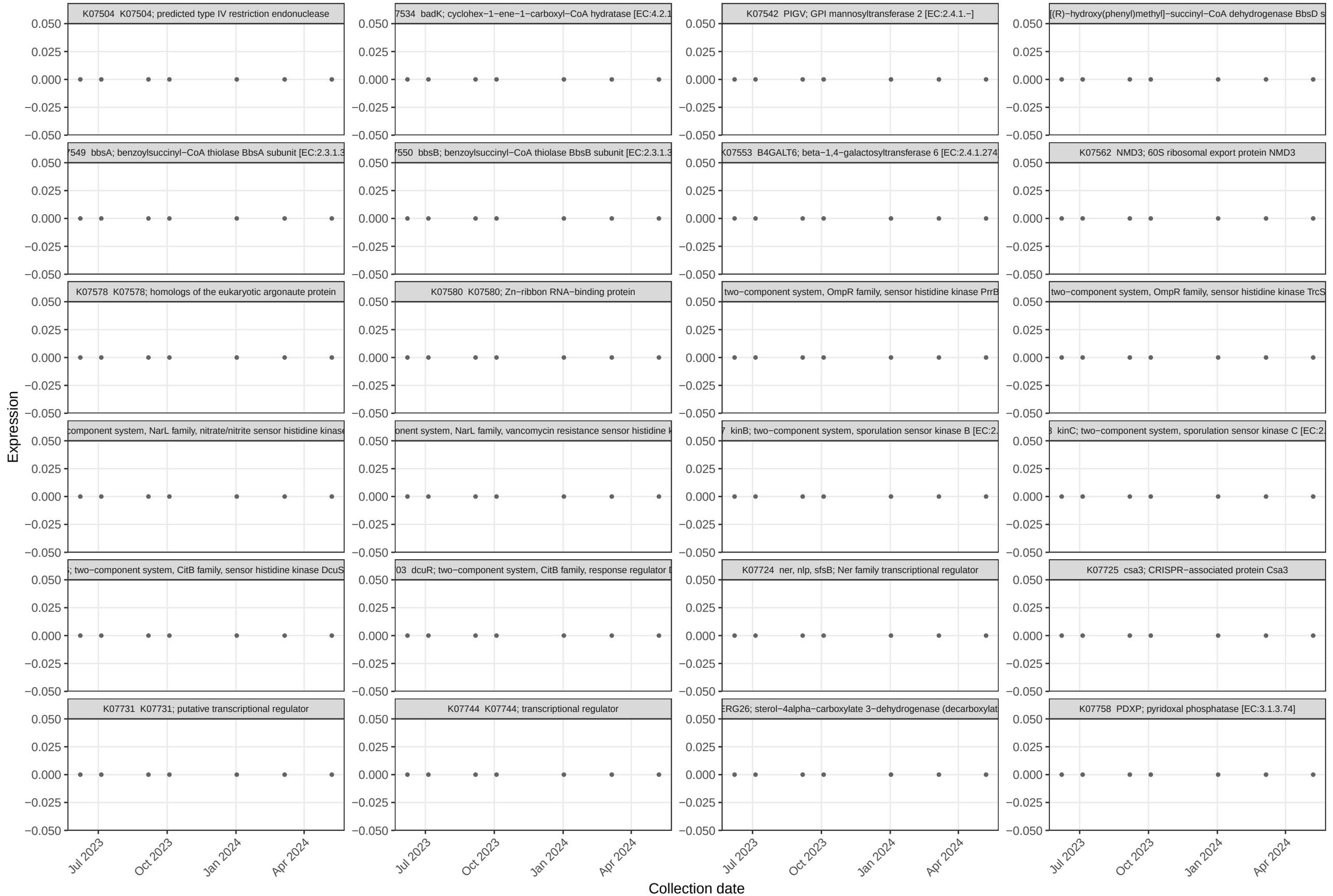
Jul 2023
Oct 2023
Jan 2024
Apr 2024

Collection date

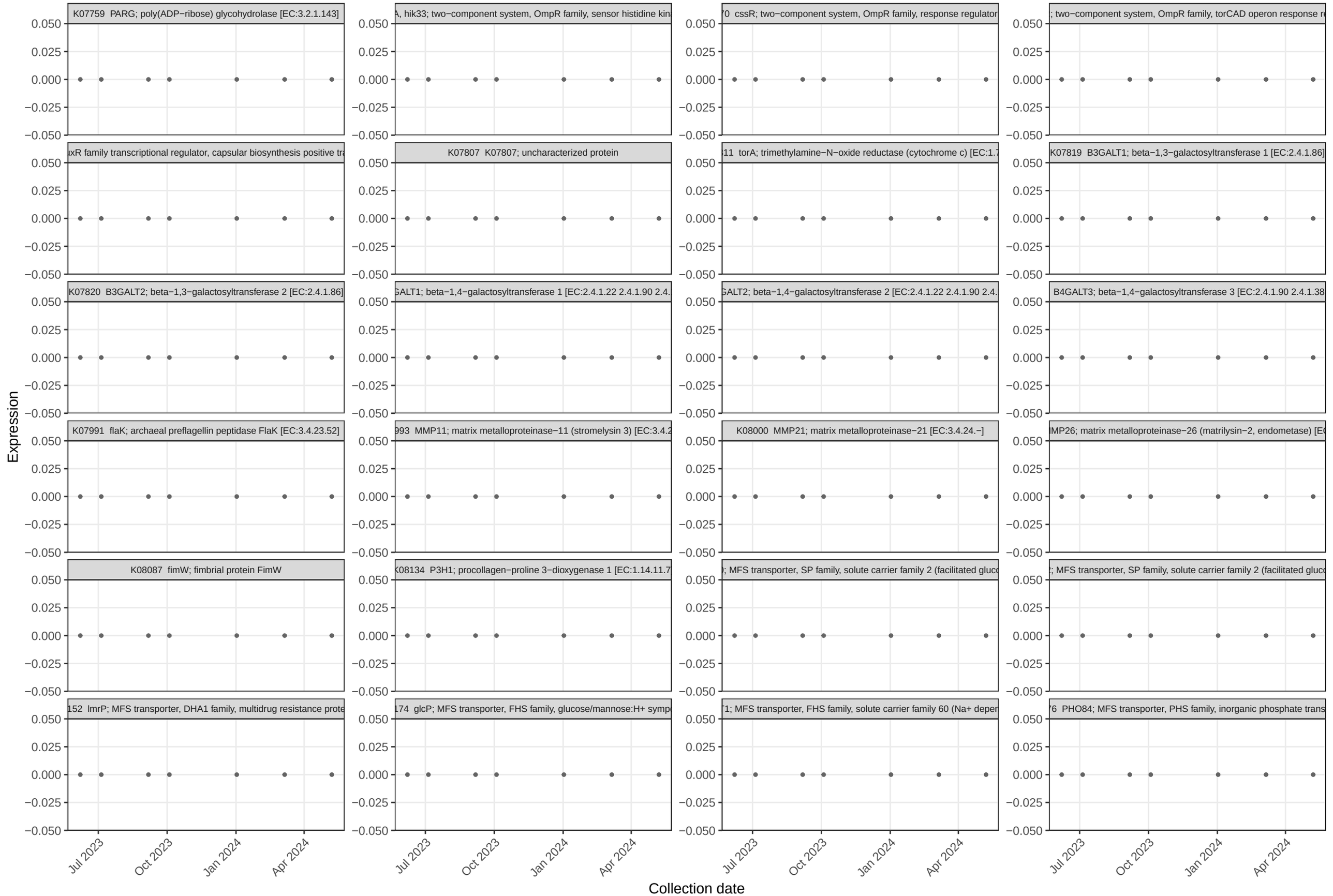
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

er, ACS family, solute carrier family 17 (sodium−dependent inorgani | sporter, OCT family, solute carrier family 22 (organic anion/cation tra | FS transporter, OCT family, solute carrier family 22 (organic anion/c | MFS transporter, OCT family, solute carrier family 22 (organic anion/

FS transporter, OCT family, solute carrier family 22 (organic cation t | FS transporter, OCT family, solute carrier family 22 (organic cation t | FS transporter, OCT family, solute carrier family 22 (organic anion t | UMF2; MFS transporter, UMF2 family, putative MFS family transport

itG, ymfD, yfmO; MFS transporter, ACDE family, multidrug resistan | K08222 yqgE; MFS transporter, YQGE family, putative transporter | 08254 E3.2.1.59; glucan endo−1,3−alpha−glucosidase [EC:3.2.1.5 | K08264 hdrD; heterodisulfide reductase subunit D [EC:1.8.98.1]

K08280 wbbJ; lipopolysaccharide O−acetyltransferase [EC:2.3.1.−] | K08285 TRIM18, MID1; midline 1 [EC:2.3.2.27] | K08286 E2.7.11.−; protein−serine/threonine kinase [EC:2.7.11.−] | K08297 caiA; crotonobetainyl−CoA dehydrogenase [EC:1.3.8.13]

gatY−kbaY; tagatose 1,6−diphosphate aldolase GatY/KbaY [EC:4 | K08315 hycI; hydrogenase 3 maturation protease [EC:3.4.23.51] | /droxybutyrate dehydrogenase / sulfolactaldehyde 3−reductase [EC | K08320 nudG; (d)CTP diphosphatase [EC:3.6.1.65]

K08349 fdnH; formate dehydrogenase−N, beta subunit | 99 LGR6; leucine−rich repeat−containing G protein−coupled recep | system, NtrC family, phosphoglycerate transport system sensor hist | component system, OmpR family, clock−associated histidine kinase

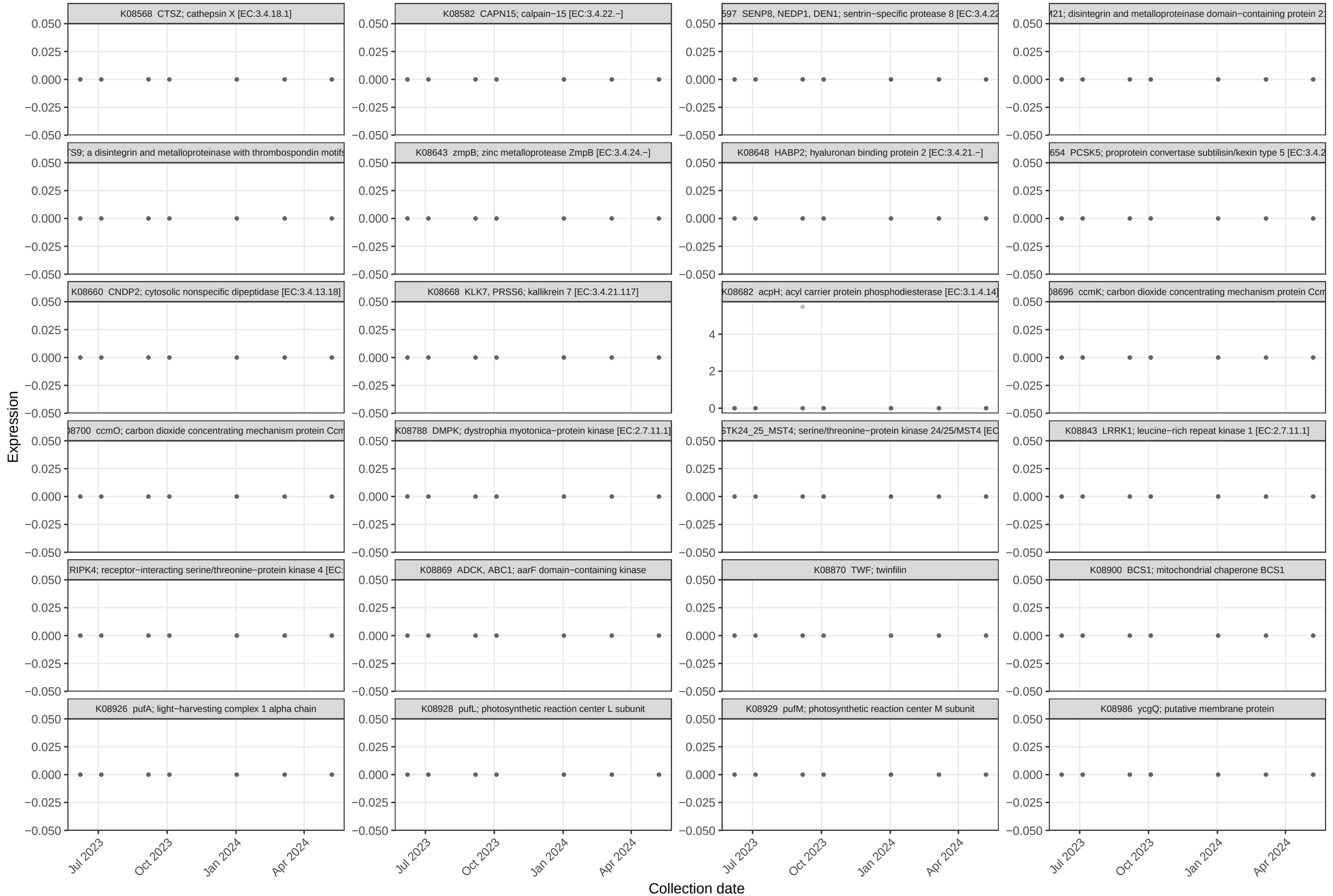
0.050 0.025 0.000 −0.025 −0.050

Jul 2023 Oct 2023 Jan 2024 Apr 2024

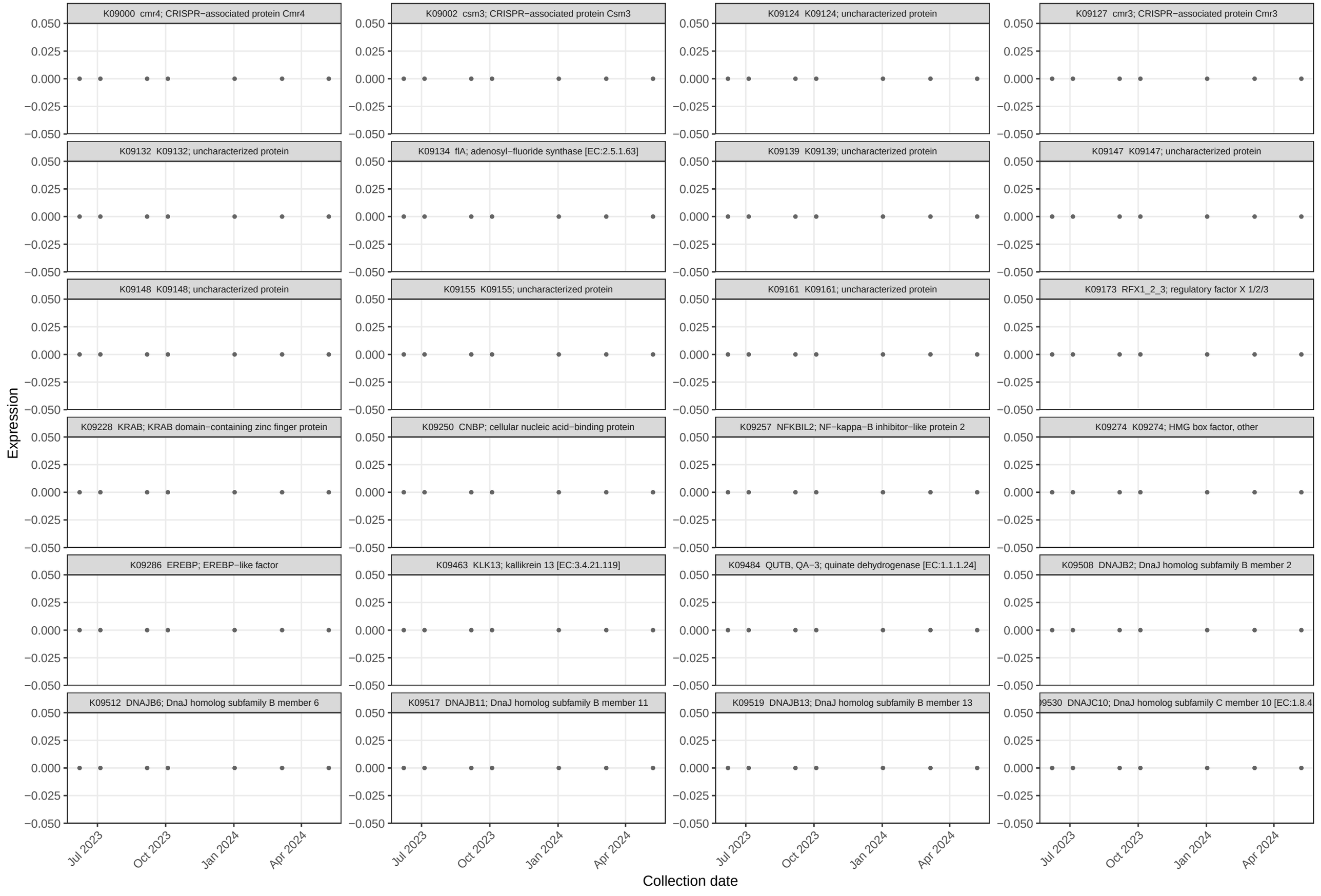
Expression

Collection date

All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

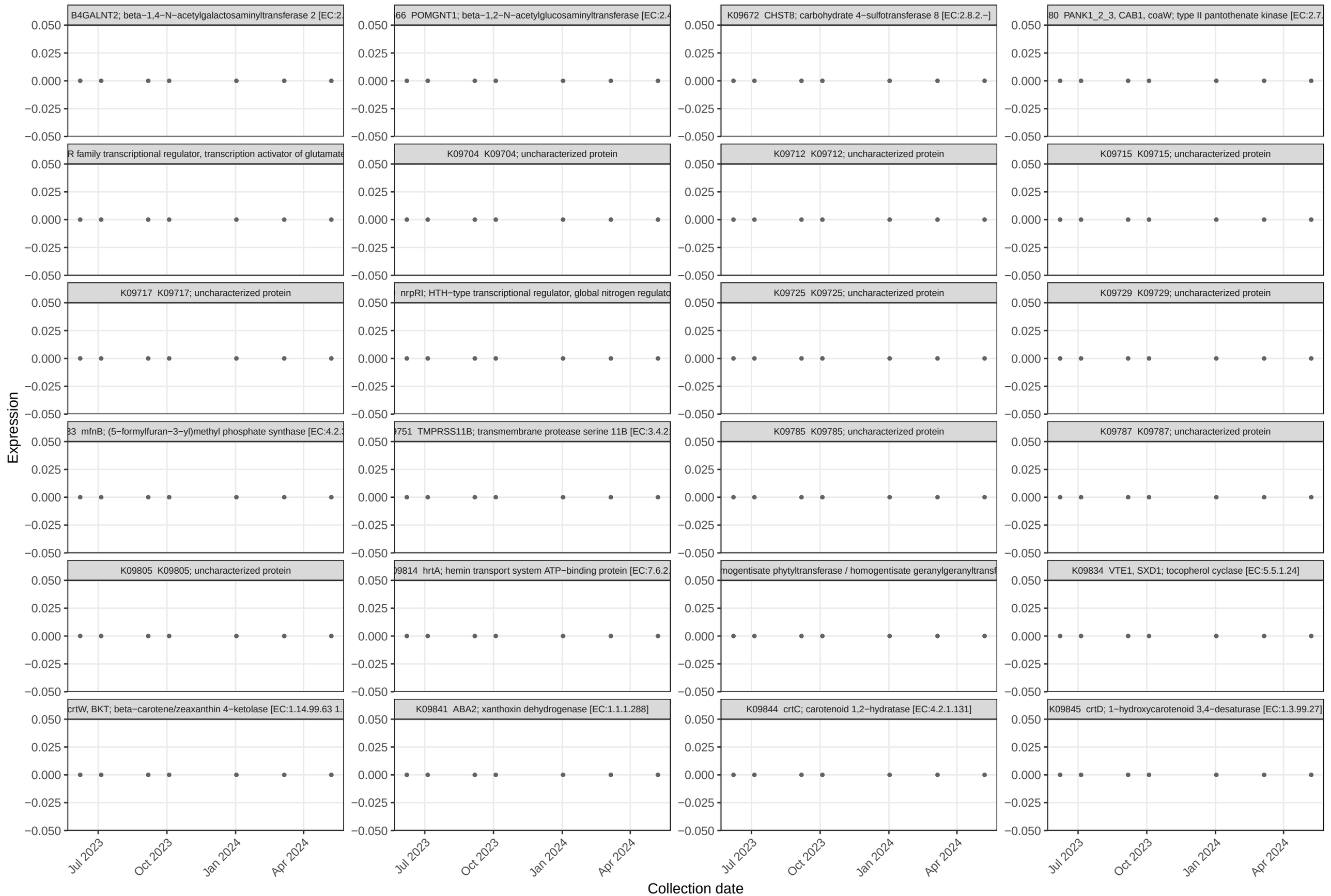


brite_D filtered temporal GAMs: page 247 of 303

All remaining BRITE classes after exclusions

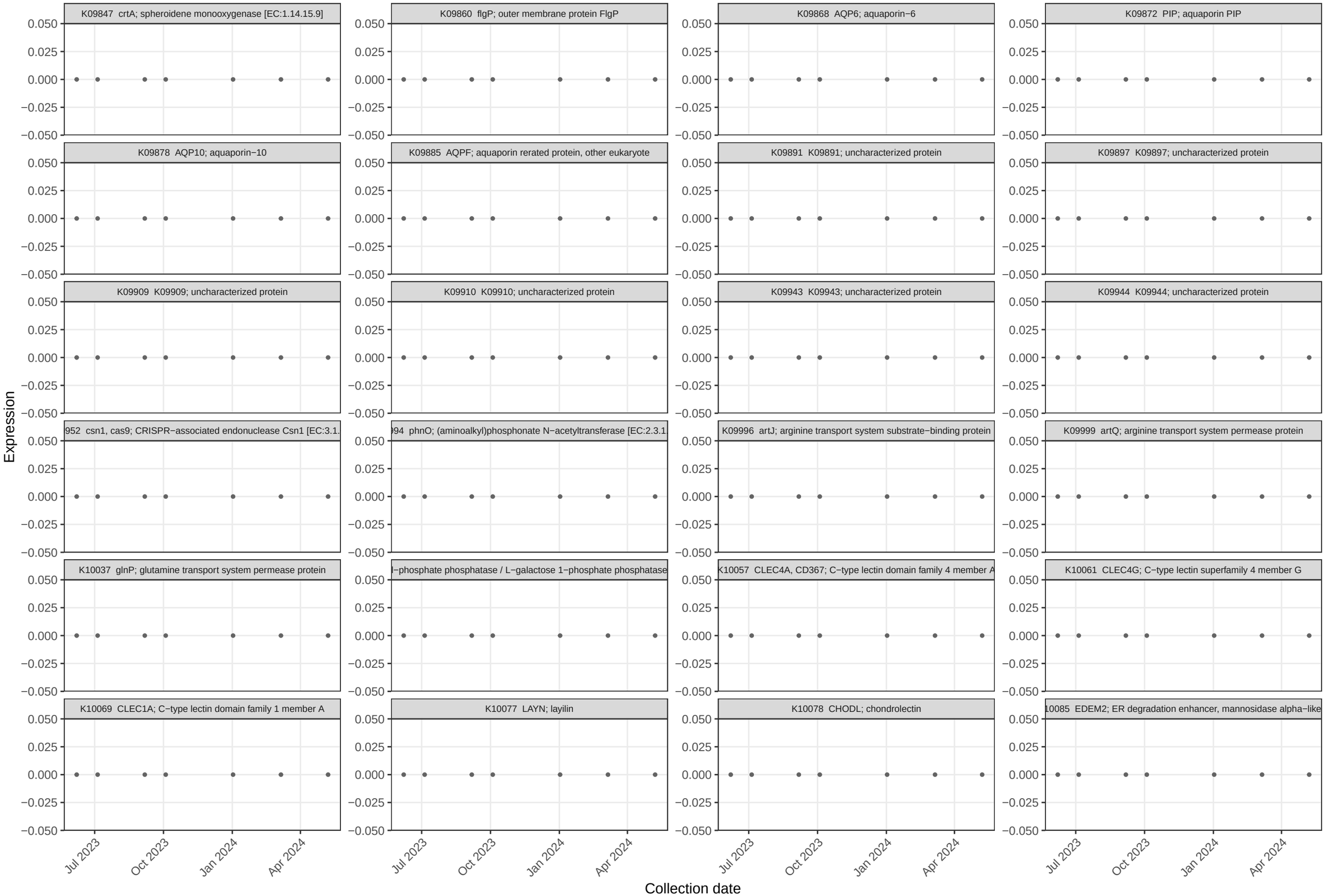
K09536 DNAJC16; DnaJ homolog subfamily C member 16
K09560 ST13; suppressor of tumorigenicity protein 13
09566 PPIG; peptidyl−prolyl isomerase G (cyclophilin G) [EC:5.2.1.
67 PPIH, CYPH; peptidyl−prolyl isomerase H (cyclophilin H) [EC:5.
K09572 FKBP6; FK506−binding protein 6 [EC:5.2.1.8]
K09577 FKBP14; FK506−binding protein 14 [EC:5.2.1.8]
K09580 PDIA1, P4HB; protein disulfide−isomerase A1 [EC:5.3.4.1]
09584 PDIA6, TXNDC7; protein disulfide−isomerase A6 [EC:5.3.4.
K09591 DET2; steroid 5−alpha−reductase [EC:1.3.1.22]
K09610 ECEL1; endothelin−converting enzyme−like 1 [EC:3.4.24.−
K09611 NPEPL1; probable aminopeptidase NPEPL1 [EC:3.4.11.−]
614 CORIN; atrial natriuretic peptide−converting enzyme [EC:3.4.2
K09615 TPSG1; tryptase gamma 1 [EC:3.4.21.−]
K09618 KLK9; kallikrein 9 [EC:3.4.21.−]
K09621 KLK12; kallikrein 12 [EC:3.4.21.−]
K09623 KLK15; kallikrein 15 [EC:3.4.21.−]
K09624 PRSS12; neurotrypsin [EC:3.4.21.−]
K09629 PRSS33; serine protease 33 [EC:3.4.21.−]
K09630 PRSS36; polyserase 2 [EC:3.4.21.−]
K09636 TMPRSS5; transmembrane protease serine 5 [EC:3.4.21.−
K09637 TMPRSS6; transmembrane protease serine 6 [EC:3.4.21.−
K09638 TMPRSS7; transmembrane protease serine 7 [EC:3.4.21.−
47 IMP1; mitochondrial inner membrane protease subunit 1 [EC:3.4
48 IMP2; mitochondrial inner membrane protease subunit 2 [EC:3.4
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Jul 2023
Oct 2023
Jan 2024
Apr 2024
Expression
Collection date

All remaining BRITE classes after exclusions



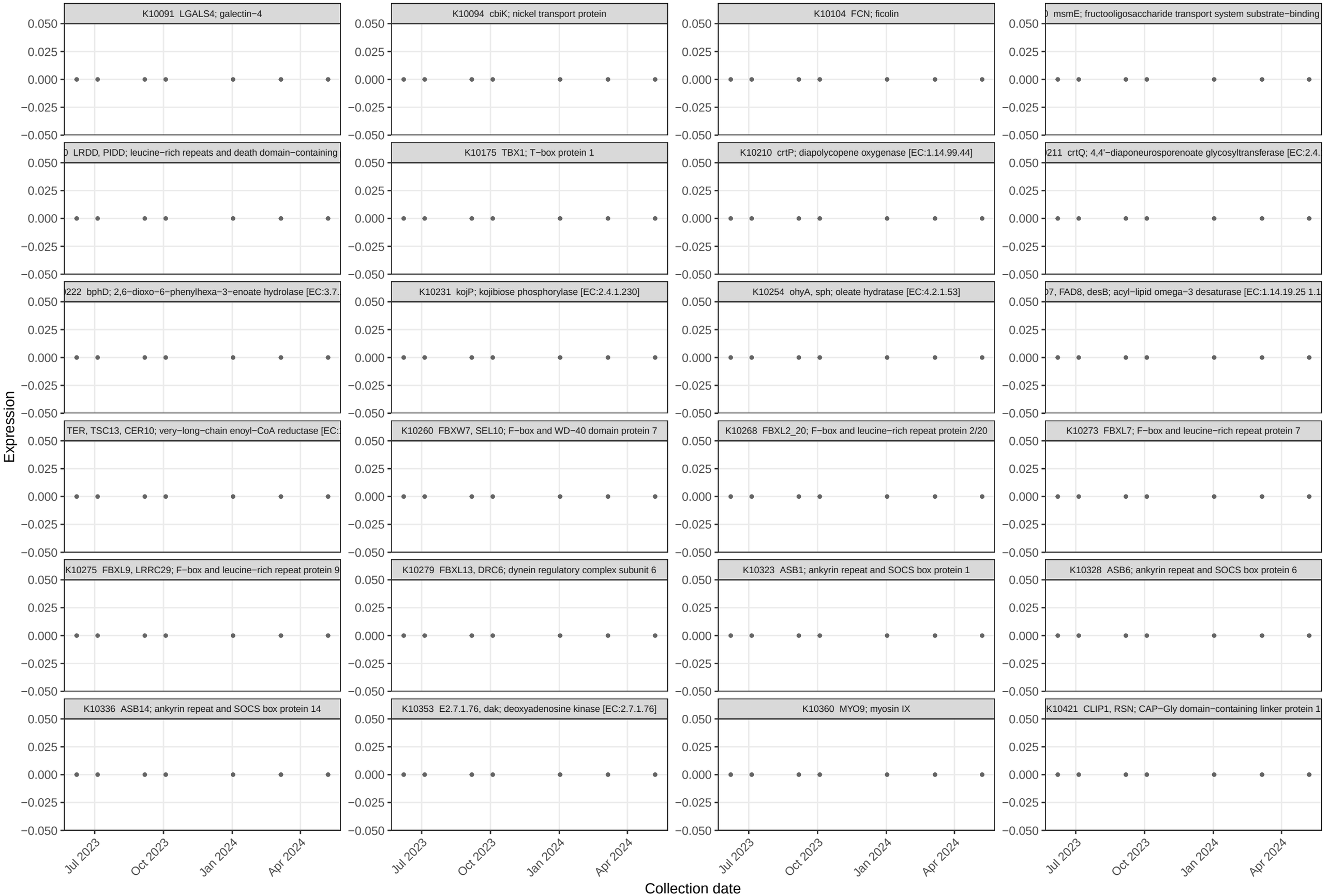
brite_D filtered temporal GAMs: page 249 of 303

All remaining BRITE classes after exclusions

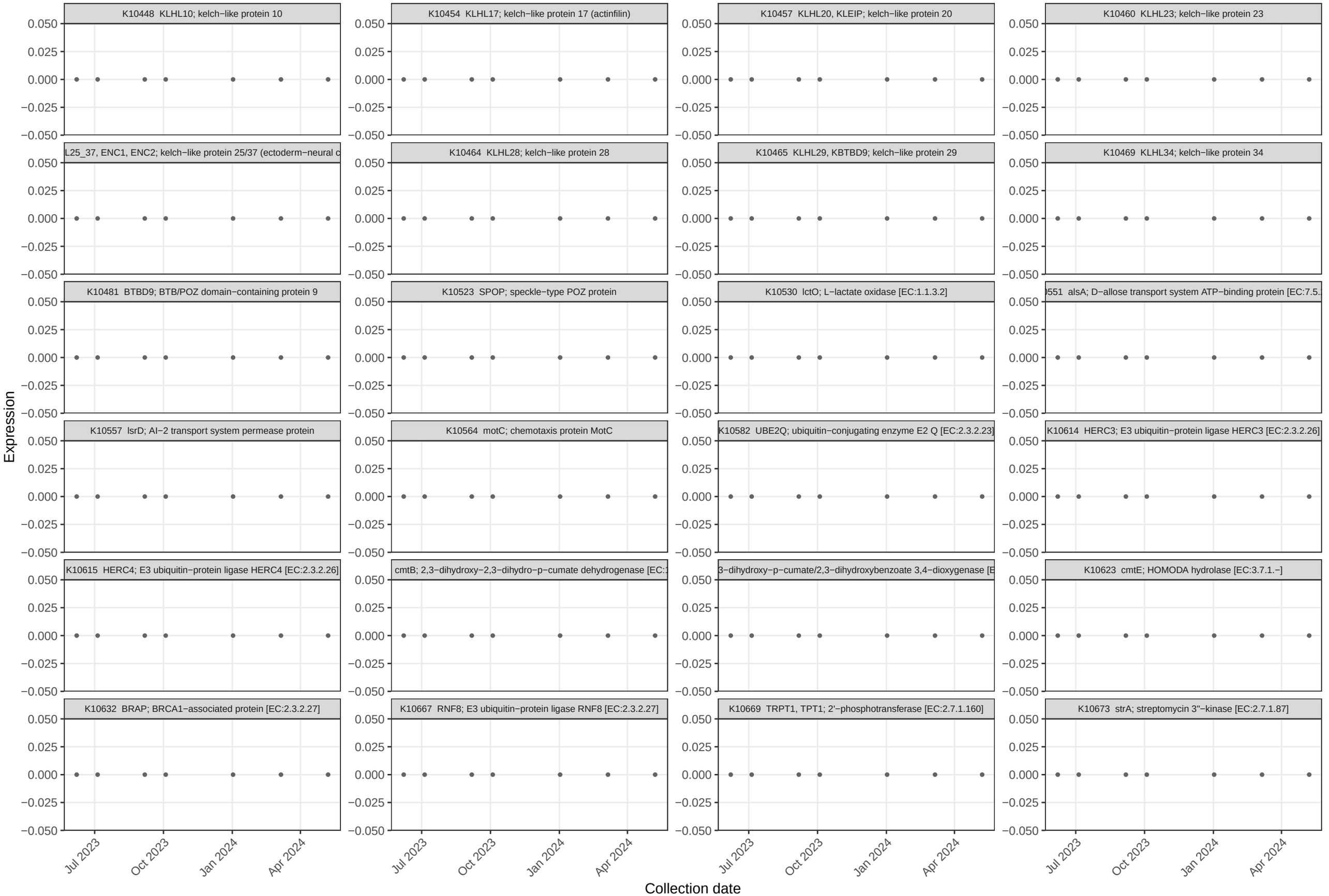


brite_D filtered temporal GAMs: page 250 of 303

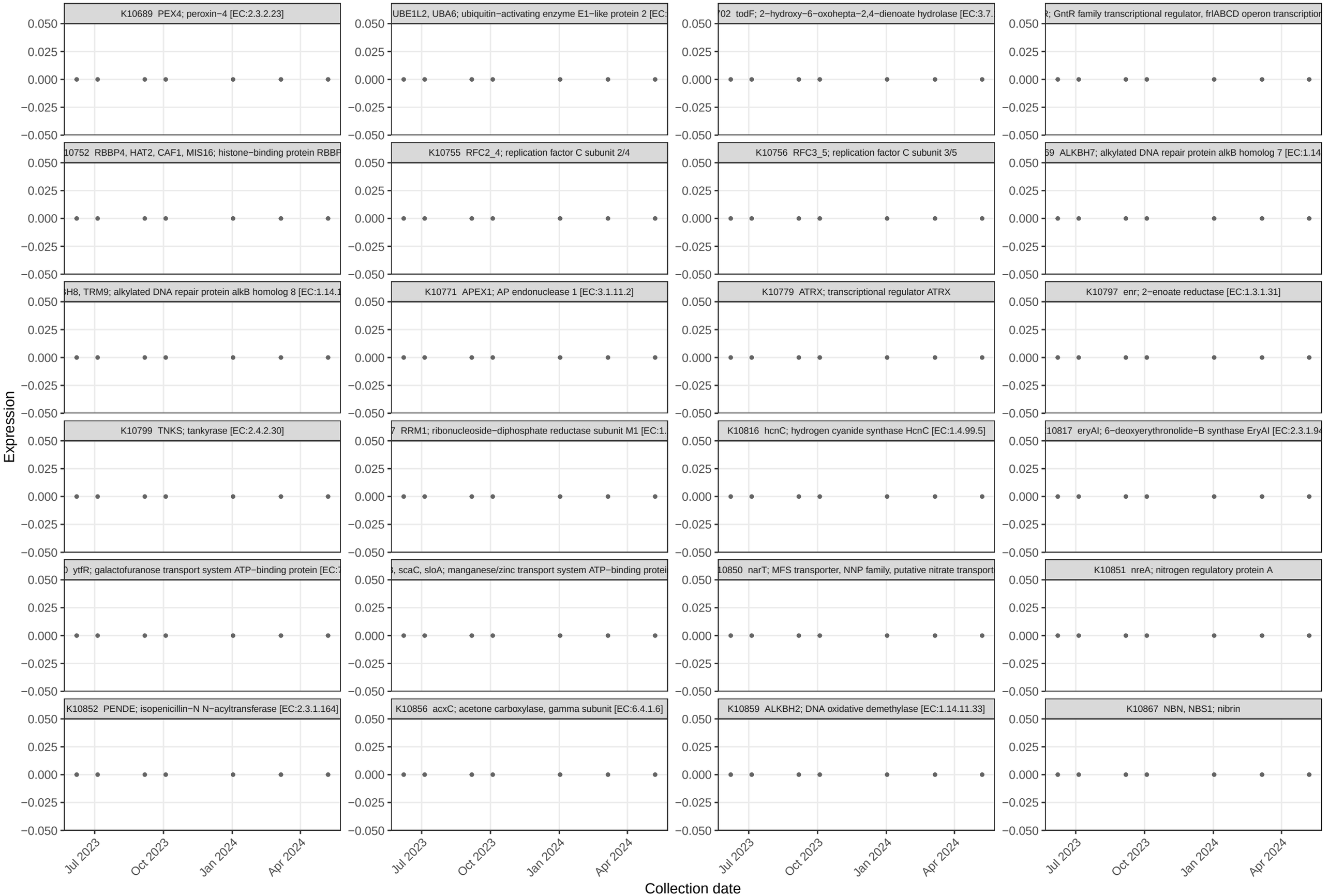
All remaining BRITE classes after exclusions



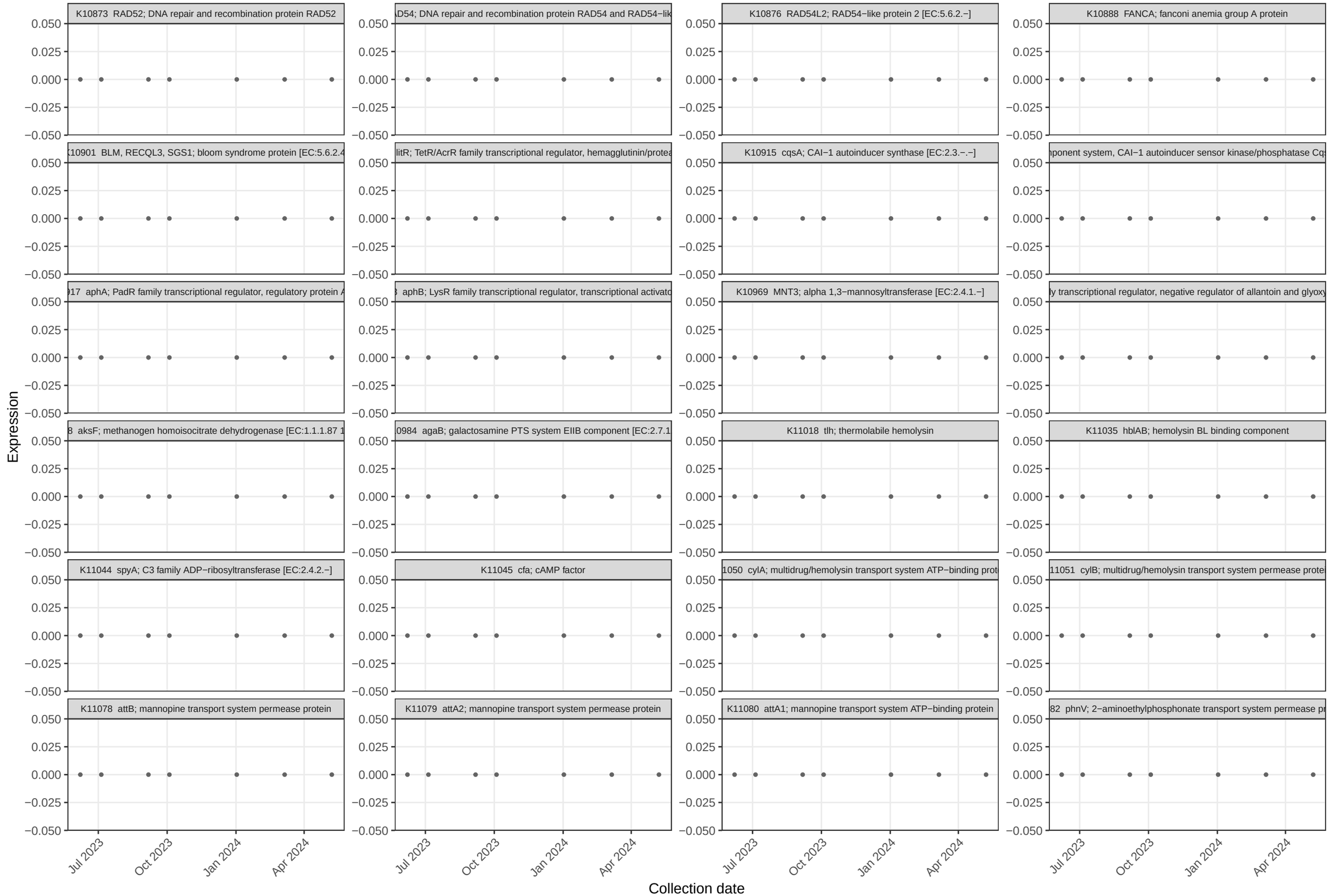
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

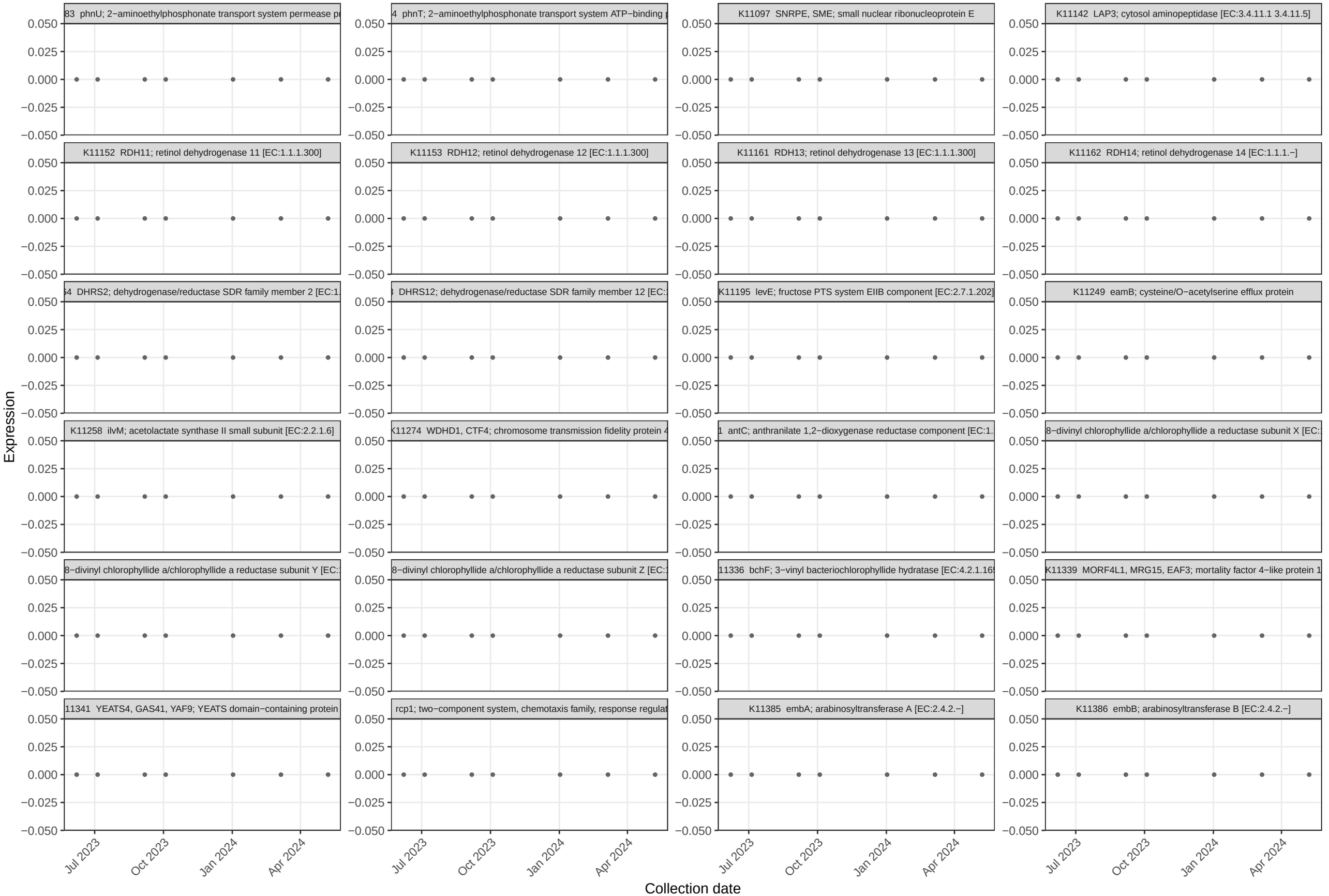


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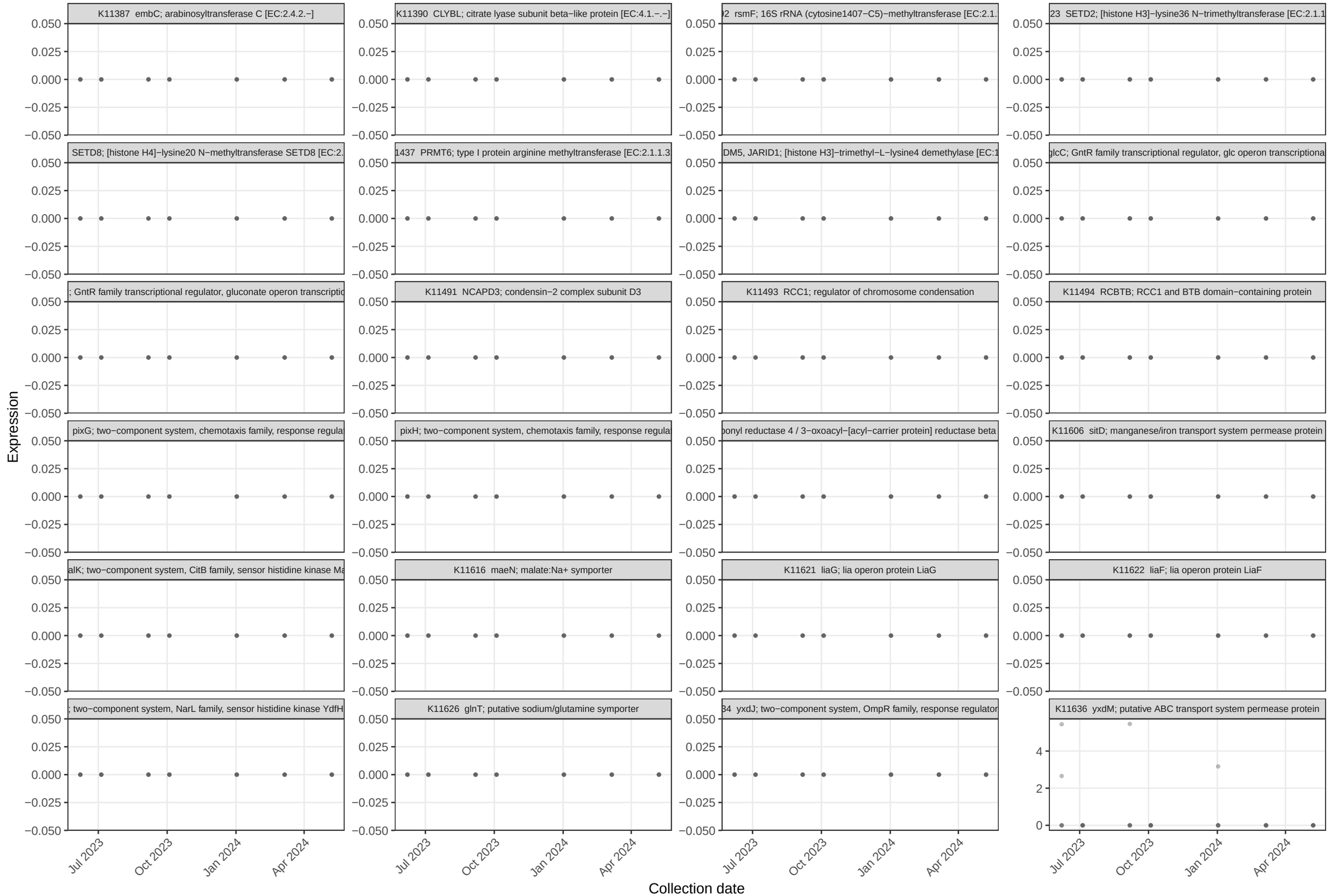


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All remaining BRITE classes after exclusions

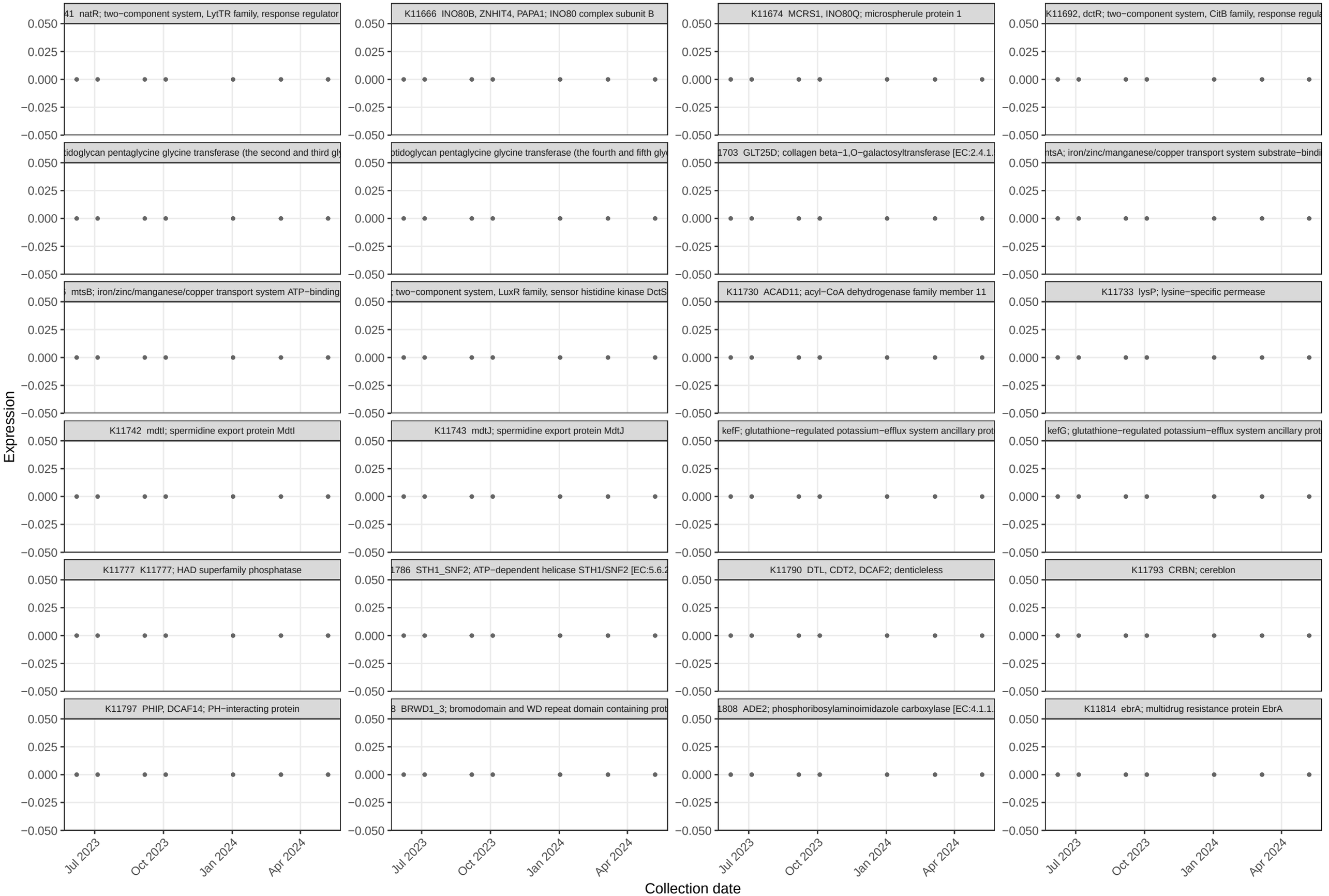


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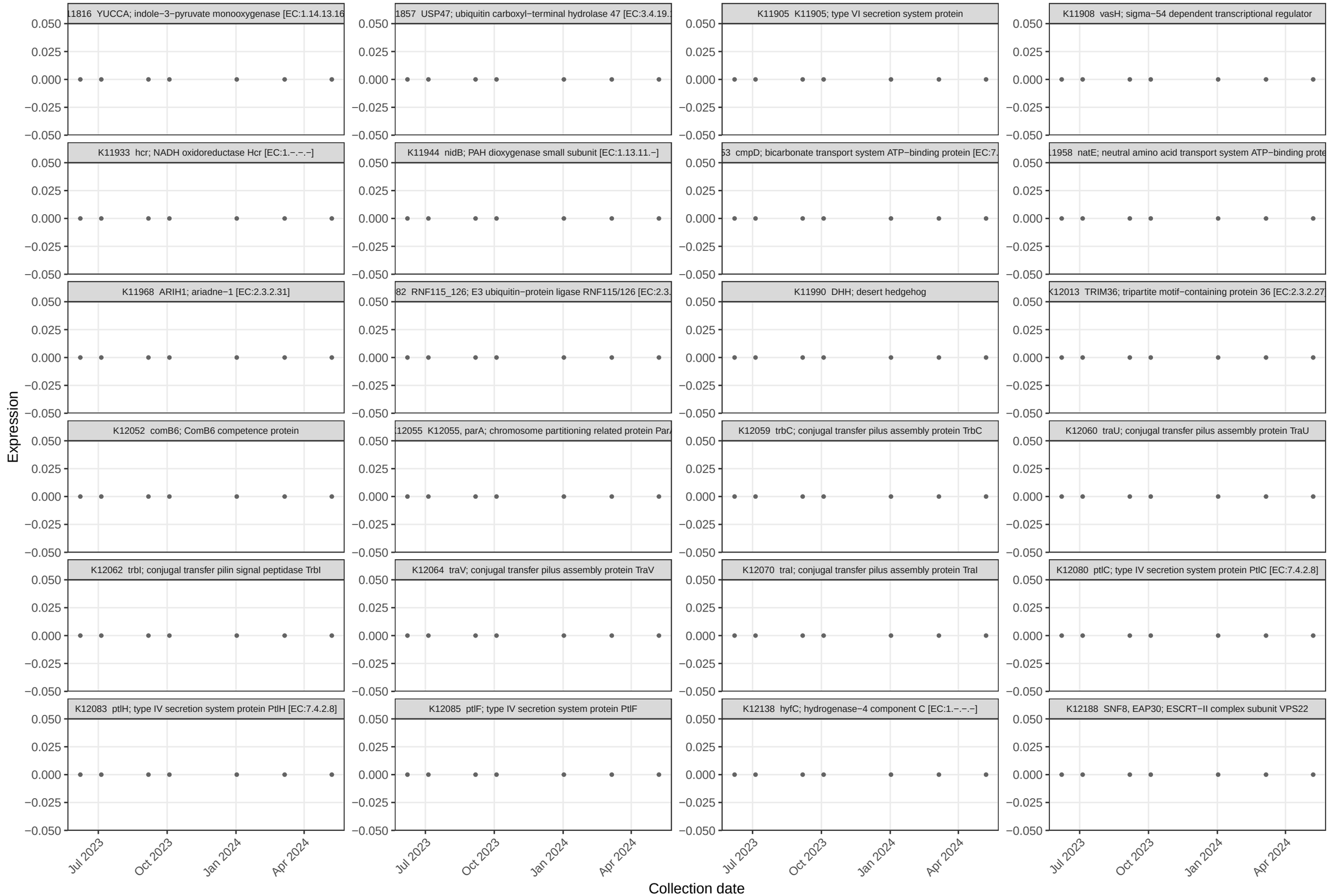


brite_D filtered temporal GAMs: page 256 of 303

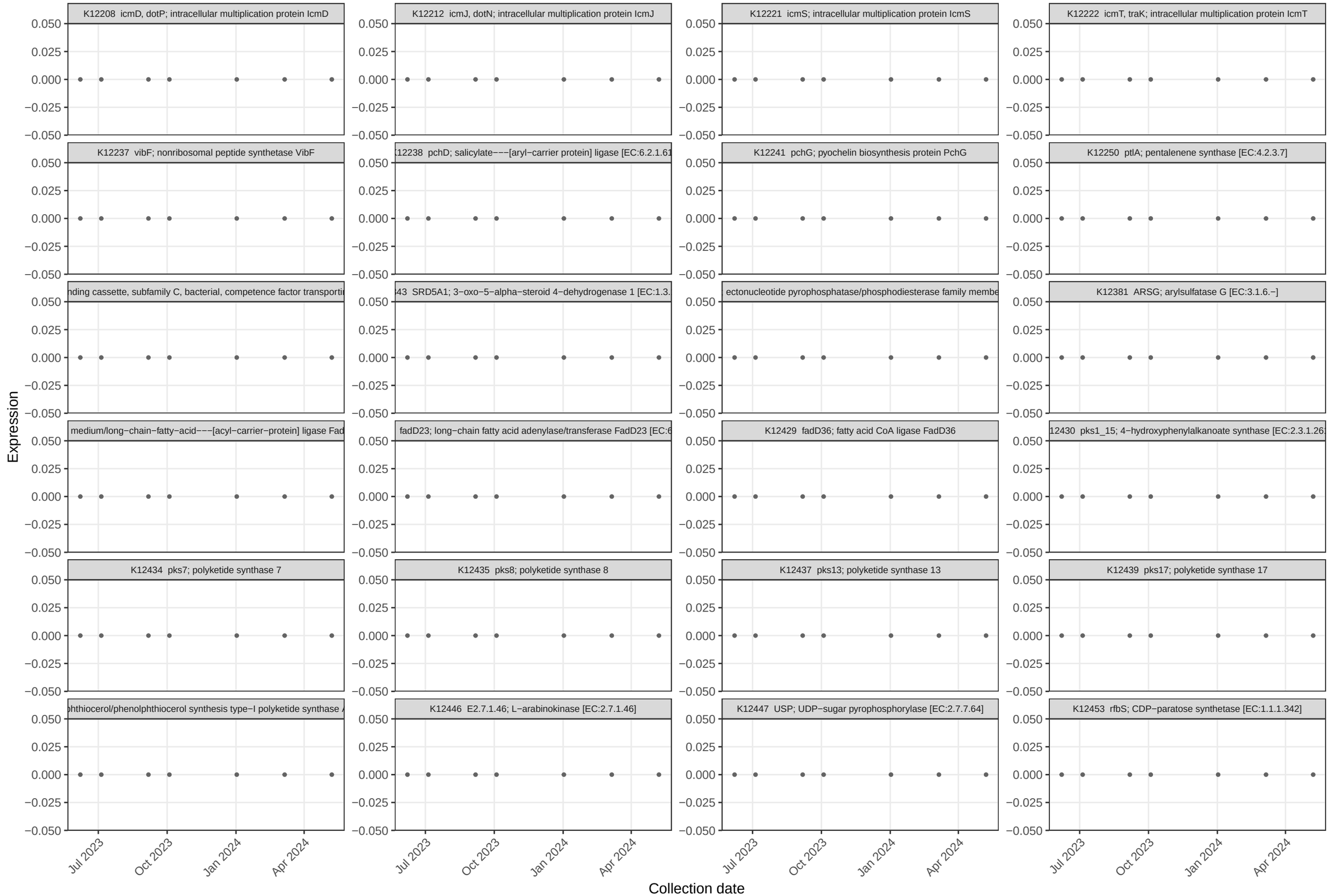
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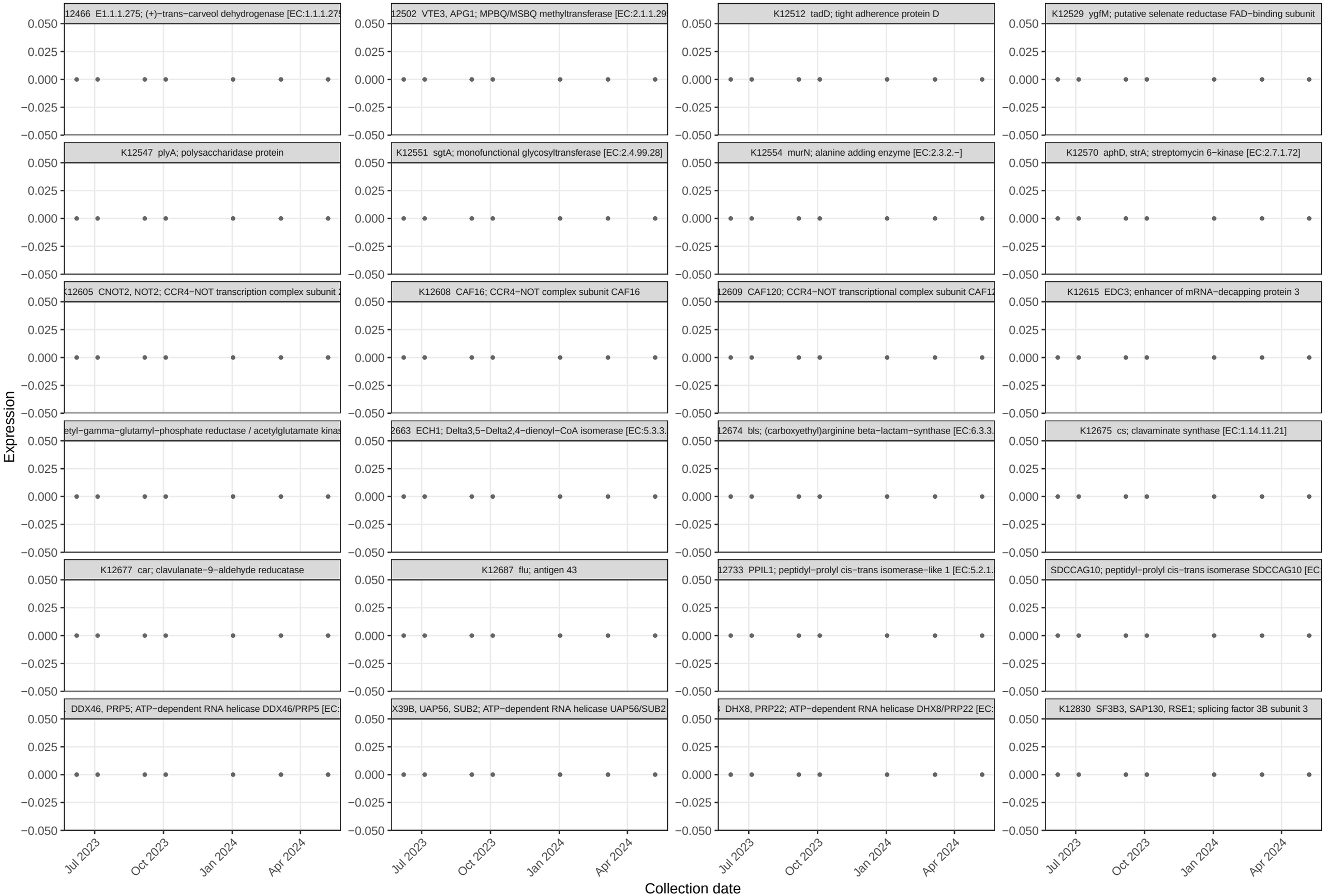
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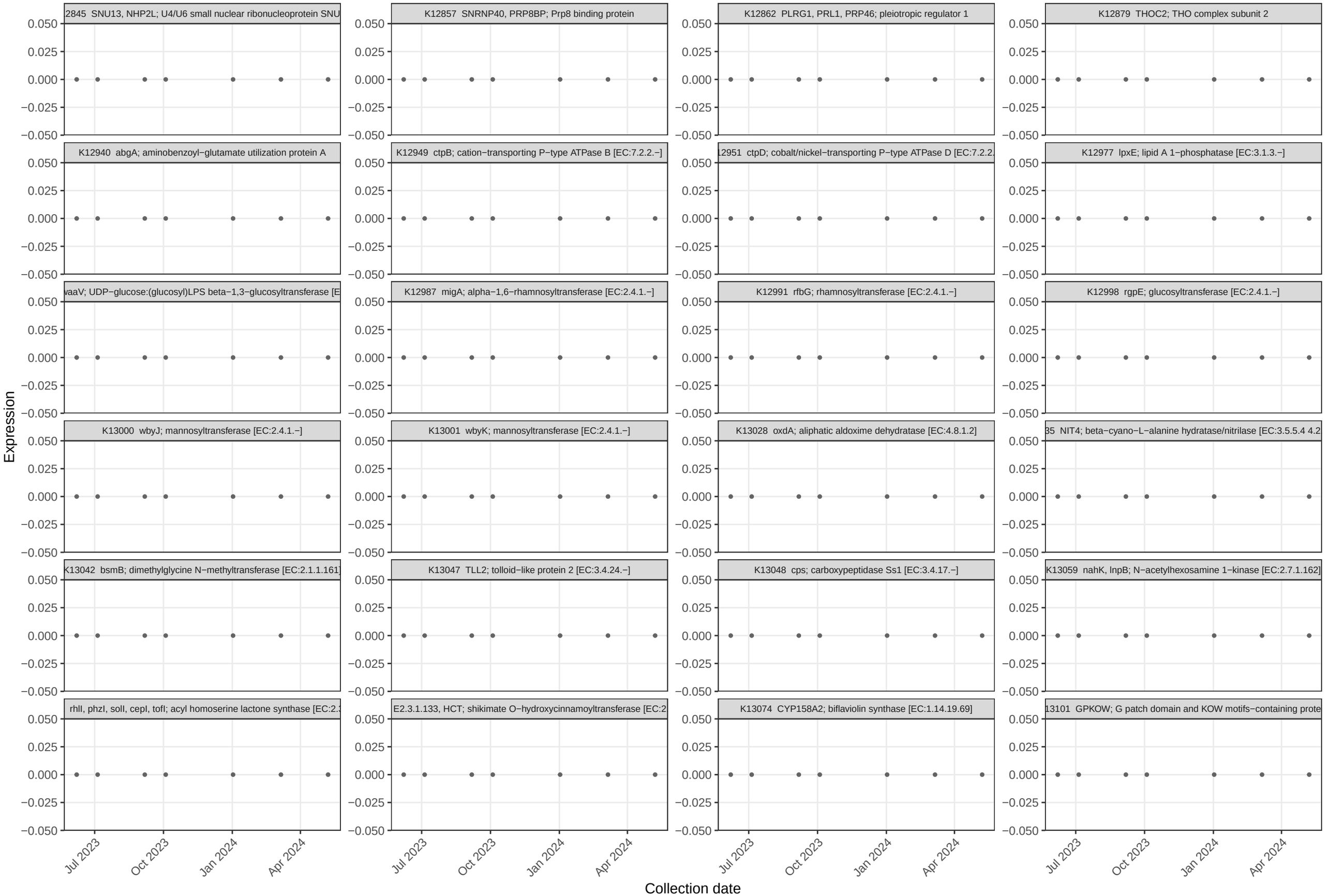
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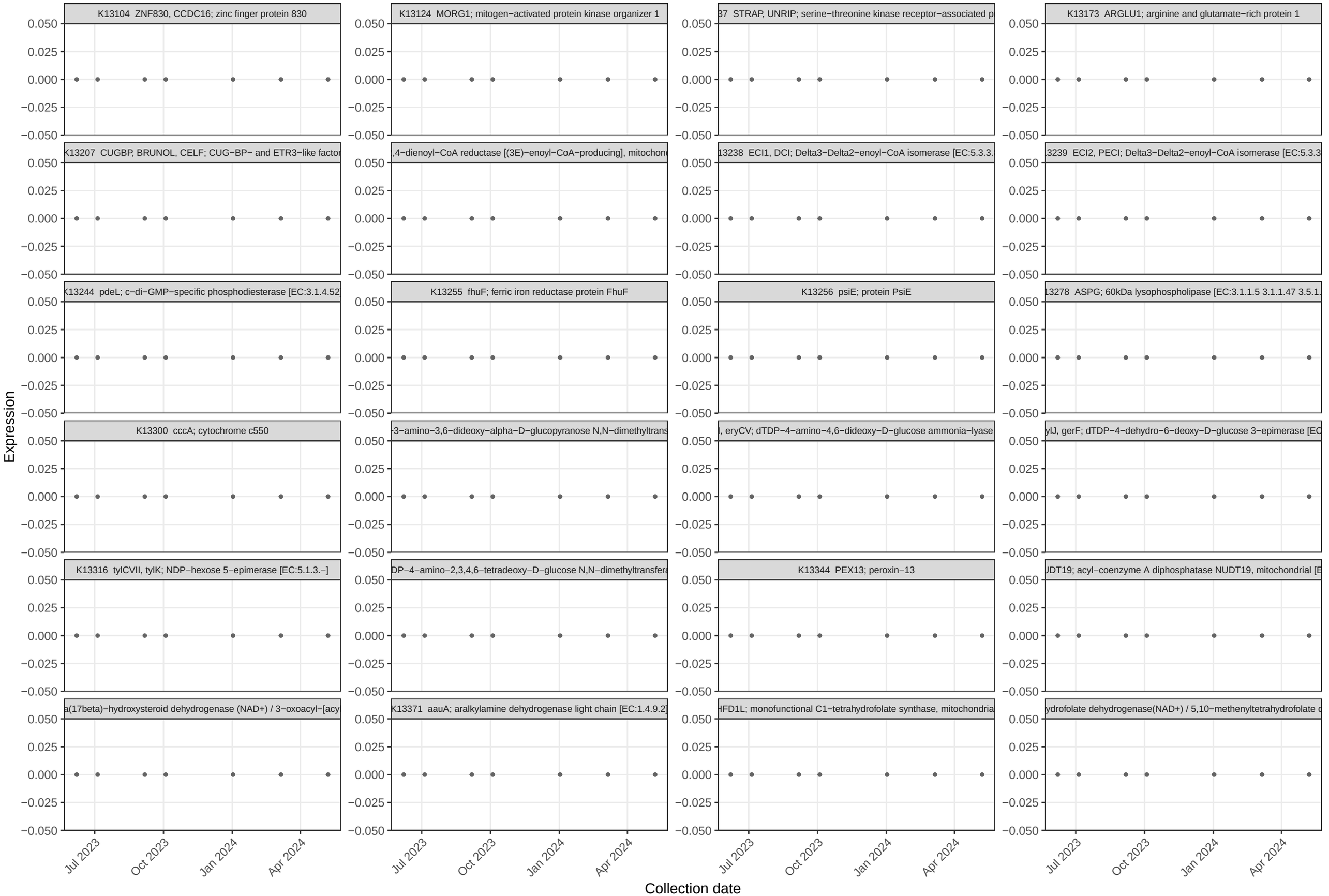
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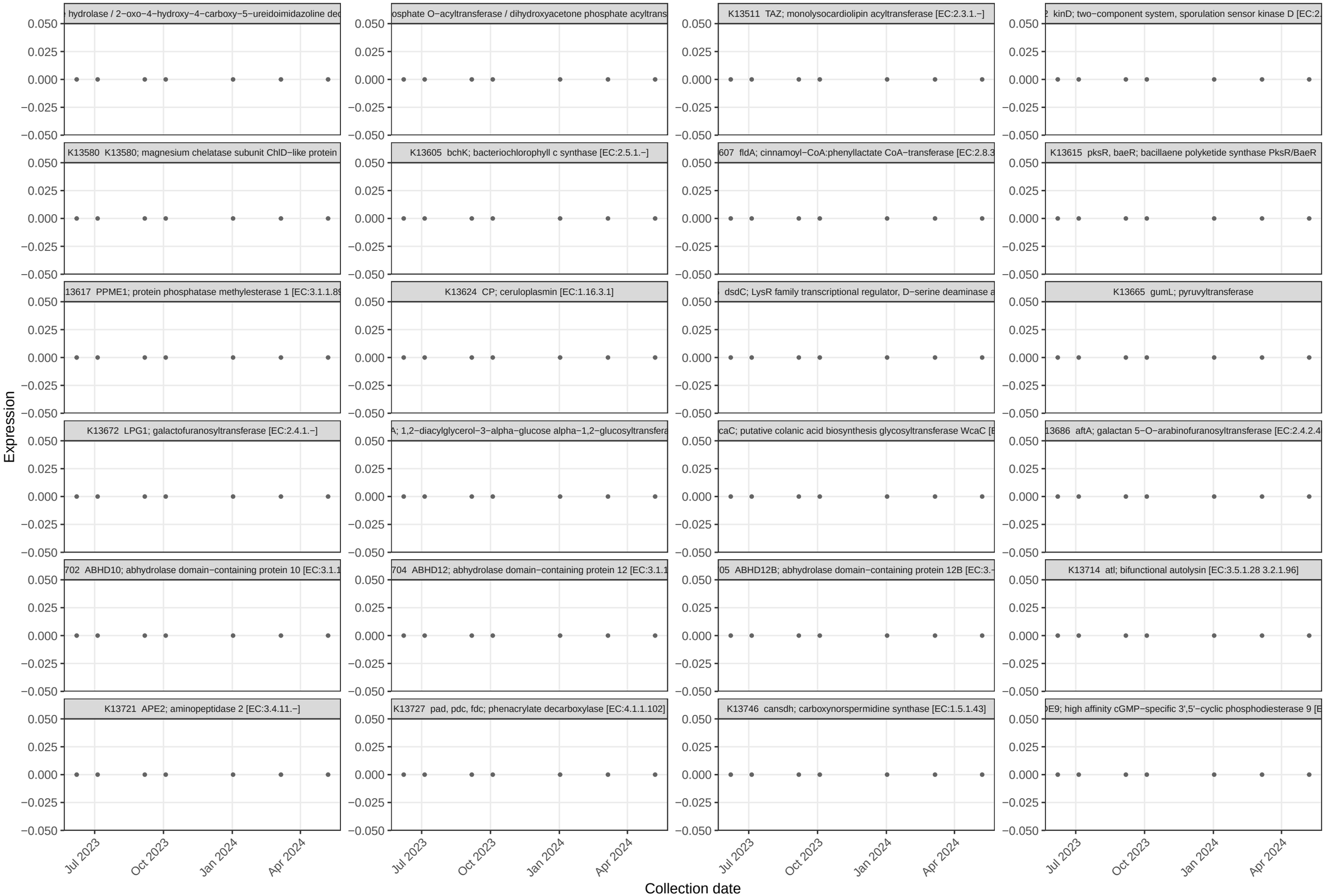
All remaining BRITE classes after exclusions



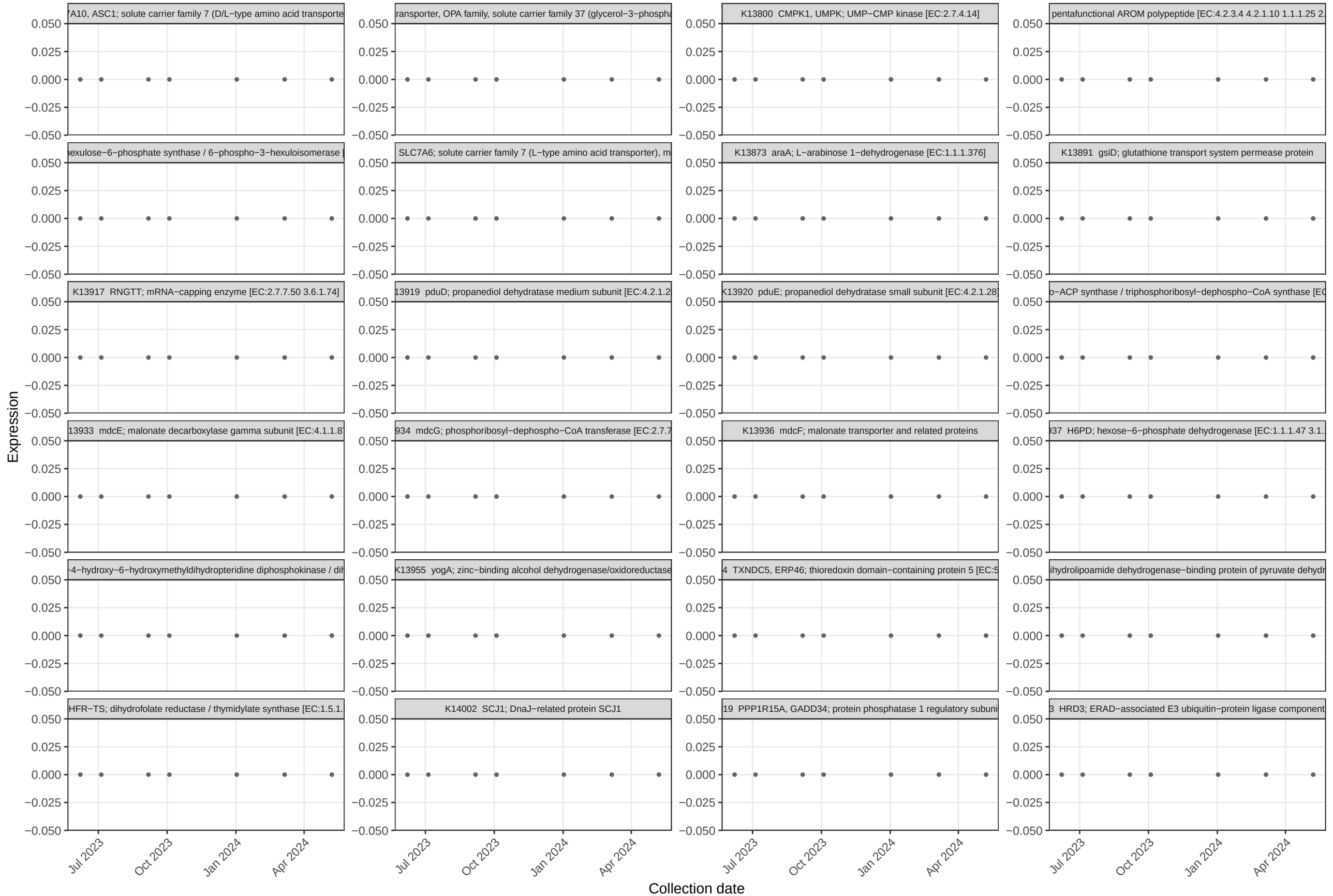
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

mdh2, mxaI; methanol dehydrogenase (cytochrome c) subunit 2 [EC

K14043 CPS4; syn−copalyl−diphosphate synthase [EC:5.5.1.14]

K14052 puuP; putrescine importer

K14089 echD; ech hydrogenase subunit D

K14091 echF; ech hydrogenase subunit F

vhuA, vhcA; F420−non−reducing hydrogenase large subunit [EC:1.1

sphohydrolase / phosphoribosyl−AMP cyclohydrolase / histidinol de

18S rRNA (adenine1779−N6/adenine1780−N6)−dimethyltransferase

K14250 oxyN, snoaM, dpsY, aknW, mtmY; cyclase

K14251 oxyF; C−methyltransferase [EC:2.1.1.−]

253 oxyE; 6−methylpretetramide 4−monooxygenase [EC:1.14.13.2

K14262 DUG2; di− and tripeptidase [EC:3.4.−.−]

-xylose 1−dehydrogenase (NADP+, D−xylono−1,5−lactone−forming

K14292 TGS1; trimethylguanosine synthase [EC:2.1.1.−]

K14325 RNPS1; RNA−binding protein with serine−rich domain 1

6 UPF1, RENT1; regulator of nonsense transcripts 1 [EC:5.6.2.5 5

K14327 UPF2, RENT2; regulator of nonsense transcripts 2

K14331 K14331; fatty aldehyde decarbonylase [EC:4.1.99.5]

K14358 SCRN; secernin

K14369 eryG; erythromycin 3"−O−methyltransferase [EC:2.1.1.254]

4370 eryK, CYP113A; erythromycin 12 hydroxylase [EC:1.14.13.15]

K14374 aveD; avermectin B 5−O−methyltransferase [EC:2.1.1.−]

K14375 aveBI; dTDP−L−oleandrosyltransferase [EC:2.4.1.−]

C5A2, SGLT2; solute carrier family 5 (sodium/glucose cotransporter

Expression

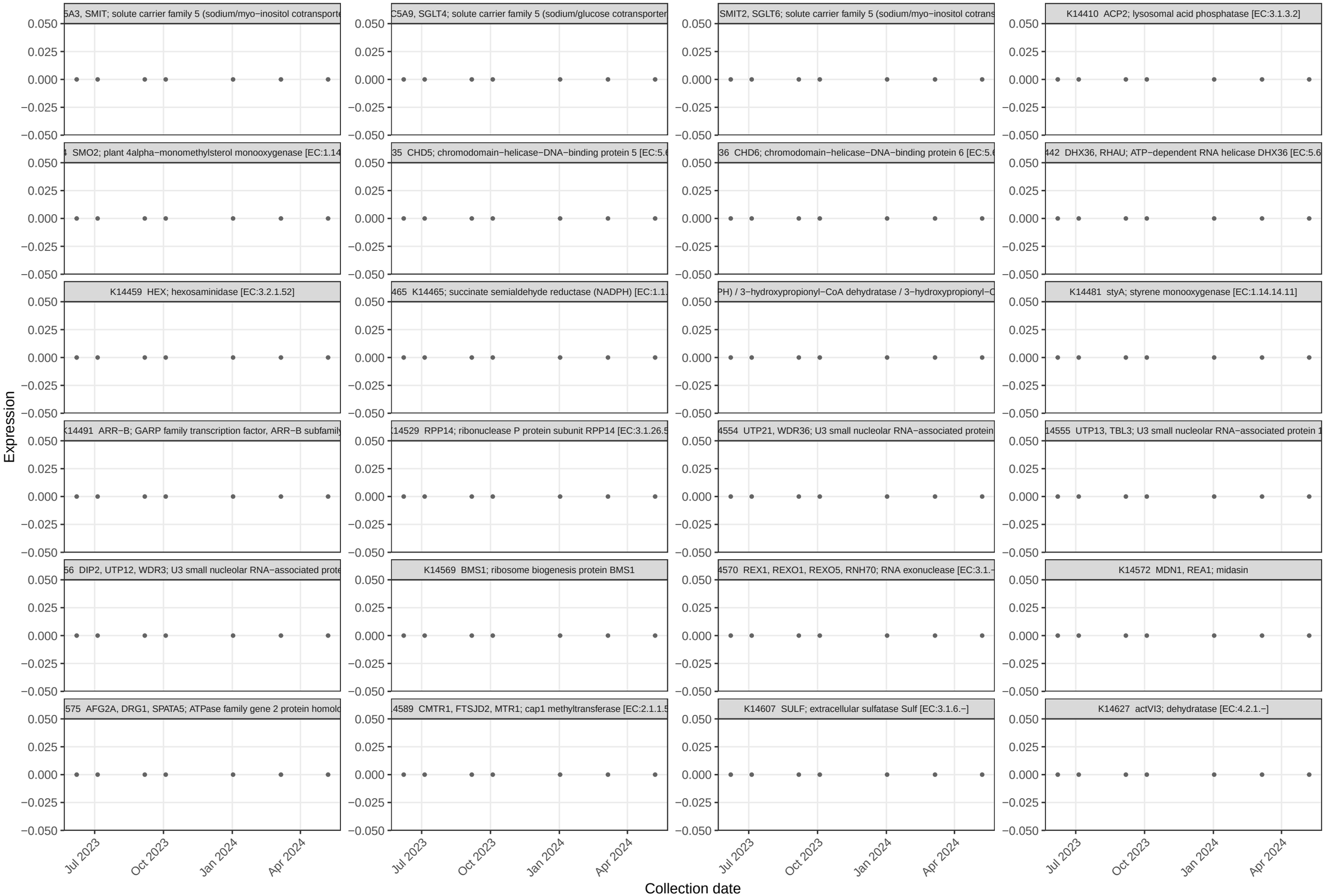
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Jul 2023
Oct 2023
Jan 2024
Apr 2024

Collection date

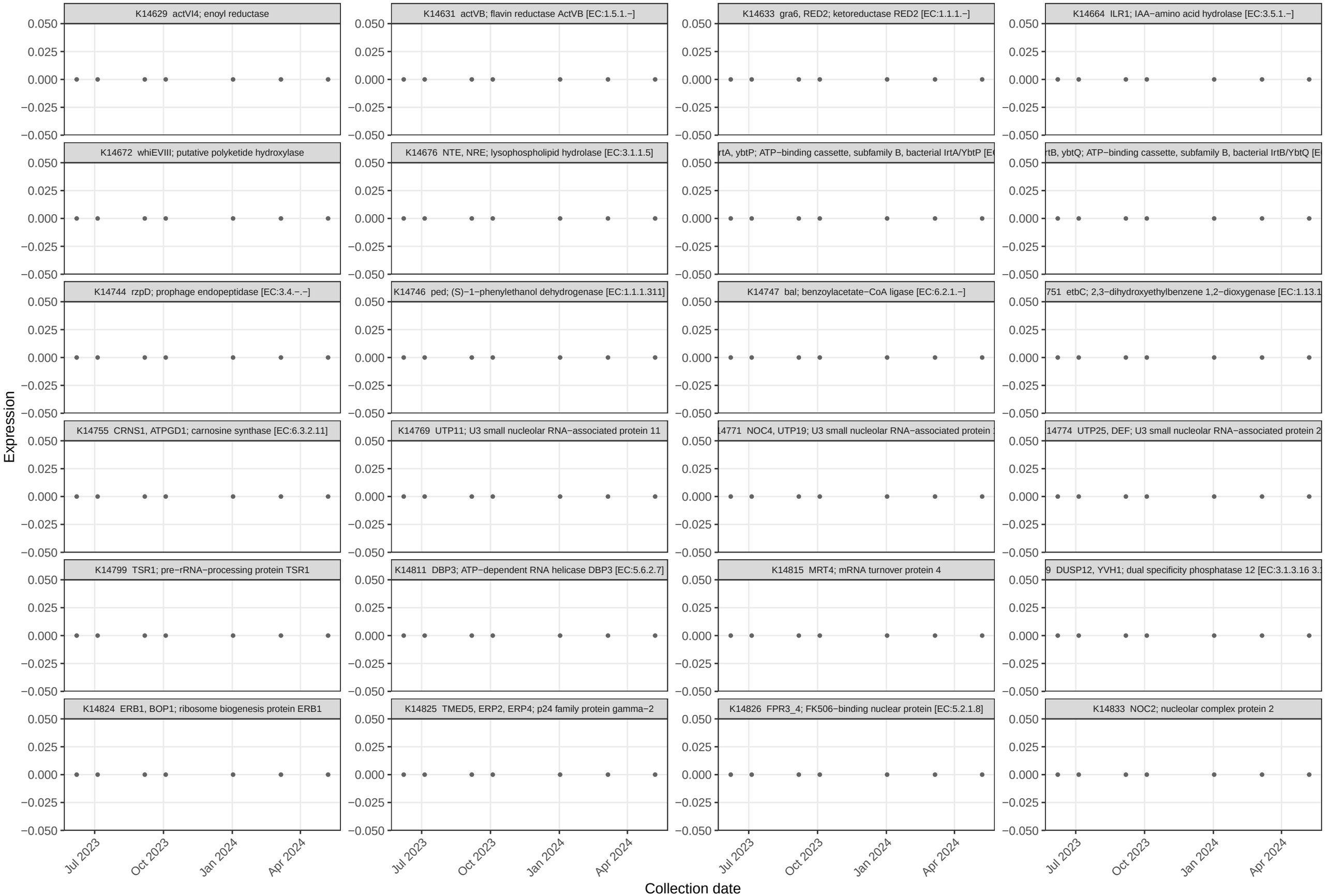
brite_D filtered temporal GAMs: page 265 of 303

All remaining BRITE classes after exclusions

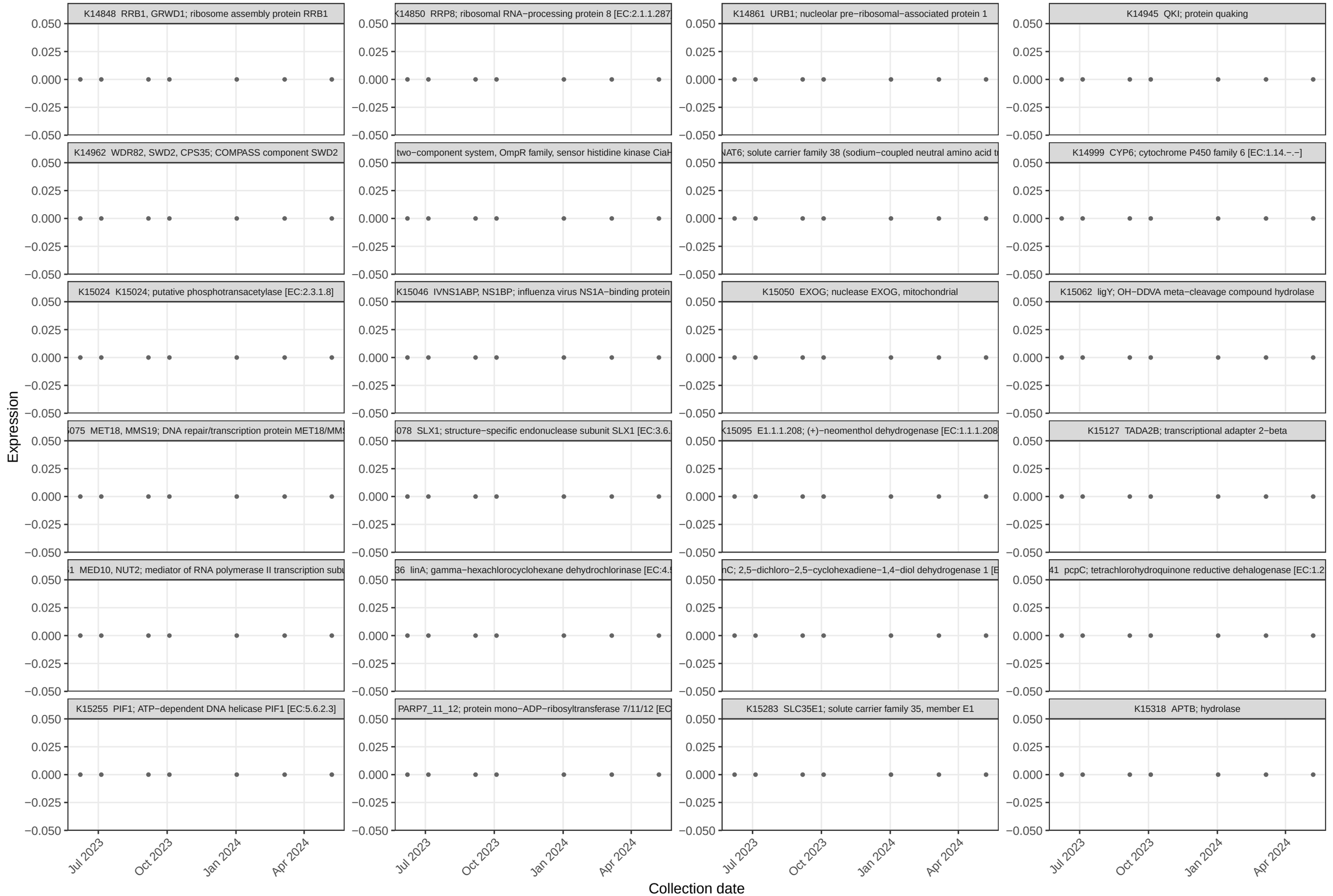


brite_D filtered temporal GAMs: page 266 of 303

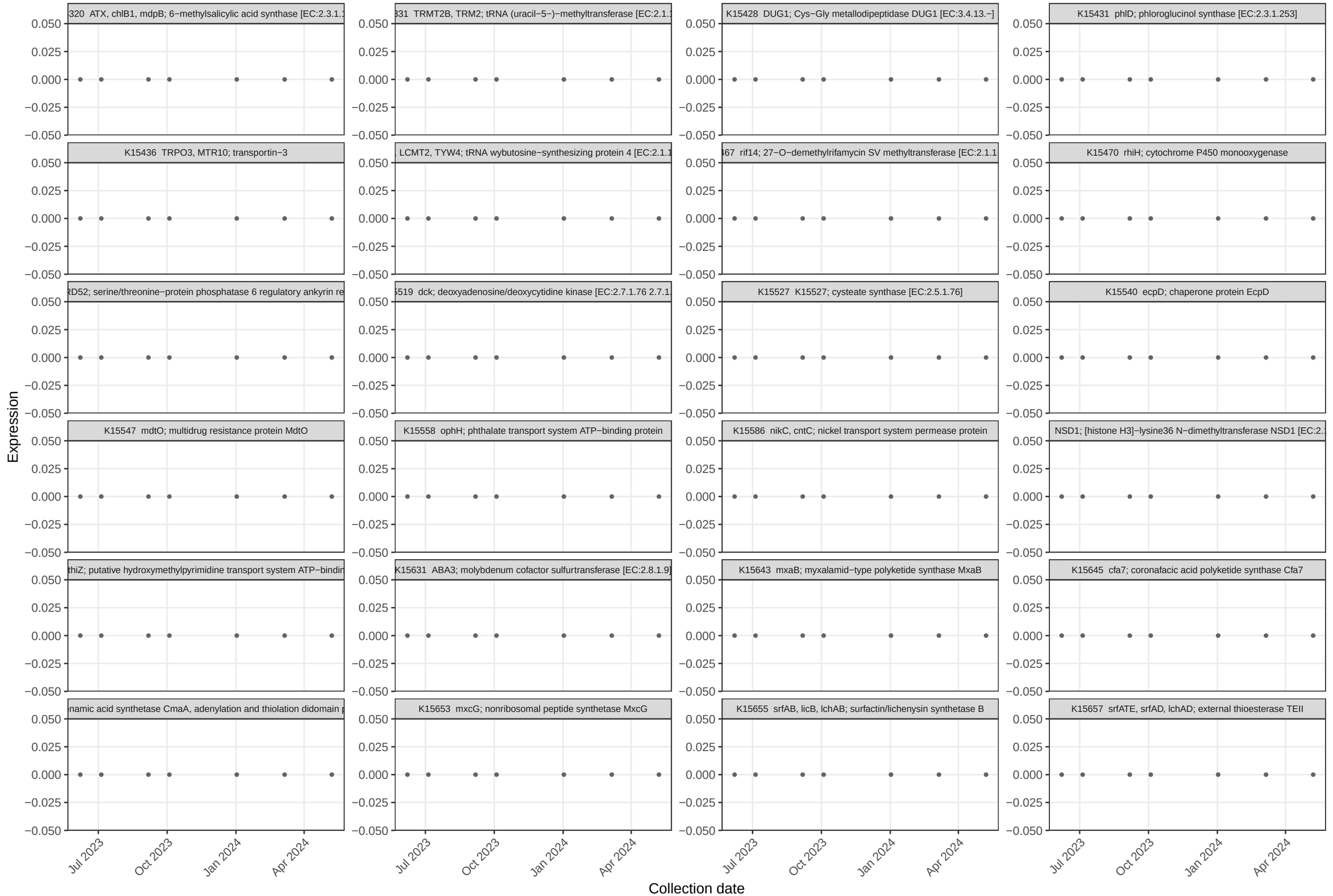
All remaining BRITE classes after exclusions



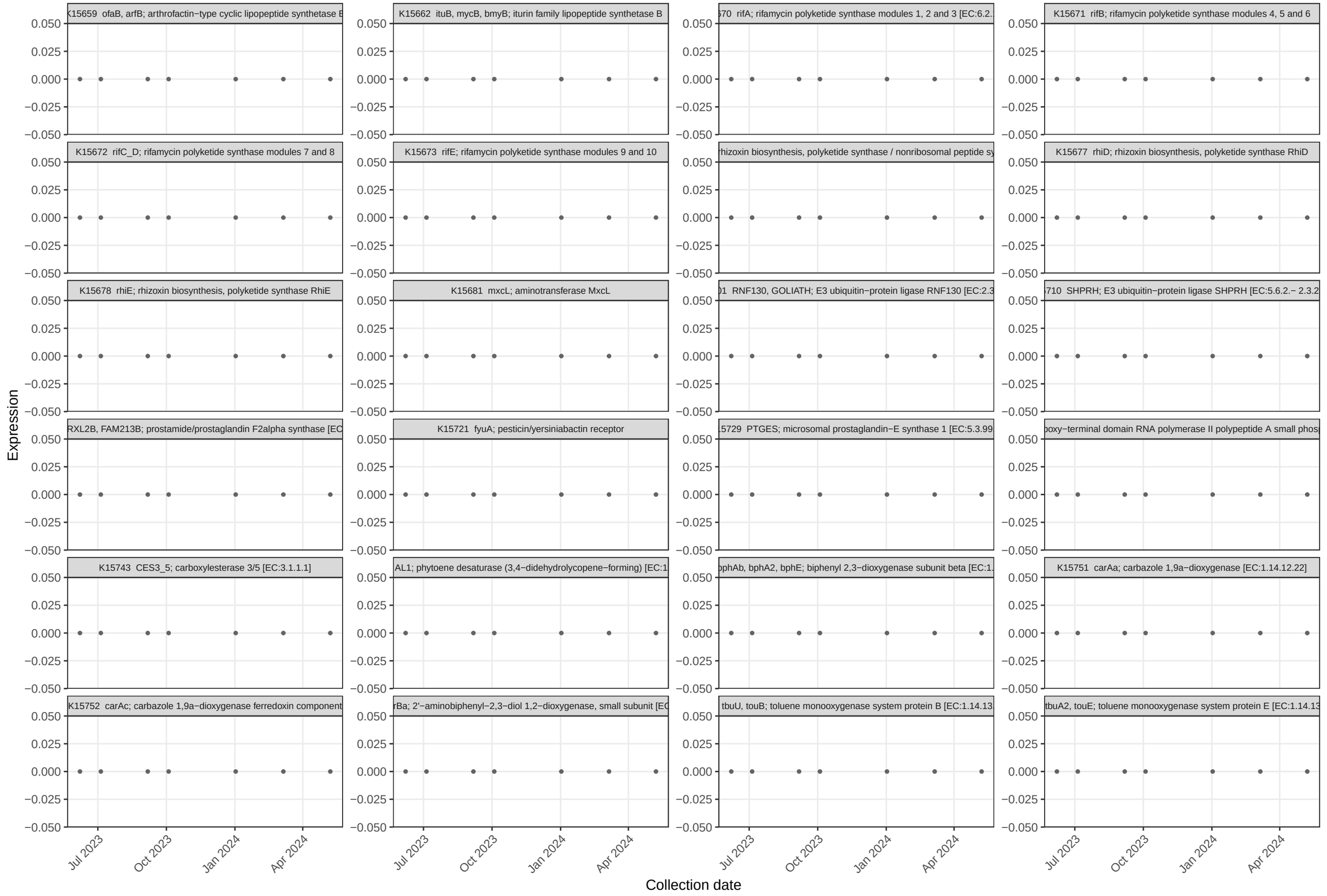
All remaining BRITE classes after exclusions



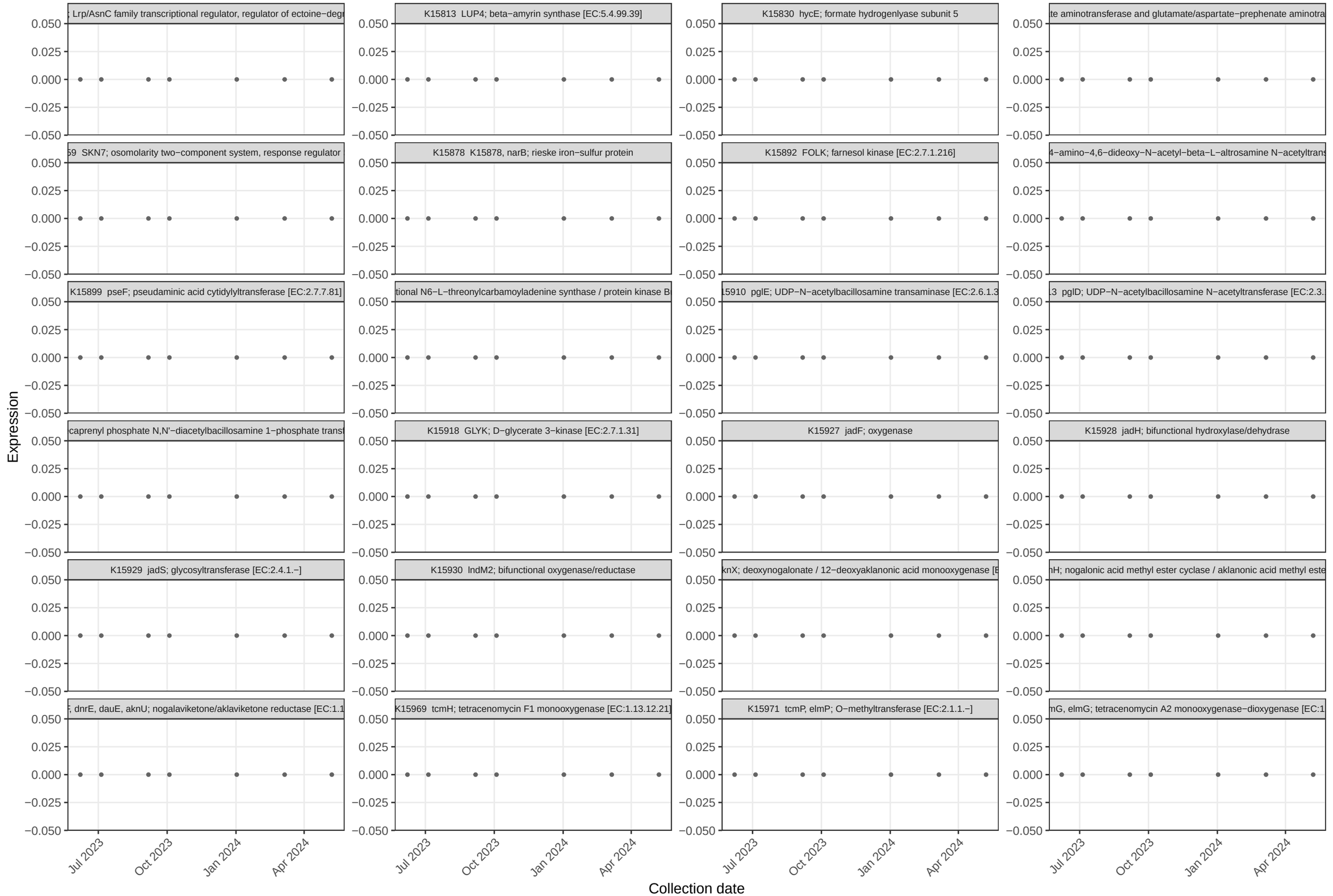
All remaining BRITE classes after exclusions



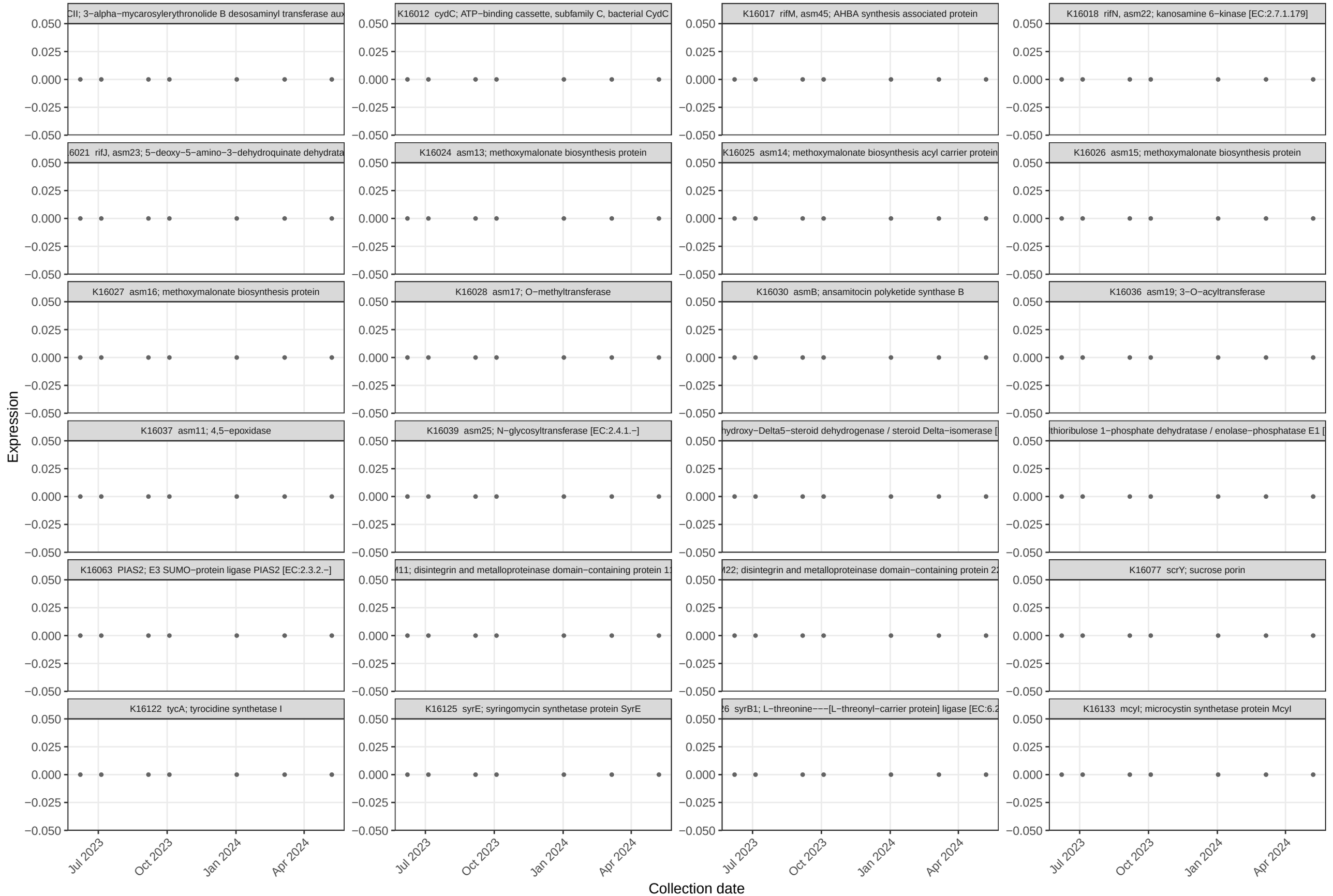
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

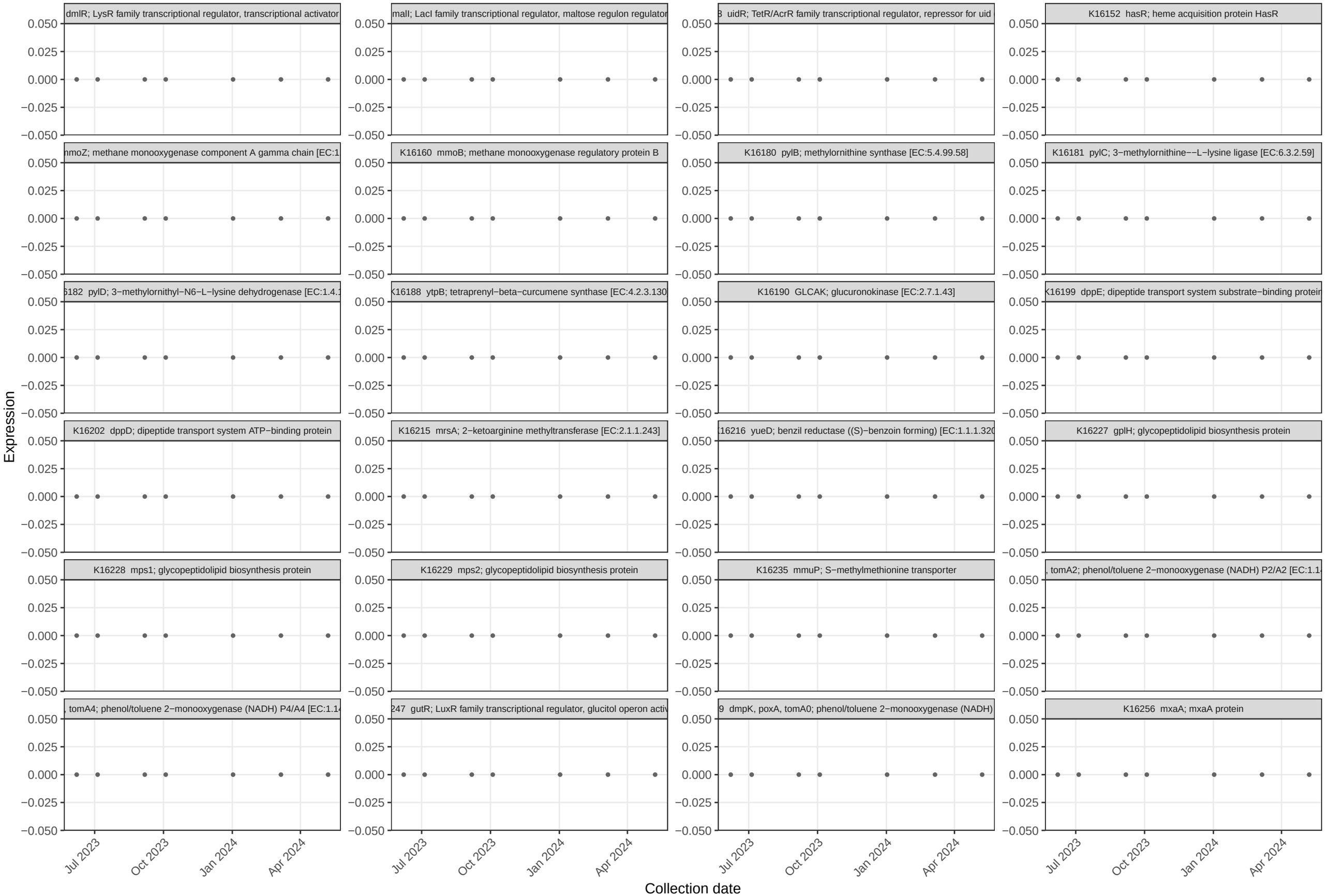


All remaining BRITE classes after exclusions

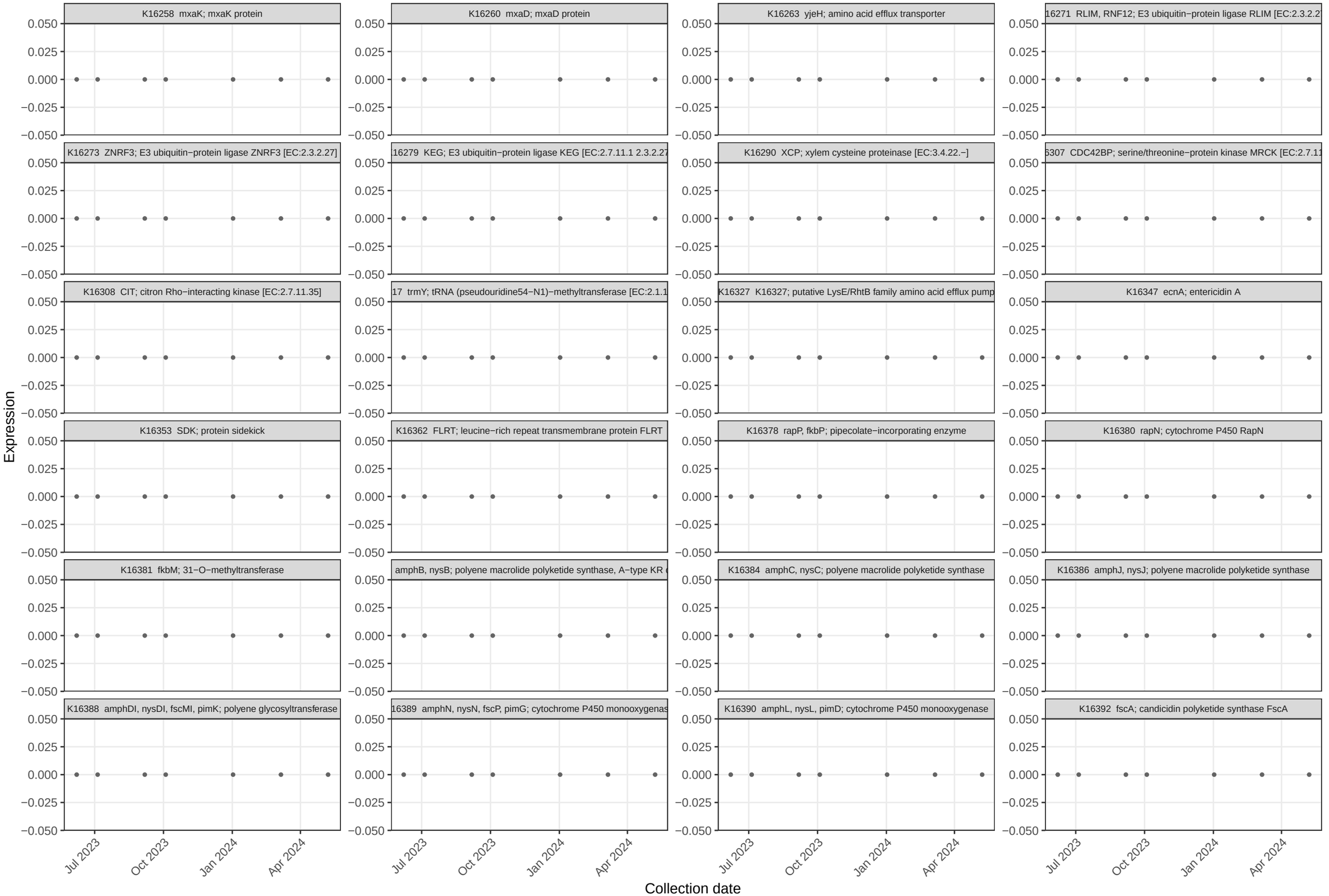


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All remaining BRITE classes after exclusions

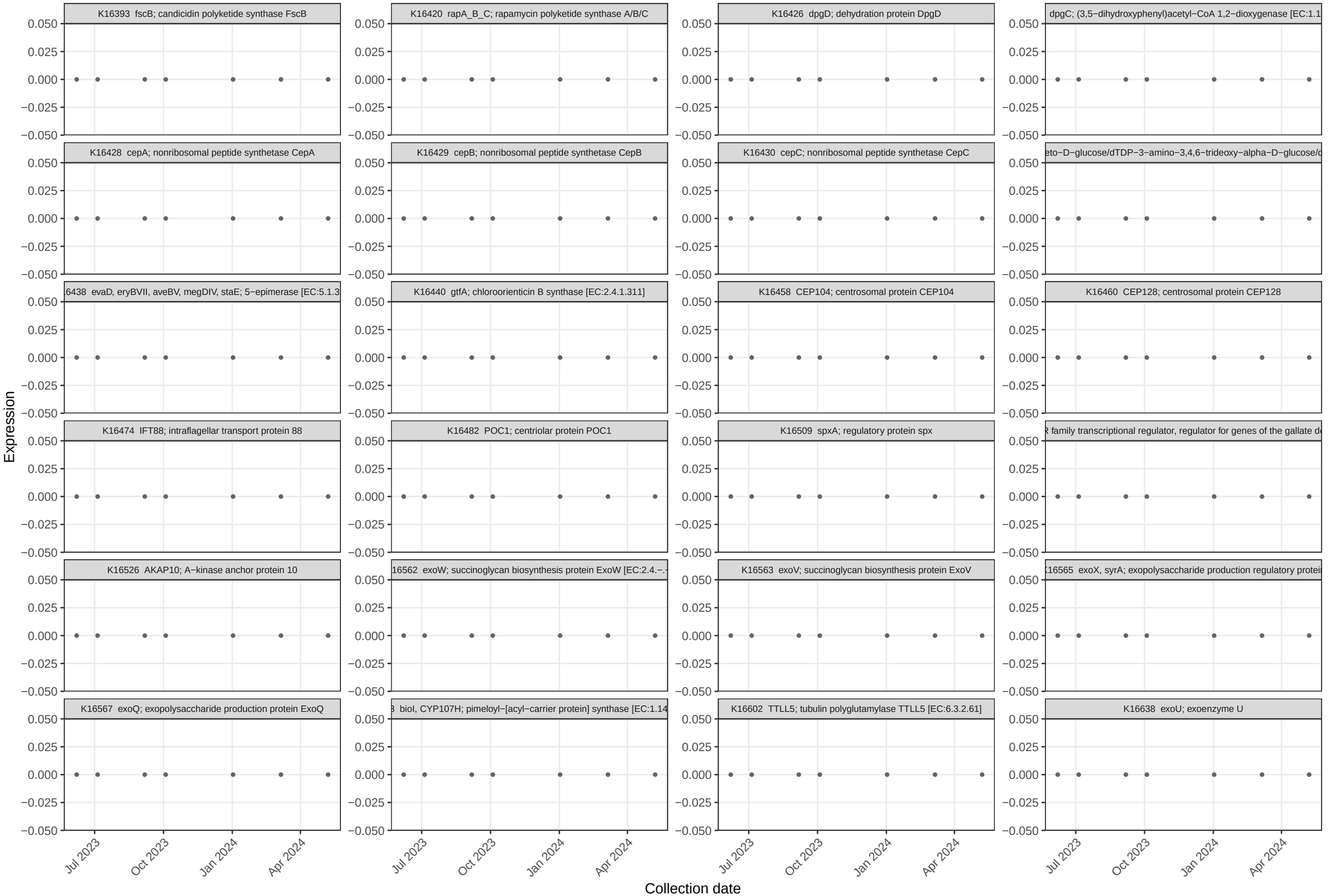


All remaining BRITE classes after exclusions



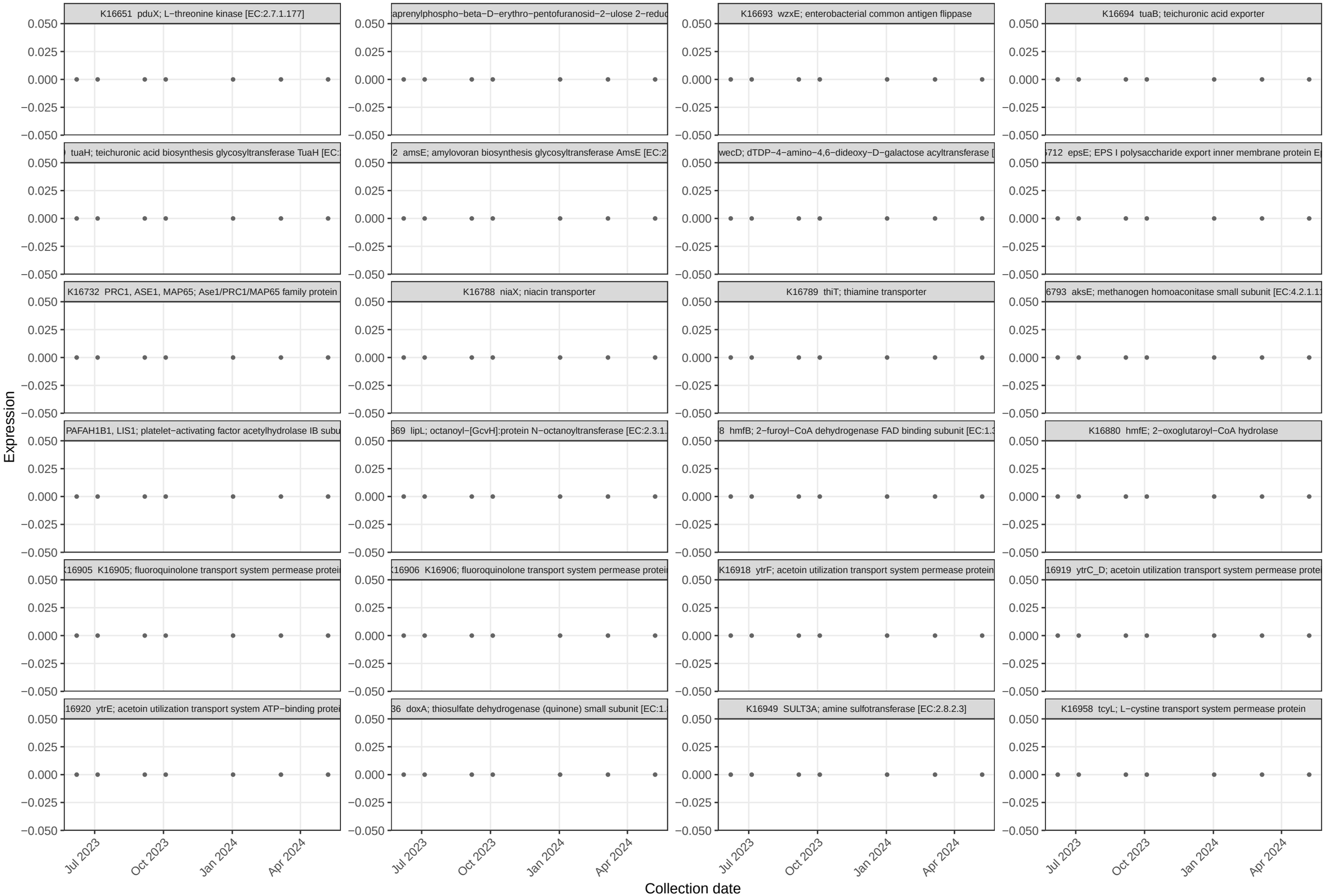
brite_D filtered temporal GAMs: page 274 of 303

All remaining BRITE classes after exclusions



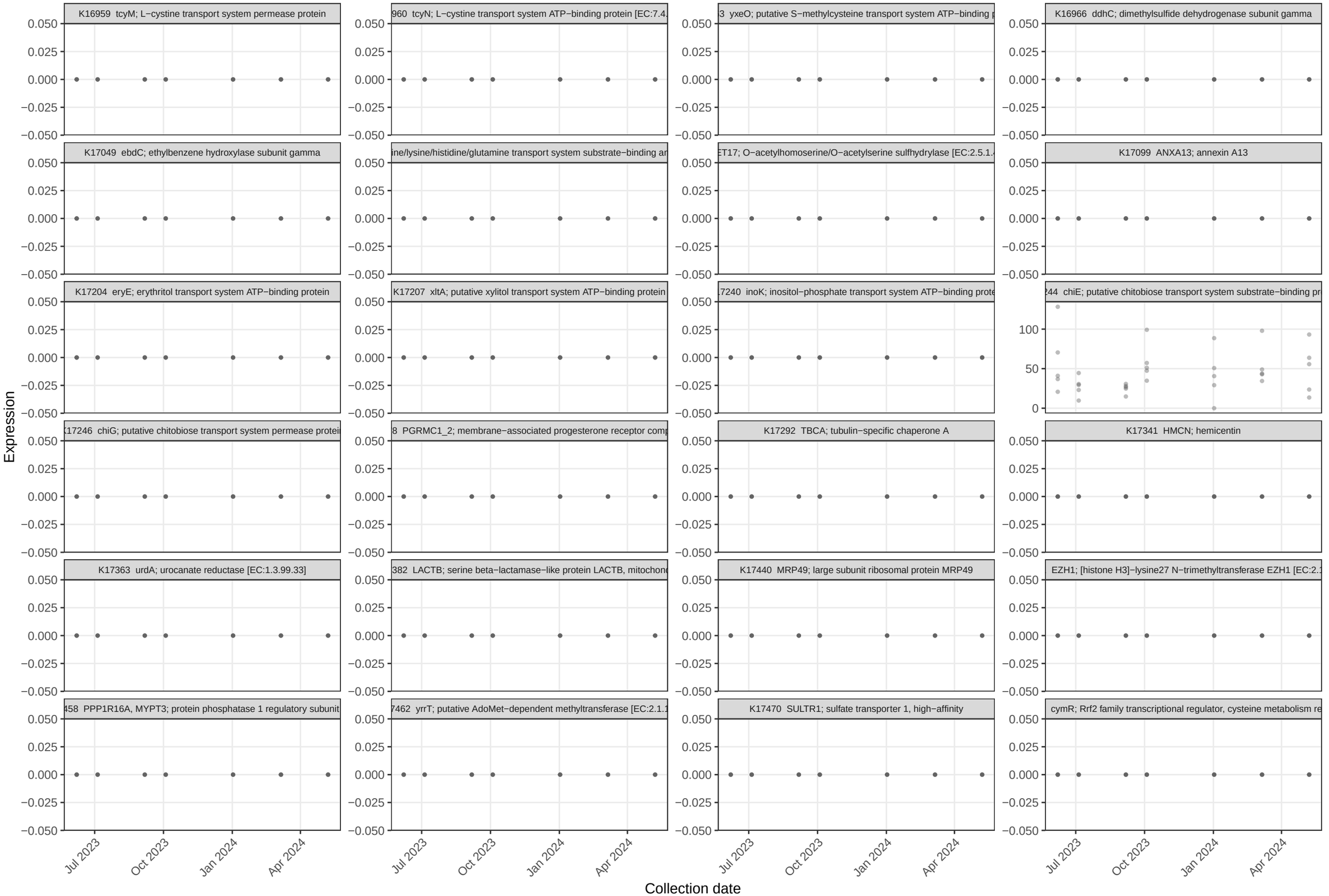
brite_D filtered temporal GAMs: page 275 of 303

All remaining BRITE classes after exclusions

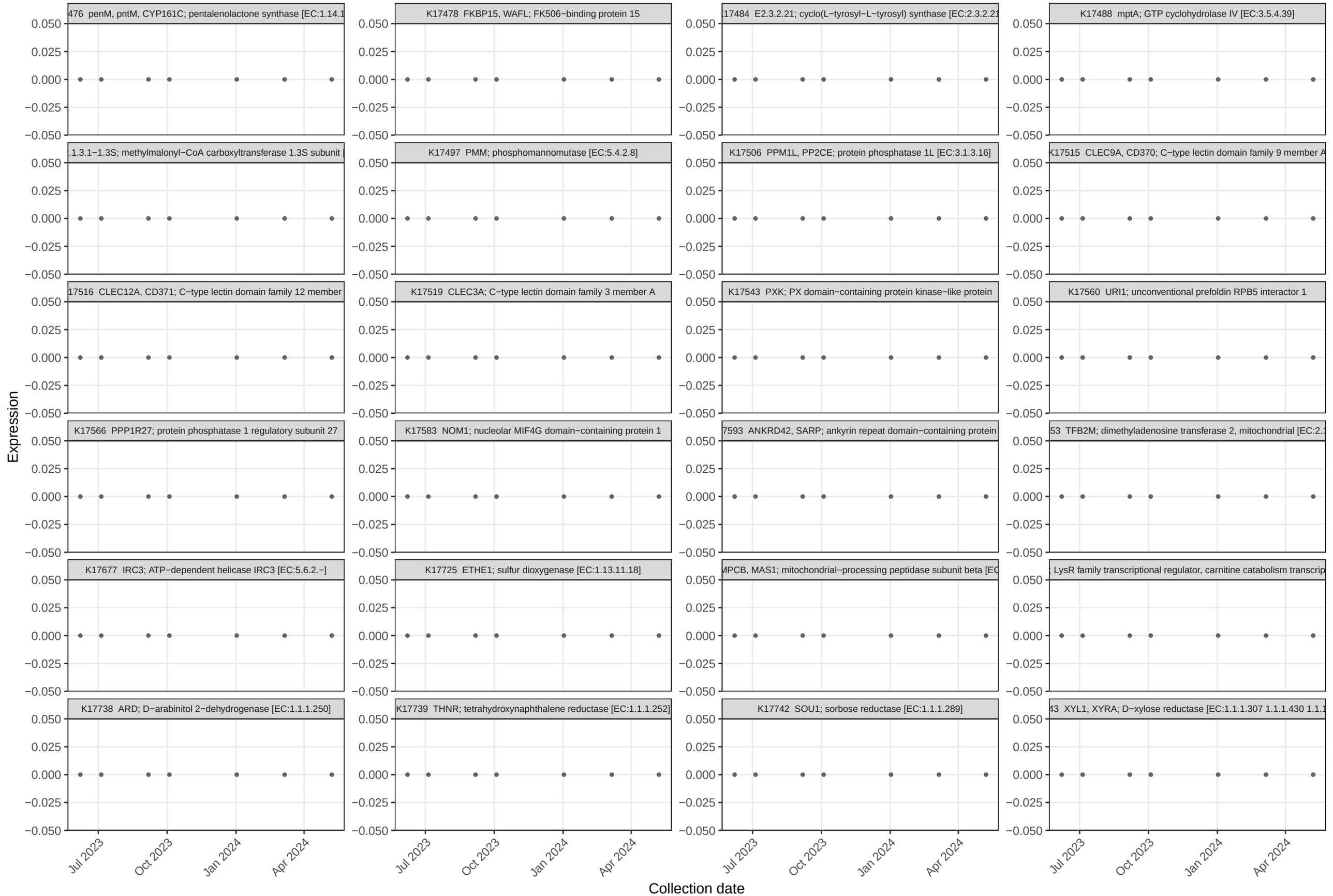


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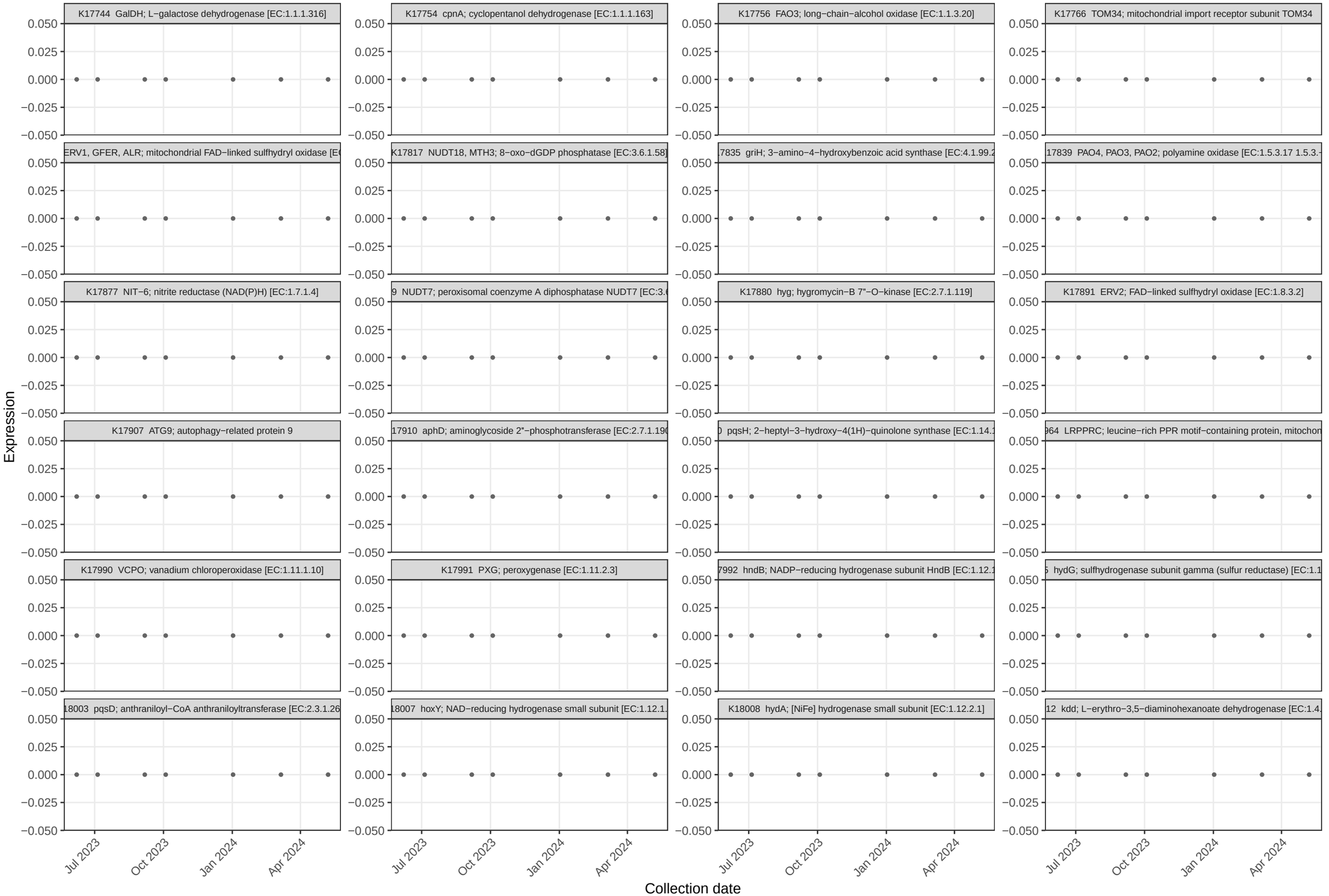
All remaining BRITE classes after exclusions



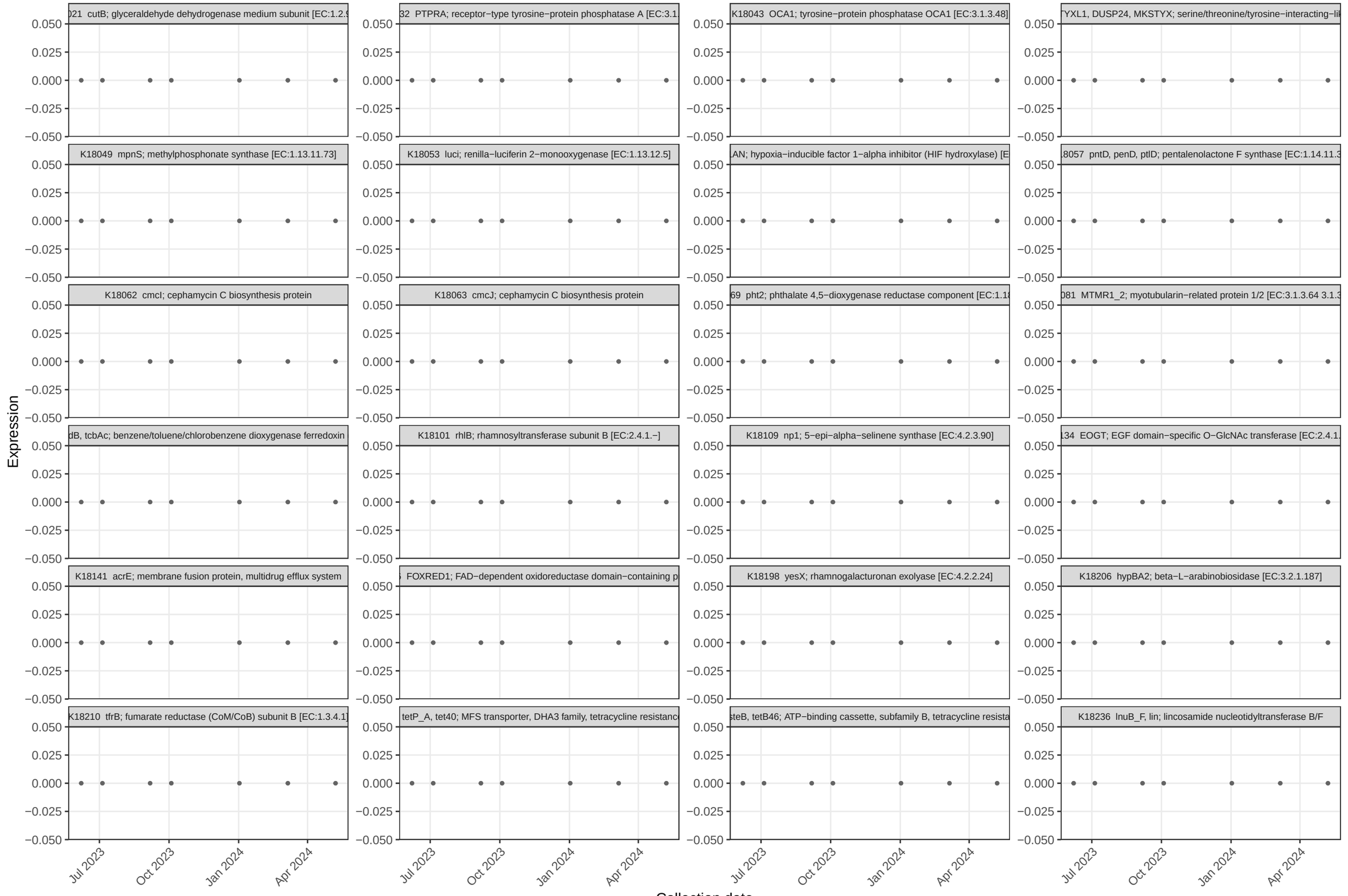
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

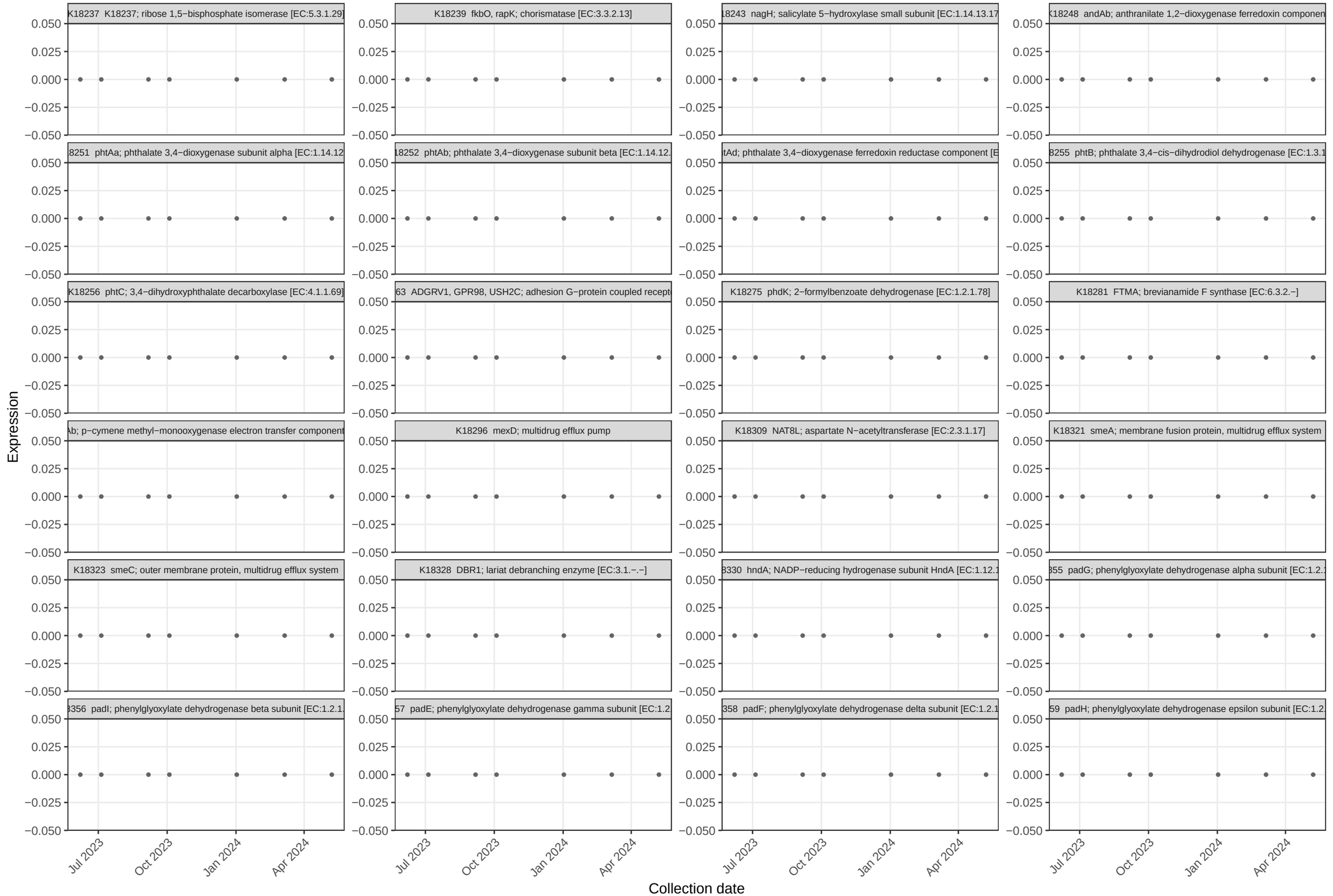


All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 280 of 303

All remaining BRITE classes after exclusions



brite_D filtered temporal GAMs: page 281 of 303

All remaining BRITE classes after exclusions

padB; phenylacetyl−CoA:acceptor oxidoreductase [EC:1.17.5.1 3.
362 padC; phenylacetyl−CoA:acceptor oxidoreductase 27−kDa sub
363 padD; phenylacetyl−CoA:acceptor oxidoreductase 26−kDa sub
K18377 hpa2; lysozyme−related protein Hpa2
K18399 BPHL; valacyclovir hydrolase [EC:3.1.−.−]
K18408 TDRD9; ATP−dependent RNA helicase TDRD9 [EC:5.6.2.5
GASZ; ankyrin repeat, SAM and basic leucine zipper domain−cont
K18416 THEX1, ERI1; 3'−5' exoribonuclease 1 [EC:3.1.−.−]
3453 NUDT23; ADP−ribose/FAD diphosphatase [EC:3.6.1.13 3.6.1.
K18471 ydjG; methylglyoxal reductase [EC:1.1.1.−]
4 fabM; trans−2−decenoyl−[acyl−carrier protein] isomerase [EC:5.3
tetR; TetR/AcrR family transcriptional regulator, tetracycline repress
K18479 yihS; sulfoquinovose isomerase [EC:5.3.1.31]
K18538 thnN; thienamycin biosynthesis protein ThnN
K18539 thnO; thienamycin biosynthesis protein ThnO
K18544 sprA; streptogrisin A [EC:3.4.21.80]
K18548 sprE; glutamyl endopeptidase II [EC:3.4.21.82]
50 ISN1; IMP and pyridine−specific 5'−nucleotidase [EC:3.1.3.99 3.
cmlR, cmx; MFS transporter, DHA1 family, chloramphenicol resistan
556 frdA; NADH−dependent fumarate reductase subunit A [EC:1.3.
K18560 frdE; NADH−dependent fumarate reductase subunit E
62 thnQ, cmmQ; 2−oxoglutarate−dependent dioxygenase [EC:1.14
hnL; 2−(S−pantetheinyl)−carbapenam−3−carboxylate synthase [EC
K18566 thnF, cmmF; N−acetyltransferase [EC:2.3.1.−]

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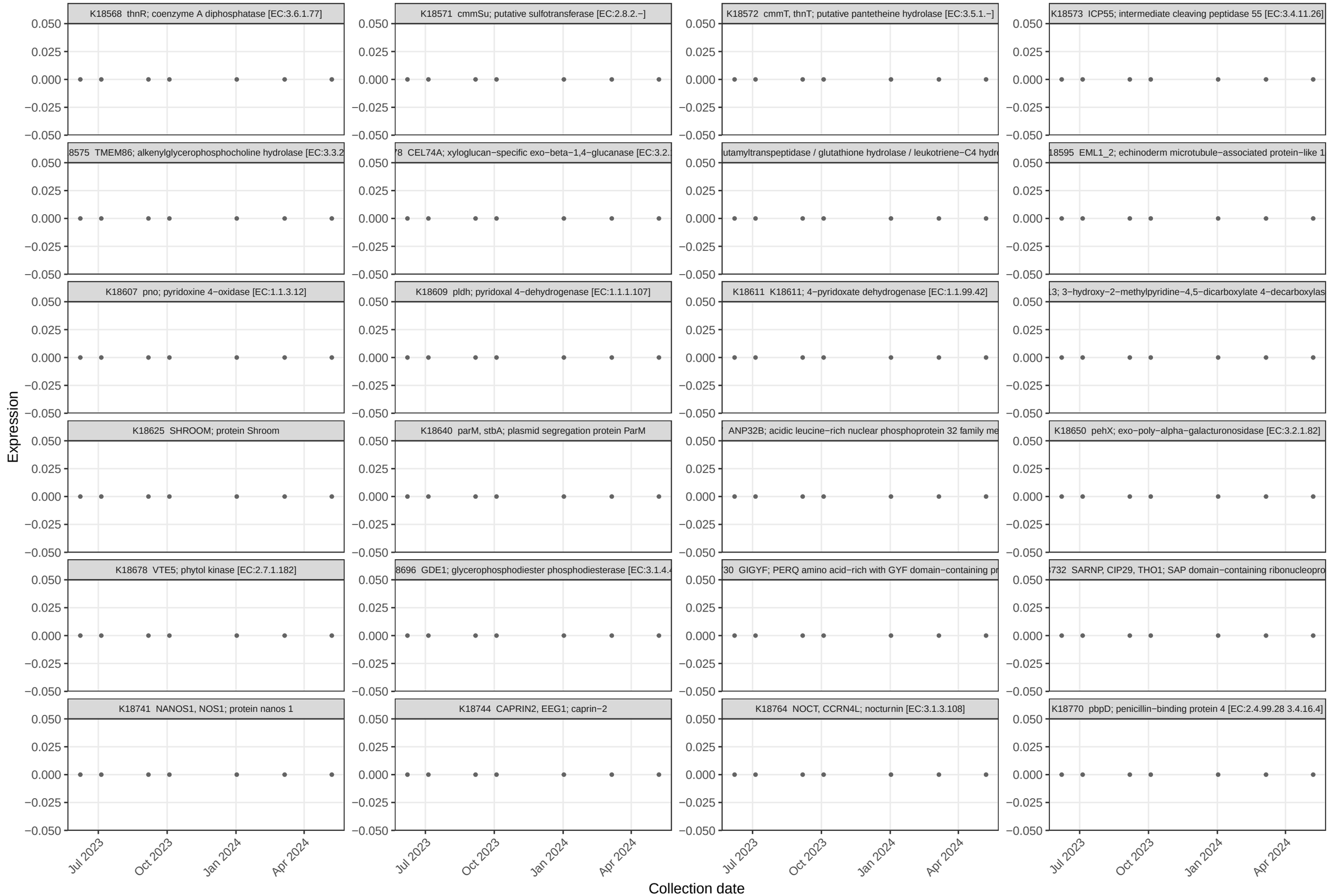
Jul 2023
Oct 2023
Jan 2024
Apr 2024

Expression

Collection date

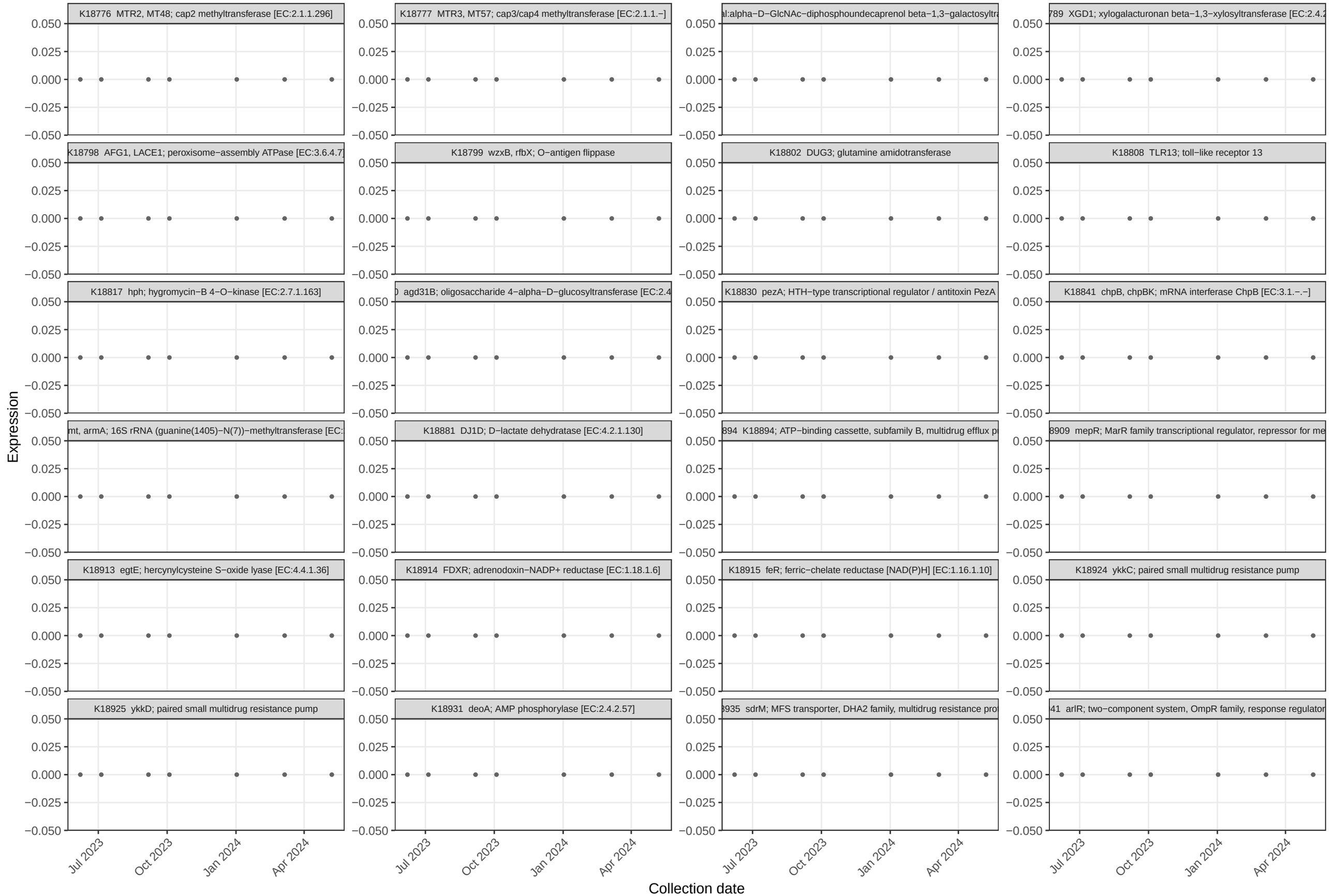
brite_D filtered temporal GAMs: page 282 of 303

All remaining BRITE classes after exclusions

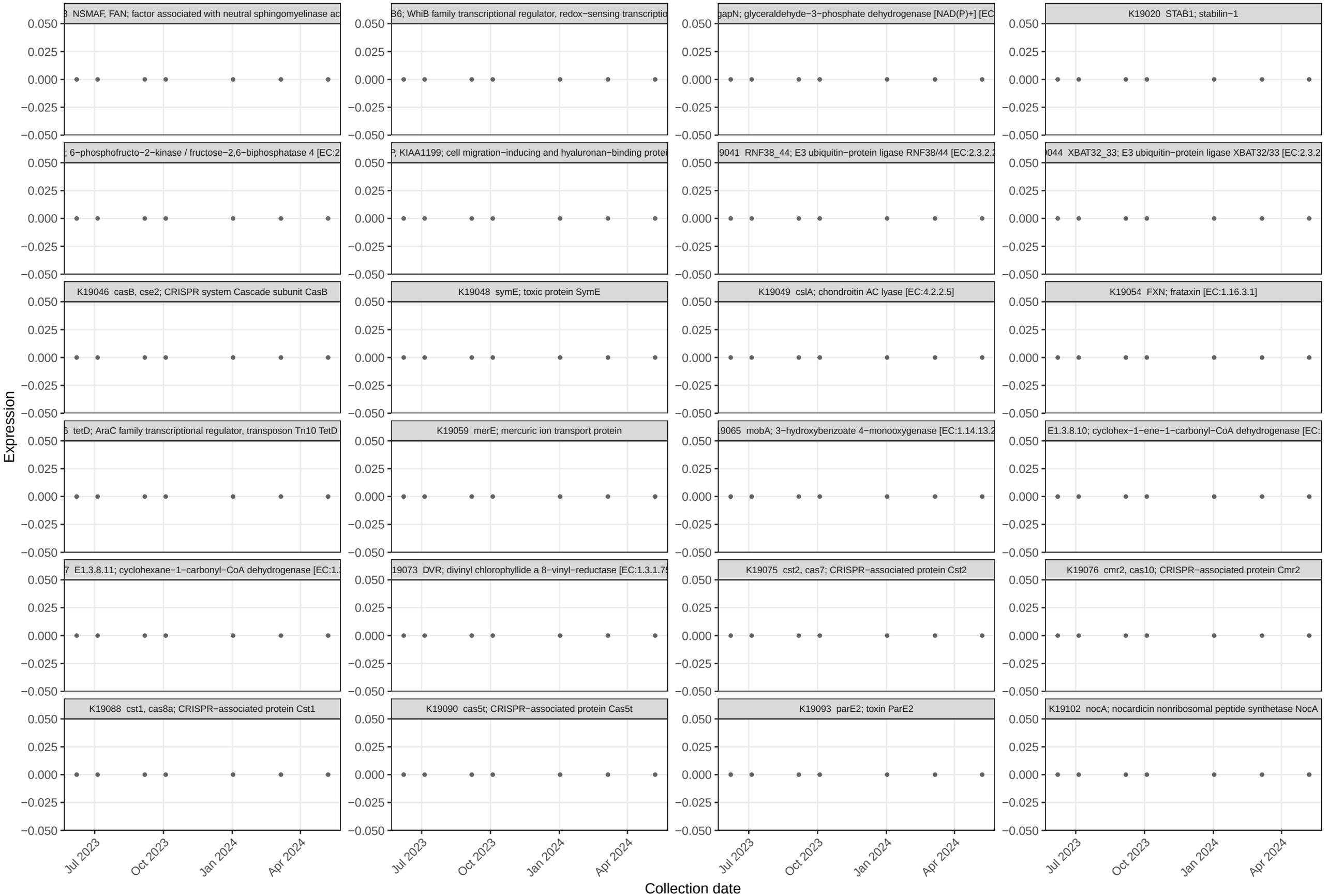


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All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions

K19103 nocB; nocardicin nonribosomal peptide synthetase NocB
K19105 nocJ; nocardicin−A epimerase [EC:5.1.1.14]
K19106 nocL; nocardicin C N−oxygenase [EC:1.14.15.43]
K19117 csd1, cas8c; CRISPR−associated protein Csd1
K19118 csd2, cas7; CRISPR−associated protein Csd2
K19119 cas5d; CRISPR−associated protein Cas5d
K19123 casA, cse1; CRISPR system Cascade subunit CasA
K19126 casE, cse3; CRISPR system Cascade subunit CasE
130 csy4, cas6f; CRISPR−associated endonuclease Csy4 [EC:3.1.
K19132 csb2; CRISPR−associated protein Csb2
K19134 csx10; CRISPR−associated protein Csx10
K19138 csm2; CRISPR−associated protein Csm2
K19139 csm4; CRISPR−associated protein Csm4
K19140 csm5; CRISPR−associated protein Csm5
K19141 cmr5; CRISPR−associated protein Cmr5
K19142 cmr6; CRISPR−associated protein Cmr6
K19144 csx3; CRISPR−associated protein Csx3
K19145 csx16; CRISPR−associated protein Csx16
K19155 yhaV; toxin YhaV [EC:3.1.−.−]
K19157 yafQ; mRNA interferase YafQ [EC:3.1.−.−]
K19176 FAAH2; fatty acid amide hydrolase 2 [EC:3.5.1.99]
hA; 6−hydroxypseudooxynicotine dehydrogenase subunit alpha [EC
dhB; 6−hydroxypseudooxynicotine dehydrogenase subunit beta [EC
L; 6−hydroxypseudooxynicotine dehydrogenase subunit gamma [E

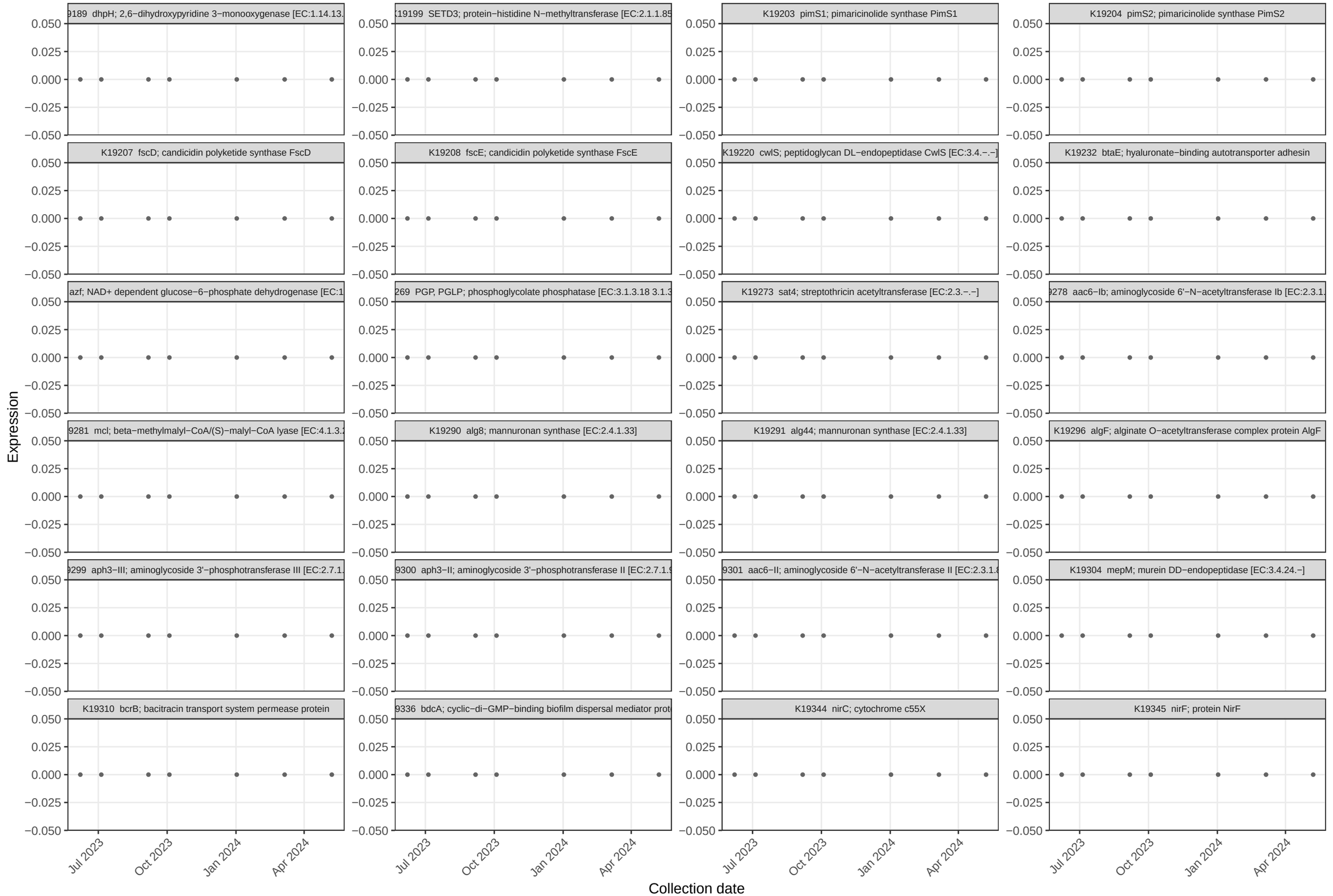
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Expression

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Apr 2024

Collection date

All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

K19370 DNAJC22; DnaJ homolog subfamily C member 22
K19371 DNAJC25; DnaJ homolog subfamily C member 25
K19373 DNAJC28; DnaJ homolog subfamily C member 28
12 argX; glutamate−−−[amino group carrier protein] ligase [EC:6.3.
KDM1B, AOF1, LSD2; lysine−specific histone demethylase 1B [EC
417 slrR; XRE family transcriptional regulator, biofilm formation regu
K19420 epsA, capA; protein tyrosine kinase modulator
K19422 epsD; glycosyltransferase EpsD [EC:2.4.−.−]
GME1, DDK1; mitochondrial genome maintenance exonuclease 1 [
sigma−54 dependent transcriptional regulator, gfr operon transcript
gfrB; fructoselysine/glucoselysine PTS system EIIB component [EC
K19510 gfrF; fructoselysine−6−phosphate deglycase
K19517 MIK; 1D−myo−inositol 3−kinase [EC:2.7.1.64]
9525 VPS13A_C; vacuolar protein sorting−associated protein 13A
K19527 VPS13D; vacuolar protein sorting−associated protein 13D
9532 LBR; Delta14−sterol reductase (lamin−B receptor) [EC:1.3.1.
1 bioJ; pimeloyl−[acyl−carrier protein] methyl ester esterase [EC:3.
K19572 CECR1, ADA2; adenosine deaminase CECR1 [EC:3.5.4.4
9576 norA; MFS transporter, DHA1 family, quinolone resistance pro
K19588 dopDH; 2,5−dioxopentanoate dehydrogenase [EC:1.2.1.26
K19599 DOME; cytokine receptor domeless
K19606 CRTAP; cartilage−associated protein
component system, sensor histidine kinase and response regulator
18 fitG; LysR family transcriptional regulator, activator of insecticida

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All remaining BRITE classes after exclusions

phcS; two−component system, sensor histidine kinase PhcS [EC:2

K19636 USH2A; usherin

K19653 geoA; geraniol dehydrogenase (NAD+) [EC:1.1.1.347]

K19659 phaJ; enoyl−CoA hydratase [EC:4.2.1.119]

; two−component system, NtrC family, sensor histidine kinase HupT

CSC; osomolarity two−component system, sensor histidine kinase N

00 sap; 3−succinoylsemialdehyde−pyridine dehydrogenase [EC:1.2

9711 E1.1.1.387; L−serine 3−dehydrogenase (NAD+) [EC:1.1.1.38

2 cobY; adenosylcobinamide−phosphate guanylyltransferase [EC:2.

peridеine−2−carboxylate/1−pyrroline−2−carboxylate reductase [NAD

K19749 RAI1; retinoic acid−induced protein 1

K19755 RSPH1; radial spoke head protein 1

19761 GGACT; gamma−glutamylaminecyclotransferase [EC:2.3.2.

K19777 hdeA; acid stress chaperone HdeA

3 HCS1; DNA polymerase alpha−associated DNA helicase A [EC:5

9785 POF1; nicotinamide−nucleotide adenylyltransferase [EC:2.7.7

K19788 OLA1; obg−like ATPase 1

mfnC; (5−formylfuran−3−yl)methyl phosphate transaminase [EC:2.

K19795 E3.5.1.95; ureidomalonase [EC:3.5.1.95]

K19803 lpxT; Kdo2−lipid A phosphotransferase [EC:2.7.4.29]

19811 hsdA; 3alpha−hydroxysteroid 3−dehydrogenase [EC:1.1.1.35

K19818 ndhA; nicotine dehydrogenase subunit A [EC:1.5.99.4]

K19819 ndhB; nicotine dehydrogenase subunit B [EC:1.5.99.4]

K19826 nctB, 6−hlno; (S)−6−hydroxynicotine oxidase [EC:1.5.3.5]

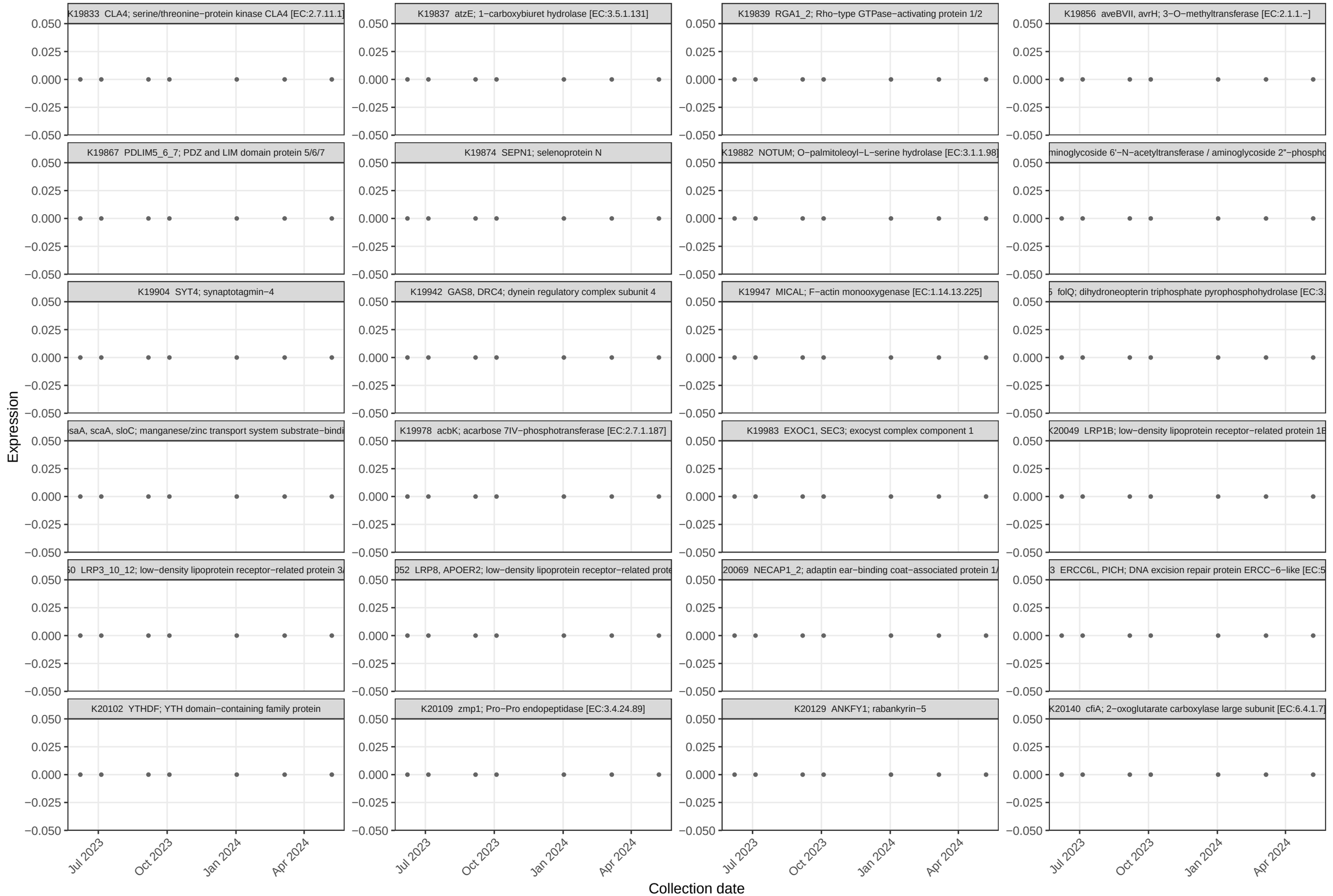
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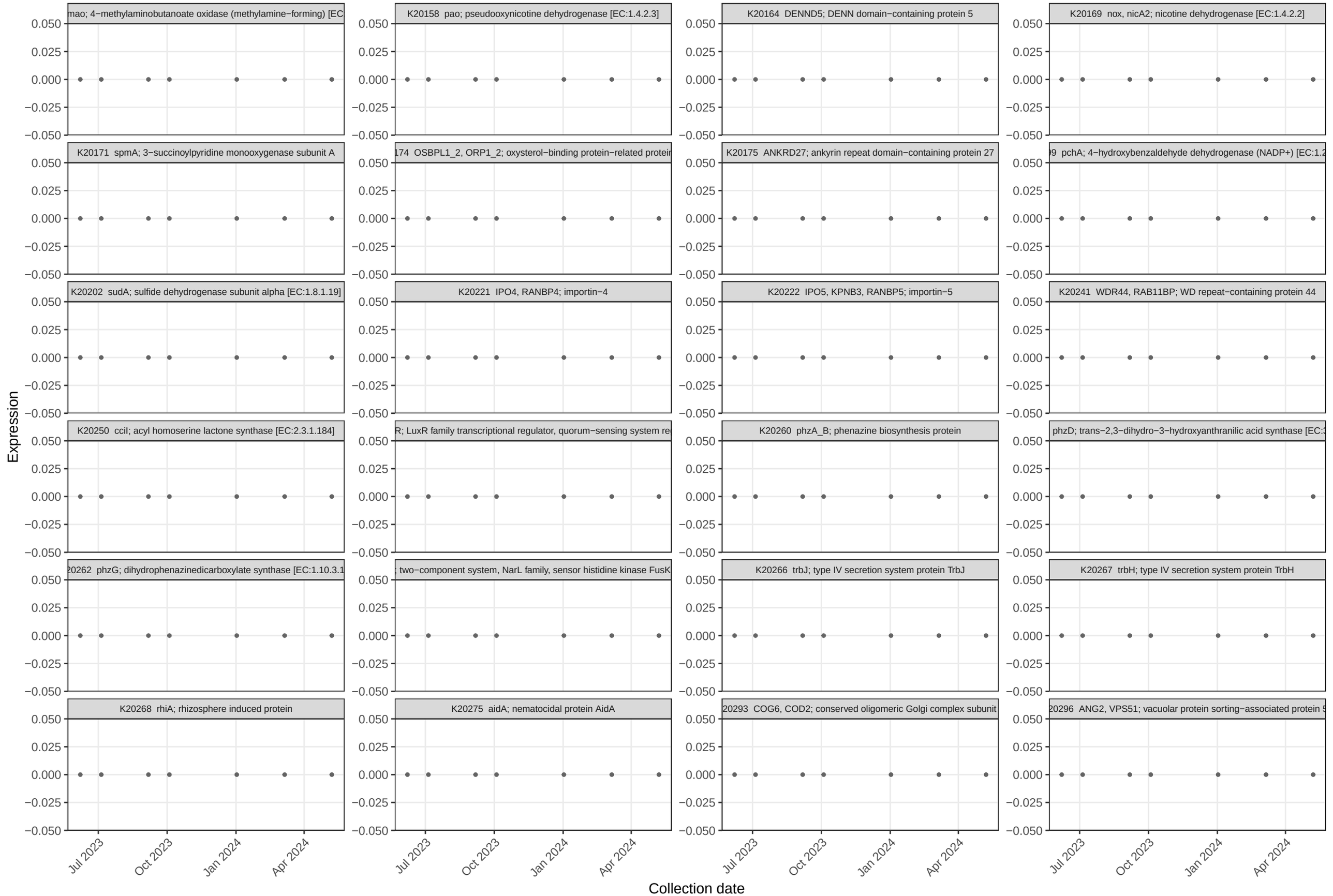
Jul 2023
Oct 2023
Jan 2024
Apr 2024

Collection date

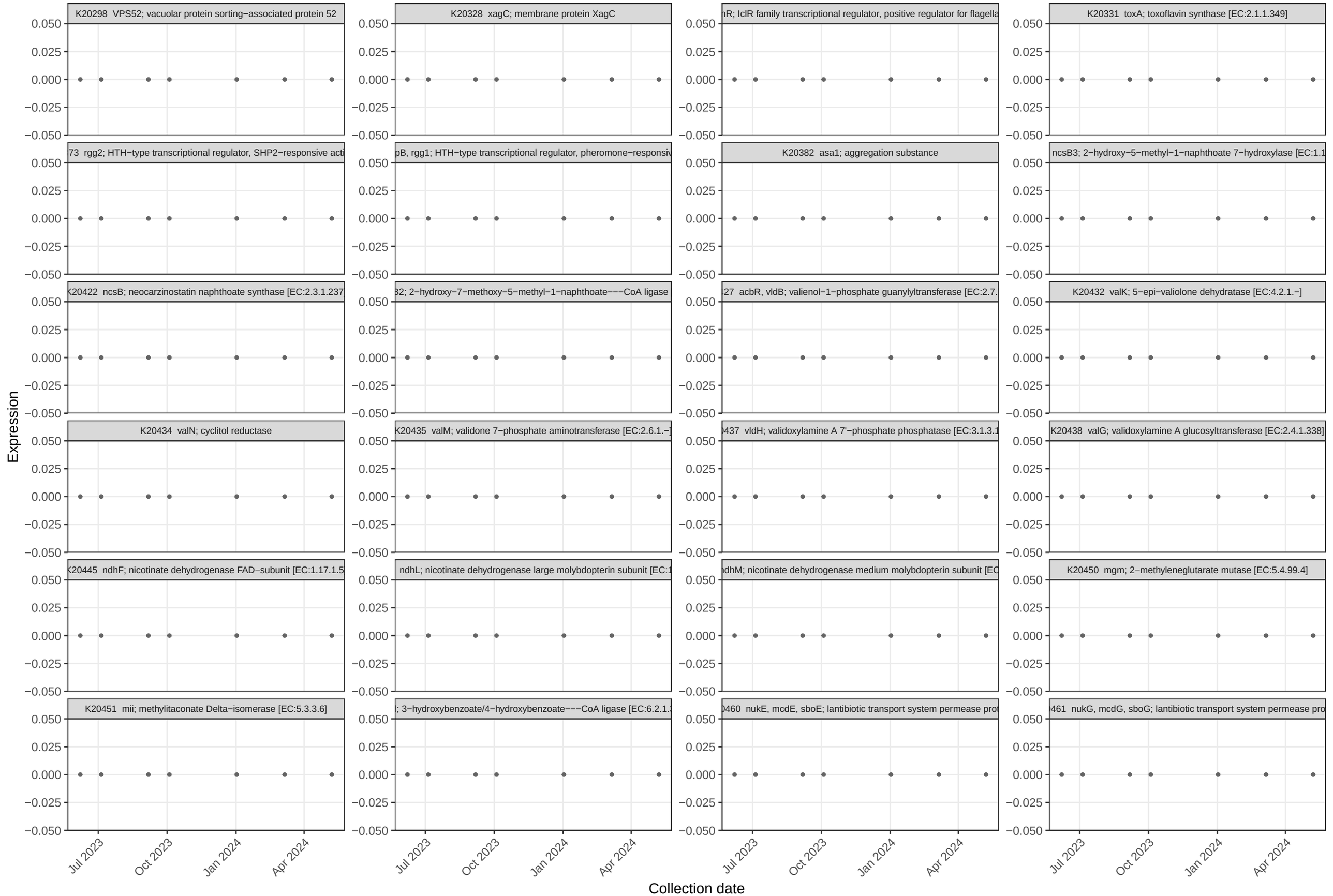
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

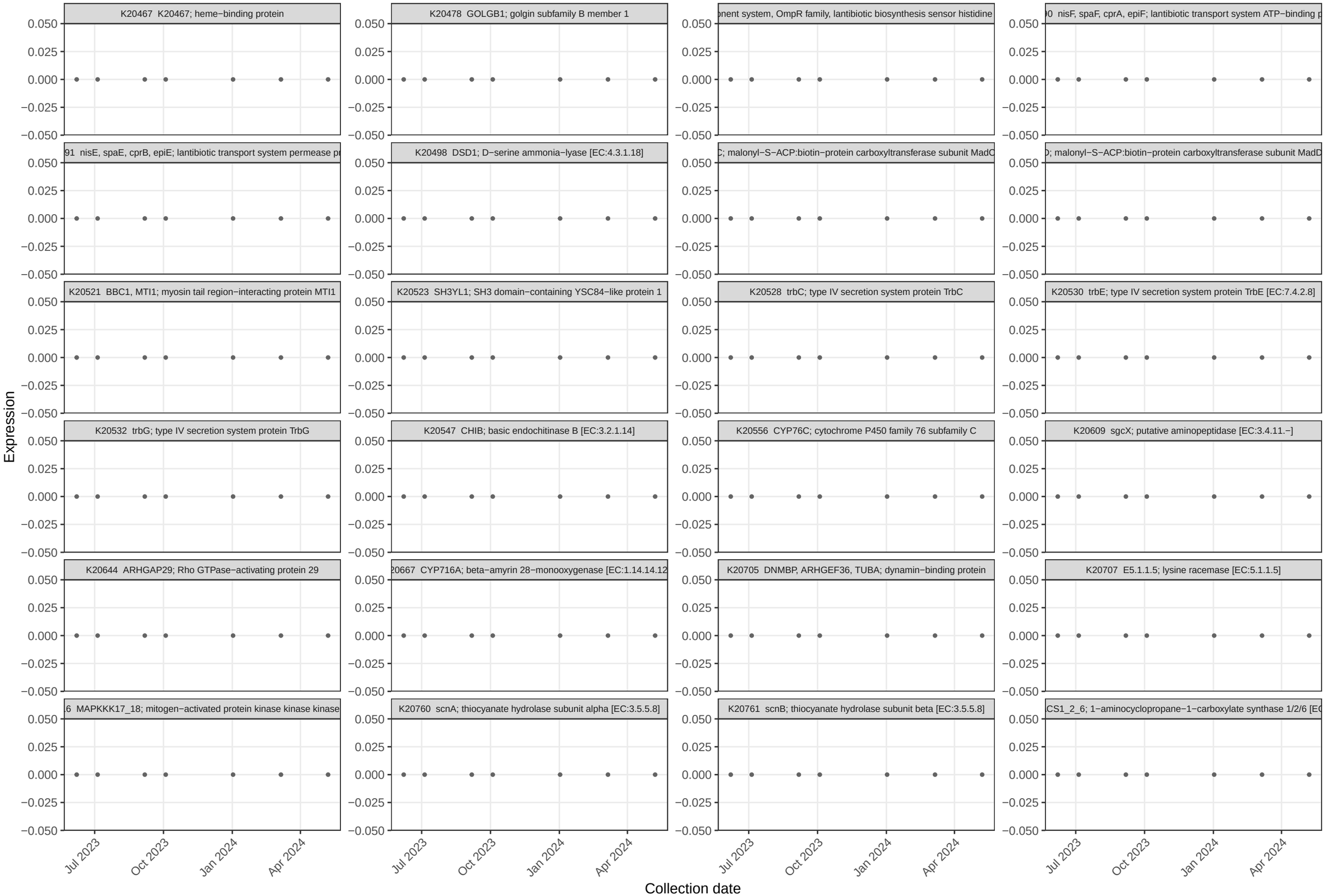


All remaining BRITE classes after exclusions

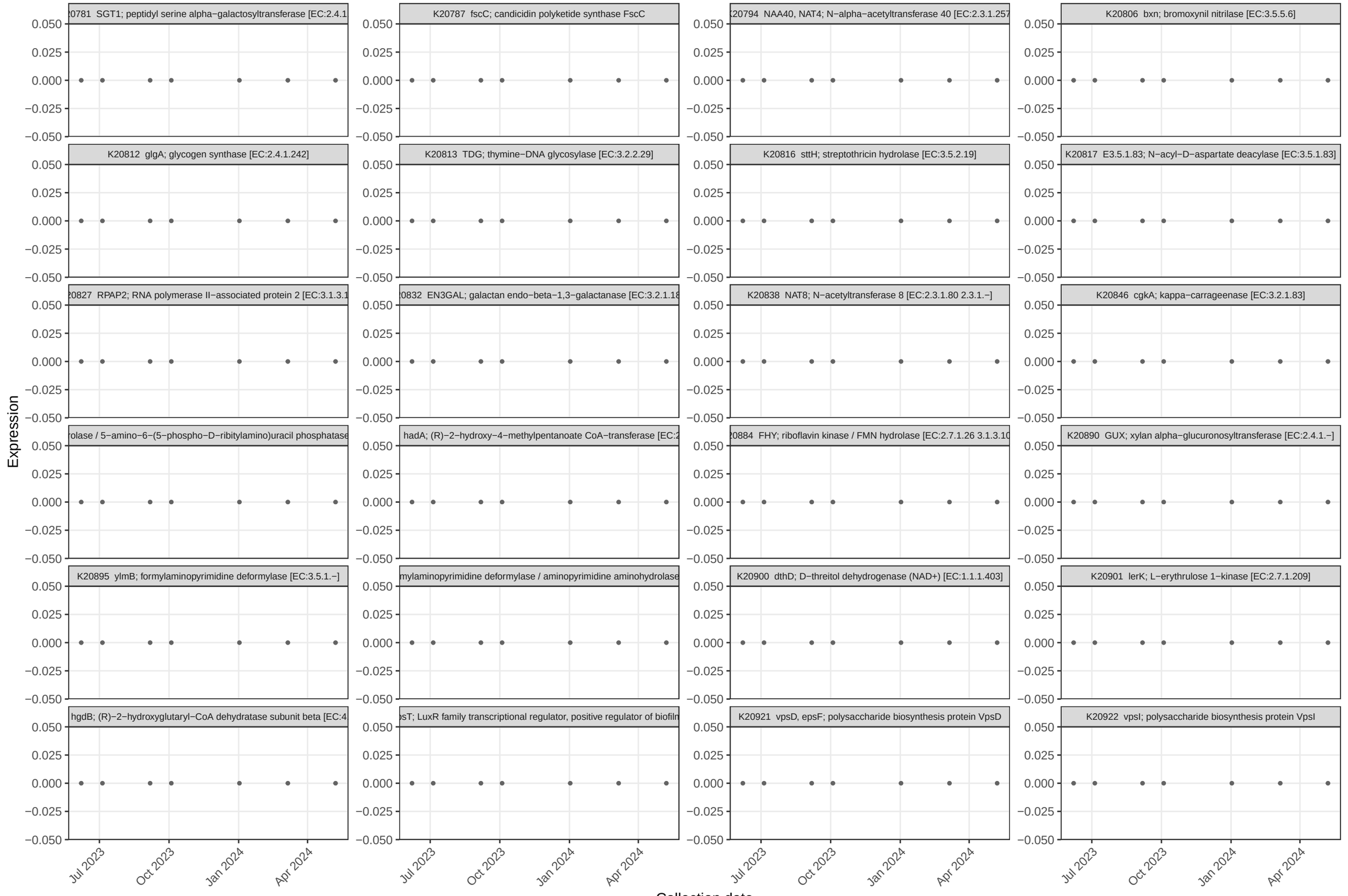


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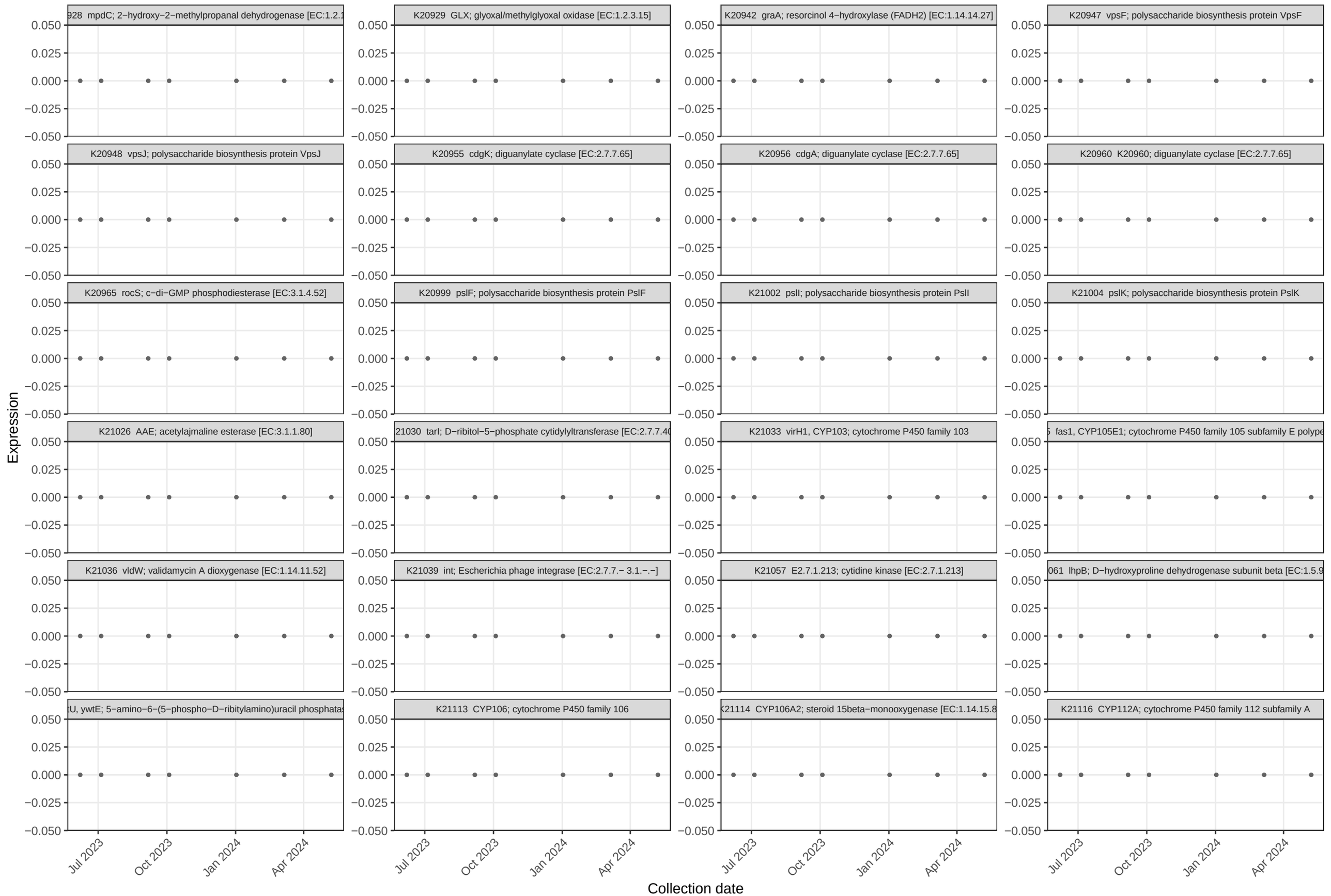
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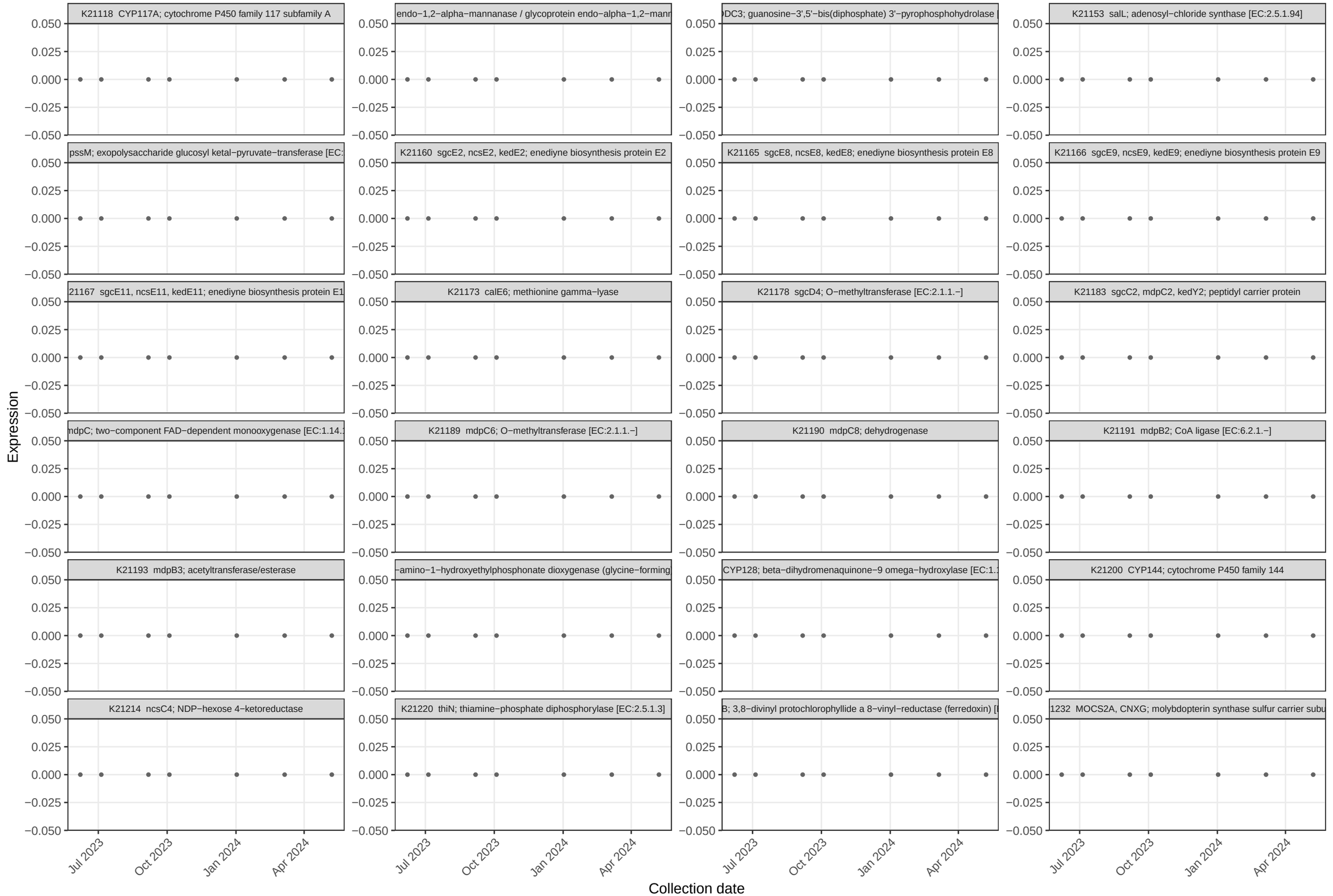
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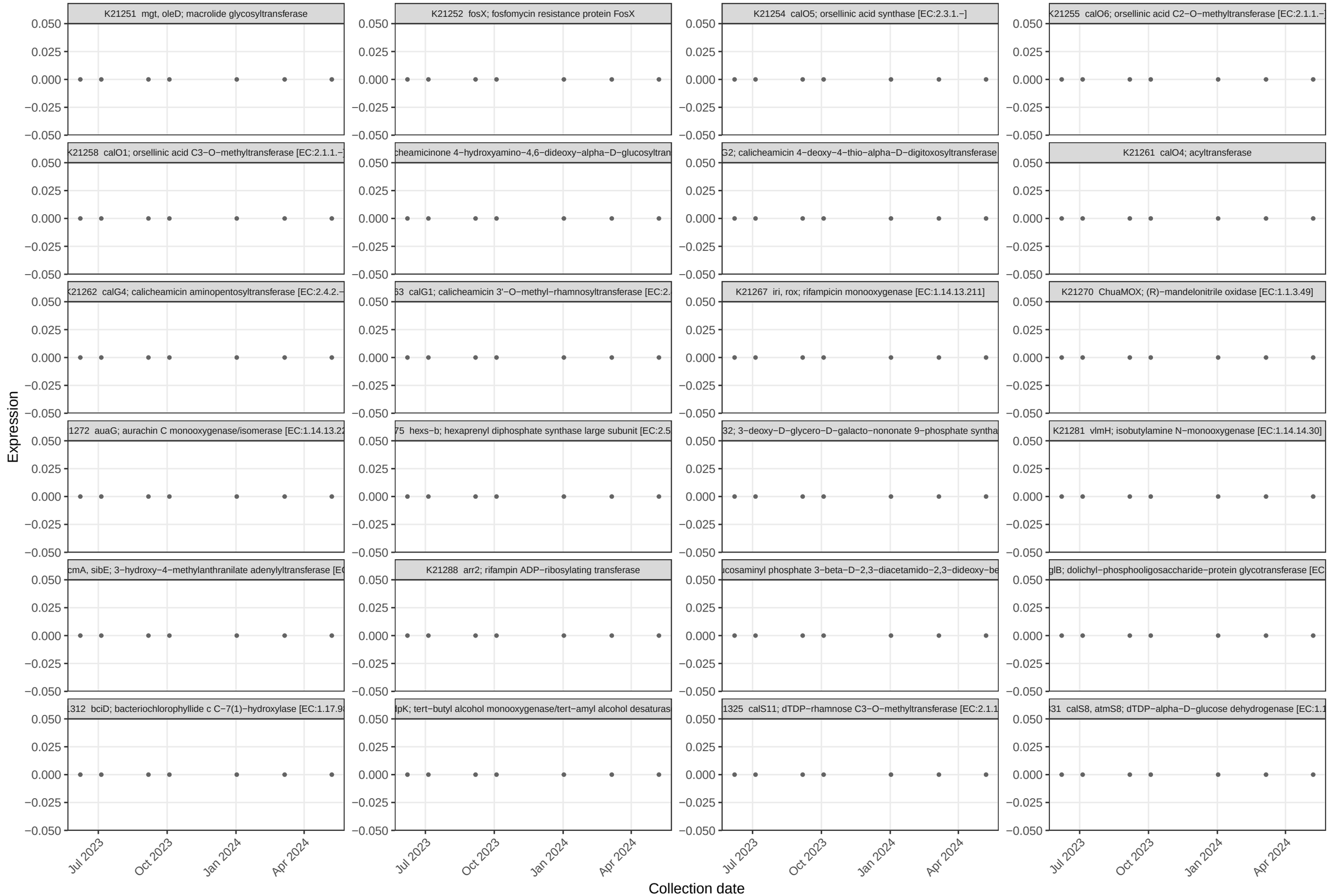
All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

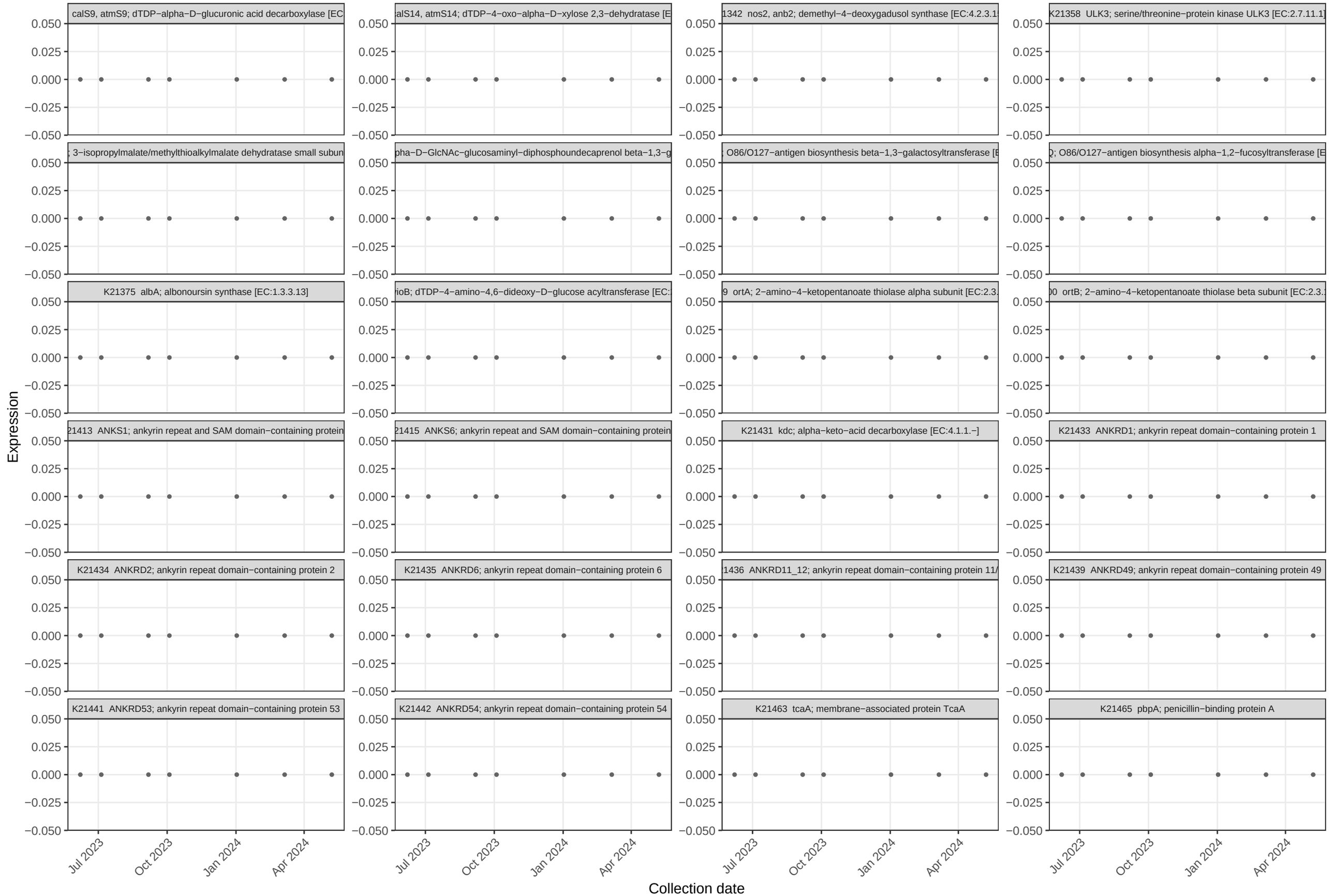


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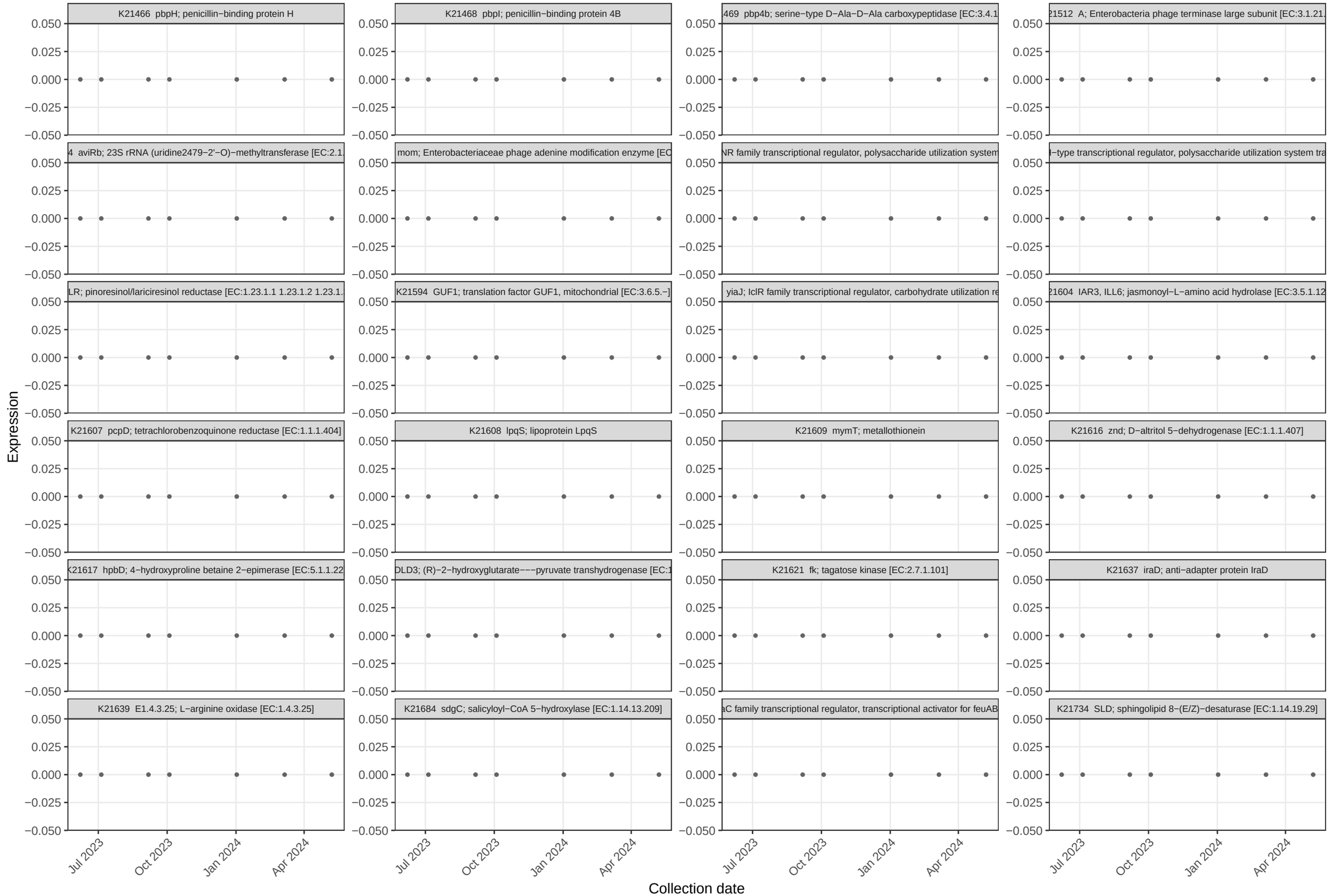


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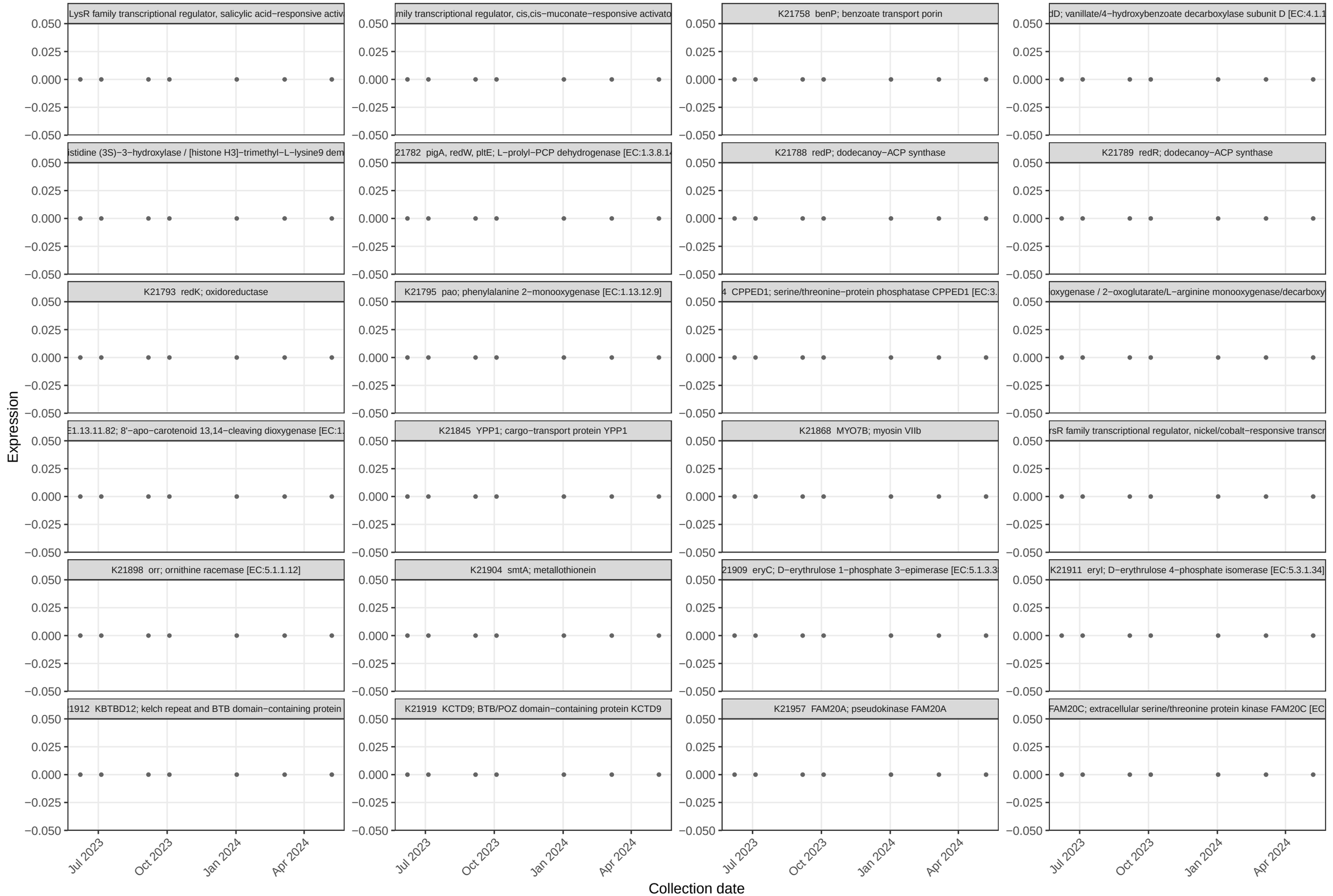
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All remaining BRITE classes after exclusions

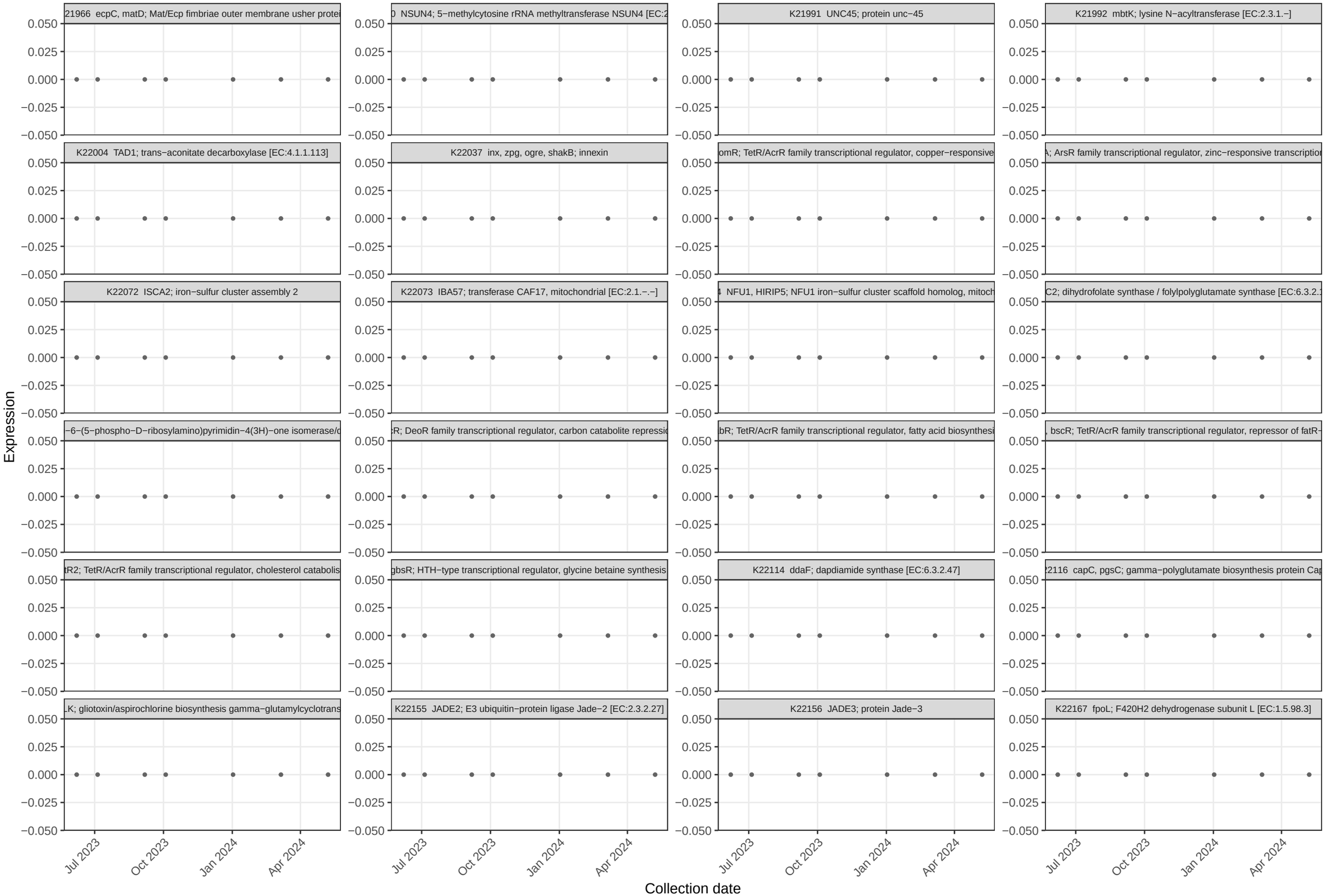


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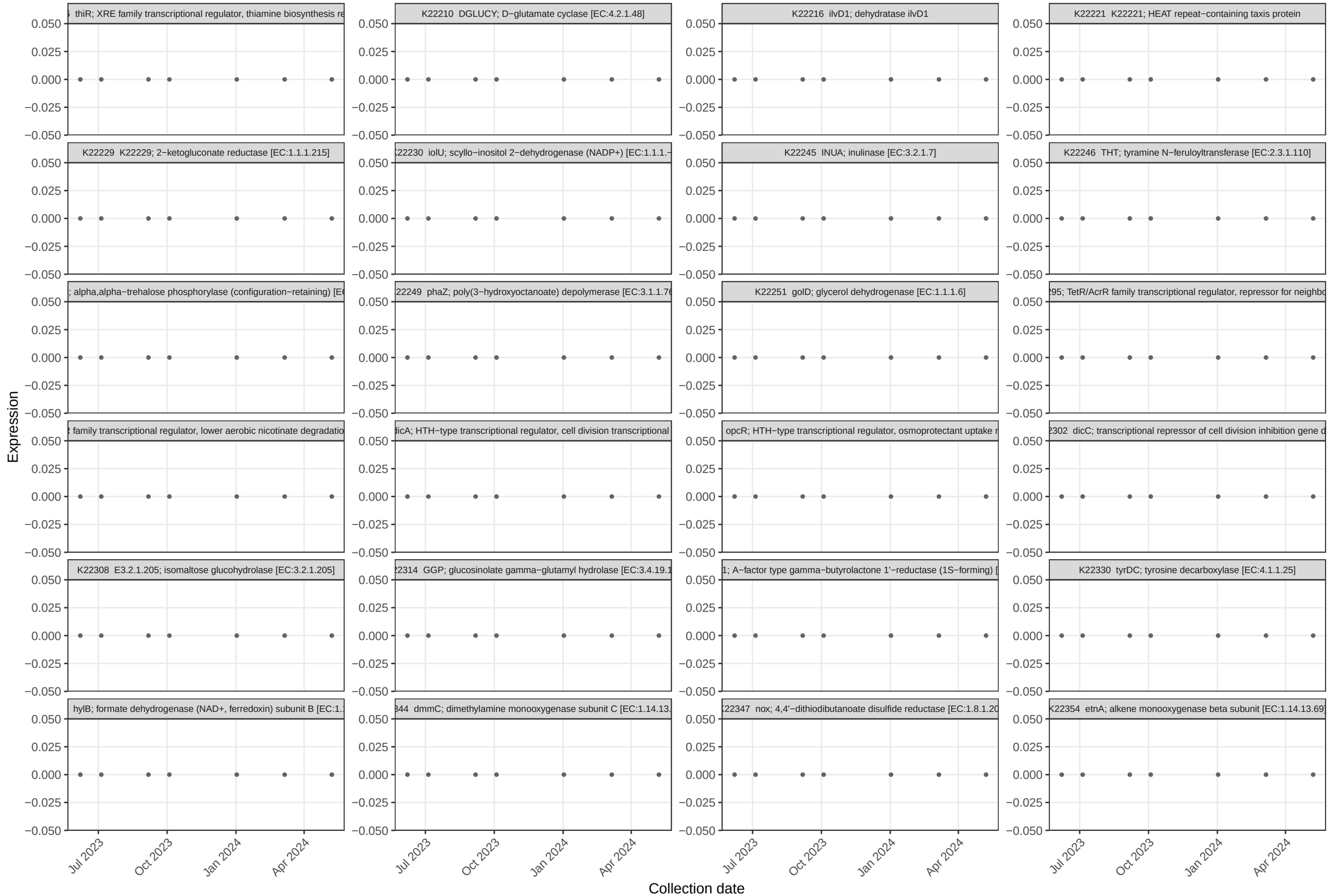
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All remaining BRITE classes after exclusions



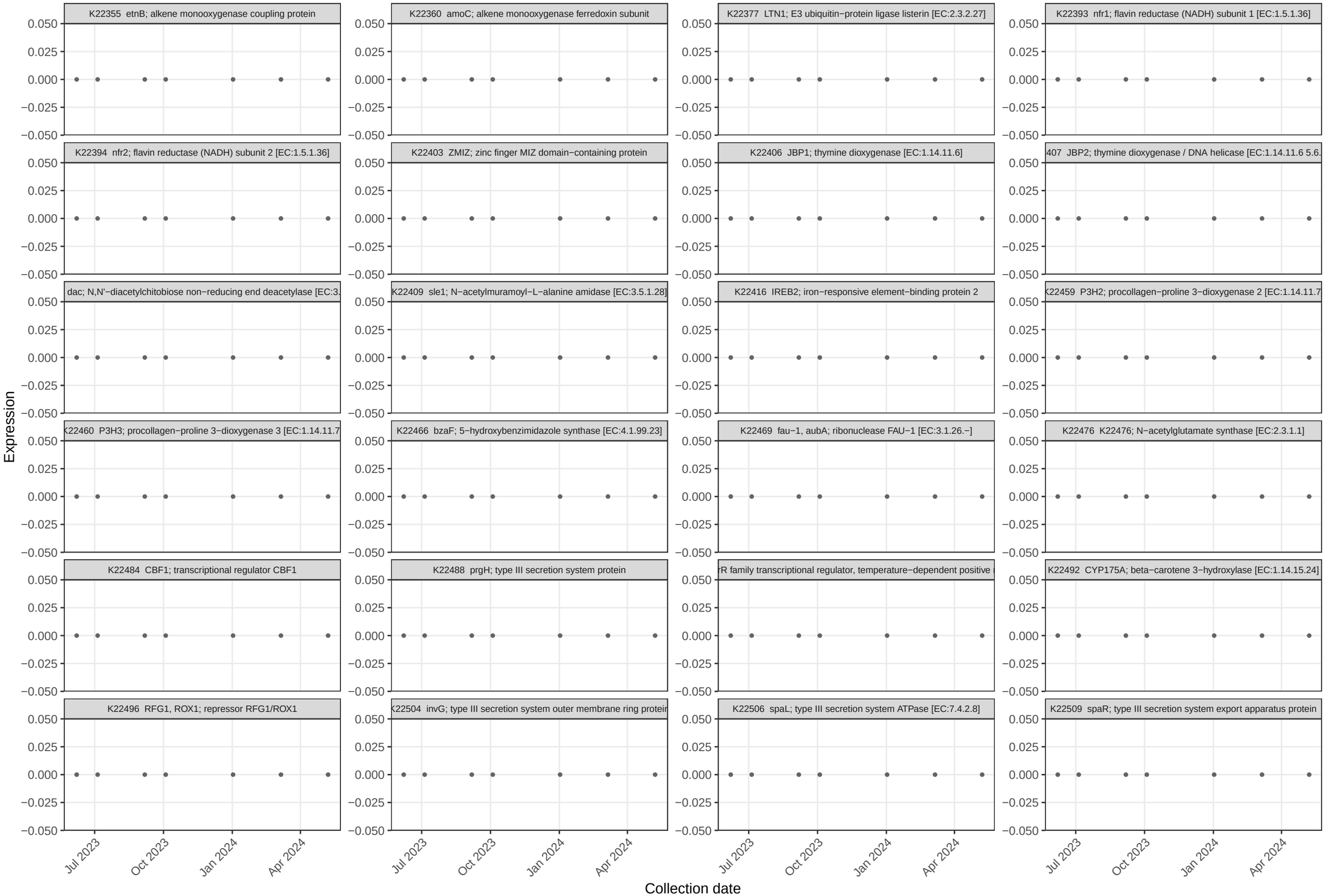
brite_D filtered temporal GAMs: page 301 of 303

All remaining BRITE classes after exclusions



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All remaining BRITE classes after exclusions



All remaining BRITE classes after exclusions

